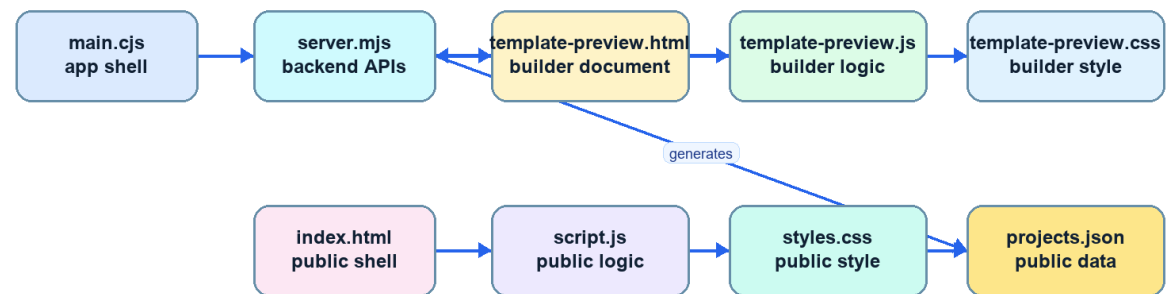


OMB Portfolio Builder Important Code Reference

A generated PDF reference for the code and control files that drive the builder, website, installer, Cloudflare AI worker, workflows, parser, compile workspace, and generated documentation.

How Key Files Communicate



Coverage summary

Item	Explanation
Important files documented	19
Selection rule	Only behavior-driving code/control files are included. Favicons, binary assets, and passive marker files are intentionally skipped here.
Categories	Configuration or structured data: 4, GitHub workflow: 2, JavaScript source: 7, NSIS installer script: 1, Python tooling: 1, Website/builder UI source: 4

File index

Item	Explanation
main.cjs	JavaScript source. Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
server.mjs	JavaScript source. Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
template-preview.js	JavaScript source. Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
template-preview.html	Website/builder UI source. Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
template-preview.css	Website/builder UI source. Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
script.js	JavaScript source. Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
index.html	Website/builder UI source. Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
styles.css	Website/builder UI source. Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.

Item	Explanation
cloudflare/portfolio-ai-worker.js	JavaScript source. Cloudflare Worker/deployment area. Used for public AI backend and Cloudflare deployment notes.
cloudflare/wrangler.toml	Configuration or structured data. Cloudflare Worker/deployment area. Used for public AI backend and Cloudflare deployment notes.
build/installer.nsh	NSIS installer script. Packaging and installer customization. Used while creating the Windows app installer.
docs/build_complete_guide.py	Python tooling. Documentation and generated reference area. Used to explain the app and source structure.
builder-rich-future-sections.js	JavaScript source. Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
electronics-search.js	JavaScript source. Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
package.json	Configuration or structured data. Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
project-templates.json	Configuration or structured data. Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
projects.json	Configuration or structured data. Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
.github/workflows/build-windows-builder.yml	GitHub workflow. GitHub automation. GitHub Actions reads this during CI/release/branch-gate runs.
.github/workflows/main-branch-gate.yml	GitHub workflow. GitHub automation. GitHub Actions reads this during CI/release/branch-gate runs.

Important Code Reference: main.cjs

Item	Explanation
File	main.cjs
Category	JavaScript source
Folder role	Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
Size	15,134 bytes
Extension	.cjs
MIME type	text/plain
Text lines	453

Why this file exists

- Electron main process: creates the desktop window, starts the local backend, handles menus, update handoff, and application lifecycle.

How it is used

Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.

How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

Imports, requires, and external dependencies

Item	Explanation
Line 3	const fs = require("node:fs");
Line 4	const fsp = require("node:fs/promises");
Line 5	const net = require("node:net");
Line 6	const path = require("node:path");

Important implementation signals

Item	Explanation
Process execution at line 2	Runs Git, compilers, installers, or helper tools outside the JavaScript runtime. Evidence: const { execFile } = require("node:child_process");
Compiler or simulator at line 3	Part of the Compile Code workspace or language/tool detection. Evidence: const fs = require("node:fs");
Compiler or simulator at line 4	Part of the Compile Code workspace or language/tool detection. Evidence: const fsp = require("node:fs/promises");
Compiler or simulator at line 5	Part of the Compile Code workspace or language/tool detection. Evidence: const net = require("node:net");
Compiler or simulator at line 6	Part of the Compile Code workspace or language/tool detection. Evidence: const path = require("node:path");
Compiler or simulator at line 7	Part of the Compile Code workspace or language/tool detection. Evidence: const { pathToFileURL } = require("node:url");
Compiler or simulator at line 8	Part of the Compile Code workspace or language/tool detection. Evidence: const { promisify } = require("node:util");
Process execution at line 13	Runs Git, compilers, installers, or helper tools outside the JavaScript runtime. Evidence: const execFileAsync = promisify(execFile);
Security or auth at line 27	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "_headers"
Cloudflare or AI at line 38	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: const directoryRefreshes = ["cloudflare"];
Security or auth at line 48	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "_headers",

Item	Explanation
Git command at line 53	Repository state, publishing, branch checks, or release automation. Evidence: process.env.GIT_EXE,
Git command at line 54	Repository state, publishing, branch checks, or release automation. Evidence: "git",
Git command at line 55	Repository state, publishing, branch checks, or release automation. Evidence: "C:\\Program Files\\Git\\cmd\\git.exe",
Git command at line 56	Repository state, publishing, branch checks, or release automation. Evidence: "C:\\Program Files\\Git\\bin\\git.exe",
Git command at line 57	Repository state, publishing, branch checks, or release automation. Evidence: "C:\\Program Files (x86)\\Git\\cmd\\git.exe",
Git command at line 58	Repository state, publishing, branch checks, or release automation. Evidence: "C:\\Program Files (x86)\\Git\\bin\\git.exe"
Installer at line 61	Windows setup, update, shortcut, or installation behavior. Evidence: function dispatchBuilderMenuAction(action) {
Installer at line 110	Windows setup, update, shortcut, or installation behavior. Evidence: { label: "New Project", accelerator: "CmdOrCtrl+N", click: () => dispatchBuilderMenuAction({ type: "new-project" }) },
Installer at line 111	Windows setup, update, shortcut, or installation behavior. Evidence: { label: "Publishing Target", click: () => dispatchBuilderMenuAction({ type: "publishing-target" }) },
Installer at line 113	Windows setup, update, shortcut, or installation behavior. Evidence: { label: "Save Draft", accelerator: "CmdOrCtrl+S", click: () => dispatchBuilderMenuAction({ type: "save-draft" }) },
Installer at line 114	Windows setup, update, shortcut, or installation behavior. Evidence: { label: "Apply to Site", accelerator: "CmdOrCtrl+Shift+S", click: () => dispatchBuilderMenuAction({ type: "apply-site" }) },
Rich text at line 127	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: { role: "paste" },
Installer at line 141	Windows setup, update, shortcut, or installation behavior. Evidence: { label: "Portfolio Preview", click: () => dispatchBuilderMenuAction({ type: "portfolio-preview" }) },
Installer at line 142	Windows setup, update, shortcut, or installation behavior. Evidence: { label: "Check Updates", click: () => dispatchBuilderMenuAction({ type: "check-updates" }) },
Installer at line 154	Windows setup, update, shortcut, or installation behavior. Evidence: { label: "Open Preferences", accelerator: "CmdOrCtrl+," , click: () => dispatchBuilderMenuAction({ type: "preferences" }) },
Installer at line 156	Windows setup, update, shortcut, or installation behavior. Evidence: { label: "Light Mode", click: () => dispatchBuilderMenuAction({ type: "set-theme", theme: "light" }) },
Installer at line 157	Windows setup, update, shortcut, or installation behavior. Evidence: { label: "Dark Mode", click: () => dispatchBuilderMenuAction({ type: "set-theme", theme: "dark" }) }
Installer at line 163	Windows setup, update, shortcut, or installation behavior. Evidence: { label: "Builder Guide", accelerator: "F1", click: () => dispatchBuilderMenuAction({ type: "builder-guide" }) },
Network request at line 164	Frontend-backend communication or public web/API calls. Evidence: { label: "GitHub Releases", click: () => shell.openExternal("https://github.com/otienomaurice/omb-portfolio-builder/releases/latest") },
File write at line 190	Writes drafts, generated portfolio output, compiler artifacts, uploaded files, or installer data. Evidence: await fsp.mkdir(path.dirname(target), { recursive: true });
File write at line 191	Writes drafts, generated portfolio output, compiler artifacts, uploaded files, or installer data. Evidence: await fsp.copyFile(source, target);
File write at line 196	Writes drafts, generated portfolio output, compiler artifacts, uploaded files, or installer data. Evidence: await fsp.mkdir(target, { recursive: true });
File write at line 212	Writes drafts, generated portfolio output, compiler artifacts, uploaded files, or installer data. Evidence: await fsp.mkdir(path.dirname(target), { recursive: true });
Git command at line 241	Repository state, publishing, branch checks, or release automation. Evidence: throw lastError new Error("Git executable was not found.");
Git command at line 250	Repository state, publishing, branch checks, or release automation. Evidence: return fileExists(path.join(workspaceRoot, ".git"));
File write at line 267	Writes drafts, generated portfolio output, compiler artifacts, uploaded files, or installer data. Evidence: await fsp.mkdir(workspaceRoot, { recursive: true });
Installer at line 279	Windows setup, update, shortcut, or installation behavior. Evidence: const defaultWorkspaceHome = process.platform === "win32" && process.env.LOCALAPPDATA
Installer at line 280	Windows setup, update, shortcut, or installation behavior. Evidence: ? path.join(process.env.LOCALAPPDATA, defaultWorkspaceHomeName)
File write at line 286	Writes drafts, generated portfolio output, compiler artifacts, uploaded files, or installer data. Evidence: await fsp.mkdir(workspaceRoot, { recursive: true });

Item	Explanation
File write at line 287	Writes drafts, generated portfolio output, compiler artifacts, uploaded files, or installer data. Evidence: <code>await fsp.mkdir(portfolioRoot, { recursive: true });</code>
Network request at line 363	Frontend-backend communication or public web/API calls. Evidence: <code>return `http://127.0.0.1:\${port}`;</code>
Rich text at line 377	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <code>backgroundColor: "#eef8fd",</code>

Important constants and variables

Item	Explanation
fs	Line 3: <code>const fs = require("node:fs");</code>
fsp	Line 4: <code>const fsp = require("node:fs/promises");</code>
net	Line 5: <code>const net = require("node:net");</code>
path	Line 6: <code>const path = require("node:path");</code>
mainWindow	Line 10: <code>let mainWindow = null;</code>
builderOrigin	Line 11: <code>let builderOrigin = "";</code>
updateShutdownStarted	Line 12: <code>let updateShutdownStarted = false;</code>
execFileAsync	Line 13: <code>const execFileAsync = promisify(execFile);</code>
defaultRepositoryUrl	Line 14: <code>const defaultRepositoryUrl = process.env.OMB_BUILDER_REPOSITORY "";</code>
defaultWorkspaceHomeName	Line 15: <code>const defaultWorkspaceHomeName = "OMB Portfolio Builder";</code>
rootFilesToRefresh	Line 17: <code>const rootFilesToRefresh = [</code>
rootFilesToSeed	Line 30: <code>const rootFilesToSeed = [</code>
directoryRefreshes	Line 38: <code>const directoryRefreshes = ["cloudflare"];</code>
directorySeeds	Line 39: <code>const directorySeeds = ["assets", "Backgrounds", "docs", ".well-known"];</code>
portfolioFilesToSeed	Line 40: <code>const portfolioFilesToSeed = [</code>
portfolioDirectoriesToSeed	Line 51: <code>const portfolioDirectoriesToSeed = ["assets", "Backgrounds", "docs", ".well-known"];</code>
gitCandidates	Line 52: <code>const gitCandidates = [</code>
payload	Line 63: <code>const payload = JSON.stringify(action);</code>
shouldFullscreen	Line 98: <code>const shouldFullscreen = typeof nextState === "boolean" ? nextState : !mainWindow.isFullScreen();</code>
entries	Line 197: <code>const entries = await fsp.readdir(source, { withFileTypes: true });</code>
sourcePath	Line 199: <code>const sourcePath = path.join(source, entry.name);</code>
targetPath	Line 200: <code>const targetPath = path.join(target, entry.name);</code>
entries	Line 218: <code>const entries = await fsp.readdir(directory);</code>
lastError	Line 226: <code>let lastError = null;</code>
current	Line 254: <code>let current = path.dirname(process.execPath);</code>
next	Line 257: <code>const next = path.dirname(current);</code>
bundledSiteRoot	Line 278: <code>const bundledSiteRoot = path.join(process.resourcesPath, "site");</code>
defaultWorkspaceHome	Line 279: <code>const defaultWorkspaceHome = process.platform === "win32" && process.env.LOCALAPPDATA</code>
workspaceHome	Line 282: <code>const workspaceHome = process.env.OMB_WORKSPACE_HOME defaultWorkspaceHome;</code>
workspaceRoot	Line 283: <code>const workspaceRoot = path.join(workspaceHome, "builder");</code>
portfolioRoot	Line 284: <code>const portfolioRoot = process.env.OMB_PORTFOLIO_WORKSPACE path.join(workspaceHome, "portfolio");</code>
envWorkspace	Line 315: <code>const envWorkspace = process.env.OMB_BUILDER_WORKSPACE;</code>
workspaceRoot	Line 323: <code>const workspaceRoot = path.resolve(__dirname);</code>
nearbyRepository	Line 329: <code>const nearbyRepository = findRepositoryNearExecutable();</code>

Item	Explanation
tester	Line 341: const tester = net.createServer();
address	Line 344: const address = tester.address();
port	Line 345: const port = typeof address === "object" && address ? address.port : 0;
port	Line 352: const port = await findFreePort();
serverPath	Line 361: const serverPath = path.join(workspaceRoot, "server.mjs");
iconPath	Line 367: const iconPath = path.join(workspaceRoot, "assets", "omb-app-icon-256.png");
refreshFullscreenMenu	Line 388: const refreshFullscreenMenu = () => {
gotLock	Line 426: const gotLock = app.requestSingleInstanceLock();

Function-by-function detail

dispatchBuilderMenuAction - lines 61-68

Item	Explanation
Source location	Lines 61-68 (8 source lines).
Purpose	Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.
Declaration	function dispatchBuilderMenuAction(action) {
Touches	filesystem
Calls detected	isDestroyed, stringify, executeJavaScript, dispatchEvent, CustomEvent
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

61: function dispatchBuilderMenuAction(action) {
62:   if (!mainWindow || mainWindow.isDestroyed()) return;
63:   const payload = JSON.stringify(action);
64:   mainWindow.webContents.executeJavaScript(
65:     `window.dispatchEvent(new CustomEvent("builder-menu-action", { detail: ${payload} }));`,
66:     true
67:   ).catch(() => {});
68: }

```

quitForBuilderUpdate - lines 70-92

Item	Explanation
Source location	Lines 70-92 (23 source lines).
Purpose	Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.
Declaration	function quitForBuilderUpdate() {
Touches	Mostly local calculations or local UI state.
Calls detected	isDestroyed, destroy, quit, exit
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

70: function quitForBuilderUpdate() {
71:   if (updateShutdownStarted) return;
72:   updateShutdownStarted = true;
73:   try {
74:     if (mainWindow && !mainWindow.isDestroyed()) {
75:       mainWindow.destroy();
76:     }
77:   } catch {
78:     // The process exit fallback below still guarantees the updater can continue.
79:   }
80:   try {
81:     app.quit();
82:   } catch {
83:     // Fall through to app.exit.
84:   }
85:   setTimeout(() => {
86:     try {
87:       app.exit(0);
88:     } catch {
89:       process.exit(0);
90:     }
91:   }, 1500);
92: }

```

setBuilderFullScreen - lines 96-103

Item	Explanation
Source location	Lines 96-103 (8 source lines).
Purpose	Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.
Declaration	function setBuilderFullScreen(nextState = null) {
Touches	Mostly local calculations or local UI state.
Calls detected	isDestroyed, isFullScreen, setFullScreen, setMenuBarVisibility, setAutoHideMenuBar, setMenu, createAppMenu
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

96: function setBuilderFullScreen(nextState = null) {
97:   if (!mainWindow || mainWindow.isDestroyed()) return;
98:   const shouldFullscreen = typeof nextState === "boolean" ? nextState : !mainWindow.isFullScreen();
99:   mainWindow.setFullScreen(shouldFullscreen);
100:  mainWindow.setMenuBarVisibility(true);
101:  mainWindow.setAutoHideMenuBar(false);
102:  mainWindow.setMenu(createAppMenu(builderOrigin));
103: }

```

createAppMenu - lines 105-184

Item	Explanation
Source location	Lines 105-184 (80 source lines).
Purpose	Creates an object, UI structure, process, payload, or generated artifact.
Declaration	function createAppMenu(origin) {
Touches	filesystem, network/API requests
Calls detected	buildFromTemplate, dispatchBuilderMenuAction, setBuilderFullScreen, openExternal, isDestroyed, isMinimized, restore, focus, fileExists, accessSync
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

105: function createAppMenu(origin) {
106:   return Menu.buildFromTemplate([
107:     {
108:       label: "File",
109:       submenu: [
110:         { label: "New Project", accelerator: "CmdOrCtrl+N", click: () => dispatchBuilderMenuAction({
type: "new-project" }) },
111:         { label: "Publishing Target", click: () => dispatchBuilderMenuAction({ type: "publishing-target"
}) },
112:         { type: "separator" },
113:         { label: "Save Draft", accelerator: "CmdOrCtrl+S", click: () => dispatchBuilderMenuAction({ type:
"save-draft" }) },
114:         { label: "Apply to Site", accelerator: "CmdOrCtrl+Shift+S", click: () =>
dispatchBuilderMenuAction({ type: "apply-site" }) },
115:         { type: "separator" },
116:         { role: "quit", label: "Exit" }
117:       ]
118:     },
119:     {
120:       label: "Edit",
121:       submenu: [
122:         { role: "undo" },
123:         { role: "redo" },
124:         { type: "separator" },
125:         { role: "cut" },
126:         { role: "copy" },
127:         { role: "paste" },
128:         { role: "selectAll" }
129:       ]
130:     },
131:     {
132:       label: "View",
133:       submenu: [
134:         { role: "reload" },
135:         { role: "forceReload" },
136:         { type: "separator" },
137:         { role: "resetZoom" },
138:         { role: "zoomIn" },
139:         { role: "zoomOut" },
140:         { type: "separator" },
141:         { label: "Portfolio Preview", click: () => dispatchBuilderMenuAction({ type: "portfolio-preview"
}) },
142:         { label: "Check Updates", click: () => dispatchBuilderMenuAction({ type: "check-updates" }) },
143:         { type: "separator" },
144:         {
145:           label: mainWindow?.isFullScreen?.() ? "Exit Full Screen" : "Toggle Full Screen",
146:           accelerator: "F11",
147:           click: () => setBuilderFullScreen()
148:         }
149:       ]
150:     },
151:     {
152:       label: "Preferences",
153:       submenu: [
154:         { label: "Open Preferences", accelerator: "CmdOrCtrl+," , click: () => dispatchBuilderMenuAction({
type: "preferences" }) },
155:         { type: "separator" },
156:         { label: "Light Mode", click: () => dispatchBuilderMenuAction({ type: "set-theme", theme: "light"
}) },
157:         { label: "Dark Mode", click: () => dispatchBuilderMenuAction({ type: "set-theme", theme: "dark"
}) }
158:       ]
159:     },
160:     {
161:       label: "Help",
162:       submenu: [
163:         { label: "Builder Guide", accelerator: "F1", click: () => dispatchBuilderMenuAction({ type:
"builder-guide" }) },
164:         { label: "GitHub Releases", click: () =>
shell.openExternal("https://github.com/otienomaurice/omb-portfolio-builder/releases/latest") },
165:         {
166:           label: "Focus Builder Window",
167:           click: () => {
168:             if (!mainWindow || mainWindow.isDestroyed()) return;
169:             if (mainWindow.isMinimized()) mainWindow.restore();
170:             mainWindow.focus();
171:           }
172:         }
173:       ]
174:     }
175:   ]);
176: }
177:
178: function fileExists(filePath) {
179:   try {
180:     fs.accessSync(filePath);
181:     return true;
182:   } catch {
183:     return false;
184:   }

```


fileExists - lines 178-185

Item	Explanation
Source location	Lines 178-185 (8 source lines).
Purpose	Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.
Declaration	function fileExists(filePath) {
Touches	filesystem
Calls detected	accessSync
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
178: function fileExists(filePath) {
179:   try {
180:     fs.accessSync(filePath);
181:     return true;
182:   } catch {
183:     return false;
184:   }
185: }
```

copyFileIfAvailable - lines 187-192

Item	Explanation
Source location	Lines 187-192 (6 source lines).
Purpose	Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.
Declaration	async function copyFileIfAvailable(source, target, overwrite = false) {
Touches	filesystem
Calls detected	fileExists, mkdir, dirname, copyFile
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
187: async function copyFileIfAvailable(source, target, overwrite = false) {
188:   if (!fileExists(source)) return;
189:   if (!overwrite && fileExists(target)) return;
190:   await fsp.mkdir(path.dirname(target), { recursive: true });
191:   await fsp.copyFile(source, target);
192: }
```

copyDirectoryMissingFiles - lines 194-207

Item	Explanation
Source location	Lines 194-207 (14 source lines).
Purpose	Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.
Declaration	async function copyDirectoryMissingFiles(source, target) {
Touches	filesystem
Calls detected	fileExists, mkdir, readdir, join, isDirectory, isFile, copyFileIfAvailable
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	4 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

194: async function copyDirectoryMissingFiles(source, target) {
195:   if (!fileExists(source)) return;
196:   await fsp.mkdir(target, { recursive: true });
197:   const entries = await fsp.readdir(source, { withFileTypes: true });
198:   for (const entry of entries) {
199:     const sourcePath = path.join(source, entry.name);
200:     const targetPath = path.join(target, entry.name);
201:     if (entry.isDirectory()) {
202:       await copyDirectoryMissingFiles(sourcePath, targetPath);
203:     } else if (entry.isFile()) {
204:       await copyFileIfAvailable(sourcePath, targetPath, false);
205:     }
206:   }
207: }

```

copyDirectoryRefresh - lines 209-214

Item	Explanation
Source location	Lines 209-214 (6 source lines).
Purpose	Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.
Declaration	async function copyDirectoryRefresh(source, target) {
Touches	filesystem
Calls detected	fileExists, rm, mkdir, dirname, cp
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	3 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

209: async function copyDirectoryRefresh(source, target) {
210:   if (!fileExists(source)) return;
211:   await fsp.rm(target, { recursive: true, force: true });
212:   await fsp.mkdir(path.dirname(target), { recursive: true });
213:   await fsp.cp(source, target, { recursive: true, force: true });
214: }

```

directoryIsEmpty - lines 216-223

Item	Explanation
Source location	Lines 216-223 (8 source lines).
Purpose	Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.
Declaration	async function directoryIsEmpty(directory) {
Touches	Mostly local calculations or local UI state.
Calls detected	readdir
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

216: async function directoryIsEmpty(directory) {
217:   try {
218:     const entries = await fsp.readdir(directory);
219:     return entries.length === 0;
220:   } catch {
221:     return true;
222:   }
223: }

```

runGit - lines 225-242

Item	Explanation
Source location	Lines 225-242 (18 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function runGit(args, cwd) {
Touches	Mostly local calculations or local UI state.
Calls detected	execFileAsync, Error
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

225: async function runGit(args, cwd) {
226:   let lastError = null;
227:   for (const candidate of gitCandidates) {
228:     try {
229:       await execFileAsync(candidate, args, {
230:         cwd,
231:         maxBuffer: 10 * 1024 * 1024,
232:         windowsHide: true
233:       });
234:       return true;
235:     } catch (error) {
236:       lastError = error;
237:       if (error.code === "ENOENT") continue;
238:       throw error;
239:     }
240:   }
241:   throw lastError || new Error("Git executable was not found.");
242: }

```

workspaceLooksUsable - lines 244-247

Item	Explanation
Source location	Lines 244-247 (4 source lines).
Purpose	Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.
Declaration	function workspaceLooksUsable(workspaceRoot) {
Touches	Mostly local calculations or local UI state.
Calls detected	fileExists, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

244: function workspaceLooksUsable(workspaceRoot) {
245:   return fileExists(path.join(workspaceRoot, "server.mjs"))
246:     && fileExists(path.join(workspaceRoot, "index.html"));
247: }

```

workspaceIsGitBacked - lines 249-251

Item	Explanation
Source location	Lines 249-251 (3 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	function workspaceIsGitBacked(workspaceRoot) {
Touches	Mostly local calculations or local UI state.
Calls detected	fileExists, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
249: function workspaceIsGitBacked(workspaceRoot) {
250:   return fileExists(path.join(workspaceRoot, ".git"));
251: }
```

findRepositoryNearExecutable - lines 253-262

Item	Explanation
Source location	Lines 253-262 (10 source lines).
Purpose	Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.
Declaration	function findRepositoryNearExecutable() {
Touches	Mostly local calculations or local UI state.
Calls detected	dirname, workspaceLooksUsable, workspaceIsGitBacked
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
253: function findRepositoryNearExecutable() {
254:   let current = path.dirname(process.execPath);
255:   for (let index = 0; index < 8; index += 1) {
256:     if (workspaceLooksUsable(current) && workspaceIsGitBacked(current)) return current;
257:     const next = path.dirname(current);
258:     if (next === current) break;
259:     current = next;
260:   }
261:   return "";
262: }
```

cloneWorkspaceIfPossible - lines 264-275

Item	Explanation
Source location	Lines 264-275 (12 source lines).
Purpose	Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.
Declaration	async function cloneWorkspaceIfPossible(workspaceRoot) {
Touches	filesystem
Calls detected	workspaceIsGitBacked, mkdir, directoryIsEmpty, runGit, dirname
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	3 await expression(s), 5 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

264: async function cloneWorkspaceIfPossible(workspaceRoot) {
265:   if (!defaultRepositoryUrl) return false;
266:   if (workspaceIsGitBacked(workspaceRoot)) return true;
267:   await fsp.mkdir(workspaceRoot, { recursive: true });
268:   if (!await directoryIsEmpty(workspaceRoot)) return false;
269:   try {
270:     await runGit(["clone", "--depth", "1", defaultRepositoryUrl, workspaceRoot],
path.dirname(workspaceRoot));
271:     return workspaceIsGitBacked(workspaceRoot);
272:   } catch {
273:     return false;
274:   }
275: }

```

preparePackagedWorkspace - lines 277-312

Item	Explanation
Source location	Lines 277-312 (36 source lines).
Purpose	Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.
Declaration	async function preparePackagedWorkspace() {
Touches	filesystem
Calls detected	join, getPath, mkdir, cloneWorkspaceIfPossible, copyFileIfAvailable, copyDirectoryRefresh, copyDirectoryMissingFiles
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	10 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

277: async function preparePackagedWorkspace() {
278:   const bundledSiteRoot = path.join(process.resourcesPath, "site");
279:   const defaultWorkspaceHome = process.platform === "win32" && process.env.LOCALAPPDATA
280:     ? path.join(process.env.LOCALAPPDATA, defaultWorkspaceHomeName)
281:     : path.join(app.getPath("userData"), "workspace");
282:   const workspaceHome = process.env.OMB_WORKSPACE_HOME || defaultWorkspaceHome;
283:   const workspaceRoot = path.join(workspaceHome, "builder");
284:   const portfolioRoot = process.env.OMB_PORTFOLIO_WORKSPACE || path.join(workspaceHome, "portfolio");
285:
286:   await fsp.mkdir(workspaceRoot, { recursive: true });
287:   await fsp.mkdir(portfolioRoot, { recursive: true });
288:   await cloneWorkspaceIfPossible(portfolioRoot);
289:
290:   for (const fileName of rootFilesToRefresh) {
291:     await copyFileIfAvailable(path.join(bundledSiteRoot, fileName), path.join(workspaceRoot, fileName),
true);
292:   }
293:   for (const fileName of rootFilesToSeed) {
294:     await copyFileIfAvailable(path.join(bundledSiteRoot, fileName), path.join(workspaceRoot, fileName),
false);
295:   }
296:   for (const directoryName of directoryRefreshes) {
297:     await copyDirectoryRefresh(path.join(bundledSiteRoot, directoryName), path.join(workspaceRoot,
directoryName));
298:   }
299:   for (const directoryName of directorySeeds) {
300:     await copyDirectoryMissingFiles(path.join(bundledSiteRoot, directoryName), path.join(workspaceRoot,
directoryName));
301:   }
302:
303:   for (const fileName of portfolioFilesToSeed) {
304:     await copyFileIfAvailable(path.join(bundledSiteRoot, fileName), path.join(portfolioRoot, fileName),
false);
305:   }
306:   await copyFileIfAvailable(path.join(bundledSiteRoot, "portfolio-README.md"), path.join(portfolioRoot,
"README.md"), false);

```

```

307:   for (const directoryName of portfolioDirectoriesToSeed) {
308:     await copyDirectoryMissingFiles(path.join(bundledSiteRoot, directoryName), path.join(portfolioRoot,
directoryName));
309:   }
310:
311:   return { workspaceRoot, portfolioRoot };
312: }

```

resolveWorkspaceRoots - lines 314-337

Item	Explanation
Source location	Lines 314-337 (24 source lines).
Purpose	Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.
Declaration	async function resolveWorkspaceRoots() {
Touches	Mostly local calculations or local UI state.
Calls detected	workspaceLooksUsable, join, dirname, resolve, findRepositoryNearExecutable, preparePackagedWorkspace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

314: async function resolveWorkspaceRoots() {
315:   const envWorkspace = process.env.OMB_BUILDER_WORKSPACE;
316:   if (envWorkspace && workspaceLooksUsable(envWorkspace)) {
317:     return {
318:       workspaceRoot: envWorkspace,
319:       portfolioRoot: process.env.OMB_PORTFOLIO_WORKSPACE || path.join(path.dirname(envWorkspace),
"portfolio")
320:     };
321:   }
322:   if (!app.isPackaged) {
323:     const workspaceRoot = path.resolve(__dirname);
324:     return {
325:       workspaceRoot,
326:       portfolioRoot: process.env.OMB_PORTFOLIO_WORKSPACE || workspaceRoot
327:     };
328:   }
329:   const nearbyRepository = findRepositoryNearExecutable();
330:   if (nearbyRepository) {
331:     return {
332:       workspaceRoot: nearbyRepository,
333:       portfolioRoot: process.env.OMB_PORTFOLIO_WORKSPACE || nearbyRepository
334:     };
335:   }
336:   return preparePackagedWorkspace();
337: }

```

findFreePort - lines 339-349

Item	Explanation
Source location	Lines 339-349 (11 source lines).
Purpose	Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.
Declaration	function findFreePort() {
Touches	Mostly local calculations or local UI state.
Calls detected	createServer, once, listen, address, close, resolve
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

339: function findFreePort() {
340:   return new Promise((resolve, reject) => {
341:     const tester = net.createServer();
342:     tester.once("error", reject);
343:     tester.listen(0, "127.0.0.1", () => {
344:       const address = tester.address();
345:       const port = typeof address === "object" && address ? address.port : 0;
346:       tester.close(() => resolve(port));
347:     });
348:   });
349: }

```

startBuilderServer - lines 351-364

Item	Explanation
Source location	Lines 351-364 (14 source lines).
Purpose	Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.
Declaration	async function startBuilderServer(workspaceRoot, portfolioRoot) {
Touches	network/API requests
Calls detected	findFreePort, getVersion, join, pathToFileURL, now
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

351: async function startBuilderServer(workspaceRoot, portfolioRoot) {
352:   const port = await findFreePort();
353:   process.env.PORT = String(port);
354:   process.env.HOST = "127.0.0.1";
355:   process.env.OMB_DESKTOP_APP = "1";
356:   process.env.OMB_BUILDER_WORKSPACE = workspaceRoot;
357:   process.env.OMB_PORTFOLIO_WORKSPACE = portfolioRoot;
358:   process.env.OMB_BUILDER_REPOSITORY = defaultRepositoryUrl;
359:   process.env.OMB_APP_VERSION = app.getVersion();
360:
361:   const serverPath = path.join(workspaceRoot, "server.mjs");
362:   await import(`${pathToFileURL(serverPath).href}?desktop=${Date.now()}`);
363:   return `http://127.0.0.1:${port}`;
364: }

```

createWindow - lines 366-423

Item	Explanation
Source location	Lines 366-423 (58 source lines).
Purpose	Creates an object, UI structure, process, payload, or generated artifact.
Declaration	function createWindow(workspaceRoot, origin) {
Touches	Mostly local calculations or local UI state.
Calls detected	join, BrowserWindow, fileExists, setMenu, createAppMenu, setMenuBarVisibility, setAutoHideMenuBar, isDestroyed, on, preventDefault, setBuilderFullScreen, isFullScreen
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

366: function createWindow(workspaceRoot, origin) {
367:   const iconPath = path.join(workspaceRoot, "assets", "omb-app-icon-256.png");
368:   nativeTheme.themeSource = "light";
369:   mainWindow = new BrowserWindow({
370:     width: 1440,
371:     height: 960,

```

```

372:     minWidth: 980,
373:     minHeight: 680,
374:     title: "OMB Portfolio Builder",
375:     autoHideMenuBar: false,
376:     icon: fileExists(iconPath) ? iconPath : undefined,
377:     backgroundColor: "#eef8fd",
378:     webPreferences: {
379:       contextIsolation: true,
380:       nodeIntegration: false,
381:       sandbox: true
382:     }
383:   });
384:   mainWindow.setMenu(createAppMenu(origin));
385:   mainWindow.setMenuBarVisibility(true);
386:   mainWindow.setAutoHideMenuBar(false);
387:
388:   const refreshFullscreenMenu = () => {
389:     if (mainWindow && !mainWindow.isDestroyed()) {
390:       mainWindow.setMenuBarVisibility(true);
391:       mainWindow.setAutoHideMenuBar(false);
392:       mainWindow.setMenu(createAppMenu(origin));
393:     }
394:   };
395:   mainWindow.on("enter-full-screen", refreshFullscreenMenu);
396:   mainWindow.on("leave-full-screen", refreshFullscreenMenu);
397:
398:   mainWindow.webContents.on("before-input-event", (event, input) => {
399:     if (input.key === "F11") {
400:       event.preventDefault();
401:       setBuilderFullScreen();
402:       return;
403:     }
404:     if (input.key === "Escape" && mainWindow.isFullScreen()) {
405:       event.preventDefault();
406:       setBuilderFullScreen(false);
407:     }
408:   });
409:
410:   mainWindow.webContents.setWindowOpenHandler(({ url }) => {
411:     if (url.startsWith(origin)) return { action: "allow" };
412:     shell.openExternal(url);
413:     return { action: "deny" };
414:   });
415:
416:   mainWindow.webContents.on("will-navigate", (event, url) => {
417:     if (url.startsWith(origin)) return;
418:     event.preventDefault();
419:     shell.openExternal(url);
420:   });
421:
422:   mainWindow.loadURL(`${origin}/template-preview.html`);
423: }

```

refreshFullscreenMenu - lines 388-394

Item	Explanation
Source location	Lines 388-394 (7 source lines).
Purpose	Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.
Declaration	const refreshFullscreenMenu = () => {
Touches	Mostly local calculations or local UI state.
Calls detected	isDestroyed, setMenuBarVisibility, setAutoHideMenuBar, setMenu, createAppMenu
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

388:   const refreshFullscreenMenu = () => {
389:     if (mainWindow && !mainWindow.isDestroyed()) {
390:       mainWindow.setMenuBarVisibility(true);
391:       mainWindow.setAutoHideMenuBar(false);
392:       mainWindow.setMenu(createAppMenu(origin));
393:     }
394:   };

```


boot - lines 425-447

Item	Explanation
Source location	Lines 425-447 (23 source lines).
Purpose	Part of Electron desktop startup, window control, workspace setup, menu behavior, or update handoff.
Declaration	async function boot() {
Touches	Mostly local calculations or local UI state.
Calls detected	requestSingleInstanceLock, quit, on, isMinimized, restore, focus, whenReady, resolveWorkspaceRoots, startBuilderServer, createWindow, showErrorBox
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	3 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
425: async function boot() {
426:   const gotLock = app.requestSingleInstanceLock();
427:   if (!gotLock) {
428:     app.quit();
429:     return;
430:   }
431:
432:   app.on("second-instance", () => {
433:     if (!mainWindow) return;
434:     if (mainWindow.isMinimized()) mainWindow.restore();
435:     mainWindow.focus();
436:   });
437:
438:   await app.whenReady();
439:   try {
440:     const { workspaceRoot, portfolioRoot } = await resolveWorkspaceRoots();
441:     builderOrigin = await startBuilderServer(workspaceRoot, portfolioRoot);
442:     createWindow(workspaceRoot, builderOrigin);
443:   } catch (error) {
444:     dialog.showErrorBox("OMB Portfolio Builder could not start", error?.message || String(error));
445:     app.quit();
446:   }
447: }
```

Representative opening snippet

```
1: const { app, BrowserWindow, Menu, dialog, nativeTheme, shell } = require("electron");
2: const { execFile } = require("node:child_process");
3: const fs = require("node:fs");
4: const fsp = require("node:fs/promises");
5: const net = require("node:net");
6: const path = require("node:path");
7: const { pathToFileURL } = require("node:url");
8: const { promisify } = require("node:util");
9:
10: let mainWindow = null;
11: let builderOrigin = "";
12: let updateShutdownStarted = false;
13: const execFileAsync = promisify(execFile);
14: const defaultRepositoryUrl = process.env.OMB_BUILDER_REPOSITORY || "";
15: const defaultWorkspaceHomeName = "OMB Portfolio Builder";
16:
17: const rootFilesToRefresh = [
18:   "server.mjs",
19:   "index.html",
20:   "styles.css",
21:   "script.js",
22:   "electronics-search.js",
23:   "builder-rich-future-sections.js",
24:   "template-preview.html",
25:   "template-preview.css",
26:   "template-preview.js",
27:   "_headers"
28: ];
29:
30: const rootFilesToSeed = [
31:   "projects.json",
32:   "project-templates.json",
33:   "README.md",
34:   ".nojekyll",
35:   "robots.txt"
```

```

36: ];
37:
38: const directoryRefreshes = ["cloudflare"];
39: const directorySeeds = ["assets", "Backgrounds", "docs", ".well-known"];
40: const portfolioFilesToSeed = [
41:   "projects.json",
42:   "index.html",
43:   "styles.css",
44:   "script.js",
45:   "electronics-search.js",
46:   ".nojekyll",
47:   "robots.txt",
48:   "_headers",
49:   "sitemap.xml"
50: ];
51: const portfolioDirectoriesToSeed = ["assets", "Backgrounds", "docs", ".well-known"];
52: const gitCandidates = [
53:   process.env.GIT_EXE,
54:   "git",
55:   "C:\\Program Files\\Git\\cmd\\git.exe",
56:   "C:\\Program Files\\Git\\bin\\git.exe",
57:   "C:\\Program Files (x86)\\Git\\cmd\\git.exe",
58:   "C:\\Program Files (x86)\\Git\\bin\\git.exe"
59: ].filter(Boolean);
60:
61: function dispatchBuilderMenuAction(action) {
62:   if (!mainWindow || mainWindow.isDestroyed()) return;
63:   const payload = JSON.stringify(action);
64:   mainWindow.webContents.executeJavaScript(
65:     `window.dispatchEvent(new CustomEvent("builder-menu-action", { detail: ${payload} }));`,
66:     true
67:   ).catch(() => {});
68: }
69:
70: function quitForBuilderUpdate() {
71:   if (updateShutdownStarted) return;
72:   updateShutdownStarted = true;

```

Safe change checklist

- Confirm which app, website, installer, workflow, or document consumes this file.
- Change it on a feature branch from development.
- Regenerate docs if the file purpose, API surface, or behavior changes.
- Run the smallest relevant smoke test before merging into development.

Important Code Reference: server.mjs

Item	Explanation
File	server.mjs
Category	JavaScript source
Folder role	Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
Size	147,116 bytes
Extension	.mjs
MIME type	text/plain
Text lines	3,632

Why this file exists

- Local backend server used by the builder for drafts, parsing, publishing, compiler execution, authentication checks, and file operations.

How it is used

Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.

How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

Imports, requires, and external dependencies

Item	Explanation
Line 1	import { createServer } from "node:http";
Line 2	import { execFile, spawn } from "node:child_process";
Line 3	import { createHash } from "node:crypto";
Line 4	import { access as fsAccess, chmod, cp, mkdir, mkdtemp, readFile, readdir, rm, writeFile } from "node:fs/promises";
Line 5	import os from "node:os";
Line 6	import path from "node:path";
Line 7	import { promisify } from "node:util";
Line 8	import { fileURLToPath } from "node:url";

API endpoints mentioned or called

Item	Explanation
/api/catalog	Line 3343. Loads project/catalog data for the builder.
/api/templates	Line 3353. Loads appearance template definitions.
/api/publish-target	Line 3358. Reads or writes the Git publishing target and authorization state.
/api/system-check	Line 3367. Checks local tool/system readiness.
/api/app-update	Line 3372. Checks or starts builder app update behavior.
/api/security-report	Line 3377. Returns security/download/auth reporting for the builder.
/api/code/tools	Line 3386. Checks compiler/runtime availability.
/api/portfolio-ai	Line 3402. Handles AI assistant questions.
/api/code/beautify	Line 3407. Formats source code for cleaner display and editing.

Item	Explanation
/api/code/save	Line 3422. Saves source code into the local compile workspace.
/api/code/compile	Line 3433. Compiles or runs a source file and returns terminal output.
/api/code/install-tools	Line 3444. Attempts to install missing compiler tools.
/api/publish-target	Line 3455. Reads or writes the Git publishing target and authorization state.
/api/install-git	Line 3465. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior.
/api/app-update/install	Line 3479. Checks or starts builder app update behavior.
/api/github-authenticate	Line 3493. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior.
/api/sync-from-publish-target	Line 3510. Reads or writes the Git publishing target and authorization state.
/api/save-draft	Line 3525. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior.
/api/apply-catalog	Line 3525. Loads project/catalog data for the builder.
/api/apply-catalog	Line 3534. Loads project/catalog data for the builder.
/api/upload	Line 3577. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior.

Important implementation signals

Item	Explanation
Compiler or simulator at line 1	Part of the Compile Code workspace or language/tool detection. Evidence: import { createServer } from "node:http";
Process execution at line 2	Runs Git, compilers, installers, or helper tools outside the JavaScript runtime. Evidence: import { execFile, spawn } from "node:child_process";
Compiler or simulator at line 3	Part of the Compile Code workspace or language/tool detection. Evidence: import { createHash } from "node:crypto";
File read at line 4	Reads local builder state, project files, website assets, or configuration. Evidence: import { access as fsAccess, chmod, cp, mkdir, mkdtemp, readFile, readdir, rm, writeFile } from "node:fs/promises";
Compiler or simulator at line 5	Part of the Compile Code workspace or language/tool detection. Evidence: import os from "node:os";
Compiler or simulator at line 6	Part of the Compile Code workspace or language/tool detection. Evidence: import path from "node:path";
Compiler or simulator at line 7	Part of the Compile Code workspace or language/tool detection. Evidence: import { promisify } from "node:util";
Compiler or simulator at line 8	Part of the Compile Code workspace or language/tool detection. Evidence: import { fileURLToPath } from "node:url";
Process execution at line 15	Runs Git, compilers, installers, or helper tools outside the JavaScript runtime. Evidence: const execFileAsync = promisify(execFile);
File read at line 16	Reads local builder state, project files, website assets, or configuration. Evidence: const packageJson = JSON.parse(await readFile(path.join(root, "package.json"), "utf8").catch(() => "{}"));
Browser DOM at line 34	Builder or website interface behavior in the browser/Electron renderer. Evidence: ".xlsx": "application/vnd.openxmlformats-officedocument.spreadsheetml.sheet",
Security or auth at line 40	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: const publishAuthCachePath = path.join(root, ".omb-publish-session.json");
Security or auth at line 51	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "_headers",
Security or auth at line 57	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: const publishAuthCacheTtlMs = 24 * 60 * 60 * 1000;
Security or auth at line 58	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: const publishAuthExtendedTtlMs = 30 * 24 * 60 * 60 * 1000;
Security or auth at line 59	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: const publishAuthHistoryWindowMs = 7 * 24 * 60 * 60 * 1000;
Security or auth at line 60	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: const publishAuthExtendedThreshold = 3;
Security or auth at line 63	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: const publishAuthorizationHelp = [
Security or auth at line 65	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "Open Publishing target, enter the repository URL, then click Authenticate target.",

Item	Explanation
Git command at line 66	Repository state, publishing, branch checks, or release automation. Evidence: "A GitHub/Git Credential Manager browser sign-in may open. Sign in with an account that has write access to the selected Pages repository.",
Security or auth at line 67	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "Authenticate target only verifies write access and remembers it for the matching repository and branch.",
Security or auth at line 68	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "Load from target only imports compatible website files from the authenticated repository into the local builder workspace.",
Git command at line 72	Repository state, publishing, branch checks, or release automation. Evidence: process.env.GIT_EXE,
Git command at line 73	Repository state, publishing, branch checks, or release automation. Evidence: "git",
Git command at line 74	Repository state, publishing, branch checks, or release automation. Evidence: "C:\\Program Files\\Git\\cmd\\git.exe",
Git command at line 75	Repository state, publishing, branch checks, or release automation. Evidence: "C:\\Program Files\\Git\\bin\\git.exe",
Git command at line 76	Repository state, publishing, branch checks, or release automation. Evidence: "C:\\Program Files (x86)\\Git\\cmd\\git.exe",
Git command at line 77	Repository state, publishing, branch checks, or release automation. Evidence: "C:\\Program Files (x86)\\Git\\bin\\git.exe",
Git command at line 78	Repository state, publishing, branch checks, or release automation. Evidence: process.env.LOCALAPPDATA ? path.join(process.env.LOCALAPPDATA, "Programs", "Git", "cmd", "git.exe") : "",
Git command at line 79	Repository state, publishing, branch checks, or release automation. Evidence: process.env.LOCALAPPDATA ? path.join(process.env.LOCALAPPDATA, "Programs", "Git", "bin", "git.exe") : ""
Compiler or simulator at line 87	Part of the Compile Code workspace or language/tool detection. Evidence: primaryTools: ["gcc"],
Compiler or simulator at line 97	Part of the Compile Code workspace or language/tool detection. Evidence: verilog: {
Compiler or simulator at line 100	Part of the Compile Code workspace or language/tool detection. Evidence: label: "Verilog",
Compiler or simulator at line 101	Part of the Compile Code workspace or language/tool detection. Evidence: primaryTools: ["iverilog", "vvp"],
Compiler or simulator at line 102	Part of the Compile Code workspace or language/tool detection. Evidence: winget: ["Icarus.Verilog"]
Compiler or simulator at line 104	Part of the Compile Code workspace or language/tool detection. Evidence: systemverilog: {
Compiler or simulator at line 107	Part of the Compile Code workspace or language/tool detection. Evidence: label: "SystemVerilog",
Compiler or simulator at line 108	Part of the Compile Code workspace or language/tool detection. Evidence: primaryTools: ["iverilog", "vvp"],
Compiler or simulator at line 109	Part of the Compile Code workspace or language/tool detection. Evidence: winget: ["Icarus.Verilog"]
Compiler or simulator at line 118	Part of the Compile Code workspace or language/tool detection. Evidence: java: {
Compiler or simulator at line 119	Part of the Compile Code workspace or language/tool detection. Evidence: defaultFile: "Main.java",
Compiler or simulator at line 120	Part of the Compile Code workspace or language/tool detection. Evidence: extensions: [".java"],
Compiler or simulator at line 121	Part of the Compile Code workspace or language/tool detection. Evidence: label: "Java",
Compiler or simulator at line 122	Part of the Compile Code workspace or language/tool detection. Evidence: primaryTools: ["javac", "java"],
Compiler or simulator at line 129	Part of the Compile Code workspace or language/tool detection. Evidence: primaryTools: ["node"],
Compiler or simulator at line 132	Part of the Compile Code workspace or language/tool detection. Evidence: python: {
Compiler or simulator at line 135	Part of the Compile Code workspace or language/tool detection. Evidence: label: "Python",

Item	Explanation
Compiler or simulator at line 136	Part of the Compile Code workspace or language/tool detection. Evidence: primaryTools: ["python"],
Compiler or simulator at line 149	Part of the Compile Code workspace or language/tool detection. Evidence: gcc: [
Compiler or simulator at line 150	Part of the Compile Code workspace or language/tool detection. Evidence: "gcc",
Compiler or simulator at line 151	Part of the Compile Code workspace or language/tool detection. Evidence: process.env.LOCALAPPDATA ? path.join(process.env.LOCALAPPDATA, "Microsoft", "WinGet", "Packages", "BrechtSanders.WinLibs.POSIX.UCRT_Microsoft.Winget.Source_8wekyb3d8bbwe", "mingw64", "bin", "gcc.exe") : ""
Installer at line 156	Windows setup, update, shortcut, or installation behavior. Evidence: process.env.LOCALAPPDATA ? path.join(process.env.LOCALAPPDATA, "Microsoft", "WinGet", "Packages", "BrechtSanders.WinLibs.POSIX.UCRT_Microsoft.Winget.Source_8wekyb3d8bbwe", "mingw64", "bin", "g++.exe") : ""
Compiler or simulator at line 161	Part of the Compile Code workspace or language/tool detection. Evidence: javac: [
Compiler or simulator at line 162	Part of the Compile Code workspace or language/tool detection. Evidence: "javac",
Compiler or simulator at line 163	Part of the Compile Code workspace or language/tool detection. Evidence: "C:\\Program Files\\Eclipse Adoptium\\jdk-21.0.11.10-hotspot\\bin\\javac.exe"
Compiler or simulator at line 165	Part of the Compile Code workspace or language/tool detection. Evidence: java: [
Compiler or simulator at line 166	Part of the Compile Code workspace or language/tool detection. Evidence: "java",
Compiler or simulator at line 167	Part of the Compile Code workspace or language/tool detection. Evidence: "C:\\Program Files\\Eclipse Adoptium\\jdk-21.0.11.10-hotspot\\bin\\java.exe"
Compiler or simulator at line 169	Part of the Compile Code workspace or language/tool detection. Evidence: node: [
Compiler or simulator at line 170	Part of the Compile Code workspace or language/tool detection. Evidence: /(?:^[\\V])node(?:\\.exe)?\$/i.test(process.execPath "") ? process.execPath : "",
Compiler or simulator at line 171	Part of the Compile Code workspace or language/tool detection. Evidence: "node",
Compiler or simulator at line 172	Part of the Compile Code workspace or language/tool detection. Evidence: "C:\\Program Files\\nodejs\\node.exe",
Compiler or simulator at line 173	Part of the Compile Code workspace or language/tool detection. Evidence: process.env.LOCALAPPDATA ? path.join(process.env.LOCALAPPDATA, "Programs", "nodejs", "node.exe") : ""
Compiler or simulator at line 175	Part of the Compile Code workspace or language/tool detection. Evidence: python: [process.env.PYTHON, "python", "py"],
Compiler or simulator at line 176	Part of the Compile Code workspace or language/tool detection. Evidence: iverilog: ["iverilog", "C:\\iverilog\\bin\\iverilog.exe"],
Compiler or simulator at line 177	Part of the Compile Code workspace or language/tool detection. Evidence: vvp: ["vvp", "C:\\iverilog\\bin\\vvp.exe"],
Installer at line 180	Windows setup, update, shortcut, or installation behavior. Evidence: "C:\\Program Files\\ADI\\LTspice\\LTspice.exe",
Installer at line 181	Windows setup, update, shortcut, or installation behavior. Evidence: "C:\\Program Files\\LTC\\LTspiceXVII\\XVIIx64.exe",
Installer at line 182	Windows setup, update, shortcut, or installation behavior. Evidence: "C:\\Program Files\\Analog Devices\\LTspice\\LTspice.exe",
Installer at line 183	Windows setup, update, shortcut, or installation behavior. Evidence: process.env.LOCALAPPDATA ? path.join(process.env.LOCALAPPDATA, "Programs", "ADI", "LTspice", "LTspice.exe") : ""
Local storage at line 186	Local persistence, draft state, auth cache, update snooze, or browser-side caching. Evidence: const compileToolCache = new Map();
Local storage at line 187	Local persistence, draft state, auth cache, update snooze, or browser-side caching. Evidence: const compileToolVersionCache = new Map();
Cloudflare or AI at line 192	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "Answer the visitor's question first in a concise ChatGPT-like format, then add portfolio links or context only when they help.",
Security or auth at line 210	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "Do not invent portfolio projects, credentials, employers, files, or test results that are not in the context.",

Item	Explanation
Security or auth at line 218	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: function securityHeaders(extra = {}) {
Security or auth at line 222	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "Permissions-Policy": "camera=(), microphone=(), geolocation=(), payment=(), usb=()",
Security or auth at line 230	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: ...securityHeaders(),
Local storage at line 232	Local persistence, draft state, auth cache, update snooze, or browser-side caching. Evidence: "Cache-Control": "no-store, no-cache, must-revalidate"
Compiler or simulator at line 263	Part of the Compile Code workspace or language/tool detection. Evidence: systemverilog: "systemverilog",
Compiler or simulator at line 264	Part of the Compile Code workspace or language/tool detection. Evidence: sv: "systemverilog",

Important constants and variables

Item	Explanation
root	Line 10: const root = path.dirname(fileURLToPath(import.meta.url));
portfolioRoot	Line 11: const portfolioRoot = path.resolve(process.env.OMB_PORTFOLIO_WORKSPACE root);
compileRoot	Line 12: const compileRoot = path.resolve(process.env.OMB_CODE_WORKSPACE path.join(path.dirname(root), "compile-code"));
port	Line 13: const port = Number(process.env.PORT 8080);
host	Line 14: const host = process.env.HOST "0.0.0.0";
execFileAsync	Line 15: const execFileAsync = promisify(execFile);
packageJson	Line 16: const packageJson = JSON.parse(await readFile(path.join(root, "package.json"), "utf8").catch(() => "{}"));
types	Line 18: const types = {
draftPath	Line 38: const draftPath = path.join(root, "projects.local.json");
catalogPath	Line 39: const catalogPath = path.join(root, "projects.json");
publishAuthCachePath	Line 40: const publishAuthCachePath = path.join(root, ".omb-publish-session.json");
publishPaths	Line 41: const publishPaths = [
publishAuthCacheTtlMs	Line 57: const publishAuthCacheTtlMs = 24 * 60 * 60 * 1000;
publishAuthExtendedTtlMs	Line 58: const publishAuthExtendedTtlMs = 30 * 24 * 60 * 60 * 1000;
publishAuthHistoryWindowMs	Line 59: const publishAuthHistoryWindowMs = 7 * 24 * 60 * 60 * 1000;
publishAuthExtendedThreshold	Line 60: const publishAuthExtendedThreshold = 3;
defaultSiteRepository	Line 61: const defaultSiteRepository = process.env.OMB_BUILDER_REPOSITORY "";
blockedAppUpdateVersions	Line 62: const blockedAppUpdateVersions = new Set(["0.2.16"]);
publishAuthorizationHelp	Line 63: const publishAuthorizationHelp = [
gitCandidates	Line 71: const gitCandidates = [
compileLanguageProfiles	Line 82: const compileLanguageProfiles = {
compileToolCandidates	Line 148: const compileToolCandidates = {
compileToolCache	Line 186: const compileToolCache = new Map();
compileToolVersionCache	Line 187: const compileToolVersionCache = new Map();
portfolioAiInstructions	Line 189: const portfolioAiInstructions = [
clean	Line 257: const clean = String(value "").trim().toLowerCase().replace(/[_\s-]+/g, "");
aliases	Line 258: const aliases = {
text	Line 283: const text = String(source "");

Item	Explanation
text	Line 289: const text = String(source "");
ext	Line 295: const ext = path.extname(String(fileName "").toLowerCase());
byName	Line 305: const byName = languageFromFileName(fileName, code);
source	Line 307: const source = String(code "");
profile	Line 322: const profile = compileLanguageProfiles[normalizeCodeLanguage(language)] compileLanguageProfiles.javascript;
fallback	Line 323: const fallback = profile.defaultFile;
parsed	Line 324: const parsed = path.parse(String(value fallback));
safeName	Line 325: const safeName = safeSegment(parsed.name, path.parse(fallback).name);
ext	Line 326: let ext = String(parsed.ext path.extname(fallback)).toLowerCase();
depth	Line 332: let depth = 0;
line	Line 337: const line = rawLine.trim();
formatted	Line 340: const formatted = `\${" ".repeat(depth)}\${line}`;
opens	Line 341: const opens = (line.match(/(?:/g) []).length;
closes	Line 342: const closes = (line.match(/(?:\)/g) []).length;
normalized	Line 351: const normalized = String(code "").replace(/r\n?/g, "\n").replace(/t/g, " ");
lang	Line 352: const lang = normalizeCodeLanguage(language);
entries	Line 376: let entries = [];
fullPath	Line 383: const fullPath = path.join(folder, entry.name);
found	Line 388: const found = await findExecutableUnder(path.join(folder, entry.name), names, maxDepth - 1);
candidates	Line 396: const candidates = (compileToolCandidates[toolName] [toolName]).filter(Boolean);
command	Line 406: const command = process.platform === "win32" ? "where" : "command";
args	Line 407: const args = process.platform === "win32" ? [candidate] : ["-v", candidate];
result	Line 408: const result = await execFileAsync(command, args, { timeout: 5000, windowsHide: true });
found	Line 409: const found = String(result.stdout "").split(/\r?\n/).map((line) => line.trim()).find(Boolean);
found	Line 420: const found = await findExecutableUnder(
found	Line 430: const found = await findExecutableUnder("C:\\Program Files\\Eclipse Adoptium", [`\${toolName}.exe`], 5);
elapsedSeconds	Line 444: const elapsedSeconds = Number.isFinite(result.elapsedMs) ? (result.elapsedMs / 1000).toFixed(2) : "";
elapsed	Line 445: const elapsed = elapsedSeconds ? ` in \${elapsedSeconds}s` : "";
status	Line 446: const status = result.timedOut
output	Line 449: const output = [result.stdout, result.stderr].map((part) => String(part "").trimEnd()).filter(Boolean).join("\n");
clean	Line 454: const clean = String(value "");
output	Line 464: let output = String(text "");
from	Line 466: const from = replacement?.from "";
to	Line 467: const to = replacement?.to "";
isCpp	Line 481: const isCpp = language === "cpp";
ext	Line 482: const ext = path.extname(String(fileName "").toLowerCase());
header	Line 483: const header = [".h", ".hpp", ".hh", ".hxx"].includes(ext);
parsed	Line 504: const parsed = path.parse(String(fileName "program"));
result	Line 511: const result = await runProcess(toolPath, ["--version"], { timeoutMs: 7000 });
version	Line 512: const version = [result.stdout, result.stderr]

Item	Explanation
elapsedSeconds	Line 524: const elapsedSeconds = Number.isFinite(result.elapsedMs) ? (result.elapsedMs / 1000).toFixed(2) : "";
status	Line 525: const status = result.timedOut
output	Line 528: const output = [result.stdout, result.stderr].map((part) => String(part "").trimEnd()).filter(Boolean).join("\n");
raw	Line 533: const raw = [result.stdout, result.stderr].map((part) => String(part "").trimEnd()).filter(Boolean).join("\n");
lines	Line 534: const lines = raw.split(/\r?\n/).map((line) => line.trimEnd()).filter(Boolean);
finishLines	Line 535: const finishLines = [];
signalLines	Line 536: const signalLines = [];
elapsedSeconds	Line 544: const elapsedSeconds = Number.isFinite(result.elapsedMs) ? (result.elapsedMs / 1000).toFixed(2) : "";
status	Line 545: const status = result.timedOut
body	Line 548: const body = signalLines.length
suffix	Line 555: const suffix = signalLines.length && finishLines.length ? ` \n\${finishLines.join("\n")}` : "";
timeoutMs	Line 560: const timeoutMs = options.timeoutMs 20000;

JSON/YAML-style keys detected

Item	Explanation
.css	Line 19: ".css": "text/css",
.csv	Line 20: ".csv": "text/csv",
.html	Line 21: ".html": "text/html",
.ico	Line 22: ".ico": "image/x-icon",
.js	Line 23: ".js": "application/javascript",
.json	Line 24: ".json": "application/json",
.md	Line 25: ".md": "text/markdown",
.pdf	Line 26: ".pdf": "application/pdf",
.png	Line 27: ".png": "image/png",
.jpg	Line 28: ".jpg": "image/jpeg",
.jpeg	Line 29: ".jpeg": "image/jpeg",
.svg	Line 30: ".svg": "image/svg+xml",
.txt	Line 31: ".txt": "text/plain",
.webp	Line 32: ".webp": "image/webp",
.xml	Line 33: ".xml": "application/xml",
.xlsx	Line 34: ".xlsx": "application/vnd.openxmlformats-officedocument.spreadsheetml.sheet",
.zip	Line 35: ".zip": "application/zip"
g++	Line 153: "g++": [
clang++	Line 159: "clang++": ["clang++"],
X-Content-Type-Options	Line 220: "X-Content-Type-Options": "nosniff",
Referrer-Policy	Line 221: "Referrer-Policy": "strict-origin-when-cross-origin",
Permissions-Policy	Line 222: "Permissions-Policy": "camera=(), microphone=(), geolocation=(), payment=(), usb=()",
Cross-Origin-Opener-Policy	Line 223: "Cross-Origin-Opener-Policy": "same-origin",
Content-Type	Line 231: "Content-Type": "application/json",
Cache-Control	Line 232: "Cache-Control": "no-store, no-cache, must-revalidate"
c++	Line 262: "c++": "cpp",

Item	Explanation
Accept	Line 1386: "Accept": accept,
User-Agent	Line 1387: "User-Agent": "OMB-Portfolio-BUILDER-AI"
Authorization	Line 3314: "Authorization": `Bearer \${apiKey}`,
Content-Type	Line 3315: "Content-Type": "application/json"
Content-Type	Line 3621: "Content-Type": types[path.extname(filePath)] "application/octet-stream",
Cache-Control	Line 3622: "Cache-Control": "no-store, no-cache, must-revalidate"

Function-by-function detail

securityHeaders - lines 218-226

Item	Explanation
Source location	Lines 218-226 (9 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function securityHeaders(extra = {}) {
Touches	filesystem, authentication/security data
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

218: function securityHeaders(extra = {}) {
219:   return {
220:     "X-Content-Type-Options": "nosniff",
221:     "Referrer-Policy": "strict-origin-when-cross-origin",
222:     "Permissions-Policy": "camera=(), microphone=(), geolocation=(), payment=(), usb=()",
223:     "Cross-Origin-Opener-Policy": "same-origin",
224:     ...extra
225:   };
226: }

```

sendJson - lines 228-235

Item	Explanation
Source location	Lines 228-235 (8 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function sendJson(response, status, data) {
Touches	Mostly local calculations or local UI state.
Calls detected	writeHead, securityHeaders, end, stringify
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

228: function sendJson(response, status, data) {
229:   response.writeHead(status, {
230:     ...securityHeaders(),
231:     "Content-Type": "application/json",

```

```

232:     "Cache-Control": "no-store, no-cache, must-revalidate"
233:   });
234:   response.end(JSON.stringify(data, null, 2));
235: }

```

clampText - lines 237-239

Item	Explanation
Source location	Lines 237-239 (3 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function clampText(value, maxLength = 12000) {
Touches	Mostly local calculations or local UI state.
Calls detected	slice
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

237: function clampText(value, maxLength = 12000) {
238:   return String(value || "").slice(0, maxLength);
239: }

```

stripHtmlToText - lines 241-254

Item	Explanation
Source location	Lines 241-254 (14 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function stripHtmlToText(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	replace, trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

241: function stripHtmlToText(value = "") {
242:   return String(value || "")
243:     .replace(/<script[\s\S]*?<\script>/gi, " ")
244:     .replace(/<style[\s\S]*?<\style>/gi, " ")
245:     .replace(/[<^>]+>/g, " ")
246:     .replace(/&nbsp;/gi, " ")
247:     .replace(/&amp;/gi, "&")
248:     .replace(/&lt;/gi, "<")
249:     .replace(/&gt;/gi, ">")
250:     .replace(/&quot;/gi, '"')
251:     .replace(/&#39;/gi, "'")
252:     .replace(/\\s+/g, " ")
253:     .trim();
254: }

```

normalizeCodeLanguage - lines 256-280

Item	Explanation
Source location	Lines 256-280 (25 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.

Item	Explanation
Declaration	function normalizeCodeLanguage(value = "") {
Touches	filesystem
Calls detected	trim, toLowerCase, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

256: function normalizeCodeLanguage(value = "") {
257:   const clean = String(value || "").trim().toLowerCase().replace(/[_\s-]+/g, "");
258:   const aliases = {
259:     c: "c",
260:     cpp: "cpp",
261:     cplusplus: "cpp",
262:     "c++": "cpp",
263:     systemverilog: "systemverilog",
264:     sv: "systemverilog",
265:     verilog: "verilog",
266:     v: "verilog",
267:     ltspice: "ltspice",
268:     spice: "ltspice",
269:     cir: "ltspice",
270:     java: "java",
271:     javascript: "javascript",
272:     js: "javascript",
273:     node: "javascript",
274:     python: "python",
275:     py: "python",
276:     html: "html",
277:     htm: "html"
278:   };
279:   return aliases[clean] || (compileLanguageProfiles[clean] ? clean : "javascript");
280: }

```

sourceLooksCpp - lines 282-286

Item	Explanation
Source location	Lines 282-286 (5 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function sourceLooksCpp(source = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

282: function sourceLooksCpp(source = "") {
283:   const text = String(source || "");
284:   return
285:   /#include\s*<(?:iostream|string|vector|array|map|unordered_map|memory|algorithm|optional|variant)>/.test(text)
  ||
  285:   /\b(std::|cout|cin|cerr|namespace|template\s*<|class\s+\w+|new\s+\w|delete\s+|constexpr|nullptr|using
\s+namespace)\b/.test(text);
286: }

```

sourceLooksC - lines 288-292

Item	Explanation
Source location	Lines 288-292 (5 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function sourceLooksC(source = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

288: function sourceLooksC(source = "") {
289:   const text = String(source || "");
290:   return /#include\s*<(?:stdio|stdlib|string|stdint|stdbool|math)\.h>/.test(text) ||
291:     /\b(printf|scanf|malloc|calloc|realloc|free|struct\s+\w+|typedef\s+struct)\b/.test(text);
292: }

```

languageFromFileName - lines 294-302

Item	Explanation
Source location	Lines 294-302 (9 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function languageFromFileName(fileName = "", code = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	extname, toLowerCase, sourceLooksCpp, sourceLooksC, entries, find, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

294: function languageFromFileName(fileName = "", code = "") {
295:   const ext = path.extname(String(fileName || "").toLowerCase());
296:   if (ext === ".h") {
297:     if (sourceLooksCpp(code)) return "cpp";
298:     if (sourceLooksC(code)) return "c";
299:     return "c";
300:   }
301:   return Object.entries(compileLanguageProfiles).find(([ profile ]) =>
profile.extensions.includes(ext))?.[0] || "";
302: }

```

detectCodeLanguageFromSource - lines 304-319

Item	Explanation
Source location	Lines 304-319 (16 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function detectCodeLanguageFromSource(code = "", fileName = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	languageFromFileName, b, sourceLooksCpp, sourceLooksC
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 10 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

304: function detectCodeLanguageFromSource(code = "", fileName = "") {
305:   const byName = languageFromFileName(fileName, code);
306:   if (byName) return byName;
307:   const source = String(code || "");
308:   if (/<\?[a-z][\s\S]*?>/i.test(source) || /<!doctype\s+html/i.test(source)) return "html";
309:   if (/\\b(always_ff|always_comb|always_latch|logic|interface|covergroup|assert\s+property|typedef\s+enum|
class\s+w+)\b/.test(source)) return "systemverilog";
310:   if (/\\b(module|endmodule|always|assign|reg|wire|initial|posedge|negedge)\b/.test(source)) return
"verilog";
311:   if (/^\\s*\\.?(tran|ac|dc|op|model|subckt|ends|param)\b/im.test(source) ||
/bv\\w+\\s+w+\\s+w+\\s+(?:DC|SIN|PULSE)?/i.test(source)) return "ltspice";
312:   if (/\\b(def|elif|from\\s+w+\\s+import|self|None|True|False)\b/.test(source)) return "python";
313:   if (/\\b(public\\s+class|static\\s+void\\s+main|System\\.out)\b/.test(source)) return "java";
314:   if (sourceLooksCpp(source) || sourceLooksC(source) ||
/b(#include|main\\s*\\(|puts\\s*\\(|fprintf|sizeof\\s*\\(|\\b/.test(source)) {
315:     return sourceLooksCpp(source) ? "cpp" : "c";
316:   }
317:   if (/\\b(const|let|var|function|=>|console\\.log|document\\.|window\\.)\b/.test(source)) return
"javascript";
318:   return "javascript";
319: }

```

safeCodeFileName - lines 321-329

Item	Explanation
Source location	Lines 321-329 (9 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function safeCodeFileName(value = "", language = "javascript") {
Touches	filesystem
Calls detected	normalizeCodeLanguage, parse, safeSegment, extname, toLowerCase, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

321: function safeCodeFileName(value = "", language = "javascript") {
322:   const profile = compileLanguageProfiles[normalizeCodeLanguage(language)] ||
compileLanguageProfiles.javascript;
323:   const fallback = profile.defaultFile;
324:   const parsed = path.parse(String(value || fallback));
325:   const safeName = safeSegment(parsed.name, path.parse(fallback).name);
326:   let ext = String(parsed.ext || path.extname(fallback)).toLowerCase();
327:   if (!profile.extensions.includes(ext)) ext = path.extname(fallback);
328:   return `${safeName}${ext}`;
329: }

```

indentBraceCode - lines 331-345

Item	Explanation
Source location	Lines 331-345 (15 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function indentBraceCode(code = "") {
Touches	filesystem
Calls detected	replace, split, map, trim, max, repeat, match

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

331: function indentBraceCode(code = "") {
332:   let depth = 0;
333:   return String(code || "")
334:     .replace(/\r\n?/g, "\n")
335:     .split("\n")
336:     .map((rawLine) => {
337:       const line = rawLine.trim();
338:       if (!line) return "";
339:       if (/^[{}]\]/.test(line)) depth = Math.max(0, depth - 1);
340:       const formatted = `${"  ".repeat(depth)}${line}`;
341:       const opens = (line.match(/[{\[()]/g) || []).length;
342:       const closes = (line.match(/[}\]()/g) || []).length;
343:       depth = Math.max(0, depth + opens - closes);
344:       return formatted;
345:     })

```

beautifyCode - lines 350-372

Item	Explanation
Source location	Lines 350-372 (23 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function beautifyCode(code = "", language = "javascript") {
Touches	filesystem
Calls detected	replace, normalizeCodeLanguage, includes, indentBraceCode, split, map, trim, join, trimEnd
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

350: function beautifyCode(code = "", language = "javascript") {
351:   const normalized = String(code || "").replace(/\r\n?/g, "\n").replace(/\t/g, "  ");
352:   const lang = normalizeCodeLanguage(language);
353:   if (["c", "cpp", "java", "javascript", "verilog", "systemverilog"].includes(lang)) {
354:     return indentBraceCode(normalized)
355:       .replace(/[ \t]+$/gm, "")
356:       .replace(/\n{4,}/g, "\n\n\n");
357:   }
358:   if (lang === "html") {
359:     return normalized
360:       .replace(/>\s+</g, ">\n<")
361:       .split("\n")
362:       .map((line) => line.trim())
363:       .join("\n")
364:       .trim();
365:   }
366:   return normalized
367:     .split("\n")
368:     .map((line) => line.trimEnd())
369:     .join("\n")
370:     .replace(/\n{4,}/g, "\n\n\n")
371:     .trimEnd();
372: }

```

findExecutableUnder - lines 374-392

Item	Explanation
Source location	Lines 374-392 (19 source lines).

Item	Explanation
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	async function findExecutableUnder(folder, names = [], maxDepth = 5) {
Touches	Mostly local calculations or local UI state.
Calls detected	pathExists, readdir, join, isFile, some, toLowerCase, isDirectory
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	3 await expression(s), 5 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

374: async function findExecutableUnder(folder, names = [], maxDepth = 5) {
375:   if (!folder || maxDepth < 0 || !(await pathExists(folder))) return "";
376:   let entries = [];
377:   try {
378:     entries = await readdir(folder, { withFileTypes: true });
379:   } catch {
380:     return "";
381:   }
382:   for (const entry of entries) {
383:     const fullPath = path.join(folder, entry.name);
384:     if (entry.isFile() && names.some((name) => entry.name.toLowerCase() === name.toLowerCase())) return
fullPath;
385:   }
386:   for (const entry of entries) {
387:     if (!entry.isDirectory()) continue;
388:     const found = await findExecutableUnder(path.join(folder, entry.name), names, maxDepth - 1);
389:     if (found) return found;
390:   }
391:   return "";
392: }

```

findTool - lines 394-437

Item	Explanation
Source location	Lines 394-437 (44 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	async function findTool(toolName) {
Touches	filesystem
Calls detected	has, get, filter, isAbsolute, pathExists, set, execFileAsync, split, map, trim, find, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	4 await expression(s), 6 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

394: async function findTool(toolName) {
395:   if (compileToolCache.has(toolName)) return compileToolCache.get(toolName);
396:   const candidates = (compileToolCandidates[toolName] || [toolName]).filter(Boolean);
397:   for (const candidate of candidates) {
398:     if (path.isAbsolute(candidate)) {
399:       if (await pathExists(candidate)) {
400:         compileToolCache.set(toolName, candidate);
401:         return candidate;
402:       }
403:       continue;
404:     }
405:     try {
406:       const command = process.platform === "win32" ? "where" : "command";
407:       const args = process.platform === "win32" ? [candidate] : ["-v", candidate];
408:       const result = await execFileAsync(command, args, { timeout: 5000, windowsHide: true });
409:       const found = String(result.stdout || "").split(/\r?\n/).map((line) => line.trim()).find(Boolean);
410:       if (found) {
411:         compileToolCache.set(toolName, found);

```



```

412:         return found;
413:     }
414: } catch {
415:     // Try the next candidate.
416: }
417: }
418:
419: if (["gcc", "g++"].includes(toolName) && process.env.LOCALAPPDATA) {
420:     const found = await findExecutableUnder(
421:         path.join(process.env.LOCALAPPDATA, "Microsoft", "WinGet", "Packages"),
422:         [toolName === "gcc" ? "gcc.exe" : "g++.exe"],
423:         7
424:     );
425:     compileToolCache.set(toolName, found || "");
426:     return found || "";
427: }
428:
429: if (["javac", "java"].includes(toolName)) {
430:     const found = await findExecutableUnder("C:\\Program Files\\Eclipse Adoptium", [`${toolName}.exe`],
5);
431:     compileToolCache.set(toolName, found || "");
432:     return found || "";
433: }
434:
435: compileToolCache.set(toolName, "");
436: return "";
437: }

```

terminalLine - lines 439-441

Item	Explanation
Source location	Lines 439-441 (3 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function terminalLine(label, text = "") {
Touches	filesystem
Calls detected	trim, filter, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

439: function terminalLine(label, text = "") {
440:     return [`$ ${label}`, String(text || "").trim()].filter(Boolean).join("\n");
441: }

```

processTerminalText - lines 443-451

Item	Explanation
Source location	Lines 443-451 (9 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function processTerminalText(result = {}) {
Touches	filesystem
Calls detected	isFinite, toFixed, map, trimEnd, filter, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

443: function processTerminalText(result = {}) {
444:     const elapsedSeconds = Number.isFinite(result.elapsedMs) ? (result.elapsedMs / 1000).toFixed(2) : "";

```

```

445:   const elapsed = elapsedSeconds ? ` in ${elapsedSeconds}s` : "";
446:   const status = result.timedOut
447:     ? `Timed out after ${elapsedSeconds} ${elapsedSeconds}s` : "the configured timeout"`
448:     : `Exit code ${result.code ?? "unknown"}${elapsed}`;
449:   const output = [result.stdout, result.stderr].map((part) => String(part ||
450:     "").trimEnd()).filter(Boolean).join("\n");
451:   return [status, output || (result.ok ? "Completed without diagnostic output." : "No compiler output was
returned.")]
451: }

```

pathVariantsForReplacement - lines 453-461

Item	Explanation
Source location	Lines 453-461 (9 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function pathVariantsForReplacement(value = "") {
Touches	filesystem
Calls detected	from, Set, normalize, replace, filter
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

453: function pathVariantsForReplacement(value = "") {
454:   const clean = String(value || "");
455:   return Array.from(new Set([
456:     clean,
457:     path.normalize(clean),
458:     clean.replace(/\\g, "/"),
459:     path.normalize(clean).replace(/\\g, "/")
460:   ]).filter(Boolean));
461: }

```

replacePathReferences - lines 463-474

Item	Explanation
Source location	Lines 463-474 (12 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function replacePathReferences(text = "", replacements = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	pathVariantsForReplacement, split, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

463: function replacePathReferences(text = "", replacements = []) {
464:   let output = String(text || "");
465:   for (const replacement of replacements) {
466:     const from = replacement?.from || "";
467:     const to = replacement?.to || "";
468:     if (!from || !to) continue;
469:     for (const variant of pathVariantsForReplacement(from)) {
470:       output = output.split(variant).join(to);
471:     }
472:   }
473:   return output;
474: }

```

processTerminalTextWithPaths - lines 476-478

Item	Explanation
Source location	Lines 476-478 (3 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function processTerminalTextWithPaths(result = {}, replacements = []) {
Touches	filesystem
Calls detected	replacePathReferences, processTerminalText
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
476: function processTerminalTextWithPaths(result = {}, replacements = []) {
477:   return replacePathReferences(processTerminalText(result), replacements);
478: }
```

cFamilyCompileProfile - lines 480-501

Item	Explanation
Source location	Lines 480-501 (22 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function cFamilyCompileProfile(language = "c", fileName = "main.c") {
Touches	Mostly local calculations or local UI state.
Calls detected	extname, toLowerCase, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
480: function cFamilyCompileProfile(language = "c", fileName = "main.c") {
481:   const isCpp = language === "cpp";
482:   const ext = path.extname(String(fileName || "").toLowerCase());
483:   const header = [".h", ".hpp", ".hh", ".hxx"].includes(ext);
484:   return {
485:     compiler: isCpp ? "g++" : "gcc",
486:     standard: isCpp ? "-std=c++20" : "-std=c17",
487:     languageFlag: isCpp ? "c++" : "c",
488:     label: isCpp ? "C++" : "C",
489:     header,
490:     flags: [
491:       isCpp ? "-std=c++20" : "-std=c17",
492:       "-Wall",
493:       "-Wextra",
494:       "-Wpedantic",
495:       "-Wshadow",
496:       "-O0",
497:       "-g3",
498:       "-fdiagnostics-color=never"
499:     ]
500:   };
501: }
```

cFamilyBinaryName - lines 503-506

Item	Explanation
Source location	Lines 503-506 (4 source lines).

Item	Explanation
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function cFamilyBinaryName(fileName = "program") {
Touches	Mostly local calculations or local UI state.
Calls detected	parse, safeSegment
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

503: function cFamilyBinaryName(fileName = "program") {
504:   const parsed = path.parse(String(fileName || "program"));
505:   return `${safeSegment(parsed.name, "program")}.exe`;
506: }

```

toolVersionLine - lines 508-521

Item	Explanation
Source location	Lines 508-521 (14 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	async function toolVersionLine(toolPath = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	has, get, runProcess, map, trim, filter, join, split, find, basename, set
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

508: async function toolVersionLine(toolPath = "") {
509:   if (!toolPath) return "";
510:   if (compileToolVersionCache.has(toolPath)) return compileToolVersionCache.get(toolPath);
511:   const result = await runProcess(toolPath, ["-version"], { timeoutMs: 7000 });
512:   const version = [result.stdout, result.stderr]
513:     .map((part) => String(part || "").trim())
514:     .filter(Boolean)
515:     .join("\n")
516:     .split(/\r?\n/)
517:     .map((line) => line.trim())
518:     .find(Boolean) || path.basename(toolPath);
519:   compileToolVersionCache.set(toolPath, version);
520:   return version;
521: }

```

cFamilyRunOutput - lines 523-530

Item	Explanation
Source location	Lines 523-530 (8 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function cFamilyRunOutput(result = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	isFinite, toFixed, map, trimEnd, filter, join
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

523: function cFamilyRunOutput(result = {}) {
524:   const elapsedSeconds = Number.isFinite(result.elapsedMs) ? (result.elapsedMs / 1000).toFixed(2) : "";
525:   const status = result.timedOut
526:     ? `Program timed out after ${elapsedSeconds} ? ${elapsedSeconds}s` : "the configured timeout"`
527:     : `Program exit code ${result.code ?? "unknown"}${elapsedSeconds} ? ` in ${elapsedSeconds}s` : ""`;
528:   const output = [result.stdout, result.stderr].map((part) => String(part ||
"")).trimEnd().filter(Boolean).join("\n");
529:   return [status, output || (result.ok ? "Program completed without output." : "No program output was
returned.")]>.join("\n");
530: }

```

cleanHdlSimulationOutput - lines 532-557

Item	Explanation
Source location	Lines 532-557 (26 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function cleanHdlSimulationOutput(result = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	map, trimEnd, filter, join, split, push, replace, isFinite, toFixed
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

532: function cleanHdlSimulationOutput(result = {}) {
533:   const raw = [result.stdout, result.stderr].map((part) => String(part ||
"")).trimEnd().filter(Boolean).join("\n");
534:   const lines = raw.split(/\r?\n/).map((line) => line.trimEnd()).filter(Boolean);
535:   const finishLines = [];
536:   const signalLines = [];
537:   for (const line of lines) {
538:     if (/^$finish called at/i.test(line)) {
539:       finishLines.push(line.replace(/^.*?:\d+:\s*/, ""));
540:     } else {
541:       signalLines.push(line);
542:     }
543:   }
544:   const elapsedSeconds = Number.isFinite(result.elapsedMs) ? (result.elapsedMs / 1000).toFixed(2) : "";
545:   const status = result.timedOut
546:     ? `Simulation timed out after ${elapsedSeconds} ? ${elapsedSeconds}s` : "the configured timeout"`
547:     : `Simulation exit code ${result.code ?? "unknown"}${elapsedSeconds} ? ` in ${elapsedSeconds}s` :
""`;
548:   const body = signalLines.length
549:     ? signalLines.join("\n")
550:     : finishLines.length
551:     ? finishLines.join("\n")
552:     : result.ok
553:     ? "Simulation completed without printed output."
554:     : "No simulator output was returned.";
555:   const suffix = signalLines.length && finishLines.length ? ` \n${finishLines.join("\n")}` : "";
556:   return `${status}\n${body}${suffix}`;
557: }

```

runProcess - lines 559-599

Item	Explanation
Source location	Lines 559-599 (41 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.

Item	Explanation
Declaration	async function runProcess(command, args = [], options = {}) {
Touches	filesystem, external processes
Calls detected	now, spawn, execFile, kill, on, toString, call, write, end, resolve, trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

559: async function runProcess(command, args = [], options = {}) {
560:   const timeoutMs = options.timeoutMs || 20000;
561:   const cwd = options.cwd || compileRoot;
562:   return new Promise((resolve) => {
563:     const startedAt = Date.now();
564:     const child = spawn(command, args, {
565:       cwd,
566:       env: { ...process.env, ...(options.env || {}) },
567:       shell: false,
568:       windowsHide: true
569:     });
570:     let stdout = "";
571:     let stderr = "";
572:     let timedOut = false;
573:     const timer = setTimeout(() => {
574:       timedOut = true;
575:       if (process.platform === "win32" && child.pid) {
576:         execFile("taskkill", ["/pid", String(child.pid), "/T", "/F"], { windowsHide: true }, () => {});
577:       }
578:       child.kill("SIGKILL");
579:     }, timeoutMs);
580:     child.stdout?.on("data", (chunk) => {
581:       stdout += chunk.toString();
582:     });
583:     child.stderr?.on("data", (chunk) => {
584:       stderr += chunk.toString();
585:     });
586:     if (Object.prototype.hasOwnProperty.call(options, "input")) {
587:       child.stdin?.write(String(options.input || ""));
588:     }
589:     child.stdin?.end();
590:     child.on("error", (error) => {
591:       clearTimeout(timer);
592:       resolve({ ok: false, code: -1, stdout, stderr: `${stderr}\n${error.message}`.trim(), timedOut,
elapsedMs: Date.now() - startedAt });
593:     });
594:     child.on("close", (code) => {
595:       clearTimeout(timer);
596:       resolve({ ok: !timedOut && code === 0, code, stdout, stderr, timedOut, elapsedMs: Date.now() -
startedAt });
597:     });
598:   });
599: }

```

compileToolStatus - lines 601-620

Item	Explanation
Source location	Lines 601-620 (20 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	async function compileToolStatus() {
Touches	Mostly local calculations or local UI state.
Calls detected	Set, values, flatMap, findTool, entries, every, fromEntries, map
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
601: async function compileToolStatus() {
602:   const tools = {};
603:   const uniqueTools = [...new Set(Object.values(compileLanguageProfiles).flatMap((profile) =>
profile.primaryTools))];
604:   for (const tool of uniqueTools) {
605:     tools[tool] = await findTool(tool);
606:   }
607:   const languages = {};
608:   for (const [id, profile] of Object.entries(compileLanguageProfiles)) {
609:     const required = profile.primaryTools;
610:     languages[id] = {
611:       label: profile.label,
612:       defaultFile: profile.defaultFile,
613:       extensions: profile.extensions,
614:       ready: required.every((tool) => tools[tool]),
615:       tools: Object.fromEntries(required.map((tool) => [tool, tools[tool] || "missing"])),
616:       installHints: profile.winget || []
617:     };
618:   }
619:   return { compileRoot, languages, tools };
620: }
```

isHdlLanguage - lines 622-624

Item	Explanation
Source location	Lines 622-624 (3 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function isHdlLanguage(language = "") {
Touches	filesystem
Calls detected	includes, normalizeCodeLanguage
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
622: function isHdlLanguage(language = "") {
623:   return ["verilog", "systemverilog"].includes(normalizeCodeLanguage(language));
624: }
```

inferCompileFileRole - lines 626-633

Item	Explanation
Source location	Lines 626-633 (8 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function inferCompileFileRole(fileName = "", code = "", language = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	isHdlLanguage, toLowerCase, b, tb
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
626: function inferCompileFileRole(fileName = "", code = "", language = "") {
627:   if (!isHdlLanguage(language)) return "source";
628:   const haystack = `${fileName}\n${code}`.toLowerCase();
629:   if (/\\b(tb|testbench)\\b|(^[\\-])tb([\\-]|\\.)|test[\\-]?bench|\\$dumpfile|\\$dumpvars|module\\s+tb[_a-z0-9]*\\/i.test(haystack)) {
630:     return "testbench";
631:   }
632:   return "unknown";
633: }
```

```

631:   }
632:   return "design";
633: }

```

normalizeCompileFileRole - lines 635-639

Item	Explanation
Source location	Lines 635-639 (5 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function normalizeCompileFileRole(value = "", language = "") {
Touches	filesystem
Calls detected	isHdlLanguage, trim, toLowerCase, replace, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

635: function normalizeCompileFileRole(value = "", language = "") {
636:   if (!isHdlLanguage(language)) return "source";
637:   const clean = String(value || "").trim().toLowerCase().replace(/[_\s-]+/g, "");
638:   return ["tb", "testbench", "bench"].includes(clean) ? "testbench" : "design";
639: }

```

saveCompileSource - lines 641-662

Item	Explanation
Source location	Lines 641-662 (22 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	async function saveCompileSource({ projectId, fileId, title, fileName, language, role = "", code, stdin = "" }) {
Touches	filesystem
Calls detected	normalizeCodeLanguage, detectCodeLanguageFromSource, safeSegment, safeCodeFileName, parse, resolveInsideCompileRoot, mkdir, writeFile, clampText, normalizeCompileFileRole, inferCompileFileRole, toISOString
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	4 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

641: async function saveCompileSource({ projectId, fileId, title, fileName, language, role = "", code, stdin =
"" }) {
642:   const lang = normalizeCodeLanguage(language || detectCodeLanguageFromSource(code, fileName));
643:   const projectFolder = safeSegment(projectId, "project");
644:   const codeFileName = safeCodeFileName(fileName, lang);
645:   const fileFolder = safeSegment(fileId || path.parse(codeFileName).name, path.parse(codeFileName).name);
646:   const folder = resolveInsideCompileRoot(projectFolder, fileFolder);
647:   const sourcePath = resolveInsideCompileRoot(projectFolder, fileFolder, codeFileName);
648:   await mkdir(folder, { recursive: true });
649:   await writeFile(sourcePath, String(code || ""), "utf8");
650:   if (stdin) await writeFile(resolveInsideCompileRoot(projectFolder, fileFolder, "stdin.txt"),
String(stdin), "utf8");
651:   const metadata = {
652:     id: fileFolder,
653:     title: clampText(title || codeFileName, 180),
654:     fileName: codeFileName,
655:     language: lang,
656:     role: normalizeCompileFileRole(role || inferCompileFileRole(codeFileName, code, lang), lang),
657:     sourcePath,
658:     savedAt: new Date().toISOString()
659:   };
660:   await writeFile(resolveInsideCompileRoot(projectFolder, fileFolder, "compile-meta.json"),
`${JSON.stringify(metadata, null, 2)}\n`, "utf8");

```



```

661:   return metadata;
662: }

```

javaMainClassName - lines 664-670

Item	Explanation
Source location	Lines 664-670 (7 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function javaMainClassName(code = "", fileName = "Main.java") {
Touches	Mostly local calculations or local UI state.
Calls detected	match, parse
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

664: function javaMainClassName(code = "", fileName = "Main.java") {
665:   const source = String(code || "");
666:   const publicMatch = source.match(/bpublic\s+class\s+([A-Za-z_$][\w$]*)/);
667:   if (publicMatch) return publicMatch[1];
668:   const classMatch = source.match(/bclass\s+([A-Za-z_$][\w$]*)/);
669:   return classMatch?.[1] || path.parse(fileName).name || "Main";
670: }

```

compileCacheKey - lines 672-677

Item	Explanation
Source location	Lines 672-677 (6 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function compileCacheKey(parts = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	createHash, update, stringify, digest, slice
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

672: function compileCacheKey(parts = {}) {
673:   return createHash("sha256")
674:     .update(JSON.stringify(parts))
675:     .digest("hex")
676:     .slice(0, 24);
677: }

```

compileCacheDirectory - lines 679-681

Item	Explanation
Source location	Lines 679-681 (3 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function compileCacheDirectory(projectId, language, key) {
Touches	Mostly local calculations or local UI state.
Calls detected	resolveInsideCompileRoot, safeSegment

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

679: function compileCacheDirectory(projectId, language, key) {
680:   return resolveInsideCompileRoot(safeSegment(projectId, "project"), ".build-cache",
safeSegment(language, "language"), key);
681: }

```

cachedBuildLine - lines 683-685

Item	Explanation
Source location	Lines 683-685 (3 source lines).
Purpose	Reads, writes, or validates cached state.
Declaration	function cachedBuildLine(type, outputPath) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

683: function cachedBuildLine(type, outputPath) {
684:   return `${type} cache hit: using existing artifact\n${outputPath}`;
685: }

```

hdlModuleNames - lines 687-689

Item	Explanation
Source location	Lines 687-689 (3 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function hdlModuleNames(code = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	matchAll, map
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

687: function hdlModuleNames(code = "") {
688:   return [...String(code || "").matchAll(/bmodule\s+([A-Za-z_$][\w$]*)/g)].map((match) => match[1]);
689: }

```

hdlHasWaveDump - lines 691-694

Item	Explanation
Source location	Lines 691-694 (4 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function hdlHasWaveDump(code = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

691: function hdlHasWaveDump(code = "") {
692:   const source = String(code || "");
693:   return /\$dumpfile\s*\(/.test(source) && /\$dumpvars\s*\(/.test(source);
694: }

```

hdlFilesFromPayload - lines 696-727

Item	Explanation
Source location	Lines 696-727 (32 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function hdlFilesFromPayload(payload = {}, activeFileName = "", activeLanguage = "verilog") {
Touches	filesystem
Calls detected	isArray, Map, normalizeCodeLanguage, detectCodeLanguageFromSource, isHdlLanguage, trim, safeCodeFileName, safeSegment, parse, set, clampText, normalizeCompileFileRole
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

696: function hdlFilesFromPayload(payload = {}, activeFileName = "", activeLanguage = "verilog") {
697:   const incoming = Array.isArray(payload.workspaceFiles) ? payload.workspaceFiles : [];
698:   const byId = new Map();
699:   const addFile = (item = {}, fallbackId = "") => {
700:     const code = String(item.code || "");
701:     const language = normalizeCodeLanguage(item.language || detectCodeLanguageFromSource(code,
item.fileName));
702:     if (!isHdlLanguage(language) || !code.trim()) return;
703:     const fileName = safeCodeFileName(item.fileName || activeFileName ||
compileLanguageProfiles[language].defaultFile, language);
704:     const id = safeSegment(item.id || fallbackId || fileName, path.parse(fileName).name);
705:     byId.set(id, {
706:       id,
707:       title: clampText(item.title || fileName, 180),
708:       fileName,
709:       language,
710:       role: normalizeCompileFileRole(item.role || inferCompileFileRole(fileName, code, language),
language),
711:       code
712:     });
713:   };
714:   incoming.forEach((item, index) => addFile(item, `hdl-${index + 1}`));
715:   addFile({
716:     id: payload.fileId,
717:     title: payload.title,
718:     fileName: activeFileName,
719:     language: activeLanguage,
720:     role: payload.role,
721:     code: payload.code
722:   }, "active");
723:   return [...byId.values()].sort((a, b) => {

```

```

724:     if (a.role === b.role) return a.fileName.localeCompare(b.fileName);
725:     return a.role === "testbench" ? 1 : -1;
726:   });
727: }

```

compileActionFromPayload - lines 729-733

Item	Explanation
Source location	Lines 729-733 (5 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function compileActionFromPayload(value = "", language = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, toLowerCase, replace, includes, isHdlLanguage
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

729: function compileActionFromPayload(value = "", language = "") {
730:   const clean = String(value || "").trim().toLowerCase().replace(/[_\s-]+/g, "-");
731:   if (["compile", "build", "run", "simulate"].includes(clean)) return clean;
732:   return isHdlLanguage(language) ? "simulate" : "run";
733: }

```

compileActionLabel - lines 735-740

Item	Explanation
Source location	Lines 735-740 (6 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function compileActionLabel(action = "run", language = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	isHdlLanguage
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

735: function compileActionLabel(action = "run", language = "") {
736:   if (action === "compile") return "Compile";
737:   if (action === "build") return "Build project";
738:   if (action === "simulate") return "Simulate";
739:   return isHdlLanguage(language) ? "Simulate" : "Run";
740: }

```

compileWorkspaceFilesFromPayload - lines 742-770

Item	Explanation
Source location	Lines 742-770 (29 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function compileWorkspaceFilesFromPayload(payload = {}, activeFileName = "", activeLanguage = "javascript") {
Touches	filesystem

Item	Explanation
Calls detected	isArray, Map, trim, normalizeCodeLanguage, detectCodeLanguageFromSource, safeCodeFileName, safeSegment, parse, set, clampText, normalizeCompileFileRole, inferCompileFileRole
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

742: function compileWorkspaceFilesFromPayload(payload = {}, activeFileName = "", activeLanguage =
"javascript") {
743:   const incoming = Array.isArray(payload.workspaceFiles) ? payload.workspaceFiles : [];
744:   const byId = new Map();
745:   const addFile = (item = {}, fallbackId = "") => {
746:     const code = String(item.code || "");
747:     if (!code.trim()) return;
748:     const language = normalizeCodeLanguage(item.language || detectCodeLanguageFromSource(code,
item.fileName));
749:     const fileName = safeCodeFileName(item.fileName || activeFileName ||
compileLanguageProfiles[language]?.defaultFile || "source.txt", language);
750:     const id = safeSegment(item.id || fallbackId || fileName, path.parse(fileName).name);
751:     byId.set(id, {
752:       id,
753:       title: clampText(item.title || fileName, 180),
754:       fileName,
755:       language,
756:       role: normalizeCompileFileRole(item.role || inferCompileFileRole(fileName, code, language),
language),
757:       code
758:     });
759:   };
760:   incoming.forEach((item, index) => addFile(item, `workspace-${index + 1}`));
761:   addFile({
762:     id: payload.fileId,
763:     title: payload.title,
764:     fileName: activeFileName,
765:     language: activeLanguage,
766:     role: payload.role,
767:     code: payload.code
768:   }, "active");
769:   return [...byId.values()].sort((a, b) => a.fileName.localeCompare(b.fileName));
770: }

```

writeCompileWorkspaceSources - lines 772-788

Item	Explanation
Source location	Lines 772-788 (17 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	async function writeCompileWorkspaceSources(files = [], targetDir = "", options = {}) {
Touches	filesystem
Calls detected	mkdir, Map, includes, normalizeCodeLanguage, extname, toLowerCase, parse, get, set, join, writeFile, push
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

772: async function writeCompileWorkspaceSources(files = [], targetDir = "", options = {}) {
773:   await mkdir(targetDir, { recursive: true });
774:   const seen = new Map();
775:   const written = [];
776:   for (const file of files) {
777:     if (options.languages?.length && !options.languages.includes(normalizeCodeLanguage(file.language)))
continue;
778:     if (options.extensions?.length &&
!options.extensions.includes(path.extname(file.fileName).toLowerCase())) continue;
779:     const parsed = path.parse(file.fileName);
780:     const count = seen.get(file.fileName) || 0;

```

```

781:     seen.set(file.fileName, count + 1);
782:     const uniqueName = count ? `${parsed.name}_${count}${parsed.ext}` : file.fileName;
783:     const sourcePath = path.join(targetDir, uniqueName);
784:     await writeFile(sourcePath, file.code, "utf8");
785:     written.push({ ...file, uniqueName, sourcePath });
786: }
787: return written;
788: }

```

sourceDisplayName - lines 790-792

Item	Explanation
Source location	Lines 790-792 (3 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function sourceDisplayName(file = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	basename
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

790: function sourceDisplayName(file = {}) {
791:   return file.uniqueName || file.fileName || path.basename(file.sourcePath || "source");
792: }

```

cFamilyWorkspaceSources - lines 794-803

Item	Explanation
Source location	Lines 794-803 (10 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function cFamilyWorkspaceSources(files = [], language = "c", activeFileName = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	Set, filter, has, extname, toLowerCase, some, push
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

794: function cFamilyWorkspaceSources(files = [], language = "c", activeFileName = "") {
795:   const exts = language === "cpp"
796:     ? new Set([".cpp", ".cc", ".cxx", ".c++", ".hpp", ".hh", ".hxx", ".h"])
797:     : new Set([".c", ".h"]);
798:   const selected = files.filter((file) => exts.has(path.extname(file.fileName).toLowerCase()));
799:   if (!selected.some((file) => file.fileName === activeFileName)) {
800:     selected.push(...files.filter((file) => file.fileName === activeFileName));
801:   }
802:   return selected;
803: }

```

writeHdlSimulationSources - lines 805-821

Item	Explanation
Source location	Lines 805-821 (17 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.

Item	Explanation
Declaration	async function writeHdlSimulationSources(files = [], cacheDir = "") {
Touches	filesystem
Calls detected	join, rm, mkdir, Map, parse, get, set, writeFile, push
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	3 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

805: async function writeHdlSimulationSources(files = [], cacheDir = "") {
806:   const sourceDir = path.join(cacheDir, "sources");
807:   await rm(sourceDir, { recursive: true, force: true });
808:   await mkdir(sourceDir, { recursive: true });
809:   const seen = new Map();
810:   const written = [];
811:   for (const file of files) {
812:     const parsed = path.parse(file.fileName);
813:     const count = seen.get(file.fileName) || 0;
814:     seen.set(file.fileName, count + 1);
815:     const uniqueName = count ? `${parsed.name}_${count}${parsed.ext}` : file.fileName;
816:     const sourcePath = path.join(sourceDir, uniqueName);
817:     await writeFile(sourcePath, file.code, "utf8");
818:     written.push({ ...file, uniqueName, sourcePath });
819:   }
820:   return written;
821: }

```

findFilesByExtension - lines 823-836

Item	Explanation
Source location	Lines 823-836 (14 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	async function findFilesByExtension(folder = "", extension = ".vcd", maxDepth = 3) {
Touches	Mostly local calculations or local UI state.
Calls detected	pathExists, readdir, join, isDirectory, push, isFile, toLowerCase, endsWith
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	3 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

823: async function findFilesByExtension(folder = "", extension = ".vcd", maxDepth = 3) {
824:   if (!folder || maxDepth < 0 || !(await pathExists(folder))) return [];
825:   const entries = await readdir(folder, { withFileTypes: true }).catch(() => []);
826:   const matches = [];
827:   for (const entry of entries) {
828:     const fullPath = path.join(folder, entry.name);
829:     if (entry.isDirectory()) {
830:       matches.push(...await findFilesByExtension(fullPath, extension, maxDepth - 1));
831:     } else if (entry.isFile() && entry.name.toLowerCase().endsWith(extension.toLowerCase())) {
832:       matches.push(fullPath);
833:     }
834:   }
835:   return matches;
836: }

```

normalizeVcdValue - lines 838-844

Item	Explanation
Source location	Lines 838-844 (7 source lines).

Item	Explanation
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeVcdValue(raw = "") {
Touches	filesystem
Calls detected	trim, toLowerCase, slice, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

838: function normalizeVcdValue(raw = "") {
839:   const clean = String(raw || "").trim();
840:   if (!clean) return "x";
841:   if (/^[01xz]$/i.test(clean)) return clean.toLowerCase();
842:   if (/^[br]^[01xz_]+$/i.test(clean)) return clean.slice(1).replace(/_/g, "").toLowerCase();
843:   return clean.toLowerCase();
844: }

```

parseVcdScopeText - lines 846-933

Item	Explanation
Source location	Lines 846-933 (88 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function parseVcdScopeText(text = "", source = "waveform.vcd") {
Touches	filesystem, authentication/security data
Calls detected	split, Map, trim, startsWith, replace, push, includes, match, pop, has, set, filter
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

846: function parseVcdScopeText(text = "", source = "waveform.vcd") {
847:   const lines = String(text || "").split(/\r?\n/);
848:   const scopes = [];
849:   const signalsByCode = new Map();
850:   const timeScaleParts = [];
851:   let inTimescale = false;
852:   let time = 0;
853:   let maxTime = 0;
854:   let definitionDone = false;
855:   for (const rawLine of lines) {
856:     const line = rawLine.trim();
857:     if (!line) continue;
858:     if (line.startsWith("$timescale")) {
859:       inTimescale = true;
860:       const inline = line.replace("$timescale", "").replace("$end", "").trim();
861:       if (inline) timeScaleParts.push(inline);
862:       continue;
863:     }
864:     if (inTimescale) {
865:       if (line === "$end") {
866:         inTimescale = false;
867:       } else {
868:         timeScaleParts.push(line.replace("$end", "").trim());
869:         if (line.includes("$end")) inTimescale = false;
870:       }
871:       continue;
872:     }
873:     const scopeMatch = line.match(/^\$scope\s+\$S+\$s+(.?)\$s+\$end$/);
874:     if (scopeMatch) {
875:       scopes.push(scopeMatch[1]);
876:       continue;

```



```

877:     }
878:     if (/^\$supscope\b/.test(line)) {
879:         scopes.pop();
880:         continue;
881:     }
882:     const varMatch = line.match(/^\$var\s+\$+\s+(\d+)\s+(\$+)\s+(.?)\s+\$end$/);
883:     if (varMatch) {
884:         const [, width, code, reference] = varMatch;
885:         if (!signalsByCode.has(code) && signalsByCode.size < 64) {
886:             const cleanReference = reference.replace(/s+\[^\s\]+\$/ , "");
887:             signalsByCode.set(code, {
888:                 code,
889:                 name: [...scopes, cleanReference].filter(Boolean).join("."),
890:                 reference: cleanReference,
891:                 width: Number(width) || 1,
892:                 changes: []
893:             });
894:         }
895:         continue;
896:     }
897:     if (line.startsWith("$enddefinitions")) {
898:         definitionDone = true;
899:         continue;
900:     }
901:     if (!definitionDone) continue;
902:     if (line.startsWith("#")) {
903:         time = Number(line.slice(1)) || 0;
904:         maxTime = Math.max(maxTime, time);
905:         continue;
906:     }
907:     let code = "";
908:     let value = "";
909:     const vectorMatch = line.match(/^(br)[01xz_]+\s+(\$+)\$/i);
910:     if (vectorMatch) {
911:         value = normalizeVcdValue(vectorMatch[1]);
912:         code = vectorMatch[2];
913:     } else if (/^[01xz]/i.test(line)) {
914:         value = normalizeVcdValue(line[0]);
915:         code = line.slice(1).trim();
916:     }
917:     const signal = signalsByCode.get(code);
918:     if (!signal || !value) continue;
919:     const previous = signal.changes[signal.changes.length - 1];
920:     if (!previous || previous.value !== value || previous.time !== time) {
921:         if (signal.changes.length < 600) signal.changes.push({ time, value });
922:     }
923: }
924: const signals = [...signalsByCode.values()]
925:     .filter((signal) => signal.changes.length)
926:     .slice(0, 32);
927: return {
928:     source: path.basename(source),
929:     timeScale: timeScaleParts.join(" ").replace(/s+/g, " ").trim() || "ticks",
930:     maxTime,
931:     signals
932: };
933: }

```

readHdlWaveform - lines 935-942

Item	Explanation
Source location	Lines 935-942 (8 source lines).
Purpose	Loads data from disk, network, browser storage, or a service.
Declaration	async function readHdlWaveform(cacheDir = "") {
Touches	filesystem, authentication/security data
Calls detected	findFilesByExtension, sort, readFile, slice, parseVcdScopeText
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

935: async function readHdlWaveform(cacheDir = "") {
936:     const vcdFiles = await findFilesByExtension(cacheDir, ".vcd", 4);
937:     if (!vcdFiles.length) return null;
938:     const selected = vcdFiles.sort((a, b) => a.length - b.length)[0];
939:     const text = await readFile(selected, "utf8");

```

```

940:   const limited = text.length > 6_000_000 ? text.slice(0, 6_000_000) : text;
941:   return parseVcdScopeText(limited, selected);
942: }

```

clearHdlWaveforms - lines 944-947

Item	Explanation
Source location	Lines 944-947 (4 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	async function clearHdlWaveforms(cacheDir = "") {
Touches	filesystem
Calls detected	findFilesByExtension, all, map, rm
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

944: async function clearHdlWaveforms(cacheDir = "") {
945:   const vcdFiles = await findFilesByExtension(cacheDir, ".vcd", 4);
946:   await Promise.all(vcdFiles.map((filePath) => rm(filePath, { force: true })).catch(() => {}));
947: }

```

compileAndRunCode - lines 949-1319

Item	Explanation
Source location	Lines 949-1319 (371 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	async function compileAndRunCode(payload = {}) {
Touches	filesystem, authentication/security data
Calls detected	trim, toLowerCase, normalizeCodeLanguage, detectCodeLanguageFromSource, safeCodeFileName, saveCompileSource, safeSegment, resolveInsideCompileRoot, compileActionFromPayload, compileWorkspaceFilesFromPayload, async, now
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	44 await expression(s), 15 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

949: async function compileAndRunCode(payload = {}) {
950:   const requestedLanguage = String(payload.language || "").trim().toLowerCase();
951:   const language = requestedLanguage && requestedLanguage !== "auto"
952:     ? normalizeCodeLanguage(requestedLanguage)
953:     : detectCodeLanguageFromSource(payload.code, payload.fileName);
954:   const profile = compileLanguageProfiles[language] || compileLanguageProfiles.javascript;
955:   const fileName = safeCodeFileName(payload.fileName, language);
956:   const saved = await saveCompileSource({ ..payload, language, fileName });
957:   const projectFolder = safeSegment(payload.projectId, "project");
958:   const sourcePath = resolveInsideCompileRoot(projectFolder, safeSegment(saved.id), saved.fileName);
959:   const sourceCode = String(payload.code || "");
960:   const action = compileActionFromPayload(payload.action, language);
961:   const allWorkspaceFiles = compileWorkspaceFilesFromPayload(payload, saved.fileName, language);
962:   let runDir = "";
963:   let runSourcePath = "";
964:   let runWorkspaceSources = [];
965:   const ensureRunSource = async (options = {}) => {
966:     if (runDir && runSourcePath) return { runDir, runSourcePath, writtenSources: runWorkspaceSources };
967:     const runId = `${Date.now()}-${Math.random().toString(16).slice(2)}`;
968:     runDir = resolveInsideCompileRoot(projectFolder, ".runs", runId);
969:     await mkdir(runDir, { recursive: true });
970:     const filesToWrite = Array.isArray(options.files) && options.files.length ? options.files :
allWorkspaceFiles;

```

```

971:     runWorkspaceSources = await writeCompileWorkspaceSources(filesToWrite, runDir, {
972:       languages: options.languages,
973:       extensions: options.extensions
974:     });
975:     const active = runWorkspaceSources.find((file) => file.id === saved.id || file.fileName ===
saved.fileName);
976:     runSourcePath = active?.sourcePath || path.join(runDir, saved.fileName);
977:     if (!active) {
978:       await cp(sourcePath, runSourcePath, { force: true });
979:       runWorkspaceSources.push({
980:         id: saved.id,
981:         title: saved.title,
982:         fileName: saved.fileName,
983:         language,
984:         role: saved.role,
985:         code: sourceCode,
986:         uniqueName: saved.fileName,
987:         sourcePath: runSourcePath
988:       });
989:     }
990:     return { runDir, runSourcePath, writtenSources: runWorkspaceSources };
991:   };
992:   const stdin = String(payload.stdin || "");
993:   const forceRebuild = payload.forceRebuild === true;
994:   const terminal = [];
995:   terminal.push(`${compileActionLabel(action, language)} request for ${saved.fileName}`);
996:   const append = (label, result) => {
997:     terminal.push(terminalLine(label, processTerminalText(result)));
998:   };
999:   const missing = async (tools) => {
1000:     const found = {};
1001:     for (const tool of tools) found[tool] = await findTool(tool);
1002:     const missingTools = tools.filter((tool) => !found[tool]);
1003:     return { found, missingTools };
1004:   };
1005:
1006:   if (language === "html") {
1007:     const openTags = (String(payload.code || "").match(/<[a-z][\w:-]*\b[^>*/>/gi) || []).length;
1008:     const closeTags = (String(payload.code || "").match(/<\/[a-z][\w:-]*>/gi) || []).length;
1009:     const ok = openTags >= closeTags;
1010:     return {
1011:       ok,
1012:       language,
1013:       saved,
1014:       terminal: [
1015:         "HTML validation pass",
1016:         ok ? "No obvious tag-balance issue was detected." : "The file may have more closing tags than
opening tags."
1017:       ],
1018:       `Saved source: ${saved.sourcePath}`
1019:     }.join("\n");
1020:   }
1021:
1022:   const stdinOptions = stdin ? { input: stdin } : {};
1023:   let ok = false;
1024:
1025:   if (language === "javascript") {
1026:     const node = await findTool("node");
1027:     if (!node) return { ok: false, language, saved, terminal: "Node.js was not found. Install Node.js to
run JavaScript." };
1028:     const jsFiles = allWorkspaceFiles.filter((file) => normalizeCodeLanguage(file.language) ===
"javascript");
1029:     const runSource = await ensureRunSource({ files: jsFiles.length ? jsFiles : allWorkspaceFiles,
languages: ["javascript"] });
1030:     const targets = action === "build"
? runSource.writtenSources.filter((file) => normalizeCodeLanguage(file.language) === "javascript")
: runSource.writtenSources.filter((file) => file.sourcePath === runSource.runSourcePath);
1031:     ok = true;
1032:     for (const target of targets.length ? targets : [{ sourcePath: runSource.runSourcePath, uniqueName:
saved.fileName }]) {
1033:       const check = await runProcess(node, ["--check", target.sourcePath], { cwd: runSource.runDir,
timeoutMs: 15000 });
1034:       terminal.push(terminalLine(`${path.basename(node)} --check ${sourceDisplayName(target)}`,
processTerminalText(check)));
1035:       ok = ok && check.ok;
1036:     }
1037:     if (action === "compile" || action === "build") {
1038:       terminal.push(ok ? "JavaScript syntax check completed." : "JavaScript syntax check found errors.");
1039:     } else if (ok) {
1040:       const run = await runProcess(node, [runSource.runSourcePath], { cwd: runSource.runDir, timeoutMs:
20000, ...stdinOptions });
1041:       append(`${path.basename(node)} ${saved.fileName}`, run);
1042:       ok = run.ok;
1043:     }
1044:   } else if (language === "python") {
1045:     const python = await findTool("python");
1046:     if (!python) return { ok: false, language, saved, terminal: "Python was not found. Install Python to
run Python code." };
1047:     const pyFiles = allWorkspaceFiles.filter((file) => normalizeCodeLanguage(file.language) ===
"python");
1048:     const runSource = await ensureRunSource({ files: pyFiles.length ? pyFiles : allWorkspaceFiles,
languages: ["python"] });

```

```

1051:     const targets = action === "build"
1052:       ? runSource.writtenSources.filter((file) => normalizeCodeLanguage(file.language) === "python")
1053:       : runSource.writtenSources.filter((file) => file.sourcePath === runSource.runSourcePath);
1054:     ok = true;
1055:     for (const target of targets.length ? targets : [{ sourcePath: runSource.runSourcePath, uniqueName:
saved.fileName }]) {
1056:       const check = await runProcess(python, ["-m", "py_compile", target.sourcePath], { cwd:
runSource.runDir, timeoutMs: 15000 });
1057:       terminal.push(terminalLine(`${path.basename(python)} -m py_compile ${sourceDisplayName(target)}`,
processTerminalText(check)));
1058:       ok = ok && check.ok;
1059:     }
1060:     if (action === "compile" || action === "build") {
1061:       terminal.push(ok ? "Python bytecode compile check completed." : "Python compile check found
errors.");
1062:     } else if (ok) {
1063:       const run = await runProcess(python, [runSource.runSourcePath], { cwd: runSource.runDir, timeoutMs:
20000, ...stdinOptions });
1064:       append(`${path.basename(python)} ${saved.fileName}`, run);
1065:       ok = run.ok;
1066:     }
1067:   } else if (language === "c" || language === "cpp") {
1068:     const cProfile = cFamilyCompileProfile(language, saved.fileName);
... excerpt truncated after 120 lines. Open the source file for the rest of this function.

```

append - lines 996-998

Item	Explanation
Source location	Lines 996-998 (3 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	const append = (label, result) => {
Touches	filesystem
Calls detected	push, terminalLine, processTerminalText
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

996:   const append = (label, result) => {
997:     terminal.push(terminalLine(label, processTerminalText(result)));
998:   };

```

missing - lines 999-1004

Item	Explanation
Source location	Lines 999-1004 (6 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	const missing = async (tools) => {
Touches	Mostly local calculations or local UI state.
Calls detected	async, findTool, filter
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

999:   const missing = async (tools) => {
1000:     const found = {};
1001:     for (const tool of tools) found[tool] = await findTool(tool);
1002:     const missingTools = tools.filter((tool) => !found[tool]);
1003:     return { found, missingTools };

```

```
1004:    };
```

installCompilerTools - lines 1321-1352

Item	Explanation
Source location	Lines 1321-1352 (32 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	async function installCompilerTools(language = "") {
Touches	filesystem
Calls detected	normalizeCodeLanguage, findTool, runProcess, push, terminalLine, join, processTerminalText, clear, compileToolStatus
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	3 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1321: async function installCompilerTools(language = "") {
1322:   const lang = normalizeCodeLanguage(language);
1323:   const profile = compileLanguageProfiles[lang] || compileLanguageProfiles.javascript;
1324:   if (!profile.winget?.length) {
1325:     return {
1326:       ok: false,
1327:       language: lang,
1328:       terminal: `${profile.label} does not have an automatic Winget compiler install configured. Install
the required tools manually, then restart the builder.`
1329:     };
1330:   }
1331:   const winget = await findTool("winget") || "winget";
1332:   const terminal = [];
1333:   let ok = true;
1334:   for (const packageId of profile.winget) {
1335:     const args = [
1336:       "install",
1337:       "--source",
1338:       "winget",
1339:       "--accept-source-agreements",
1340:       "--accept-package-agreements",
1341:       "--silent",
1342:       "--id",
1343:       packageId,
1344:       "--exact"
1345:     ];
1346:     const result = await runProcess(winget, args, { cwd: root, timeoutMs: 600000 });
1347:     terminal.push(terminalLine(`winget ${args.join(" ")}`, processTerminalText(result)));
1348:     ok = ok && result.ok;
1349:   }
1350:   compileToolCache.clear();
1351:   return { ok, language: lang, terminal: terminal.join("\n\n"), tools: await compileToolStatus() };
1352: }
```

sourceLooksTextual - lines 1354-1360

Item	Explanation
Source location	Lines 1354-1360 (7 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function sourceLooksTextual(source = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	toLowerCase, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1354: function sourceLooksTextual(source = {}) {
1355:   const url = String(source.url || "");
1356:   const kind = String(source.kind || "").toLowerCase();
1357:   if (["text", "webpage", "github", "linkedin", "public_profile", "resume_text", "code"].includes(kind))
return true;
1358:   return /\. (txt|md|markdown|csv|json|xml|log|c|h|cpp|hpp|py|js|mjs|ts|tsx|v|sv|vhdl?|spice|cir|net|asc|s
ch|kicad_sch|kicad_pcb|ino|xdc|sdc|tcl)$/.test(url)
1359:   || /github\.com|raw\.githubusercontent\.com/.test(url);
1360: }
```

sourceUrlAllowed - lines 1362-1379

Item	Explanation
Source location	Lines 1362-1379 (18 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function sourceUrlAllowed(url) {
Touches	Mostly local calculations or local UI state.
Calls detected	URL, toLowerCase, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1362: function sourceUrlAllowed(url) {
1363:   try {
1364:     const parsed = new URL(url);
1365:     const hostName = parsed.hostname.toLowerCase();
1366:     return [
1367:       "localhost",
1368:       "127.0.0.1",
1369:       "github.com",
1370:       "api.github.com",
1371:       "raw.githubusercontent.com",
1372:       "gist.githubusercontent.com",
1373:       "linkedin.com",
1374:       "www.linkedin.com"
1375:     ].includes(hostName);
1376:   } catch {
1377:     return false;
1378:   }
1379: }
```

gitHubHeaders - lines 1384-1389

Item	Explanation
Source location	Lines 1384-1389 (6 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	function gitHubHeaders(accept = "application/vnd.github+json") {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1384: function gitHubHeaders(accept = "application/vnd.github+json") {
1385:   return {
1386:     "Accept": accept,
```

```

1387:     "User-Agent": "OMB-Portfolio-Builder-AI"
1388:   };
1389: }

```

parseGitHubSourceUrl - lines 1391-1418

Item	Explanation
Source location	Lines 1391-1418 (28 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	function parseGitHubSourceUrl(url) {
Touches	Mostly local calculations or local UI state.
Calls detected	URL, toLowerCase, split, filter, slice, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 7 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1391: function parseGitHubSourceUrl(url) {
1392:   try {
1393:     const parsed = new URL(url);
1394:     const host = parsed.hostname.toLowerCase();
1395:     const parts = parsed.pathname.split("/").filter(Boolean);
1396:     if (host === "raw.githubusercontent.com" && parts.length >= 4) {
1397:       return {
1398:         type: "file",
1399:         owner: parts[0],
1400:         repo: parts[1],
1401:         branch: parts[2],
1402:         filePath: parts.slice(3).join("/")
1403:       };
1404:     }
1405:     if (host !== "github.com" || !parts.length) return null;
1406:     if (parts.length === 1) return { type: "profile", owner: parts[0] };
1407:     const base = { owner: parts[0], repo: parts[1] };
1408:     if (parts[2] === "blob" && parts.length >= 5) {
1409:       return { ...base, type: "file", branch: parts[3], filePath: parts.slice(4).join("/") };
1410:     }
1411:     if (parts[2] === "tree" && parts.length >= 4) {
1412:       return { ...base, type: "repo", branch: parts[3] };
1413:     }
1414:     return { ...base, type: "repo" };
1415:   } catch {
1416:     return null;
1417:   }
1418: }

```

fetchGitHubJson - lines 1420-1424

Item	Explanation
Source location	Lines 1420-1424 (5 source lines).
Purpose	Loads data from disk, network, browser storage, or a service.
Declaration	async function fetchGitHubJson(url) {
Touches	network/API requests
Calls detected	fetch, gitHubHeaders, json
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1420: async function fetchGitHubJson(url) {

```

```

1421:   const response = await fetch(url, { headers: gitHubHeaders() });
1422:   if (!response.ok) return null;
1423:   return response.json().catch(() => null);
1424: }

```

fetchLimitedText - lines 1426-1434

Item	Explanation
Source location	Lines 1426-1434 (9 source lines).
Purpose	Loads data from disk, network, browser storage, or a service.
Declaration	async function fetchLimitedText(url, maxLength = 12000) {
Touches	network/API requests
Calls detected	fetch, gitHubHeaders, clampText, text, get, stripHtmlToText
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1426: async function fetchLimitedText(url, maxLength = 12000) {
1427:   const response = await fetch(url, {
1428:     headers: gitHubHeaders("text/plain,text/markdown,text/html,application/json;q=0.8,*/*;q=0.1")
1429:   });
1430:   if (!response.ok) return "";
1431:   const rawText = clampText(await response.text(), maxLength);
1432:   const contentType = response.headers.get("Content-Type") || "";
1433:   return /html/i.test(contentType) ? stripHtmlToText(rawText) : rawText;
1434: }

```

githubQuestionTokens - lines 1436-1442

Item	Explanation
Source location	Lines 1436-1442 (7 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	function githubQuestionTokens(question = "") {
Touches	authentication/security data
Calls detected	Set, toLowerCase, replace, split, filter, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1436: function githubQuestionTokens(question = "") {
1437:   return [...new Set(String(question || "").
1438:     .toLowerCase()
1439:     .replace(/[\^a-z0-9_#\+\.\s-]/g, " ")
1440:     .split(/\s+/)
1441:     .filter((token) => token.length >= 3 && ![
1442:       "the", "and", "with", "from", "show", "code", "file",

```

scoreGitHubFile - lines 1444-1455

Item	Explanation
Source location	Lines 1444-1455 (12 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.

Item	Explanation
Declaration	function scoreGitHubFile(filePath = "", question = "") {
Touches	filesystem, authentication/security data
Calls detected	toLowerCase, githubQuestionTokens, b, forEach, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1444: function scoreGitHubFile(filePath = "", question = "") {
1445:   const cleanPath = filePath.toLowerCase();
1446:   const tokens = githubQuestionTokens(question);
1447:   let score = githubTextFilePattern.test(cleanPath) ? 10 : 0;
1448:   if (/readme\.md$/i.test(cleanPath)) score += 55;
1449:   if (/\.(\.c|h|cpp|hpp|py|js|mjs|ts|v|sv|vhd|?|ino|tcl|xdc|sdc)$/i.test(cleanPath)) score += 28;
1450:   if (/b(test|tb|bench|sim|simulation|src|firmware|hardware|rtl|driver|main)\b/i.test(cleanPath)) score
+= 12;
1451:   tokens.forEach((token) => {
1452:     if (cleanPath.includes(token)) score += 18;
1453:   });
1454:   return score;
1455: }

```

wantsGitHubCode - lines 1457-1459

Item	Explanation
Source location	Lines 1457-1459 (3 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	function wantsGitHubCode(question = "") {
Touches	filesystem
Calls detected	b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1457: function wantsGitHubCode(question = "") {
1458:   return /b(code|source|snippet|implementation|firmware|driver|module|verilog|vhd|python|javascript|typ
escript|c\+\+|cpp|c\s+code|pull|show|display)\b/i.test(question);
1459: }

```

scoreGitHubRepo - lines 1461-1481

Item	Explanation
Source location	Lines 1461-1481 (21 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	function scoreGitHubRepo(repo = {}, question = "") {
Touches	filesystem, authentication/security data
Calls detected	githubQuestionTokens, isArray, join, toLowerCase, forEach, includes, b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1461: function scoreGitHubRepo(repo = {}, question = "") {
1462:   const tokens = githubQuestionTokens(question);
1463:   const haystack = [
1464:     repo.full_name,
1465:     repo.name,
1466:     repo.description,
1467:     repo.language,
1468:     ...(Array.isArray(repo.topics) ? repo.topics : [])
1469:   ].join(" ").toLowerCase();
1470:   let score = repo.fork ? -10 : 12;
1471:   if (!repo.archived) score += 6;
1472:   if (repo.language) score += 4;
1473:   if (repo.name && !/\.github\.io$/i.test(repo.name)) score += 3;
1474:   tokens.forEach((token) => {
1475:     if (haystack.includes(token)) score += 22;
1476:   });
1477:   if (/\\b(vco|oscillator|pwm|analog|mixed|pcb|firmware|stm32|fpga|asic|verilog|embedded|signal|design)\\b/
i.test(haystack)) {
1478:     score += 12;
1479:   }
1480:   return score;
1481: }

```

fetchGitHubProfileSource - lines 1483-1543

Item	Explanation
Source location	Lines 1483-1543 (61 source lines).
Purpose	Loads data from disk, network, browser storage, or a service.
Declaration	async function fetchGitHubProfileSource(source = {}, parsed = {}) {
Touches	network/API requests
Calls detected	all, fetchGitHubJson, isArray, wantsGitHubCode, sort, scoreGitHubRepo, slice, fetchGitHubRepositorySource, trim, push, map, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1483: async function fetchGitHubProfileSource(source = {}, parsed = {}) {
1484:   const owner = parsed.owner;
1485:   const [profile, repos] = await Promise.all([
1486:     fetchGitHubJson(`https://api.github.com/users/${encodeURIComponent(owner)}`),
1487:     fetchGitHubJson(`https://api.github.com/users/${encodeURIComponent(owner)}/repos?per_page=30&sort=updated`)
1488:   ]);
1489:   const repoList = Array.isArray(repos) ? repos : [];
1490:   const includeCode = wantsGitHubCode(source.question || "");
1491:   const selectedRepos = includeCode
1492:     ? [...repoList]
1493:       .sort((a, b) => scoreGitHubRepo(b, source.question || "") - scoreGitHubRepo(a, source.question || ""))
1494:       .slice(0, 3)
1495:       : [];
1496:   const repoBlocks = [];
1497:   for (const repo of selectedRepos) {
1498:     if (!repo?.name || !repo?.html_url) continue;
1499:     const repoSource = await fetchGitHubRepositorySource({
1500:       ...source,
1501:       context: `Public GitHub repository selected from ${owner}'s profile`,
1502:       label: repo.full_name,
1503:       maxCodeFiles: 3,
1504:       maxFileTextLength: 3600,
1505:       maxRepositoryTextLength: 9000,
1506:       url: repo.html_url
1507:     }, {
1508:       owner,
1509:       repo: repo.name,
1510:     }

```

```

1511:         type: "repo",
1512:         branch: repo.default_branch
1513:     });
1514:     if (repoSource?.text?.trim()) repoBlocks.push(repoSource.text.trim());
1515: }
1516:
1517: const lines = [
1518:     `GitHub public profile: ${owner}`,
1519:     profile?.name ? `Name: ${profile.name}` : "",
1520:     profile?.bio ? `Bio: ${profile.bio}` : "",
1521:     profile?.html_url ? `Profile URL: ${profile.html_url}` : source.url,
1522:     "",
1523:     "Public repositories visible from the profile:",
1524:     ...repoList.slice(0, 20).map((repo) => [
1525:         `- ${repo.full_name}`,
1526:         repo.description ? `: ${repo.description}` : "",
1527:         repo.language ? ` | Language: ${repo.language}` : "",
1528:         repo.html_url ? ` | URL: ${repo.html_url}` : "",
1529:         repo.updated_at ? ` | Updated: ${repo.updated_at}` : ""
1530:     ]).join("")),
1531:     includeCode && selectedRepos.length ? "" : "",
1532:     includeCode && selectedRepos.length ? "Selected public repositories inspected for source code:" : "",
1533:     ...selectedRepos.map((repo) => `- ${repo.full_name}${repo.language ? ` | Language: ${repo.language}` : ""}
: ""}${repo.description ? ` | ${repo.description}` : ""}`),
1534:     ...repoBlocks.flatMap((block) => ["", "---", block])
1535: ].filter(Boolean);
1536: return {
1537:     context: source.context || "Public GitHub profile",
1538:     title: source.title || source.label || `GitHub profile: ${owner}`,
1539:     type: includeCode ? "public GitHub profile and selected repository code" : "public GitHub profile",
1540:     url: source.url,
1541:     text: clampText(lines.join("\n"), includeCode ? 30000 : 9000)
1542: };
1543: }

```

fetchGitHubRepositorySource - lines 1545-1612

Item	Explanation
Source location	Lines 1545-1612 (68 source lines).
Purpose	Loads data from disk, network, browser storage, or a service.
Declaration	async function fetchGitHubRepositorySource(source = {}, parsed = {}) {
Touches	network/API requests
Calls detected	fetchGitHubJson, wantsGitHubCode, max, min, isFinite, filter, fetchLimitedText, trim, push, clampText, join, isArray
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	5 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1545: async function fetchGitHubRepositorySource(source = {}, parsed = {}) {
1546:     const { owner, repo } = parsed;
1547:     const metadata = await
fetchGitHubJson(`https://api.github.com/repos/${encodeURIComponent(owner)}/${encodeURIComponent(repo)}`);
1548:     const branch = parsed.branch || metadata?.default_branch || "main";
1549:     const question = source.question || "";
1550:     const requestedMaxCodeFiles = Number(source.maxCodeFiles || (wantsGitHubCode(question) ? 6 : 3));
1551:     const maxCodeFiles = Math.max(1, Math.min(8, Number.isFinite(requestedMaxCodeFiles) ?
requestedMaxCodeFiles : 3));
1552:     const requestedFileTextLength = Number(source.maxFileTextLength || (wantsGitHubCode(question) ? 7000 :
3000));
1553:     const maxFileTextLength = Math.max(1000, Math.min(10000, Number.isFinite(requestedFileTextLength) ?
requestedFileTextLength : 3000));
1554:     const requestedRepoTextLength = Number(source.maxRepositoryTextLength || (wantsGitHubCode(question) ?
22000 : 14000));
1555:     const maxRepositoryTextLength = Math.max(5000, Math.min(26000, Number.isFinite(requestedRepoTextLength)
? requestedRepoTextLength : 14000));
1556:     const lines = [
1557:         `GitHub repository: ${owner}/${repo}`,
1558:         metadata?.description ? `Description: ${metadata.description}` : "",
1559:         metadata?.language ? `Primary language: ${metadata.language}` : "",
1560:         metadata?.html_url ? `Repository URL: ${metadata.html_url}` : source.url,
1561:         metadata?.default_branch ? `Default branch: ${metadata.default_branch}` : "",
1562:         ""
1563:     ].filter(Boolean);
1564:
1565:     if (parsed.type === "file" && parsed.filePath) {
1566:         const rawUrl = `https://raw.githubusercontent.com/${owner}/${repo}/${branch}/${parsed.filePath}`;

```

```

1567:     const text = await fetchLimitedText(rawUrl, 14000);
1568:     if (text.trim()) lines.push(`Source file: ${parsed.filePath}`, text.trim());
1569:     return {
1570:       context: source.context || "Public GitHub source file",
1571:       title: source.title || source.label || `${owner}/${repo}/${parsed.filePath}`,
1572:       type: "public GitHub source file",
1573:       url: source.url,
1574:       text: clampText(lines.join("\n"), 14000)
1575:     };
1576:   }
1577:
1578:   const tree = await fetchGitHubJson(`https://api.github.com/repos/${encodeURIComponent(owner)}/${encodeURIComponent(repo)}/git/trees/${encodeURIComponent(branch)}?recursive=1`);
1579:   const files = Array.isArray(tree?.tree)
1580:     ? tree.tree
1581:     : [];
1582:   .filter((item) => item.type === "blob" && item.path && githubTextFilePattern.test(item.path) && !githubSkipFilePattern.test(item.path))
1583:   .sort((a, b) => scoreGitHubFile(b.path, question) - scoreGitHubFile(a.path, question))
1584:   : [];
1585:   const readmePath = files.find((file) => /^(^|\/)readme\.md$/i.test(file.path))?.path || "README.md";
1586:   const readmeText = await
1587:   fetchLimitedText(`https://raw.githubusercontent.com/${owner}/${repo}/${branch}/${readmePath}`, 9000).catch(() =>
1588:   "");
1589:   if (readmeText.trim()) lines.push(`README excerpt from ${readmePath}:`, readmeText.trim(), "");
1590:   if (files.length) {
1591:     lines.push("Repository text/code file list:", ...files.slice(0, 30).map((file) => `- ${file.path}`),
1592:     "");
1593:   }
1594:   const includeCode = wantsGitHubCode(question);
1595:   const selectedFiles = files
1596:     .filter((file) => !/readme\.md$/i.test(file.path))
1597:     .slice(0, maxCodeFiles);
1598:   for (const file of selectedFiles) {
1599:     const rawUrl = `https://raw.githubusercontent.com/${owner}/${repo}/${branch}/${file.path}`;
1600:     const fileText = await fetchLimitedText(rawUrl, maxFileTextLength).catch(() => "");
1601:     if (!fileText.trim()) continue;
1602:     lines.push(`Source file: ${file.path}`, fileText.trim(), "");
1603:   }
1604:
1605:   return {
1606:     context: source.context || "Public GitHub repository",
1607:     title: source.title || source.label || `${owner}/${repo}`,
1608:     type: includeCode ? "public GitHub repository and code" : "public GitHub repository",
1609:     url: source.url,
1610:     text: clampText(lines.join("\n"), maxRepositoryTextLength)
1611:   };
1612: }

```

fetchGitHubSourceText - lines 1614-1619

Item	Explanation
Source location	Lines 1614-1619 (6 source lines).
Purpose	Loads data from disk, network, browser storage, or a service.
Declaration	async function fetchGitHubSourceText(source = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	parseGitHubSourceUrl, fetchGitHubProfileSource, fetchGitHubRepositorySource
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1614: async function fetchGitHubSourceText(source = {}) {
1615:   const parsed = parseGitHubSourceUrl(source.url || "");
1616:   if (!parsed) return null;
1617:   if (parsed.type === "profile") return fetchGitHubProfileSource(source, parsed);
1618:   return fetchGitHubRepositorySource(source, parsed);
1619: }

```

readLocalSourceText - lines 1621-1639

Item	Explanation
Source location	Lines 1621-1639 (19 source lines).
Purpose	Loads data from disk, network, browser storage, or a service.
Declaration	async function readLocalSourceText(url) {
Touches	filesystem
Calls detected	URL, Set, has, toLowerCase, replace, includes, extname, readFile, resolveInsideRoot
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 6 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1621: async function readLocalSourceText(url) {
1622:   let parsed;
1623:   try {
1624:     parsed = new URL(url);
1625:   } catch {
1626:     return "";
1627:   }
1628:   const localHosts = new Set(["localhost", "127.0.0.1"]);
1629:   if (!localHosts.has(parsed.hostname.toLowerCase())) return "";
1630:   const pathname = decodeURIComponent(parsed.pathname.replace(/^\/+/, ""));
1631:   if (!pathname || pathname.includes("..")) return "";
1632:   const ext = path.extname(pathname).toLowerCase();
1633:   if (![".txt", ".md", ".json", ".csv", ".log", ".xml", ".js", ".mjs", ".css", ".c", ".h", ".cpp",
".hpp", ".py", ".v", ".sv", ".spice", ".cir", ".net", ".asc", ".sch", ".kicad_sch", ".kicad_pcb"].includes(ext))
return "";
1634:   try {
1635:     return await readFile(resolveInsideRoot(pathname), "utf8");
1636:   } catch {
1637:     return "";
1638:   }
1639: }

```

fetchSourceText - lines 1641-1679

Item	Explanation
Source location	Lines 1641-1679 (39 source lines).
Purpose	Loads data from disk, network, browser storage, or a service.
Declaration	async function fetchSourceText(source = {}) {
Touches	network/API requests
Calls detected	sourceLooksTextual, sourceUrlAllowed, fetchGitHubSourceText, trim, readLocalSourceText, clampText, replace, fetch, get, text, stripHtmlToText
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	4 await expression(s), 8 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1641: async function fetchSourceText(source = {}) {
1642:   const url = String(source.url || "");
1643:   if (!url || !sourceLooksTextual(source) || !sourceUrlAllowed(url)) return null;
1644:   if (/github\.com|raw\.githubusercontent\.com/i.test(url)) {
1645:     const githubText = await fetchGitHubSourceText(source);
1646:     if (githubText?.text?.trim()) return githubText;
1647:   }
1648:   const localText = await readLocalSourceText(url);
1649:   if (localText.trim()) {
1650:     return {
1651:       context: source.context || "",
1652:       title: source.title || source.label || url,
1653:       type: source.type || source.kind || "text source",
1654:       url,

```

```

1655:     text: clampText(localText.replace(/\s+\n/g, "\n").trim(), 9000)
1656:   };
1657: }
1658:
1659: try {
1660:   const response = await fetch(url, {
1661:     headers: { "Accept": "text/plain,text/markdown,text/html,application/json;q=0.8,*/*;q=0.1" }
1662:   });
1663:   if (!response.ok) return null;
1664:   const contentType = response.headers.get("Content-Type") || "";
1665:   if (!/text|json|xml|html|markdown/i.test(contentType)) return null;
1666:   const rawText = clampText(await response.text(), 12000);
1667:   const text = /html/i.test(contentType) ? stripHtmlToText(rawText) : rawText;
1668:   if (!text.trim()) return null;
1669:   return {
1670:     context: source.context || "",
1671:     title: source.title || source.label || url,
1672:     type: source.type || source.kind || "public source",
1673:     url,
1674:     text: clampText(text.trim(), 9000)
1675:   };
1676: } catch {
1677:   return null;
1678: }
1679: }

```

enrichPortfolioContext - lines 1681-1692

Item	Explanation
Source location	Lines 1681-1692 (12 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	async function enrichPortfolioContext(context = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	isArray, slice, map, all, filter
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1681: async function enrichPortfolioContext(context = {}) {
1682:   const question = String(context.question || "");
1683:   const sources = Array.isArray(context.sourceFetches)
1684:     ? context.sourceFetches.slice(0, 10).map((source) => ({ ...source, question: source.question ||
question }))
1685:     : [];
1686:   const sourceExcerpts = (await Promise.all(sources.map(fetchSourceText))).filter(Boolean);
1687:   return {
1688:     ...context,
1689:     sourceExcerpts,
1690:     sourceFetchPolicy: "Fetched excerpts are limited to safe text-like same-site, GitHub/raw GitHub,
LinkedIn, and public portfolio URLs. GitHub repository links can be expanded into repository metadata, README
text, and selected public source files when a question asks for code. PDFs/images are represented by captions,
metadata, filenames, and companion text files when available."
1691:   };
1692: }

```

cleanConversationHistory - lines 1694-1703

Item	Explanation
Source location	Lines 1694-1703 (10 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function cleanConversationHistory(value) {
Touches	Mostly local calculations or local UI state.
Calls detected	isArray, slice, map, clampText, trim, filter
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1694: function cleanConversationHistory(value) {
1695:   if (!Array.isArray(value)) return [];
1696:   return value
1697:     .slice(-8)
1698:     .map((item) => ({
1699:       role: item?.role === "assistant" ? "assistant" : "user",
1700:       content: clampText(item?.content, 1400).trim()
1701:     }))
1702:     .filter((item) => item.content);
1703: }

```

extractOpenAiText - lines 1705-1717

Item	Explanation
Source location	Lines 1705-1717 (13 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function extractOpenAiText(data) {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, isArray, forEach, push, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1705: function extractOpenAiText(data) {
1706:   if (typeof data?.output_text === "string" && data.output_text.trim()) return data.output_text.trim();
1707:   const output = Array.isArray(data?.output) ? data.output : [];
1708:   const chunks = [];
1709:
1710:   output.forEach((item) => {
1711:     (item.content || []).forEach((content) => {
1712:       if (typeof content.text === "string" && content.text.trim()) chunks.push(content.text.trim());
1713:     });
1714:   });
1715:
1716:   return chunks.join("\n\n").trim();
1717: }

```

extractOllamaText - lines 1719-1723

Item	Explanation
Source location	Lines 1719-1723 (5 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function extractOllamaText(data = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1719: function extractOllamaText(data = {}) {
1720:   if (typeof data.message?.content === "string") return data.message.content.trim();
1721:   if (typeof data.response === "string") return data.response.trim();
1722:   return "";
1723: }

```

callOllamaPortfolioAi - lines 1725-1773

Item	Explanation
Source location	Lines 1725-1773 (49 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	async function callOllamaPortfolioAi({ question, intent, conversation, context, allowWebSearch }) {
Touches	network/API requests
Calls detected	AbortController, abort, fetch, replace, stringify, clampText, join, json, extractOllamaText
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1725: async function callOllamaPortfolioAi({ question, intent, conversation, context, allowWebSearch }) {
1726:   const host = process.env.OLLAMA_HOST || "http://127.0.0.1:11434";
1727:   const model = process.env.OLLAMA_MODEL || "llama3.2";
1728:   const controller = new AbortController();
1729:   const timeout = setTimeout(() => controller.abort(), Number(process.env.OLLAMA_TIMEOUT_MS || 45000));
1730:   try {
1731:     const response = await fetch(`${host.replace(/\/+$/, "").}/api/chat`, {
1732:       method: "POST",
1733:       headers: { "Content-Type": "application/json" },
1734:       signal: controller.signal,
1735:       body: JSON.stringify({
1736:         model,
1737:         stream: false,
1738:         messages: [
1739:           { role: "system", content: portfolioAiInstructions },
1740:           {
1741:             role: "user",
1742:             content: [
1743:               `Visitor question: ${question}`,
1744:               `Question intent: ${intent}`,
1745:               `Web/public lookup requested by browser: ${allowWebSearch ? "yes" : "no"}`,
1746:               "",
1747:               "Recent conversation JSON:",
1748:               clampText(JSON.stringify(conversation, null, 2), 8000),
1749:               "",
1750:               "Portfolio context JSON:",
1751:               clampText(JSON.stringify(context, null, 2), 18000)
1752:             ].join("\n")
1753:           }
1754:         ],
1755:         options: {
1756:           temperature: Number(process.env.OLLAMA_TEMPERATURE || 0.35)
1757:         }
1758:       })
1759:     });
1760:     const data = await response.json().catch(() => ({}));
1761:     if (!response.ok) {
1762:       return { ok: false, error: data?.error || `Ollama returned HTTP ${response.status}` };
1763:     }
1764:     const answer = extractOllamaText(data);
1765:     return answer
1766:       ? { ok: true, answer, model }
1767:       : { ok: false, error: "Ollama did not return an answer." };
1768:   } catch (error) {
1769:     return { ok: false, error: error?.name === "AbortError" ? "Ollama timed out." : "Ollama is not reachable on this machine." };
1770:   } finally {
1771:     clearTimeout(timeout);

```



```

1772:   }
1773: }

```

ruleBasedConversationAnswer - lines 1775-1786

Item	Explanation
Source location	Lines 1775-1786 (12 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function ruleBasedConversationAnswer(question = "") {
Touches	filesystem
Calls detected	toLowerCase, b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1775: function ruleBasedConversationAnswer(question = "") {
1776:   const clean = String(question || "").toLowerCase();
1777:   if (/^(\b(thanks|thank you|appreciate it)\b/.test(clean)) {
1778:     return "You're welcome. I can keep helping with this portfolio, project evidence, resume links, or
related electronics and embedded-systems questions.";
1779:   }
1780:
1781:   if (/^(\b(who are you|what are you)\b/.test(clean)) {
1782:     return "I am this portfolio's assistant. I can help visitors explore the saved engineering work,
explain project details, summarize portfolio evidence, and answer related electronics, embedded, analog,
digital, FPGA, ASIC, PCB, and firmware questions.";
1783:   }
1784:
1785:   return "Hi. I can help you explore this portfolio, explain projects, summarize project evidence, open
relevant sections, or answer related electronics and embedded-systems questions. You can ask about a specific
project, a tool like KiCad or STM32CubeIDE, or a general topic such as embedded systems, op amps, FPGA design,
ASICs, or PCB testing.";
1786: }

```

isLocalRequest - lines 1788-1791

Item	Explanation
Source location	Lines 1788-1791 (4 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function isLocalRequest(request) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1788: function isLocalRequest(request) {
1789:   const address = request.socket.remoteAddress || "";
1790:   return address === "127.0.0.1" || address === "::1" || address === "::ffff:127.0.0.1";
1791: }

```

readJsonFile - lines 1793-1795

Item	Explanation
Source location	Lines 1793-1795 (3 source lines).
Purpose	Loads data from disk, network, browser storage, or a service.
Declaration	async function readJsonFile(filePath) {
Touches	filesystem
Calls detected	parse, readFile
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1793: async function readJsonFile(filePath) {
1794:   return JSON.parse(await readFile(filePath, "utf8"));
1795: }

```

readRequestJson - lines 1797-1810

Item	Explanation
Source location	Lines 1797-1810 (14 source lines).
Purpose	Loads data from disk, network, browser storage, or a service.
Declaration	async function readRequestJson(request) {
Touches	Mostly local calculations or local UI state.
Calls detected	Error, push, parse, concat, toString, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1797: async function readRequestJson(request) {
1798:   const chunks = [];
1799:   let size = 0;
1800:
1801:   for await (const chunk of request) {
1802:     size += chunk.length;
1803:     if (size > 60 * 1024 * 1024) {
1804:       throw new Error("Request body is too large.");
1805:     }
1806:     chunks.push(chunk);
1807:   }
1808:
1809:   return JSON.parse((Buffer.concat(chunks).toString("utf8") || "{}").replace(/^\uFEFF/, ""));
1810: }

```

safeSegment - lines 1812-1819

Item	Explanation
Source location	Lines 1812-1819 (8 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function safeSegment(value, fallback = "item") {
Touches	Mostly local calculations or local UI state.
Calls detected	toLowerCase, trim, replace
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1812: function safeSegment(value, fallback = "item") {
1813:   const segment = String(value || fallback)
1814:     .toLowerCase()
1815:     .trim()
1816:     .replace(/^[^a-z0-9._-]+/g, "-")
1817:     .replace(/^-+|-+$/g, "");
1818:   return segment || fallback;
1819: }

```

safeFileName - lines 1821-1826

Item	Explanation
Source location	Lines 1821-1826 (6 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function safeFileName(value) {
Touches	Mostly local calculations or local UI state.
Calls detected	parse, safeSegment, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1821: function safeFileName(value) {
1822:   const parsed = path.parse(String(value || "file"));
1823:   const name = safeSegment(parsed.name, "file");
1824:   const ext = safeSegment(parsed.ext.replace(".", ""), "");
1825:   return ext ? `${name}.${ext}` : name;
1826: }

```

resolveInsideRoot - lines 1828-1834

Item	Explanation
Source location	Lines 1828-1834 (7 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function resolveInsideRoot(...segments) {
Touches	filesystem
Calls detected	normalize, join, startsWith, Error
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1828: function resolveInsideRoot(...segments) {
1829:   const target = path.normalize(path.join(root, ...segments));
1830:   if (!target.startsWith(root)) {
1831:     throw new Error("Resolved path is outside the workspace.");
1832:   }

```

```

1833:   return target;
1834: }

```

resolveInsidePortfolioRoot - lines 1836-1843

Item	Explanation
Source location	Lines 1836-1843 (8 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function resolveInsidePortfolioRoot(...segments) {
Touches	Mostly local calculations or local UI state.
Calls detected	resolve, relative, startsWith, isAbsolute, Error
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1836: function resolveInsidePortfolioRoot(...segments) {
1837:   const target = path.resolve(portfolioRoot, ...segments);
1838:   const relative = path.relative(portfolioRoot, target);
1839:   if (relative.startsWith(".") || path.isAbsolute(relative)) {
1840:     throw new Error("Resolved path is outside the portfolio workspace.");
1841:   }
1842:   return target;
1843: }

```

resolveInsideCompileRoot - lines 1845-1852

Item	Explanation
Source location	Lines 1845-1852 (8 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function resolveInsideCompileRoot(...segments) {
Touches	Mostly local calculations or local UI state.
Calls detected	resolve, relative, startsWith, isAbsolute, Error
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1845: function resolveInsideCompileRoot(...segments) {
1846:   const target = path.resolve(compileRoot, ...segments);
1847:   const relative = path.relative(compileRoot, target);
1848:   if (relative.startsWith(".") || path.isAbsolute(relative)) {
1849:     throw new Error("Resolved path is outside the compile workspace.");
1850:   }
1851:   return target;
1852: }

```

samePath - lines 1854-1856

Item	Explanation
Source location	Lines 1854-1856 (3 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function samePath(left, right) {

Item	Explanation
Touches	Mostly local calculations or local UI state.
Calls detected	resolve, toLowerCase
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1854: function samePath(left, right) {
1855:   return path.resolve(left).toLowerCase() === path.resolve(right).toLowerCase();
1856: }
```

pathExists - lines 1858-1865

Item	Explanation
Source location	Lines 1858-1865 (8 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	async function pathExists(filePath) {
Touches	Mostly local calculations or local UI state.
Calls detected	fsAccess
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1858: async function pathExists(filePath) {
1859:   try {
1860:     await fsAccess(filePath);
1861:     return true;
1862:   } catch {
1863:     return false;
1864:   }
1865: }
```

publishAccessError - lines 1867-1877

Item	Explanation
Source location	Lines 1867-1877 (11 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	function publishAccessError(message, details = "", extra = {}) {
Touches	authentication/security data
Calls detected	Error
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1867: function publishAccessError(message, details = "", extra = {}) {
1868:   const error = new Error(message);
```

```

1869:   error.code = "PUBLISH_AUTHORIZATION_REQUIRED";
1870:   error.details = details || publishAuthorizationHelp;
1871:   error.publishAccess = {
1872:     authorizationRequired: true,
1873:     help: publishAuthorizationHelp,
1874:     ...extra
1875:   };
1876:   return error;
1877: }

```

gitFailureText - lines 1879-1885

Item	Explanation
Source location	Lines 1879-1885 (7 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	function gitFailureText(error) {
Touches	Mostly local calculations or local UI state.
Calls detected	filter, join, trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1879: function gitFailureText(error) {
1880:   return [
1881:     error?.stderr,
1882:     error?.stdout,
1883:     error?.message
1884:   ].filter(Boolean).join("\n").trim();
1885: }

```

remoteUrlForDisplay - lines 1887-1891

Item	Explanation
Source location	Lines 1887-1891 (5 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	function remoteUrlForDisplay(remoteUrl = "") {
Touches	network/API requests
Calls detected	replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1887: function remoteUrlForDisplay(remoteUrl = "") {
1888:   return String(remoteUrl || "")
1889:     .replace(/^https:\/\/([^\:\/]+):[^\:\/]+@/i, "https://$1:***@")
1890:     .replace(/^https:\/\/[^\:\/]+@/i, "https://");
1891: }

```

parseGitHubRemote - lines 1893-1907

Item	Explanation
Source location	Lines 1893-1907 (15 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.

Item	Explanation
Declaration	function parseGitHubRemote(remoteUrl = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, match, replace, URL, split
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 5 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1893: function parseGitHubRemote(remoteUrl = "") {
1894:   const trimmed = String(remoteUrl || "").trim();
1895:   const sshMatch = trimmed.match(/^git@github\.com:([^\s]+)\.(.+)?(?:\.git)?$/i);
1896:   if (sshMatch) return { owner: sshMatch[1], repo: sshMatch[2].replace(/\.git$/i, "") };
1897:
1898:   try {
1899:     const parsed = new URL(trimmed.replace(/^git\+/, ""));
1900:     if (!/github\.com$/i.test(parsed.hostname)) return null;
1901:     const parts = parsed.pathname.replace(/^\/+/, "").split("/");
1902:     if (parts.length < 2) return null;
1903:     return { owner: parts[0], repo: parts[1].replace(/\.git$/i, "") };
1904:   } catch {
1905:     return null;
1906:   }
1907: }

```

validatePublishRemoteUrl - lines 1909-1927

Item	Explanation
Source location	Lines 1909-1927 (19 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	function validatePublishRemoteUrl(value = "") {
Touches	network/API requests
Calls detected	trim, URL, toLowerCase, Error, replace, split, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1909: function validatePublishRemoteUrl(value = "") {
1910:   const remoteUrl = String(value || "").trim();
1911:   if (!remoteUrl) return "";
1912:   if (/^git@github\.com:[A-Za-z0-9_.-]+\.[A-Za-z0-9_.-]+(?:\.git)?$/i.test(remoteUrl)) return remoteUrl;
1913:   try {
1914:     const parsed = new URL(remoteUrl);
1915:     if (parsed.protocol !== "https:" || parsed.hostname.toLowerCase() !== "github.com") {
1916:       throw new Error("Only GitHub HTTPS repository URLs are supported by the guided setup.");
1917:     }
1918:     const parts = parsed.pathname.replace(/^\/+/, "").split("/");
1919:     if (parts.length < 2 || !parts[0] || !parts[1]) {
1920:       throw new Error("GitHub repository URL must include owner and repository name.");
1921:     }
1922:     return `https://github.com/${parts[0]}/${parts[1].replace(/\.git$/i, "")}.git`;
1923:   } catch (error) {
1924:     if (error.message?.includes("Only GitHub") || error.message?.includes("owner")) throw error;
1925:     throw new Error("Enter a GitHub repository URL such as https://github.com/USERNAME/USERNAME.github.io.git.");
1926:   }
1927: }

```

validateCustomDomain - lines 1929-1936

Item	Explanation
Source location	Lines 1929-1936 (8 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function validateCustomDomain(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, toLowerCase, Error
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1929: function validateCustomDomain(value = "") {
1930:   const domain = String(value || "").trim().toLowerCase();
1931:   if (!domain) return "";
1932:   if (!/^(?:[a-z0-9-]{0,61}[a-z0-9])?\.\.[a-z]{2,63}$/i.test(domain)) {
1933:     throw new Error("Custom domain must look like example.com or portfolio.example.com.");
1934:   }
1935:   return domain;
1936: }

```

compareVersions - lines 1938-1947

Item	Explanation
Source location	Lines 1938-1947 (10 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function compareVersions(left = "", right = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	split, map, max
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1938: function compareVersions(left = "", right = "") {
1939:   const leftParts = String(left || "0").split(/[.-]/).map((part) => Number.parseInt(part, 10) || 0);
1940:   const rightParts = String(right || "0").split(/[.-]/).map((part) => Number.parseInt(part, 10) || 0);
1941:   const length = Math.max(leftParts.length, rightParts.length);
1942:   for (let index = 0; index < length; index += 1) {
1943:     const delta = (leftParts[index] || 0) - (rightParts[index] || 0);
1944:     if (delta !== 0) return delta;
1945:   }
1946:   return 0;
1947: }

```

bumpPublishedAssetVersions - lines 1949-1965

Item	Explanation
Source location	Lines 1949-1965 (17 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function bumpPublishedAssetVersions(baseRoot = portfolioRoot) {
Touches	filesystem
Calls detected	join, readFile, now, toString, replace, writeFile

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1949: async function bumpPublishedAssetVersions(baseRoot = portfolioRoot) {
1950:   let html = "";
1951:   const indexPath = path.join(baseRoot, "index.html");
1952:   try {
1953:     html = await readFile(indexPath, "utf8");
1954:   } catch {
1955:     return false;
1956:   }
1957:   const version = Date.now().toString();
1958:   const next = html
1959:     .replace(/(href="styles\.css)(?:\?v=[^"]*)?"/g, ` $1?v=${version}`)
1960:     .replace(/(src="script\.js)(?:\?v=[^"]*)?"/g, ` $1?v=${version}`)
1961:     .replace(/(src="electronics-search\.js)(?:\?v=[^"]*)?"/g, ` $1?v=${version}`);
1962:   if (next === html) return false;
1963:   await writeFile(indexPath, next, "utf8");
1964:   return true;
1965: }

```

workspaceHasCompatibleSiteFiles - lines 1967-1976

Item	Explanation
Source location	Lines 1967-1976 (10 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	async function workspaceHasCompatibleSiteFiles(baseRoot = portfolioRoot) {
Touches	filesystem
Calls detected	readFile, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1967: async function workspaceHasCompatibleSiteFiles(baseRoot = portfolioRoot) {
1968:   for (const fileName of ["index.html", "styles.css", "script.js"]) {
1969:     try {
1970:       await readFile(path.join(baseRoot, fileName));
1971:     } catch {
1972:       return false;
1973:     }
1974:   }
1975:   return true;
1976: }

```

getPublishTargetInfo - lines 1978-2044

Item	Explanation
Source location	Lines 1978-2044 (67 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function getPublishTargetInfo() {
Touches	filesystem, authentication/security data
Calls detected	samePath, workspaceHasCompatibleSiteFiles, runPublishGit, trim, remoteUrlForDisplay, parseGitHubRemote, readFile, resolveInsidePortfolioRoot, readPublishAuthCache, publishAuthCachelsFresh, publishAuthCacheExpiresAt

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	6 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1978: async function getPublishTargetInfo() {
1979:   const info = {
1980:     workspace: root,
1981:     portfolioWorkspace: portfolioRoot,
1982:     separatedWorkspace: !samePath(root, portfolioRoot),
1983:     defaultRepository: defaultSiteRepository,
1984:     gitBacked: false,
1985:     compatible: await workspaceHasCompatibleSiteFiles(),
1986:     remote: "",
1987:     branch: "",
1988:     repository: "",
1989:     customDomain: ""
1990:   };
1991:
1992:   try {
1993:     info.gitBacked = (await runPublishGit(["rev-parse", "--is-inside-work-tree"])).stdout.trim() ===
"true";
1994:   } catch {
1995:     return info;
1996:   }
1997:
1998:   try {
1999:     info.remote = remoteUrlForDisplay((await runPublishGit(["remote", "get-url",
"origin"])).stdout.trim());
2000:   } catch {
2001:     info.remote = "";
2002:   }
2003:
2004:   try {
2005:     info.branch = (await runPublishGit(["branch", "--show-current"])).stdout.trim();
2006:   } catch {
2007:     info.branch = "";
2008:   }
2009:
2010:   const parsedRemote = parseGitHubRemote(info.remote);
2011:   info.repository = parsedRemote ? `${parsedRemote.owner}/${parsedRemote.repo}` : info.remote;
2012:
2013:   try {
2014:     info.customDomain = (await readFile(resolveInsidePortfolioRoot("CNAME"), "utf8")).trim();
2015:   } catch {
2016:     info.customDomain = "";
2017:   }
2018:
2019:   const accessShape = {
2020:     branch: info.branch,
2021:     remote: info.remote,
2022:     repository: info.repository
2023:   };
2024:   const cached = await readPublishAuthCache();
2025:   if (publishAuthCacheIsFresh(cached, accessShape)) {
2026:     info.authorizationChecked = true;
2027:     info.authorizationCached = true;
2028:     info.checkedAt = cached.checkedAt;
2029:     info.expiresAt = publishAuthCacheExpiresAt(cached);
2030:     info.extendedTrust = Boolean(cached.extendedTrust);
2031:     info.successCountLastWeek = cached.successCountLastWeek || 0;
2032:     info.trustDays = cached.trustDays || (cached.extendedTrust ? 30 : 1);
2033:   } else {
2034:     info.authorizationChecked = false;
2035:     info.authorizationCached = false;
2036:     info.checkedAt = "";
2037:     info.expiresAt = "";
2038:     info.extendedTrust = false;
2039:     info.successCountLastWeek = cached?.successCountLastWeek || 0;
2040:     info.trustDays = 0;
2041:   }
2042:
2043:   return info;
2044: }

```

runGit - lines 2046-2064

Item	Explanation
Source location	Lines 2046-2064 (19 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function runGit(args, options = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	execFileAsync, Error
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2046: async function runGit(args, options = {}) {
2047:   let lastError = null;
2048:   for (const candidate of gitCandidates) {
2049:     try {
2050:       const result = await execFileAsync(candidate, args, {
2051:         cwd: options.cwd || root,
2052:         maxBuffer: options.maxBuffer || 10 * 1024 * 1024,
2053:         timeout: options.timeout || 0,
2054:         windowsHide: options.windowsHide !== false
2055:       });
2056:       return { ...result, git: candidate };
2057:     } catch (error) {
2058:       lastError = error;
2059:       if (error.code === "ENOENT") continue;
2060:       throw error;
2061:     }
2062:   }
2063:   throw lastError || new Error("Git executable was not found.");
2064: }

```

runPublishGit - lines 2066-2068

Item	Explanation
Source location	Lines 2066-2068 (3 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function runPublishGit(args, options = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	runGit
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2066: async function runPublishGit(args, options = {}) {
2067:   return runGit(args, { ...options, cwd: options.cwd || portfolioRoot });
2068: }

```

runOptionalCommand - lines 2070-2087

Item	Explanation
Source location	Lines 2070-2087 (18 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	async function runOptionalCommand(command, args = [], options = {}) {

Item	Explanation
Touches	Mostly local calculations or local UI state.
Calls detected	execFileAsync
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2070: async function runOptionalCommand(command, args = [], options = {}) {
2071:   try {
2072:     const result = await execFileAsync(command, args, {
2073:       cwd: options.cwd || root,
2074:       maxBuffer: options.maxBuffer || 2 * 1024 * 1024,
2075:       timeout: options.timeout || 15000,
2076:       windowsHide: options.windowsHide !== false
2077:     });
2078:     return { ok: true, stdout: result.stdout || "", stderr: result.stderr || "" };
2079:   } catch (error) {
2080:     return {
2081:       ok: false,
2082:       stdout: error.stdout || "",
2083:       stderr: error.stderr || "",
2084:       error: error.message || String(error)
2085:     };
2086:   }
2087: }

```

getGitStatus - lines 2089-2136

Item	Explanation
Source location	Lines 2089-2136 (48 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function getGitStatus() {
Touches	authentication/security data
Calls detected	runGit, trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2089: async function getGitStatus() {
2090:   let git = { ok: false, stdout: "", stderr: "", error: "Git for Windows was not found." };
2091:   try {
2092:     const gitResult = await runGit(["--version"]);
2093:     git = { ok: true, stdout: gitResult.stdout || "", stderr: gitResult.stderr || "" };
2094:   } catch (error) {
2095:     git = { ok: false, stdout: error.stdout || "", stderr: error.stderr || "", error: error.message ||
"Git for Windows was not found." };
2096:   }
2097:   let credentialManager = { ok: false, version: "", error: "Git Credential Manager was not found." };
2098:   if (git.ok) {
2099:     let gcm = { ok: false, stdout: "", stderr: "", error: "Git Credential Manager was not found." };
2100:     try {
2101:       const gcmResult = await runGit(["credential-manager", "--version"]);
2102:       gcm = { ok: true, stdout: gcmResult.stdout || "", stderr: gcmResult.stderr || "" };
2103:     } catch (error) {
2104:       gcm = { ok: false, stdout: error.stdout || "", stderr: error.stderr || "", error: error.message ||
"Git Credential Manager was not found." };
2105:     }
2106:     credentialManager = {
2107:       ok: gcm.ok,
2108:       version: (gcm.stdout || gcm.stderr || "").trim(),
2109:       error: gcm.ok ? "" : (gcm.stderr || gcm.error || "Git Credential Manager was not found.")
2110:     };
2111:   }

```

```

2112:
2113:   return {
2114:     git: {
2115:       ok: git.ok,
2116:       version: (git.stdout || git.stderr || "").trim(),
2117:       error: git.ok ? "" : (git.stderr || git.error || "Git for Windows was not found.")
2118:     },
2119:     credentialManager,
2120:     nodeRuntime: {
2121:       ok: true,
2122:       version: process.versions.node,
2123:       bundled: true,
2124:       note: "Bundled inside the desktop app. No manual Node.js install is required."
2125:     },
2126:     pnpm: {
2127:       ok: true,
2128:       required: false,
2129:       note: "pnpm is only needed by developers building the installer from source."
2130:     },
2131:     app: {
2132:       version: process.env.OMB_APP_VERSION || packageJson.version || "0.0.0",
2133:       desktopApp: process.env.OMB_DESKTOP_APP === "1"
2134:     }
2135:   };
2136: }

```

publishPathIsStageable - lines 2138-2152

Item	Explanation
Source location	Lines 2138-2152 (15 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function publishPathIsStageable(relativePath) {
Touches	Mostly local calculations or local UI state.
Calls detected	fsAccess, resolveInsidePortfolioRoot, runPublishGit
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2138: async function publishPathIsStageable(relativePath) {
2139:   try {
2140:     await fsAccess(resolveInsidePortfolioRoot(relativePath));
2141:     return true;
2142:   } catch {
2143:     // Missing files can still be stageable if Git already tracks them and they were deleted.
2144:   }
2145:
2146:   try {
2147:     await runPublishGit(["ls-files", "--error-unmatch", "--", relativePath]);
2148:     return true;
2149:   } catch {
2150:     return false;
2151:   }
2152: }

```

stageablePublishPaths - lines 2154-2160

Item	Explanation
Source location	Lines 2154-2160 (7 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function stageablePublishPaths(pathsToCheck) {
Touches	Mostly local calculations or local UI state.
Calls detected	publishPathIsStageable, push
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	1 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2154: async function stageablePublishPaths(pathsToCheck) {
2155:   const stageable = [];
2156:   for (const relativePath of pathsToCheck) {
2157:     if (await publishPathIsStageable(relativePath)) stageable.push(relativePath);
2158:   }
2159:   return stageable;
2160: }

```

runGitWithInput - lines 2162-2204

Item	Explanation
Source location	Lines 2162-2204 (43 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function runGitWithInput(args, input, options = {}) {
Touches	external processes
Calls detected	spawn, on, toString, once, resolve, Error, reject, end
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2162: async function runGitWithInput(args, input, options = {}) {
2163:   let lastError = null;
2164:   for (const candidate of gitCandidates) {
2165:     try {
2166:       return await new Promise((resolve, reject) => {
2167:         const child = spawn(candidate, args, {
2168:           cwd: options.cwd || portfolioRoot,
2169:           windowsHide: true,
2170:           stdio: ["pipe", "pipe", "pipe"]
2171:         });
2172:         let stdout = "";
2173:         let stderr = "";
2174:
2175:         child.stdout.on("data", (chunk) => {
2176:           stdout += chunk.toString();
2177:         });
2178:         child.stderr.on("data", (chunk) => {
2179:           stderr += chunk.toString();
2180:         });
2181:         child.once("error", reject);
2182:         child.once("close", (code) => {
2183:           if (code === 0) {
2184:             resolve({ stdout, stderr, git: candidate });
2185:             return;
2186:           }
2187:           const error = new Error(stderr || stdout || `Git exited with code ${code}.`);
2188:           error.stdout = stdout;
2189:           error.stderr = stderr;
2190:           error.code = code;
2191:           error.git = candidate;
2192:           reject(error);
2193:         });
2194:         child.stdin.end(input);
2195:       });
2196:     } catch (error) {
2197:       lastError = error;
2198:       if (error.code === "ENOENT") continue;
2199:       throw error;
2200:     }
2201:   }
2202: }
2203: throw lastError || new Error("Git executable was not found.");
2204: }

```

storeGitCredentials - lines 2206-2237

Item	Explanation
Source location	Lines 2206-2237 (32 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	async function storeGitCredentials(remoteUrl, username, password) {
Touches	authentication/security data
Calls detected	trim, Error, URL, toLowerCase, replace, runPublishGit, join, runGitWithInput
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2206: async function storeGitCredentials(remoteUrl, username, password) {
2207:   const cleanUsername = String(username || "").trim();
2208:   const cleanPassword = String(password || "");
2209:   if (!cleanUsername && !cleanPassword) return { stored: false };
2210:   if (!cleanUsername || !cleanPassword) {
2211:     throw new Error("Both username and password/token are required when credentials are provided.");
2212:   }
2213:
2214:   let parsed = null;
2215:   try {
2216:     parsed = new URL(remoteUrl);
2217:   } catch {
2218:     throw new Error("Username and password/token credentials require a GitHub HTTPS repository URL.");
2219:   }
2220:
2221:   if (parsed.protocol !== "https:" || parsed.hostname.toLowerCase() !== "github.com") {
2222:     throw new Error("Username and password/token credentials require a GitHub HTTPS repository URL.");
2223:   }
2224:
2225:   const pathName = parsed.pathname.replace(/^\/+/, "");
2226:   await runPublishGit(["config", "--local", "credential.useHttpPath", "true"]);
2227:   const credentialInput = [
2228:     "protocol=https",
2229:     "host=github.com",
2230:     `path=${pathName}`,
2231:     `username=${cleanUsername}`,
2232:     `password=${cleanPassword}`,
2233:     ""
2234:   ].join("\n");
2235:   await runGitWithInput(["credential", "approve"], credentialInput, { cwd: portfolioRoot });
2236:   return { stored: true, username: cleanUsername };
2237: }
```

validateCredentialPair - lines 2239-2246

Item	Explanation
Source location	Lines 2239-2246 (8 source lines).
Purpose	Part of authentication, authorization, or credential checking.
Declaration	function validateCredentialPair(username, password) {
Touches	authentication/security data
Calls detected	trim, Error
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2239: function validateCredentialPair(username, password) {
```

```

2240:   const cleanUsername = String(username || "").trim();
2241:   const cleanPassword = String(password || "");
2242:   if ((cleanUsername && !cleanPassword) || (!cleanUsername && cleanPassword)) {
2243:     throw new Error("Both username and password/token are required when credentials are provided.");
2244:   }
2245:   return { cleanUsername, cleanPassword };
2246: }

```

normalizeGitBranchName - lines 2248-2327

Item	Explanation
Source location	Lines 2248-2327 (80 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	function normalizeGitBranchName(value = "") {
Touches	filesystem, authentication/security data
Calls detected	trim, includes, endsWith, startsWith, detectRemoteDefaultBranch, runPublishGit, match, originRemoteExists, setOriginRemote, checkoutPublishBranch, verifyPublishWriteAccess, safeSegment
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	10 await expression(s), 10 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2248: function normalizeGitBranchName(value = "") {
2249:   const branch = String(value || "").trim();
2250:   if (
2251:     !branch ||
2252:     branch.length > 180 ||
2253:     branch.includes(".") ||
2254:     branch.endsWith(".") ||
2255:     branch.includes("@") ||
2256:     /[\\:\s~^?*[\]\x00-\x1f]/.test(branch) ||
2257:     branch.startsWith("/") ||
2258:     branch.endsWith("/") ||
2259:     branch.includes("//")
2260:   ) {
2261:     return "";
2262:   }
2263:   return branch;
2264: }
2265:
2266: async function detectRemoteDefaultBranch(remoteUrl = "") {
2267:   if (!remoteUrl) return "";
2268:   try {
2269:     const result = await runPublishGit(["ls-remote", "--symref", remoteUrl, "HEAD"], { timeout: 45000 });
2270:     const match = String(result.stdout || "").match(/^ref:\s+refs\/heads\/(.+)\s+HEAD$/m);
2271:     return normalizeGitBranchName(match?.[1] || "");
2272:   } catch {
2273:     return "";
2274:   }
2275: }
2276:
2277: async function originRemoteExists() {
2278:   try {
2279:     await runPublishGit(["remote", "get-url", "origin"]);
2280:     return true;
2281:   } catch {
2282:     return false;
2283:   }
2284: }
2285:
2286: async function setOriginRemote(remoteUrl) {
2287:   if (await originRemoteExists()) {
2288:     await runPublishGit(["remote", "set-url", "origin", remoteUrl]);
2289:   } else {
2290:     await runPublishGit(["remote", "add", "origin", remoteUrl]);
2291:   }
2292: }
2293:
2294: async function checkoutPublishBranch(branch) {
2295:   const cleanBranch = normalizeGitBranchName(branch) || "main";
2296:   const current = (await runPublishGit(["branch", "--show-current"])).stdout.trim();
2297:   if (current === cleanBranch) return cleanBranch;
2298:   try {
2299:     await runPublishGit(["checkout", cleanBranch]);
2300:   } catch {
2301:     await runPublishGit(["checkout", "-B", cleanBranch]);

```



```

2302:   }
2303:   return cleanBranch;
2304: }
2305:
2306: async function verifyPublishWriteAccess(access = {}) {
2307:   const checkBranch = `omb-auth-check-${safeSegment(os.hostname(), "device")}-${Date.now()}`;
2308:   try {
2309:     await runPublishGit(["push", "--dry-run", "origin", `HEAD:refs/heads/${checkBranch}`], {
2310:       timeout: 60 * 1000
2311:     });
2312:     return {
2313:       method: "temporary-dry-run-ref",
2314:       ref: checkBranch
2315:     };
2316:   } catch (error) {
2317:     throw publishAccessError(
2318:       "GitHub authorization is required before this website can be changed.",
2319:       gitFailureText(error),
2320:       access
2321:     );
2322:   }
2323: }
2324:
2325: async function ensurePublishHeadForWriteCheck() {
2326:   try {
2327:     await runPublishGit(["rev-parse", "--verify", "HEAD"]);

```

detectRemoteDefaultBranch - lines 2266-2275

Item	Explanation
Source location	Lines 2266-2275 (10 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function detectRemoteDefaultBranch(remoteUrl = "") {
Touches	filesystem
Calls detected	runPublishGit, match, normalizeGitBranchName
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2266: async function detectRemoteDefaultBranch(remoteUrl = "") {
2267:   if (!remoteUrl) return "";
2268:   try {
2269:     const result = await runPublishGit(["ls-remote", "--symref", remoteUrl, "HEAD"], { timeout: 45000 });
2270:     const match = String(result.stdout || "").match(/^ref:\s+refs\/heads\/(.+)\s+HEAD$/m);
2271:     return normalizeGitBranchName(match?.[1] || "");
2272:   } catch {
2273:     return "";
2274:   }
2275: }

```

originRemoteExists - lines 2277-2284

Item	Explanation
Source location	Lines 2277-2284 (8 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function originRemoteExists() {
Touches	Mostly local calculations or local UI state.
Calls detected	runPublishGit
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 2 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2277: async function originRemoteExists() {
2278:   try {
2279:     await runPublishGit(["remote", "get-url", "origin"]);
2280:     return true;
2281:   } catch {
2282:     return false;
2283:   }
2284: }

```

setOriginRemote - lines 2286-2292

Item	Explanation
Source location	Lines 2286-2292 (7 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function setOriginRemote(remoteUrl) {
Touches	Mostly local calculations or local UI state.
Calls detected	originRemoteExists, runPublishGit
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	3 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2286: async function setOriginRemote(remoteUrl) {
2287:   if (await originRemoteExists()) {
2288:     await runPublishGit(["remote", "set-url", "origin", remoteUrl]);
2289:   } else {
2290:     await runPublishGit(["remote", "add", "origin", remoteUrl]);
2291:   }
2292: }

```

checkoutPublishBranch - lines 2294-2304

Item	Explanation
Source location	Lines 2294-2304 (11 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function checkoutPublishBranch(branch) {
Touches	filesystem
Calls detected	normalizeGitBranchName, runPublishGit, trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	3 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2294: async function checkoutPublishBranch(branch) {
2295:   const cleanBranch = normalizeGitBranchName(branch) || "main";
2296:   const current = (await runPublishGit(["branch", "--show-current"])).stdout.trim();
2297:   if (current === cleanBranch) return cleanBranch;
2298:   try {
2299:     await runPublishGit(["checkout", cleanBranch]);
2300:   } catch {
2301:     await runPublishGit(["checkout", "-B", cleanBranch]);

```

```

2302:   }
2303:   return cleanBranch;
2304: }

```

verifyPublishWriteAccess - lines 2306-2323

Item	Explanation
Source location	Lines 2306-2323 (18 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function verifyPublishWriteAccess(access = {}) {
Touches	authentication/security data
Calls detected	safeSegment, hostname, now, runPublishGit, publishAccessError, gitFailureText
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2306: async function verifyPublishWriteAccess(access = {}) {
2307:   const checkBranch = `omb-auth-check-${safeSegment(os.hostname(), "device")}-${Date.now()}`;
2308:   try {
2309:     await runPublishGit(["push", "--dry-run", "origin", `HEAD:refs/heads/${checkBranch}`], {
2310:       timeout: 60 * 1000
2311:     });
2312:     return {
2313:       method: "temporary-dry-run-ref",
2314:       ref: checkBranch
2315:     };
2316:   } catch (error) {
2317:     throw publishAccessError(
2318:       "GitHub authorization is required before this website can be changed.",
2319:       gitFailureText(error),
2320:       access
2321:     );
2322:   }
2323: }

```

ensurePublishHeadForWriteCheck - lines 2325-2342

Item	Explanation
Source location	Lines 2325-2342 (18 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function ensurePublishHeadForWriteCheck() {
Touches	authentication/security data
Calls detected	runPublishGit
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2325: async function ensurePublishHeadForWriteCheck() {
2326:   try {
2327:     await runPublishGit(["rev-parse", "--verify", "HEAD"]);
2328:     return { created: false };
2329:   } catch {
2330:     await runPublishGit([
2331:       "-c",
2332:       "user.name=OMB Portfolio Builder",
2333:       "-c",
2334:       "user.email=omb-builder@localhost",
2335:       "commit",

```

```

2336:         "--allow-empty",
2337:         "-m",
2338:         "Initialize publish authorization check"
2339:     });
2340:     return { created: true };
2341: }
2342: }

```

synchronizePublishBranchFromRemote - lines 2344-2373

Item	Explanation
Source location	Lines 2344-2373 (30 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function synchronizePublishBranchFromRemote(branch, access = {}) {
Touches	filesystem
Calls detected	normalizeGitBranchName, runPublishGit, trim, checkoutPublishBranch, publishAccessError, gitFailureText
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	4 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2344: async function synchronizePublishBranchFromRemote(branch, access = {}) {
2345:     const cleanBranch = normalizeGitBranchName(branch) || "main";
2346:     try {
2347:         const remoteBranch = await runPublishGit(["ls-remote", "--heads", "origin", cleanBranch], {
2348:             timeout: 45 * 1000
2349:         });
2350:         if (!String(remoteBranch.stdout || "").trim()) {
2351:             return {
2352:                 branch: cleanBranch,
2353:                 remoteBranchExists: false,
2354:                 synchronized: false
2355:             };
2356:         }
2357:
2358:         await runPublishGit(["fetch", "origin", cleanBranch], { timeout: 2 * 60 * 1000 });
2359:         await checkoutPublishBranch(cleanBranch);
2360:         await runPublishGit(["reset", "--hard", `origin/${cleanBranch}`], { timeout: 60 * 1000 });
2361:         return {
2362:             branch: cleanBranch,
2363:             remoteBranchExists: true,
2364:             synchronized: true
2365:         };
2366:     } catch (error) {
2367:         throw publishAccessError(
2368:             "The selected website target could not be synchronized from GitHub.",
2369:             gitFailureText(error),
2370:             { ...access, branch: cleanBranch }
2371:         );
2372:     }
2373: }

```

capturePublishTargetState - lines 2375-2403

Item	Explanation
Source location	Lines 2375-2403 (29 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function capturePublishTargetState() {
Touches	filesystem
Calls detected	runPublishGit, trim, readFile, resolveInsidePortfolioRoot
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	3 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2375: async function capturePublishTargetState() {
2376:   const snapshot = {
2377:     branch: "",
2378:     customDomainExists: false,
2379:     customDomainText: "",
2380:     remote: "",
2381:     remoteExists: false
2382:   };
2383:   try {
2384:     snapshot.remote = (await runPublishGit(["remote", "get-url", "origin"])).stdout.trim();
2385:     snapshot.remoteExists = true;
2386:   } catch {
2387:     snapshot.remote = "";
2388:     snapshot.remoteExists = false;
2389:   }
2390:   try {
2391:     snapshot.branch = (await runPublishGit(["branch", "--show-current"])).stdout.trim();
2392:   } catch {
2393:     snapshot.branch = "";
2394:   }
2395:   try {
2396:     snapshot.customDomainText = await readFile(resolveInsidePortfolioRoot("CNAME"), "utf8");
2397:     snapshot.customDomainExists = true;
2398:   } catch {
2399:     snapshot.customDomainText = "";
2400:     snapshot.customDomainExists = false;
2401:   }
2402:   return snapshot;
2403: }

```

restorePublishTargetState - lines 2405-2429

Item	Explanation
Source location	Lines 2405-2429 (25 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function restorePublishTargetState(snapshot = {}) {
Touches	filesystem
Calls detected	setOriginRemote, originRemoteExists, runPublishGit, checkoutPublishBranch, writeFile, resolveInsidePortfolioRoot, rm
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	6 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2405: async function restorePublishTargetState(snapshot = {}) {
2406:   try {
2407:     if (snapshot.remoteExists && snapshot.remote) {
2408:       await setOriginRemote(snapshot.remote);
2409:     } else if (await originRemoteExists()) {
2410:       await runPublishGit(["remote", "remove", "origin"]);
2411:     }
2412:   } catch {
2413:     // Best-effort rollback; preserve the original error for the caller.
2414:   }
2415:   try {
2416:     if (snapshot.branch) await checkoutPublishBranch(snapshot.branch);
2417:   } catch {
2418:     // Best-effort rollback; preserve the original error for the caller.
2419:   }
2420:   try {
2421:     if (snapshot.customDomainExists) {
2422:       await writeFile(resolveInsidePortfolioRoot("CNAME"), snapshot.customDomainText, "utf8");
2423:     } else {
2424:       await rm(resolveInsidePortfolioRoot("CNAME"), { force: true });
2425:     }
2426:   } catch {
2427:     // Best-effort rollback; preserve the original error for the caller.
2428:   }

```

2429: }

writeTargetCustomDomain - lines 2431-2439

Item	Explanation
Source location	Lines 2431-2439 (9 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	async function writeTargetCustomDomain(domain, customDomainProvided) {
Touches	filesystem
Calls detected	resolveInsidePortfolioRoot, samePath, push, resolveInsideRoot, writeFile, rm
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2431: async function writeTargetCustomDomain(domain, customDomainProvided) {
2432:   const targetPaths = [resolveInsidePortfolioRoot("CNAME")];
2433:   if (!samePath(root, portfolioRoot)) targetPaths.push(resolveInsideRoot("CNAME"));
2434:   if (customDomainProvided && domain) {
2435:     for (const targetPath of targetPaths) await writeFile(targetPath, `${domain}\n`, "utf8");
2436:   } else if (customDomainProvided && !domain) {
2437:     for (const targetPath of targetPaths) await rm(targetPath, { force: true });
2438:   }
2439: }
```

ensureGitRepository - lines 2441-2454

Item	Explanation
Source location	Lines 2441-2454 (14 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function ensureGitRepository() {
Touches	filesystem
Calls detected	mkdir, runPublishGit, trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	5 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2441: async function ensureGitRepository() {
2442:   await mkdir(portfolioRoot, { recursive: true });
2443:   try {
2444:     const insideWorkTree = (await runPublishGit(["rev-parse", "--is-inside-work-tree"])).stdout.trim();
2445:     if (insideWorkTree === "true") return;
2446:   } catch {
2447:     await runPublishGit(["init"]);
2448:   }
2449:
2450:   const branch = (await runPublishGit(["branch", "--show-current"])).stdout.trim();
2451:   if (!branch) {
2452:     await runPublishGit(["checkout", "-B", "main"]);
2453:   }
2454: }
```

configurePublishTarget - lines 2456-2486

Item	Explanation
Source location	Lines 2456-2486 (31 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function configurePublishTarget(options = {}) {
Touches	authentication/security data
Calls detected	call, validatePublishRemoteUrl, validateCustomDomain, validateCredentialPair, ensureGitRepository, setOriginRemote, detectRemoteDefaultBranch, checkoutPublishBranch, writeTargetCustomDomain, getPublishTargetInfo, storeGitCredentials
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	8 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2456: async function configurePublishTarget(options = {}) {
2457:   const {
2458:     repositoryUrl = "",
2459:     customDomain = "",
2460:     authUsername = "",
2461:     authPassword = ""
2462:   } = options;
2463:   const customDomainProvided = Object.prototype.hasOwnProperty.call(options, "customDomain");
2464:   const remoteUrl = validatePublishRemoteUrl(repositoryUrl);
2465:   const domain = validateCustomDomain(customDomain);
2466:   validateCredentialPair(authUsername, authPassword);
2467:
2468:   await ensureGitRepository();
2469:
2470:   if (remoteUrl) {
2471:     await setOriginRemote(remoteUrl);
2472:     const defaultBranch = await detectRemoteDefaultBranch(remoteUrl);
2473:     if (defaultBranch) await checkoutPublishBranch(defaultBranch);
2474:   }
2475:
2476:   await writeTargetCustomDomain(domain, customDomainProvided);
2477:
2478:   const currentRemote = remoteUrl || (await getPublishTargetInfo()).remote;
2479:   const credentials = await storeGitCredentials(currentRemote, authUsername, authPassword);
2480:   const target = await getPublishTargetInfo();
2481:   return {
2482:     ...target,
2483:     credentialsStored: credentials.stored,
2484:     credentialUsername: credentials.username || ""
2485:   };
2486: }

```

readPublishAuthCache - lines 2488-2494

Item	Explanation
Source location	Lines 2488-2494 (7 source lines).
Purpose	Loads data from disk, network, browser storage, or a service.
Declaration	async function readPublishAuthCache() {
Touches	filesystem, authentication/security data
Calls detected	parse, readFile
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2488: async function readPublishAuthCache() {
2489:   try {
2490:     return JSON.parse(await readFile(publishAuthCachePath, "utf8"));

```

```

2491:   } catch {
2492:     return null;
2493:   }
2494: }

```

writePublishAuthCache - lines 2496-2535

Item	Explanation
Source location	Lines 2496-2535 (40 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	async function writePublishAuthCache(access = {}) {
Touches	filesystem, authentication/security data
Calls detected	readPublishAuthCache, getTime, isArray, map, parsePublishAuthTimestamp, filter, isFinite, toISOString, publishAuthCacheScope, writeFile, stringify, chmod
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	3 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2496: async function writePublishAuthCache(access = {}) {
2497:   const previousCache = await readPublishAuthCache();
2498:   const checkedAt = new Date();
2499:   const checkedAtMs = checkedAt.getTime();
2500:   const sameTarget = previousCache
2501:     && previousCache.remote === access.remote
2502:     && previousCache.branch === access.branch
2503:     && previousCache.repository === access.repository;
2504:   const previousHistory = sameTarget && Array.isArray(previousCache.successHistory)
2505:     ? previousCache.successHistory
2506:     : [];
2507:   const successTimestamps = [
2508:     ...previousHistory
2509:       .map((entry) => parsePublishAuthTimestamp(entry))
2510:       .filter((timestamp) => Number.isFinite(timestamp))
2511:       .filter((timestamp) => checkedAtMs - timestamp < publishAuthExtendedTtlMs),
2512:     checkedAtMs
2513:   ];
2514:   const successHistory = successTimestamps.map((timestamp) => new Date(timestamp).toISOString());
2515:   const successCountLastWeek = successTimestamps
2516:     .filter((timestamp) => checkedAtMs - timestamp < publishAuthHistoryWindowMs)
2517:     .length;
2518:   const extendedTrust = successCountLastWeek > publishAuthExtendedThreshold;
2519:   const ttlMs = extendedTrust ? publishAuthExtendedTtlMs : publishAuthCacheTtlMs;
2520:   const cache = {
2521:     branch: access.branch || "",
2522:     checkedAt: checkedAt.toISOString(),
2523:     expiresAt: new Date(checkedAtMs + ttlMs).toISOString(),
2524:     extendedTrust,
2525:     remote: access.remote || "",
2526:     repository: access.repository || "",
2527:     scope: publishAuthCacheScope(),
2528:     successCountLastWeek,
2529:     successHistory,
2530:     trustDays: extendedTrust ? 30 : 1
2531:   };
2532:   await writeFile(publishAuthCachePath, `${JSON.stringify(cache, null, 2)}\n`, "utf8");
2533:   await chmod(publishAuthCachePath, 0o600).catch(() => {});
2534:   return cache;
2535: }

```

publishAuthCacheScope - lines 2537-2552

Item	Explanation
Source location	Lines 2537-2552 (16 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	function publishAuthCacheScope() {
Touches	authentication/security data

Item	Explanation
Calls detected	userInfo, createHash, update, hostname, join, digest
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2537: function publishAuthCacheScope() {
2538:   let username = process.env.USERNAME || process.env.USER || "";
2539:   try {
2540:     username = os.userInfo().username || username;
2541:   } catch {
2542:     // Environment username is a sufficient fallback for cache scoping.
2543:   }
2544:   return createHash("sha256")
2545:     .update([
2546:       os.hostname(),
2547:       username,
2548:       root,
2549:       portfolioRoot
2550:     ].join("\n"))
2551:     .digest("hex");
2552: }

```

parsePublishAuthTimestamp - lines 2554-2559

Item	Explanation
Source location	Lines 2554-2559 (6 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	function parsePublishAuthTimestamp(value = "") {
Touches	filesystem, authentication/security data
Calls detected	parse, isFinite, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2554: function parsePublishAuthTimestamp(value = "") {
2555:   const direct = Date.parse(String(value || ""));
2556:   if (Number.isFinite(direct)) return direct;
2557:   const normalized = String(value || "").replace(/(\.d{3})\d+(Z|[-]\d{2}:?\d{2})$/i, "$1$2");
2558:   return Date.parse(normalized);
2559: }

```

publishAuthCacheExpiresAt - lines 2561-2567

Item	Explanation
Source location	Lines 2561-2567 (7 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	function publishAuthCacheExpiresAt(cache) {
Touches	authentication/security data
Calls detected	parsePublishAuthTimestamp, isFinite, toISOString
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2561: function publishAuthCacheExpiresAt(cache) {
2562:   const explicitExpiresAt = parsePublishAuthTimestamp(cache?.expiresAt || "");
2563:   if (Number.isFinite(explicitExpiresAt)) return new Date(explicitExpiresAt).toISOString();
2564:   const checkedAt = parsePublishAuthTimestamp(cache?.checkedAt || "");
2565:   if (!Number.isFinite(checkedAt)) return "";
2566:   return new Date(checkedAt + publishAuthCacheTtlMs).toISOString();
2567: }

```

publishAuthCachelsFresh - lines 2569-2576

Item	Explanation
Source location	Lines 2569-2576 (8 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	function publishAuthCachelsFresh(cache, access) {
Touches	authentication/security data
Calls detected	publishAuthCacheScope, parse, publishAuthCacheExpiresAt, isFinite, now
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 5 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2569: function publishAuthCacheIsFresh(cache, access) {
2570:   if (!cache || !access) return false;
2571:   if (cache.remote !== access.remote || cache.branch !== access.branch || cache.repository !==
access.repository) return false;
2572:   if (cache.scope !== publishAuthCacheScope()) return false;
2573:   const expiresAt = Date.parse(publishAuthCacheExpiresAt(cache));
2574:   if (!Number.isFinite(expiresAt)) return false;
2575:   return Date.now() < expiresAt;
2576: }

```

assertPublishAccess - lines 2578-2669

Item	Explanation
Source location	Lines 2578-2669 (92 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function assertPublishAccess(options = {}) {
Touches	authentication/security data
Calls detected	runPublishGit, trim, publishAccessError, gitFailureText, remoteUrlForDisplay, workspaceHasCompatibleSiteFiles, parseGitHubRemote, readPublishAuthCache, publishAuthCachelsFresh, publishAuthCacheExpiresAt, synchronizePublishBranchFromRemote, ensurePublishHeadForWriteCheck
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	9 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2578: async function assertPublishAccess(options = {}) {
2579:   let insideWorkTree = "";
2580:   try {

```

```

2581:     insideWorkTree = (await runPublishGit(["rev-parse", "--is-inside-work-tree"])).stdout.trim();
2582:   } catch (error) {
2583:     throw publishAccessError(
2584:       "This workspace is not connected to a Git repository.",
2585:       gitFailureText(error),
2586:       { repository: defaultSiteRepository }
2587:     );
2588:   }
2589:
2590:   if (insideWorkTree !== "true") {
2591:     throw publishAccessError(
2592:       "This workspace is local-only and cannot publish yet.",
2593:       "Clone or associate a GitHub Pages/static-site repository before applying to site.",
2594:       { repository: defaultSiteRepository }
2595:     );
2596:   }
2597:
2598:   let remote = "";
2599:   let branch = "";
2600:   try {
2601:     remote = (await runPublishGit(["remote", "get-url", "origin"])).stdout.trim();
2602:     branch = (await runPublishGit(["branch", "--show-current"])).stdout.trim();
2603:   } catch (error) {
2604:     throw publishAccessError(
2605:       "A publish remote or branch is missing.",
2606:       gitFailureText(error),
2607:       { repository: defaultSiteRepository }
2608:     );
2609:   }
2610:
2611:   if (!remote || !branch) {
2612:     throw publishAccessError(
2613:       "A publish remote or branch is missing.",
2614:       "Set the origin remote and use a named branch before applying to site.",
2615:       { remote: remoteUrlForDisplay(remote), branch }
2616:     );
2617:   }
2618:
2619:   if (options.requireCompatible !== false && !await workspaceHasCompatibleSiteFiles()) {
2620:     throw publishAccessError(
2621:       "This repository does not look like a compatible static portfolio website.",
2622:       "The workspace must include index.html, styles.css, and script.js before it can be used as a
publish target.",
2623:       { remote: remoteUrlForDisplay(remote), branch }
2624:     );
2625:   }
2626:
2627:   const parsedRemote = parseGitHubRemote(remote);
2628:   const repository = parsedRemote ? `${parsedRemote.owner}/${parsedRemote.repo}` :
remoteUrlForDisplay(remote);
2629:
2630:   const access = {
2631:     branch,
2632:     remote: remoteUrlForDisplay(remote),
2633:     repository,
2634:     authorizationChecked: false
2635:   };
2636:
2637:   const cached = await readPublishAuthCache();
2638:   if (!options.force && publishAuthCacheIsFresh(cached, access)) {
2639:     return {
2640:       ...access,
2641:       authorizationCached: true,
2642:       authorizationChecked: true,
2643:       checkedAt: cached.checkedAt,
2644:       expiresAt: publishAuthCacheExpiresAt(cached),
2645:       extendedTrust: Boolean(cached.extendedTrust),
2646:       successCountLastWeek: cached.successCountLastWeek || 0,
2647:       trustDays: cached.trustDays || (cached.extendedTrust ? 30 : 1)
2648:     };
2649:   }
2650:
2651:   const remoteSync = await synchronizePublishBranchFromRemote(branch, access);
2652:   const localHead = await ensurePublishHeadForWriteCheck();
2653:   const writeCheck = await verifyPublishWriteAccess(access);
2654:
2655:   const authCache = await writePublishAuthCache(access);
2656:   return {
2657:     ...access,
2658:     remoteSync,
2659:     localHead,
2660:     writeCheck,
2661:     authorizationChecked: true,
2662:     authorizationCached: false,
2663:     checkedAt: authCache.checkedAt,
2664:     expiresAt: authCache.expiresAt,
2665:     extendedTrust: Boolean(authCache.extendedTrust),
2666:     successCountLastWeek: authCache.successCountLastWeek || 0,
2667:     trustDays: authCache.trustDays || 1
2668:   };
2669: }

```

syncFromPublishTarget - lines 2671-2745

Item	Explanation
Source location	Lines 2671-2745 (75 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function syncFromPublishTarget() {
Touches	filesystem
Calls detected	assertPublishAccess, runPublishGit, trim, mkdtemp, join, tmpdir, runGit, fsAccess, push, includes, publishAccessError, resolveInsideRoot
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	14 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2671: async function syncFromPublishTarget() {
2672:   const publishAccess = await assertPublishAccess({ requireCompatible: false });
2673:   const branch = publishAccess.branch || "main";
2674:   const remote = (await runPublishGit(["remote", "get-url", "origin])).stdout.trim();
2675:   const importRoot = await mkdtemp(path.join(os.tmpdir(), "omb-publish-target-"));
2676:   const cloneRoot = path.join(importRoot, "target");
2677:   const importPaths = [
2678:     "projects.json",
2679:     "assets",
2680:     "docs",
2681:     "Backgrounds",
2682:     "CNAME",
2683:     ".nojekyll",
2684:     "robots.txt",
2685:     "sitemap.xml"
2686:   ];
2687:
2688:   try {
2689:     await runGit(["clone", "--depth", "1", "--branch", branch, remote, cloneRoot], {
2690:       cwd: importRoot,
2691:       timeout: 3 * 60 * 1000
2692:     });
2693:
2694:     const availablePaths = [];
2695:     for (const importPath of importPaths) {
2696:       try {
2697:         await fsAccess(path.join(cloneRoot, importPath));
2698:         availablePaths.push(importPath);
2699:       } catch {
2700:         // Optional path is absent in the selected site repository.
2701:       }
2702:     }
2703:
2704:     if (!availablePaths.includes("projects.json")) {
2705:       throw publishAccessError(
2706:         "The selected repository does not contain a projects.json portfolio catalog.",
2707:         "The target repository must contain a compatible OMB Portfolio Builder catalog before it can be
imported.",
2708:         publishAccess
2709:       );
2710:     }
2711:
2712:     const backupRoot = resolveInsideRoot(
2713:       ".omb-backups",
2714:       `load-target-${new Date().toISOString().replace(/[:.]/g, "-")}`
2715:     );
2716:     await mkdir(backupRoot, { recursive: true });
2717:
2718:     for (const importPath of availablePaths) {
2719:       const sourcePath = path.join(cloneRoot, importPath);
2720:       const targetPath = resolveInsideRoot(importPath);
2721:       try {
2722:         await fsAccess(targetPath);
2723:         await cp(targetPath, path.join(backupRoot, importPath), { recursive: true, force: true });
2724:         await rm(targetPath, { recursive: true, force: true });
2725:       } catch {
2726:         // Nothing local to back up for this path.
2727:       }
2728:       await cp(sourcePath, targetPath, { recursive: true, force: true });
2729:     }
2730:
2731:     const importedCatalog = await readFile(path.join(cloneRoot, "projects.json"), "utf8");
```

```

2732:     await writeFile(draftPath, importedCatalog, "utf8");
2733:     const portfolioSync = await syncPortfolioPublishFiles({ removeMissing: false });
2734:
2735:     return {
2736:       ...publishAccess,
2737:       imported: availablePaths,
2738:       portfolioSync,
2739:       backup: path.relative(root, backupRoot),
2740:       draftUpdated: true
2741:     };
2742:   } finally {
2743:     await rm(importRoot, { recursive: true, force: true });
2744:   }
2745: }

```

authenticateGitHubForTarget - lines 2747-2843

Item	Explanation
Source location	Lines 2747-2843 (97 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function authenticateGitHubForTarget(options = {}) {
Touches	authentication/security data
Calls detected	call, validatePublishRemoteUrl, validateCustomDomain, validateCredentialPair, publishAccessError, getPublishTargetInfo, ensureGitRepository, capturePublishTargetState, setOriginRemote, getGitStatus, detectRemoteDefaultBranch, checkoutPublishBranch
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	19 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2747: async function authenticateGitHubForTarget(options = {}) {
2748:   const {
2749:     repositoryUrl = "",
2750:     customDomain = "",
2751:     authUsername = "",
2752:     authPassword = ""
2753:   } = options;
2754:   const customDomainProvided = Object.prototype.hasOwnProperty.call(options, "customDomain");
2755:   const remoteUrl = validatePublishRemoteUrl(repositoryUrl);
2756:   const domain = validateCustomDomain(customDomain);
2757:   const credentials = validateCredentialPair(authUsername, authPassword);
2758:   if (!remoteUrl) {
2759:     throw publishAccessError(
2760:       "A GitHub repository URL is required before authentication.",
2761:       "Enter the GitHub Pages or compatible static-site repository URL, then click Authenticate target.",
2762:       await getPublishTargetInfo()
2763:     );
2764:   }
2765:
2766:   await ensureGitRepository();
2767:   const snapshot = await capturePublishTargetState();
2768:
2769:   try {
2770:     await setOriginRemote(remoteUrl);
2771:     const status = await getGitStatus();
2772:     const preliminaryTarget = await getPublishTargetInfo();
2773:     if (!status.git.ok) {
2774:       throw publishAccessError(
2775:         "Git for Windows is required for publishing.",
2776:         "Install Git for Windows, then reopen the builder and authenticate again.",
2777:         { ...preliminaryTarget, system: status }
2778:       );
2779:     }
2780:
2781:     const targetBranch = await detectRemoteDefaultBranch(remoteUrl) || "main";
2782:     await checkoutPublishBranch(targetBranch);
2783:
2784:     const targetBeforeAuth = await getPublishTargetInfo();
2785:     const cached = await readPublishAuthCache();
2786:     const cachedAccess = {
2787:       branch: targetBeforeAuth.branch,
2788:       remote: targetBeforeAuth.remote,
2789:       repository: targetBeforeAuth.repository
2790:     };
2791:     if (!credentials.cleanUsername && publishAuthCacheIsFresh(cached, cachedAccess)) {
2792:       await writeTargetCustomDomain(domain, customDomainProvided);

```

```

2793:     const target = await getPublishTargetInfo();
2794:     return {
2795:       ...cachedAccess,
2796:       branch: target.branch || cachedAccess.branch,
2797:       authorizationCached: true,
2798:       authorizationChecked: true,
2799:       checkedAt: cached.checkedAt,
2800:       expiresAt: publishAuthCacheExpiresAt(cached),
2801:       extendedTrust: Boolean(cached.extendedTrust),
2802:       successCountLastWeek: cached.successCountLastWeek || 0,
2803:       trustDays: cached.trustDays || (cached.extendedTrust ? 30 : 1),
2804:       credentialsStored: false,
2805:       credentialUsername: "",
2806:       system: status,
2807:       target
2808:     };
2809:   }
2810:
2811:   if (!status.credentialManager.ok && !credentials.cleanUsername) {
2812:     throw publishAccessError(
2813:       "Git Credential Manager is required for guided GitHub sign-in.",
2814:       "Install or repair Git for Windows with Git Credential Manager enabled, or provide a GitHub
username and personal access token.",
2815:       { ...targetBeforeAuth, system: status }
2816:     );
2817:   }
2818:
2819:   const storedCredentials = await storeGitCredentials(remoteUrl, authUsername, authPassword);
2820:   if (!storedCredentials.stored) {
2821:     await runGit(["credential-manager", "github", "login"], {
2822:       cwd: portfolioRoot,
2823:       timeout: 5 * 60 * 1000,
2824:       windowsHide: false
2825:     });
2826:   }
2827:
2828:   const access = await assertPublishAccess({ force: true, requireCompatible: false });
2829:   await writeTargetCustomDomain(domain, customDomainProvided);
2830:   const target = await getPublishTargetInfo();
2831:   return {
2832:     ...access,
2833:     branch: target.branch || access.branch,
2834:     credentialsStored: storedCredentials.stored,
2835:     credentialUsername: storedCredentials.username || "",
2836:     system: await getGitStatus(),
2837:     target
2838:   };
2839: } catch (error) {
2840:   await restorePublishTargetState(snapshot);
2841:   throw error;
2842: }
2843: }

```

installGitForWindows - lines 2845-2868

Item	Explanation
Source location	Lines 2845-2868 (24 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function installGitForWindows() {
Touches	external processes, network/API requests, authentication/security data
Calls detected	runOptionalCommand, spawn,_unref
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2845: async function installGitForWindows() {
2846:   const winget = await runOptionalCommand("winget", ["--version"]);
2847:   const downloadUrl = "https://git-scm.com/download/win";
2848:   if (!winget.ok) {
2849:     return {
2850:       launched: false,
2851:       downloadUrl,
2852:       message: "winget was not found. Open the Git for Windows download page and install Git with Git
Credential Manager enabled."
2853:     };

```

```

2854:   }
2855:
2856:   const child = spawn("winget", ["install", "--id", "Git.Git", "-e", "--source", "winget"], {
2857:     detached: true,
2858:     shell: true,
2859:     stdio: "ignore",
2860:     windowsHide: false
2861:   });
2862:   child.unref();
2863:   return {
2864:     launched: true,
2865:     downloadUrl,
2866:     message: "The Git for Windows installer was launched through winget. Follow the installer prompts,
then reopen the builder."
2867:   };
2868: }

```

getUpdateInfo - lines 2870-2906

Item	Explanation
Source location	Lines 2870-2906 (37 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	async function getUpdateInfo() {
Touches	network/API requests
Calls detected	fetch, Error, json, replace, find, has, compareVersions
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2870: async function getUpdateInfo() {
2871:   const currentVersion = process.env.OMB_APP_VERSION || packageJson.version || "0.0.0";
2872:   const owner = "otienomaurice";
2873:   const repo = "omb-portfolio-builder";
2874:   try {
2875:     const response = await fetch(`https://api.github.com/repos/${owner}/${repo}/releases/latest`, {
2876:       headers: { "Accept": "application/vnd.github+json", "User-Agent": "OMB-Portfolio-Builder" }
2877:     });
2878:     if (!response.ok) throw new Error(`GitHub returned ${response.status}`);
2879:     const release = await response.json();
2880:     const latestVersion = String(release.tag_name || "").replace(/^builder-v/i, "");
2881:     const installer = (release.assets || []).find((asset) => /Setup-.*\.exe$/i.test(asset.name));
2882:     const portable = (release.assets || []).find((asset) => /Portable-.*\.exe$/i.test(asset.name));
2883:     const updateBlocked = blockedAppUpdateVersions.has(latestVersion);
2884:     return {
2885:       ok: true,
2886:       currentVersion,
2887:       latestVersion,
2888:       updateAvailable: !updateBlocked && compareVersions(latestVersion, currentVersion) > 0,
2889:       updateBlocked,
2890:       blockedReason: updateBlocked
? `Builder ${latestVersion} is skipped because its installer can hang while running the old
uninstaller during in-app updates. Wait for builder 0.2.17 or newer, then run Update again.`
2892:       : "",
2893:       releaseUrl: release.html_url || "",
2894:       installerUrl: installer?.browser_download_url || "",
2895:       installerName: installer?.name || "",
2896:       portableUrl: portable?.browser_download_url || ""
2897:     };
2898:   } catch (error) {
2899:     return {
2900:       ok: false,
2901:       currentVersion,
2902:       latestVersion: currentVersion,
2903:       updateAvailable: false,
2904:       error: error.message || "Could not check for updates."
2905:     };
2906:   }

```

installer - lines 2881-2897

Item	Explanation
Source location	Lines 2881-2897 (17 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	const installer = (release.assets []).find((asset) => /Setup-.*\.exe\$/i.test(asset.name));
Touches	Mostly local calculations or local UI state.
Calls detected	find, has, compareVersions
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2881:     const installer = (release.assets || []).find((asset) => /Setup-.*\.exe$/i.test(asset.name));
2882:     const portable = (release.assets || []).find((asset) => /Portable-.*\.exe$/i.test(asset.name));
2883:     const updateBlocked = blockedAppUpdateVersions.has(latestVersion);
2884:     return {
2885:       ok: true,
2886:       currentVersion,
2887:       latestVersion,
2888:       updateAvailable: !updateBlocked && compareVersions(latestVersion, currentVersion) > 0,
2889:       updateBlocked,
2890:       blockedReason: updateBlocked
2891:       ? `Builder ${latestVersion} is skipped because its installer can hang while running the old
uninstaller during in-app updates. Wait for builder 0.2.17 or newer, then run Update again.`
2892:       : "",
2893:       releaseUrl: release.html_url || "",
2894:       installerUrl: installer?.browser_download_url || "",
2895:       installerName: installer?.name || "",
2896:       portableUrl: portable?.browser_download_url || ""
2897:     };

```

portable - lines 2882-2897

Item	Explanation
Source location	Lines 2882-2897 (16 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	const portable = (release.assets []).find((asset) => /Portable-.*\.exe\$/i.test(asset.name));
Touches	Mostly local calculations or local UI state.
Calls detected	find, has, compareVersions
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2882:     const portable = (release.assets || []).find((asset) => /Portable-.*\.exe$/i.test(asset.name));
2883:     const updateBlocked = blockedAppUpdateVersions.has(latestVersion);
2884:     return {
2885:       ok: true,
2886:       currentVersion,
2887:       latestVersion,
2888:       updateAvailable: !updateBlocked && compareVersions(latestVersion, currentVersion) > 0,
2889:       updateBlocked,
2890:       blockedReason: updateBlocked
2891:       ? `Builder ${latestVersion} is skipped because its installer can hang while running the old
uninstaller during in-app updates. Wait for builder 0.2.17 or newer, then run Update again.`
2892:       : "",
2893:       releaseUrl: release.html_url || "",
2894:       installerUrl: installer?.browser_download_url || "",
2895:       installerName: installer?.name || "",
2896:       portableUrl: portable?.browser_download_url || ""
2897:     };

```


getBuilderReleaseDownloadReport - lines 2909-2914

Item	Explanation
Source location	Lines 2909-2914 (6 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	async function getBuilderReleaseDownloadReport() {
Touches	network/API requests
Calls detected	fetch
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2909: async function getBuilderReleaseDownloadReport() {
2910:   const owner = "otienomaurice";
2911:   const repo = "omb-portfolio-builder";
2912:   const response = await fetch(`https://api.github.com/repos/${owner}/${repo}/releases/latest`, {
2913:     headers: { "Accept": "application/vnd.github+json", "User-Agent": "OMB-Portfolio-Builder" }
2914:   });
```

assets - lines 2917-2923

Item	Explanation
Source location	Lines 2917-2923 (7 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	const assets = (release.assets []).map((asset) => ({
Touches	Mostly local calculations or local UI state.
Calls detected	map
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2917:   const assets = (release.assets || []).map((asset) => ({
2918:     name: asset.name || "",
2919:     downloadCount: Number(asset.download_count || 0),
2920:     sizeBytes: Number(asset.size || 0),
2921:     updatedAt: asset.updated_at || "",
2922:     url: asset.browser_download_url || ""
2923:   }));
```

getSecurityReport - lines 2933-2977

Item	Explanation
Source location	Lines 2933-2977 (45 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	async function getSecurityReport() {
Touches	authentication/security data
Calls detected	allSettled, getBuilderReleaseDownloadReport, readPublishAuthCache, getPublishTargetInfo, toISOString, publishAuthCachesFresh, publishAuthCacheExpiresAt
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2933: async function getSecurityReport() {
2934:   const [downloadResult, authCacheResult, targetResult] = await Promise.allSettled([
2935:     getBuilderReleaseDownloadReport(),
2936:     readPublishAuthCache(),
2937:     getPublishTargetInfo()
2938:   ]);
2939:   const authCache = authCacheResult.status === "fulfilled" ? authCacheResult.value : null;
2940:   const target = targetResult.status === "fulfilled" ? targetResult.value : null;
2941:   return {
2942:     ok: true,
2943:     generatedAt: new Date().toISOString(),
2944:     appDownloads: downloadResult.status === "fulfilled"
2945:       ? { ok: true, ...downloadResult.value }
2946:       : { ok: false, error: downloadResult.reason?.message || "GitHub release download counts were not
available." },
2947:     publishingAuth: {
2948:       targetRepository: target?.repository || "",
2949:       branch: target?.branch || "",
2950:       authenticated: Boolean(authCache && publishAuthCacheIsFresh(authCache, {
2951:         remote: target?.remote || "",
2952:         repository: target?.repository || "",
2953:         branch: target?.branch || ""
2954:       })),
2955:       expiresAt: publishAuthCacheExpiresAt(authCache) || "",
2956:       trustDays: Number(authCache?.trustDays || 0),
2957:       successCountLastWeek: Number(authCache?.successCountLastWeek || 0)
2958:     },
2959:     websiteAccess: {
2960:       available: false,
2961:       summary: "Visitor counts and visitor IP addresses are not available from a static GitHub Pages site
or GitHub release download counts.",
2962:       recommendedSetup: [
2963:         "Proxy the custom domain through Cloudflare instead of DNS-only.",
2964:         "Enable Cloudflare Web Analytics for aggregate page-view counts.",
2965:         "Use Cloudflare Logpush, Analytics Engine, or a Worker-backed endpoint if you need IP-level
security logs.",
2966:         "Publish a privacy notice before collecting raw IP addresses or persistent identifiers."
2967:       ]
2968:     },
2969:     securityControls: [
2970:       "Local builder write APIs are restricted to local requests.",
2971:       "Publishing target details are only returned to the local machine.",
2972:       "GitHub publishing authorization is cached for one day by default, or longer after repeated
successful authorizations.",
2973:       "The desktop app runs with Electron nodeIntegration disabled, contextIsolation enabled, and sandbox
enabled.",
2974:       "Deployable website security headers and security.txt metadata are included in the publishable site
files."
2975:     ]
2976:   };
2977: }

```

safeUpdateFileSegment - lines 2979-2984

Item	Explanation
Source location	Lines 2979-2984 (6 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function safeUpdateFileSegment(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	replace, slice
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2979: function safeUpdateFileSegment(value = "") {
2980:   return String(value || "")
2981:   .replace(/[^a-z0-9._-]+/gi, "-")
2982:   .replace(/^-+|-+$/g, "")
2983:   .slice(0, 80) || "latest";
2984: }
```

powershellSingleQuoted - lines 2986-2988

Item	Explanation
Source location	Lines 2986-2988 (3 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	function powershellSingleQuoted(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	replaceAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2986: function powershellSingleQuoted(value = "") {
2987:   return `${String(value).replaceAll("'", "'')}';
2988: }
```

downloadAndLaunchAppUpdate - lines 2990-3165

Item	Explanation
Source location	Lines 2990-3165 (176 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	async function downloadAndLaunchAppUpdate() {
Touches	filesystem, external processes, network/API requests
Calls detected	getUpdateInfo, Error, join, tmpdir, mkdir, safeUpdateFileSegment, fetch, from, arrayBuffer, writeFile, dirname, homedir
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	7 await expression(s), 6 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2990: async function downloadAndLaunchAppUpdate() {
2991:   const update = await getUpdateInfo();
2992:   if (!update.ok) throw new Error(update.error || "Could not check for updates.");
2993:   if (update.updateBlocked) {
2994:     throw new Error(update.blockedReason || `Builder ${update.latestVersion} is temporarily blocked from
automatic updates.`);
2995:   }
2996:   if (!update.updateAvailable) {
2997:     throw new Error(`OMB Portfolio Builder ${update.currentVersion || ""} is already up to date.`);
2998:   }
2999:   if (!update.installerUrl) {
3000:     throw new Error("The latest GitHub Release does not include a Windows setup installer.");
3001:   }
3002:   if (process.platform !== "win32") {
3003:     throw new Error("Automatic installer updates are currently available only on Windows.");
3004:   }
3005:
3006:   const updateRoot = path.join(os.tmpdir(), "omb-portfolio-builder-updates");
3007:   await mkdir(updateRoot, { recursive: true });
3008:   const installerName = update.installerName && /\.exe$/i.test(update.installerName)
3009:     ? update.installerName
```

```

3010:     : `OMB-Portfolio-Builder-Setup-${safeUpdateFileSegment(update.latestVersion)}-x64.exe`;
3011:     const installerPath = path.join(updateRoot, installerName);
3012:
3013:     const response = await fetch(update.installerUrl, {
3014:       headers: { "Accept": "application/octet-stream", "User-Agent": "OMB-Portfolio-Builder" }
3015:     });
3016:     if (!response.ok) throw new Error(`GitHub returned ${response.status} while downloading the
installer.`);
3017:     const installerBuffer = Buffer.from(await response.arrayBuffer());
3018:     if (installerBuffer.length < 1024 * 1024) {
3019:       throw new Error("The downloaded installer was unexpectedly small, so the update was stopped.");
3020:     }
3021:     await writeFile(installerPath, installerBuffer);
3022:
3023:     const updateLogPath = path.join(updateRoot,
`update-${safeUpdateFileSegment(update.latestVersion)}.log`);
3024:     const currentExecutable = /OMB Portfolio Builder\.exe$/i.test(process.execPath || "") ?
process.execPath : "";
3025:     const currentInstallDirectory = currentExecutable ? path.dirname(currentExecutable) : "";
3026:     const localAppDataExecutable = process.env.LOCALAPPDATA
3027:       ? path.join(process.env.LOCALAPPDATA, "Programs", "OMB Portfolio Builder", "OMB Portfolio
Builder.exe")
3028:       : "";
3029:     const legacyManagedApplicationExecutable = path.join(os.homedir(), "OMB", "application", "OMB Portfolio
Builder", "OMB Portfolio Builder.exe");
3030:     const legacyFlatApplicationExecutable = path.join(os.homedir(), "OMB", "application", "OMB Portfolio
Builder.exe");
3031:     const relaunchCandidates = Array.from(new Set([
3032:       localAppDataExecutable,
3033:       currentExecutable,
3034:       currentInstallDirectory ? path.join(currentInstallDirectory, "OMB Portfolio Builder.exe") : "",
3035:       legacyFlatApplicationExecutable,
3036:       legacyManagedApplicationExecutable,
3037:       process.env.ProgramFiles ? path.join(process.env.ProgramFiles, "OMB Portfolio Builder", "OMB
Portfolio Builder.exe") : "",
3038:       process.env["ProgramFiles(x86)"] ? path.join(process.env["ProgramFiles(x86)"], "OMB Portfolio
Builder", "OMB Portfolio Builder.exe") : ""
3039:     ]).filter(Boolean));
3040:     const relaunchCandidatesPs = `@(${relaunchCandidates.map(powershellSingleQuoted).join(", ")});`;
3041:     const launcherPath = path.join(updateRoot, `run-
update-${safeUpdateFileSegment(update.latestVersion)}.ps1`);
3042:     const launcherCommandPath = path.join(updateRoot, `run-
update-${safeUpdateFileSegment(update.latestVersion)}.cmd`);
3043:     const launcherScript = [
3044:       "$ErrorActionPreference = 'Continue'",
3045:       "`$parentPid = ${process.pid}`",
3046:       "`$installer = ${powershellSingleQuoted(installerPath)}",
3047:       "`$log = ${powershellSingleQuoted(updateLogPath)}",
3048:       "`$candidates = ${relaunchCandidatesPs}",
3049:       "function Write-UpdateLog([string]$Message) {",
3050:       "  try { Add-Content -LiteralPath $log -Value ((Get-Date).ToString('s') + ' ' + $Message) } catch
{}",
3051:       "}",
3052:       "function Wait-ForExecutable([string[]]$Paths, [int]$Seconds) {",
3053:       "  $deadline = (Get-Date).AddSeconds($Seconds)",
3054:       "  do {",
3055:       "    foreach ($candidate in $Paths) {",
3056:       "      if ($candidate -and (Test-Path -LiteralPath $candidate)) { return $candidate }",
3057:       "    }",
3058:       "    Start-Sleep -Seconds 1",
3059:       "  } while ((Get-Date) -lt $deadline)",
3060:       "  return $null",
3061:       "}",
3062:       "function Stop-BUILDERProcesses([int]$ProcessId, [string[]]$Paths) {",
3063:       "  try {",
3064:       "    $normalizedPaths = @($Paths | Where-Object { $_ } | ForEach-Object {",
3065:       "      try { [System.IO.Path]::GetFullPath($_).ToLowerInvariant() } catch { $_.ToLowerInvariant() }",
3066:       "    }",
3067:       "    $processes = Get-CimInstance Win32_Process | Where-Object {",
3068:       "      $_.ProcessId -eq $ProcessId -or",
3069:       "      $_.Name -eq 'OMB Portfolio Builder.exe' -or",
3070:       "      ($_.ExecutablePath -and ($normalizedPaths -contains
([System.IO.Path]::GetFullPath($_.ExecutablePath).ToLowerInvariant()))",
3071:       "    } | Sort-Object ProcessId -Unique",
3072:       "    foreach ($process in $processes) {",
3073:       "      try {",
3074:       "        Write-UpdateLog ('Stopping previous builder process PID ' + $process.ProcessId + ' at ' +
$process.ExecutablePath)",
3075:       "        Stop-Process -Id $process.ProcessId -Force -ErrorAction SilentlyContinue",
3076:       "      } catch { Write-UpdateLog ('Could not stop PID ' + $process.ProcessId + ': ' +
$_.Exception.Message) }",
3077:       "    }",
3078:       "    Start-Sleep -Seconds 2",
3079:       "  } catch { Write-UpdateLog ('Could not stop previous builder processes: ' + $_.Exception.Message)
}",
3080:       "}",
3081:       "function Run-Installer([switch]$Elevated) {",
3082:       "  $process = $null",
3083:       "  if ($Elevated) {",
3084:       "    Write-UpdateLog 'Starting installer with Windows elevation prompt.'",
3085:       "    $process = Start-Process -FilePath $installer -ArgumentList @('/S') -Verb RunAs -PassThru",
3086:       "  } else {",

```

```

3087:      "      Write-UpdateLog 'Starting installer silently.'",
3088:      "      $process = Start-Process -FilePath $installer -ArgumentList @('/S') -WindowStyle Hidden
-PassThru",
3089:      "    }",
3090:      "    if (-not $process) { return 0 }",
3091:      "    if (-not $process.WaitForExit(900000)) {",
3092:      "      Write-UpdateLog ('Installer timed out after 15 minutes. PID: ' + $process.Id)",
3093:      "      try { Stop-Process -Id $process.Id -Force -ErrorAction SilentlyContinue } catch { Write-
UpdateLog ('Could not stop timed-out installer: ' + $_.Exception.Message) }",
3094:      "      return 124",
3095:      "    }",
3096:      "    return $process.ExitCode",
3097:      "  }",
3098:      "Write-UpdateLog 'Update launcher started.'",
3099:      "Start-Sleep -Seconds 3",
3100:      "try { Wait-Process -Id $parentPid -Timeout 8; Write-UpdateLog 'Previous app process exited.' } catch
{ Write-UpdateLog ('Previous app wait finished: ' + $_.Exception.Message) }",
3101:      "Stop-BuilderProcesses $parentPid $candidates",
3102:      "Start-Sleep -Seconds 1",
3103:      "$installerProcess = $null",
3104:      "$installerExit = 1",
3105:      "try {",
3106:      "    Write-UpdateLog ('Starting installer: ' + $installer)",
3107:      "    $installerExit = Run-Installer",
3108:      "    Write-UpdateLog ('Installer exit code: ' + $installerExit)",
3109:      "} catch {",
... excerpt truncated after 120 lines. Open the source file for the rest of this function.

```

syncPortfolioPublishFiles - lines 3167-3200

Item	Explanation
Source location	Lines 3167-3200 (34 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function syncPortfolioPublishFiles(options = {}) {
Touches	filesystem
Calls detected	samePath, mkdir, resolveInsideRoot, resolveInsidePortfolioRoot, pathExists, rm, dirname, cp, push
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	6 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3167: async function syncPortfolioPublishFiles(options = {}) {
3168:   if (samePath(root, portfolioRoot)) {
3169:     return { copied: [], skipped: [], workspace: portfolioRoot, separatedWorkspace: false };
3170:   }
3171:
3172:   const { removeMissing = false } = options;
3173:   const copied = [];
3174:   const skipped = [];
3175:
3176:   await mkdir(portfolioRoot, { recursive: true });
3177:
3178:   for (const relativePath of publishPaths) {
3179:     const sourcePath = resolveInsideRoot(relativePath);
3180:     const targetPath = resolveInsidePortfolioRoot(relativePath);
3181:     if (await pathExists(sourcePath)) {
3182:       await rm(targetPath, { recursive: true, force: true });
3183:       await mkdir(path.dirname(targetPath), { recursive: true });
3184:       await cp(sourcePath, targetPath, { recursive: true, force: true });
3185:       copied.push(relativePath);
3186:     } else if (removeMissing) {
3187:       await rm(targetPath, { recursive: true, force: true });
3188:       skipped.push(relativePath);
3189:     } else {
3190:       skipped.push(relativePath);
3191:     }
3192:   }
3193:
3194:   return {
3195:     copied,
3196:     skipped,
3197:     workspace: portfolioRoot,
3198:     separatedWorkspace: true
3199:   };
3200: }

```

publishSiteChanges - lines 3202-3238

Item	Explanation
Source location	Lines 3202-3238 (37 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function publishSiteChanges(publishAccess = null) {
Touches	Mostly local calculations or local UI state.
Calls detected	assertPublishAccess, runPublishGit, trim, synchronizePublishBranchFromRemote, syncPortfolioPublishFiles, bumpPublishedAssetVersions, stageablePublishPaths, Error, toISOString, slice
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	10 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
3202: async function publishSiteChanges(publishAccess = null) {
3203:   const access = publishAccess || await assertPublishAccess();
3204:   const branch = access.branch || (await runPublishGit(["branch", "--show-current"])).stdout.trim();
3205:   const remoteSync = await synchronizePublishBranchFromRemote(branch, access);
3206:   const portfolioSync = await syncPortfolioPublishFiles({ removeMissing: true });
3207:   await bumpPublishedAssetVersions(portfolioRoot);
3208:
3209:   const stageablePaths = await stageablePublishPaths(publishPaths);
3210:   if (!stageablePaths.length) {
3211:     throw new Error("No compatible site files were available to publish.");
3212:   }
3213:
3214:   await runPublishGit(["add", "-A", "--", ...stageablePaths]);
3215:   const status = await runPublishGit(["status", "--porcelain", "--", ...stageablePaths]);
3216:   const hasChanges = status.stdout.trim().length > 0;
3217:
3218:   let commit = null;
3219:   if (hasChanges) {
3220:     const message = `Update portfolio site ${new Date().toISOString().slice(0, 10)}`;
3221:     commit = await runPublishGit(["commit", "-m", message]);
3222:   }
3223:
3224:   const pushArgs = branch ? ["push", "origin", branch] : ["push"];
3225:   const push = await runPublishGit(pushArgs);
3226:
3227:   return {
3228:     ...access,
3229:     portfolioWorkspace: portfolioRoot,
3230:     remoteSync,
3231:     portfolioSync,
3232:     branch: branch || "current branch",
3233:     committed: hasChanges,
3234:     commitOutput: commit?.stdout || commit?.stderr || "",
3235:     pushed: true,
3236:     pushOutput: push.stdout || push.stderr || ""
3237:   };
3238: }
```

handlePortfolioAi - lines 3240-3322

Item	Explanation
Source location	Lines 3240-3322 (83 source lines).
Purpose	Handles an event, request, command, or user action.
Declaration	async function handlePortfolioAi(request, response) {
Touches	network/API requests, authentication/security data
Calls detected	readRequestJson, clampText, trim, cleanConversationHistory, Set, has, sendJson, enrichPortfolioContext, callOllamaPortfolioAi, toLowerCase, stringify, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	5 await expression(s), 4 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3240: async function handlePortfolioAi(request, response) {
3241:   const body = await readRequestJson(request);
3242:   const question = clampText(body.question, 1200).trim();
3243:   const conversation = cleanConversationHistory(body.conversation);
3244:   const validIntents = new Set(["general_conversation", "general_engineering", "general_knowledge",
"portfolio_specific"]);
3245:   const intent = validIntents.has(body.intent) ? body.intent : "portfolio_specific";
3246:   const allowWebSearch = body.allowWebSearch === true;
3247:
3248:   if (!question) {
3249:     sendJson(response, 400, { error: "Question is required." });
3250:     return;
3251:   }
3252:
3253:   const context = await enrichPortfolioContext({ ...(body.context || {}), question });
3254:
3255:   const apiKey = process.env.OPENAI_API_KEY || "";
3256:   if (!apiKey) {
3257:     const ollama = await callOllamaPortfolioAi({ question, intent, conversation, context, allowWebSearch
});
3258:     if (ollama.ok) {
3259:       sendJson(response, 200, {
3260:         answer: ollama.answer,
3261:         model: ollama.model,
3262:         provider: "ollama",
3263:         usedWebSearch: false
3264:       });
3265:       return;
3266:     }
3267:     sendJson(response, 503, { error: ollama.error || "No local Ollama model or OPENAI_API_KEY is
available for the local backend." });
3268:     return;
3269:   }
3270:
3271:   const model = process.env.OPENAI_MODEL || "gpt-5.4";
3272:   const fallbackModel = process.env.OPENAI_FALLBACK_MODEL || "gpt-4.1";
3273:   const webSearchMode = String(process.env.OPENAI_ENABLE_WEB_SEARCH || "auto").toLowerCase();
3274:   const enableWebSearch = webSearchMode === "true" || (webSearchMode !== "false" && allowWebSearch);
3275:   const buildPayload = (selectedModel) => ({
3276:     model: selectedModel,
3277:     input: [
3278:       {
3279:         role: "developer",
3280:         content: [{ type: "input_text", text: portfolioAiInstructions }]
3281:       },
3282:       {
3283:         role: "user",
3284:         content: [{
3285:           type: "input_text",
3286:           text: [
3287:             `Visitor question: ${question}`,
3288:             `Question intent: ${intent}`,
3289:             `Web search allowed for this question: ${enableWebSearch ? "yes" : "no"}`,
3290:             "",
3291:             "Recent conversation JSON:",
3292:             clampText(JSON.stringify(conversation, null, 2), 8000),
3293:             "",
3294:             "Portfolio context JSON:",
3295:             clampText(JSON.stringify(context, null, 2), 18000)
3296:           ].join("\n")
3297:         ]
3298:       }
3299:     ],
3300:     max_output_tokens: Number(process.env.OPENAI_MAX_OUTPUT_TOKENS || 1100),
3301:     reasoning: { effort: process.env.OPENAI_REASONING_EFFORT || "medium" },
3302:     text: { verbosity: process.env.OPENAI_VERBOSITY || "medium" }
3303:   });
3304:
3305:   const callModel = async (selectedModel) => {
3306:     const payload = buildPayload(selectedModel);
3307:     if (enableWebSearch) {
3308:       payload.tools = [{ type: "web_search", search_context_size:
process.env.OPENAI_WEB_SEARCH_CONTEXT_SIZE || "low" }];
3309:     }
3310:
3311:     const openAiResponse = await fetch("https://api.openai.com/v1/responses", {
3312:       method: "POST",
3313:       headers: {
3314:         "Authorization": `Bearer ${apiKey}`,
3315:         "Content-Type": "application/json"
3316:       },
3317:       body: JSON.stringify(payload)
3318:     });

```

```

3319:
3320:     const data = await openAiResponse.json().catch(() => ({}));
3321:     return { data, openAiResponse, selectedModel };
3322: };

```

buildPayload - lines 3275-3303

Item	Explanation
Source location	Lines 3275-3303 (29 source lines).
Purpose	Creates an object, UI structure, process, payload, or generated artifact.
Declaration	const buildPayload = (selectedModel) => ({
Touches	authentication/security data
Calls detected	clampText, stringify, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3275: const buildPayload = (selectedModel) => ({
3276:   model: selectedModel,
3277:   input: [
3278:     {
3279:       role: "developer",
3280:       content: [{ type: "input_text", text: portfolioAiInstructions }]
3281:     },
3282:     {
3283:       role: "user",
3284:       content: [{
3285:         type: "input_text",
3286:         text: [
3287:           `Visitor question: ${question}`,
3288:           `Question intent: ${intent}`,
3289:           `Web search allowed for this question: ${enableWebSearch ? "yes" : "no"}`,
3290:           "",
3291:           "Recent conversation JSON:",
3292:           clampText(JSON.stringify(conversation, null, 2), 8000),
3293:           "",
3294:           "Portfolio context JSON:",
3295:           clampText(JSON.stringify(context, null, 2), 18000)
3296:         ].join("\n")
3297:       }
3298:     ]
3299:   },
3300:   max_output_tokens: Number(process.env.OPENAI_MAX_OUTPUT_TOKENS || 1100),
3301:   reasoning: { effort: process.env.OPENAI_REASONING_EFFORT || "medium" },
3302:   text: { verbosity: process.env.OPENAI_VERBOSITY || "medium" }
3303: });

```

callModel - lines 3305-3318

Item	Explanation
Source location	Lines 3305-3318 (14 source lines).
Purpose	Part of the local backend API, filesystem, compiler, Git, AI, update, or publish workflow.
Declaration	const callModel = async (selectedModel) => {
Touches	network/API requests, authentication/security data
Calls detected	async, buildPayload, fetch, stringify
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:


```

3305:   const callModel = async (selectedModel) => {
3306:     const payload = buildPayload(selectedModel);
3307:     if (enableWebSearch) {
3308:       payload.tools = [{ type: "web_search", search_context_size:
process.env.OPENAI_WEB_SEARCH_CONTEXT_SIZE || "low" }];
3309:     }
3310:
3311:     const openAiResponse = await fetch("https://api.openai.com/v1/responses", {
3312:       method: "POST",
3313:       headers: {
3314:         "Authorization": `Bearer ${apiKey}`,
3315:         "Content-Type": "application/json"
3316:       },
3317:       body: JSON.stringify(payload)
3318:     });

```

handleApi - lines 3342-3602

Item	Explanation
Source location	Lines 3342-3602 (261 source lines).
Purpose	Handles an event, request, command, or user action.
Declaration	async function handleApi(request, response, url) {
Touches	filesystem, network/API requests, authentication/security data
Calls detected	readJsonFile, sendJson, join, isLocalRequest, getPublishTargetInfo, getGitStatus, getUpdateInfo, getSecurityReport, compileToolStatus, handlePortfolioAi, readRequestJson, normalizeCodeLanguage
API endpoints	/api/app-update, /api/app-update/install, /api/apply-catalog, /api/catalog, /api/code/beautify, /api/code/compile, /api/code/install-tools, /api/code/save, /api/code/tools, /api/github-authenticate, /api/install-git, /api/portfolio-ai, /api/publish-target, /api/save-draft, /api/security-report, /api/sync-from-publish-target, /api/system-check, /api/templates, /api/upload
Storage keys	No local/session storage keys detected.
Async/return clues	35 await expression(s), 29 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3342: async function handleApi(request, response, url) {
3343:   if (request.method === "GET" && url.pathname === "/api/catalog") {
3344:     try {
3345:       const catalog = await readJsonFile(draftPath);
3346:       sendJson(response, 200, { source: "draft", catalog });
3347:     } catch {
3348:       sendJson(response, 200, { source: "published", catalog: await readJsonFile(catalogPath) });
3349:     }
3350:     return true;
3351:   }
3352:
3353:   if (request.method === "GET" && url.pathname === "/api/templates") {
3354:     sendJson(response, 200, await readJsonFile(path.join(root, "project-templates.json")));
3355:     return true;
3356:   }
3357:
3358:   if (request.method === "GET" && url.pathname === "/api/publish-target") {
3359:     if (!isLocalRequest(request)) {
3360:       sendJson(response, 403, { error: "Publishing target details are only available from this computer."
});
3361:       return true;
3362:     }
3363:     sendJson(response, 200, { ok: true, target: await getPublishTargetInfo() });
3364:     return true;
3365:   }
3366:
3367:   if (request.method === "GET" && url.pathname === "/api/system-check") {
3368:     sendJson(response, 200, { ok: true, system: await getGitStatus(), target: await
getPublishTargetInfo() });
3369:     return true;
3370:   }
3371:
3372:   if (request.method === "GET" && url.pathname === "/api/app-update") {
3373:     sendJson(response, 200, await getUpdateInfo());
3374:     return true;
3375:   }
3376:
3377:   if (request.method === "GET" && url.pathname === "/api/security-report") {
3378:     if (!isLocalRequest(request)) {
3379:       sendJson(response, 403, { error: "Security reports are only available from this computer." });
3380:       return true;
3381:     }

```

```

3382:     sendJson(response, 200, await getSecurityReport());
3383:     return true;
3384: }
3385:
3386: if (request.method === "GET" && url.pathname === "/api/code/tools") {
3387:   if (!isLocalRequest(request)) {
3388:     sendJson(response, 403, { error: "Compiler details are only available from this computer." });
3389:     return true;
3390:   }
3391:   sendJson(response, 200, { ok: true, tools: await compileToolStatus() });
3392:   return true;
3393: }
3394:
3395: if (request.method !== "POST") return false;
3396:
3397: if (!isLocalRequest(request)) {
3398:   sendJson(response, 403, { error: "Write actions are only allowed from this computer." });
3399:   return true;
3400: }
3401:
3402: if (url.pathname === "/api/portfolio-ai") {
3403:   await handlePortfolioAi(request, response);
3404:   return true;
3405: }
3406:
3407: if (url.pathname === "/api/code/beautify") {
3408:   try {
3409:     const body = await readRequestJson(request);
3410:     const language = normalizeCodeLanguage(body.language || detectCodeLanguageFromSource(body.code,
body.fileName));
3411:     sendJson(response, 200, {
3412:       ok: true,
3413:       language,
3414:       code: beautifyCode(body.code || "", language)
3415:     });
3416:   } catch (error) {
3417:     sendJson(response, 400, { ok: false, error: error.message || "Code could not be beautified." });
3418:   }
3419:   return true;
3420: }
3421:
3422: if (url.pathname === "/api/code/save") {
3423:   try {
3424:     const body = await readRequestJson(request);
3425:     const saved = await saveCompileSource(body);
3426:     sendJson(response, 200, { ok: true, saved });
3427:   } catch (error) {
3428:     sendJson(response, 400, { ok: false, error: error.message || "Code could not be saved." });
3429:   }
3430:   return true;
3431: }
3432:
3433: if (url.pathname === "/api/code/compile") {
3434:   try {
3435:     const body = await readRequestJson(request);
3436:     const result = await compileAndRunCode(body);
3437:     sendJson(response, 200, { ok: result.ok, result });
3438:   } catch (error) {
3439:     sendJson(response, 400, { ok: false, error: error.message || "Code could not be compiled.", result:
null });
3440:   }
3441:   return true;
3442: }
3443:
3444: if (url.pathname === "/api/code/install-tools") {
3445:   try {
3446:     const body = await readRequestJson(request);
3447:     const result = await installCompilerTools(body.language || "");
3448:     sendJson(response, 200, result);
3449:   } catch (error) {
3450:     sendJson(response, 400, { ok: false, error: error.message || "Compiler tools could not be
installed." });
3451:   }
3452:   return true;
3453: }
3454:
3455: if (url.pathname === "/api/publish-target") {
3456:   await readRequestJson(request).catch(() => ({}));
3457:   sendJson(response, 400, {
3458:     ok: false,
3459:     error: "Publishing target setup now requires GitHub authentication.",
3460:     details: "Use Authenticate target. The builder keeps the previous target until GitHub write access
is verified."
3461:   });
... excerpt truncated after 120 lines. Open the source file for the rest of this function.

```

Representative opening snippet

```

1: import { createServer } from "node:http";
2: import { execFile, spawn } from "node:child_process";

```

```

3: import { createHash } from "node:crypto";
4: import { access as fsAccess, chmod, cp, mkdir, mkdtemp, readFile, readdir, rm, writeFile } from
"node:fs/promises";
5: import os from "node:os";
6: import path from "node:path";
7: import { promisify } from "node:util";
8: import { fileURLToPath } from "node:url";
9:
10: const root = path.dirname(fileURLToPath(import.meta.url));
11: const portfolioRoot = path.resolve(process.env.OMB_PORTFOLIO_WORKSPACE || root);
12: const compileRoot = path.resolve(process.env.OMB_CODE_WORKSPACE || path.join(path.dirname(root),
"compile-code"));
13: const port = Number(process.env.PORT || 8080);
14: const host = process.env.HOST || "0.0.0.0";
15: const execFileAsync = promisify(execFile);
16: const packageJson = JSON.parse(await readFile(path.join(root, "package.json"), "utf8").catch(() =>
"{}"));
17:
18: const types = {
19:   ".css": "text/css",
20:   ".csv": "text/csv",
21:   ".html": "text/html",
22:   ".ico": "image/x-icon",
23:   ".js": "application/javascript",
24:   ".json": "application/json",
25:   ".md": "text/markdown",
26:   ".pdf": "application/pdf",
27:   ".png": "image/png",
28:   ".jpg": "image/jpeg",
29:   ".jpeg": "image/jpeg",
30:   ".svg": "image/svg+xml",
31:   ".txt": "text/plain",
32:   ".webp": "image/webp",
33:   ".xml": "application/xml",
34:   ".xlsx": "application/vnd.openxmlformats-officedocument.spreadsheetml.sheet",
35:   ".zip": "application/zip"
36: };
37:
38: const draftPath = path.join(root, "projects.local.json");
39: const catalogPath = path.join(root, "projects.json");
40: const publishAuthCachePath = path.join(root, ".omb-publish-session.json");
41: const publishPaths = [
42:   "projects.json",
43:   "docs",
44:   "assets",
45:   "index.html",
46:   "styles.css",
47:   "script.js",
48:   "electronics-search.js",
49:   "Backgrounds",
50:   ".nojekyll",
51:   "_headers",
52:   ".well-known",
53:   "robots.txt",
54:   "sitemap.xml",
55:   "CNAME"
56: ];
57: const publishAuthCacheTtlMs = 24 * 60 * 60 * 1000;
58: const publishAuthExtendedTtlMs = 30 * 24 * 60 * 60 * 1000;
59: const publishAuthHistoryWindowMs = 7 * 24 * 60 * 60 * 1000;
60: const publishAuthExtendedThreshold = 3;
61: const defaultSiteRepository = process.env.OMB_BUILDER_REPOSITORY || "";
62: const blockedAppUpdateVersions = new Set(["0.2.16"]);
63: const publishAuthorizationHelp = [
64:   "Publishing was blocked before live website files were applied.",
65:   "Open Publishing target, enter the repository URL, then click Authenticate target.",
66:   "A GitHub/Git Credential Manager browser sign-in may open. Sign in with an account that has write
access to the selected Pages repository.",
67:   "Authenticate target only verifies write access and remembers it for the matching repository and
branch.",
68:   "Load from target only imports compatible website files from the authenticated repository into the
local builder workspace.",
69:   "Until a compatible writable website repository is associated, the builder remains local-only."
70: ].join(" ");
71: const gitCandidates = [
72:   process.env.GIT_EXE,

```

Safe change checklist

- Confirm which app, website, installer, workflow, or document consumes this file.
- Change it on a feature branch from development.
- Regenerate docs if the file purpose, API surface, or behavior changes.
- Run the smallest relevant smoke test before merging into development.

Important Code Reference: template-preview.js

Item	Explanation
File	template-preview.js
Category	JavaScript source
Folder role	Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
Size	454,437 bytes
Extension	.js
MIME type	application/javascript
Text lines	10,826

Why this file exists

- Builder frontend brain: state handling, rich editors, project builder, previews, parser triggers, publishing UI, compile workspace, and guide windows.

How it is used

Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.

How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

API endpoints mentioned or called

Item	Explanation
/api/save-draft	Line 1637. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior.
/api/upload	Line 5430. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior.
/api/upload	Line 8413. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior.
/api/upload	Line 8473. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior.
/api/apply-catalog	Line 8613. Loads project/catalog data for the builder.
/api/apply-catalog	Line 8637. Loads project/catalog data for the builder.
/api/apply-catalog	Line 8647. Loads project/catalog data for the builder.
/api/apply-catalog	Line 8662. Loads project/catalog data for the builder.
/api/apply-catalog	Line 8674. Loads project/catalog data for the builder.
/api/save-draft	Line 8685. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior.
/api/save-draft	Line 10770. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior.
/api/apply-catalog	Line 10772. Loads project/catalog data for the builder.
/api/save-draft	Line 10796. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior.

Important implementation signals

Item	Explanation
Browser DOM at line 1	Builder or website interface behavior in the browser/Electron renderer. Evidence: const builderGuideOpen = document.querySelector("#builder-guide-open");
Browser DOM at line 2	Builder or website interface behavior in the browser/Electron renderer. Evidence: const builderGuideDialog = document.querySelector("#builder-guide-dialog");

Item	Explanation
Browser DOM at line 3	Builder or website interface behavior in the browser/Electron renderer. Evidence: const builderGuideClose = document.querySelector("#builder-guide-close");
Browser DOM at line 4	Builder or website interface behavior in the browser/Electron renderer. Evidence: const builderGuideSearch = document.querySelector("#builder-guide-search");
Browser DOM at line 5	Builder or website interface behavior in the browser/Electron renderer. Evidence: const builderGuideResults = document.querySelector("#builder-guide-results");
Browser DOM at line 6	Builder or website interface behavior in the browser/Electron renderer. Evidence: const builderGuideSections = document.querySelector("#builder-guide-sections");
Browser DOM at line 7	Builder or website interface behavior in the browser/Electron renderer. Evidence: const builderGuideTopicDialog = document.querySelector("#builder-guide-topic-dialog");
Browser DOM at line 8	Builder or website interface behavior in the browser/Electron renderer. Evidence: const builderGuideTopicTitle = document.querySelector("#builder-guide-topic-title");
Browser DOM at line 9	Builder or website interface behavior in the browser/Electron renderer. Evidence: const builderGuideTopicBody = document.querySelector("#builder-guide-topic-body");
Browser DOM at line 10	Builder or website interface behavior in the browser/Electron renderer. Evidence: const builderGuideTopicClose = document.querySelector("#builder-guide-topic-close");
Browser DOM at line 11	Builder or website interface behavior in the browser/Electron renderer. Evidence: const templateSelect = document.querySelector("#template-select");
Browser DOM at line 12	Builder or website interface behavior in the browser/Electron renderer. Evidence: const newCategorySelect = document.querySelector("#new-category-select");
Browser DOM at line 13	Builder or website interface behavior in the browser/Electron renderer. Evidence: const placementSelect = document.querySelector("#placement-select");
Browser DOM at line 14	Builder or website interface behavior in the browser/Electron renderer. Evidence: const newProjectTitle = document.querySelector("#new-project-title");
Browser DOM at line 15	Builder or website interface behavior in the browser/Electron renderer. Evidence: const addProjectButton = document.querySelector("#add-project");
Browser DOM at line 16	Builder or website interface behavior in the browser/Electron renderer. Evidence: const addCategoryButton = document.querySelector("#add-category");
Browser DOM at line 17	Builder or website interface behavior in the browser/Electron renderer. Evidence: const addSiteSectionButton = document.querySelector("#add-site-section");
Browser DOM at line 18	Builder or website interface behavior in the browser/Electron renderer. Evidence: const funFactsInput = document.querySelector("#fun-facts-input");
Browser DOM at line 19	Builder or website interface behavior in the browser/Electron renderer. Evidence: const funFactsCount = document.querySelector("#fun-facts-count");
Browser DOM at line 20	Builder or website interface behavior in the browser/Electron renderer. Evidence: const saveFunFactsButton = document.querySelector("#save-fun-facts");
Browser DOM at line 21	Builder or website interface behavior in the browser/Electron renderer. Evidence: const clearFunFactsButton = document.querySelector("#clear-fun-facts");
Browser DOM at line 22	Builder or website interface behavior in the browser/Electron renderer. Evidence: const siteHeroEyebrowInput = document.querySelector("#site-hero-eyebrow");
Browser DOM at line 23	Builder or website interface behavior in the browser/Electron renderer. Evidence: const siteHeroTitleInput = document.querySelector("#site-hero-title");
Browser DOM at line 24	Builder or website interface behavior in the browser/Electron renderer. Evidence: const siteHeroCopyInput = document.querySelector("#site-hero-copy");
Browser DOM at line 25	Builder or website interface behavior in the browser/Electron renderer. Evidence: const saveSiteContentButton = document.querySelector("#save-site-content");
Browser DOM at line 26	Builder or website interface behavior in the browser/Electron renderer. Evidence: const profileDisplayNameInput = document.querySelector("#profile-display-name");
Browser DOM at line 27	Builder or website interface behavior in the browser/Electron renderer. Evidence: const profilePortfolioLabelInput = document.querySelector("#profile-portfolio-label");
Browser DOM at line 28	Builder or website interface behavior in the browser/Electron renderer. Evidence: const profileContactIntroInput = document.querySelector("#profile-contact-intro");
Browser DOM at line 29	Builder or website interface behavior in the browser/Electron renderer. Evidence: const profileEmailInput = document.querySelector("#profile-email");

Item	Explanation
Browser DOM at line 30	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const profilePhoneInput = document.querySelector("#profile-phone");</code>
Browser DOM at line 31	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const profileGithubInput = document.querySelector("#profile-github");</code>
Browser DOM at line 32	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const profileLinkedInInput = document.querySelector("#profile-linkedin");</code>
Browser DOM at line 33	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const profileWebsiteInput = document.querySelector("#profile-website");</code>
Browser DOM at line 34	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const saveProfileContentButton = document.querySelector("#save-profile-content");</code>
Browser DOM at line 35	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const profileImageUploadButton = document.querySelector("#profile-image-upload");</code>
Browser DOM at line 36	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const profileHeroUploadButton = document.querySelector("#profile-hero-upload");</code>
Browser DOM at line 37	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const profileResumeUploadButton = document.querySelector("#profile-resume-upload");</code>
Browser DOM at line 38	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const profileBrandUploadButton = document.querySelector("#profile-brand-upload");</code>
Browser DOM at line 39	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const profileAssetStatus = document.querySelector("#profile-asset-status");</code>
Browser DOM at line 40	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const categoryDropdown = document.querySelector("#category-dropdown");</code>
Browser DOM at line 41	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const siteSectionList = document.querySelector("#site-section-list");</code>
Browser DOM at line 42	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectCreateDialog = document.querySelector("#project-create-dialog");</code>
Browser DOM at line 43	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectCreateForm = document.querySelector("#project-create-form");</code>
Browser DOM at line 44	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectCreateCategory = document.querySelector("#project-create-category");</code>
Browser DOM at line 45	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectCreateCancel = document.querySelector("#project-create-cancel");</code>
Browser DOM at line 46	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const categoryDialog = document.querySelector("#category-dialog");</code>
Browser DOM at line 47	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const categoryForm = document.querySelector("#category-form");</code>
Browser DOM at line 48	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const categoryTitle = document.querySelector("#category-title");</code>
Browser DOM at line 49	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const categoryDescription = document.querySelector("#category-description");</code>
Browser DOM at line 50	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const categoryAccent = document.querySelector("#category-accent");</code>
Browser DOM at line 51	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const categoryCancel = document.querySelector("#category-cancel");</code>
Browser DOM at line 52	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectTree = document.querySelector("#project-tree");</code>
Browser DOM at line 53	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectSearchInput = document.querySelector("#project-search");</code>
Browser DOM at line 54	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectSearchClear = document.querySelector("#project-search-clear");</code>
Browser DOM at line 55	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectSearchStatus = document.querySelector("#project-search-status");</code>
Browser DOM at line 56	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectDialog = document.querySelector("#project-dialog");</code>

Item	Explanation
Browser DOM at line 57	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectWindowTitle = document.querySelector("#project-window-title");</code>
Browser DOM at line 58	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectWindowClose = document.querySelector("#project-window-close");</code>
Browser DOM at line 59	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectWindowDelete = document.querySelector("#project-window-delete");</code>
Browser DOM at line 60	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const viewProjectPreviewButton = document.querySelector("#view-project-preview");</code>
Browser DOM at line 61	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const saveProjectButton = document.querySelector("#save-project");</code>
Browser DOM at line 62	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const saveProjectCloseButton = document.querySelector("#save-project-close");</code>
Browser DOM at line 63	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectTitleMenu = document.querySelector("#project-title-menu");</code>
Browser DOM at line 64	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const titleRenameActions = document.querySelector("#title-rename-actions");</code>
Browser DOM at line 65	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const titleRenameSave = document.querySelector("#title-rename-save");</code>
Browser DOM at line 66	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const titleRenameCancel = document.querySelector("#title-rename-cancel");</code>
Browser DOM at line 67	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectMetaDialog = document.querySelector("#project-meta-dialog");</code>
Browser DOM at line 68	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectMetaTitle = document.querySelector("#project-meta-title");</code>
Browser DOM at line 69	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectMetaGrid = document.querySelector("#project-meta-grid");</code>
Browser DOM at line 70	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectMetaClose = document.querySelector("#project-meta-close");</code>
Browser DOM at line 71	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectFields = document.querySelector("#project-fields");</code>
Browser DOM at line 72	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const sectionTabs = document.querySelector("#section-tabs");</code>
Browser DOM at line 73	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const sectionContent = document.querySelector("#section-content");</code>
Browser DOM at line 74	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectWindowContent = document.querySelector(".project-window-content");</code>
Browser DOM at line 75	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectPreviewDialog = document.querySelector("#project-preview-dialog");</code>
Browser DOM at line 76	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectPreviewTitle = document.querySelector("#project-preview-title");</code>
Browser DOM at line 77	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectPreviewContent = document.querySelector("#project-preview-content");</code>
Browser DOM at line 78	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectPreviewClose = document.querySelector("#project-preview-close");</code>
Browser DOM at line 79	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectPreviewBack = document.querySelector("#project-preview-back");</code>
Browser DOM at line 80	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const projectPreviewForward = document.querySelector("#project-preview-forward");</code>

Important constants and variables

Item	Explanation
builderGuideOpen	Line 1: <code>const builderGuideOpen = document.querySelector("#builder-guide-open");</code>

Item	Explanation
builderGuideDialog	Line 2: const builderGuideDialog = document.querySelector("#builder-guide-dialog");
builderGuideClose	Line 3: const builderGuideClose = document.querySelector("#builder-guide-close");
builderGuideSearch	Line 4: const builderGuideSearch = document.querySelector("#builder-guide-search");
builderGuideResults	Line 5: const builderGuideResults = document.querySelector("#builder-guide-results");
builderGuideSections	Line 6: const builderGuideSections = document.querySelector("#builder-guide-sections");
builderGuideTopicDialog	Line 7: const builderGuideTopicDialog = document.querySelector("#builder-guide-topic-dialog");
builderGuideTopicTitle	Line 8: const builderGuideTopicTitle = document.querySelector("#builder-guide-topic-title");
builderGuideTopicBody	Line 9: const builderGuideTopicBody = document.querySelector("#builder-guide-topic-body");
builderGuideTopicClose	Line 10: const builderGuideTopicClose = document.querySelector("#builder-guide-topic-close");
templateSelect	Line 11: const templateSelect = document.querySelector("#template-select");
newCategorySelect	Line 12: const newCategorySelect = document.querySelector("#new-category-select");
placementSelect	Line 13: const placementSelect = document.querySelector("#placement-select");
newProjectTitle	Line 14: const newProjectTitle = document.querySelector("#new-project-title");
addProjectButton	Line 15: const addProjectButton = document.querySelector("#add-project");
addCategoryButton	Line 16: const addCategoryButton = document.querySelector("#add-category");
addSiteSectionButton	Line 17: const addSiteSectionButton = document.querySelector("#add-site-section");
funFactsInput	Line 18: const funFactsInput = document.querySelector("#fun-facts-input");
funFactsCount	Line 19: const funFactsCount = document.querySelector("#fun-facts-count");
saveFunFactsButton	Line 20: const saveFunFactsButton = document.querySelector("#save-fun-facts");
clearFunFactsButton	Line 21: const clearFunFactsButton = document.querySelector("#clear-fun-facts");
siteHeroEyebrowInput	Line 22: const siteHeroEyebrowInput = document.querySelector("#site-hero-eyebrow");
siteHeroTitleInput	Line 23: const siteHeroTitleInput = document.querySelector("#site-hero-title");
siteHeroCopyInput	Line 24: const siteHeroCopyInput = document.querySelector("#site-hero-copy");
saveSiteContentButton	Line 25: const saveSiteContentButton = document.querySelector("#save-site-content");
profileDisplayNameInput	Line 26: const profileDisplayNameInput = document.querySelector("#profile-display-name");
profilePortfolioLabelInput	Line 27: const profilePortfolioLabelInput = document.querySelector("#profile-portfolio-label");
profileContactIntroInput	Line 28: const profileContactIntroInput = document.querySelector("#profile-contact-intro");
profileEmailInput	Line 29: const profileEmailInput = document.querySelector("#profile-email");
profilePhoneInput	Line 30: const profilePhoneInput = document.querySelector("#profile-phone");
profileGithubInput	Line 31: const profileGithubInput = document.querySelector("#profile-github");
profileLinkedInInput	Line 32: const profileLinkedInInput = document.querySelector("#profile-linkedin");
profileWebsiteInput	Line 33: const profileWebsiteInput = document.querySelector("#profile-website");
saveProfileContentButton	Line 34: const saveProfileContentButton = document.querySelector("#save-profile-content");
profileImageUploadButton	Line 35: const profileImageUploadButton = document.querySelector("#profile-image-upload");
profileHeroUploadButton	Line 36: const profileHeroUploadButton = document.querySelector("#profile-hero-upload");
profileResumeUploadButton	Line 37: const profileResumeUploadButton = document.querySelector("#profile-resume-upload");
profileBrandUploadButton	Line 38: const profileBrandUploadButton = document.querySelector("#profile-brand-upload");
profileAssetStatus	Line 39: const profileAssetStatus = document.querySelector("#profile-asset-status");
categoryDropdown	Line 40: const categoryDropdown = document.querySelector("#category-dropdown");
siteSectionList	Line 41: const siteSectionList = document.querySelector("#site-section-list");
projectCreateDialog	Line 42: const projectCreateDialog = document.querySelector("#project-create-dialog");

Item	Explanation
projectcreateForm	Line 43: const projectcreateForm = document.querySelector("#project-create-form");
projectCreateCategory	Line 44: const projectCreateCategory = document.querySelector("#project-create-category");
projectCreateCancel	Line 45: const projectCreateCancel = document.querySelector("#project-create-cancel");
categoryDialog	Line 46: const categoryDialog = document.querySelector("#category-dialog");
categoryForm	Line 47: const categoryForm = document.querySelector("#category-form");
categoryTitle	Line 48: const categoryTitle = document.querySelector("#category-title");
categoryDescription	Line 49: const categoryDescription = document.querySelector("#category-description");
categoryAccent	Line 50: const categoryAccent = document.querySelector("#category-accent");
categoryCancel	Line 51: const categoryCancel = document.querySelector("#category-cancel");
projectTree	Line 52: const projectTree = document.querySelector("#project-tree");
projectSearchInput	Line 53: const projectSearchInput = document.querySelector("#project-search");
projectSearchClear	Line 54: const projectSearchClear = document.querySelector("#project-search-clear");
projectSearchStatus	Line 55: const projectSearchStatus = document.querySelector("#project-search-status");
projectDialog	Line 56: const projectDialog = document.querySelector("#project-dialog");
projectWindowTitle	Line 57: const projectWindowTitle = document.querySelector("#project-window-title");
projectWindowClose	Line 58: const projectWindowClose = document.querySelector("#project-window-close");
projectWindowDelete	Line 59: const projectWindowDelete = document.querySelector("#project-window-delete");
viewProjectPreviewButton	Line 60: const viewProjectPreviewButton = document.querySelector("#view-project-preview");
saveProjectButton	Line 61: const saveProjectButton = document.querySelector("#save-project");
saveProjectCloseButton	Line 62: const saveProjectCloseButton = document.querySelector("#save-project-close");
projectTitleMenu	Line 63: const projectTitleMenu = document.querySelector("#project-title-menu");
titleRenameActions	Line 64: const titleRenameActions = document.querySelector("#title-rename-actions");
titleRenameSave	Line 65: const titleRenameSave = document.querySelector("#title-rename-save");
titleRenameCancel	Line 66: const titleRenameCancel = document.querySelector("#title-rename-cancel");
projectMetaDialog	Line 67: const projectMetaDialog = document.querySelector("#project-meta-dialog");
projectMetaTitle	Line 68: const projectMetaTitle = document.querySelector("#project-meta-title");
projectMetaGrid	Line 69: const projectMetaGrid = document.querySelector("#project-meta-grid");
projectMetaClose	Line 70: const projectMetaClose = document.querySelector("#project-meta-close");
projectFields	Line 71: const projectFields = document.querySelector("#project-fields");
sectionTabs	Line 72: const sectionTabs = document.querySelector("#section-tabs");
sectionContent	Line 73: const sectionContent = document.querySelector("#section-content");
projectWindowContent	Line 74: const projectWindowContent = document.querySelector(".project-window-content");
projectPreviewDialog	Line 75: const projectPreviewDialog = document.querySelector("#project-preview-dialog");
projectPreviewTitle	Line 76: const projectPreviewTitle = document.querySelector("#project-preview-title");
projectPreviewContent	Line 77: const projectPreviewContent = document.querySelector("#project-preview-content");
projectPreviewClose	Line 78: const projectPreviewClose = document.querySelector("#project-preview-close");
projectPreviewBack	Line 79: const projectPreviewBack = document.querySelector("#project-preview-back");
projectPreviewForward	Line 80: const projectPreviewForward = document.querySelector("#project-preview-forward");

JSON/YAML-style keys detected

Item	Explanation
skin-light-blue	Line 433: "skin-light-blue": "appearance-light-blue-red-click",
skin-clean-white	Line 434: "skin-clean-white": "appearance-white-blue-click",
skin-deep-navy	Line 435: "skin-deep-navy": "appearance-deep-navy-cyan-click",
skin-red-warm	Line 436: "skin-red-warm": "appearance-warm-red-pale-click",
skin-emerald-instrument	Line 437: "skin-emerald-instrument": "appearance-emerald-mint-click",
skin-amber-power	Line 438: "skin-amber-power": "appearance-amber-cream-click",
skin-violet-mixed	Line 439: "skin-violet-mixed": "appearance-violet-lilac-click",
skin-graphite-asic	Line 440: "skin-graphite-asic": "appearance-graphite-white-click",
analog-opamp-topology	Line 441: "analog-opamp-topology": "appearance-light-blue-red-click",
analog-power-charger	Line 442: "analog-power-charger": "appearance-amber-cream-click",
analog-filter-front-end	Line 443: "analog-filter-front-end": "appearance-light-blue-red-click",
analog-sensor-interface	Line 444: "analog-sensor-interface": "appearance-emerald-mint-click",
analog-mixed-signal-timing	Line 445: "analog-mixed-signal-timing": "appearance-violet-lilac-click",
digital-fpga-pipeline	Line 446: "digital-fpga-pipeline": "appearance-deep-navy-cyan-click",
digital-asic-block	Line 447: "digital-asic-block": "appearance-graphite-white-click",
digital-verification-suite	Line 448: "digital-verification-suite": "appearance-graphite-white-click",
digital-interface-controller	Line 449: "digital-interface-controller": "appearance-deep-navy-cyan-click",
digital-signal-processing	Line 450: "digital-signal-processing": "appearance-violet-lilac-click",
embedded-stm32-sensor	Line 451: "embedded-stm32-sensor": "appearance-emerald-mint-click",
embedded-rtos-control	Line 452: "embedded-rtos-control": "appearance-graphite-white-click",
embedded-power-monitor	Line 453: "embedded-power-monitor": "appearance-amber-cream-click",
embedded-wireless-node	Line 454: "embedded-wireless-node": "appearance-deep-navy-cyan-click",
embedded-motor-control	Line 455: "embedded-motor-control": "appearance-violet-lilac-click"
engineering-file	Line 2512: "engineering-file": "Engineering file link",
google-drive	Line 2515: "google-drive": "Google Drive link",
Brief	Line 2656: "Brief": "Overview",
Project Brief	Line 2657: "Project Brief": "Overview",
Design Brief	Line 2658: "Design Brief": "Design Overview",
Simulation Brief	Line 2659: "Simulation Brief": "Simulation Overview"
--responsive-section-card-min	Line 2901: "--responsive-section-card-min": profile.sectionCardMin,
--responsive-content-max	Line 2902: "--responsive-content-max": profile.contentMax,
--responsive-media-max	Line 2903: "--responsive-media-max": profile.mediaMax,
--responsive-touch-target	Line 2904: "--responsive-touch-target": profile.touchTarget
--accent	Line 2919: "--accent": accent,
--template-bg	Line 2920: "--template-bg": visual.background,
--template-panel	Line 2921: "--template-panel": visual.panel,
--template-accent	Line 2922: "--template-accent": visual.accent,
--template-hover	Line 2923: "--template-hover": visual.hover,
--template-click	Line 2924: "--template-click": visual.click,
--template-click-text	Line 2925: "--template-click-text": visual.clickText,
--template-line	Line 2926: "--template-line": visual.line,

Item	Explanation
--template-text	Line 2927: "--template-text": visual.text,
--accent	Line 2952: "--accent": accent,
--template-bg	Line 2953: "--template-bg": visual.background,
--template-panel	Line 2954: "--template-panel": visual.panel,
--template-accent	Line 2955: "--template-accent": visual.accent,
--template-hover	Line 2956: "--template-hover": visual.hover,
--template-click	Line 2957: "--template-click": visual.click,
--template-click-text	Line 2958: "--template-click-text": visual.clickText,
--template-line	Line 2959: "--template-line": visual.line,
--template-text	Line 2960: "--template-text": visual.text,
compile-code	Line 7449: "compile-code": () => renderCompileCodeSection(project),

Event handlers and user interactions

Item	Explanation
Line 1048	document.addEventListener("pointerdown", beginDialogResize);
Line 1049	document.addEventListener("pointerdown", beginDialogDrag);
Line 1050	document.addEventListener("pointermove", moveDialogResize);
Line 1051	document.addEventListener("pointermove", moveDialogDrag);
Line 1052	document.addEventListener("pointerup", endDialogResize);
Line 1053	document.addEventListener("pointerup", endDialogDrag);
Line 1054	document.addEventListener("pointercancel", endDialogResize);
Line 1055	document.addEventListener("pointercancel", endDialogDrag);
Line 1056	document.addEventListener("click", (event) => {
Line 1061	document.addEventListener("click", (event) => {
Line 1067	dialog.addEventListener("close", () => {
Line 1076	window.addEventListener("resize", () => {
Line 1319	document.addEventListener("visibilitychange", () => {
Line 6166	siteLinkDialog.addEventListener("close", onClose);
Line 8325	dialog.addEventListener("close", onClose);
Line 8747	templateSelect.addEventListener("change", () => {
Line 8752	templatePreviewClose.addEventListener("click", () => {
Line 8880	builderGuideOpen.addEventListener("click", () => {
Line 8887	builderGuideClose.addEventListener("click", () => {
Line 8891	builderGuideTopicClose?.addEventListener("click", () => {
Line 8895	builderGuideTopicDialog?.addEventListener("close", () => {
Line 8899	builderGuideSearch?.addEventListener("input", filterBuilderGuide);
Line 8900	builderGuideDialog?.addEventListener("click", (event) => {
Line 8915	builderGuideDialog?.addEventListener("keydown", (event) => {
Line 8923	projectWindowTitle.addEventListener("dblclick", (event) => {
Line 8929	projectWindowTitle.addEventListener("keydown", (event) => {
Line 8946	projectWindowTitle.addEventListener("contextmenu", (event) => {

Item	Explanation
Line 8954	projectTitleMenu.addEventListener("click", async (event) => {
Line 8991	titleRenameSave.addEventListener("click", () => endTitleRename(true));
Line 8992	titleRenameCancel.addEventListener("click", () => endTitleRename(false));
Line 8993	projectMetaClose.addEventListener("click", () => closeDialogElement(projectMetaDialog));
Line 8995	viewProjectPreviewButton.addEventListener("click", openProjectPortfolioPreview);
Line 8996	projectPreviewClose.addEventListener("click", closeProjectPreviewWindow);
Line 8997	projectPreviewDialog.addEventListener("close", () => {
Line 9001	projectPreviewBack.addEventListener("click", projectPreviewBackStep);
Line 9002	projectPreviewForward.addEventListener("click", projectPreviewForwardStep);
Line 9004	projectPreviewContent.addEventListener("click", (event) => {
Line 9012	saveProjectButton.addEventListener("click", saveSelectedProjectToPortfolio);
Line 9013	saveProjectCloseButton.addEventListener("click", saveSelectedProjectAndClose);
Line 9014	openPortfolioPreviewButton.addEventListener("click", openPortfolioPreview);
Line 9015	saveAllSectionsButton.addEventListener("click", saveAllSections);
Line 9016	portfolioPreviewClose.addEventListener("click", () => {
Line 9019	portfolioPreviewDialog.addEventListener("close", () => {
Line 9022	publishTargetOpen?.addEventListener("click", openPublishTargetDialog);
Line 9023	publishTargetClose?.addEventListener("click", () => {
Line 9026	publishTargetCancel?.addEventListener("click", () => {
Line 9029	publishTargetForm?.addEventListener("submit", savePublishTarget);
Line 9030	publishTargetSync?.addEventListener("click", syncFromPublishTarget);
Line 9031	publishTargetCheck?.addEventListener("click", loadSystemReadiness);
Line 9032	publishTargetInstallGit?.addEventListener("click", installGitForPublishing);
Line 9033	publishTargetRepository?.addEventListener("input", () => {
Line 9036	publishTargetDomain?.addEventListener("input", () => {
Line 9040	publishTargetUsername?.addEventListener("input", markPublishTargetNeedsAuthentication);
Line 9041	publishTargetPassword?.addEventListener("input", markPublishTargetNeedsAuthentication);
Line 9042	publishResultClose.addEventListener("click", () => {
Line 9046	appUpdateClose?.addEventListener("click", () => {
Line 9050	appUpdateLater?.addEventListener("click", () => {
Line 9058	appUpdateSkip?.addEventListener("click", () => {
Line 9065	appUpdateApply?.addEventListener("click", () => {
Line 9069	appUpdateCheckButton?.addEventListener("click", () => {
Line 9073	securityReportOpen?.addEventListener("click", openSecurityReport);
Line 9074	securityReportRefresh?.addEventListener("click", openSecurityReport);
Line 9075	securityReportClose?.addEventListener("click", () => {
Line 9078	securityReportOk?.addEventListener("click", () => {
Line 9082	addProjectButton.addEventListener("click", () => {
Line 9086	projectSearchInput?.addEventListener("input", () => {
Line 9091	projectSearchClear?.addEventListener("click", () => {
Line 9100	quickOpenSelected?.addEventListener("click", () => {

Item	Explanation
Line 9106	quickPreviewSelected?.addEventListener("click", () => {
Line 9112	addCategoryButton?.addEventListener("click", openCategoryDialog);
Line 9113	categoryCancel?.addEventListener("click", () => closeDialogElement(categoryDialog, "cancel"));
Line 9114	categoryForm?.addEventListener("submit", (event) => {
Line 9120	addSiteSectionButton.addEventListener("click", addSiteSection);
Line 9122	funFactsInput?.addEventListener("input", () => {
Line 9128	saveFunFactsButton?.addEventListener("click", () => {
Line 9135	clearFunFactsButton?.addEventListener("click", () => {
Line 9146	input.addEventListener("input", () => {
Line 9154	saveSiteContentButton?.addEventListener("click", () => {
Line 9171	input.addEventListener("input", () => {
Line 9179	saveProfileContentButton?.addEventListener("click", () => {

Function-by-function detail

isHdlLanguage - lines 240-242

Item	Explanation
Source location	Lines 240-242 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function isHdlLanguage(language = "") {
Touches	filesystem
Calls detected	includes, normalizeCodeLanguage
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

240: function isHdlLanguage(language = "") {
241:   return ["verilog", "systemverilog"].includes(normalizeCodeLanguage(language));
242: }

```

inferCompileFileRole - lines 244-251

Item	Explanation
Source location	Lines 244-251 (8 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function inferCompileFileRole(fileName = "", code = "", language = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	isHdlLanguage, toLowerCase, b, tb
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
244: function inferCompileFileRole(fileName = "", code = "", language = "") {
245:   if (!isHdlLanguage(language)) return "source";
246:   const haystack = `${fileName}\n${code}`.toLowerCase();
247:   if (/\\b(tb|testbench)\\b|^|[_-]|tb|_|\\.|test|_|?bench|\\$dumpfile|\\$dumpvars|module\\s+tb|_a-z0-
9]*\\/i.test(haystack)) {
248:     return "testbench";
249:   }
250:   return "design";
251: }
```

normalizeCompileFileRole - lines 253-257

Item	Explanation
Source location	Lines 253-257 (5 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function normalizeCompileFileRole(value = "", language = "") {
Touches	filesystem
Calls detected	isHdlLanguage, trim, toLowerCase, replace, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
253: function normalizeCompileFileRole(value = "", language = "") {
254:   if (!isHdlLanguage(language)) return "source";
255:   const clean = String(value || "").trim().toLowerCase().replace(/[_\s-]+/g, "");
256:   return ["tb", "testbench", "bench"].includes(clean) ? "testbench" : "design";
257: }
```

compileFileRoleLabel - lines 259-263

Item	Explanation
Source location	Lines 259-263 (5 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function compileFileRoleLabel(role = "source") {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
259: function compileFileRoleLabel(role = "source") {
260:   if (role === "testbench") return "Testbench";
261:   if (role === "design") return "Design";
262:   return "Program source";
263: }
```

setStatus - lines 509-511

Item	Explanation
Source location	Lines 509-511 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.

Item	Explanation
Declaration	function setStatus(message) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
509: function setStatus(message) {
510:   builderStatus.textContent = message;
511: }
```

setSaveState - lines 513-517

Item	Explanation
Source location	Lines 513-517 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function setSaveState(state, message) {
Touches	browser/Electron DOM
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
513: function setSaveState(state, message) {
514:   if (!builderSaveState) return;
515:   builderSaveState.dataset.state = state || "loaded";
516:   builderSaveState.textContent = message || "Loaded";
517: }
```

projectSearchText - lines 519-521

Item	Explanation
Source location	Lines 519-521 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function projectSearchText(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	toLowerCase, replace, trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
519: function projectSearchText(value = "") {
520:   return String(value || "").toLowerCase().replace(/\s+/g, " ").trim();
```

521: }

regexEscape - lines 523-525

Item	Explanation
Source location	Lines 523-525 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function regexEscape(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
523: function regexEscape(value = "") {
524:   return String(value).replace(/[\.*+?^${}()|[\]\$\\"/g, "\\$&");
525: }
```

highlightSearchText - lines 527-533

Item	Explanation
Source location	Lines 527-533 (7 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function highlightSearchText(value = "", query = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	projectSearchText, escapeHtml, regexEscape, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
527: function highlightSearchText(value = "", query = "") {
528:   const text = String(value || "");
529:   const cleanQuery = projectSearchText(query);
530:   if (!cleanQuery) return escapeHtml(text);
531:   const matcher = new RegExp(`${regexEscape(cleanQuery)}`, "ig");
532:   return escapeHtml(text).replace(matcher, "<mark>$1</mark>");
533: }
```

projectSearchHaystack - lines 535-555

Item	Explanation
Source location	Lines 535-555 (21 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function projectSearchHaystack(project = {}, category = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	projectSearchText, stringify, filter, join
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

535: function projectSearchHaystack(project = {}, category = {}) {
536:   return projectSearchText([
537:     category.label,
538:     category.description,
539:     project.title,
540:     project.summary,
541:     project.status,
542:     project.category,
543:     project.templateLabel,
544:     ...(project.focus || []),
545:     ...(project.highlights || []),
546:     JSON.stringify(project.sections || []),
547:     JSON.stringify(project.design || {}),
548:     JSON.stringify(project.documents || []),
549:     JSON.stringify(project.tests || []),
550:     JSON.stringify(project.tools || []),
551:     JSON.stringify(project.links || []),
552:     JSON.stringify(project.media || []),
553:     JSON.stringify(project.compileCode || {}),
554:   ]).filter(Boolean).join(" ");
555: }

```

projectMatchesSearch - lines 557-561

Item	Explanation
Source location	Lines 557-561 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function projectMatchesSearch(project, category, query) {
Touches	Mostly local calculations or local UI state.
Calls detected	projectSearchText, projectSearchHaystack, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

557: function projectMatchesSearch(project, category, query) {
558:   const cleanQuery = projectSearchText(query);
559:   if (!cleanQuery) return true;
560:   return projectSearchHaystack(project, category).includes(cleanQuery);
561: }

```

updateProjectSearchStatus - lines 563-575

Item	Explanation
Source location	Lines 563-575 (13 source lines).
Purpose	Updates UI, state, cache, or stored data.
Declaration	function updateProjectSearchStatus(matchCount = catalog.projects.length) {
Touches	browser/Electron DOM
Calls detected	projectSearchText
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

563: function updateProjectSearchStatus(matchCount = catalog.projects.length) {
564:   const cleanQuery = projectSearchText(projectSearchQuery);
565:   if (projectSearchInput && document.activeElement !== projectSearchInput) {
566:     projectSearchInput.value = projectSearchQuery;
567:   }
568:   if (projectSearchClear) projectSearchClear.hidden = !cleanQuery;
569:   if (!projectSearchStatus) return;
570:   if (!cleanQuery) {
571:     projectSearchStatus.textContent = "Showing all projects.";
572:     return;
573:   }
574:   projectSearchStatus.textContent = `${matchCount} matching project${matchCount === 1 ? "" : "s"} for
"${projectSearchQuery}"`;
575: }

```

updateBuilderWorkflow - lines 577-597

Item	Explanation
Source location	Lines 577-597 (21 source lines).
Purpose	Updates UI, state, cache, or stored data.
Declaration	function updateBuilderWorkflow() {
Touches	Mostly local calculations or local UI state.
Calls detected	selectedProject, Set, map, filter, has, forEach
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

577: function updateBuilderWorkflow() {
578:   const project = selectedProject();
579:   const savedIds = new Set((savedPortfolioCatalog.projects || []).map((item) => item.id));
580:   const savedCount = (catalog.projects || []).filter((projectItem) =>
savedIds.has(projectItem.id)).length;
581:   const visibleSections = (catalog.siteSections || []).filter(siteSectionRenderable).length;
582:   if (workflowSelectedProject) {
583:     workflowSelectedProject.textContent = project ? project.title || "Untitled project" : "No project
selected";
584:   }
585:   if (workflowTotalProjects) {
586:     workflowTotalProjects.textContent = `${catalog.projects.length} project${catalog.projects.length ===
1 ? "" : "s"}`;
587:   }
588:   if (workflowSavedProjects) {
589:     workflowSavedProjects.textContent = `${savedCount} parsed`;
590:   }
591:   if (workflowVisibleSections) {
592:     workflowVisibleSections.textContent = `${visibleSections} live section${visibleSections === 1 ? "" :
"s"}`;
593:   }
594:   [quickOpenSelected, quickPreviewSelected].forEach((button) => {
595:     if (button) button.disabled = !project;
596:   });
597: }

```

savedCount - lines 580-584

Item	Explanation
Source location	Lines 580-584 (5 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.

Item	Explanation
Declaration	<code>const savedCount = (catalog.projects []).filter((projectItem) => savedIds.has(projectItem.id)).length;</code>
Touches	Mostly local calculations or local UI state.
Calls detected	filter, has
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

580:   const savedCount = (catalog.projects || []).filter((projectItem) =>
savedIds.has(projectItem.id)).length;
581:   const visibleSections = (catalog.siteSections || []).filter(siteSectionRenderable).length;
582:   if (workflowSelectedProject) {
583:     workflowSelectedProject.textContent = project ? project.title || "Untitled project" : "No project
selected";
584:   }

```

dialogDragHandles - lines 599-604

Item	Explanation
Source location	Lines 599-604 (6 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	<code>function dialogDragHandles(dialog) {</code>
Touches	browser/Electron DOM, filesystem
Calls detected	querySelectorAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

599: function dialogDragHandles(dialog) {
600:   return [
601:     ...dialog.querySelectorAll(".project-dialog-heading, .project-preview-heading, .template-dialog-
heading"),
602:     ...dialog.querySelectorAll(".asset-dialog form > div:first-child")
603:   ];
604: }

```

dialogWindowActionTarget - lines 606-608

Item	Explanation
Source location	Lines 606-608 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	<code>function dialogWindowActionTarget(handle) {</code>
Touches	browser/Electron DOM
Calls detected	querySelector
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
606: function dialogWindowActionTarget(handle) {
607:   return handle.querySelector(".project-window-actions, .section-view-actions") || handle;
608: }
```

updateDialogWindowButtons - lines 610-635

Item	Explanation
Source location	Lines 610-635 (26 source lines).
Purpose	Updates UI, state, cache, or stored data.
Declaration	function updateDialogWindowButtons(dialog) {
Touches	browser/Electron DOM
Calls detected	contains, querySelectorAll, forEach, setAttribute, dialogCanStepBack, dialogCanStepForward
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
610: function updateDialogWindowButtons(dialog) {
611:   const isMinimized = dialog.classList.contains("is-minimized-dialog");
612:   const isHidden = dialog.classList.contains("is-hidden-dialog");
613:   const isMaximized = dialog.classList.contains("is-maximized-dialog");
614:   dialog.querySelectorAll("[data-dialog-minimize='true']").forEach((button) => {
615:     button.textContent = isMinimized ? "+" : "\u2212";
616:     button.title = isMinimized ? "Restore window" : "Minimize window";
617:     button.setAttribute("aria-label", isMinimized ? "Restore window" : "Minimize window");
618:   });
619:   dialog.querySelectorAll("[data-dialog-hide='true']").forEach((button) => {
620:     button.textContent = isHidden ? "\u25b4" : "\u25be";
621:     button.title = isHidden ? "Show window" : "Hide window";
622:     button.setAttribute("aria-label", isHidden ? "Show window" : "Hide window");
623:   });
624:   dialog.querySelectorAll("[data-dialog-maximize='true']").forEach((button) => {
625:     button.textContent = isMaximized ? "\u2750" : "\u25a1";
626:     button.title = isMaximized ? "Restore size" : "Maximize window";
627:     button.setAttribute("aria-label", isMaximized ? "Restore size" : "Maximize window");
628:   });
629:   dialog.querySelectorAll("[data-dialog-back='true']").forEach((button) => {
630:     button.disabled = !dialogCanStepBack(dialog);
631:   });
632:   dialog.querySelectorAll("[data-dialog-forward='true']").forEach((button) => {
633:     button.disabled = !dialogCanStepForward(dialog);
634:   });
635: }
```

dialogTitleForDock - lines 637-639

Item	Explanation
Source location	Lines 637-639 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function dialogTitleForDock(dialog) {
Touches	browser/Electron DOM
Calls detected	querySelector, trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

637: function dialogTitleForDock(dialog) {
638:   return dialog.querySelector("h2")?.textContent?.trim() || "Window";
639: }

```

makeDialogControl - lines 641-650

Item	Explanation
Source location	Lines 641-650 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function makeDialogControl(action, label, title) {
Touches	browser/Electron DOM
Calls detected	createElement, setAttribute
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

641: function makeDialogControl(action, label, title) {
642:   const button = document.createElement("button");
643:   button.type = "button";
644:   button.className = `dialog-window-control dialog-window-${action}-control`;
645:   button.dataset.dialogAction = action;
646:   button.textContent = label;
647:   button.title = title;
648:   button.setAttribute("aria-label", title);
649:   return button;
650: }

```

ensureDialogLeftControls - lines 652-674

Item	Explanation
Source location	Lines 652-674 (23 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function ensureDialogLeftControls(dialog, handle) {
Touches	browser/Electron DOM
Calls detected	querySelector, createElement, makeDialogControl, append, prepend, forEach, toUpperCase, slice
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

652: function ensureDialogLeftControls(dialog, handle) {
653:   if (handle.querySelector(".dialog-window-left-controls")) return;
654:   const titlebar = handle.querySelector(".section-view-titlebar");
655:   const controls = document.createElement("div");
656:   controls.className = "dialog-window-left-controls";
657:
658:   if (titlebar?.querySelector(".section-view-back, .section-view-forward")) {
659:     const refresh = makeDialogControl("refresh", "\u21bb", "Refresh window");
660:     refresh.dataset.dialogRefresh = "true";
661:     controls.append(refresh);

```

```

662:     titlebar.prepend(controls);
663:   } else {
664:     [
665:       makeDialogControl("back", "\u2190", "Back"),
666:       makeDialogControl("forward", "\u2192", "Forward"),
667:       makeDialogControl("refresh", "\u21bb", "Refresh window")
668:     ].forEach((button) => {
669:       button.dataset[`dialog${button.dataset.dialogAction[0].toUpperCase()}${button.dataset.dialogAction.slice(1)}`] =
        "true";
670:       controls.append(button);
671:     });
672:     handle.prepend(controls);
673:   }
674: }

```

ensureDialogWindowControls - lines 676-720

Item	Explanation
Source location	Lines 676-720 (45 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function ensureDialogWindowControls(dialog, handle) {
Touches	browser/Electron DOM, filesystem
Calls detected	querySelectorAll, forEach, remove, ensureDialogLeftControls, dialogWindowActionTarget, querySelector, createElement, append, makeDialogControl, find, closest, add
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

676: function ensureDialogWindowControls(dialog, handle) {
677:   dialog.querySelectorAll(".dialog-minimize-button").forEach((button) => button.remove());
678:   ensureDialogLeftControls(dialog, handle);
679:   const target = dialogWindowActionTarget(handle);
680:   let cluster = target.querySelector(".dialog-window-control-cluster");
681:   if (!cluster) {
682:     cluster = document.createElement("div");
683:     cluster.className = "dialog-window-control-cluster";
684:     target.append(cluster);
685:   }
686:
687:   if (!cluster.querySelector("[data-dialog-hide='true']")) {
688:     const hide = makeDialogControl("hide", "\u25be", "Hide window");
689:     hide.dataset.dialogHide = "true";
690:     cluster.append(hide);
691:   }
692:   if (!cluster.querySelector("[data-dialog-minimize='true']")) {
693:     const minimize = makeDialogControl("minimize", "\u2212", "Minimize window");
694:     minimize.dataset.dialogMinimize = "true";
695:     cluster.append(minimize);
696:   }
697:   if (!cluster.querySelector("[data-dialog-maximize='true']")) {
698:     const maximize = makeDialogControl("maximize", "\u25a1", "Maximize window");
699:     maximize.dataset.dialogMaximize = "true";
700:     cluster.append(maximize);
701:   }
702:
703:   let closeButton = cluster.querySelector(".project-window-close, .close-control, [data-dialog-close='true']");
704:   if (!closeButton) {
705:     closeButton = [...dialog.querySelectorAll(".project-window-close, .close-control")]
706:       .find((button) => button.closest(".project-dialog-heading, .project-preview-heading, .template-dialog-heading, .asset-dialog form > div:first-child") === handle);
707:   }
708:   if (closeButton) {
709:     closeButton.classList.add("dialog-window-control", "dialog-close-button");
710:     closeButton.dataset.dialogAction = "close";
711:     closeButton.dataset.dialogClose = "true";
712:     cluster.append(closeButton);
713:   } else {
714:     const cloneClose = makeDialogControl("close", "\u00d7", "Close window");
715:     cloneClose.classList.add("project-window-close", "dialog-close-button");
716:     cloneClose.dataset.dialogClose = "true";
717:     cluster.append(cloneClose);
718:   }
719:   updateDialogWindowButtons(dialog);

```

720: }

ensureDialogResizeHandles - lines 722-731

Item	Explanation
Source location	Lines 722-731 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function ensureDialogResizeHandles(dialog) {
Touches	browser/Electron DOM
Calls detected	querySelector, forEach, createElement, setAttribute, append
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
722: function ensureDialogResizeHandles(dialog) {
723:   if (dialog.querySelector("[data-dialog-resize-handle]")) return;
724:   ["n", "e", "s", "w", "ne", "nw", "se", "sw"].forEach((direction) => {
725:     const handle = document.createElement("div");
726:     handle.className = `dialog-resize-handle resize-${direction}`;
727:     handle.dataset.dialogResizeHandle = direction;
728:     handle.setAttribute("aria-hidden", "true");
729:     dialog.append(handle);
730:   });
731: }
```

markDialogDragHandles - lines 733-740

Item	Explanation
Source location	Lines 733-740 (8 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function markDialogDragHandles(dialog) {
Touches	browser/Electron DOM
Calls detected	dialogDragHandles, forEach, ensureDialogWindowControls, ensureDialogResizeHandles
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
733: function markDialogDragHandles(dialog) {
734:   dialogDragHandles(dialog).forEach((handle) => {
735:     handle.dataset.dialogDragHandle = "true";
736:     handle.title = handle.title || "Drag to move this window";
737:     ensureDialogWindowControls(dialog, handle);
738:   });
739:   ensureDialogResizeHandles(dialog);
740: }
```

draggableDialogFromEvent - lines 742-751

Item	Explanation
Source location	Lines 742-751 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function draggableDialogFromEvent(event) {

Item	Explanation
Touches	browser/Electron DOM
Calls detected	closest, contains
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 6 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

742: function draggableDialogFromEvent(event) {
743:   if (event.target.closest("[data-dialog-resize-handle]")) return null;
744:   const handle = event.target.closest("[data-dialog-drag-handle='true']");
745:   if (!handle) return null;
746:   const dialog = handle.closest("dialog");
747:   if (!dialog?.open) return null;
748:   if (dialog.classList.contains("is-hidden-dialog")) return null;
749:   if (event.target.closest("button, a, input, textarea, select, label, summary,
[contenteditable='true']")) return null;
750:   return dialog;
751: }

```

clampDialogPosition - lines 753-762

Item	Explanation
Source location	Lines 753-762 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function clampDialogPosition(left, top, dialog) {
Touches	Mostly local calculations or local UI state.
Calls detected	getBoundingClientRect, max, min
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

753: function clampDialogPosition(left, top, dialog) {
754:   const rect = dialog.getBoundingClientRect();
755:   const margin = 12;
756:   const maxLeft = Math.max(margin, window.innerWidth - rect.width - margin);
757:   const maxTop = Math.max(margin, window.innerHeight - rect.height - margin);
758:   return {
759:     left: Math.min(Math.max(left, margin), maxLeft),
760:     top: Math.min(Math.max(top, margin), maxTop)
761:   };
762: }

```

anchorDialogForDrag - lines 764-773

Item	Explanation
Source location	Lines 764-773 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function anchorDialogForDrag(dialog) {
Touches	browser/Electron DOM
Calls detected	getBoundingClientRect, clampDialogPosition, add
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

764: function anchorDialogForDrag(dialog) {
765:   const rect = dialog.getBoundingClientRect();
766:   const position = clampDialogPosition(rect.left, rect.top, dialog);
767:   dialog.classList.add("is-draggable-dialog");
768:   dialog.style.left = `${position.left}px`;
769:   dialog.style.top = `${position.top}px`;
770:   dialog.style.right = "auto";
771:   dialog.style.bottom = "auto";
772:   dialog.style.margin = "0";
773: }

```

saveDialogRestoreState - lines 775-788

Item	Explanation
Source location	Lines 775-788 (14 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	function saveDialogRestoreState(dialog) {
Touches	browser/Electron DOM
Calls detected	getBoundingClientRect, stringify
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

775: function saveDialogRestoreState(dialog) {
776:   if (dialog.dataset.restoreWindowState) return;
777:   const rect = dialog.getBoundingClientRect();
778:   dialog.dataset.restoreWindowState = JSON.stringify({
779:     bottom: dialog.style.bottom || "",
780:     height: dialog.style.height || `${rect.height}px`,
781:     left: dialog.style.left || `${rect.left}px`,
782:     margin: dialog.style.margin || "",
783:     maxHeight: dialog.style.maxHeight || "",
784:     right: dialog.style.right || "",
785:     top: dialog.style.top || `${rect.top}px`,
786:     width: dialog.style.width || `${rect.width}px`
787:   });
788: }

```

restoreDialogWindowState - lines 790-805

Item	Explanation
Source location	Lines 790-805 (16 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function restoreDialogWindowState(dialog) {
Touches	browser/Electron DOM
Calls detected	parse, removeAttribute, entries, forEach
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
790: function restoreDialogWindowState(dialog) {
791:   let state = null;
792:   try {
793:     state = JSON.parse(dialog.dataset.restoreWindowState || "null");
794:   } catch {
795:     state = null;
796:   }
797:   delete dialog.dataset.restoreWindowState;
798:   if (!state) {
799:     dialog.removeAttribute("style");
800:     return;
801:   }
802:   Object.entries(state).forEach(([key, value]) => {
803:     dialog.style[key] = value;
804:   });
805: }
```

anchorDialogForResize - lines 807-814

Item	Explanation
Source location	Lines 807-814 (8 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function anchorDialogForResize(dialog) {
Touches	browser/Electron DOM
Calls detected	anchorDialogForDrag, getBoundingClientRect, add
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
807: function anchorDialogForResize(dialog) {
808:   anchorDialogForDrag(dialog);
809:   const rect = dialog.getBoundingClientRect();
810:   dialog.classList.add("is-resized-dialog");
811:   dialog.style.width = `${rect.width}px`;
812:   dialog.style.height = `${rect.height}px`;
813:   dialog.style.maxHeight = "none";
814: }
```

resizeDialogBounds - lines 816-843

Item	Explanation
Source location	Lines 816-843 (28 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function resizeDialogBounds(state, event) {
Touches	Mostly local calculations or local UI state.
Calls detected	min, max, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
816: function resizeDialogBounds(state, event) {
817:   const margin = 12;
818:   const direction = state.direction;
819:   const minWidth = Math.min(320, Math.max(180, window.innerWidth - margin * 2));
820:   const minHeight = Math.min(170, Math.max(120, window.innerHeight - margin * 2));
821:   let { left, top, width, height } = state.rect;
```

```

822:   const dx = event.clientX - state.startX;
823:   const dy = event.clientY - state.startY;
824:
825:   if (direction.includes("e")) {
826:     width = Math.min(Math.max(state.rect.width + dx, minWidth), window.innerWidth - left - margin);
827:   }
828:   if (direction.includes("s")) {
829:     height = Math.min(Math.max(state.rect.height + dy, minHeight), window.innerHeight - top - margin);
830:   }
831:   if (direction.includes("w")) {
832:     const right = state.rect.left + state.rect.width;
833:     left = Math.min(Math.max(state.rect.left + dx, margin), right - minWidth);
834:     width = right - left;
835:   }
836:   if (direction.includes("n")) {
837:     const bottom = state.rect.top + state.rect.height;
838:     top = Math.min(Math.max(state.rect.top + dy, margin), bottom - minHeight);
839:     height = bottom - top;
840:   }
841:
842:   return { height, left, top, width };
843: }

```

beginDialogResize - lines 845-863

Item	Explanation
Source location	Lines 845-863 (19 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function beginDialogResize(event) {
Touches	browser/Electron DOM
Calls detected	closest, contains, anchorDialogForResize, getBoundingClientRect, add, preventDefault
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

845: function beginDialogResize(event) {
846:   if (event.button !== 0) return;
847:   const handle = event.target.closest("[data-dialog-resize-handle]");
848:   const dialog = handle?.closest("dialog");
849:   if (!handle || !dialog?.open || dialog.classList.contains("is-minimized-dialog") ||
dialog.classList.contains("is-hidden-dialog")) return;
850:   anchorDialogForResize(dialog);
851:   activeDialogResize = {
852:     dialog,
853:     direction: handle.dataset.dialogResizeHandle,
854:     pointerId: event.pointerId,
855:     pointerTarget: event.target,
856:     rect: dialog.getBoundingClientRect(),
857:     startX: event.clientX,
858:     startY: event.clientY
859:   };
860:   event.target.setPointerCapture?.(event.pointerId);
861:   dialog.classList.add("is-resizing-dialog");
862:   event.preventDefault();
863: }

```

moveDialogResize - lines 865-872

Item	Explanation
Source location	Lines 865-872 (8 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function moveDialogResize(event) {
Touches	Mostly local calculations or local UI state.
Calls detected	resizeDialogBounds
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

865: function moveDialogResize(event) {
866:   if (!activeDialogResize) return;
867:   const next = resizeDialogBounds(activeDialogResize, event);
868:   activeDialogResize.dialog.style.left = `${next.left}px`;
869:   activeDialogResize.dialog.style.top = `${next.top}px`;
870:   activeDialogResize.dialog.style.width = `${next.width}px`;
871:   activeDialogResize.dialog.style.height = `${next.height}px`;
872: }

```

endDialogResize - lines 874-879

Item	Explanation
Source location	Lines 874-879 (6 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function endDialogResize() {
Touches	browser/Electron DOM
Calls detected	remove
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

874: function endDialogResize() {
875:   if (!activeDialogResize) return;
876:   activeDialogResize.pointerTarget?.releasePointerCapture?.(activeDialogResize.pointerId);
877:   activeDialogResize.dialog.classList.remove("is-resizing-dialog");
878:   activeDialogResize = null;
879: }

```

beginDialogDrag - lines 881-897

Item	Explanation
Source location	Lines 881-897 (17 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function beginDialogDrag(event) {
Touches	browser/Electron DOM
Calls detected	draggableDialogFromEvent, anchorDialogForDrag, getBoundingClientRect, add, preventDefault
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

881: function beginDialogDrag(event) {
882:   if (event.button !== 0) return;
883:   const dialog = draggableDialogFromEvent(event);
884:   if (!dialog) return;
885:   anchorDialogForDrag(dialog);

```

```

886:   const rect = dialog.getBoundingClientRect();
887:   activeDialogDrag = {
888:     dialog,
889:     offsetX: event.clientX - rect.left,
890:     offsetY: event.clientY - rect.top,
891:     pointerId: event.pointerId,
892:     pointerTarget: event.target
893:   };
894:   event.target.setPointerCapture?.(event.pointerId);
895:   dialog.classList.add("is-dragging-dialog");
896:   event.preventDefault();
897: }

```

moveDialogDrag - lines 899-908

Item	Explanation
Source location	Lines 899-908 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function moveDialogDrag(event) {
Touches	Mostly local calculations or local UI state.
Calls detected	clampDialogPosition
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

899: function moveDialogDrag(event) {
900:   if (!activeDialogDrag) return;
901:   const next = clampDialogPosition(
902:     event.clientX - activeDialogDrag.offsetX,
903:     event.clientY - activeDialogDrag.offsetY,
904:     activeDialogDrag.dialog
905:   );
906:   activeDialogDrag.dialog.style.left = `${next.left}px`;
907:   activeDialogDrag.dialog.style.top = `${next.top}px`;
908: }

```

endDialogDrag - lines 910-915

Item	Explanation
Source location	Lines 910-915 (6 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function endDialogDrag() {
Touches	browser/Electron DOM
Calls detected	remove
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

910: function endDialogDrag() {
911:   if (!activeDialogDrag) return;
912:   activeDialogDrag.pointerTarget?.releasePointerCapture?.(activeDialogDrag.pointerId);
913:   activeDialogDrag.dialog.classList.remove("is-dragging-dialog");
914:   activeDialogDrag = null;
915: }

```

toggleDialogMinimized - lines 917-927

Item	Explanation
Source location	Lines 917-927 (11 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function toggleDialogMinimized(dialog) {
Touches	browser/Electron DOM
Calls detected	contains, toggleDialogHidden, anchorDialogForDrag, remove, toggle, updateDialogWindowButtons
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

917: function toggleDialogMinimized(dialog) {
918:   if (!dialog) return;
919:   if (dialog.classList.contains("is-hidden-dialog")) {
920:     toggleDialogHidden(dialog);
921:     return;
922:   }
923:   anchorDialogForDrag(dialog);
924:   dialog.classList.remove("is-maximized-dialog");
925:   dialog.classList.toggle("is-minimized-dialog");
926:   updateDialogWindowButtons(dialog);
927: }

```

toggleDialogMaximized - lines 929-949

Item	Explanation
Source location	Lines 929-949 (21 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function toggleDialogMaximized(dialog) {
Touches	browser/Electron DOM
Calls detected	contains, remove, restoreDialogWindowState, saveDialogRestoreState, add, updateDialogWindowButtons
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

929: function toggleDialogMaximized(dialog) {
930:   if (!dialog) return;
931:   if (dialog.classList.contains("is-hidden-dialog")) dialog.classList.remove("is-hidden-dialog");
932:   if (dialog.classList.contains("is-maximized-dialog")) {
933:     dialog.classList.remove("is-maximized-dialog");
934:     restoreDialogWindowState(dialog);
935:   } else {
936:     saveDialogRestoreState(dialog);
937:     dialog.classList.remove("is-minimized-dialog");
938:     dialog.classList.add("is-draggable-dialog", "is-maximized-dialog");
939:     dialog.style.left = "0";
940:     dialog.style.top = "0";
941:     dialog.style.right = "auto";
942:     dialog.style.bottom = "auto";
943:     dialog.style.margin = "0";
944:     dialog.style.width = "100vw";
945:     dialog.style.height = "100dvh";
946:     dialog.style.maxHeight = "none";
947:   }
948:   updateDialogWindowButtons(dialog);
949: }

```

toggleDialogHidden - lines 951-970

Item	Explanation
Source location	Lines 951-970 (20 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function toggleDialogHidden(dialog) {
Touches	browser/Electron DOM
Calls detected	contains, remove, restoreDialogWindowState, saveDialogRestoreState, add, min, calc, updateDialogWindowButtons
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

951: function toggleDialogHidden(dialog) {
952:   if (!dialog) return;
953:   if (dialog.classList.contains("is-hidden-dialog")) {
954:     dialog.classList.remove("is-hidden-dialog");
955:     restoreDialogWindowState(dialog);
956:   } else {
957:     saveDialogRestoreState(dialog);
958:     dialog.classList.remove("is-minimized-dialog", "is-maximized-dialog");
959:     dialog.classList.add("is-draggable-dialog", "is-hidden-dialog");
960:     dialog.style.left = "auto";
961:     dialog.style.top = "auto";
962:     dialog.style.right = "12px";
963:     dialog.style.bottom = "10px";
964:     dialog.style.margin = "0";
965:     dialog.style.width = "min(300px, calc(100vw - 24px))";
966:     dialog.style.height = "32px";
967:     dialog.style.maxHeight = "32px";
968:   }
969:   updateDialogWindowButtons(dialog);
970: }

```

dialogCanStepBack - lines 972-976

Item	Explanation
Source location	Lines 972-976 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function dialogCanStepBack(dialog) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

972: function dialogCanStepBack(dialog) {
973:   if (dialog === projectDialog) return projectWindowBackStack.length > 0;
974:   if (dialog === projectPreviewDialog) return activeProjectPreviewSectionIndex >= 0;
975:   return false;
976: }

```

dialogCanStepForward - lines 978-982

Item	Explanation
Source location	Lines 978-982 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.

Item	Explanation
Declaration	function dialogCanStepForward(dialog) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

978: function dialogCanStepForward(dialog) {
979:   if (dialog === projectDialog) return projectWindowForwardStack.length > 0;
980:   if (dialog === projectPreviewDialog) return activeProjectPreviewForwardStack.length > 0;
981:   return false;
982: }

```

renderProjectWindowSectionFromHistory - lines 984-990

Item	Explanation
Source location	Lines 984-990 (7 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderProjectWindowSectionFromHistory(sectionId) {
Touches	Mostly local calculations or local UI state.
Calls detected	renderAll, updateDialogWindowButtons
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

984: function renderProjectWindowSectionFromHistory(sectionId) {
985:   suppressProjectWindowHistory = true;
986:   activeSectionId = sectionId || "brief";
987:   renderAll();
988:   suppressProjectWindowHistory = false;
989:   updateDialogWindowButtons(projectDialog);
990: }

```

stepProjectWindowBack - lines 992-996

Item	Explanation
Source location	Lines 992-996 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function stepProjectWindowBack() {
Touches	Mostly local calculations or local UI state.
Calls detected	push, renderProjectWindowSectionFromHistory, pop
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
992: function stepProjectWindowBack() {
993:   if (!projectWindowBackStack.length) return;
994:   projectWindowForwardStack.push(activeSectionId);
995:   renderProjectWindowSectionFromHistory(projectWindowBackStack.pop());
996: }
```

stepProjectWindowForward - lines 998-1002

Item	Explanation
Source location	Lines 998-1002 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function stepProjectWindowForward() {
Touches	Mostly local calculations or local UI state.
Calls detected	push, renderProjectWindowSectionFromHistory, pop
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
998: function stepProjectWindowForward() {
999:   if (!projectWindowForwardStack.length) return;
1000:   projectWindowBackStack.push(activeSectionId);
1001:   renderProjectWindowSectionFromHistory(projectWindowForwardStack.pop());
1002: }
```

refreshDialogWindow - lines 1004-1015

Item	Explanation
Source location	Lines 1004-1015 (12 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function refreshDialogWindow(dialog) {
Touches	Mostly local calculations or local UI state.
Calls detected	saveVisibleRichEditors, refreshOpenPreviews, renderPreview, renderAll, setStatus, updateDialogWindowButtons
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1004: function refreshDialogWindow(dialog) {
1005:   saveVisibleRichEditors();
1006:   if (dialog === projectPreviewDialog) {
1007:     refreshOpenPreviews();
1008:   } else if (dialog === portfolioPreviewDialog) {
1009:     renderPreview();
1010:   } else if (dialog === projectDialog) {
1011:     renderAll();
1012:   }
1013:   setStatus("Window refreshed.");
1014:   updateDialogWindowButtons(dialog);
1015: }
```

closeDialogFromControl - lines 1017-1025

Item	Explanation
Source location	Lines 1017-1025 (9 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function closeDialogFromControl(dialog, button) {
Touches	browser/Electron DOM
Calls detected	querySelectorAll, find, contains, click, closeDialogElement
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1017: function closeDialogFromControl(dialog, button) {
1018:   const originalClose = [...dialog.querySelectorAll(".project-window-close, .close-control")]
1019:     .find((item) => item !== button && !item.dataset.dialogClose);
1020:   if (originalClose && !button.classList.contains("close-control") && !button.id) {
1021:     originalClose.click();
1022:     return;
1023:   }
1024:   closeDialogElement(dialog, "close");
1025: }

```

handleDialogWindowAction - lines 1027-1044

Item	Explanation
Source location	Lines 1027-1044 (18 source lines).
Purpose	Handles an event, request, command, or user action.
Declaration	function handleDialogWindowAction(button) {
Touches	browser/Electron DOM
Calls detected	closest, toggleDialogHidden, toggleDialogMinimized, toggleDialogMaximized, refreshDialogWindow, stepProjectWindowBack, projectPreviewBackStep, stepProjectWindowForward, projectPreviewForwardStep, closeDialogFromControl
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1027: function handleDialogWindowAction(button) {
1028:   const dialog = button.closest("dialog");
1029:   if (!dialog?.open) return;
1030:   const action = button.dataset.dialogAction;
1031:   if (action === "hide") toggleDialogHidden(dialog);
1032:   if (action === "minimize") toggleDialogMinimized(dialog);
1033:   if (action === "maximize") toggleDialogMaximized(dialog);
1034:   if (action === "refresh") refreshDialogWindow(dialog);
1035:   if (action === "back") {
1036:     if (dialog === projectDialog) stepProjectWindowBack();
1037:     else if (dialog === projectPreviewDialog) projectPreviewBackStep();
1038:   }
1039:   if (action === "forward") {
1040:     if (dialog === projectDialog) stepProjectWindowForward();
1041:     else if (dialog === projectPreviewDialog) projectPreviewForwardStep();
1042:   }
1043:   if (action === "close") closeDialogFromControl(dialog, button);
1044: }

```

enableDraggableDialogs - lines 1046-1084

Item	Explanation
Source location	Lines 1046-1084 (39 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function enableDraggableDialogs() {
Touches	browser/Electron DOM, filesystem
Calls detected	querySelectorAll, forEach, addEventListener, closest, handleDialogWindowAction, toggleDialogHidden, remove, updateDialogWindowButtons, getBoundingClientRect, clampDialogPosition
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1046: function enableDraggableDialogs() {
1047:   document.querySelectorAll("dialog").forEach(markDialogDragHandles);
1048:   document.addEventListener("pointerdown", beginDialogResize);
1049:   document.addEventListener("pointerdown", beginDialogDrag);
1050:   document.addEventListener("pointermove", moveDialogResize);
1051:   document.addEventListener("pointermove", moveDialogDrag);
1052:   document.addEventListener("pointerup", endDialogResize);
1053:   document.addEventListener("pointerup", endDialogDrag);
1054:   document.addEventListener("pointercancel", endDialogResize);
1055:   document.addEventListener("pointercancel", endDialogDrag);
1056:   document.addEventListener("click", (event) => {
1057:     const button = event.target.closest("[data-dialog-action]");
1058:     if (!button) return;
1059:     handleDialogWindowAction(button);
1060:   });
1061:   document.addEventListener("click", (event) => {
1062:     const hiddenDialog = event.target.closest("dialog.is-hidden-dialog");
1063:     if (!hiddenDialog || event.target.closest("button")) return;
1064:     toggleDialogHidden(hiddenDialog);
1065:   });
1066:   document.querySelectorAll("dialog").forEach((dialog) => {
1067:     dialog.addEventListener("close", () => {
1068:       dialog.classList.remove("is-minimized-dialog");
1069:       dialog.classList.remove("is-hidden-dialog");
1070:       dialog.classList.remove("is-maximized-dialog");
1071:       dialog.classList.remove("is-resizing-dialog");
1072:       delete dialog.dataset.restoreWindowState;
1073:       updateDialogWindowButtons(dialog);
1074:     });
1075:   });
1076:   window.addEventListener("resize", () => {
1077:     document.querySelectorAll("dialog.is-draggable-dialog[open]").forEach((dialog) => {
1078:       const rect = dialog.getBoundingClientRect();
1079:       const next = clampDialogPosition(rect.left, rect.top, dialog);
1080:       dialog.style.left = `${next.left}px`;
1081:       dialog.style.top = `${next.top}px`;
1082:     });
1083:   });
1084: }

```

markDraftNeedsSave - lines 1086-1089

Item	Explanation
Source location	Lines 1086-1089 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function markDraftNeedsSave() {
Touches	Mostly local calculations or local UI state.
Calls detected	setSaveState
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1086: function markDraftNeedsSave() {
1087:   draftSavedSinceChanges = false;
1088:   setSaveState("dirty", "Save draft needed");
1089: }

```

showBuilderError - lines 1091-1097

Item	Explanation
Source location	Lines 1091-1097 (7 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function showBuilderError(title, message, details = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	showModal
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1091: function showBuilderError(title, message, details = "") {
1092:   publishResultEyebrow.textContent = "Builder check";
1093:   publishResultTitle.textContent = title;
1094:   publishResultMessage.textContent = message;
1095:   publishResultOutput.textContent = details;
1096:   publishResultDialog.showModal();
1097: }

```

appUpdateWasSnoozed - lines 1105-1110

Item	Explanation
Source location	Lines 1105-1110 (6 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function appUpdateWasSnoozed(version = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	getItem, now
API endpoints	None detected inside this function body.
Storage keys	omb-snoozed-update-at, omb-snoozed-update-version
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1105: function appUpdateWasSnoozed(version = "") {
1106:   if (!version) return false;
1107:   const snoozedVersion = localStorage.getItem("omb-snoozed-update-version") || "";
1108:   const snoozedAt = Number(localStorage.getItem("omb-snoozed-update-at") || 0);
1109:   return snoozedVersion === version && Date.now() - snoozedAt < APP_UPDATE_SNOOZE_MS;
1110: }

```

shouldShowUpdateDialog - lines 1112-1117

Item	Explanation
Source location	Lines 1112-1117 (6 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function shouldShowUpdateDialog(update = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	getItem, appUpdateWasSnoozed
API endpoints	None detected inside this function body.
Storage keys	omb-dismissed-update-version, omb-skipped-update-version
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1112: function shouldShowUpdateDialog(update = {}) {
1113:   if (!update.updateAvailable || !update.latestVersion) return false;
1114:   const skippedVersion = localStorage.getItem("omb-skipped-update-version") || localStorage.getItem("omb-dismissed-update-version") || "";
1115:   if (skippedVersion === update.latestVersion) return false;
1116:   return !appUpdateWasSnoozed(update.latestVersion);
1117: }

```

showUpdateDialog - lines 1119-1158

Item	Explanation
Source location	Lines 1119-1158 (40 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function showUpdateDialog(update = {}, options = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	shouldShowUpdateDialog, escapeHtml, filter, join, showModal, show
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1119: function showUpdateDialog(update = {}, options = {}) {
1120:   const force = Boolean(options.force);
1121:   if (!appUpdateDialog || (!force && !shouldShowUpdateDialog(update))) return;
1122:   pendingAppUpdate = update;
1123:   const hasInstaller = Boolean(update.installerUrl);
1124:   const updateBlocked = Boolean(update.updateBlocked);
1125:   const hasUpdate = Boolean(update.updateAvailable && update.latestVersion);
1126:   appUpdateTitle.textContent = updateBlocked ? "Update paused" : hasUpdate ? "Update available" :
"Builder is up to date";
1127:   appUpdateMessage.textContent = updateBlocked
1128:   ? `OMB Portfolio Builder ${update.latestVersion || "this release"} is available, but this version is
paused for automatic updates.`
1129:   : hasUpdate
1130:   ? `OMB Portfolio Builder ${update.latestVersion} is available. You are running
${update.currentVersion || "an older version"}.`
1131:   : `You are running OMB Portfolio Builder ${update.currentVersion || "the current installed
version"}.`;
1132:   appUpdateDetails.textContent = updateBlocked
1133:   ? (update.blockedReason || "This release is temporarily blocked from automatic updates. Check again
after the next release is available.")
1134:   : hasUpdate
1135:   ? "Choose Update to download the latest installer, close this app, and install the new version
automatically. Remind me later postpones the prompt; Skip this version ignores this release."
1136:   : "No newer released builder was found on GitHub Releases.";
1137:   appUpdateReleaseMeta.innerHTML = (hasUpdate || updateBlocked)
1138:   ? [
1139:     `<span>Current version: ${escapeHtml(update.currentVersion || "unknown")}</span>`,
1140:     `<span>Available version: ${escapeHtml(update.latestVersion || "unknown")}</span>`,
1141:     `update.releaseUrl ? `<span>Release source: GitHub Releases</span>` : ""

```

```

1142:     ].filter(Boolean).join("")
1143:     : [
1144:       `<span>Current version: ${escapeHtml(update.currentVersion || "unknown")}</span>`,
1145:       update.checkedAt ? `<span>Checked: ${escapeHtml(update.checkedAt)}</span>` : ""
1146:     ].filter(Boolean).join("");
1147: appUpdateLater.textContent = hasUpdate ? "Remind me later" : "Close";
1148: appUpdateSkip.hidden = !hasUpdate;
1149: appUpdateApply.hidden = !hasUpdate;
1150: appUpdateApply.disabled = hasUpdate && !hasInstaller;
1151: appUpdateApply.textContent = hasInstaller ? "Update" : "Open release page";
1152: if (appUpdateDialog.open) return;
1153: try {
1154:   appUpdateDialog.showModal();
1155: } catch {
1156:   appUpdateDialog.show();
1157: }
1158: }

```

startAppUpdateInstall - lines 1160-1193

Item	Explanation
Source location	Lines 1160-1193 (34 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	async function startAppUpdateInstall() {
Touches	network/API requests
Calls detected	fetch, now, stringify, json, Error, showBuilderError
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1160: async function startAppUpdateInstall() {
1161:   const update = pendingAppUpdate || {};
1162:   appUpdateApply.disabled = true;
1163:   appUpdateLater.disabled = true;
1164:   appUpdateSkip.disabled = true;
1165:   appUpdateApply.textContent = "Updating...";
1166:   appUpdateDetails.textContent = "Downloading the installer. The builder will close automatically, then
the installer will update the app.";
1167:   try {
1168:     const response = await fetch(`/api/app-update/install?t=${Date.now()}`, {
1169:       method: "POST",
1170:       cache: "no-store",
1171:       headers: { "Content-Type": "application/json" },
1172:       body: JSON.stringify({ latestVersion: update.latestVersion || "" })
1173:     });
1174:     const result = await response.json();
1175:     if (!response.ok || !result.ok) throw new Error(result.error || "The update could not be started.");
1176:     appUpdateMessage.textContent = "Update started";
1177:     appUpdateDetails.textContent = "The installer is ready. This window will close so the new version can
be installed.";
1178:   } catch (error) {
1179:     appUpdateApply.disabled = false;
1180:     appUpdateLater.disabled = false;
1181:     appUpdateSkip.disabled = false;
1182:     appUpdateApply.textContent = "Update";
1183:     appUpdateDetails.textContent = error.message || "The update could not be started.";
1184:     const fallback = update.releaseUrl || update.installerUrl || update.portableUrl || "";
1185:     if (fallback) {
1186:       showBuilderError(
1187:         "Automatic update could not start",
1188:         "The app could not run the installer automatically. The release page can still be opened
manually.",
1189:         fallback
1190:       );
1191:     }
1192:   }
1193: }

```

checkForAppUpdates - lines 1195-1235

Item	Explanation
Source location	Lines 1195-1235 (41 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	async function checkForAppUpdates(options = {}) {
Touches	network/API requests
Calls detected	now, fetch, json, showUpdateDialog, toLocaleString
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1195: async function checkForAppUpdates(options = {}) {
1196:   const force = Boolean(options.force);
1197:   const manual = Boolean(options.manual);
1198:   if (!force && Date.now() - lastAppUpdateCheckAt < 5 * 60 * 1000) return;
1199:   lastAppUpdateCheckAt = Date.now();
1200:   if (manual) {
1201:     if (appUpdateCheckButton) {
1202:       appUpdateCheckButton.disabled = true;
1203:       appUpdateCheckButton.textContent = "Checking...";
1204:     }
1205:   }
1206:   try {
1207:     const response = await fetch(`/api/app-update?t=${Date.now()}`, { cache: "no-store" });
1208:     const update = await response.json();
1209:     if (response.ok && update.ok !== false) {
1210:       showUpdateDialog({ ...update, checkedAt: new Date().toLocaleString() }, { force: manual });
1211:     } else if (manual) {
1212:       showUpdateDialog({
1213:         currentVersion: update.currentVersion || "unknown",
1214:         updateAvailable: false,
1215:         checkedAt: new Date().toLocaleString()
1216:       }, { force: true });
1217:     }
1218:   } catch {
1219:     if (manual) {
1220:       showUpdateDialog({
1221:         currentVersion: "unknown",
1222:         updateAvailable: false,
1223:         checkedAt: new Date().toLocaleString()
1224:       }, { force: true });
1225:     }
1226:     // Update checks should never interrupt builder work.
1227:   } finally {
1228:     if (manual) {
1229:       if (appUpdateCheckButton) {
1230:         appUpdateCheckButton.disabled = false;
1231:         appUpdateCheckButton.textContent = "Check updates";
1232:       }
1233:     }
1234:   }
1235: }

```

formatReportDate - lines 1237-1240

Item	Explanation
Source location	Lines 1237-1240 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function formatReportDate(value = "") {
Touches	filesystem
Calls detected	isNaN, getTime, toLocaleString
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1237: function formatReportDate(value = "") {
1238:   const date = new Date(value);
1239:   return Number.isNaN(date.getTime()) ? "" : date.toLocaleString();
1240: }

```

renderSecurityReport - lines 1242-1294

Item	Explanation
Source location	Lines 1242-1294 (53 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderSecurityReport(report = {}) {
Touches	filesystem, authentication/security data
Calls detected	isArray, map, escapeHtml, toLocaleString, join, formatReportDate
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1242: function renderSecurityReport(report = {}) {
1243:   if (!securityReportBody) return;
1244:   const downloads = report.appDownloads || {};
1245:   const assets = Array.isArray(downloads.assets) ? downloads.assets : [];
1246:   const auth = report.publishingAuth || {};
1247:   const access = report.websiteAccess || {};
1248:   const controls = Array.isArray(report.securityControls) ? report.securityControls : [];
1249:   const assetRows = assets.length
1250:     ? assets.map((asset) => `
1251:       <li>
1252:         <span>${escapeHtml(asset.name || "Release asset")}</span>
1253:         <span>${Number(asset.downloadCount || 0).toLocaleString()} downloads</span>
1254:       </li>
1255:     `).join("")
1256:   : `<li><span>No release assets found</span><span>0 downloads</span></li>`;
1257:   securityReportBody.innerHTML = `
1258:     <div class="security-report-grid">
1259:       <section class="security-report-card">
1260:         <h3>Builder downloads</h3>
1261:         <strong>${downloads.ok ? Number(downloads.totalDownloads || 0).toLocaleString() :
"Unavailable"}</strong>
1262:         <p>${downloads.ok
1263:           ? `Latest release: ${escapeHtml(downloads.releaseTag || downloads.releaseName || "latest")}`
1264:           : escapeHtml(downloads.error || "GitHub release counts are not available.")}</p>
1265:         <ul class="security-report-assets">${assetRows}</ul>
1266:       </section>
1267:       <section class="security-report-card">
1268:         <h3>Publishing authorization</h3>
1269:         <strong>${auth.authenticated ? "Active" : "Not active"}</strong>
1270:         <p>${auth.authenticated
1271:           ? `This builder has a remembered GitHub publishing session until
${escapeHtml(formatReportDate(auth.expiresAt) || auth.expiresAt || "the cache expires")}.`
1272:           : "No fresh GitHub publishing session is cached for the current target."}</p>
1273:         <ul>
1274:           <li>Target: ${escapeHtml(auth.targetRepository || "No publishing target saved")}</li>
1275:           <li>Branch: ${escapeHtml(auth.branch || "Not set")}</li>
1276:           <li>Successful authorizations this week: ${Number(auth.successCountLastWeek ||
0).toLocaleString()}</li>
1277:         </ul>
1278:       </section>
1279:       <section class="security-report-card">
1280:         <h3>Website access reporting</h3>
1281:         <strong>${access.available ? "Configured" : "Needs Cloudflare"}</strong>
1282:         <p>${escapeHtml(access.summary || "Visitor and IP reporting needs a server-side analytics
source.")}</p>
1283:         <ul>${(access.recommendedSetup || []).map((item) =>
1284:           <li>${escapeHtml(item)}</li>`)}</ul>
1285:       </section>
1286:     </div>

```



```

1286:         <h3>Security controls</h3>
1287:         <strong>${controls.length.toLocaleString()}</strong>
1288:         <p>Controls currently enabled or prepared for publishing.</p>
1289:         <ul>${controls.map((item) => `<li>${escapeHtml(item)}</li>`).join("")}</ul>
1290:     </section>
1291: </div>
1292: <p>Generated ${escapeHtml(formatReportDate(report.generatedAt) || "just now")}</p>
1293: `;
1294: }

```

openSecurityReport - lines 1296-1314

Item	Explanation
Source location	Lines 1296-1314 (19 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	async function openSecurityReport() {
Touches	network/API requests
Calls detected	showModal, updateDialogWindowButtons, fetch, now, json, Error, renderSecurityReport, escapeHtml
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1296: async function openSecurityReport() {
1297:   if (!securityReportDialog || !securityReportBody) return;
1298:   securityReportBody.innerHTML = "<p>Loading security report...</p>";
1299:   if (!securityReportDialog.open) securityReportDialog.showModal();
1300:   updateDialogWindowButtons(securityReportDialog);
1301:   try {
1302:     const response = await fetch(`/api/security-report?t=${Date.now()}`, { cache: "no-store" });
1303:     const report = await response.json();
1304:     if (!response.ok || report.ok === false) throw new Error(report.error || "Security report could not
be loaded.");
1305:     renderSecurityReport(report);
1306:   } catch (error) {
1307:     securityReportBody.innerHTML = `
1308:       <div class="security-report-card">
1309:         <h3>Security report unavailable</h3>
1310:         <p>${escapeHtml(error.message || "The report could not be loaded.")}</p>
1311:       </div>
1312:     `;
1313:   }
1314: }

```

scheduleAppUpdateChecks - lines 1316-1325

Item	Explanation
Source location	Lines 1316-1325 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function scheduleAppUpdateChecks() {
Touches	browser/Electron DOM
Calls detected	checkForAppUpdates, addEventListener, now
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1316: function scheduleAppUpdateChecks() {
1317:   checkForAppUpdates({ force: true });
1318:   window.setInterval(() => checkForAppUpdates({ force: true }), APP_UPDATE_CHECK_INTERVAL_MS);

```

```

1319: document.addEventListener("visibilitychange", () => {
1320:   if (document.visibilityState !== "visible") return;
1321:   if (Date.now() - lastAppUpdateCheckAt >= APP_UPDATE_FOCUS_RECHECK_MS) {
1322:     checkForAppUpdates({ force: true });
1323:   }
1324: });
1325: }

```

renderPublishTargetInfo - lines 1327-1353

Item	Explanation
Source location	Lines 1327-1353 (27 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderPublishTargetInfo(target = {}) {
Touches	browser/Electron DOM, authentication/security data
Calls detected	toLocaleString, map, escapeHtml, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1327: function renderPublishTargetInfo(target = {}) {
1328:   if (!publishTargetCurrent) return;
1329:   currentPublishTarget = target;
1330:   const checkedText = target.checkedAt ? ` at ${new Date(target.checkedAt).toLocaleString()}` : "";
1331:   const expiresText = target.expiresAt ? `; remembered until ${new
Date(target.expiresAt).toLocaleString()}` : "";
1332:   const trustText = target.extendedTrust ? " (30-day trusted target)" : target.authorizationCached ? "
(daily cache)" : "";
1333:   const authorizationStatus = target.authorizationChecked
1334:     ? `Verified${checkedText}${trustText}${expiresText}`
1335:     : "Not verified for this repository and branch";
1336:   const rows = [
1337:     ["Builder workspace", target.workspace || "Not available"],
1338:     ["Portfolio workspace", target.portfolioWorkspace || target.workspace || "Not available"],
1339:     ["Repository", target.repository || target.remote || "No repository connected"],
1340:     ["Branch", target.branch || "No branch selected"],
1341:     ["GitHub authorization", authorizationStatus],
1342:     ["Compatible site", target.compatible ? "Yes" : "No"],
1343:     ["Git-backed", target.gitBacked ? "Yes" : "No"]
1344:   ];
1345:   publishTargetCurrent.innerHTML = rows
1346:     .map(([label, value]) =>
`<div><strong>${escapeHtml(label)}</strong><span>${escapeHtml(value)}</span></div>`)
1347:     .join("");
1348:   if (publishTargetRepository) {
1349:     publishTargetRepository.value = target.remote || "";
1350:     publishTargetRepository.dataset.loadedValue = target.remote || "";
1351:   }
1352:   if (publishTargetSync) publishTargetSync.disabled = !target.authorizationChecked;
1353: }

```

renderSystemReadiness - lines 1355-1394

Item	Explanation
Source location	Lines 1355-1394 (40 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderSystemReadiness(system = {}, target = {}) {
Touches	authentication/security data
Calls detected	toLocaleString, map, escapeHtml, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1355: function renderSystemReadiness(system = {}, target = {}) {
1356:   if (!publishTargetReadinessList) return;
1357:   const rows = [
1358:     {
1359:       ok: true,
1360:       text: `Electron/Node runtime: bundled with the app${system.nodeRuntime?.version ? `
1361: ($ {system.nodeRuntime.version})` : ""}.`
1362:     },
1363:     {
1364:       ok: system.pnpm?.required === false,
1365:       text: "pnpm: not required for installed users."
1366:     },
1367:     {
1368:       ok: Boolean(system.git?.ok),
1369:       text: system.git?.ok
1370:       ? `Git for Windows: ${system.git.version || "installed"}.`
1371:       : "Git for Windows: missing. Install it before loading or publishing a target."
1372:     },
1373:     {
1374:       ok: Boolean(system.credentialManager?.ok),
1375:       text: system.credentialManager?.ok
1376:       ? `Git Credential Manager: ${system.credentialManager.version || "installed"}.`
1377:       : "Git Credential Manager: missing. Install Git for Windows with Git Credential Manager enabled."
1378:     },
1379:     {
1380:       ok: Boolean(target?.remote),
1381:       text: target?.remote
1382:       ? `Publishing target: ${target.repository || target.remote}.`
1383:       : "Publishing target: enter a repository URL and click Authenticate target."
1384:     },
1385:     {
1386:       ok: Boolean(target?.authorizationChecked),
1387:       text: target?.authorizationChecked
1388:       ? `GitHub authorization: verified for ${target.branch || "the selected branch"}${target.expiresAt
1389: ? ` until ${new Date(target.expiresAt).toLocaleString()}` : ""}.`
1390:       : "GitHub authorization: required before Load from target or Apply to site."
1391:     }
1392:   ];
1393:   publishTargetReadinessList.innerHTML = rows
1394:     .map((row) => `<li class="${row.ok ? "ready" : "blocked"}">${escapeHtml(row.ok ? `OK - ${row.text}` :
1395: `Needs action - ${row.text}`)}</li>`)
1396:     .join("");
1397: }

```

markPublishTargetNeedsAuthentication - lines 1396-1405

Item	Explanation
Source location	Lines 1396-1405 (10 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	function markPublishTargetNeedsAuthentication() {
Touches	authentication/security data
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1396: function markPublishTargetNeedsAuthentication() {
1397:   if (publishTargetSync) publishTargetSync.disabled = true;
1398:   if (currentPublishTarget) {
1399:     currentPublishTarget = {
1400:       ...currentPublishTarget,
1401:       authorizationChecked: false,
1402:       authorizationCached: false
1403:     };
1404:   }
1405: }

```

loadSystemReadiness - lines 1407-1418

Item	Explanation
Source location	Lines 1407-1418 (12 source lines).
Purpose	Loads data from disk, network, browser storage, or a service.
Declaration	async function loadSystemReadiness() {
Touches	network/API requests
Calls detected	fetch, now, json, Error, renderSystemReadiness, escapeHtml
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1407: async function loadSystemReadiness() {
1408:   if (!publishTargetReadinessList) return;
1409:   publishTargetReadinessList.innerHTML = "<li>Checking local publishing tools...</li>";
1410:   try {
1411:     const response = await fetch(`/api/system-check?t=${Date.now()}`, { cache: "no-store" });
1412:     const result = await response.json();
1413:     if (!response.ok || !result.ok) throw new Error(result.error || "System check failed.");
1414:     renderSystemReadiness(result.system || {}, result.target || currentPublishTarget || {});
1415:   } catch (error) {
1416:     publishTargetReadinessList.innerHTML = `<li class="blocked">${escapeHtml(error.message || "System
check failed.")}</li>`;
1417:   }
1418: }
```

setPublishTargetProgress - lines 1420-1426

Item	Explanation
Source location	Lines 1420-1426 (7 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	function setPublishTargetProgress(title, steps = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	escapeHtml, toLocaleTimeString, map, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1420: function setPublishTargetProgress(title, steps = []) {
1421:   if (!publishTargetCurrent) return;
1422:   publishTargetCurrent.innerHTML =
1423:     <div><strong>${escapeHtml(title)}</strong><span>${escapeHtml(new
Date().toLocaleTimeString())}</span></div>
1424:     ${steps.map((step) =>
`<div><strong>${escapeHtml(step.label)}</strong><span>${escapeHtml(step.value)}</span></div>`).join("")}
1425:   `;
1426: }
```

loadPublishTargetInfo - lines 1428-1439

Item	Explanation
Source location	Lines 1428-1439 (12 source lines).
Purpose	Loads data from disk, network, browser storage, or a service.

Item	Explanation
Declaration	async function loadPublishTargetInfo() {
Touches	network/API requests
Calls detected	fetch, now, json, Error, renderPublishTargetInfo
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1428: async function loadPublishTargetInfo() {
1429:   if (!publishTargetDialog) return;
1430:   publishTargetCurrent.textContent = "Current target is loading.";
1431:   try {
1432:     const response = await fetch(`/api/publish-target?t=${Date.now()}`, { cache: "no-store" });
1433:     const result = await response.json();
1434:     if (!response.ok || !result.ok) throw new Error(result.error || "Publishing target could not be
loaded.");
1435:     renderPublishTargetInfo(result.target || {});
1436:   } catch (error) {
1437:     publishTargetCurrent.textContent = error.message || "Publishing target could not be loaded.";
1438:   }
1439: }

```

openPublishTargetDialog - lines 1441-1445

Item	Explanation
Source location	Lines 1441-1445 (5 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function openPublishTargetDialog() {
Touches	Mostly local calculations or local UI state.
Calls detected	loadPublishTargetInfo, loadSystemReadiness, showModal
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1441: async function openPublishTargetDialog() {
1442:   await loadPublishTargetInfo();
1443:   await loadSystemReadiness();
1444:   publishTargetDialog.showModal();
1445: }

```

savePublishTarget - lines 1447-1450

Item	Explanation
Source location	Lines 1447-1450 (4 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	async function savePublishTarget(event) {
Touches	authentication/security data
Calls detected	preventDefault, authenticatePublishTarget
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	1 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1447: async function savePublishTarget(event) {
1448:   event.preventDefault();
1449:   await authenticatePublishTarget({ fromSave: true });
1450: }

```

currentPublishTargetPayload - lines 1452-1456

Item	Explanation
Source location	Lines 1452-1456 (5 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	function currentPublishTargetPayload() {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1452: function currentPublishTargetPayload() {
1453:   return {
1454:     repositoryUrl: publishTargetRepository?.value || ""
1455:   };
1456: }

```

installGitForPublishing - lines 1458-1481

Item	Explanation
Source location	Lines 1458-1481 (24 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function installGitForPublishing() {
Touches	network/API requests, authentication/security data
Calls detected	fetch, now, json, Error, open, loadSystemReadiness, showBuilderError
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	3 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1458: async function installGitForPublishing() {
1459:   publishTargetCurrent.textContent = "Starting Git for Windows setup...";
1460:   try {
1461:     const response = await fetch(`/api/install-git?t=${Date.now()}`, {
1462:       method: "POST",
1463:       cache: "no-store",
1464:       headers: { "Content-Type": "application/json" },
1465:       body: "{}"
1466:     });
1467:     const result = await response.json();
1468:     if (!response.ok || !result.ok) throw new Error(result.error || "Git installer could not be started.");

```

```

1469:     if (!result.install?.launched && result.install?.downloadUrl) {
1470:         window.open(result.install.downloadUrl, "_blank", "noreferrer");
1471:     }
1472:     await loadSystemReadiness();
1473:     showBuilderError(
1474:         result.install?.launched ? "Git setup started" : "Open Git download",
1475:         result.install?.message || "Install Git for Windows with Git Credential Manager enabled, then
reopen the builder.",
1476:         result.install?.downloadUrl || ""
1477:     );
1478: } catch (error) {
1479:     publishTargetCurrent.textContent = error.message || "Git installer could not be started.";
1480: }
1481: }

```

authenticatePublishTarget - lines 1483-1536

Item	Explanation
Source location	Lines 1483-1536 (54 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function authenticatePublishTarget(options = {}) {
Touches	network/API requests, authentication/security data
Calls detected	setPublishTargetProgress, fetch, now, stringify, currentPublishTargetPayload, json, Error, filter, join, renderPublishTargetInfo, renderSystemReadiness, showBuilderError
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1483: async function authenticatePublishTarget(options = {}) {
1484:     const fromSave = Boolean(options.fromSave);
1485:     if (publishTargetSync) publishTargetSync.disabled = true;
1486:     setPublishTargetProgress(
1487:         fromSave ? "Authenticating publishing target" : "Checking GitHub authorization",
1488:         [
1489:             { label: "Step 1", value: "Validating the repository URL and local Git tools." },
1490:             { label: "Step 2", value: "Using the saved daily authorization if it is still valid." },
1491:             { label: "Step 3", value: "Finding the target repository default branch." },
1492:             { label: "Step 4", value: "Opening GitHub sign-in only when a fresh authorization is required." }
1493:         ]
1494:     );
1495:     try {
1496:         const response = await fetch(`/api/github-authenticate?t=${Date.now()}`, {
1497:             method: "POST",
1498:             cache: "no-store",
1499:             headers: { "Content-Type": "application/json" },
1500:             body: JSON.stringify(currentPublishTargetPayload())
1501:         });
1502:         const result = await response.json();
1503:         if (!response.ok || !result.ok) {
1504:             const message = result.error || "GitHub authentication did not complete.";
1505:             throw new Error([message, result.details].filter(Boolean).join("\n\n"));
1506:         }
1507:         renderPublishTargetInfo(result.target || {});
1508:         renderSystemReadiness(result.auth?.system || {}, result.target || {});
1509:         if (publishTargetPassword) publishTargetPassword.value = "";
1510:         const usedCachedAuth = Boolean(result.auth?.authorizationCached ||
result.target?.authorizationCached);
1511:         setPublishTargetProgress(
1512:             usedCachedAuth ? "Target authenticated from saved authorization" : "Target authenticated",
1513:             [
1514:                 { label: "Repository", value: result.target?.repository || result.auth?.repository || "Selected
target" },
1515:                 { label: "Branch", value: result.target?.branch || result.auth?.branch || "Detected target
branch" },
1516:                 { label: "Authorization", value: usedCachedAuth ? "A valid saved authorization was reused; no new
GitHub sign-in was needed." : "GitHub write access was verified now." },
1517:                 { label: "Next step", value: "Click Load from target when you want to import the latest
compatible website files into this builder." }
1518:             ]
1519:         );
1520:         if (publishTargetSync) publishTargetSync.disabled = false;
1521:     } catch (error) {
1522:         if (publishTargetSync) publishTargetSync.disabled = true;
1523:         setPublishTargetProgress(
1524:             "Target was not saved",

```

```

1525:     [
1526:       { label: "Reason", value: error.message || "GitHub authentication did not complete." },
1527:       { label: "Status", value: "The previous publishing target was kept. Fix the repository or sign-
in, then try Authenticate target again." }
1528:     ]
1529:   );
1530:   showBuilderError(
1531:     "Target authentication failed",
1532:     "The builder did not save this publishing target because GitHub write access was not verified.",
1533:     error.message || "GitHub authentication did not complete."
1534:   );
1535: }
1536: }

```

syncFromPublishTarget - lines 1538-1584

Item	Explanation
Source location	Lines 1538-1584 (47 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function syncFromPublishTarget(options = {}) {
Touches	network/API requests, authentication/security data
Calls detected	fetch, now, json, Error, filter, join, renderPublishTargetInfo, closeDialogElement, loadData, showBuilderError
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	3 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1538: async function syncFromPublishTarget(options = {}) {
1539:   if (!publishTargetDialog) return;
1540:   const afterAuth = Boolean(options.afterAuth);
1541:   publishTargetCurrent.textContent = afterAuth
1542:     ? "GitHub authentication was accepted. Importing the target portfolio now..."
1543:     : "Checking GitHub authorization and loading compatible portfolio files from the target
repository...";
1544:   try {
1545:     const response = await fetch(`/api/sync-from-publish-target?t=${Date.now()}`, {
1546:       method: "POST",
1547:       cache: "no-store",
1548:       headers: { "Content-Type": "application/json" },
1549:       body: "{}"
1550:     });
1551:     const result = await response.json();
1552:     if (!response.ok || !result.ok) {
1553:       throw new Error([
1554:         result.error || "Publishing target could not be imported.",
1555:         result.details || "Click Authenticate target, complete the browser sign-in if prompted, then try
Load from target again."
1556:       ].filter(Boolean).join("\n\n"));
1557:     }
1558:     renderPublishTargetInfo(result.target || {});
1559:     closeDialogElement(publishTargetDialog);
1560:     await loadData();
1561:     draftSavedSinceChanges = true;
1562:     showBuilderError(
1563:       afterAuth ? "GitHub connected and target loaded" : "Target content loaded",
1564:       afterAuth
1565:         ? "Authentication succeeded, and compatible portfolio files were loaded into this local builder
workspace. The local draft now matches the imported target catalog."
1566:         : "Compatible portfolio files were loaded from the authenticated publishing target into this
local builder workspace. The local draft now matches the imported target catalog.",
1567:       [
1568:         `Imported: ${result.sync?.imported || []}.join(", ")`,
1569:         result.sync?.backup ? `Backup: ${result.sync.backup}` : ""
1570:       ].filter(Boolean).join("\n")
1571:     );
1572:   } catch (error) {
1573:     publishTargetCurrent.textContent = afterAuth
1574:       ? `Authentication completed, but target loading failed. ${error.message || "Publishing target could
not be imported."}`
1575:       : error.message || "Publishing target could not be imported.";
1576:     if (afterAuth) {
1577:       showBuilderError(
1578:         "Target saved, loading failed",
1579:         "GitHub authorization succeeded, but the builder could not import compatible portfolio files from
that target.",
1580:         error.message || "Publishing target could not be imported."

```



```

1581:         });
1582:     }
1583: }
1584: }

```

showPublishResult - lines 1586-1619

Item	Explanation
Source location	Lines 1586-1619 (34 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	function showPublishResult(result) {
Touches	authentication/security data
Calls detected	filter, join, showModal
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1586: function showPublishResult(result) {
1587:   const publish = result.publish || {};
1588:   const pushed = Boolean(publish.pushed);
1589:   const authorizationRequired = Boolean(publish.authorizationRequired);
1590:   publishResultEyebrow.textContent = "GitHub publish";
1591:   publishResultTitle.textContent = pushed
1592:     ? "Successfully pushed to GitHub"
1593:     : authorizationRequired
1594:     ? "Publishing access required"
1595:     : "Site applied, push failed";
1596:   publishResultMessage.textContent = pushed
1597:     ? `Your portfolio changes were committed and pushed to ${publish.branch || "GitHub"}.`
1598:     : authorizationRequired
1599:     ? "No live website files were applied. Open Publishing target, click Authenticate with GitHub,
complete the browser sign-in, then try Apply to site again."
1600:     : "The site was applied locally, but Git push did not complete.";
1601:
1602:   const output = [
1603:     pushed ? "PUSH STATUS: SUCCESS" : "PUSH STATUS: FAILED",
1604:     `FILE: ${result.file || "projects.json"}`,
1605:     authorizationRequired ? "AUTHORIZATION: REQUIRED BEFORE WEBSITE CHANGES" : "",
1606:     publish.repository ? `REPOSITORY: ${publish.repository}` : "",
1607:     publish.remote ? `REMOTE: ${publish.remote}` : "",
1608:     publish.branch ? `BRANCH: ${publish.branch}` : "",
1609:     publish.committed === false ? "COMMIT: No file changes to commit." : "",
1610:     publish.commitOutput ? `COMMIT OUTPUT:\n${publish.commitOutput}` : "",
1611:     publish.pushOutput ? `PUSH OUTPUT:\n${publish.pushOutput}` : "",
1612:     publish.error ? `ERROR:\n${publish.error}` : "",
1613:     publish.details ? `DETAILS:\n${publish.details}` : "",
1614:     publish.help ? `NEXT STEP:\n${publish.help}` : "",
1615:   ].filter(Boolean).join("\n\n");
1616:
1617:   publishResultOutput.textContent = output || "No Git output was returned.";
1618:   publishResultDialog.showModal();
1619: }

```

scheduleAutosave - lines 1621-1627

Item	Explanation
Source location	Lines 1621-1627 (7 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function scheduleAutosave(delay = 900) {
Touches	Mostly local calculations or local UI state.
Calls detected	markDraftNeedsSave, autosaveDraft
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1621: function scheduleAutosave(delay = 900) {
1622:   markDraftNeedsSave();
1623:   clearTimeout(autosaveTimer);
1624:   autosaveTimer = setTimeout(() => {
1625:     autosaveDraft();
1626:   }, delay);
1627: }

```

autosaveDraft - lines 1629-1659

Item	Explanation
Source location	Lines 1629-1659 (31 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	async function autosaveDraft() {
Touches	network/API requests
Calls detected	setSaveState, fetch, stringify, json, setStatus, scheduleAutosave
API endpoints	/api/save-draft
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1629: async function autosaveDraft() {
1630:   if (autosaveInFlight) {
1631:     autosaveQueued = true;
1632:     return;
1633:   }
1634:   autosaveInFlight = true;
1635:   setSaveState("saving", "Autosaving...");
1636:   try {
1637:     const response = await fetch("/api/save-draft", {
1638:       method: "POST",
1639:       headers: { "Content-Type": "application/json" },
1640:       body: JSON.stringify({ catalog })
1641:     });
1642:     const result = await response.json();
1643:     if (!response.ok) {
1644:       setStatus(result.error || "Autosave failed.");
1645:       setSaveState("error", "Autosave failed");
1646:     } else if (!draftSavedSinceChanges) {
1647:       setSaveState("dirty", "Autosaved locally");
1648:     }
1649:   } catch {
1650:     setStatus("Autosave failed. Make sure the local server is running.");
1651:     setSaveState("error", "Autosave failed");
1652:   } finally {
1653:     autosaveInFlight = false;
1654:     if (autosaveQueued) {
1655:       autosaveQueued = false;
1656:       scheduleAutosave(300);
1657:     }
1658:   }
1659: }

```

schedulePreviewRender - lines 1661-1667

Item	Explanation
Source location	Lines 1661-1667 (7 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function schedulePreviewRender() {

Item	Explanation
Touches	Mostly local calculations or local UI state.
Calls detected	renderPreview
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1661: function schedulePreviewRender() {
1662:   if (!portfolioPreviewDialog?.open) return;
1663:   clearTimeout(previewRenderTimer);
1664:   previewRenderTimer = setTimeout(() => {
1665:     renderPreview();
1666:   }, 220);
1667: }

```

scheduleChromeRender - lines 1669-1676

Item	Explanation
Source location	Lines 1669-1676 (8 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function scheduleChromeRender() {
Touches	Mostly local calculations or local UI state.
Calls detected	renderTree, renderSectionTabs, selectedProject, renderPreview
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1669: function scheduleChromeRender() {
1670:   clearTimeout(chromeRenderTimer);
1671:   chromeRenderTimer = setTimeout(() => {
1672:     renderTree();
1673:     renderSectionTabs(selectedProject());
1674:     if (portfolioPreviewDialog?.open) renderPreview();
1675:   }, 260);
1676: }

```

updateAssetDialogVisibility - lines 1678-1691

Item	Explanation
Source location	Lines 1678-1691 (14 source lines).
Purpose	Updates UI, state, cache, or stored data.
Declaration	function updateAssetDialogVisibility() {
Touches	browser/Electron DOM
Calls detected	querySelector, trim, labelForUrlAssetKind, inferUrlAssetKind
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1678: function updateAssetDialogVisibility() {
1679:   const isLocal = assetSource.value === "local";
1680:   const isCaptionFile = captionSource.value === "file";
1681:   document.querySelector(".asset-local-field").hidden = !isLocal;
1682:   document.querySelector(".asset-url-field").hidden = isLocal;
1683:   document.querySelector(".caption-text-field").hidden = isCaptionFile;
1684:   document.querySelector(".caption-file-field").hidden = !isCaptionFile;
1685:   const url = assetUrl?.value?.trim() || "";
1686:   if (assetDetectedNote) {
1687:     assetDetectedNote.textContent = isLocal || !url
1688:       ? ""
1689:       : `Detected: ${labelForUrlAssetKind(inferUrlAssetKind(url, assetSource.value))}`;
1690:   }
1691: }
```

slugify - lines 1693-1699

Item	Explanation
Source location	Lines 1693-1699 (7 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function slugify(value) {
Touches	Mostly local calculations or local UI state.
Calls detected	toLowerCase, trim, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1693: function slugify(value) {
1694:   return String(value || "").
1695:     .toLowerCase()
1696:     .trim()
1697:     .replace(/[^\a-z0-9]+/g, "-")
1698:     .replace(/^-+|-+$/g, "");
1699: }
```

wordCount - lines 1701-1703

Item	Explanation
Source location	Lines 1701-1703 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function wordCount(value) {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, split, filter
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1701: function wordCount(value) {
1702:   return String(value || "").trim().split(/\s+/).filter(Boolean).length;
1703: }
```

limitWords - lines 1705-1708

Item	Explanation
Source location	Lines 1705-1708 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function limitWords(value, maxWords = 1000) {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, split, filter, slice, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1705: function limitWords(value, maxWords = 1000) {
1706:   const words = String(value || "").trim().split(/\s+/).filter(Boolean);
1707:   return words.length > maxWords ? words.slice(0, maxWords).join(" ") : value;
1708: }

```

normalizeFunFacts - lines 1710-1716

Item	Explanation
Source location	Lines 1710-1716 (7 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeFunFacts(value) {
Touches	filesystem
Calls detected	isArray, split, map, replace, trim, filter, slice
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1710: function normalizeFunFacts(value) {
1711:   const items = Array.isArray(value) ? value : String(value || "").split(/\r?\n/);
1712:   return items
1713:     .map((item) => String(item || "").replace(/\s+/g, " ").trim())
1714:     .filter(Boolean)
1715:     .slice(0, 3);
1716: }

```

syncFunFactsFromInput - lines 1718-1726

Item	Explanation
Source location	Lines 1718-1726 (9 source lines).
Purpose	Keeps two places aligned, such as local data and target files.
Declaration	function syncFunFactsFromInput() {
Touches	filesystem
Calls detected	extractRichSummary, normalizeFunFacts, plainTextFromRich
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1718: function syncFunFactsFromInput() {
1719:   if (!funFactsInput) return [];
1720:   const rich = extractRichSummary(funFactsInput);
1721:   const facts = normalizeFunFacts(plainTextFromRich(rich));
1722:   catalog.funFacts = facts;
1723:   catalog.funFactsRich = rich;
1724:   if (funFactsCount) funFactsCount.textContent = `${facts.length}/3`;
1725:   return facts;
1726: }

```

renderFunFactsEditor - lines 1728-1734

Item	Explanation
Source location	Lines 1728-1734 (7 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderFunFactsEditor() {
Touches	browser/Electron DOM, filesystem
Calls detected	normalizeFunFacts, populateStandaloneRichEditor, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1728: function renderFunFactsEditor() {
1729:   if (!funFactsInput) return;
1730:   const facts = normalizeFunFacts(catalog.funFacts || []);
1731:   if (funFactsCount) funFactsCount.textContent = `${facts.length}/3`;
1732:   if (document.activeElement === funFactsInput) return;
1733:   populateStandaloneRichEditor(funFactsInput, catalog.funFactsRich, facts.join("\n"));
1734: }

```

populateTextOnlyRichField - lines 1736-1740

Item	Explanation
Source location	Lines 1736-1740 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function populateTextOnlyRichField(field, rich, fallbackText = "") {
Touches	browser/Electron DOM
Calls detected	populateStandaloneRichEditor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1736: function populateTextOnlyRichField(field, rich, fallbackText = "") {
1737:   if (!field) return;
1738:   populateStandaloneRichEditor(field, rich, fallbackText);
1739:   field.dataset.richTextOnly = "true";
1740: }

```

richFieldValue - lines 1742-1754

Item	Explanation
Source location	Lines 1742-1754 (13 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function richFieldValue(field, fallbackText = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, textBlocksFromPlainText, extractRichSummary, plainTextFromRich
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1742: function richFieldValue(field, fallbackText = "") {
1743:   if (!field) {
1744:     return {
1745:       plain: String(fallbackText || "").trim(),
1746:       rich: { blocks: textBlocksFromPlainText(fallbackText) }
1747:     };
1748:   }
1749:   const rich = extractRichSummary(field);
1750:   return {
1751:     plain: plainTextFromRich(rich, "\n").trim(),
1752:     rich
1753:   };
1754: }
```

mergeRichFieldMaps - lines 1756-1761

Item	Explanation
Source location	Lines 1756-1761 (6 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function mergeRichFieldMaps(current = {}, fallback = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	clone
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1756: function mergeRichFieldMaps(current = {}, fallback = {}) {
1757:   return {
1758:     ...clone(fallback || {}),
1759:     ...clone(current || {})
1760:   };
1761: }
```

parsedSiteContentRich - lines 1763-1765

Item	Explanation
Source location	Lines 1763-1765 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function parsedSiteContentRich() {

Item	Explanation
Touches	Mostly local calculations or local UI state.
Calls detected	mergeRichFieldMaps
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1763: function parsedSiteContentRich() {
1764:   return mergeRichFieldMaps(catalog.siteContentRich, savedPortfolioCatalog.siteContentRich);
1765: }
```

parsedProfileRich - lines 1767-1769

Item	Explanation
Source location	Lines 1767-1769 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function parsedProfileRich() {
Touches	Mostly local calculations or local UI state.
Calls detected	mergeRichFieldMaps
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1767: function parsedProfileRich() {
1768:   return mergeRichFieldMaps(catalog.profileRich, savedPortfolioCatalog.profileRich);
1769: }
```

parsedFunFactsRich - lines 1771-1775

Item	Explanation
Source location	Lines 1771-1775 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function parsedFunFactsRich() {
Touches	Mostly local calculations or local UI state.
Calls detected	call, clone
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1771: function parsedFunFactsRich() {
1772:   return Object.prototype.hasOwnProperty.call(catalog, "funFactsRich")
1773:     ? clone(catalog.funFactsRich || null)
1774:     : clone(savedPortfolioCatalog.funFactsRich || null);
1775: }
```


syncPortfolioTextInputsForParsing - lines 1777-1781

Item	Explanation
Source location	Lines 1777-1781 (5 source lines).
Purpose	Keeps two places aligned, such as local data and target files.
Declaration	function syncPortfolioTextInputsForParsing() {
Touches	Mostly local calculations or local UI state.
Calls detected	syncFunFactsFromInput, syncSiteContentFromInputs, syncProfileFromInputs
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1777: function syncPortfolioTextInputsForParsing() {
1778:   if (funFactsInput) syncFunFactsFromInput();
1779:   if (siteHeroTitleInput || siteHeroCopyInput || siteHeroEyebrowInput) syncSiteContentFromInputs();
1780:   if (profileDisplayNameInput || profileEmailInput || profilePhoneInput) syncProfileFromInputs();
1781: }
```

parsedPortfolioGlobals - lines 1783-1796

Item	Explanation
Source location	Lines 1783-1796 (14 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function parsedPortfolioGlobals() {
Touches	filesystem
Calls detected	syncPortfolioTextInputsForParsing, clone, normalizeFieldStyles, normalizeFunFacts, parsedFunFactsRich, normalizeProfile, parsedProfileRich, normalizeSiteContent, parsedSiteContentRich, filter, map
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1783: function parsedPortfolioGlobals() {
1784:   syncPortfolioTextInputsForParsing();
1785:   return {
1786:     categories: clone(catalog.categories || savedPortfolioCatalog.categories || []),
1787:     fieldStyles: clone(normalizeFieldStyles(catalog.fieldStyles || savedPortfolioCatalog.fieldStyles || {})),
1788:     funFacts: normalizeFunFacts(catalog.funFacts || savedPortfolioCatalog.funFacts || []),
1789:     funFactsRich: parsedFunFactsRich(),
1790:     profile: clone(normalizeProfile(catalog.profile || savedPortfolioCatalog.profile || {})),
1791:     profileRich: parsedProfileRich(),
1792:     siteContent: clone(normalizeSiteContent(catalog.siteContent || savedPortfolioCatalog.siteContent || {})),
1793:     siteContentRich: parsedSiteContentRich(),
1794:     siteSections: (catalog.siteSections || savedPortfolioCatalog.siteSections || []).filter(siteSectionRenderable).map(clone)
1795:   };
1796: }
```

syncParsedPortfolioGlobalsIntoSaved - lines 1798-1800

Item	Explanation
Source location	Lines 1798-1800 (3 source lines).
Purpose	Keeps two places aligned, such as local data and target files.

Item	Explanation
Declaration	function syncParsedPortfolioGlobalsIntoSaved() {
Touches	Mostly local calculations or local UI state.
Calls detected	assign, parsedPortfolioGlobals
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1798: function syncParsedPortfolioGlobalsIntoSaved() {
1799:   Object.assign(savedPortfolioCatalog, parsedPortfolioGlobals());
1800: }
```

siteContentRichValue - lines 1802-1804

Item	Explanation
Source location	Lines 1802-1804 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function siteContentRichValue(key) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1802: function siteContentRichValue(key) {
1803:   return catalog.siteContentRich?.[key] || savedPortfolioCatalog.siteContentRich?.[key] || null;
1804: }
```

profileRichValue - lines 1806-1808

Item	Explanation
Source location	Lines 1806-1808 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function profileRichValue(key) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1806: function profileRichValue(key) {
1807:   return catalog.profileRich?.[key] || savedPortfolioCatalog.profileRich?.[key] || null;
1808: }
```

syncSiteRichField - lines 1810-1823

Item	Explanation
Source location	Lines 1810-1823 (14 source lines).
Purpose	Keeps two places aligned, such as local data and target files.
Declaration	function syncSiteRichField(editor) {
Touches	browser/Electron DOM, filesystem
Calls detected	richFieldValue, normalizeSiteContent, markDraftNeedsSave, scheduleAutosave, schedulePreviewRender
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1810: function syncSiteRichField(editor) {
1811:   const key = editor?.dataset.siteField;
1812:   if (!key) return;
1813:   catalog.siteContentRich = catalog.siteContentRich || {};
1814:   const { plain, rich } = richFieldValue(editor, catalog.siteContent?.[key] || "");
1815:   catalog.siteContent = normalizeSiteContent({
1816:     ...(catalog.siteContent || {}),
1817:     [key]: plain
1818:   });
1819:   catalog.siteContentRich[key] = rich;
1820:   markDraftNeedsSave();
1821:   scheduleAutosave();
1822:   schedulePreviewRender();
1823: }
```

syncProfileRichField - lines 1825-1838

Item	Explanation
Source location	Lines 1825-1838 (14 source lines).
Purpose	Keeps two places aligned, such as local data and target files.
Declaration	function syncProfileRichField(editor) {
Touches	browser/Electron DOM, filesystem
Calls detected	richFieldValue, normalizeProfile, markDraftNeedsSave, scheduleAutosave, schedulePreviewRender
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1825: function syncProfileRichField(editor) {
1826:   const key = editor?.dataset.profileField;
1827:   if (!key) return;
1828:   catalog.profileRich = catalog.profileRich || {};
1829:   const { plain, rich } = richFieldValue(editor, catalog.profile?.[key] || "");
1830:   catalog.profile = normalizeProfile({
1831:     ...(catalog.profile || {}),
1832:     [key]: plain
1833:   });
1834:   catalog.profileRich[key] = rich;
1835:   markDraftNeedsSave();
1836:   scheduleAutosave();
1837:   schedulePreviewRender();
1838: }
```

renderSiteContentEditor - lines 1840-1851

Item	Explanation
Source location	Lines 1840-1851 (12 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderSiteContentEditor() {
Touches	browser/Electron DOM, filesystem
Calls detected	normalizeSiteContent, populateTextOnlyRichField, siteContentRichValue
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1840: function renderSiteContentEditor() {
1841:   const content = normalizeSiteContent(catalog.siteContent || {});
1842:   if (siteHeroEyebrowInput && document.activeElement !== siteHeroEyebrowInput) {
1843:     populateTextOnlyRichField(siteHeroEyebrowInput, siteContentRichValue("heroEyebrow"),
content.heroEyebrow);
1844:   }
1845:   if (siteHeroTitleInput && document.activeElement !== siteHeroTitleInput) {
1846:     populateTextOnlyRichField(siteHeroTitleInput, siteContentRichValue("heroTitle"), content.heroTitle);
1847:   }
1848:   if (siteHeroCopyInput && document.activeElement !== siteHeroCopyInput) {
1849:     populateTextOnlyRichField(siteHeroCopyInput, siteContentRichValue("heroCopy"), content.heroCopy);
1850:   }
1851: }

```

renderProfileEditor - lines 1853-1881

Item	Explanation
Source location	Lines 1853-1881 (29 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderProfileEditor() {
Touches	browser/Electron DOM, filesystem
Calls detected	normalizeProfile, forEach, populateTextOnlyRichField, profileRichValue, filter, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1853: function renderProfileEditor() {
1854:   const profile = normalizeProfile(catalog.profile || {});
1855:   const fields = [
1856:     [profileDisplayNameInput, profile.displayName],
1857:     [profilePortfolioLabelInput, profile.portfolioLabel],
1858:     [profileContactIntroInput, profile.contactIntro],
1859:     [profileEmailInput, profile.email],
1860:     [profilePhoneInput, profile.phone],
1861:     [profileGithubInput, profile.githubUrl],
1862:     [profileLinkedInInput, profile.linkedinUrl],
1863:     [profileWebsiteInput, profile.websiteUrl]
1864:   ];
1865:   fields.forEach(([field, value]) => {
1866:     if (field && document.activeElement !== field) {
1867:       populateTextOnlyRichField(field, profileRichValue(field.dataset.profileField), value);
1868:     }
1869:   });
1870:   if (profileAssetStatus) {
1871:     const assetLabels = [
1872:       profile.profileImage ? "profile photo" : "",
1873:       profile.heroImage ? "main background" : "",
1874:       profile.resumeUrl ? "resume" : "",
1875:       profile.brandImage ? "brand icon" : ""

```

```

1876:     ].filter(Boolean);
1877:     profileAssetStatus.textContent = assetLabels.length
1878:     ? `Added: ${assetLabels.join(", ")}`.
1879:     : "No profile photo, resume, brand icon, or main background added yet.";
1880:   }
1881: }

```

syncSiteContentFromInputs - lines 1883-1897

Item	Explanation
Source location	Lines 1883-1897 (15 source lines).
Purpose	Keeps two places aligned, such as local data and target files.
Declaration	function syncSiteContentFromInputs() {
Touches	filesystem
Calls detected	richFieldValue, normalizeSiteContent
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1883: function syncSiteContentFromInputs() {
1884:   catalog.siteContentRich = catalog.siteContentRich || {};
1885:   const eyebrow = richFieldValue(siteHeroEyebrowInput, catalog.siteContent?.heroEyebrow || "");
1886:   const title = richFieldValue(siteHeroTitleInput, catalog.siteContent?.heroTitle || "");
1887:   const copy = richFieldValue(siteHeroCopyInput, catalog.siteContent?.heroCopy || "");
1888:   catalog.siteContent = normalizeSiteContent({
1889:     heroCopy: copy.plain,
1890:     heroEyebrow: eyebrow.plain,
1891:     heroTitle: title.plain
1892:   });
1893:   catalog.siteContentRich.heroCopy = copy.rich;
1894:   catalog.siteContentRich.heroEyebrow = eyebrow.rich;
1895:   catalog.siteContentRich.heroTitle = title.rich;
1896:   return catalog.siteContent;
1897: }

```

syncProfileFromInputs - lines 1899-1926

Item	Explanation
Source location	Lines 1899-1926 (28 source lines).
Purpose	Keeps two places aligned, such as local data and target files.
Declaration	function syncProfileFromInputs() {
Touches	filesystem
Calls detected	richFieldValue, normalizeProfile, entries, forEach
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1899: function syncProfileFromInputs() {
1900:   catalog.profileRich = catalog.profileRich || {};
1901:   const fields = {
1902:     contactIntro: richFieldValue(profileContactIntroInput, catalog.profile?.contactIntro || ""),
1903:     displayName: richFieldValue(profileDisplayNameInput, catalog.profile?.displayName || ""),
1904:     email: richFieldValue(profileEmailInput, catalog.profile?.email || ""),
1905:     githubUrl: richFieldValue(profileGithubInput, catalog.profile?.githubUrl || ""),
1906:     linkedinUrl: richFieldValue(profileLinkedInInput, catalog.profile?.linkedinUrl || ""),
1907:     phone: richFieldValue(profilePhoneInput, catalog.profile?.phone || ""),
1908:     portfolioLabel: richFieldValue(profilePortfolioLabelInput, catalog.profile?.portfolioLabel || ""),
1909:     websiteUrl: richFieldValue(profileWebsiteInput, catalog.profile?.websiteUrl || "")

```

```

1910:   };
1911:   catalog.profile = normalizeProfile({
1912:     ...(catalog.profile || {}),
1913:     contactIntro: fields.contactIntro.plain,
1914:     displayName: fields.displayName.plain,
1915:     email: fields.email.plain,
1916:     githubUrl: fields.githubUrl.plain,
1917:     linkedinUrl: fields.linkedinUrl.plain,
1918:     phone: fields.phone.plain,
1919:     portfolioLabel: fields.portfolioLabel.plain,
1920:     websiteUrl: fields.websiteUrl.plain
1921:   });
1922:   Object.entries(fields).forEach(([key, value]) => {
1923:     catalog.profileRich[key] = value.rich;
1924:   });
1925:   return catalog.profile;
1926: }

```

clone - lines 1928-1930

Item	Explanation
Source location	Lines 1928-1930 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function clone(value) {
Touches	Mostly local calculations or local UI state.
Calls detected	parse, stringify
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1928: function clone(value) {
1929:   return JSON.parse(JSON.stringify(value));
1930: }

```

closeDialogElement - lines 1932-1948

Item	Explanation
Source location	Lines 1932-1948 (17 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function closeDialogElement(dialog, returnValue = "close") {
Touches	Mostly local calculations or local UI state.
Calls detected	hasAttribute, close, removeAttribute, dispatchEvent, Event
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1932: function closeDialogElement(dialog, returnValue = "close") {
1933:   if (!dialog) return;
1934:   const closeValue = String(returnValue || "close");
1935:   const wasOpen = Boolean(dialog.open || dialog.hasAttribute("open"));
1936:   if (typeof dialog.close === "function") {
1937:     try {
1938:       if (dialog.open) dialog.close(closeValue);
1939:     } catch {
1940:       // Fall through to manual cleanup if the browser rejects the native close call.
1941:     }
1942:     if (!dialog.open && !dialog.hasAttribute("open")) return;
1943:   }

```

```

1944:   dialog.returnValue = closeValue;
1945:   dialog.open = false;
1946:   dialog.removeAttribute("open");
1947:   if (wasOpen) dialog.dispatchEvent(new Event("close"));
1948: }

```

getByPath - lines 1950-1952

Item	Explanation
Source location	Lines 1950-1952 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function getByPath(target, pathValue) {
Touches	Mostly local calculations or local UI state.
Calls detected	split, filter, reduce
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1950: function getByPath(target, pathValue) {
1951:   return String(pathValue || "").split(".").filter(Boolean).reduce((current, key) => current?.[key],
target);
1952: }

```

setByPath - lines 1954-1963

Item	Explanation
Source location	Lines 1954-1963 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function setByPath(target, pathValue, value) {
Touches	Mostly local calculations or local UI state.
Calls detected	split, filter, pop, forEach
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1954: function setByPath(target, pathValue, value) {
1955:   const keys = String(pathValue || "").split(".").filter(Boolean);
1956:   const last = keys.pop();
1957:   let current = target;
1958:   keys.forEach((key) => {
1959:     current[key] = current[key] || {};
1960:     current = current[key];
1961:   });
1962:   if (last) current[last] = value;
1963: }

```

defaultDesignModel - lines 1965-1971

Item	Explanation
Source location	Lines 1965-1971 (7 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function defaultDesignModel() {

Item	Explanation
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1965: function defaultDesignModel() {
1966:     return {
1967:         brief: { summary: "", summaryRich: null, files: [] },
1968:         documentation: { summary: "", summaryRich: null, files: [], references: [], mathAnalysis: [] },
1969:         simulation: { summary: "", summaryRich: null, files: [], results: [] }
1970:     };
1971: }

```

isAnalogProject - lines 1973-1975

Item	Explanation
Source location	Lines 1973-1975 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function isAnalogProject(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1973: function isAnalogProject(project) {
1974:     return project?.category === "analog-mixed-signal";
1975: }

```

ensureDesignModel - lines 1977-1992

Item	Explanation
Source location	Lines 1977-1992 (16 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function ensureDesignModel(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	defaultDesignModel
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1977: function ensureDesignModel(project) {
1978:     if (!project) return null;
1979:     const defaults = defaultDesignModel();

```



```

1980:   project.design = project.design || {};
1981:   project.design.brief = { ...defaults.brief, ...(project.design.brief || {}) };
1982:   project.design.documentation = { ...defaults.documentation, ...(project.design.documentation || {}) };
1983:   project.design.documentation.summaryRich = project.design.documentation.summaryRich || null;
1984:   project.design.simulation = { ...defaults.simulation, ...(project.design.simulation || {}) };
1985:   project.design.brief.files = project.design.brief.files || [];
1986:   project.design.documentation.files = project.design.documentation.files || [];
1987:   project.design.documentation.references = project.design.documentation.references || [];
1988:   project.design.documentation.mathAnalysis = project.design.documentation.mathAnalysis || [];
1989:   project.design.simulation.files = project.design.simulation.files || [];
1990:   project.design.simulation.results = project.design.simulation.results || [];
1991:   return project.design;
1992: }

```

ensureCompileCode - lines 1994-2034

Item	Explanation
Source location	Lines 1994-2034 (41 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function ensureCompileCode(project) {
Touches	filesystem
Calls detected	isArray, map, normalizeCodeLanguage, detectCodeLanguage, safeClientCodeFileName, defaultCodeFileName, normalizeCompileFileRole, inferCompileFileRole, slugify, now, some, compileAppendDestinations
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1994: function ensureCompileCode(project) {
1995:   if (!project) return null;
1996:   project.compileCode = project.compileCode || {};
1997:   project.compileCode.files = Array.isArray(project.compileCode.files) ? project.compileCode.files : [];
1998:   project.compileCode.files = project.compileCode.files.map((file, index) => {
1999:     const language = normalizeCodeLanguage(file.language || detectCodeLanguage(file.code || "",
file.fileName || ""));
2000:     const fileName = safeClientCodeFileName(file.fileName || defaultCodeFileName(language), language);
2001:     const role = normalizeCompileFileRole(file.role || inferCompileFileRole(fileName, file.code || "",
language), language);
2002:     return {
2003:       id: file.id || slugify(`${fileName}-${index}-${Date.now()}`),
2004:       title: file.title || fileName,
2005:       fileName,
2006:       language,
2007:       role,
2008:       code: String(file.code || ""),
2009:       stdin: String(file.stdin || ""),
2010:       savedPath: file.savedPath || file.workspacePath || "",
2011:       savedAt: file.savedAt || "",
2012:       waveform: file.waveform || file.lastResult?.waveform || null,
2013:       lastResult: file.lastResult || null,
2014:       dirty: Boolean(file.dirty)
2015:     };
2016:   });
2017:   if (!project.compileCode.activeFileId || !project.compileCode.files.some((file) => file.id ===
project.compileCode.activeFileId)) {
2018:     project.compileCode.activeFileId = project.compileCode.files[0]?.id || "";
2019:   }
2020:   project.compileCode.terminal = String(project.compileCode.terminal || "");
2021:   const appendDestinations = compileAppendDestinations(project);
2022:   if (!project.compileCode.appendTargetId || !appendDestinations.some((destination) => destination.id ===
project.compileCode.appendTargetId)) {
2023:     project.compileCode.appendTargetId = appendDestinations[0]?.id || "";
2024:   }
2025:   project.compileCode.messages = Array.isArray(project.compileCode.messages)
2026:     ? project.compileCode.messages.map((message) => ({
2027:       id: message.id || slugify(`message-${Date.now()}-${Math.random().toString(16).slice(2)}`),
2028:       at: message.at || new Date().toISOString(),
2029:       level: ["info", "success", "warning", "error"].includes(message.level) ? message.level : "info",
2030:       text: String(message.text || "")
2031:     })).filter((message) => message.text)
2032:     : [];
2033:   return project.compileCode;
2034: }

```

escapeHtml - lines 2036-2043

Item	Explanation
Source location	Lines 2036-2043 (8 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function escapeHtml(value) {
Touches	Mostly local calculations or local UI state.
Calls detected	replaceAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2036: function escapeHtml(value) {
2037:   return String(value || "")
2038:     .replaceAll("&", "&amp;")
2039:     .replaceAll("<", "&lt;")
2040:     .replaceAll(">", "&gt;")
2041:     .replaceAll("'", "&quot;")
2042:     .replaceAll('"', "&#039;");
2043: }
```

escapeCodeHtml - lines 2045-2051

Item	Explanation
Source location	Lines 2045-2051 (7 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function escapeCodeHtml(value) {
Touches	Mostly local calculations or local UI state.
Calls detected	replaceAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2045: function escapeCodeHtml(value) {
2046:   return String(value || "")
2047:     .replaceAll("&", "&amp;")
2048:     .replaceAll("<", "&lt;")
2049:     .replaceAll(">", "&gt;")
2050:     .replaceAll("'", "&quot;");
2051: }
```

normalizeCodeLanguage - lines 2053-2059

Item	Explanation
Source location	Lines 2053-2059 (7 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeCodeLanguage(value = "") {
Touches	filesystem
Calls detected	trim, toLowerCase, replace, find, includes

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2053: function normalizeCodeLanguage(value = "") {
2054:   const clean = String(value || "").trim().toLowerCase().replace(/[_-]+/g, " ");
2055:   const match = supportedCodeLanguages.find((language) =>
2056:     language.id === clean || language.aliases.includes(clean)
2057:   );
2058:   return match?.id || "javascript";
2059: }

```

codeLanguageProfile - lines 2061-2063

Item	Explanation
Source location	Lines 2061-2063 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function codeLanguageProfile(value = "") {
Touches	filesystem
Calls detected	find, normalizeCodeLanguage
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2061: function codeLanguageProfile(value = "") {
2062:   return supportedCodeLanguages.find((language) => language.id === normalizeCodeLanguage(value)) ||
supportedCodeLanguages.find((language) => language.id === "javascript");
2063: }

```

codeLanguageLabel - lines 2065-2067

Item	Explanation
Source location	Lines 2065-2067 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function codeLanguageLabel(value = "") {
Touches	filesystem
Calls detected	find, normalizeCodeLanguage
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2065: function codeLanguageLabel(value = "") {
2066:   return supportedCodeLanguages.find((language) => language.id === normalizeCodeLanguage(value))?.label
|| "Code";
2067: }

```

sourceLooksCpp - lines 2069-2073

Item	Explanation
Source location	Lines 2069-2073 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function sourceLooksCpp(source = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2069: function sourceLooksCpp(source = "") {
2070:   const text = String(source || "");
2071:   return
#include\s*<(?:iostream|string|vector|array|map|unordered_map|memory|algorithm|optional|variant)>/.test(text)
||
2072:   /\b(std::|cout|cin|cerr|namespace|template\s*<|class\s+\w+|new\s+\w|delete\s+|constexpr|nullptr|using
\s+namespace)\b/.test(text);
2073: }
```

sourceLooksC - lines 2075-2079

Item	Explanation
Source location	Lines 2075-2079 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function sourceLooksC(source = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2075: function sourceLooksC(source = "") {
2076:   const text = String(source || "");
2077:   return #include\s*<(?:stdio|stdlib|string|stdint|stdbool|math)\.h>/.test(text) ||
2078:   /\b(printf|scanf|malloc|calloc|realloc|free|struct\s+\w+|typedef\s+struct)\b/.test(text);
2079: }
```

languageFromFileName - lines 2081-2090

Item	Explanation
Source location	Lines 2081-2090 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function languageFromFileName(fileName = "", source = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	toLowerCase, includes, split, pop, sourceLooksCpp, sourceLooksC, find
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2081: function languageFromFileName(fileName = "", source = "") {
2082:   const clean = String(fileName || "").toLowerCase();
2083:   const extension = clean.includes(".") ? `${clean.split(".").pop()}` : "";
2084:   if (extension === ".h") {
2085:     if (sourceLooksCpp(source)) return "cpp";
2086:     if (sourceLooksC(source)) return "c";
2087:     return "c";
2088:   }
2089:   return supportedCodeLanguages.find((language) => language.extensions?.includes(extension))?.id || "";
2090: }

```

defaultCodeFileName - lines 2092-2094

Item	Explanation
Source location	Lines 2092-2094 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function defaultCodeFileName(language = "javascript") {
Touches	Mostly local calculations or local UI state.
Calls detected	codeLanguageProfile
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2092: function defaultCodeFileName(language = "javascript") {
2093:   return codeLanguageProfile(language)?.defaultFile || "main.js";
2094: }

```

safeClientCodeFileName - lines 2096-2104

Item	Explanation
Source location	Lines 2096-2104 (9 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function safeClientCodeFileName(fileName = "", language = "javascript") {
Touches	Mostly local calculations or local UI state.
Calls detected	codeLanguageProfile, trim, replace, match, toLowerCase, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2096: function safeClientCodeFileName(fileName = "", language = "javascript") {
2097:   const profile = codeLanguageProfile(language);
2098:   const fallback = profile.defaultFile || "main.js";
2099:   const clean = String(fileName || fallback).trim().replace(/[\\/:?*"<>|]+/g, "-");
2100:   const namePart = clean.replace(/\.([^.]+)$/, "").replace(/^[a-zA-Z0-9_-]+/g, "-").replace(/^-+|-+$/g,
  "")) || fallback.replace(/\.([^.]+)$/, "");

```

```

2101:   const extMatch = clean.match(/(\.[a-zA-Z0-9_-]+)$/);
2102:   const ext = (extMatch?.[1] || fallback.match(/(\.[^.]+)$/)?.[1] || ".js").toLowerCase();
2103:   return `${namePart}${profile.extensions?.includes(ext) ? ext : fallback.match(/(\.[^.]+)$/)?.[1] ||
".js"}`;
2104: }

```

normalizeCodePasteMode - lines 2106-2108

Item	Explanation
Source location	Lines 2106-2108 (3 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeCodePasteMode(value = "") {
Touches	filesystem
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2106: function normalizeCodePasteMode(value = "") {
2107:   return value === "source" ? "source" : "basic";
2108: }

```

normalizeCodeText - lines 2110-2119

Item	Explanation
Source location	Lines 2110-2119 (10 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeCodeText(value = "", pasteMode = "source") {
Touches	filesystem
Calls detected	replace, normalizeCodePasteMode, split, map, trimEnd, join, trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2110: function normalizeCodeText(value = "", pasteMode = "source") {
2111:   const normalized = String(value || "").replace(/\r\n?/g, "\n").replace(/\u00a0/g, " ");
2112:   if (normalizeCodePasteMode(pasteMode) === "source") return normalized.replace(/\t/g, "
").replace(/\s+$/g, "");
2113:   return normalized
2114:     .split("\n")
2115:     .map((line) => line.trimEnd())
2116:     .join("\n")
2117:     .replace(/\n{4,}/g, "\n\n\n")
2118:     .trim();
2119: }

```

detectCodeLanguage - lines 2121-2136

Item	Explanation
Source location	Lines 2121-2136 (16 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function detectCodeLanguage(value = "", fileName = "") {

Item	Explanation
Touches	Mostly local calculations or local UI state.
Calls detected	languageFromFileName, b, sourceLooksCpp, sourceLooksC
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 10 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2121: function detectCodeLanguage(value = "", fileName = "") {
2122:   const byFile = languageFromFileName(fileName, value);
2123:   if (byFile) return byFile;
2124:   const code = String(value || "");
2125:   if (/<\/?[a-z][\s\S]*?>/i.test(code) || /<!doctype\s+html/i.test(code)) return "html";
2126:   if (/\\b(always_ff|always_comb|always_latch|logic|interface|covergroup|assert\s+property|typedef\s+enum|
class\s+w+)\b/.test(code)) return "systemverilog";
2127:   if (/\\b(module|endmodule|always|assign|reg|wire|initial|posedge|negedge)\b/.test(code)) return
"verilog";
2128:   if (/^\\s*\\.?(tran|ac|dc|op|model|subckt|ends|param)\b/im.test(code) ||
/bv\\w+\\s+\\w+\\s+\\w+\\s+(?:DC|SIN|PULSE)?/i.test(code)) return "ltspice";
2129:   if (/\\b(def|elif|import\\s+\\w+|from\\s+\\w+\\s+import|self|None|True|False)\\b/.test(code)) return "python";
2130:   if (/\\b(public\\s+class|private\\s+|protected\\s+|static\\s+void\\s+main|System\\.out)\\b/.test(code)) return
"java";
2131:   if (sourceLooksCpp(code) || sourceLooksC(code) ||
/b(#include|main\\s*\\(|puts\\s*\\(|fprintf|sizeof\\s*\\(|\\b/.test(code)) {
2132:     return sourceLooksCpp(code) ? "cpp" : "c";
2133:   }
2134:   if (/\\b(const|let|var|function|=>|console\\.log|document\\.|window\\.|\\b/.test(code)) return "javascript";
2135:   return "javascript";
2136: }

```

tokenizedCodeHtml - lines 2138-2192

Item	Explanation
Source location	Lines 2138-2192 (55 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function tokenizedCodeHtml(code = "", language = "javascript") {
Touches	filesystem, authentication/security data
Calls detected	escapeCodeHtml, fromCharCode, floor, replace, tokenName, push, protect, includes, normalizeCodeLanguage, b, forEach, replaceAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 9 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2138: function tokenizedCodeHtml(code = "", language = "javascript") {
2139:   let html = escapeCodeHtml(code);
2140:   const tokens = [];
2141:   const tokenName = (index) => {
2142:     let value = Number(index) + 1;
2143:     let output = "";
2144:     while (value > 0) {
2145:       value -= 1;
2146:       output = String.fromCharCode(65 + (value % 26)) + output;
2147:       value = Math.floor(value / 26);
2148:     }
2149:     return `uE000${output}uE001`;
2150:   };
2151:   const protect = (pattern, className) => {
2152:     html = html.replace(pattern, (match) => {
2153:       const token = tokenName(tokens.length);
2154:       tokens.push({ token, html: `<span class="${className}">${match}</span>` });
2155:       return token;
2156:     });
2157:   };

```

```

2158:
2159:   if (language === "html") {
2160:     protect(/&lt;!--[\s\S]*?--&gt;/g, "code-token-comment");
2161:     protect(/&lt;\/?([a-zA-Z0-9:-]+)([\s\S]*?)(\/?&gt;)/g, "code-token-tag");
2162:   } else {
2163:     const commentPattern = ["python", "ltspice"].includes(normalizeCodeLanguage(language))
2164:       ? /\^\/\^*[\s\S]*?*\^\/\^\/\^\/[\^\\n]*#[\^\\n]*/g
2165:       : /\^\/\^*[\s\S]*?*\^\/\^\/\^\/[\^\\n]*/g;
2166:     protect(commentPattern, "code-token-comment");
2167:     protect(/"(?:\\\.|[\^"\\])*"'|'(?:\\\.|[\^"\\])*`"/g, "code-token-string");
2168:     if (["c", "cpp"].includes(normalizeCodeLanguage(language))) {
2169:       protect(/#\s*w+[\^\\n]*/gm, "code-token-preprocessor");
2170:     }
2171:   }
2172:
2173:   protect(/\\b(?:0x[\\da-f]+|\\d+(?:\\.\\d+)?(?:e[+-]?\\d+)?)/gi, "code-token-number");
2174:
2175:   const keywordMap = {
2176:     c: "_Alignas|_Alignof|_Atomic|_Bool|_Complex|_Generic|_Imaginary|_Noreturn|_Static_assert|_Thread_loc
al|auto|break|case|char|const|continue|default|do|double|else|enum|extern|float|for|goto|if|inline|int|long|regi
ster|restrict|return|short|signed|sizeof|static|struct|switch|typedef|union|unsigned|void|volatile|while",
2177:     cpp: "alignas|alignof|and|and_eq|asm|auto|bitand|bitor|bool|break|case|catch|char|char8_t|char16_t|ch
ar32_t|class|compl|concept|const|constexpr|constinit|const_cast|continue|co_await|co_return|co_yield|d
ecltype|default|delete|do|double|dynamic_cast|else|enum|explicit|export|extern|false|float|for|friend|if|inline|
int|long|mutable|namespace|new|noexcept|not|not_eq|nullptr|operator|or|or_eq|private|protected|public|register|r
einterpret_cast|requires|return|short|signed|sizeof|static|static_assert|static_cast|struct|switch|template|this
|thread_local|throw|true|try|typedef|typeid|typename|union|unsigned|using|virtual|void|volatile|wchar_t|while|xor
r|xor_eq",
2178:     verilog: "always|and|assign|begin|buf|case|casez|casez|deassign|default|defparam|disable|edge|else|en
d|endcase|endfunction|endmodule|endprimitive|endspecify|endtable|endtask|event|for|force|forever|fork|function|g
enerate|genvar|if|initial|inout|input|integer|join|module|nand|negedge|nor|not|or|output|parameter|posedge|primi
tive|reg|release|repeat|signed|specify|supply0|supply1|table|task|tri|wand|while|wire|wor|xnor|xor",
2179:     systemverilog: "always|always_comb|always_ff|always_latch|assign|automatic|begin|bit|case|class|clock
ing|covergroup|default|disable|do|else|end|endcase|endclass|endclocking|endfunction|endmodule|endpackage|endtask
|enum|for|forever|function|generate|genvar|if|initial|input|int|interface|logic|module|negedge|output|package|pa
rameter|posedge|reg|return|signed|task|typedef|wire",
2180:     ltspice: "ac|dc|end|ends|four|func|global|ic|include|lib|meas|model|nodeset|op|options|param|plot|pro
be|save|step|subckt|temp|tran",
2181:     java: "abstract|assert|boolean|break|byte|case|catch|char|class|const|continue|default|do|double|else
|enum|extends|final|finally|float|for|if|implements|import|instanceof|int|interface|long|native|new|null|package
|private|protected|public|return|short|static|strictfp|super|switch|synchronized|this|throw|throws|transient|try
|void|volatile|while",
2182:     javascript: "async|await|break|case|catch|class|const|continue|debugger|default|delete|do|else|export
|extends|false|finally|for|from|function|if|import|in|instanceof|let|new|null|of|return|static|super|switch|this
|throw|true|try|typeof|undefined|var|void|while|yield",
2183:     python: "and|as|assert|async|await|break|class|continue|def|del|elif|else|except|False|finally|for|fr
om|global|if|import|in|is|lambda|None|nonlocal|not|or|pass|raise|return|True|try|while|with|yield",
2184:     html: "html|head|body|main|section|article|div|span|script|style|link|meta|title|header|footer|nav|bu
tton|input|form|label|img|a|p|pre|code"
2185:   };
2186:   const keywords = keywordMap[normalizeCodeLanguage(language)] || keywordMap.javascript;
2187:   html = html.replace(new RegExp(`\\b(${keywords})\\b`, "g"), '<span class="code-token-
keyword">$1</span>');
2188:   tokens.forEach((token) => {
2189:     html = html.replaceAll(token.token, token.html);
2190:   });
2191:   return html;
2192: }

```

tokenName - lines 2141-2150

Item	Explanation
Source location	Lines 2141-2150 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const tokenName = (index) => {
Touches	authentication/security data
Calls detected	fromCharCode, floor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2141:   const tokenName = (index) => {
2142:     let value = Number(index) + 1;
2143:     let output = "";
2144:     while (value > 0) {

```



```

2145:     value -= 1;
2146:     output = String.fromCharCode(65 + (value % 26)) + output;
2147:     value = Math.floor(value / 26);
2148: }
2149: return ``\uE000${output}\uE001`;
2150: };

```

protect - lines 2151-2157

Item	Explanation
Source location	Lines 2151-2157 (7 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const protect = (pattern, className) => {
Touches	authentication/security data
Calls detected	replace, tokenName, push
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2151: const protect = (pattern, className) => {
2152:   html = html.replace(pattern, (match) => {
2153:     const token = tokenName(tokens.length);
2154:     tokens.push({ token, html: `<span class="${className}">${match}</span>` });
2155:     return token;
2156:   });
2157: };

```

renderRichCodeBlock - lines 2194-2204

Item	Explanation
Source location	Lines 2194-2204 (11 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderRichCodeBlock(block = {}) {
Touches	filesystem, authentication/security data
Calls detected	normalizeCodeLanguage, detectCodeLanguage, normalizeCodeText, codeLanguageLabel, escapeHtml, tokenizedCodeHtml
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2194: function renderRichCodeBlock(block = {}) {
2195:   const language = normalizeCodeLanguage(block.language || detectCodeLanguage(block.code || ""));
2196:   const code = normalizeCodeText(block.code || "", block.pasteMode || "source");
2197:   const label = codeLanguageLabel(language);
2198:   return
2199:     <figure class="rich-code-block rich-code-language-${language}">
2200:       <figcaption><span>&lt;&gt;</span> ${escapeHtml(label)}</figcaption>
2201:       <pre><code>${tokenizedCodeHtml(code, language)}</code></pre>
2202:     </figure>
2203:   `;
2204: }

```

renderPlainMultiline - lines 2206-2208

Item	Explanation
Source location	Lines 2206-2208 (3 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderPlainMultiline(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	escapeHtml, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2206: function renderPlainMultiline(value = "") {
2207:   return escapeHtml(value).replace(/\r\n?|\n/g, "<br>");
2208: }

```

normalizeSiteContent - lines 2210-2216

Item	Explanation
Source location	Lines 2210-2216 (7 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeSiteContent(content = {}) {
Touches	filesystem
Calls detected	trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2210: function normalizeSiteContent(content = {}) {
2211:   return {
2212:     heroCopy: String(content.heroCopy || "").trim() || defaultSiteContent.heroCopy,
2213:     heroEyebrow: String(content.heroEyebrow || "").trim() || defaultSiteContent.heroEyebrow,
2214:     heroTitle: String(content.heroTitle || "").trim() || defaultSiteContent.heroTitle
2215:   };
2216: }

```

normalizeProfile - lines 2218-2234

Item	Explanation
Source location	Lines 2218-2234 (17 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeProfile(profile = {}) {
Touches	filesystem
Calls detected	trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2218: function normalizeProfile(profile = {}) {
2219:   return {
2220:     brandImage: String(profile.brandImage || "").trim(),
2221:     brandText: String(profile.brandText || "").trim() || String(profile.displayName || "").trim() ||
defaultProfile.brandText,
2222:     contactIntro: String(profile.contactIntro || "").trim(),
2223:     displayName: String(profile.displayName || "").trim(),
2224:     email: String(profile.email || "").trim(),
2225:     githubUrl: String(profile.githubUrl || "").trim(),
2226:     heroImage: String(profile.heroImage || "").trim(),
2227:     linkedinUrl: String(profile.linkedinUrl || "").trim(),
2228:     phone: String(profile.phone || "").trim(),
2229:     portfolioLabel: String(profile.portfolioLabel || "").trim() || defaultProfile.portfolioLabel,
2230:     profileImage: String(profile.profileImage || "").trim(),
2231:     resumeUrl: String(profile.resumeUrl || "").trim(),
2232:     websiteUrl: String(profile.websiteUrl || "").trim()
2233:   };
2234: }

```

normalizePlainFieldStyle - lines 2236-2250

Item	Explanation
Source location	Lines 2236-2250 (15 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizePlainFieldStyle(style = {}) {
Touches	filesystem
Calls detected	cleanFontFamily, normalizeFontPx, normalizeTextColor, rgbColorToHex
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2236: function normalizePlainFieldStyle(style = {}) {
2237:   const fontFamily = cleanFontFamily(style.fontFamily || "");
2238:   const fontPx = normalizeFontPx(style.fontPx || style.fontSize || "");
2239:   const color = normalizeTextColor(style.color || "");
2240:   const normalized = {};
2241:
2242:   if (fontFamily) normalized.fontFamily = fontFamily;
2243:   if (fontPx) normalized.fontPx = fontPx;
2244:   if (color) normalized.color = rgbColorToHex(color);
2245:   if (style.bold === true || style.bold === "true") normalized.bold = true;
2246:   if (style.italic === true || style.italic === "true") normalized.italic = true;
2247:   if (style.underline === true || style.underline === "true") normalized.underline = true;
2248:
2249:   return normalized;
2250: }

```

normalizeFieldStyles - lines 2252-2259

Item	Explanation
Source location	Lines 2252-2259 (8 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeFieldStyles(styles = {}) {
Touches	filesystem
Calls detected	fromEntries, entries, filter, has, map, normalizePlainFieldStyle, keys
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2252: function normalizeFieldStyles(styles = {}) {
2253:   return Object.fromEntries(
2254:     Object.entries(styles || {})
2255:       .filter(([fieldId]) => persistentPlainStyleFieldIds.has(fieldId))
2256:       .map(([fieldId, style]) => [fieldId, normalizePlainFieldStyle(style)])
2257:       .filter(([ , style]) => Object.keys(style).length)
2258:   );
2259: }

```

loadBuilderPreferences - lines 2261-2273

Item	Explanation
Source location	Lines 2261-2273 (13 source lines).
Purpose	Loads data from disk, network, browser storage, or a service.
Declaration	function loadBuilderPreferences() {
Touches	Mostly local calculations or local UI state.
Calls detected	parse, getItem, includes, applyBuilderPreferences
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2261: function loadBuilderPreferences() {
2262:   try {
2263:     const stored = JSON.parse(localStorage.getItem(preferenceStorageKey) || "{}");
2264:     builderPreferences = {
2265:       ...defaultBuilderPreferences,
2266:       ...stored,
2267:       theme: ["light", "dark"].includes(stored.theme) ? stored.theme : defaultBuilderPreferences.theme
2268:     };
2269:   } catch {
2270:     builderPreferences = { ...defaultBuilderPreferences };
2271:   }
2272:   applyBuilderPreferences();
2273: }

```

applyBuilderPreferences - lines 2275-2280

Item	Explanation
Source location	Lines 2275-2280 (6 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function applyBuilderPreferences() {
Touches	browser/Electron DOM
Calls detected	includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2275: function applyBuilderPreferences() {
2276:   const theme = ["light", "dark"].includes(builderPreferences.theme) ? builderPreferences.theme :
"light";
2277:   document.documentElement.dataset.builderTheme = theme;
2278:   document.body.dataset.builderTheme = theme;
2279:   if (preferenceTheme) preferenceTheme.value = theme;
2280: }
```

setBuilderTheme - lines 2282-2287

Item	Explanation
Source location	Lines 2282-2287 (6 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function setBuilderTheme(theme = "light") {
Touches	Mostly local calculations or local UI state.
Calls detected	includes, setItem, stringify, applyBuilderPreferences, setStatus
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2282: function setBuilderTheme(theme = "light") {
2283:   builderPreferences.theme = ["light", "dark"].includes(theme) ? theme : "light";
2284:   localStorage.setItem(preferenceStorageKey, JSON.stringify(builderPreferences));
2285:   applyBuilderPreferences();
2286:   setStatus(`Builder ${builderPreferences.theme} mode enabled.`);
2287: }
```

openPreferencesDialog - lines 2289-2292

Item	Explanation
Source location	Lines 2289-2292 (4 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function openPreferencesDialog() {
Touches	Mostly local calculations or local UI state.
Calls detected	applyBuilderPreferences, showModal
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2289: function openPreferencesDialog() {
2290:   applyBuilderPreferences();
2291:   if (!preferencesDialog?.open) preferencesDialog.showModal();
2292: }
```

saveBuilderPreferencesFromDialog - lines 2294-2297

Item	Explanation
Source location	Lines 2294-2297 (4 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	function saveBuilderPreferencesFromDialog() {

Item	Explanation
Touches	Mostly local calculations or local UI state.
Calls detected	setBuilderTheme, setStatus
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2294: function saveBuilderPreferencesFromDialog() {
2295:   setBuilderTheme(preferenceTheme?.value || "light");
2296:   setStatus("Preferences saved.");
2297: }

```

plainFieldStyleToCss - lines 2299-2309

Item	Explanation
Source location	Lines 2299-2309 (11 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function plainFieldStyleToCss(style = {}) {
Touches	filesystem
Calls detected	normalizePlainFieldStyle, push, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2299: function plainFieldStyleToCss(style = {}) {
2300:   const normalized = normalizePlainFieldStyle(style);
2301:   const rules = [];
2302:   if (normalized.fontFamily) rules.push(`font-family: ${normalized.fontFamily}`);
2303:   if (normalized.fontPx) rules.push(`font-size: ${normalized.fontPx}px`);
2304:   if (normalized.color) rules.push(`color: ${normalized.color}`);
2305:   if (normalized.bold) rules.push("font-weight: 700");
2306:   if (normalized.italic) rules.push("font-style: italic");
2307:   if (normalized.underline) rules.push("text-decoration: underline");
2308:   return rules.join("; ");
2309: }

```

plainFieldStyleAttribute - lines 2311-2314

Item	Explanation
Source location	Lines 2311-2314 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function plainFieldStyleAttribute(styles = {}, fieldId = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	plainFieldStyleToCss, escapeHtml
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2311: function plainFieldStyleAttribute(styles = {}, fieldId = "") {
2312:   const css = plainFieldStyleToCss(styles[fieldId]);
2313:   return css ? `style="${escapeHtml(css)}"` : "";
2314: }
```

plainStyleFromControl - lines 2316-2326

Item	Explanation
Source location	Lines 2316-2326 (11 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function plainStyleFromControl(control) {
Touches	browser/Electron DOM, filesystem
Calls detected	normalizePlainFieldStyle
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2316: function plainStyleFromControl(control) {
2317:   if (!control) return {};
2318:   return normalizePlainFieldStyle({
2319:     bold: control.dataset.builderBold,
2320:     color: control.dataset.builderColor,
2321:     fontFamily: control.dataset.builderFontFamily,
2322:     fontPx: control.dataset.builderFontPx,
2323:     italic: control.dataset.builderItalic,
2324:     underline: control.dataset.builderUnderline
2325:   });
2326: }
```

applyPlainFieldStyleToControl - lines 2328-2343

Item	Explanation
Source location	Lines 2328-2343 (16 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function applyPlainFieldStyleToControl(control, style = {}) {
Touches	browser/Electron DOM, filesystem
Calls detected	normalizePlainFieldStyle
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2328: function applyPlainFieldStyleToControl(control, style = {}) {
2329:   if (!control) return;
2330:   const normalized = normalizePlainFieldStyle(style);
2331:   control.dataset.builderFontFamily = normalized.fontFamily || "";
2332:   control.dataset.builderFontPx = normalized.fontPx || "";
2333:   control.dataset.builderColor = normalized.color || "";
2334:   control.dataset.builderBold = normalized.bold ? "true" : "false";
2335:   control.dataset.builderItalic = normalized.italic ? "true" : "false";
2336:   control.dataset.builderUnderline = normalized.underline ? "true" : "false";
2337:   control.style.fontFamily = normalized.fontFamily || "";
2338:   control.style.fontSize = normalized.fontPx ? `${normalized.fontPx}px` : "";
2339:   control.style.color = normalized.color || "";
2340:   control.style.fontWeight = normalized.bold ? "700" : "";
2341:   control.style.fontStyle = normalized.italic ? "italic" : "";
2342:   control.style.textDecoration = normalized.underline ? "underline" : "";
```

2343: }

applyStoredPlainFieldStyle - lines 2345-2349

Item	Explanation
Source location	Lines 2345-2349 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function applyStoredPlainFieldStyle(control) {
Touches	filesystem
Calls detected	has, normalizeFieldStyles, applyPlainFieldStyleToControl
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2345: function applyStoredPlainFieldStyle(control) {
2346:   if (!control?.id || !persistentPlainStyleFieldIds.has(control.id)) return;
2347:   const styles = normalizeFieldStyles(catalog.fieldStyles || {});
2348:   applyPlainFieldStyleToControl(control, styles[control.id] || {});
2349: }
```

storePlainControlStyle - lines 2351-2365

Item	Explanation
Source location	Lines 2351-2365 (15 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	function storePlainControlStyle(control) {
Touches	filesystem
Calls detected	has, normalizeFieldStyles, plainStyleFromControl, keys, clone, markDraftNeedsSave, scheduleAutosave, schedulePreviewRender
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2351: function storePlainControlStyle(control) {
2352:   if (!control?.id || !persistentPlainStyleFieldIds.has(control.id)) return;
2353:   const nextStyles = normalizeFieldStyles(catalog.fieldStyles || {});
2354:   const style = plainStyleFromControl(control);
2355:   if (Object.keys(style).length) {
2356:     nextStyles[control.id] = style;
2357:   } else {
2358:     delete nextStyles[control.id];
2359:   }
2360:   catalog.fieldStyles = nextStyles;
2361:   savedPortfolioCatalog.fieldStyles = clone(nextStyles);
2362:   markDraftNeedsSave();
2363:   scheduleAutosave();
2364:   schedulePreviewRender();
2365: }
```

mailComposeLink - lines 2367-2370

Item	Explanation
Source location	Lines 2367-2370 (4 source lines).

Item	Explanation
Purpose	Part of the private builder frontend and editing experience.
Declaration	function mailComposeLink(email = "") {
Touches	network/API requests
Calls detected	trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2367: function mailComposeLink(email = "") {
2368:   const clean = String(email || "").trim();
2369:   return clean ? `https://mail.google.com/mail/?view=cm&fs=1&to=${encodeURIComponent(clean)}` : "";
2370: }

```

phoneLink - lines 2372-2375

Item	Explanation
Source location	Lines 2372-2375 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function phoneLink(phone = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2372: function phoneLink(phone = "") {
2373:   const clean = String(phone || "").trim();
2374:   return clean ? `tel:${clean.replace(/^[^d+]/g, "")}` : "";
2375: }

```

textBlocksFromPlainText - lines 2377-2385

Item	Explanation
Source location	Lines 2377-2385 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function textBlocksFromPlainText(text) {
Touches	filesystem
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2377: function textBlocksFromPlainText(text) {
2378:   return [{
2379:     align: "left",
2380:     fontFamily: "Arial",
2381:     fontSize: "normal",
2382:     text: String(text || ""),
2383:     type: "paragraph"
2384:   }];
2385: }

```

richBlocksForProject - lines 2387-2389

Item	Explanation
Source location	Lines 2387-2389 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function richBlocksForProject(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	clone, textBlocksFromPlainText
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2387: function richBlocksForProject(project) {
2388:   return project?.summaryRich?.blocks?.length ? clone(project.summaryRich.blocks) :
textBlocksFromPlainText(project?.summary || "");
2389: }

```

plainTextFromRich - lines 2391-2402

Item	Explanation
Source location	Lines 2391-2402 (12 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function plainTextFromRich(rich, separator = "\n\n") {
Touches	filesystem
Calls detected	map, filter, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 6 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2391: function plainTextFromRich(rich, separator = "\n\n") {
2392:   return (rich?.blocks || [])
2393:     .map((block) => {
2394:       if (block.type === "paragraph") return block.text || "";
2395:       if (block.type === "formula") return block.formula || "";
2396:       if (block.type === "code") return block.code || "";
2397:       if (block.type === "image") return [block.title, block.caption].filter(Boolean).join(" ");
2398:       return "";
2399:     })
2400:     .filter(Boolean)
2401:     .join(separator);
2402: }

```

unwrapFormula - lines 2404-2417

Item	Explanation
Source location	Lines 2404-2417 (14 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function unwrapFormula(value) {
Touches	filesystem
Calls detected	trim, match
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2404: function unwrapFormula(value) {
2405:   const text = String(value || "").trim();
2406:   const wrappers = [
2407:     [/^\$\$([\s\S]+\)\$\$$/, 1],
2408:     [/^\$\$([\s\S]+\)\$\$$/, 1],
2409:     [/^\$\$([\s\S]+\)\$\$$/, 1],
2410:     [/^\$([\s\S]+\)\$\$$/, 1]
2411:   ];
2412:   for (const [pattern, group] of wrappers) {
2413:     const match = text.match(pattern);
2414:     if (match) return match[group].trim();
2415:   }
2416:   return text;
2417: }

```

isFormulaOnly - lines 2419-2426

Item	Explanation
Source location	Lines 2419-2426 (8 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function isFormulaOnly(value) {
Touches	filesystem
Calls detected	trim, looksLikeWebOrContactText
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2419: function isFormulaOnly(value) {
2420:   const text = String(value || "").trim();
2421:   if (!text) return false;
2422:   if (looksLikeWebOrContactText(text)) return false;
2423:   if (/^\$\$([\s\S]+\)\$\$$/ .test(text) || /^\$\$([\s\S]+\)\$\$$/ .test(text) || /^\$([\s\S]+\)\$\$$/ .test(text))
return true;
2424:   const mathSignals = /\frac|\sqrt|\sum|\int|\Delta|\omega|\pi|[\^_]/[\Sigma\Omega]/;
2425:   return text.length <= 180 && mathSignals.test(text) && !/[.!?]\s*$/ .test(text);
2426: }

```

renderInlineMath - lines 2428-2437

Item	Explanation
Source location	Lines 2428-2437 (10 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderInlineMath(text) {

Item	Explanation
Touches	filesystem
Calls detected	escapeHtml, replace, b, match, slice, normalizeLinkTarget
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2428: function renderInlineMath(text) {
2429:   const escaped = escapeHtml(text);
2430:   const withMath = escaped.replace(/\$([\^$]+)\$/g, '<span class="rich-inline-formula">$1</span>');
2431:   return withMath.replace(/\b(https?:\/\/[\^s<]+|www\.[\^s<]+|(?:(?:github|linkedin|gitlab|bitbucket|drive|d
ocs)\.com\/[\^s<]+))/gi, (match) => {
2432:     const trailing = match.match(/[(),.;:!?]+\$/)?.[0] || "";
2433:     const clean = trailing ? match.slice(0, -trailing.length) : match;
2434:     const href = normalizeLinkTarget(clean, { assumeWeb: true });
2435:     return `<a href="${href}" target="_blank" rel="noreferrer">${clean}</a>${trailing}`;
2436:   });
2437: }

```

renderMultilineInlineText - lines 2439-2441

Item	Explanation
Source location	Lines 2439-2441 (3 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderMultilineInlineText(text = "") {
Touches	filesystem
Calls detected	renderInlineMath, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2439: function renderMultilineInlineText(text = "") {
2440:   return renderInlineMath(text).replace(/\r\n?|\n/g, "<br>");
2441: }

```

looksLikeBareWebAddress - lines 2443-2450

Item	Explanation
Source location	Lines 2443-2450 (8 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function looksLikeBareWebAddress(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, split, toLowerCase
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2443: function looksLikeBareWebAddress(value = "") {
2444:   const clean = String(value || "").trim();
2445:   if (!clean || /\s/.test(clean) || /^(\.?\.\.?\/|#|mailto:|tel:)/i.test(clean)) return false;
2446:   if (/^[a-z][a-z0-9+.-]*:/i.test(clean)) return false;
2447:   const host = clean.split(/[/?#]/)[0].toLowerCase();
2448:   if (!/^[a-z0-9-]+\.(?![a-z0-9-]+)$/.test(host)) return false;
2449:   return !/\.(pdf|docx?|xlsx?|pptx?|zip|7z|rar|png|jpe?g|gif|svg|webp|txt|md|csv|json|xml|log|c|h|cpp|hpp|py|js|mjs|ts|v|sv|vhd|l?|spice|cir|net|asc|sch|kicad_sch|kicad_pcb)$/i.test(host);
2450: }
```

looksLikeWebOrContactText - lines 2452-2460

Item	Explanation
Source location	Lines 2452-2460 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function looksLikeWebOrContactText(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, b, looksLikeBareWebAddress, split, some, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 6 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2452: function looksLikeWebOrContactText(value = "") {
2453:   const clean = String(value || "").trim();
2454:   if (!clean) return false;
2455:   if (/^b(?:https?:\/\/|www\.)[^s<>"]+$/i.test(clean)) return true;
2456:   if (/^b[a-z0-9._%+~]@[a-z0-9.-]+\.[a-z]{2,}\b/i.test(clean)) return true;
2457:   if (/^b(?:github|linkedin|gitlab|bitbucket|drive|docs)\.com\/[^s<>"]+$/i.test(clean)) return true;
2458:   if (looksLikeBareWebAddress(clean)) return true;
2459:   return clean.split(/\s+/).some((word) => looksLikeBareWebAddress(word.replace(/^[("'+|[]"'.,;:!?]+$/g, "")));
2460: }
```

normalizeLinkTarget - lines 2462-2469

Item	Explanation
Source location	Lines 2462-2469 (8 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeLinkTarget(target = "", options = {}) {
Touches	filesystem, network/API requests
Calls detected	trim, looksLikeBareWebAddress
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 5 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2462: function normalizeLinkTarget(target = "", options = {}) {
2463:   const clean = String(target || "").trim();
2464:   if (!clean) return "";
2465:   if (/^\/\//.test(clean)) return `https:${clean}`;
2466:   if (/^www\./i.test(clean)) return `https://${clean}`;
2467:   if (options.assumeWeb && looksLikeBareWebAddress(clean)) return `https://${clean}`;
2468:   return clean;
2469: }
```

isWebsiteLinkItem - lines 2471-2479

Item	Explanation
Source location	Lines 2471-2479 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function isWebsiteLinkItem(item = {}, target = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, filter, join, toLowerCase, b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2471: function isWebsiteLinkItem(item = {}, target = "") {
2472:   const clean = String(target || "").trim();
2473:   if (/^(https?:)?\:\/\/.i.test(clean) || /^www\..i.test(clean)) return true;
2474:   const typeText = [item.kind, item.type, item.status, item.meta, item.label, item.title]
2475:     .filter(Boolean)
2476:     .join(" ")
2477:     .toLowerCase();
2478:   return /\b(link|website|web link|url|external|github|linkedin|drive|social|profile)\b/.test(typeText);
2479: }
```

extensionFromUrl - lines 2481-2485

Item	Explanation
Source location	Lines 2481-2485 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function extensionFromUrl(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	split, match, toLowerCase
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2481: function extensionFromUrl(value = "") {
2482:   const clean = String(value || "").split(/[?#]/)[0];
2483:   const match = clean.match(/(\.[a-z0-9_]+)$/i);
2484:   return match ? match[1].toLowerCase() : "";
2485: }
```

urlLooksLikeDirectImage - lines 2487-2491

Item	Explanation
Source location	Lines 2487-2491 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function urlLooksLikeDirectImage(value = "") {
Touches	filesystem
Calls detected	normalizeLinkTarget
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2487: function urlLooksLikeDirectImage(value = "") {
2488:   const clean = normalizeLinkTarget(value, { assumeweb: true });
2489:   return /^(data:image\/|blob:)/i.test(clean) ||
2490:     /\.(png|jpe?g|webp|gif|svg|bmp|avif)([?#].*)?$/i.test(clean);
2491: }

```

inferUrlAssetKind - lines 2493-2506

Item	Explanation
Source location	Lines 2493-2506 (14 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function inferUrlAssetKind(value = "", source = "web") {
Touches	filesystem
Calls detected	normalizeLinkTarget, extensionFromUrl, urlLooksLikeDirectImage, includes, looksLikeBareWebAddress
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 10 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2493: function inferUrlAssetKind(value = "", source = "web") {
2494:   const clean = normalizeLinkTarget(value, { assumeweb: true });
2495:   const extension = extensionFromUrl(clean);
2496:   if (source === "drive" || /^https?:\/\/?(\drive|docs)\.google\.com\/\//i.test(clean)) return "google-
drive";
2497:   if (/github\.com|raw\.githubusercontent\.com/i.test(clean)) return "github";
2498:   if (/linkedin\.com/i.test(clean)) return "linkedin";
2499:   if (urlLooksLikeDirectImage(clean)) return "image";
2500:   if ([".pdf"].includes(extension)) return "pdf";
2501:   if ([".doc", ".docx", ".ppt", ".pptx", ".xls", ".xlsx", ".csv", ".txt", ".md"].includes(extension))
return "document";
2502:   if ([".zip", ".7z", ".rar"].includes(extension)) return "archive";
2503:   if ([".c", ".h", ".cpp", ".hpp", ".py", ".js", ".mjs", ".ts", ".v", ".sv", ".vhdl", ".spice",
".cir", ".net", ".asc", ".sch", ".kicad_sch", ".kicad_pcb"].includes(extension)) return "engineering-file";
2504:   if (/^https?:\/\//i.test(clean) || /^www\./i.test(value) || looksLikeBareWebAddress(value)) return
"webpage";
2505:   return extension ? "file" : "link";
2506: }

```

labelForUrlAssetKind - lines 2508-2522

Item	Explanation
Source location	Lines 2508-2522 (15 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function labelForUrlAssetKind(kind = "link") {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2508: function labelForUrlAssetKind(kind = "link") {
2509:   return {
2510:     archive: "Archive link",
2511:     document: "Document link",
2512:     "engineering-file": "Engineering file link",
2513:     file: "File link",
2514:     github: "GitHub link",
2515:     "google-drive": "Google Drive link",
2516:     image: "Image URL",
2517:     linkedin: "LinkedIn link",
2518:     link: "Link",
2519:     pdf: "PDF link",
2520:     webpage: "Website link"
2521:   }[kind] || "Link";
2522: }

```

displayNameFromUrl - lines 2524-2533

Item	Explanation
Source location	Lines 2524-2533 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function displayNameFromUrl(value = "", fallback = "Linked asset") {
Touches	filesystem
Calls detected	normalizeLinkTarget, URL, split, filter, pop, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2524: function displayNameFromUrl(value = "", fallback = "Linked asset") {
2525:   const clean = normalizeLinkTarget(value, { assumeWeb: true });
2526:   try {
2527:     const parsed = new URL(clean);
2528:     const lastPath = parsed.pathname.split("/").filter(Boolean).pop();
2529:     return decodeURIComponent(lastPath || parsed.hostname.replace(/^www\./, "")) || fallback;
2530:   } catch {
2531:     return String(value || "").split("/").filter(Boolean).pop() || fallback;
2532:   }
2533: }

```

linkifyRichTextNodes - lines 2535-2570

Item	Explanation
Source location	Lines 2535-2570 (36 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function linkifyRichTextNodes(root) {
Touches	browser/Electron DOM, filesystem
Calls detected	createTreeWalker, acceptNode, closest, nextNode, push, forEach, createDocumentFragment, replace, b, append, createTextNode, slice
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2535: function linkifyRichTextNodes(root) {
2536:   const walker = document.createTreeWalker(root, NodeFilter.SHOW_TEXT, {
2537:     acceptNode(node) {
2538:       if (!node.textContent ||
!/(https?:\/\/|www\.(?:github|linkedin|gitlab|bitbucket|drive|docs)\.com\/)/i.test(node.textContent)) return
NodeFilter.FILTER_REJECT;
2539:       return node.parentElement?.closest("a") ? NodeFilter.FILTER_REJECT : NodeFilter.FILTER_ACCEPT;
2540:     }
2541:   });
2542:   const nodes = [];
2543:   let node = walker.nextNode();
2544:   while (node) {
2545:     nodes.push(node);
2546:     node = walker.nextNode();
2547:   }
2548:
2549:   nodes.forEach((textNode) => {
2550:     const text = textNode.textContent || "";
2551:     const fragment = document.createDocumentFragment();
2552:     let cursor = 0;
2553:     text.replace(/\b(https?:\/\/|www\.([^\s<]+|(?:(?:github|linkedin|gitlab|bitbucket|drive|docs)\.com
\/[^\s<]+)/gi, (match, _url, offset) => {
2554:       if (offset > cursor) fragment.append(document.createTextNode(text.slice(cursor, offset)));
2555:       const trailing = match.match(/[.,;!?]+$/)?.[0] || "";
2556:       const clean = trailing ? match.slice(0, -trailing.length) : match;
2557:       const link = document.createElement("a");
2558:       link.href = normalizeLinkTarget(clean, { assumeWeb: true });
2559:       link.target = "_blank";
2560:       link.rel = "noreferrer";
2561:       link.textContent = clean;
2562:       fragment.append(link);
2563:       if (trailing) fragment.append(document.createTextNode(trailing));
2564:       cursor = offset + match.length;
2565:       return match;
2566:     });
2567:     if (cursor < text.length) fragment.append(document.createTextNode(text.slice(cursor)));
2568:     textNode.replaceWith(fragment);
2569:   });
2570: }
```

inlineStyleSignature - lines 2572-2583

Item	Explanation
Source location	Lines 2572-2583 (12 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function inlineStyleSignature(element) {
Touches	Mostly local calculations or local UI state.
Calls detected	join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2572: function inlineStyleSignature(element) {
2573:   const style = element?.style;
2574:   if (!style) return "";
2575:   return [
2576:     style.fontFamily || "",
2577:     style.fontSize || "",
2578:     style.color || "",
2579:     style.fontWeight || "",
2580:     style.fontStyle || "",
2581:     style.textDecoration || ""
2582:   ].join("|");
2583: }
```

normalizeInlineStyleSpans - lines 2585-2605

Item	Explanation
Source location	Lines 2585-2605 (21 source lines).

Item	Explanation
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeInlineStyleSpans(root) {
Touches	browser/Electron DOM, filesystem
Calls detected	querySelectorAll, forEach, remove, getAttribute, replaceWith, normalize, matches, inlineStyleSignature, append
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2585: function normalizeInlineStyleSpans(root) {
2586:   if (!root) return;
2587:   root.querySelectorAll("span.rich-inline-style, span[style]").forEach((span) => {
2588:     span.style.display = "inline";
2589:     if (!span.textContent) {
2590:       span.remove();
2591:       return;
2592:     }
2593:     if (!span.getAttribute("style")) span.replaceWith(...span.childNodes);
2594:   });
2595:   root.normalize();
2596:   root.querySelectorAll("span.rich-inline-style, span[style]").forEach((span) => {
2597:     let next = span.nextSibling;
2598:     while (next?.nodeType === Node.ELEMENT_NODE && next.matches("span.rich-inline-style, span[style]") &&
inlineStyleSignature(next) === inlineStyleSignature(span)) {
2599:       span.append(...next.childNodes);
2600:       const remove = next;
2601:       next = next.nextSibling;
2602:       remove.remove();
2603:     }
2604:   });
2605: }

```

sanitizeRichInlineHtml - lines 2607-2651

Item	Explanation
Source location	Lines 2607-2651 (45 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function sanitizeRichInlineHtml(value = "") {
Touches	browser/Electron DOM, filesystem
Calls detected	createElement, Set, querySelectorAll, forEach, has, replaceWith, getAttribute, trim, normalizeLinkTarget, cleanFontFamily, normalizeFontPx, normalizeTextColor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2607: function sanitizeRichInlineHtml(value = "") {
2608:   const template = document.createElement("template");
2609:   template.innerHTML = String(value || "");
2610:   const allowedTags = new Set(["A", "B", "BR", "EM", "I", "SPAN", "STRONG", "U"]);
2611:
2612:   [...template.content.querySelectorAll("*")].forEach((node) => {
2613:     if (!allowedTags.has(node.tagName)) {
2614:       node.replaceWith(...node.childNodes);
2615:       return;
2616:     }
2617:
2618:     const href = node.tagName === "A" ? String(node.getAttribute("href") || "").trim() : "";
2619:     const normalizedHref = normalizeLinkTarget(href, { assumeweab: true });
2620:     const safeHref = /^((https?:|mailto:|tel:|)|\s)/i.test(normalizedHref) ? normalizedHref : "";
2621:     const fontFamily = cleanFontFamily(node.style.fontFamily || "");

```

```

2622:     const fontPx = normalizeFontPx(parseFloat(node.style.fontSize || ""));
2623:     const color = normalizeTextColor(node.style.color || "");
2624:     const rawFontWeight = node.style.fontWeight || "";
2625:     const fontWeight = /^(bold|[6-9]00)$/i.test(rawFontWeight)
2626:       ? "700"
2627:       : /^(normal|[1-5]00)$/i.test(rawFontWeight)
2628:       ? "400"
2629:       : "";
2630:     const fontStyle = node.style.fontStyle === "italic" ? "italic" : "";
2631:     const textDecoration = node.style.textDecoration.includes("underline") ? "underline" : "";
2632:
2633:     [...node.attributes].forEach((attribute) => node.removeAttribute(attribute.name));
2634:
2635:     if (node.tagName === "A" && safeHref) {
2636:       node.setAttribute("href", safeHref);
2637:       node.setAttribute("target", "_blank");
2638:       node.setAttribute("rel", "noreferrer");
2639:     }
2640:     if (fontFamily) node.style.fontFamily = fontFamily;
2641:     if (fontPx) node.style.fontSize = `${fontPx}px`;
2642:     if (color) node.style.color = color;
2643:     if (fontWeight) node.style.fontWeight = fontWeight;
2644:     if (fontStyle) node.style.fontStyle = fontStyle;
2645:     if (textDecoration) node.style.textDecoration = textDecoration;
2646:   });
2647:
2648:   linkifyRichTextNodes(template.content);
2649:   normalizeInlineStyleSpans(template.content);
2650:   return template.innerHTML;
2651: }

```

displayTitle - lines 2653-2662

Item	Explanation
Source location	Lines 2653-2662 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function displayTitle(value, fallback = "Untitled item") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2653: function displayTitle(value, fallback = "Untitled item") {
2654:   const title = String(value || fallback).trim();
2655:   const replacements = {
2656:     "Brief": "Overview",
2657:     "Project Brief": "Overview",
2658:     "Design Brief": "Design Overview",
2659:     "Simulation Brief": "Simulation Overview"
2660:   };
2661:   return replacements[title] || title;
2662: }

```

cleanFontFamily - lines 2664-2671

Item	Explanation
Source location	Lines 2664-2671 (8 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function cleanFontFamily(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	split, map, replace, trim, filter, slice, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2664: function cleanFontFamily(value = "") {
2665:   return String(value)
2666:     .split(",")
2667:     .map((item) => item.replace(/^[^w\s-]/g, "").trim())
2668:     .filter(Boolean)
2669:     .slice(0, 3)
2670:     .join(", ");
2671: }

```

normalizeFontPx - lines 2673-2676

Item	Explanation
Source location	Lines 2673-2676 (4 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeFontPx(value) {
Touches	filesystem
Calls detected	isFinite
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2673: function normalizeFontPx(value) {
2674:   const size = Number(value);
2675:   return Number.isFinite(size) && size >= 8 && size <= 24 ? String(size) : "";
2676: }

```

normalizeTextColor - lines 2677-2684

Item	Explanation
Source location	Lines 2677-2684 (8 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeTextColor(value = "") {
Touches	filesystem
Calls detected	trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2677: function normalizeTextColor(value = "") {
2678:   const color = String(value || "").trim();
2679:   return /^(#[0-9a-f]{3}|[0-9a-f]{6})$/i.test(color) ||
2680:     /^rgba?\(/i.test(color) ||
2681:     /^hsla?\(/i.test(color)
2682:     ? color
2683:     : "";
2684: }

```

summaryRichPathFromTextPath - lines 2686-2688

Item	Explanation
Source location	Lines 2686-2688 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function summaryRichPathFromTextPath(pathValue = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2686: function summaryRichPathFromTextPath(pathValue = "") {
2687:   return String(pathValue || "").replace(/\.summary$/, ".summaryRich");
2688: }
```

overviewFolderFromPath - lines 2690-2695

Item	Explanation
Source location	Lines 2690-2695 (6 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function overviewFolderFromPath(pathValue = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2690: function overviewFolderFromPath(pathValue = "") {
2691:   if (pathValue.includes("design.brief")) return "design-overview";
2692:   if (pathValue.includes("design.documentation")) return "documentation-overview";
2693:   if (pathValue.includes("design.simulation")) return "simulation-overview";
2694:   return "overview";
2695: }
```

richBlocksFromValue - lines 2697-2699

Item	Explanation
Source location	Lines 2697-2699 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function richBlocksFromValue(rich, fallbackText = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	clone, textBlocksFromPlainText
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2697: function richBlocksFromValue(rich, fallbackText = "") {
2698:   return rich?.blocks?.length ? clone(rich.blocks) : textBlocksFromPlainText(fallbackText);
2699: }

```

richTextStyle - lines 2700-2714

Item	Explanation
Source location	Lines 2700-2714 (15 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function richTextStyle(block = {}) {
Touches	filesystem
Calls detected	cleanFontFamily, normalizeFontPx, normalizeTextColor, push, escapeHtml, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2700: function richTextStyle(block = {}) {
2701:   const styles = [];
2702:   const fontFamily = cleanFontFamily(block.fontFamily || "Arial") || "Arial";
2703:   const fontPx = normalizeFontPx(block.fontPx);
2704:   const color = normalizeTextColor(block.color || "");
2705:
2706:   if (fontFamily) styles.push(`font-family: ${fontFamily}`);
2707:   if (fontPx) styles.push(`font-size: ${fontPx}px`);
2708:   if (color) styles.push(`color: ${color}`);
2709:   if (block.bold) styles.push("font-weight: 700");
2710:   if (block.italic) styles.push("font-style: italic");
2711:   if (block.underline) styles.push("text-decoration: underline");
2712:
2713:   return styles.length ? ` style="${escapeHtml(styles.join("; "))}"` : "";
2714: }

```

normalizeCropAspect - lines 2716-2718

Item	Explanation
Source location	Lines 2716-2718 (3 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeCropAspect(value = "") {
Touches	filesystem
Calls detected	includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2716: function normalizeCropAspect(value = "") {
2717:   return ["1 / 1", "4 / 3", "16 / 9", "3 / 4"].includes(value) ? value : "original";
2718: }

```

normalizeCropZoom - lines 2720-2723

Item	Explanation
Source location	Lines 2720-2723 (4 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeCropZoom(value = 1) {
Touches	filesystem
Calls detected	isFinite, min, max
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2720: function normalizeCropZoom(value = 1) {
2721:   const zoom = Number(value);
2722:   return Number.isFinite(zoom) ? Math.min(3, Math.max(1, zoom)) : 1;
2723: }
```

normalizeCropPosition - lines 2725-2728

Item	Explanation
Source location	Lines 2725-2728 (4 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeCropPosition(value = 50) {
Touches	filesystem
Calls detected	isFinite, min, max
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2725: function normalizeCropPosition(value = 50) {
2726:   const position = Number(value);
2727:   return Number.isFinite(position) ? Math.min(100, Math.max(0, position)) : 50;
2728: }
```

richImageCropStyle - lines 2730-2738

Item	Explanation
Source location	Lines 2730-2738 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function richImageCropStyle(block = {}) {
Touches	filesystem
Calls detected	normalizeCropAspect, normalizeCropZoom, normalizeCropPosition, push, escapeHtml, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2730: function richImageCropStyle(block = {}) {
2731:   const aspect = normalizeCropAspect(block.cropAspect);
2732:   const zoom = normalizeCropZoom(block.cropZoom);
2733:   const x = normalizeCropPosition(block.cropX);
2734:   const y = normalizeCropPosition(block.cropY);
2735:   const styles = [ `--crop-zoom: ${zoom}`, `--crop-x: ${x}%`, `--crop-y: ${y}%` ];
2736:   if (aspect !== "original") styles.push(`--crop-aspect: ${aspect}`);
2737:   return `style="${escapeHtml(styles.join(" ");)}"`;
2738: }

```

richImageDownloadLink - lines 2740-2745

Item	Explanation
Source location	Lines 2740-2745 (6 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function richImageDownloadLink(block = {}) {
Touches	filesystem
Calls detected	escapeHtml, cleanRichImageTitle, normalizeLinkTarget, isWebsiteLinkItem, linkAttributes, downloadAttribute
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2740: function richImageDownloadLink(block = {}) {
2741:   const label = escapeHtml(cleanRichImageTitle(block) || block.caption || "Image file");
2742:   const url = block.url || "#";
2743:   const target = normalizeLinkTarget(url, { assumeWeb: isWebsiteLinkItem(block, url) });
2744:   return `<p class="rich-download-only"><a class="resource-link"
href="${escapeHtml(target)}"${linkAttributes(target, block)}${downloadAttribute(target,
block)}>${label}</a></p>`;
2745: }

```

cleanRichImageTitle - lines 2747-2750

Item	Explanation
Source location	Lines 2747-2750 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function cleanRichImageTitle(block = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2747: function cleanRichImageTitle(block = {}) {
2748:   const title = String(block.title || "").trim();
2749:   return /^pasted image\b/i.test(title) ? "" : title;
2750: }

```


renderRichContent - lines 2752-2787

Item	Explanation
Source location	Lines 2752-2787 (36 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderRichContent(rich, fallbackText = "") {
Touches	filesystem
Calls detected	textBlocksFromPlainText, map, includes, richImageDownloadLink, cleanRichImageTitle, normalizeLinkTarget, normalizeCropAspect, richImageCropStyle, escapeHtml, unwrapFormula, looksLikeWebOrContactText, renderInlineMath
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 7 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2752: function renderRichContent(rich, fallbackText = "") {
2753:   const blocks = rich?.blocks?.length ? rich.blocks : textBlocksFromPlainText(fallbackText);
2754:   return `
2755:     <div class="rich-content">
2756:       ${blocks.map((block) => {
2757:         const align = ["left", "center", "right"].includes(block.align) ? block.align : "left";
2758:         if (block.type === "image") {
2759:           if (block.display === "download") return richImageDownloadLink(block);
2760:           const title = cleanRichImageTitle(block);
2761:           const imageSrc = normalizeLinkTarget(block.url, { assumeWeb: true });
2762:           return `
2763:             <figure class="rich-image justify-${align}">
2764:               <span class="rich-image-viewport crop-${normalizeCropAspect(block.cropAspect)} ===
"original" ? "original" : "active"${richImageCropStyle(block)}>
2765:                 
2766:               </span>
2767:               ${((title || block.caption) ? `<figcaption>${title ? `<strong>${escapeHtml(title)}</strong>`
: ""}${block.caption ? `<span>${escapeHtml(block.caption)}</span>` : ""}</figcaption>` : ""}
2768:             </figure>
2769:           `;
2770:         }
2771:         if (block.type === "formula") {
2772:           const formula = unwrapFormula(block.formula);
2773:           if (looksLikeWebOrContactText(formula)) {
2774:             return `<p class="rich-paragraph rich-text-normal
text-${align}">${renderInlineMath(formula)}</p>`;
2775:           }
2776:           return `<div class="rich-formula justify-${align}">${escapeHtml(formula)}</div>`;
2777:         }
2778:         if (block.type === "code") return renderRichCodeBlock(block);
2779:         const size = ["small", "normal", "large"].includes(block.fontSize) ? block.fontSize : "normal";
2780:         const content = block.html
          ? sanitizeRichInlineHtml(block.html)
2781:          : renderInlineMath(block.text || "");
2782:         return `<p class="rich-paragraph rich-text-${size}
text-${align}"${richTextStyle(block)}>${content}</p>`;
2783:       }).join("")}
2784:     </div>
2785:   `;
2786: }
2787: }
```

renderRichFieldContent - lines 2789-2799

Item	Explanation
Source location	Lines 2789-2799 (11 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderRichFieldContent(rich, fallbackText = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	textBlocksFromPlainText, map, includes, sanitizeRichInlineHtml, renderInlineMath, richTextStyle, filter, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2789: function renderRichFieldContent(rich, fallbackText = "") {
2790:   const blocks = rich?.blocks?.length ? rich.blocks : textBlocksFromPlainText(fallbackText);
2791:   return blocks.map((block) => {
2792:     if (block.type !== "paragraph") return "";
2793:     const align = ["left", "center", "right"].includes(block.align) ? block.align : "left";
2794:     const content = block.html
2795:       ? sanitizeRichInlineHtml(block.html)
2796:       : renderInlineMath(block.text || "");
2797:     return `<span class="rich-field-line text-${align}"${richTextStyle(block)}>${content}</span>`;
2798:   }).filter(Boolean).join("");
2799: }

```

saveSelectedProjectToPortfolio - lines 2801-2817

Item	Explanation
Source location	Lines 2801-2817 (17 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	function saveSelectedProjectToPortfolio() {
Touches	Mostly local calculations or local UI state.
Calls detected	selectedProject, parseProjectForPortfolio, syncParsedPortfolioGlobalsIntoSaved, filter, Set, map, has, findIndex, max, splice, markDraftNeedsSave, refreshOpenPreviews
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2801: function saveSelectedProjectToPortfolio() {
2802:   const project = selectedProject();
2803:   if (!project) return false;
2804:   const parsedProject = parseProjectForPortfolio(project);
2805:   syncParsedPortfolioGlobalsIntoSaved();
2806:   const nextProjects = (savedPortfolioCatalog.projects || []).filter((item) => item.id !== project.id);
2807:   const workingIds = new Set((catalog.projects || []).map((item) => item.id));
2808:   savedPortfolioCatalog.projects = nextProjects.filter((item) => workingIds.has(item.id));
2809:   const workingIndex = catalog.projects.findIndex((item) => item.id === project.id);
2810:   const insertAt = Math.max(0, workingIndex);
2811:   savedPortfolioCatalog.projects.splice(insertAt, 0, parsedProject);
2812:   markDraftNeedsSave();
2813:   refreshOpenPreviews();
2814:   setStatus(`${project.title} saved into the portfolio preview.`);
2815:   updateBuilderWorkflow();
2816:   return true;
2817: }

```

nextProjects - lines 2806-2814

Item	Explanation
Source location	Lines 2806-2814 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const nextProjects = (savedPortfolioCatalog.projects []).filter((item) => item.id !== project.id);
Touches	Mostly local calculations or local UI state.
Calls detected	filter, Set, map, has, findIndex, max, splice, markDraftNeedsSave, refreshOpenPreviews, setStatus
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2806: const nextProjects = (savedPortfolioCatalog.projects || []).filter((item) => item.id !== project.id);
2807: const workingIds = new Set((catalog.projects || []).map((item) => item.id));
2808: savedPortfolioCatalog.projects = nextProjects.filter((item) => workingIds.has(item.id));
2809: const workingIndex = catalog.projects.findIndex((item) => item.id === project.id);
2810: const insertAt = Math.max(0, workingIndex);
2811: savedPortfolioCatalog.projects.splice(insertAt, 0, parsedProject);
2812: markDraftNeedsSave();
2813: refreshOpenPreviews();
2814: setStatus(`${project.title} saved into the portfolio preview.`);

```

removeProjectFromPortfolio - lines 2819-2821

Item	Explanation
Source location	Lines 2819-2821 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function removeProjectFromPortfolio(projectId) {
Touches	Mostly local calculations or local UI state.
Calls detected	filter
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2819: function removeProjectFromPortfolio(projectId) {
2820:   savedPortfolioCatalog.projects = (savedPortfolioCatalog.projects || []).filter((item) => item.id !==
projectId);
2821: }

```

deleteProjectById - lines 2823-2836

Item	Explanation
Source location	Lines 2823-2836 (14 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function deleteProjectById(projectId) {
Touches	browser/Electron DOM
Calls detected	find, filter, removeProjectFromPortfolio, closeDialogElement, setStatus, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2823: function deleteProjectById(projectId) {
2824:   const project = catalog.projects.find((item) => item.id === projectId);
2825:   if (!project) return;
2826:   catalog.projects = catalog.projects.filter((item) => item.id !== project.id);
2827:   removeProjectFromPortfolio(project.id);
2828:   selectedProjectId = catalog.projects[0]?.id || "";
2829:   activeSectionId = "brief";
2830:   if (projectDialog?.open && project.id === projectWindowTitle.dataset.projectId) {

```

```

2831:     closeModalElement(projectDialog);
2832:   }
2833:   setStatus("Project deleted locally.");
2834:   scheduleAutosave();
2835:   renderAll();
2836: }

```

openDeleteConfirm - lines 2838-2843

Item	Explanation
Source location	Lines 2838-2843 (6 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function openDeleteConfirm({ title = "Delete project", message, onConfirm }) {
Touches	filesystem
Calls detected	showModal
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2838: function openDeleteConfirm({ title = "Delete project", message, onConfirm }) {
2839:   pendingDeleteAction = onConfirm;
2840:   deleteConfirmTitle.textContent = title;
2841:   deleteConfirmMessage.textContent = message;
2842:   deleteConfirmDialog.showModal();
2843: }

```

requestProjectDelete - lines 2845-2853

Item	Explanation
Source location	Lines 2845-2853 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function requestProjectDelete(projectId) {
Touches	filesystem
Calls detected	find, openDeleteConfirm, deleteProjectById
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2845: function requestProjectDelete(projectId) {
2846:   const project = catalog.projects.find((item) => item.id === projectId);
2847:   if (!project) return;
2848:   openDeleteConfirm({
2849:     title: "Delete project",
2850:     message: "Are you sure you want to delete this project?",
2851:     onConfirm: () => deleteProjectById(project.id)
2852:   });
2853: }

```

parserItem - lines 2855-2866

Item	Explanation
Source location	Lines 2855-2866 (12 source lines).
Purpose	Part of the private builder frontend and editing experience.

Item	Explanation
Declaration	function parserItem(title, description = "", url = "", meta = "", kind = "text", rich = null, children = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	filter, clone
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2855: function parserItem(title, description = "", url = "", meta = "", kind = "text", rich = null, children =
[]) {
2856:   const item = {
2857:     children: children.filter((child) => child && (child.title || child.description || child.url ||
child.children?.length)),
2858:     description: description || "",
2859:     kind,
2860:     meta: meta || "",
2861:     title: title || "Untitled item",
2862:     url: url || ""
2863:   };
2864:   if (rich?.blocks?.length) item.rich = clone(rich);
2865:   return item;
2866: }

```

canonicalTemplateId - lines 2868-2870

Item	Explanation
Source location	Lines 2868-2870 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function canonicalTemplateId(id) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2868: function canonicalTemplateId(id) {
2869:   return legacyTemplateSkins[id] || id || "";
2870: }

```

projectTemplateId - lines 2872-2874

Item	Explanation
Source location	Lines 2872-2874 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function projectTemplateId(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	canonicalTemplateId
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2872: function projectTemplateId(project) {
2873:   return canonicalTemplateId(project?.portfolioView?.template?.id || project?.templateId || "");
2874: }
```

projectTemplateFor - lines 2876-2879

Item	Explanation
Source location	Lines 2876-2879 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function projectTemplateFor(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	projectTemplateId, find
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2876: function projectTemplateFor(project) {
2877:   const templateId = projectTemplateId(project);
2878:   return templates.find((template) => template.id === templateId) || {};
2879: }
```

projectTemplateClass - lines 2881-2884

Item	Explanation
Source location	Lines 2881-2884 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function projectTemplateClass(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	projectTemplateId
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2881: function projectTemplateClass(project) {
2882:   const templateId = projectTemplateId(project);
2883:   return templateId ? `project-template project-template-${templateId}` : "project-template-white";
2884: }
```

responsiveClassName - lines 2886-2891

Item	Explanation
Source location	Lines 2886-2891 (6 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function responsiveClassName(value, fallback) {

Item	Explanation
Touches	Mostly local calculations or local UI state.
Calls detected	toLowerCase, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2886: function responsiveClassName(value, fallback) {
2887:   return String(value || fallback || "")
2888:     .toLowerCase()
2889:     .replace(/^[^a-z0-9]+/g, "-")
2890:     .replace(/^-|-$/g, "") || fallback;
2891: }

```

projectResponsiveClass - lines 2893-2896

Item	Explanation
Source location	Lines 2893-2896 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function projectResponsiveClass(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	projectResponsiveProfile, responsiveClassName
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2893: function projectResponsiveClass(project) {
2894:   const profile = projectResponsiveProfile(project);
2895:   return `responsive-project responsive-density-${responsiveClassName(profile.density, "balanced")}
responsive-card-${responsiveClassName(profile.cardLayout, "single")}`;
2896: }

```

responsiveStyleValues - lines 2898-2906

Item	Explanation
Source location	Lines 2898-2906 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function responsiveStyleValues(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	projectResponsiveProfile
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2898: function responsiveStyleValues(project) {
2899:   const profile = projectResponsiveProfile(project);

```

```

2900:   return {
2901:     "--responsive-section-card-min": profile.sectionCardMin,
2902:     "--responsive-content-max": profile.contentMax,
2903:     "--responsive-media-max": profile.mediaMax,
2904:     "--responsive-touch-target": profile.touchTarget
2905:   };
2906: }

```

hasPublicTemplate - lines 2908-2910

Item	Explanation
Source location	Lines 2908-2910 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function hasPublicTemplate(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	projectTemplateId
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2908: function hasPublicTemplate(project) {
2909:   return Boolean(projectTemplateId(project));
2910: }

```

projectTemplateVisual - lines 2912-2914

Item	Explanation
Source location	Lines 2912-2914 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function projectTemplateVisual(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	projectTemplateFor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2912: function projectTemplateVisual(project) {
2913:   return project?.portfolioView?.template?.visual || project?.templateVisual ||
projectTemplateFor(project).visual || null;
2914: }

```

projectTemplateStyle - lines 2916-2934

Item	Explanation
Source location	Lines 2916-2934 (19 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function projectTemplateStyle(project, accent) {
Touches	Mostly local calculations or local UI state.
Calls detected	projectTemplateVisual, responsiveStyleValues, entries, filter, map, join

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2916: function projectTemplateStyle(project, accent) {
2917:   const visual = projectTemplateVisual(project) || {};
2918:   const values = {
2919:     "--accent": accent,
2920:     "--template-bg": visual.background,
2921:     "--template-panel": visual.panel,
2922:     "--template-accent": visual.accent,
2923:     "--template-hover": visual.hover,
2924:     "--template-click": visual.click,
2925:     "--template-click-text": visual.clickText,
2926:     "--template-line": visual.line,
2927:     "--template-text": visual.text,
2928:     ...responsiveStyleValues(project)
2929:   };
2930:   return Object.entries(values)
2931:     .filter(([, value]) => value)
2932:     .map(([key, value]) => `${key}: ${value}`)
2933:     .join("; ");
2934: }

```

applyProjectTemplateToElement - lines 2936-2970

Item	Explanation
Source location	Lines 2936-2970 (35 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function applyProjectTemplateToElement(element, project, accent) {
Touches	browser/Electron DOM
Calls detected	filter, startsWith, forEach, remove, projectTemplateClass, split, add, projectResponsiveClass, projectTemplateVisual, responsiveStyleValues, entries, setProperty
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2936: function applyProjectTemplateToElement(element, project, accent) {
2937:   if (!element) return;
2938:   [...element.classList]
2939:     .filter((className) =>
2940:       className === "project-template" ||
2941:       className === "project-template-white" ||
2942:       className === "responsive-project" ||
2943:       className.startsWith("project-template-") ||
2944:       className.startsWith("responsive-density-") ||
2945:       className.startsWith("responsive-card-")
2946:     )
2947:     .forEach((className) => element.classList.remove(className));
2948:   projectTemplateClass(project).split(" ").forEach((className) => element.classList.add(className));
2949:   projectResponsiveClass(project).split(" ").forEach((className) => element.classList.add(className));
2950:   const visual = projectTemplateVisual(project) || {};
2951:   const values = {
2952:     "--accent": accent,
2953:     "--template-bg": visual.background,
2954:     "--template-panel": visual.panel,
2955:     "--template-accent": visual.accent,
2956:     "--template-hover": visual.hover,
2957:     "--template-click": visual.click,
2958:     "--template-click-text": visual.clickText,
2959:     "--template-line": visual.line,
2960:     "--template-text": visual.text,
2961:     ...responsiveStyleValues(project)
2962:   };

```

```

2963:   Object.entries(values).forEach(([key, value]) => {
2964:     if (value) {
2965:       element.style.setProperty(key, value);
2966:     } else {
2967:       element.style.removeProperty(key);
2968:     }
2969:   });
2970: }

```

resourcesFrom - lines 2972-2988

Item	Explanation
Source location	Lines 2972-2988 (17 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function resourcesFrom(project, key) {
Touches	Mostly local calculations or local UI state.
Calls detected	map, parserItem
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 6 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2972: function resourcesFrom(project, key) {
2973:   return (project[key] || []).map((item) => {
2974:     if (key === "tests") {
2975:       return parserItem(item.name, item.result || item.method, item.artifact, item.method || item.status
|| "Test", "test", item.richDescription);
2976:     }
2977:     if (key === "pcbs") {
2978:       return parserItem(item.name, item.revision || item.status, item.artifact, item.status || "PCB",
"pcb", item.richDescription);
2979:     }
2980:     if (key === "media") {
2981:       return parserItem(item.title, item.caption, item.url, item.status || "Image", "image",
item.richDescription);
2982:     }
2983:     if (key === "links") {
2984:       return parserItem(item.label, item.description || "", item.url, "Link", "link",
item.richDescription);
2985:     }
2986:     return parserItem(item.title || item.name || item.label, item.description || item.caption, item.url
|| item.artifact, item.type || item.status || "Document", "document", item.richDescription);
2987:   });
2988: }

```

resourcesFromItems - lines 2990-3003

Item	Explanation
Source location	Lines 2990-3003 (14 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function resourcesFromItems(items = [], fallbackKind = "document") {
Touches	Mostly local calculations or local UI state.
Calls detected	map, parserItem
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2990: function resourcesFromItems(items = [], fallbackKind = "document") {
2991:   return (items || []).map((item) => {

```

```

2992:     const url = item.url || item.artifact || "";
2993:     const kind = item.kind || fallbackKind;
2994:     return parserItem(
2995:         item.title || item.name || item.label || "Untitled item",
2996:         item.description || item.caption || item.result || item.method || "",
2997:         url,
2998:         item.type || item.status || item.method || "",
2999:         kind,
3000:         item.richDescription
3001:     );
3002: });
3003: }

```

parsedSubsection - lines 3005-3007

Item	Explanation
Source location	Lines 3005-3007 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function parsedSubsection(title, description = "", items = [], rich = null) {
Touches	Mostly local calculations or local UI state.
Calls detected	parserItem
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3005: function parsedSubsection(title, description = "", items = [], rich = null) {
3006:   return parserItem(title, description, "", "Subsection", "subsection", rich, items);
3007: }

```

richHasContent - lines 3009-3018

Item	Explanation
Source location	Lines 3009-3018 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function richHasContent(rich) {
Touches	filesystem
Calls detected	some
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3009: function richHasContent(rich) {
3010:   return Boolean(rich?.blocks?.some((block) =>
3011:     block.type === "image" && block.url ||
3012:     block.type === "formula" && block.formula ||
3013:     block.type === "code" && block.code ||
3014:     block.text ||
3015:     block.title ||
3016:     block.caption
3017:   ));
3018: }

```

parsedItemHasContent - lines 3020-3031

Item	Explanation
Source location	Lines 3020-3031 (12 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function parsedItemHasContent(item) {
Touches	Mostly local calculations or local UI state.
Calls detected	richHasContent, some, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 5 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3020: function parsedItemHasContent(item) {
3021:   if (!item) return false;
3022:   if (item.kind === "summary") return Boolean(item.description || richHasContent(item.rich));
3023:   const children = item.children || item.items || [];
3024:   if (item.kind === "subsection") {
3025:     return Boolean(item.description || item.url || richHasContent(item.rich) ||
children.some(parsedItemHasContent));
3026:   }
3027:   if (["tool", "language"].includes(item.kind)) {
3028:     return Boolean(item.title || item.description || item.url || richHasContent(item.rich) ||
children.some(parsedItemHasContent));
3029:   }
3030:   return Boolean(item.description || item.url || richHasContent(item.rich) ||
children.some(parsedItemHasContent));
3031: }

```

parsedSectionHasContent - lines 3033-3035

Item	Explanation
Source location	Lines 3033-3035 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function parsedSectionHasContent(section) {
Touches	Mostly local calculations or local UI state.
Calls detected	richHasContent, some
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3033: function parsedSectionHasContent(section) {
3034:   return Boolean(section?.description || richHasContent(section?.rich) || (section?.items ||
[]).some(parsedItemHasContent));
3035: }

```

responsiveChildren - lines 3037-3039

Item	Explanation
Source location	Lines 3037-3039 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function responsiveChildren(node) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
3037: function responsiveChildren(node) {
3038:   return node?.items || node?.children || [];
3039: }
```

richContainsMedia - lines 3041-3043

Item	Explanation
Source location	Lines 3041-3043 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function richContainsMedia(rich) {
Touches	Mostly local calculations or local UI state.
Calls detected	some
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
3041: function richContainsMedia(rich) {
3042:   return Boolean(rich?.blocks?.some((block) => block.type === "image" && block.url));
3043: }
```

responsiveNodeHasMedia - lines 3045-3051

Item	Explanation
Source location	Lines 3045-3051 (7 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function responsiveNodeHasMedia(node) {
Touches	Mostly local calculations or local UI state.
Calls detected	richContainsMedia, responsiveChildren, some
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
3045: function responsiveNodeHasMedia(node) {
3046:   if (!node) return false;
3047:   if (node.kind === "image" || richContainsMedia(node.rich)) return true;
3048:   const url = String(node.url || "");
3049:   if (/\. (apng|avif|gif|jpe?g|png|svg|webp)(\?|#|$)/i.test(url)) return true;
3050:   return responsiveChildren(node).some(responsiveNodeHasMedia);
3051: }
```

responsiveNodeCount - lines 3053-3056

Item	Explanation
Source location	Lines 3053-3056 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function responsiveNodeCount(node) {
Touches	Mostly local calculations or local UI state.
Calls detected	parsedItemHasContent, responsiveChildren, reduce
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3053: function responsiveNodeCount(node) {
3054:   if (!node || !parsedItemHasContent(node)) return 0;
3055:   return 1 + responsiveChildren(node).reduce((count, child) => count + responsiveNodeCount(child), 0);
3056: }

```

responsiveNodeDepth - lines 3058-3063

Item	Explanation
Source location	Lines 3058-3063 (6 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function responsiveNodeDepth(node) {
Touches	Mostly local calculations or local UI state.
Calls detected	parsedItemHasContent, responsiveChildren, filter, max, map
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3058: function responsiveNodeDepth(node) {
3059:   if (!node || !parsedItemHasContent(node)) return 0;
3060:   const children = responsiveChildren(node).filter(parsedItemHasContent);
3061:   if (!children.length) return 1;
3062:   return 1 + Math.max(...children.map(responsiveNodeDepth));
3063: }

```

buildResponsiveProfile - lines 3065-3096

Item	Explanation
Source location	Lines 3065-3096 (32 source lines).
Purpose	Creates an object, UI structure, process, payload, or generated artifact.
Declaration	function buildResponsiveProfile(project, parsedSections = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	filter, parsedSectionHasContent, reduce, responsiveNodeCount, max, map, some, responsiveNodeHasMedia, hasPublicTemplate
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3065: function buildResponsiveProfile(project, parsedSections = []) {
3066:   const visibleSections = (parsedSections || []).filter((section) => section?.id !== "brief" &&
parsedSectionHasContent(section));
3067:   const itemCount = visibleSections.reduce(
3068:     (count, section) => count + (section.items || []).reduce((sum, item) => sum +
responsiveNodeCount(item), 0),
3069:     0
3070:   );
3071:   const maxDepth = visibleSections.reduce(
3072:     (depth, section) => Math.max(depth, ...(section.items || []).map(responsiveNodeDepth), 1),
3073:     1
3074:   );
3075:   const hasMedia = visibleSections.some((section) => responsiveNodeHasMedia(section));
3076:   const density = itemCount >= 18 || visibleSections.length >= 6 || maxDepth >= 4
3077:     ? "dense"
3078:     : itemCount <= 6 && visibleSections.length <= 2
3079:     ? "simple"
3080:     : "balanced";
3081:   return {
3082:     version: 1,
3083:     breakpoints: {
3084:       mobile: 640,
3085:       tablet: 900,
3086:       desktop: 1180
3087:     },
3088:     cardLayout: hasPublicTemplate(project) ? "single" : "split",
3089:     contentMax: hasMedia || maxDepth >= 3 ? "1180px" : "980px",
3090:     density,
3091:     mediaMax: hasMedia ? "860px" : "720px",
3092:     sectionCardMin: density === "dense" ? "210px" : density === "simple" ? "280px" : "240px",
3093:     touchTarget: "44px",
3094:     windowMode: "full-screen"
3095:   };
3096: }

```

visibleSections - lines 3066-3095

Item	Explanation
Source location	Lines 3066-3095 (30 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const visibleSections = (parsedSections []).filter((section) => section?.id !== "brief" && parsedSectionHasContent(section));
Touches	Mostly local calculations or local UI state.
Calls detected	filter, parsedSectionHasContent, reduce, responsiveNodeCount, max, map, some, responsiveNodeHasMedia, hasPublicTemplate
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3066:   const visibleSections = (parsedSections || []).filter((section) => section?.id !== "brief" &&
parsedSectionHasContent(section));
3067:   const itemCount = visibleSections.reduce(
3068:     (count, section) => count + (section.items || []).reduce((sum, item) => sum +
responsiveNodeCount(item), 0),
3069:     0
3070:   );
3071:   const maxDepth = visibleSections.reduce(
3072:     (depth, section) => Math.max(depth, ...(section.items || []).map(responsiveNodeDepth), 1),
3073:     1
3074:   );
3075:   const hasMedia = visibleSections.some((section) => responsiveNodeHasMedia(section));
3076:   const density = itemCount >= 18 || visibleSections.length >= 6 || maxDepth >= 4
3077:     ? "dense"
3078:     : itemCount <= 6 && visibleSections.length <= 2
3079:     ? "simple"
3080:     : "balanced";

```

```

3081:   return {
3082:     version: 1,
3083:     breakpoints: {
3084:       mobile: 640,
3085:       tablet: 900,
3086:       desktop: 1180
3087:     },
3088:     cardLayout: hasPublicTemplate(project) ? "single" : "split",
3089:     contentMax: hasMedia || maxDepth >= 3 ? "1180px" : "980px",
3090:     density,
3091:     mediaMax: hasMedia ? "860px" : "720px",
3092:     sectionCardMin: density === "dense" ? "210px" : density === "simple" ? "280px" : "240px",
3093:     touchTarget: "44px",
3094:     windowMode: "full-screen"
3095:   };

```

projectResponsiveProfile - lines 3098-3107

Item	Explanation
Source location	Lines 3098-3107 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function projectResponsiveProfile(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	buildResponsiveProfile
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3098: function projectResponsiveProfile(project) {
3099:   const savedProfile = project?.portfolioView?.responsive;
3100:   if (savedProfile?.version) {
3101:     return {
3102:       ...buildResponsiveProfile(project, project?.portfolioView?.sections || []),
3103:       ...savedProfile
3104:     };
3105:   }
3106:   return buildResponsiveProfile(project, project?.portfolioView?.sections || []);
3107: }

```

parsedSection - lines 3109-3118

Item	Explanation
Source location	Lines 3109-3118 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function parsedSection(id, title, items = [], description = "", rich = null) {
Touches	Mostly local calculations or local UI state.
Calls detected	filter, richHasContent, clone
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3109: function parsedSection(id, title, items = [], description = "", rich = null) {
3110:   const section = {
3111:     description,
3112:     id,
3113:     items: items.filter(parsedItemHasContent),
3114:     title
3115:   };

```



```

3116:   if (richHasContent(rich)) section.rich = clone(rich);
3117:   return section;
3118: }

```

parsedDesignSection - lines 3120-3146

Item	Explanation
Source location	Lines 3120-3146 (27 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function parsedDesignSection(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	isAnalogProject, ensureDesignModel, resourcesFrom, resourcesFromItems, parsedSubsection, parsedSection
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3120: function parsedDesignSection(project) {
3121:   if (isAnalogProject(project)) {
3122:     const design = ensureDesignModel(project);
3123:     const documentationItems = [
3124:       ...resourcesFrom(project, "documents"),
3125:       ...resourcesFromItems(design.documentation.files, "document"),
3126:       parsedSubsection("References", "", resourcesFromItems(design.documentation.references,
"reference")),
3127:       parsedSubsection("Math / Analysis", "", resourcesFromItems(design.documentation.mathAnalysis,
"analysis"))
3128:     ];
3129:     const boardMediaItems = [
3130:       ...resourcesFrom(project, "pcbs"),
3131:       ...resourcesFrom(project, "media")
3132:     ];
3133:     return parsedSection("design", "Design", [
3134:       parsedSubsection("Design Overview", design.brief.summary || "", [
3135:         ...resourcesFromItems(design.brief.files, "document")
3136:       ], design.brief.summaryRich),
3137:       parsedSubsection("Documentation", design.documentation.summary || "", documentationItems,
design.documentation.summaryRich),
3138:       parsedSubsection("PCB and Visual Evidence", "", boardMediaItems)
3139:     ]);
3140:   }
3141:   return parsedSection("design", "Design", [
3142:     ...resourcesFrom(project, "documents"),
3143:     ...resourcesFrom(project, "pcbs"),
3144:     ...resourcesFrom(project, "media")
3145:   ]);
3146: }

```

parsedSimulationSection - lines 3148-3156

Item	Explanation
Source location	Lines 3148-3156 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function parsedSimulationSection(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	isAnalogProject, ensureDesignModel, parsedSection, parsedSubsection, resourcesFromItems
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3148: function parsedSimulationSection(project) {
3149:   if (!isAnalogProject(project)) return null;
3150:   const design = ensureDesignModel(project);
3151:   return parsedSection("simulation", "Simulation", [
3152:     parsedSubsection("Simulation Overview", design.simulation.summary || "", [],
design.simulation.summaryRich),
3153:     parsedSubsection("Simulation Files", "", resourcesFromItems(design.simulation.files, "simulation-
file")),
3154:     parsedSubsection("Simulation Results", "", resourcesFromItems(design.simulation.results, "simulation-
result"))
3155:   ]);
3156: }

```

parsedToolsSection - lines 3158-3169

Item	Explanation
Source location	Lines 3158-3169 (12 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function parsedToolsSection(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	map, parserItem, toolName, toolDescription, itemTitle, parsedSection, resourcesFrom
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3158: function parsedToolsSection(project) {
3159:   const toolItems = (project.tools || []).map((tool) => parserItem(toolName(tool), toolDescription(tool),
"", "Tool", "tool", tool?.richDescription));
3160:   const languageItems = (project.languages || []).map((language) => {
3161:     const label = typeof language === "string" ? language : itemTitle(language);
3162:     return parserItem(label, "", "", "Language", "language");
3163:   });
3164:   return parsedSection("tools", "Tools", [
3165:     ...toolItems,
3166:     ...languageItems,
3167:     ...resourcesFrom(project, "links")
3168:   ]);
3169: }

```

toolItems - lines 3159-3163

Item	Explanation
Source location	Lines 3159-3163 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const toolItems = (project.tools []).map((tool) => parserItem(toolName(tool), toolDescription(tool), "", "Tool", "tool", tool?.richDescription));
Touches	Mostly local calculations or local UI state.
Calls detected	map, parserItem, toolName, toolDescription, itemTitle
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3159:   const toolItems = (project.tools || []).map((tool) => parserItem(toolName(tool), toolDescription(tool),
"", "Tool", "tool", tool?.richDescription));
3160:   const languageItems = (project.languages || []).map((language) => {
3161:     const label = typeof language === "string" ? language : itemTitle(language);
3162:     return parserItem(label, "", "", "Language", "language");
3163:   });

```

languageItems - lines 3160-3163

Item	Explanation
Source location	Lines 3160-3163 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const languageItems = (project.languages []).map((language) => {
Touches	Mostly local calculations or local UI state.
Calls detected	map, itemTitle, parserItem
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
3160:   const languageItems = (project.languages || []).map((language) => {
3161:     const label = typeof language === "string" ? language : itemTitle(language);
3162:     return parserItem(label, "", "", "Language", "language");
3163:   });
```

parseProjectForPortfolio - lines 3171-3219

Item	Explanation
Source location	Lines 3171-3219 (49 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function parseProjectForPortfolio(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	projectTemplateFor, sectionOptions, flatMap, parsedSection, parserItem, parsedDesignSection, parsedSimulationSection, resourcesFrom, parsedToolsSection, startsWith, replace, find
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 8 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
3171: function parseProjectForPortfolio(project) {
3172:   const template = projectTemplateFor(project);
3173:   const parsedSections = sectionOptions(project).flatMap((section) => {
3174:     if (section.id === "brief") {
3175:       return parsedSection("brief", "Overview", [
3176:         parserItem("Overview", project.summary || "", "", "Overview", "summary", project.summaryRich)
3177:       ]);
3178:     }
3179:     if (section.id === "design") return parsedDesignSection(project);
3180:     if (section.id === "simulation") return parsedSimulationSection(project);
3181:     if (section.id === "tests") return parsedSection("tests", "Tests", resourcesFrom(project, "tests"));
3182:     if (section.id === "tools") return parsedToolsSection(project);
3183:     if (section.id.startsWith("custom:")) {
3184:       const sectionId = section.id.replace("custom:", "");
3185:       const custom = (project.sections || []).find((item) => item.id === sectionId);
3186:       return parsedSection(
3187:         `custom:${sectionId}`,
3188:         custom?.title || section.label,
3189:         (custom?.items || []).map((item) => item.url
3190:           ? parserItem(item.title, item.description, item.url, item.type || item.status || "File",
3191:             "file", item.richDescription)
3192:           : parsedSubsection(
3193:             item.title,
3194:             item.description,
3195:             resourcesFromItems(item.children || item.items || item.files || [], "file"),
3196:             item.richDescription
3197:           )),
3198:         custom?.description || "",
```

```

3198:         custom?.richDescription
3199:     );
3200: }
3201: return null;
3202: }).filter(parsedSectionHasContent);
3203:
3204: return {
3205:     ...clone(project),
3206:     portfolioView: {
3207:         builtAt: new Date().toISOString(),
3208:         template: {
3209:             group: "Appearance",
3210:             id: projectTemplateId(project),
3211:             label: template.label || project.templateLabel || "",
3212:             visual: template.visual ? clone(template.visual) : projectTemplateVisual(project)
3213:         },
3214:         responsive: buildResponsiveProfile(project, parsedSections),
3215:         title: project.title,
3216:         sections: parsedSections
3217:     }
3218: };
3219: }

```

custom - lines 3185-3187

Item	Explanation
Source location	Lines 3185-3187 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	<code>const custom = (project.sections []).find((item) => item.id === sectionId);</code>
Touches	Mostly local calculations or local UI state.
Calls detected	<code>find</code> , <code>parsedSection</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3185:         const custom = (project.sections || []).find((item) => item.id === sectionId);
3186:         return parsedSection(
3187:             `custom:${sectionId}`,

```

refreshOpenPreviews - lines 3221-3230

Item	Explanation
Source location	Lines 3221-3230 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	<code>function refreshOpenPreviews() {</code>
Touches	Mostly local calculations or local UI state.
Calls detected	<code>selectedProject</code> , <code>find</code> , <code>renderProjectWebsitePreview</code> , <code>renderPreview</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3221: function refreshOpenPreviews() {
3222:     if (projectPreviewDialog?.open) {
3223:         const project = selectedProject();
3224:         const savedProject = (savedPortfolioCatalog.projects || []).find((item) => item.id === project?.id);
3225:         renderProjectWebsitePreview(savedProject || project);
3226:     }
3227:     if (portfolioPreviewDialog?.open) {

```

```

3228:     renderPreview();
3229:   }
3230: }

```

savedProject - lines 3224-3227

Item	Explanation
Source location	Lines 3224-3227 (4 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	const savedProject = (savedPortfolioCatalog.projects []).find((item) => item.id === project?.id);
Touches	Mostly local calculations or local UI state.
Calls detected	find, renderProjectWebsitePreview
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3224:     const savedProject = (savedPortfolioCatalog.projects || []).find((item) => item.id === project?.id);
3225:     renderProjectWebsitePreview(savedProject || project);
3226:   }
3227:   if (portfolioPreviewDialog?.open) {

```

saveSelectedProjectAndClose - lines 3232-3236

Item	Explanation
Source location	Lines 3232-3236 (5 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	function saveSelectedProjectAndClose() {
Touches	Mostly local calculations or local UI state.
Calls detected	saveSelectedProjectToPortfolio, closeDialogElement
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3232: function saveSelectedProjectAndClose() {
3233:   if (saveSelectedProjectToPortfolio()) {
3234:     closeDialogElement(projectDialog);
3235:   }
3236: }

```

commitActiveProjectEdits - lines 3238-3246

Item	Explanation
Source location	Lines 3238-3246 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	async function commitActiveProjectEdits() {
Touches	Mostly local calculations or local UI state.
Calls detected	saveVisibleRichEditors, saveRichSummary
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3238: async function commitActiveProjectEdits() {
3239:   saveVisibleRichEditors();
3240:
3241:   if (summaryEditorProjectId) {
3242:     return saveRichSummary();
3243:   }
3244:
3245:   return true;
3246: }

```

saveAllSections - lines 3248-3259

Item	Explanation
Source location	Lines 3248-3259 (12 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	async function saveAllSections(options = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	commitActiveProjectEdits, syncParsedPortfolioGlobalsIntoSaved, map, parseProjectForPortfolio, markDraftNeedsSave, closeDialogElement, refreshOpenPreviews, setStatus, updateBuilderWorkflow
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3248: async function saveAllSections(options = {}) {
3249:   const settings = options && options.currentTarget ? {} : options;
3250:   if (!(await commitActiveProjectEdits())) return false;
3251:   syncParsedPortfolioGlobalsIntoSaved();
3252:   savedPortfolioCatalog.projects = (catalog.projects || []).map((project) =>
parseProjectForPortfolio(project));
3253:   markDraftNeedsSave();
3254:   if (projectDialog?.open) closeDialogElement(projectDialog);
3255:   if (settings.refreshOpenPreviews !== false) refreshOpenPreviews();
3256:   setStatus(`Saved ${savedPortfolioCatalog.projects.length}
project${savedPortfolioCatalog.projects.length === 1 ? "" : "s"} into the portfolio preview.`);
3257:   updateBuilderWorkflow();
3258:   return true;
3259: }

```

selectedProject - lines 3261-3263

Item	Explanation
Source location	Lines 3261-3263 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function selectedProject() {
Touches	Mostly local calculations or local UI state.
Calls detected	find
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3261: function selectedProject() {
3262:   return catalog.projects.find((project) => project.id === selectedProjectId) || catalog.projects[0];
3263: }

```

sectionWindowId - lines 3265-3269

Item	Explanation
Source location	Lines 3265-3269 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function sectionWindowId(sectionId) {
Touches	Mostly local calculations or local UI state.
Calls detected	includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3265: function sectionWindowId(sectionId) {
3266:   if (["documents", "pcbs", "media"].includes(sectionId)) return "design";
3267:   if (["links", "highlights", "tools", "languages"].includes(sectionId)) return "tools";
3268:   return sectionId || "brief";
3269: }

```

resetProjectWindowScroll - lines 3271-3279

Item	Explanation
Source location	Lines 3271-3279 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function resetProjectWindowScroll() {
Touches	Mostly local calculations or local UI state.
Calls detected	requestAnimationFrame
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3271: function resetProjectWindowScroll() {
3272:   projectWindowContent.scrollTop = 0;
3273:   requestAnimationFrame(() => {
3274:     projectWindowContent.scrollTop = 0;
3275:     requestAnimationFrame(() => {
3276:       projectWindowContent.scrollTop = 0;
3277:     });
3278:   });
3279: }

```

openProjectWindow - lines 3281-3302

Item	Explanation
Source location	Lines 3281-3302 (22 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function openProjectWindow(projectId, sectionId = "brief") {
Touches	browser/Electron DOM
Calls detected	sectionWindowId, removeAttribute, remove, renderAll, add, showModal, resetProjectWindowScroll, updateDialogWindowButtons
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3281: function openProjectWindow(projectId, sectionId = "brief") {
3282:   selectedProjectId = projectId;
3283:   projectWindowTitle.dataset.projectId = projectId;
3284:   activeSectionId = sectionWindowId(sectionId);
3285:   projectWindowBackStack = [];
3286:   projectWindowForwardStack = [];
3287:   projectTitleMenu.hidden = true;
3288:   projectDialog.removeAttribute("style");
3289:   projectDialog.classList.remove(
3290:     "is-draggable-dialog",
3291:     "is-hidden-dialog",
3292:     "is-maximized-dialog",
3293:     "is-minimized-dialog",
3294:     "is-resized-dialog"
3295:   );
3296:   delete projectDialog.dataset.restoreWindowState;
3297:   renderAll();
3298:   document.body.classList.add("project-window-open");
3299:   if (!projectDialog.open) projectDialog.showModal();
3300:   resetProjectWindowScroll();
3301:   updateDialogWindowButtons(projectDialog);
3302: }

```

categoryById - lines 3304-3306

Item	Explanation
Source location	Lines 3304-3306 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function categoryById(id) {
Touches	Mostly local calculations or local UI state.
Calls detected	find
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3304: function categoryById(id) {
3305:   return catalog.categories.find((category) => category.id === id) || {};
3306: }

```

normalizeCategory - lines 3308-3318

Item	Explanation
Source location	Lines 3308-3318 (11 source lines).

Item	Explanation
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeCategory(category = {}) {
Touches	filesystem
Calls detected	trim, slugify, normalizeTextColor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3308: function normalizeCategory(category = {}) {
3309:   const label = String(category.label || category.title || "New category").trim() || "New category";
3310:   const id = slugify(category.id || label) || "category";
3311:   const accent = normalizeTextColor(category.accent || "") || "#1677a8";
3312:   return {
3313:     accent,
3314:     description: String(category.description || "").trim(),
3315:     id,
3316:     label
3317:   };
3318: }

```

refreshCategoryControls - lines 3320-3327

Item	Explanation
Source location	Lines 3320-3327 (8 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function refreshCategoryControls() {
Touches	filesystem
Calls detected	map, clone, escapeHtml, join, renderCategoryDropdown
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3320: function refreshCategoryControls() {
3321:   catalog.categories = (catalog.categories || []).map(normalizeCategory);
3322:   savedPortfolioCatalog.categories = clone(catalog.categories || []);
3323:   if (newCategorySelect) {
3324:     newCategorySelect.innerHTML = catalog.categories.map((category) => `<option
value="${escapeHtml(category.id)}">${escapeHtml(category.label)}</option>`).join("");
3325:   }
3326:   renderCategoryDropdown();
3327: }

```

linkAttributes - lines 3329-3332

Item	Explanation
Source location	Lines 3329-3332 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function linkAttributes(url, item = {}) {
Touches	filesystem
Calls detected	normalizeLinkTarget, isWebsiteLinkItem

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3329: function linkAttributes(url, item = {}) {
3330:   const target = normalizeLinkTarget(url, { assumeWeb: isWebsiteLinkItem(item, url) });
3331:   return /^https?:\/\//i.test(target) ? ' target="_blank" rel="noreferrer"' : "";
3332: }

```

isLocalDownloadTarget - lines 3334-3337

Item	Explanation
Source location	Lines 3334-3337 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function isLocalDownloadTarget(target = "") {
Touches	filesystem
Calls detected	normalizeLinkTarget
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3334: function isLocalDownloadTarget(target = "") {
3335:   const value = normalizeLinkTarget(target, { assumeWeb: true });
3336:   return Boolean(value) && !/^(https?:)?\/\//i.test(value) && !/^(mailto:|tel:|#)/i.test(value);
3337: }

```

downloadAttribute - lines 3339-3342

Item	Explanation
Source location	Lines 3339-3342 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function downloadAttribute(target = "", item = {}) {
Touches	filesystem
Calls detected	normalizeLinkTarget, isWebsiteLinkItem, isLocalDownloadTarget
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3339: function downloadAttribute(target = "", item = {}) {
3340:   const value = normalizeLinkTarget(target, { assumeWeb: isWebsiteLinkItem(item, target) });
3341:   return isLocalDownloadTarget(value) ? " download" : "";
3342: }

```

resourceLink - lines 3344-3353

Item	Explanation
Source location	Lines 3344-3353 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function resourceLink(item, label = item.label item.title item.name) {
Touches	filesystem
Calls detected	normalizeLinkTarget, isWebsiteLinkItem, escapeHtml, linkAttributes, downloadAttribute
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3344: function resourceLink(item, label = item.label || item.title || item.name) {
3345:   const rawTarget = item.url || item.artifact || item.href || item.file || item.path || item.src || "";
3346:   const target = normalizeLinkTarget(rawTarget, { assumeWeb: isWebsiteLinkItem(item, rawTarget) });
3347:
3348:   if (!target || item.status === "planned") {
3349:     return `<span class="resource-link muted">${label}</span>`;
3350:   }
3351:
3352:   return `<a class="resource-link" href="${escapeHtml(target)}"${linkAttributes(target,
item)}${downloadAttribute(target, item)}>${escapeHtml(label)}</a>`;
3353: }

```

pillList - lines 3355-3360

Item	Explanation
Source location	Lines 3355-3360 (6 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function pillList(items, className = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	map, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3355: function pillList(items, className = "") {
3356:   return (items || []).map((item) => {
3357:     const label = typeof item === "string" ? item : item.name || item.title || item.label || "";
3358:     return `<span class="tag ${className}">${label}</span>`;
3359:   }).join("");
3360: }

```

evidenceList - lines 3362-3365

Item	Explanation
Source location	Lines 3362-3365 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function evidenceList(items, renderItem, emptyMessage) {
Touches	Mostly local calculations or local UI state.
Calls detected	map, join
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3362: function evidenceList(items, renderItem, emptyMessage) {
3363:   if (!items || !items.length) return `

${emptyMessage}</p>`;
3364:   return `


```

detailBlock - lines 3367-3374

Item	Explanation
Source location	Lines 3367-3374 (8 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function detailBlock(title, className, content) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3367: function detailBlock(title, className, content) {
3368:   return `
3369:     <details class="${className}">
3370:       <summary>${title}</summary>
3371:       ${content}
3372:     </details>
3373:   `;
3374: }

```

mediaGrid - lines 3376-3398

Item	Explanation
Source location	Lines 3376-3398 (23 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function mediaGrid(items) {
Touches	filesystem
Calls detected	map, escapeHtml, normalizeLinkTarget, isWebsiteLinkItem, linkAttributes, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3376: function mediaGrid(items) {
3377:   if (!items || !items.length) return `
```

```

3382:         <figure>
3383:             ${item.status === "planned" ? `
3384:             <div class="media-placeholder">${item.title}</div>
3385:             ` : `
3386:             <a href="${escapeHtml(normalizeLinkTarget(item.url, { assumeWeb: isWebsiteLinkItem(item,
item.url) })))"${linkAttributes(item.url, item)}>
3387:             
3388:             </a>
3389:             `}
3390:             <figcaption>
3391:                 <strong>${escapeHtml(item.title)}</strong>
3392:                 <span>${escapeHtml(item.caption || "")}</span>
3393:             </figcaption>
3394:         </figure>
3395:     `).join("")}
3396: </div>
3397: `;
3398: }

```

itemTitle - lines 3400-3403

Item	Explanation
Source location	Lines 3400-3403 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function itemTitle(item) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3400: function itemTitle(item) {
3401:   if (typeof item === "string") return item;
3402:   return item?.title || item?.name || item?.label || "Untitled item";
3403: }

```

itemUrl - lines 3405-3407

Item	Explanation
Source location	Lines 3405-3407 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function itemUrl(item) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3405: function itemUrl(item) {
3406:   return item?.url || item?.artifact || "";
3407: }

```

itemDescription - lines 3409-3411

Item	Explanation
Source location	Lines 3409-3411 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function itemDescription(item) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
3409: function itemDescription(item) {
3410:   return item?.description || item?.caption || item?.result || item?.method || "";
3411: }
```

isImageUrl - lines 3413-3415

Item	Explanation
Source location	Lines 3413-3415 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function isImageUrl(url) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
3413: function isImageUrl(url) {
3414:   return /\.(png|jpe?g|webp|gif|svg)(\?.*)?$/i.test(url || "");
3415: }
```

portfolioLink - lines 3417-3421

Item	Explanation
Source location	Lines 3417-3421 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function portfolioLink(url, label) {
Touches	filesystem
Calls detected	normalizeLinkTarget, escapeHtml, linkAttributes, downloadAttribute
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3417: function portfolioLink(url, label) {
3418:   if (!url) return `<span>${label}</span>`;
3419:   const target = normalizeLinkTarget(url, { assumeWeb: true });
3420:   return `<a
href="${escapeHtml(target)}"${linkAttributes(target)}${downloadAttribute(target)}>${escapeHtml(label)}</a>`;
3421: }

```

renderPortfolioResourceList - lines 3423-3444

Item	Explanation
Source location	Lines 3423-3444 (22 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderPortfolioResourceList(title, items, emptyMessage) {
Touches	Mostly local calculations or local UI state.
Calls detected	map, itemTitle, itemUrl, itemDescription, portfolioLink, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3423: function renderPortfolioResourceList(title, items, emptyMessage) {
3424:   return `
3425:     <section class="portfolio-preview-section">
3426:       <h3>${title}</h3>
3427:       ${items || []}.length ? `
3428:         <div class="portfolio-resource-list">
3429:           ${items.map(item => {
3430:             const titleText = itemTitle(item);
3431:             const url = itemUrl(item);
3432:             const description = itemDescription(item);
3433:             return `
3434:               <article class="portfolio-resource">
3435:                 <strong>${portfolioLink(url, titleText)}</strong>
3436:                 ${description ? `<p>${description}</p>` : ""}
3437:               </article>
3438:             `;
3439:           }).join("")}
3440:         </div>
3441:       ` : `<p class="evidence-empty">${emptyMessage}</p>`;
3442:     </section>
3443:   `;
3444: }

```

renderPortfolioMedia - lines 3446-3468

Item	Explanation
Source location	Lines 3446-3468 (23 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderPortfolioMedia(items) {
Touches	Mostly local calculations or local UI state.
Calls detected	map, itemUrl, isImageUrl, itemTitle, itemDescription, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3446: function renderPortfolioMedia(items) {
3447:   return `
3448:     <section class="portfolio-preview-section">
3449:       <h3>Images</h3>

```

```

3450:         ${items || []}.length ? `
3451:         <div class="portfolio-media-grid">
3452:             ${items.map((item) => {
3453:                 const url = itemUrl(item);
3454:                 return `
3455:                 <figure>
3456:                     ${isImageUrl(url) ? `` : `<div class="media-
placeholder">${itemTitle(item)}</div>`}
3457:                     <figcaption>
3458:                         <strong>${itemTitle(item)}</strong>
3459:                         ${itemDescription(item) ? `<span>${itemDescription(item)}</span>` : ""}
3460:                     </figcaption>
3461:                 </figure>
3462:                 `;
3463:             }).join("")}
3464:         </div>
3465:         : `<p class="evidence-empty">No images have been added yet.</p>`
3466:     </section>
3467: `;
3468: }

```

renderPortfolioTools - lines 3470-3486

Item	Explanation
Source location	Lines 3470-3486 (17 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderPortfolioTools(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	map, toolName, toolDescription, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3470: function renderPortfolioTools(project) {
3471:     return `
3472:     <section class="portfolio-preview-section">
3473:         <h3>Tools</h3>
3474:         ${project.tools || []}.length ? `
3475:         <div class="portfolio-tool-list">
3476:             ${project.tools.map((tool) => `
3477:             <article>
3478:                 <strong>${toolName(tool)}</strong>
3479:                 ${toolDescription(tool) ? `<p>${toolDescription(tool)}</p>` : ""}
3480:             </article>
3481:             `).join("")}
3482:         </div>
3483:         : `<p class="evidence-empty">No tools have been added yet.</p>`
3484:     </section>
3485: `;
3486: }

```

renderPortfolioCustomSections - lines 3488-3505

Item	Explanation
Source location	Lines 3488-3505 (18 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderPortfolioCustomSections(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	map, portfolioLink, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3488: function renderPortfolioCustomSections(project) {
3489:   return (project.sections || []).map((section) => `
3490:     <section class="portfolio-preview-section">
3491:       <h3>${section.title}</h3>
3492:       ${section.description ? `<p>${section.description}</p>` : ""}
3493:       ${section.items || []}.length ? `
3494:       <div class="portfolio-resource-list">
3495:         ${section.items.map((item) => `
3496:           <article class="portfolio-resource">
3497:             <strong>${portfolioLink(item.url, item.title || "Untitled subsection")}</strong>
3498:             ${item.description ? `<p>${item.description}</p>` : ""}
3499:           </article>
3500:         `).join("")}
3501:       </div>
3502:       ` : `<p class="evidence-empty">No content has been added yet.</p>`}
3503:     </section>
3504:   `).join("");
3505: }

```

publicCustomSectionBlocks - lines 3507-3518

Item	Explanation
Source location	Lines 3507-3518 (12 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function publicCustomSectionBlocks(project) {
Touches	filesystem
Calls detected	map, detailBlock, renderMultilineInlineText, evidenceList, resourceLink, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3507: function publicCustomSectionBlocks(project) {
3508:   return (project.sections || []).map((section) => detailBlock(section.title, "evidence-block evidence-
wide", `
3509:     ${section.description ? `<p class="evidence-
empty">${renderMultilineInlineText(section.description)}</p>` : ""}
3510:     ${evidenceList(section.items || [], (item) => `
3511:       <li>
3512:         ${item.url ? resourceLink(item, item.title) : `<strong>${item.title}</strong>`}
3513:         <span>${item.type || "Section item"} &middot; ${item.status || "tracked"}</span>
3514:         ${item.description ? `<p>${renderMultilineInlineText(item.description)}</p>` : ""}
3515:       </li>
3516:     `, "No content has been added yet.")).join("")}
3517:   `).join("");
3518: }

```

renderProjectPortfolioPreview - lines 3520-3556

Item	Explanation
Source location	Lines 3520-3556 (37 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderProjectPortfolioPreview(project) {
Touches	filesystem
Calls detected	categoryById, find, filter, previewSectionHasRenderableContent, projectTemplateClass, projectResponsiveClass, projectTemplateStyle, renderParsedBriefBlock, map, parsedPublicSection, join, renderMultilineInlineText
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3520: function renderProjectPortfolioPreview(project) {
3521:   if (!project) return `
```

briefSection - lines 3525-3528

Item	Explanation
Source location	Lines 3525-3528 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const briefSection = (project.portfolioView.sections []).find((section) => section.id === "brief");
Touches	Mostly local calculations or local UI state.
Calls detected	find, filter, previewSectionHasRenderableContent, projectTemplateClass, projectResponsiveClass, projectTemplateStyle
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3525:     const briefSection = (project.portfolioView.sections || []).find((section) => section.id ===
"brief");
3526:     const otherSections = (project.portfolioView.sections || []).filter((section) => section.id !==
"brief" && previewSectionHasRenderableContent(section));

```

```

3527:     return `
3528:       <article class="portfolio-preview-article ${projectTemplateClass(project)}
${projectResponsiveClass(project)}" style="${projectTemplateStyle(project, accent)}">

```

otherSections - lines 3526-3528

Item	Explanation
Source location	Lines 3526-3528 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const otherSections = (project.portfolioView.sections []).filter((section) => section.id !== "brief" && previewSectionHasRenderableContent(section));
Touches	Mostly local calculations or local UI state.
Calls detected	filter, previewSectionHasRenderableContent, projectTemplateClass, projectResponsiveClass, projectTemplateStyle
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3526:     const otherSections = (project.portfolioView.sections || []).filter((section) => section.id !==
"brief" && previewSectionHasRenderableContent(section));
3527:     return `
3528:       <article class="portfolio-preview-article ${projectTemplateClass(project)}
${projectResponsiveClass(project)}" style="${projectTemplateStyle(project, accent)}">

```

fullProjectPreviewHtmlExact - lines 3558-3685

Item	Explanation
Source location	Lines 3558-3685 (128 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function fullProjectPreviewHtmlExact(project) {
Touches	filesystem
Calls detected	parseProjectForPortfolio, categoryById, normalizeCategory, displayTitle, parsedPortfolioGlobals, stringify, replaceAll, escapeHtml, var, min, calc, media
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3558: function fullProjectPreviewHtmlExact(project) {
3559:   const baseHref = `${window.location.origin}/`;
3560:   if (!project) {
3561:     return `<!DOCTYPE html>
3562: <html lang="en">
3563: <head>
3564:   <meta charset="UTF-8" />
3565:   <meta name="viewport" content="width=device-width, initial-scale=1.0" />
3566:   <base href="${baseHref}" />
3567:   <link rel="stylesheet" href="styles.css" />
3568:   <title>Project preview</title>
3569: </head>
3570: <body class="project-isolated-preview">
3571:   <main class="project-isolated-shell">
3572:     <div class="empty-state">
3573:       <h3>No saved project preview</h3>
3574:       <p>Click Save project before viewing the parsed project preview.</p>
3575:     </div>
3576:   </main>
3577: </body>
3578: </html>`;
3579:   }

```

```

3580: const parsedProject = project.portfolioView ? project : parseProjectForPortfolio(project);
3581: const categorySource = categoryById(parsedProject.category);
3582: const previewCategory = normalizeCategory(categorySource.id ? categorySource : {
3583:   id: parsedProject.category || "project",
3584:   label: displayTitle(parsedProject.category || "Project", "Project")
3585: });
3586: const parsedGlobals = parsedPortfolioGlobals();
3587: const previewDataObject = {
3588:   categories: [previewCategory],
3589:   fieldStyles: parsedGlobals.fieldStyles,
3590:   funFacts: [],
3591:   funFactsRich: null,
3592:   profile: parsedGlobals.profile,
3593:   profileRich: parsedGlobals.profileRich,
3594:   projects: [parsedProject],
3595:   siteContent: parsedGlobals.siteContent,
3596:   siteContentRich: parsedGlobals.siteContentRich,
3597:   siteSections: []
3598: };
3599: const previewData = JSON.stringify(previewDataObject).replaceAll("</", "<\\\/");
3600: const title = parsedProject.portfolioView?.title || parsedProject.title || "Project preview";
3601: return `<!DOCTYPE html>
3602: <html lang="en">
3603: <head>
3604:   <meta charset="UTF-8" />
3605:   <meta name="viewport" content="width=device-width, initial-scale=1.0" />
3606:   <base href="${baseHref}" />
3607:   <title>${escapeHtml(title)} | Project preview</title>
3608:   <link rel="stylesheet" href="styles.css" />
3609:   <style>
3610:     body.project-isolated-preview {
3611:       min-height: 100vh;
3612:       margin: 0;
3613:       background: var(--paper, #f8fbff);
3614:       color: var(--ink, #172636);
3615:     }
3616:
3617:     .project-isolated-shell {
3618:       width: min(1180px, calc(100vw - 32px));
3619:       margin: 0 auto;
3620:       padding: 28px 0 36px;
3621:     }
3622:
3623:     .project-isolated-support {
3624:       display: none !important;
3625:     }
3626:
3627:     body.project-isolated-preview .project-grid {
3628:       display: block;
3629:     }
3630:
3631:     body.project-isolated-preview .category-section {
3632:       padding: 0;
3633:       border: 0;
3634:       background: transparent;
3635:     }
3636:
3637:     body.project-isolated-preview .category-heading,
3638:     body.project-isolated-preview .category-description {
3639:       display: none;
3640:     }
3641:
3642:     body.project-isolated-preview .category-projects {
3643:       display: block;
3644:     }
3645:
3646:     body.project-isolated-preview .project-card {
3647:       margin: 0;
3648:     }
3649:
3650:     @media (max-width: 720px) {
3651:       .project-isolated-shell {
3652:         width: min(100vw - 20px, 100%);
3653:         padding: 14px 0 24px;
3654:       }
3655:     }
3656:   </style>
3657: </head>
3658: <body class="project-isolated-preview">
3659:   <main class="project-isolated-shell" aria-label="Isolated project preview">
3660:     <div class="project-isolated-support" aria-hidden="true">
3661:       <label class="search-wrap" for="project-search">
3662:         <span>Search</span>
3663:         <input id="project-search" type="search" tabindex="-1" />
3664:       </label>
3665:       <div id="project-filters">
3666:         <button class="filter-button active" type="button" data-filter="all" tabindex="-1">All</button>
3667:       </div>
3668:       <span id="project-count">0</span>
3669:       <span id="project-track-count">0 Tracks</span>
3670:       <span id="project-track-labels">project categories</span>
3671:       <span id="year">${new Date().getFullYear()}</span>

```

```

3672:         </div>
3673:
3674:         <section class="section project-isolated-section" id="projects" aria-labelledby="project-isolated-
title">
3675:             <h1 class="sr-only" id="project-isolated-title">${escapeHtml(title)}</h1>
3676:             <div class="project-grid" id="project-grid" aria-live="polite"></div>
3677:         </section>
... excerpt truncated after 120 lines. Open the source file for the rest of this function.

```

renderProjectWebsitePreview - lines 3687-3705

Item	Explanation
Source location	Lines 3687-3705 (19 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderProjectWebsitePreview(project) {
Touches	browser/Electron DOM
Calls detected	categoryById, applyProjectTemplateToElement, updateProjectPreviewNavigation, add, escapeHtml, querySelector, fullProjectPreviewHtmlExact
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3687: function renderProjectWebsitePreview(project) {
3688:   const category = categoryById(project?.category);
3689:   applyProjectTemplateToElement(projectPreviewDialog, project, category.accent || "#117c7a");
3690:   projectPreviewTitle.textContent = project?.portfolioView?.title || project?.title || "Project preview";
3691:   activeProjectPreviewProject = project || null;
3692:   activeProjectPreviewSectionIndex = -1;
3693:   activeProjectPreviewPath = [];
3694:   activeProjectPreviewForwardStack = [];
3695:   updateProjectPreviewNavigation();
3696:   projectPreviewContent.classList.add("is-website-preview");
3697:   projectPreviewContent.innerHTML = `
3698:     <iframe
3699:       class="portfolio-preview-frame project-preview-frame"
3700:       title="${escapeHtml(project?.title || "Project website preview")}"
3701:     ></iframe>
3702:   `;
3703:   const frame = projectPreviewContent.querySelector("iframe");
3704:   if (frame) frame.srcdoc = fullProjectPreviewHtmlExact(project);
3705: }

```

openProjectPortfolioPreview - lines 3707-3715

Item	Explanation
Source location	Lines 3707-3715 (9 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function openProjectPortfolioPreview() {
Touches	browser/Electron DOM
Calls detected	selectedProject, find, renderProjectWebsitePreview, add, showModal
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3707: function openProjectPortfolioPreview() {
3708:   const project = selectedProject();
3709:   const savedProject = (savedPortfolioCatalog.projects || []).find((item) => item.id === project?.id);
3710:   const previewProject = savedProject || project;

```

```

3711:   renderProjectWebsitePreview(previewProject);
3712:   document.body.classList.add("full-window-open");
3713:   projectPreviewDialog.showModal();
3714:   projectPreviewDialog.scrollTop = 0;
3715: }

```

savedProject - lines 3709-3714

Item	Explanation
Source location	Lines 3709-3714 (6 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	<code>const savedProject = (savedPortfolioCatalog.projects []).find((item) => item.id === project?.id);</code>
Touches	browser/Electron DOM
Calls detected	find, renderProjectWebsitePreview, add, showModal
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3709:   const savedProject = (savedPortfolioCatalog.projects || []).find((item) => item.id === project?.id);
3710:   const previewProject = savedProject || project;
3711:   renderProjectWebsitePreview(previewProject);
3712:   document.body.classList.add("full-window-open");
3713:   projectPreviewDialog.showModal();
3714:   projectPreviewDialog.scrollTop = 0;

```

updateProjectPreviewNavigation - lines 3717-3721

Item	Explanation
Source location	Lines 3717-3721 (5 source lines).
Purpose	Updates UI, state, cache, or stored data.
Declaration	<code>function updateProjectPreviewNavigation() {</code>
Touches	Mostly local calculations or local UI state.
Calls detected	updateDialogWindowButtons
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3717: function updateProjectPreviewNavigation() {
3718:   projectPreviewBack.hidden = activeProjectPreviewSectionIndex < 0;
3719:   projectPreviewForward.hidden = !activeProjectPreviewForwardStack.length;
3720:   updateDialogWindowButtons(projectPreviewDialog);
3721: }

```

openProjectPreviewNode - lines 3723-3745

Item	Explanation
Source location	Lines 3723-3745 (23 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	<code>function openProjectPreviewNode(sectionIndex, path = [], options = {}) {</code>
Touches	Mostly local calculations or local UI state.

Item	Explanation
Calls detected	filter, previewSectionHasRenderableContent, previewNodeAtPath, parsedItemHasContent, updateProjectPreviewNavigation, displayTitle, parsedPublicSectionContent
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3723: function openProjectPreviewNode(sectionIndex, path = [], options = {}) {
3724:   if (!activeProjectPreviewProject?.portfolioView) return;
3725:   const sections = (activeProjectPreviewProject.portfolioView.sections || []).filter((section) =>
section.id !== "brief" && previewSectionHasRenderableContent(section));
3726:   const section = sections[Number(sectionIndex)];
3727:   if (!section) return;
3728:   const node = path.length ? previewNodeAtPath(section, path) : section;
3729:   if (!node) return;
3730:   if (path.length ? !parsedItemHasContent(node) : !previewSectionHasRenderableContent(section)) return;
3731:
3732:   activeProjectPreviewSectionIndex = Number(sectionIndex);
3733:   activeProjectPreviewPath = path;
3734:   if (!options.preserveForward) activeProjectPreviewForwardStack = [];
3735:   updateProjectPreviewNavigation();
3736:   projectPreviewTitle.textContent = displayTitle(node.title || section.title, "Section");
3737:   projectPreviewContent.innerHTML = `
3738:     <article class="portfolio-preview-article">
3739:       <section class="portfolio-preview-section">
3740:         ${parsedPublicSectionContent(section, Number(sectionIndex), path)}
3741:       </section>
3742:     </article>
3743:   `;
3744:   projectPreviewDialog.scrollTop = 0;
3745: }

```

sections - lines 3725-3730

Item	Explanation
Source location	Lines 3725-3730 (6 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const sections = (activeProjectPreviewProject.portfolioView.sections []).filter((section) => section.id !== "brief" && previewSectionHasRenderableContent(section));
Touches	Mostly local calculations or local UI state.
Calls detected	filter, previewSectionHasRenderableContent, previewNodeAtPath, parsedItemHasContent
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3725:   const sections = (activeProjectPreviewProject.portfolioView.sections || []).filter((section) =>
section.id !== "brief" && previewSectionHasRenderableContent(section));
3726:   const section = sections[Number(sectionIndex)];
3727:   if (!section) return;
3728:   const node = path.length ? previewNodeAtPath(section, path) : section;
3729:   if (!node) return;
3730:   if (path.length ? !parsedItemHasContent(node) : !previewSectionHasRenderableContent(section)) return;

```

projectPreviewBackStep - lines 3747-3769

Item	Explanation
Source location	Lines 3747-3769 (23 source lines).
Purpose	Part of the private builder frontend and editing experience.

Item	Explanation
Declaration	function projectPreviewBackStep() {
Touches	Mostly local calculations or local UI state.
Calls detected	push, slice, renderProjectPortfolioPreview, updateProjectPreviewNavigation, openProjectPreviewNode
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3747: function projectPreviewBackStep() {
3748:   if (!activeProjectPreviewProject) return;
3749:   if (!activeProjectPreviewPath.length) {
3750:     if (activeProjectPreviewSectionIndex >= 0) {
3751:       activeProjectPreviewForwardStack.push({
3752:         path: activeProjectPreviewPath.slice(),
3753:         sectionIndex: activeProjectPreviewSectionIndex
3754:       });
3755:     }
3756:     projectPreviewTitle.textContent = activeProjectPreviewProject.title || "Project preview";
3757:     projectPreviewContent.innerHTML = renderProjectPortfolioPreview(activeProjectPreviewProject);
3758:     activeProjectPreviewSectionIndex = -1;
3759:     activeProjectPreviewPath = [];
3760:     updateProjectPreviewNavigation();
3761:     return;
3762:   }
3763:   activeProjectPreviewForwardStack.push({
3764:     path: activeProjectPreviewPath.slice(),
3765:     sectionIndex: activeProjectPreviewSectionIndex
3766:   });
3767:   const nextPath = activeProjectPreviewPath.slice(0, -1);
3768:   openProjectPreviewNode(activeProjectPreviewSectionIndex, nextPath, { preserveForward: true });
3769: }

```

projectPreviewForwardStep - lines 3771-3775

Item	Explanation
Source location	Lines 3771-3775 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function projectPreviewForwardStep() {
Touches	Mostly local calculations or local UI state.
Calls detected	pop, openProjectPreviewNode
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3771: function projectPreviewForwardStep() {
3772:   if (!activeProjectPreviewProject || !activeProjectPreviewForwardStack.length) return;
3773:   const next = activeProjectPreviewForwardStack.pop();
3774:   openProjectPreviewNode(next.sectionIndex, next.path, { preserveForward: true });
3775: }

```

closeOrStepBackProjectPreview - lines 3777-3783

Item	Explanation
Source location	Lines 3777-3783 (7 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function closeOrStepBackProjectPreview() {

Item	Explanation
Touches	Mostly local calculations or local UI state.
Calls detected	projectPreviewBackStep, closeProjectPreviewWindow
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3777: function closeOrStepBackProjectPreview() {
3778:   if (activeProjectPreviewProject && (activeProjectPreviewSectionIndex >= 0 ||
activeProjectPreviewPath.length)) {
3779:     projectPreviewBackStep();
3780:     return;
3781:   }
3782:   closeProjectPreviewWindow();
3783: }

```

closeProjectPreviewWindow - lines 3785-3791

Item	Explanation
Source location	Lines 3785-3791 (7 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function closeProjectPreviewWindow() {
Touches	Mostly local calculations or local UI state.
Calls detected	closeDialogElement
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3785: function closeProjectPreviewWindow() {
3786:   activeProjectPreviewProject = null;
3787:   activeProjectPreviewSectionIndex = -1;
3788:   activeProjectPreviewPath = [];
3789:   activeProjectPreviewForwardStack = [];
3790:   closeDialogElement(projectPreviewDialog);
3791: }

```

publicProjectCard - lines 3793-3856

Item	Explanation
Source location	Lines 3793-3856 (64 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function publicProjectCard(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	parsedPublicProjectCard, categoryById, projectTemplateClass, projectResponsiveClass, projectTemplateStyle, detailBlock, map, itemTitle, join, evidenceList, resourceLink, mediaGrid
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3793: function publicProjectCard(project) {
3794:   if (project.portfolioView) return parsedPublicProjectCard(project);
3795:   const category = categoryById(project.category);
3796:   const accent = category.accent || "#117c7a";
3797:
3798:   return `
3799:     <article class="project-card catalog-card ${projectTemplateClass(project)}
${projectResponsiveClass(project)}" id="${project.id}" style="${projectTemplateStyle(project, accent)}">
3800:       <div class="project-body">
3801:         <h3>${project.title}</h3>
3802:         <p>${project.summary || ""}</p>
3803:
3804:         ${project.highlights && project.highlights.length ? detailBlock("Project highlights", "project-
drawer", `
3805:           <ul class="highlight-list">
3806:             ${project.highlights.map((item) => `<li>${typeof item === "string" ? item :
itemTitle(item)}</li>`).join("")}
3807:           </ul>
3808:           `) : ""}
3809:
3810:         <div class="evidence-grid" aria-label="${project.title} evidence blocks">
3811:           ${detailBlock("Documents", "evidence-block", evidenceList(project.documents, (item) => `
3812:             <li>
3813:               ${resourceLink(item, item.title)}
3814:               <span>${item.type || "Document"} &middot; ${item.status || "tracked"}</span>
3815:             </li>
3816:             ` , "No document artifact has been added yet."))}
3817:
3818:           ${detailBlock("Tests and results", "evidence-block", evidenceList(project.tests, (item) => `
3819:             <li>
3820:               ${resourceLink({ url: item.artifact, status: item.status }, item.name)}
3821:               <span>${item.method || "Validation"} &middot; ${item.status || "tracked"}</span>
3822:               ${item.result ? `<p>${item.result}</p>` : ""}
3823:             </li>
3824:             ` , "No test artifact has been added yet."))}
3825:
3826:           ${detailBlock("PCBs built", "evidence-block", evidenceList(project.pcb, (item) => `
3827:             <li>
3828:               ${resourceLink({ url: item.artifact, status: item.status }, item.name)}
3829:               <span>${item.revision || "Revision"} &middot; ${item.status || "tracked"}</span>
3830:             </li>
3831:             ` , "No PCB build has been added yet."))}
3832:
3833:           ${detailBlock("Images", "evidence-block evidence-wide", mediaGrid(project.media))}
3834:           ${publicCustomSectionBlocks(project)}
3835:         </div>
3836:
3837:         ${detailBlock("Tools and implementation files", "project-drawer", `
3838:           <div class="project-tooling">
3839:             <div>
3840:               <h4>Tools Used</h4>
3841:               <div class="project-meta">${pillList(project.tools, "tool-tag")}</div>
3842:             </div>
3843:             <div>
3844:               <h4>Languages</h4>
3845:               <div class="project-meta">${pillList(project.languages, "language-tag")}</div>
3846:             </div>
3847:           </div>
3848:         `)}
3849:
3850:         <div class="resource-list">
3851:           ${project.links || []}.map((item) => resourceLink(item)).join("")
3852:         </div>
3853:       </div>
3854:     </article>
3855:   `;
3856: }

```

renderParsedBriefBlock - lines 3858-3868

Item	Explanation
Source location	Lines 3858-3868 (11 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderParsedBriefBlock(section, fallbackSummary = "") {
Touches	Mostly local calculations or local UI state.

Item	Explanation
Calls detected	renderRichContent
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3858: function renderParsedBriefBlock(section, fallbackSummary = "") {
3859:   const briefItem = section?.items?.[0] || {};
3860:   const briefText = briefItem.description || fallbackSummary || "";
3861:   if (!briefItem.rich?.blocks?.length && !briefText) return "";
3862:   return `
3863:     <details class="project-brief-default parsed-summary" open>
3864:       <summary>Overview</summary>
3865:       ${renderRichContent(briefItem.rich, briefText)}
3866:     </details>
3867:   `;
3868: }

```

previewPathToString - lines 3870-3872

Item	Explanation
Source location	Lines 3870-3872 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function previewPathToString(path = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3870: function previewPathToString(path = []) {
3871:   return path.join(".");
3872: }

```

previewPathFromString - lines 3874-3877

Item	Explanation
Source location	Lines 3874-3877 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function previewPathFromString(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	split, map, filter, isInteger
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3874: function previewPathFromString(value = "") {

```

```

3875:   if (!value) return [];
3876:   return value.split(".").map((item) => Number(item)).filter((item) => Number.isInteger(item));
3877: }

```

previewNodeAtPath - lines 3879-3888

Item	Explanation
Source location	Lines 3879-3888 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function previewNodeAtPath(section, path = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3879: function previewNodeAtPath(section, path = []) {
3880:   let node = section;
3881:   let children = section?.items || [];
3882:   for (const index of path) {
3883:     node = children[index];
3884:     if (!node) return null;
3885:     children = node.children || [];
3886:   }
3887:   return node;
3888: }

```

previewNodeChildren - lines 3890-3892

Item	Explanation
Source location	Lines 3890-3892 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function previewNodeChildren(node) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3890: function previewNodeChildren(node) {
3891:   return node?.items || node?.children || [];
3892: }

```

previewSectionHasRenderableContent - lines 3894-3896

Item	Explanation
Source location	Lines 3894-3896 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function previewSectionHasRenderableContent(section) {
Touches	Mostly local calculations or local UI state.

Item	Explanation
Calls detected	richHasContent, some
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3894: function previewSectionHasRenderableContent(section) {
3895:   return Boolean(section?.description || richHasContent(section?.rich) || (section?.items ||
[]).some(parsedItemHasContent));
3896: }

```

previewNodeSummary - lines 3898-3906

Item	Explanation
Source location	Lines 3898-3906 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function previewNodeSummary(title, rich, text, emptyMessage = "No summary has been added yet.") {
Touches	Mostly local calculations or local UI state.
Calls detected	escapeHtml, renderRichContent
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3898: function previewNodeSummary(title, rich, text, emptyMessage = "No summary has been added yet.") {
3899:   if (!rich?.blocks?.length && !text) return "";
3900:   return `
3901:     <details class="parsed-summary" open>
3902:       <summary>${escapeHtml(title)}</summary>
3903:       ${renderRichContent(rich, text || "")}
3904:     </details>
3905:   `;
3906: }

```

previewNodeIsOverview - lines 3908-3911

Item	Explanation
Source location	Lines 3908-3911 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function previewNodeIsOverview(item = {}) {
Touches	filesystem
Calls detected	normalize, displayTitle, endsWith
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3908: function previewNodeIsOverview(item = {}) {
3909:   const title = normalize(displayTitle(item.title || item.label || ""));

```

```

3910:   return item.kind === "summary" || title === "overview" || title.endsWith(" overview");
3911: }

```

previewNodeOverviewDetails - lines 3913-3931

Item	Explanation
Source location	Lines 3913-3931 (19 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function previewNodeOverviewDetails(node, children = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	filter, push, renderRichContent, forEach, previewFileList, previewNodeChildren, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3913: function previewNodeOverviewDetails(node, children = []) {
3914:   const overviewItems = children.filter(previewNodeIsOverview);
3915:   const blocks = [];
3916:   if (node?.rich?.blocks?.length || node?.description) {
3917:     blocks.push(renderRichContent(node.rich, node.description || ""));
3918:   }
3919:   overviewItems.forEach((item) => {
3920:     if (item.rich?.blocks?.length || item.description) blocks.push(renderRichContent(item.rich,
item.description || ""));
3921:     const itemFiles = previewFileList(previewNodeChildren(item));
3922:     if (itemFiles) blocks.push(itemFiles);
3923:   });
3924:   if (!blocks.length) return "";
3925:   return
3926:     <details class="parsed-summary section-overview-default" open>
3927:       <summary>Overview</summary>
3928:       ${blocks.join("")}
3929:     </details>
3930:   `;
3931: }

```

previewNodeCard - lines 3933-3940

Item	Explanation
Source location	Lines 3933-3940 (8 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function previewNodeCard(node, sectionIndex, path) {
Touches	Mostly local calculations or local UI state.
Calls detected	displayTitle, previewPathToString, escapeHtml
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3933: function previewNodeCard(node, sectionIndex, path) {
3934:   const title = displayTitle(node.title);
3935:   return `
3936:     <button class="section-open-card subsection-open-card" type="button" data-preview-section-
index="${sectionIndex}" data-preview-resource-path="${previewPathToString(path)}">
3937:       <span>${escapeHtml(title)}</span>
3938:     </button>
3939:   `;
3940: }

```

previewChildCards - lines 3942-3950

Item	Explanation
Source location	Lines 3942-3950 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function previewChildCards(children, sectionIndex, basePath = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	filter, parsedItemHasContent, previewNodesOverview, map, previewNodeCard, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
3942: function previewChildCards(children, sectionIndex, basePath = []) {
3943:   const visibleChildren = children.filter((child) => parsedItemHasContent(child) && !child.url &&
!previewNodeIsOverview(child));
3944:   if (!visibleChildren.length) return "";
3945:   return `
3946:     <div class="subsection-grid section-content-grid">
3947:       ${children.map((child, index) => parsedItemHasContent(child) && !child.url &&
!previewNodeIsOverview(child) ? previewNodeCard(child, sectionIndex, [...basePath, index]) : "").join("")}
3948:     </div>
3949:   `;
3950: }
```

previewFileList - lines 3952-3969

Item	Explanation
Source location	Lines 3952-3969 (18 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function previewFileList(children = []) {
Touches	filesystem
Calls detected	filter, map, escapeHtml, normalizeLinkTarget, isWebsiteLinkItem, linkAttributes, downloadAttribute, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
3952: function previewFileList(children = []) {
3953:   const files = children.filter((child) => child?.url);
3954:   if (!files.length) return "";
3955:   return `
3956:     <div class="section-file-list">
3957:       ${files.map((file) => `
3958:         <a
3959:           class="section-file-link"
3960:           href="${escapeHtml(normalizeLinkTarget(file.url, { assumeWeb: isWebsiteLinkItem(file, file.url)
3961:             ${linkAttributes(file.url, file)}
3962:             ${downloadAttribute(file.url, file)}
3963:           >
3964:             ${escapeHtml(file.title || "Open link")}
3965:           </a>
3966:         `).join("")}
3967:     </div>
3968:   `;
3969: }
```

previewNodeContent - lines 3971-3983

Item	Explanation
Source location	Lines 3971-3983 (13 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function previewNodeContent(node, sectionIndex, path = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	previewFileList, previewNodeChildren, filter, previewNodeIsOverview, previewNodeOverviewDetails, previewChildCards
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3971: function previewNodeContent(node, sectionIndex, path = []) {
3972:   const isContainer = !node.kind || node.kind === "subsection";
3973:   if (!isContainer && node.url) return previewFileList([node]);
3974:   const children = previewNodeChildren(node);
3975:   const contentChildren = children.filter((child) => !previewNodeIsOverview(child));
3976:   return `
3977:     <article class="parsed-window-panel section-directory">
3978:       ${previewNodeOverviewDetails(node, children)}
3979:       ${previewChildCards(children, sectionIndex, path)}
3980:       ${previewFileList(contentChildren)}
3981:     </article>
3982:   `;
3983: }
```

parsedPublicSectionContent - lines 3985-3989

Item	Explanation
Source location	Lines 3985-3989 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function parsedPublicSectionContent(section, sectionIndex = 0, path = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	previewNodeAtPath, previewNodeContent
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3985: function parsedPublicSectionContent(section, sectionIndex = 0, path = []) {
3986:   const node = path.length ? previewNodeAtPath(section, path) : section;
3987:   if (!node) return `<p class="evidence-empty">This section could not be found.</p>`;
3988:   return previewNodeContent(node, sectionIndex, path);
3989: }
```

parsedPublicSection - lines 3991-3999

Item	Explanation
Source location	Lines 3991-3999 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function parsedPublicSection(section, index = 0) {
Touches	Mostly local calculations or local UI state.
Calls detected	filter, some, escapeHtml, displayTitle

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3991: function parsedPublicSection(section, index = 0) {
3992:   const visibleItems = (section.items || []).filter(parsedItemHasContent);
3993:   const hasImages = visibleItems.some((item) => item.kind === "image");
3994:   return `
3995:     <button class="section-open-card evidence-block ${hasImages ? "evidence-wide" : ""}" type="button"
data-preview-section-index="${index}">
3996:       <span>${escapeHtml(displayTitle(section.title, "Section"))}</span>
3997:     </button>
3998:   `;
3999: }

```

parsedPublicProjectCard - lines 4001-4019

Item	Explanation
Source location	Lines 4001-4019 (19 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function parsedPublicProjectCard(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	categoryById, find, filter, previewSectionHasRenderableContent, projectTemplateClass, projectResponsiveClass, projectTemplateStyle, renderParsedBriefBlock, map, parsedPublicSection, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4001: function parsedPublicProjectCard(project) {
4002:   const category = categoryById(project.category);
4003:   const accent = category.accent || "#117c7a";
4004:   const view = project.portfolioView;
4005:   const briefSection = (view.sections || []).find((section) => section.id === "brief");
4006:   const otherSections = (view.sections || []).filter((section) => section.id !== "brief" &&
previewSectionHasRenderableContent(section));
4007:
4008:   return `
4009:     <article class="project-card catalog-card ${projectTemplateClass(project)}
${projectResponsiveClass(project)}" id="${project.id}" style="${projectTemplateStyle(project, accent)}">
4010:       <div class="project-body">
4011:         <h3>${view.title || project.title}</h3>
4012:         ${renderParsedBriefBlock(briefSection, project.summary)}
4013:         <div class="evidence-grid" aria-label="${project.title} parsed project content">
4014:           ${otherSections.map((section, index) => parsedPublicSection(section, index)).join("")}
4015:         </div>
4016:       </div>
4017:     </article>
4018:   `;
4019: }

```

briefSection - lines 4005-4006

Item	Explanation
Source location	Lines 4005-4006 (2 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const briefSection = (view.sections []).find((section) => section.id === "brief");
Touches	Mostly local calculations or local UI state.

Item	Explanation
Calls detected	find, filter, previewSectionHasRenderableContent
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4005:   const briefSection = (view.sections || []).find((section) => section.id === "brief");
4006:   const otherSections = (view.sections || []).filter((section) => section.id !== "brief" &&
previewSectionHasRenderableContent(section));

```

otherSections - lines 4006-4006

Item	Explanation
Source location	Lines 4006-4006 (1 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const otherSections = (view.sections []).filter((section) => section.id !== "brief" && previewSectionHasRenderableContent(section));
Touches	Mostly local calculations or local UI state.
Calls detected	filter, previewSectionHasRenderableContent
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4006:   const otherSections = (view.sections || []).filter((section) => section.id !== "brief" &&
previewSectionHasRenderableContent(section));

```

publicCategorySection - lines 4021-4036

Item	Explanation
Source location	Lines 4021-4036 (16 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function publicCategorySection(category, visibleProjects) {
Touches	Mostly local calculations or local UI state.
Calls detected	map, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4021: function publicCategorySection(category, visibleProjects) {
4022:   return `
4023:     <section class="category-section ${visibleProjects.length ? "" : "empty-category-section"}" aria-
labelledby="${category.id}-preview-title">
4024:       <div class="category-heading">
4025:         <div>
4026:           <h3 id="${category.id}-preview-title">${category.label}</h3>
4027:         </div>
4028:         ${visibleProjects.length ? `<span>${visibleProjects.length} project${visibleProjects.length === 1

```

```

? "" : "s"</span>` : ""}
4029:   </div>
4030:   ${visibleProjects.length ? `<p class="category-description">${category.description}</p>` : ""}
4031:   <div class="category-projects">
4032:     ${visibleProjects.map(publicProjectCard).join("")}
4033:   </div>
4034: </section>
4035: `;
4036: }

```

renderedPublicProjects - lines 4038-4051

Item	Explanation
Source location	Lines 4038-4051 (14 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderedPublicProjects() {
Touches	Mostly local calculations or local UI state.
Calls detected	map, filter, publicCategorySection, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4038: function renderedPublicProjects() {
4039:   if (!savedPortfolioCatalog.projects.length) {
4040:     return `
4041:       <div class="empty-state">
4042:         <h3>No parsed projects saved yet.</h3>
4043:         <p>Open a project, edit it, then click Save to build it into this portfolio preview.</p>
4044:       </div>
4045:     `;
4046:   }
4047:   return savedPortfolioCatalog.categories.map((category) => {
4048:     const categoryProjects = savedPortfolioCatalog.projects.filter((project) => project.category ===
category.id);
4049:     return publicCategorySection(category, categoryProjects);
4050:   }).join("");
4051: }

```

sectionControls - lines 4053-4060

Item	Explanation
Source location	Lines 4053-4060 (8 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function sectionControls(sectionId, uploadKey = "", folder = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4053: function sectionControls(sectionId, uploadKey = "", folder = "") {
4054:   return `
4055:     <div class="local-section-controls">
4056:       <button type="button" title="Edit this section" data-preview-section="${sectionId}">Edit</button>
4057:       ${uploadKey ? `<button type="button" title="Add file or image" data-preview-upload="${uploadKey}"
data-folder="${folder}">+</button>` : ""}
4058:     </div>
4059:   `;

```

4060: }

customSectionBlocks - lines 4062-4074

Item	Explanation
Source location	Lines 4062-4074 (13 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function customSectionBlocks(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	map, detailBlock, sectionControls, evidenceList, resourceLink, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
4062: function customSectionBlocks(project) {
4063:   return (project.sections || []).map((section) => detailBlock(section.title, "evidence-block evidence-
wide", `
4064:     ${sectionControls(`custom:${section.id}`, "sections", section.id)}
4065:     ${section.description ? `
```

projectCard - lines 4076-4154

Item	Explanation
Source location	Lines 4076-4154 (79 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function projectCard(project, isSelected = false) {
Touches	Mostly local calculations or local UI state.
Calls detected	categoryById, projectTemplateClass, projectResponsiveClass, projectTemplateStyle, detailBlock, map, join, sectionControls, evidenceList, resourceLink, mediaGrid, customSectionBlocks
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
4076: function projectCard(project, isSelected = false) {
4077:   const category = categoryById(project.category);
4078:   const accent = category.accent || "#117c7a";
4079:
4080:   return `
4081:     <article class="project-card catalog-card ${projectTemplateClass(project)}
${projectResponsiveClass(project)} ${isSelected ? "preview-project-card" : ""}" id="${project.id}"
style="${projectTemplateStyle(project, accent)}">
4082:       <div class="project-body">
4083:         <div class="local-card-controls">
4084:           <button type="button" title="Open project" data-preview-project="${project.id}">Open</button>
4085:           <button type="button" title="Delete project" data-preview-
delete="${project.id}">Delete</button>
4086:         </div>
4087:         <h3>${project.title}</h3>
4088:         <p>${project.summary}</p>
```

```

4089:         ${project.highlights && project.highlights.length ? detailBlock("Project highlights", "project-
4090: drawer", `
4091:         <ul class="highlight-list">
4092:             ${project.highlights.map((item) => `<li>${item}</li>`).join("")}
4093:         </ul>
4094:         `) : ""}
4095:
4096: <div class="evidence-grid" aria-label="${project.title} evidence blocks">
4097:     ${detailBlock("Documents", "evidence-block", `
4098:         ${sectionControls("documents", "documents", "documents")}
4099:         ${evidenceList(project.documents, (item) => `
4100:             <li>
4101:                 ${resourceLink(item, item.title)}
4102:                 <span>${item.type || "Document"} &middot; ${item.status || "tracked"}</span>
4103:             </li>
4104:         ` , "No document artifact has been added yet.")}
4105:     `)}
4106:
4107:     ${detailBlock("Tests and results", "evidence-block", `
4108:         ${sectionControls("tests", "tests", "tests")}
4109:         ${evidenceList(project.tests, (item) => `
4110:             <li>
4111:                 ${resourceLink({ url: item.artifact, status: item.status }, item.name)}
4112:                 <span>${item.method || "Validation"} &middot; ${item.status || "tracked"}</span>
4113:                 ${item.result ? `<p>${item.result}</p>` : ""}
4114:             </li>
4115:         ` , "No test artifact has been added yet.")}
4116:     `)}
4117:
4118:     ${detailBlock("PCBs built", "evidence-block", `
4119:         ${sectionControls("pcbs", "pcbs", "pcbs")}
4120:         ${evidenceList(project.pcbs, (item) => `
4121:             <li>
4122:                 ${resourceLink({ url: item.artifact, status: item.status }, item.name)}
4123:                 <span>${item.revision || "Revision"} &middot; ${item.status || "tracked"}</span>
4124:             </li>
4125:         ` , "No PCB build has been added yet.")}
4126:     `)}
4127:
4128:     ${detailBlock("Images", "evidence-block evidence-wide", `
4129:         ${sectionControls("media", "media", "images")}
4130:         ${mediaGrid(project.media)}
4131:     `)}
4132:     ${customSectionBlocks(project)}
4133: </div>
4134:
4135: ${detailBlock("Tools and implementation files", "project-drawer", `
4136:     <div class="project-tooling">
4137:         <div>
4138:             <h4>Tools Used</h4>
4139:             <div class="project-meta">${pillList(project.tools, "tool-tag")}</div>
4140:         </div>
4141:         <div>
4142:             <h4>Languages</h4>
4143:             <div class="project-meta">${pillList(project.languages, "language-tag")}</div>
4144:         </div>
4145:     </div>
4146:     `)}
4147:
4148:     <div class="resource-list">
4149:         ${project.links || []}.map((item) => resourceLink(item)).join("")
4150:     </div>
4151: </div>
4152: </article>
4153: `;
4154: }

```

buildTemplateProject - lines 4156-4186

Item	Explanation
Source location	Lines 4156-4186 (31 source lines).
Purpose	Creates an object, UI structure, process, payload, or generated artifact.
Declaration	function buildTemplateProject() {
Touches	filesystem
Calls detected	find, trim, slugify, clone, defaultDesignModel
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4156: function buildTemplateProject() {
4157:   const template = templates.find((item) => item.id === templateSelect.value) || null;
4158:   const title = newProjectTitle.value.trim() || "New Hardware Project";
4159:   const id = slugify(title);
4160:   const category = newCategorySelect.value || pendingCreateCategoryId || "analog-mixed-signal";
4161:
4162:   return {
4163:     id,
4164:     title,
4165:     templateId: template?.id || "",
4166:     templateLabel: template?.label || "",
4167:     templateGroup: "Appearance",
4168:     templateVisual: template?.visual ? clone(template.visual) : null,
4169:     category,
4170:     status: "Draft",
4171:     summary: "",
4172:     summaryRich: null,
4173:     focus: [],
4174:     highlights: [],
4175:     documents: [],
4176:     tests: [],
4177:     pcbs: [],
4178:     media: [],
4179:     tools: [],
4180:     languages: [],
4181:     links: [],
4182:     sections: [],
4183:     compileCode: { files: [], activeFileId: "", terminal: "" },
4184:     design: category === "analog-mixed-signal" ? defaultDesignModel() : undefined
4185:   };
4186: }

```

insertProject - lines 4188-4204

Item	Explanation
Source location	Lines 4188-4204 (17 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function insertProject(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	filter, map, splice
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4188: function insertProject(project) {
4189:   const nextProjects = catalog.projects.filter((item) => item.id !== project.id);
4190:   const placement = placementSelect.value;
4191:
4192:   if (placement === "site-start") return [project, ...nextProjects];
4193:   if (placement === "site-end") return [...nextProjects, project];
4194:
4195:   const indexes = nextProjects
4196:     .map((item, index) => ({ item, index }))
4197:     .filter(({ item }) => item.category === project.category)
4198:     .map(({ index }) => index);
4199:
4200:   if (!indexes.length) return [...nextProjects, project];
4201:   const targetIndex = placement === "category-start" ? indexes[0] : indexes[indexes.length - 1] + 1;
4202:   nextProjects.splice(targetIndex, 0, project);
4203:   return nextProjects;
4204: }

```

renderCategoryDropdown - lines 4206-4213

Item	Explanation
Source location	Lines 4206-4213 (8 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderCategoryDropdown() {
Touches	Mostly local calculations or local UI state.
Calls detected	map, escapeHtml, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4206: function renderCategoryDropdown() {
4207:   categoryDropdown.innerHTML = (catalog.categories || []).map((category) => `
4208:     <button type="button" data-create-category="${escapeHtml(category.id)}" style="--category-accent:
${escapeHtml(category.accent || "#17c7a")}">
4209:       <span>${escapeHtml(category.label)}</span>
4210:       <small>${escapeHtml(category.description || "Project category")}</small>
4211:     </button>
4212:   `).join("");
4213: }

```

toggleCategoryDropdown - lines 4215-4219

Item	Explanation
Source location	Lines 4215-4219 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function toggleCategoryDropdown(forceOpen = null) {
Touches	Mostly local calculations or local UI state.
Calls detected	setAttribute
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4215: function toggleCategoryDropdown(forceOpen = null) {
4216:   const open = forceOpen === null ? categoryDropdown.hidden : forceOpen;
4217:   categoryDropdown.hidden = !open;
4218:   addProjectButton.setAttribute("aria-expanded", String(open));
4219: }

```

createProjectInCategory - lines 4221-4233

Item	Explanation
Source location	Lines 4221-4233 (13 source lines).
Purpose	Creates an object, UI structure, process, payload, or generated artifact.
Declaration	function createProjectInCategory(categoryId) {
Touches	Mostly local calculations or local UI state.
Calls detected	buildTemplateProject, insertProject, add, toggleCategoryDropdown, setStatus, categoryById, scheduleAutosave, renderAll, openProjectWindow
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4221: function createProjectInCategory(categoryId) {
4222:   newCategorySelect.value = categoryId;
4223:   const project = buildTemplateProject();
4224:   catalog.projects = insertProject(project);
4225:   selectedProjectId = project.id;
4226:   activeSectionId = "brief";
4227:   expandedCategories.add(categoryId);
4228:   toggleCategoryDropdown(false);
4229:   setStatus(`Project added locally in ${categoryById(categoryId).label || categoryId}. Click Save project
to include it in the portfolio preview.`);
4230:   scheduleAutosave();
4231:   renderAll();
4232:   openProjectWindow(project.id, "brief");
4233: }

```

openProjectCreateDialog - lines 4235-4245

Item	Explanation
Source location	Lines 4235-4245 (11 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function openProjectCreateDialog(categoryId) {
Touches	Mostly local calculations or local UI state.
Calls detected	categoryById, groupedTemplateOptions, toggleCategoryDropdown, showModal
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4235: function openProjectCreateDialog(categoryId) {
4236:   pendingCreateCategoryId = categoryId;
4237:   newCategorySelect.value = categoryId;
4238:   const category = categoryById(categoryId);
4239:   projectCreateCategory.textContent = category.label || "Project category";
4240:   templateSelect.innerHTML = groupedTemplateOptions();
4241:   templateSelect.value = "";
4242:   newProjectTitle.value = "";
4243:   toggleCategoryDropdown(false);
4244:   projectCreateDialog.showModal();
4245: }

```

groupedTemplateOptions - lines 4247-4253

Item	Explanation
Source location	Lines 4247-4253 (7 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function groupedTemplateOptions() {
Touches	Mostly local calculations or local UI state.
Calls detected	map, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4247: function groupedTemplateOptions() {
4248:   const emptyOption = `<option value="">No appearance - white project layout</option>`;
4249:   const options = templates
4250:     .map((template) => `<option value="${template.id}">${template.label}</option>`)
4251:     .join("");
4252:   return `${emptyOption}${options}`;
4253: }

```

showTemplatePreview - lines 4255-4276

Item	Explanation
Source location	Lines 4255-4276 (22 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function showTemplatePreview(template) {
Touches	Mostly local calculations or local UI state.
Calls detected	setProperty, map, join, showModal
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4255: function showTemplatePreview(template) {
4256:   if (!template) return;
4257:   const visual = template.visual || {};
4258:   const palette = visual.palette || ["#0f4c5c", "#7dd3c7", "#ffffff"];
4259:
4260:   templatePreviewGroup.textContent = visual.style ? `Appearance / ${visual.style}` : "Appearance";
4261:   templatePreviewTitle.textContent = template.label;
4262:   templatePreviewDescription.textContent = template.description || "";
4263:   templatePreviewCardTitle.textContent = template.label;
4264:   templatePreviewInteraction.textContent = visual.interaction || "Hover and click styles are shown in the
local builder.";
4265:   templatePreviewCard.style.setProperty("--template-primary", palette[0]);
4266:   templatePreviewCard.style.setProperty("--template-hover", palette[1]);
4267:   templatePreviewCard.style.setProperty("--template-paper", palette[2] || "#ffffff");
4268:   templatePreviewCard.style.setProperty("--template-bg", visual.background || palette[0]);
4269:   templatePreviewCard.style.setProperty("--template-panel", visual.panel || palette[2] || "#ffffff");
4270:   templatePreviewCard.style.setProperty("--template-accent", visual.accent || palette[1]);
4271:   templatePreviewCard.style.setProperty("--template-click", visual.click || palette[2] || "#ffffff");
4272:   templatePreviewCard.style.setProperty("--template-click-text", visual.clickText || visual.text ||
"#172636");
4273:   templateVisual.className = `template-visual template-visual-${visual.style || "schematic"}`;
4274:   templatePalette.innerHTML = palette.map((color) => `<span style="background:
${color}">${color}</span>`).join("");
4275:   templateDialog.showModal();
4276: }

```

renderTree - lines 4278-4319

Item	Explanation
Source location	Lines 4278-4319 (42 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderTree() {
Touches	Mostly local calculations or local UI state.
Calls detected	projectSearchText, map, filter, includes, projectMatchesSearch, has, escapeHtml, highlightSearchText, join, updateProjectSearchStatus
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4278: function renderTree() {
4279:   const cleanQuery = projectSearchText(projectSearchQuery);
4280:   let matchCount = 0;
4281:   const treeMarkup = catalog.categories.map((category) => {
4282:     const allCategoryProjects = catalog.projects.filter((project) => project.category === category.id);
4283:     const categoryHit = cleanQuery && projectSearchText(`${category.label}
${category.description}`).includes(cleanQuery);
4284:     const categoryProjects = cleanQuery
4285:       ? allCategoryProjects.filter((project) => categoryHit || projectMatchesSearch(project, category,
cleanQuery))
4286:       : allCategoryProjects;
4287:     if (cleanQuery && !categoryProjects.length && !categoryHit) return "";
4288:     matchCount += categoryProjects.length;
4289:     const isExpanded = cleanQuery ? true : expandedCategories.has(category.id);
4290:     return `
4291:       <section class="tree-category">
4292:         <div class="tree-category-heading">
4293:           <button class="tree-category-toggle" type="button" data-toggle-
category="${escapeHtml(category.id)}" aria-expanded="${isExpanded}">
4294:             <span>${highlightSearchText(category.label, projectSearchQuery)}</span>
4295:             <small>${allCategoryProjects.length} project${allCategoryProjects.length === 1 ? "" :
"s"}</small>
4296:           </button>
4297:         </div>
4298:         <div class="tree-projects" ${isExpanded ? "" : "hidden"}>
4299:           ${categoryProjects.length ? categoryProjects.map((project) => `
4300:             <div class="tree-project-row ${project.id === selectedProjectId ? "active" : ""}>
4301:               <button type="button" data-project-id="${escapeHtml(project.id)}">
4302:                 <span>${highlightSearchText(project.title, projectSearchQuery)}</span>
4303:               </button>
4304:               <button class="tree-project-delete" type="button" data-delete-
project="${escapeHtml(project.id)}" aria-label="Delete ${escapeHtml(project.title)}" title="Delete project">
4305:                 &times;
4306:               </button>
4307:             </div>
4308:           `).join("") : cleanQuery ? `<p class="tree-empty">Category matches, but no projects are in it
yet.</p>` : ""}
4309:         </div>
4310:       </section>
4311:     `;
4312:   }).join("");
4313:   projectTree.innerHTML = treeMarkup || `
4314:   <section class="tree-category tree-search-empty">
4315:     <p class="tree-empty">No project, file, tool, or section matched this search.</p>
4316:   </section>
4317:   `;
4318:   updateProjectSearchStatus(cleanQuery ? matchCount : catalog.projects.length);
4319: }

```

summaryWordCount - lines 4321-4323

Item	Explanation
Source location	Lines 4321-4323 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function summaryWordCount(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	wordCount, plainTextFromRich
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4321: function summaryWordCount(project) {

```

```

4322:   return wordCount(project?.summaryRich?.blocks?.length ? plainTextFromRich(project.summaryRich) :
project?.summary || "");
4323: }

```

renderSummaryPreview - lines 4325-4329

Item	Explanation
Source location	Lines 4325-4329 (5 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderSummaryPreview(project) {
Touches	filesystem
Calls detected	renderRichContent, renderMultilineInlineText
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4325: function renderSummaryPreview(project) {
4326:   if (project.summaryRich?.blocks?.length) return renderRichContent(project.summaryRich, project.summary
|| "");
4327:   if (project.summary) return `<p class="rich-
paragraph">${renderMultilineInlineText(project.summary)}</p>`;
4328:   return `<p class="evidence-empty">No project overview has been added yet.</p>`;
4329: }

```

renderSummaryBuilder - lines 4331-4359

Item	Explanation
Source location	Lines 4331-4359 (29 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderSummaryBuilder(project) {
Touches	filesystem
Calls detected	summaryWordCount, renderSummaryPreview
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4331: function renderSummaryBuilder(project) {
4332:   const isEditing = summaryEditorProjectId === project.id;
4333:   const summaryWords = summaryWordCount(project);
4334:   const hasSummary = summaryWords > 0 || Boolean(project.summaryRich?.blocks?.length);
4335:   return `
4336:     <div class="summary-builder wide-field">
4337:       <div class="summary-builder-heading">
4338:         <div>
4339:           <span>Project overview <small>${summaryWords}/1000 words</small></span>
4340:         </div>
4341:         <div class="summary-builder-actions">
4342:           ${isEditing
4343:             ? `<button class="button secondary" type="button" data-summary-cancel="true">Cancel
overview</button>`
4344:             : `<button class="button primary" type="button" data-summary-create="true">${hasSummary ?
"Edit overview" : "Create overview"}</button>`}
4345:           ${hasSummary && !isEditing ? `<button class="button danger-control" type="button" data-summary-
delete="true">Delete overview</button>` : ""}
4346:         </div>
4347:       </div>
4348:       ${isEditing ? `
4349:         <div class="rich-summary-panel">

```

```

4350:         <p class="rich-editor-hint">Right-click text or an inserted block for formatting, images,
formulas, code blocks, alignment, and delete controls.</p>
4351:         <div class="rich-summary-editor" data-rich-editor="summary" data-placeholder="Click anywhere
and start writing the project overview." contenteditable="true" spellcheck="true" aria-label="Rich overview
editor"></div>
4352:         <div class="summary-save-actions">
4353:             <button class="button primary" type="button" data-summary-save="true">Save overview</button>
4354:         </div>
4355:     </div>
4356:     ` : `<div class="summary-rendered-preview">${renderSummaryPreview(project)}</div>`
4357: </div>
4358: `;
4359: }

```

applyRichTextBlockStyle - lines 4361-4372

Item	Explanation
Source location	Lines 4361-4372 (12 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function applyRichTextBlockStyle(block) {
Touches	browser/Electron DOM, filesystem
Calls detected	cleanFontFamily, normalizeFontPx, normalizeTextColor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4361: function applyRichTextBlockStyle(block) {
4362:     const fontFamily = cleanFontFamily(block.dataset.fontFamily || "Arial") || "Arial";
4363:     const fontPx = normalizeFontPx(block.dataset.fontPx || "");
4364:     const color = normalizeTextColor(block.dataset.color || "");
4365:
4366:     block.style.fontFamily = fontFamily || "";
4367:     block.style.fontSize = fontPx ? `${fontPx}px` : "";
4368:     block.style.color = color || "";
4369:     block.style.fontWeight = block.dataset.bold === "true" ? "700" : "";
4370:     block.style.fontStyle = block.dataset.italic === "true" ? "italic" : "";
4371:     block.style.textDecoration = block.dataset.underline === "true" ? "underline" : "";
4372: }

```

createRichTextBlock - lines 4374-4391

Item	Explanation
Source location	Lines 4374-4391 (18 source lines).
Purpose	Creates an object, UI structure, process, payload, or generated artifact.
Declaration	function createRichTextBlock(text = "", fontSize = "normal", align = "left", fontFamily = "Arial", fontPx = "", color = "", bold = false, html = "", italic = false, underline = false) {
Touches	browser/Electron DOM, filesystem
Calls detected	createElement, cleanFontFamily, normalizeFontPx, normalizeTextColor, sanitizeRichInlineHtml, applyRichTextBlockStyle
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4374: function createRichTextBlock(text = "", fontSize = "normal", align = "left", fontFamily = "Arial", fontPx = "", color = "", bold = false, html = "", italic = false, underline = false) {
4375:     const block = document.createElement("p");
4376:     block.className = `rich-block rich-text-block rich-text-${fontSize} text-${align}`;
4377:     block.dataset.type = "paragraph";

```

```

4378:     block.dataset.fontSize = fontSize;
4379:     block.dataset.align = align;
4380:     block.dataset.fontFamily = cleanFontFamily(fontFamily || "Arial") || "Arial";
4381:     block.dataset.fontPx = normalizeFontPx(fontPx);
4382:     block.dataset.color = normalizeTextColor(color);
4383:     block.dataset.bold = bold ? "true" : "false";
4384:     block.dataset.italic = italic ? "true" : "false";
4385:     block.dataset.underline = underline ? "true" : "false";
4386:     block.spellcheck = true;
4387:     if (html) block.innerHTML = sanitizeRichInlineHtml(html);
4388:     else block.textContent = text;
4389:     applyRichTextBlockStyle(block);
4390:     return block;
4391: }

```

richBlockActions - lines 4393-4401

Item	Explanation
Source location	Lines 4393-4401 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function richBlockActions(label = "block", options = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	escapeHtml
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4393: function richBlockActions(label = "block", options = {}) {
4394:   return `
4395:     <div class="rich-block-actions">
4396:       ${options.movable ? `<button class="rich-drag-handle" type="button" draggable="true" data-rich-
drag-handle aria-label="Move ${escapeHtml(label)}">Move</button>` : ""}
4397:       <button type="button" data-rich-block-action="edit" aria-label="Edit
${escapeHtml(label)}">Edit</button>
4398:       <button class="danger-icon" type="button" data-rich-block-action="delete" aria-label="Delete
${escapeHtml(label)}">Delete</button>
4399:     </div>
4400:   `;
4401: }

```

createRichImageBlock - lines 4403-4431

Item	Explanation
Source location	Lines 4403-4431 (29 source lines).
Purpose	Creates an object, UI structure, process, payload, or generated artifact.
Declaration	function createRichImageBlock(blockData) {
Touches	browser/Electron DOM, filesystem
Calls detected	createElement, cleanRichImageTitle, normalizeCropAspect, normalizeCropZoom, normalizeCropPosition, richBlockActions, richImageCropStyle, escapeHtml, normalizeLinkTarget
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4403: function createRichImageBlock(blockData) {
4404:   const figure = document.createElement("figure");
4405:   const align = blockData.align || "center";
4406:   const title = cleanRichImageTitle(blockData);
4407:   figure.className = `rich-block rich-image-block justify-${align}`;

```

```

4408:   figure.dataset.type = "image";
4409:   figure.dataset.url = blockData.url || "";
4410:   figure.dataset.title = blockData.title || "";
4411:   figure.dataset.caption = blockData.caption || "";
4412:   figure.dataset.align = align;
4413:   figure.dataset.display = blockData.display === "download" ? "download" : "show";
4414:   figure.dataset.source = blockData.source || "local";
4415:   figure.dataset.cropAspect = normalizeCropAspect(blockData.cropAspect);
4416:   figure.dataset.cropZoom = String(normalizeCropZoom(blockData.cropZoom));
4417:   figure.dataset.cropX = String(normalizeCropPosition(blockData.cropX));
4418:   figure.dataset.cropY = String(normalizeCropPosition(blockData.cropY));
4419:   figure.contentEditable = "false";
4420:   figure.draggable = true;
4421:   figure.title = "Drag image to move it between text.";
4422:   figure.innerHTML =
4423:     `${richBlockActions("overview image", { movable: true })}
4424:     ${figure.dataset.display === "download" ? `<span class="rich-download-badge">Download only</span>` :
""}
4425:     <span class="rich-image-viewport crop-${figure.dataset.cropAspect === "original" ? "original" :
"active"}">${richImageCropStyle(blockData)}>
4426:     
4427:     </span>
4428:     ${title || blockData.caption} ? `<figcaption>${title ? `<strong>${escapeHtml(title)}</strong>` :
""}${blockData.caption ? `<span>${escapeHtml(blockData.caption)}</span>` : ""}</figcaption>` : ""}
4429:   `;
4430:   return figure;
4431: }

```

createRichFormulaBlock - lines 4433-4446

Item	Explanation
Source location	Lines 4433-4446 (14 source lines).
Purpose	Creates an object, UI structure, process, payload, or generated artifact.
Declaration	function createRichFormulaBlock(blockData) {
Touches	browser/Electron DOM, filesystem
Calls detected	createElement, unwrapFormula, richBlockActions, escapeHtml
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4433: function createRichFormulaBlock(blockData) {
4434:   const block = document.createElement("div");
4435:   const align = blockData.align || "center";
4436:   block.className = `rich-block rich-formula-block justify-${align}`;
4437:   block.dataset.type = "formula";
4438:   block.dataset.formula = unwrapFormula(blockData.formula || "");
4439:   block.dataset.align = align;
4440:   block.contentEditable = "false";
4441:   block.innerHTML = `
4442:     ${richBlockActions("formula")}
4443:     <span class="formula-text">${escapeHtml(unwrapFormula(blockData.formula || ""))}</span>
4444:   `;
4445:   return block;
4446: }

```

createRichCodeBlock - lines 4448-4465

Item	Explanation
Source location	Lines 4448-4465 (18 source lines).
Purpose	Creates an object, UI structure, process, payload, or generated artifact.
Declaration	function createRichCodeBlock(blockData = {}) {
Touches	browser/Electron DOM, filesystem, authentication/security data
Calls detected	createElement, normalizeCodePasteMode, normalizeCodeText, normalizeCodeLanguage, detectCodeLanguage, richBlockActions, escapeHtml, codeLanguageLabel, tokenizedCodeHtml

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4448: function createRichCodeBlock(blockData = {}) {
4449:   const block = document.createElement("figure");
4450:   const pasteMode = normalizeCodePasteMode(blockData.pasteMode || "source");
4451:   const code = normalizeCodeText(blockData.code || "", pasteMode);
4452:   const language = normalizeCodeLanguage(blockData.language || detectCodeLanguage(code));
4453:   block.className = `rich-block rich-code-block rich-code-language-${language}`;
4454:   block.dataset.type = "code";
4455:   block.dataset.language = language;
4456:   block.dataset.pasteMode = pasteMode;
4457:   block.dataset.code = code;
4458:   block.contentEditable = "false";
4459:   block.innerHTML = `
4460:     ${richBlockActions("code block")}
4461:     <figcaption><span>&lt;/&gt;</span> ${escapeHtml(codeLanguageLabel(language))}<button type="button"
data-rich-block-action="copy-code">Copy code</button></figcaption>
4462:     <pre><code>${tokenizedCodeHtml(code, language)}</code></pre>
4463:   `;
4464:   return block;
4465: }

```

refreshRichImageBlock - lines 4467-4497

Item	Explanation
Source location	Lines 4467-4497 (31 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function refreshRichImageBlock(block, blockData) {
Touches	browser/Electron DOM, filesystem
Calls detected	normalizeCropAspect, normalizeCropZoom, normalizeCropPosition, remove, add, cleanRichImageTitle, richBlockActions, richImageCropStyle, escapeHtml, normalizeLinkTarget
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4467: function refreshRichImageBlock(block, blockData) {
4468:   const align = blockData.align || block.dataset.align || "center";
4469:   block.dataset.url = blockData.url || block.dataset.url || "";
4470:   block.dataset.title = blockData.title || "";
4471:   block.dataset.caption = blockData.caption || "";
4472:   block.dataset.align = align;
4473:   block.dataset.display = blockData.display === "download" ? "download" : "show";
4474:   block.dataset.source = blockData.source || block.dataset.source || "local";
4475:   block.dataset.cropAspect = normalizeCropAspect(blockData.cropAspect || block.dataset.cropAspect);
4476:   block.dataset.cropZoom = String(normalizeCropZoom(blockData.cropZoom || block.dataset.cropZoom));
4477:   block.dataset.cropX = String(normalizeCropPosition(blockData.cropX ?? block.dataset.cropX));
4478:   block.dataset.cropY = String(normalizeCropPosition(blockData.cropY ?? block.dataset.cropY));
4479:   block.draggable = true;
4480:   block.title = "Drag image to move it between text.";
4481:   block.classList.remove("justify-left", "justify-center", "justify-right");
4482:   block.classList.add(`justify-${align}`);
4483:   const title = cleanRichImageTitle({ title: block.dataset.title });
4484:   block.innerHTML = `
4485:     ${richBlockActions("overview image", { movable: true })}
4486:     ${block.dataset.display === "download" ? "<span class='rich-download-badge'>Download only</span>" :
""}
4487:     <span class="rich-image-viewport crop-${block.dataset.cropAspect === "original" ? "original" :
"active"}">${richImageCropStyle({
4488:       cropAspect: block.dataset.cropAspect,
4489:       cropZoom: block.dataset.cropZoom,
4490:       cropX: block.dataset.cropX,
4491:       cropY: block.dataset.cropY

```

```

4492:     >>>
4493:     
4494:     </span>
4495:     ${title || block.dataset.caption} ? `<figcaption>${title ? `<strong>${escapeHtml(title)}</strong>` :
""}${block.dataset.caption ? `<span>${escapeHtml(block.dataset.caption)}</span>` : ""}</figcaption>` : ""}
4496: `;
4497: }

```

refreshRichFormulaBlock - lines 4499-4510

Item	Explanation
Source location	Lines 4499-4510 (12 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function refreshRichFormulaBlock(block, blockData) {
Touches	browser/Electron DOM, filesystem
Calls detected	unwrapFormula, remove, add, richBlockActions, escapeHtml
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4499: function refreshRichFormulaBlock(block, blockData) {
4500:   const align = blockData.align || block.dataset.align || "center";
4501:   const formula = unwrapFormula(blockData.formula || block.dataset.formula || "");
4502:   block.dataset.formula = formula;
4503:   block.dataset.align = align;
4504:   block.classList.remove("justify-left", "justify-center", "justify-right");
4505:   block.classList.add(`justify-${align}`);
4506:   block.innerHTML = `
4507:     ${richBlockActions("formula")}
4508:     <span class="formula-text">${escapeHtml(formula)}</span>
4509: `;
4510: }

```

refreshRichCodeBlock - lines 4512-4525

Item	Explanation
Source location	Lines 4512-4525 (14 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function refreshRichCodeBlock(block, blockData = {}) {
Touches	browser/Electron DOM, filesystem, authentication/security data
Calls detected	normalizeCodePasteMode, normalizeCodeText, normalizeCodeLanguage, detectCodeLanguage, richBlockActions, escapeHtml, codeLanguageLabel, tokenizedCodeHtml
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4512: function refreshRichCodeBlock(block, blockData = {}) {
4513:   const pasteMode = normalizeCodePasteMode(blockData.pasteMode || block.dataset.pasteMode || "source");
4514:   const code = normalizeCodeText(blockData.code || block.dataset.code || "", pasteMode);
4515:   const language = normalizeCodeLanguage(blockData.language || block.dataset.language ||
detectCodeLanguage(code));
4516:   block.dataset.language = language;
4517:   block.dataset.pasteMode = pasteMode;
4518:   block.dataset.code = code;
4519:   block.className = `rich-block rich-code-block rich-code-language-${language}`;
4520:   block.innerHTML = `
4521:     ${richBlockActions("code block")}

```



```

4522:     <figcaption><span>&lt;/span> ${escapeHtml(codeLanguageLabel(language))}<button type="button"
data-rich-block-action="copy-code">Copy code</button></figcaption>
4523:     <pre><code>${tokenizedCodeHtml(code, language)}</code></pre>
4524:     `;
4525: }

```

createRichBlockElement - lines 4527-4544

Item	Explanation
Source location	Lines 4527-4544 (18 source lines).
Purpose	Creates an object, UI structure, process, payload, or generated artifact.
Declaration	function createRichBlockElement(block) {
Touches	filesystem
Calls detected	createRichImageBlock, createRichFormulaBlock, createRichCodeBlock, createRichTextBlock
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4527: function createRichBlockElement(block) {
4528:   if (block.type === "image") return createRichImageBlock(block);
4529:   if (block.type === "formula") return createRichFormulaBlock(block);
4530:   if (block.type === "code") return createRichCodeBlock(block);
4531:
4532:   return createRichTextBlock(
4533:     block.text || "",
4534:     block.fontSize || "normal",
4535:     block.align || "left",
4536:     block.fontFamily || "Arial",
4537:     block.fontPx || "",
4538:     block.color || "",
4539:     Boolean(block.bold),
4540:     block.html || "",
4541:     Boolean(block.italic),
4542:     Boolean(block.underline)
4543:   );
4544: }

```

populateSummaryEditor - lines 4546-4550

Item	Explanation
Source location	Lines 4546-4550 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function populateSummaryEditor(project) {
Touches	browser/Electron DOM
Calls detected	querySelector, populateRichEditor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4546: function populateSummaryEditor(project) {
4547:   const editor = projectFields.querySelector("[data-rich-editor='summary']");
4548:   if (!editor) return;
4549:   populateRichEditor(editor, project);
4550: }

```

populateStandaloneRichEditor - lines 4552-4560

Item	Explanation
Source location	Lines 4552-4560 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function populateStandaloneRichEditor(editor, rich, fallbackText = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	richBlocksFromValue, forEach, append, createRichBlockElement, createRichTextBlock
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4552: function populateStandaloneRichEditor(editor, rich, fallbackText = "") {
4553:   if (!editor) return;
4554:   editor.contentEditable = "true";
4555:   editor.tabIndex = 0;
4556:   editor.spellcheck = true;
4557:   editor.innerHTML = "";
4558:   richBlocksFromValue(rich, fallbackText).forEach((block) =>
editor.append(createRichBlockElement(block)));
4559:   if (!editor.children.length) editor.append(createRichTextBlock(""));
4560: }

```

populateRichEditor - lines 4562-4592

Item	Explanation
Source location	Lines 4562-4592 (31 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function populateRichEditor(editor, project = selectedProject()) {
Touches	browser/Electron DOM
Calls detected	selectedProject, richBlocksFromValue, forEach, append, createRichBlockElement, createRichTextBlock, richBlocksForProject, getByPath
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4562: function populateRichEditor(editor, project = selectedProject()) {
4563:   if (!editor) return;
4564:   editor.contentEditable = "true";
4565:   editor.tabIndex = 0;
4566:   editor.spellcheck = true;
4567:
4568:   if (editor.dataset.richEditor === "pending-description") {
4569:     editor.innerHTML = "";
4570:     richBlocksFromValue(pendingEditor?.richDescription, pendingEditor?.description || "")
4571:       .forEach((block) => editor.append(createRichBlockElement(block)));
4572:     if (!editor.children.length) editor.append(createRichTextBlock(""));
4573:     return;
4574:   }
4575:
4576:   if (!project) return;
4577:
4578:   editor.innerHTML = "";
4579:
4580:   let blocks = [];
4581:
4582:   if (editor.dataset.richEditor === "summary" && !editor.dataset.richPath) {
4583:     blocks = richBlocksForProject(project);
4584:   } else {
4585:     const textPath = editor.dataset.richTextPath || "";
4586:     const richPath = editor.dataset.richPath || "";

```

```

4587:         blocks = richBlocksFromValue(getByPath(project, richPath), getByPath(project, textPath) || "");
4588:     }
4589:
4590:     blocks.forEach((block) => editor.append(createRichBlockElement(block)));
4591:     if (!editor.children.length) editor.append(createRichTextBlock(""));
4592: }

```

populateRichEditors - lines 4594-4596

Item	Explanation
Source location	Lines 4594-4596 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function populateRichEditors(root = document) {
Touches	browser/Electron DOM
Calls detected	querySelectorAll, forEach, populateRichEditor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4594: function populateRichEditors(root = document) {
4595:     root.querySelectorAll("[data-rich-editor]").forEach((editor) => populateRichEditor(editor));
4596: }

```

saveRichEditorToProject - lines 4598-4658

Item	Explanation
Source location	Lines 4598-4658 (61 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	function saveRichEditorToProject(editor) {
Touches	browser/Electron DOM, filesystem
Calls detected	syncFunFactsFromInput, setStatus, syncSiteRichField, syncProfileRichField, selectedProject, extractRichSummary, plainTextFromRich, wordCount, setByPath, scheduleAutosave, schedulePreviewRender
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 9 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4598: function saveRichEditorToProject(editor) {
4599:     if (!editor) return true;
4600:
4601:     if (editor.dataset.richEditor === "fun-facts") {
4602:         syncFunFactsFromInput();
4603:         setStatus("Unsaved local changes.");
4604:         return true;
4605:     }
4606:
4607:     if (editor.dataset.richEditor === "site-field") {
4608:         syncSiteRichField(editor);
4609:         setStatus("Front page text formatting updated locally. Click Save draft before applying to
site.");
4610:         return true;
4611:     }
4612:
4613:     if (editor.dataset.richEditor === "profile-field") {
4614:         syncProfileRichField(editor);
4615:         setStatus("Profile and contact formatting updated locally. Click Save draft before applying to
site.");
4616:         return true;
4617:     }

```

```

4618:
4619:     if (editor.dataset.richEditor === "standalone") {
4620:         setStatus("Unsaved dialog changes. Use the dialog save button to keep them.");
4621:         return true;
4622:     }
4623:
4624:     const project = selectedProject();
4625:
4626:     if (editor.dataset.richEditor === "pending-description" && pendingEditor) {
4627:         const rich = extractRichSummary(editor);
4628:         pendingEditor.richDescription = rich;
4629:         pendingEditor.description = plainTextFromRich(rich);
4630:         setStatus("Unsaved local changes.");
4631:         return true;
4632:     }
4633:
4634:     if (!project) return true;
4635:
4636:     const rich = extractRichSummary(editor);
4637:     const plain = plainTextFromRich(rich);
4638:
4639:     if (wordCount(plain) > 1000) {
4640:         setStatus("Overview is limited to 1000 words. Shorten the text before saving.");
4641:         return false;
4642:     }
4643:
4644:     if (editor.dataset.richEditor === "summary" && !editor.dataset.richPath) {
4645:         project.summaryRich = rich;
4646:         project.summary = plain;
4647:     } else {
4648:         const textPath = editor.dataset.richTextPath;
4649:         const richPath = editor.dataset.richPath;
4650:         if (textPath) setByPath(project, textPath, plain);
4651:         if (richPath) setByPath(project, richPath, rich);
4652:     }
4653:
4654:     setStatus("Unsaved local changes.");
4655:     scheduleAutosave();
4656:     schedulePreviewRender();
4657:     return true;
4658: }

```

saveVisibleRichEditors - lines 4660-4664

Item	Explanation
Source location	Lines 4660-4664 (5 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	function saveVisibleRichEditors() {
Touches	browser/Electron DOM
Calls detected	querySelectorAll, forEach, saveRichEditorToProject
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4660: function saveVisibleRichEditors() {
4661:     document.querySelectorAll("[data-rich-editor]").forEach((editor) => {
4662:         saveRichEditorToProject(editor);
4663:     });
4664: }

```

currentRichBlock - lines 4666-4672

Item	Explanation
Source location	Lines 4666-4672 (7 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function currentRichBlock(editor) {
Touches	browser/Electron DOM

Item	Explanation
Calls detected	getSelection, contains, querySelector, closest
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4666: function currentRichBlock(editor) {
4667:   const selection = window.getSelection();
4668:   let node = selection?.anchorNode;
4669:   if (!node || !editor.contains(node)) return editor.querySelector(".rich-block:last-child");
4670:   if (node.nodeType === Node.TEXT_NODE) node = node.parentElement;
4671:   return node.closest(".rich-block") || editor.querySelector(".rich-block:last-child");
4672: }

```

selectionRangeInsideEditor - lines 4674-4682

Item	Explanation
Source location	Lines 4674-4682 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function selectionRangeInsideEditor(editor) {
Touches	Mostly local calculations or local UI state.
Calls detected	getSelection, getRangeAt, contains, cloneRange
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4674: function selectionRangeInsideEditor(editor) {
4675:   const selection = window.getSelection();
4676:   if (!selection?.rangeCount) return null;
4677:   const range = selection.getRangeAt(0);
4678:   const container = range.commonAncestorContainer.nodeType === Node.TEXT_NODE
4679:     ? range.commonAncestorContainer.parentElement
4680:     : range.commonAncestorContainer;
4681:   return container && editor.contains(container) ? range.cloneRange() : null;
4682: }

```

captureRichSelection - lines 4684-4688

Item	Explanation
Source location	Lines 4684-4688 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function captureRichSelection(editor) {
Touches	Mostly local calculations or local UI state.
Calls detected	selectionRangeInsideEditor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
4684: function captureRichSelection(editor) {
4685:   const range = selectionRangeInsideEditor(editor);
4686:   if (range) activeRichSelectionRange = range;
4687:   return range;
4688: }
```

rememberRichFormattingSelection - lines 4690-4700

Item	Explanation
Source location	Lines 4690-4700 (11 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function rememberRichFormattingSelection(editor = activeSummaryEditor) {
Touches	filesystem
Calls detected	selectionRangeInsideEditor, rangeBelongsToEditor, cloneRange
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
4690: function rememberRichFormattingSelection(editor = activeSummaryEditor) {
4691:   if (!editor) return null;
4692:   const range = selectionRangeInsideEditor(editor)
4693:     || (rangeBelongsToEditor(activeTextSelectionRange, editor) ? activeTextSelectionRange.cloneRange() :
null)
4694:     || (rangeBelongsToEditor(activeRichSelectionRange, editor) ? activeRichSelectionRange.cloneRange() :
null);
4695:   if (!range) return null;
4696:   activeSummaryEditor = editor;
4697:   activeRichSelectionRange = range.cloneRange();
4698:   if (!range.collapsed) activeTextSelectionRange = range.cloneRange();
4699:   return range;
4700: }
```

restoreRichSelection - lines 4702-4713

Item	Explanation
Source location	Lines 4702-4713 (12 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function restoreRichSelection(editor) {
Touches	Mostly local calculations or local UI state.
Calls detected	contains, getSelection, removeAllRanges, addRange, cloneRange, showPersistentTextSelection
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
4702: function restoreRichSelection(editor) {
4703:   if (!activeRichSelectionRange) return false;
4704:   const container = activeRichSelectionRange.commonAncestorContainer.nodeType === Node.TEXT_NODE
4705:     ? activeRichSelectionRange.commonAncestorContainer.parentElement
4706:     : activeRichSelectionRange.commonAncestorContainer;
4707:   if (!container || !editor.contains(container)) return false;
4708:   const selection = window.getSelection();
4709:   selection.removeAllRanges();
4710:   selection.addRange(activeRichSelectionRange.cloneRange());
4711:   showPersistentTextSelection(activeRichSelectionRange);
4712:   return true;
4713: }
```

showPersistentTextSelection - lines 4715-4745

Item	Explanation
Source location	Lines 4715-4745 (31 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function showPersistentTextSelection(range) {
Touches	browser/Electron DOM
Calls detected	set, Highlight, cloneRange, closest, createElement, setAttribute, append, replaceChildren, getClientRects, forEach
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
4715: function showPersistentTextSelection(range) {
4716:   if (!range || range.collapsed) return;
4717:   if (window.CSS?.highlights && typeof window.Highlight === "function") {
4718:     window.CSS.highlights.set(persistentSelectionHighlightName, new
window.Highlight(range.cloneRange()));
4719:     if (persistentSelectionOverlay) persistentSelectionOverlay.hidden = true;
4720:     return;
4721:   }
4722:
4723:   const container = range.commonAncestorContainer.nodeType === Node.TEXT_NODE
4724:     ? range.commonAncestorContainer.parentElement
4725:     : range.commonAncestorContainer;
4726:   const editor = container?.closest?("[data-rich-editor]") || activeSummaryEditor;
4727:   const host = editor?.closest("dialog") || document.body;
4728:   if (!persistentSelectionOverlay) {
4729:     persistentSelectionOverlay = document.createElement("div");
4730:     persistentSelectionOverlay.className = "persistent-selection-overlay";
4731:     persistentSelectionOverlay.setAttribute("aria-hidden", "true");
4732:   }
4733:   if (persistentSelectionOverlay.parentElement !== host) host.append(persistentSelectionOverlay);
4734:   persistentSelectionOverlay.replaceChildren();
4735:   [...range.getClientRects()].forEach((rect) => {
4736:     if (rect.width < 1 || rect.height < 1) return;
4737:     const marker = document.createElement("span");
4738:     marker.style.left = `${rect.left}px`;
4739:     marker.style.top = `${rect.top}px`;
4740:     marker.style.width = `${rect.width}px`;
4741:     marker.style.height = `${rect.height}px`;
4742:     persistentSelectionOverlay.append(marker);
4743:   });
4744:   persistentSelectionOverlay.hidden = !persistentSelectionOverlay.childElementCount;
4745: }
```

clearPersistentTextSelection - lines 4747-4756

Item	Explanation
Source location	Lines 4747-4756 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function clearPersistentTextSelection(clearRanges = false) {
Touches	Mostly local calculations or local UI state.
Calls detected	replaceChildren
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
4747: function clearPersistentTextSelection(clearRanges = false) {
```

```

4748: window.CSS?.highlights?.delete?.(persistentSelectionHighlightName);
4749: if (persistentSelectionOverlay) {
4750:   persistentSelectionOverlay.replaceChildren();
4751:   persistentSelectionOverlay.hidden = true;
4752: }
4753: if (!clearRanges) return;
4754: activeRichSelectionRange = null;
4755: activeTextSelectionRange = null;
4756: }

```

refreshPersistentSelectionHighlight - lines 4758-4765

Item	Explanation
Source location	Lines 4758-4765 (8 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function refreshPersistentSelectionHighlight() {
Touches	Mostly local calculations or local UI state.
Calls detected	rangeBelongsToEditor, showPersistentTextSelection
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4758: function refreshPersistentSelectionHighlight() {
4759:   const range = rangeBelongsToEditor(activeTextSelectionRange, activeSummaryEditor)
4760:     ? activeTextSelectionRange
4761:     : rangeBelongsToEditor(activeRichSelectionRange, activeSummaryEditor)
4762:     ? activeRichSelectionRange
4763:     : null;
4764:   if (range) showPersistentTextSelection(range);
4765: }

```

rangeBelongsToEditor - lines 4767-4773

Item	Explanation
Source location	Lines 4767-4773 (7 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function rangeBelongsToEditor(range, editor) {
Touches	Mostly local calculations or local UI state.
Calls detected	contains
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4767: function rangeBelongsToEditor(range, editor) {
4768:   if (!range || range.collapsed || !editor) return false;
4769:   const container = range.commonAncestorContainer.nodeType === Node.TEXT_NODE
4770:     ? range.commonAncestorContainer.parentElement
4771:     : range.commonAncestorContainer;
4772:   return Boolean(container && editor.contains(container));
4773: }

```

editorFromRange - lines 4775-4781

Item	Explanation
Source location	Lines 4775-4781 (7 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function editorFromRange(range) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4775: function editorFromRange(range) {
4776:   if (!range || range.collapsed) return null;
4777:   const container = range.commonAncestorContainer.nodeType === Node.TEXT_NODE
4778:     ? range.commonAncestorContainer.parentElement
4779:     : range.commonAncestorContainer;
4780:   return container?.closest?("[data-rich-editor]") || null;
4781: }

```

pointInsideRange - lines 4783-4793

Item	Explanation
Source location	Lines 4783-4793 (11 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function pointInsideRange(range, x, y) {
Touches	Mostly local calculations or local UI state.
Calls detected	getClientRects, some
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4783: function pointInsideRange(range, x, y) {
4784:   if (!range || range.collapsed) return false;
4785:   return [...range.getClientRects()].some((rect) =>
4786:     rect.width > 0 &&
4787:     rect.height > 0 &&
4788:     x >= rect.left &&
4789:     x <= rect.right &&
4790:     y >= rect.top &&
4791:     y <= rect.bottom
4792:   );
4793: }

```

richEditorFromContextEvent - lines 4795-4805

Item	Explanation
Source location	Lines 4795-4805 (11 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function richEditorFromContextEvent(event) {
Touches	Mostly local calculations or local UI state.
Calls detected	rangeBelongsToEditor, pointInsideRange, editorFromRange

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4795: function richEditorFromContextEvent(event) {
4796:   const directEditor = event.target.closest?("[data-rich-editor]");
4797:   if (directEditor) return directEditor;
4798:   const savedRange = rangeBelongsToEditor(activeTextSelectionRange, activeSummaryEditor)
4799:     ? activeTextSelectionRange
4800:     : activeRichSelectionRange;
4801:   if (pointInsideRange(savedRange, event.clientX, event.clientY)) {
4802:     return editorFromRange(savedRange);
4803:   }
4804:   return null;
4805: }

```

displayRichFontName - lines 4807-4812

Item	Explanation
Source location	Lines 4807-4812 (6 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function displayRichFontName(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	split, replace, trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4807: function displayRichFontName(value = "") {
4808:   return String(value || "")
4809:     .split(",")[0]
4810:     .replace(/^[\'\\"]|[\\'\\"]$/g, "")
4811:     .trim() || "Arial";
4812: }

```

rgbColorToHex - lines 4814-4823

Item	Explanation
Source location	Lines 4814-4823 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function rgbColorToHex(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, toLowerCase, slice, split, map, join, match, min, toString, padStart
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4814: function rgbColorToHex(value = "") {
4815:   const color = String(value || "").trim();
4816:   if (/^#[0-9a-f]{6}$/i.test(color)) return color.toLowerCase();
4817:   if (/^#[0-9a-f]{3}$/i.test(color)) {
4818:     return `#${color.slice(1).split("").map((part) => part + part).join("")}`.toLowerCase();
4819:   }
4820:   const match = color.match(/^rgba?\s*(\d+)\D+(\d+)\D+(\d+)\D+(\d+)/i);
4821:   if (!match) return color;
4822:   return `#${match.slice(1, 4).map((part) => Math.min(255, Number(part)).toString(16).padStart(2,
"0")).join("")}`;
4823: }

```

textNodesInRichSelection - lines 4825-4844

Item	Explanation
Source location	Lines 4825-4844 (20 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function textNodesInRichSelection(range, editor, includeWhitespace = false) {
Touches	browser/Electron DOM
Calls detected	createTreeWalker, acceptNode, trim, closest, intersectsNode, nextNode, push
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 5 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4825: function textNodesInRichSelection(range, editor, includeWhitespace = false) {
4826:   const walker = document.createTreeWalker(editor, NodeFilter.SHOW_TEXT, {
4827:     acceptNode(node) {
4828:       if (!node.textContent || (!includeWhitespace && !node.textContent.trim())) return
NodeFilter.FILTER_REJECT;
4829:       if (node.parentElement?.closest("[contenteditable='false']")) return NodeFilter.FILTER_REJECT;
4830:       try {
4831:         return range.intersectsNode(node) ? NodeFilter.FILTER_ACCEPT : NodeFilter.FILTER_REJECT;
4832:       } catch {
4833:         return NodeFilter.FILTER_REJECT;
4834:       }
4835:     }
4836:   });
4837:   const nodes = [];
4838:   let node = walker.nextNode();
4839:   while (node) {
4840:     nodes.push(node);
4841:     node = walker.nextNode();
4842:   }
4843:   return nodes;
4844: }

```

richBlockFromRange - lines 4846-4853

Item	Explanation
Source location	Lines 4846-4853 (8 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function richBlockFromRange(range, editor) {
Touches	Mostly local calculations or local UI state.
Calls detected	contains
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4846: function richBlockFromRange(range, editor) {
4847:   if (!range || !editor) return null;
4848:   const container = range.startContainer?.nodeType === Node.TEXT_NODE
4849:     ? range.startContainer.parentElement
4850:     : range.startContainer;
4851:   const block = container?.closest?(".rich-block");
4852:   return block && editor.contains(block) ? block : null;
4853: }

```

uniformSelectionValue - lines 4855-4861

Item	Explanation
Source location	Lines 4855-4861 (7 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function uniformSelectionValue(values) {
Touches	filesystem
Calls detected	map, trim, filter, every, toLowerCase
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4855: function uniformSelectionValue(values) {
4856:   const cleanValues = values.map((value) => String(value || "").trim()).filter(Boolean);
4857:   if (!cleanValues.length) return "";
4858:   return cleanValues.every((value) => value.toLowerCase() === cleanValues[0].toLowerCase())
4859:     ? cleanValues[0]
4860:     : "";
4861: }

```

richSelectionStyleState - lines 4863-4871

Item	Explanation
Source location	Lines 4863-4871 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function richSelectionStyleState(range, editor) {
Touches	filesystem
Calls detected	textNodesInRichSelection, map, getComputedStyle, uniformSelectionValue, rgbColorToHex, displayRichFontName, round
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4863: function richSelectionStyleState(range, editor) {
4864:   const textNodes = textNodesInRichSelection(range, editor);
4865:   const styles = textNodes.map((node) => getComputedStyle(node.parentElement || editor));
4866:   return {
4867:     color: uniformSelectionValue(styles.map((style) => rgbColorToHex(style.color))),
4868:     font: uniformSelectionValue(styles.map((style) => displayRichFontName(style.fontFamily))),
4869:     size: uniformSelectionValue(styles.map((style) => String(Math.round(parseFloat(style.fontSize) ||
16))))
4870:   };
4871: }

```

closeSelectionInspectorOptions - lines 4873-4877

Item	Explanation
Source location	Lines 4873-4877 (5 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function closeSelectionInspectorOptions(except = "") {
Touches	browser/Electron DOM
Calls detected	querySelectorAll, forEach
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4873: function closeSelectionInspectorOptions(except = "") {
4874:   textSelectionInspector?.querySelectorAll("[data-selection-options]").forEach((options) => {
4875:     if (options.dataset.selectionOptions !== except) options.hidden = true;
4876:   });
4877: }

```

positionSelectionInspector - lines 4879-4900

Item	Explanation
Source location	Lines 4879-4900 (22 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function positionSelectionInspector(range) {
Touches	Mostly local calculations or local UI state.
Calls detected	getBoundingClientRect, closest, min, max
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4879: function positionSelectionInspector(range) {
4880:   if (!textSelectionInspector || selectionInspectorWasMoved) return;
4881:   const rect = range.getBoundingClientRect();
4882:   const hostDialog = textSelectionInspector.closest("dialog");
4883:   const hostRect = hostDialog?.getBoundingClientRect();
4884:   const bounds = {
4885:     bottom: Math.min(window.innerHeight - 8, (hostRect?.bottom ?? window.innerHeight) - 8),
4886:     left: Math.max(8, (hostRect?.left ?? 0) + 8),
4887:     right: Math.min(window.innerWidth - 8, (hostRect?.right ?? window.innerWidth) - 8),
4888:     top: Math.max(8, (hostRect?.top ?? 0) + 8)
4889:   };
4890:   const inspectorRect = textSelectionInspector.getBoundingClientRect();
4891:   const inspectorWidth = Math.min(inspectorRect.width || 310, bounds.right - bounds.left);
4892:   const inspectorHeight = Math.min(inspectorRect.height || 260, bounds.bottom - bounds.top);
4893:   const left = Math.max(bounds.left, Math.min(rect.left, bounds.right - inspectorWidth));
4894:   const preferredTop = rect.bottom + 10;
4895:   const top = preferredTop + inspectorHeight <= bounds.bottom
4896:     ? preferredTop
4897:     : Math.max(bounds.top, rect.top - inspectorHeight - 10);
4898:   textSelectionInspector.style.left = `${left}px`;
4899:   textSelectionInspector.style.top = `${top}px`;
4900: }

```

showTextSelectionInspector - lines 4902-4924

Item	Explanation
Source location	Lines 4902-4924 (23 source lines).

Item	Explanation
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function showTextSelectionInspector(editor, range) {
Touches	browser/Electron DOM
Calls detected	toString, trim, closest, append, cloneRange, showPersistentTextSelection, richSelectionStyleState, slice, positionSelectionInspector
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4902: function showTextSelectionInspector(editor, range) {
4903:   if (!textSelectionInspector || !editor || !range || range.collapsed) return;
4904:   const selectedText = range.toString().trim();
4905:   if (!selectedText) return;
4906:   const inspectorHost = editor.closest("dialog") || document.body;
4907:   if (textSelectionInspector.parentElement !== inspectorHost)
inspectorHost.append(textSelectionInspector);
4908:   activeSummaryEditor = editor;
4909:   activeRichSelectionRange = range.cloneRange();
4910:   activeTextSelectionRange = range.cloneRange();
4911:   showPersistentTextSelection(range);
4912:   const state = richSelectionStyleState(range, editor);
4913:   selectionTextPreview.textContent = selectedText.length > 120 ? `${selectedText.slice(0, 117)}...` :
selectedText;
4914:   selectionTextPreview.title = selectedText;
4915:   selectionFontInput.value = state.font;
4916:   selectionColorInput.value = state.color;
4917:   selectionSizeInput.value = state.size;
4918:   selectionFontInput.placeholder = state.font ? "" : "Mixed";
4919:   selectionColorInput.placeholder = state.color ? "" : "Mixed";
4920:   selectionSizeInput.placeholder = state.size ? "" : "Mixed";
4921:   if (state.color && /^[0-9a-f]{6}$/i.test(state.color)) selectionColorPicker.value = state.color;
4922:   textSelectionInspector.hidden = false;
4923:   positionSelectionInspector(range);
4924: }

```

hideTextSelectionInspector - lines 4926-4931

Item	Explanation
Source location	Lines 4926-4931 (6 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function hideTextSelectionInspector() {
Touches	Mostly local calculations or local UI state.
Calls detected	closeSelectionInspectorOptions
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4926: function hideTextSelectionInspector() {
4927:   if (!textSelectionInspector) return;
4928:   textSelectionInspector.hidden = true;
4929:   closeSelectionInspectorOptions();
4930:   selectionInspectorWasMoved = false;
4931: }

```

refreshTextSelectionInspector - lines 4933-4953

Item	Explanation
Source location	Lines 4933-4953 (21 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function refreshTextSelectionInspector() {
Touches	Mostly local calculations or local UI state.
Calls detected	matches, getSelection, showTextSelectionInspector, hideTextSelectionInspector, getRangeAt
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4933: function refreshTextSelectionInspector() {
4934:   if (!textSelectionInspector || selectionInspectorPointerInside ||
textSelectionInspector.matches(":focus-within")) return;
4935:   const selection = window.getSelection();
4936:   if (!selection?.rangeCount || selection.isCollapsed) {
4937:     if (activeTextSelectionRange && !activeTextSelectionRange.collapsed && activeSummaryEditor) {
4938:       showTextSelectionInspector(activeSummaryEditor, activeTextSelectionRange);
4939:       return;
4940:     }
4941:     hideTextSelectionInspector();
4942:     return;
4943:   }
4944:   const range = selection.getRangeAt(0);
4945:   let node = range.commonAncestorContainer;
4946:   if (node.nodeType === Node.TEXT_NODE) node = node.parentElement;
4947:   const editor = node?.closest?("[data-rich-editor]");
4948:   if (!editor) {
4949:     hideTextSelectionInspector();
4950:     return;
4951:   }
4952:   showTextSelectionInspector(editor, range);
4953: }

```

scheduleTextSelectionInspectorRefresh - lines 4955-4958

Item	Explanation
Source location	Lines 4955-4958 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function scheduleTextSelectionInspectorRefresh(delay = 0) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4955: function scheduleTextSelectionInspectorRefresh(delay = 0) {
4956:   clearTimeout(selectionInspectorRefreshTimer);
4957:   selectionInspectorRefreshTimer = setTimeout(refreshTextSelectionInspector, delay);
4958: }

```

applySelectionInspectorFormat - lines 4960-4990

Item	Explanation
Source location	Lines 4960-4990 (31 source lines).

Item	Explanation
Purpose	Part of the private builder frontend and editing experience.
Declaration	function applySelectionInspectorFormat(kind, rawValue) {
Touches	filesystem
Calls detected	cloneRange, cleanFontFamily, applyRichInlineCommand, normalizeTextColor, rgbColorToHex, normalizeFontPx, closeSelectionInspectorOptions, requestAnimationFrame, showTextSelectionInspector
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4960: function applySelectionInspectorFormat(kind, rawValue) {
4961:   if (!activeSummaryEditor || !activeTextSelectionRange) return;
4962:   activeRichSelectionRange = activeTextSelectionRange.cloneRange();
4963:   if (kind === "font") {
4964:     const font = cleanFontFamily(rawValue);
4965:     if (!font) return;
4966:     selectionFontInput.value = font;
4967:     applyRichInlineCommand(activeSummaryEditor, "fontName", font);
4968:   }
4969:   if (kind === "color") {
4970:     const color = normalizeTextColor(rawValue);
4971:     if (!color) return;
4972:     const normalized = rgbColorToHex(color);
4973:     selectionColorInput.value = normalized;
4974:     if (/^[0-9a-f]{6}$/i.test(normalized)) selectionColorPicker.value = normalized;
4975:     applyRichInlineCommand(activeSummaryEditor, "foreColor", normalized);
4976:   }
4977:   if (kind === "size") {
4978:     const size = normalizeFontPx(rawValue);
4979:     if (!size) return;
4980:     selectionSizeInput.value = size;
4981:     applyRichInlineCommand(activeSummaryEditor, "fontSize", size);
4982:   }
4983:   closeSelectionInspectorOptions();
4984:   requestAnimationFrame(() => {
4985:     if (activeRichSelectionRange && activeSummaryEditor) {
4986:       activeTextSelectionRange = activeRichSelectionRange.cloneRange();
4987:       showTextSelectionInspector(activeSummaryEditor, activeRichSelectionRange);
4988:     }
4989:   });
4990: }

```

renderSelectionInspectorOptions - lines 4992-5007

Item	Explanation
Source location	Lines 4992-5007 (16 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderSelectionInspectorOptions() {
Touches	browser/Electron DOM
Calls detected	querySelector, map, escapeHtml, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

4992: function renderSelectionInspectorOptions() {
4993:   const fontOptions = textSelectionInspector?.querySelector("[data-selection-options='font']");
4994:   const sizeOptions = textSelectionInspector?.querySelector("[data-selection-options='size']");
4995:   if (fontOptions) {
4996:     fontOptions.innerHTML = commonRichFonts.map((font) => `<button type="button" data-selection-
value="font" data-value="${escapeHtml(font)}" style="font-family:

```



```

${escapeHtml(font)}">${escapeHtml(font)}</button>`).join("");
4997:   }
4998:   if (sizeOptions) {
4999:     sizeOptions.innerHTML = commonRichSizes.map((size) => `<button type="button" data-selection-
value="size" data-value="${size}">${size} px</button>`).join("");
5000:   }
5001:   if (selectionColorSwatches) {
5002:     selectionColorSwatches.innerHTML = commonRichColors.map((color) => `<button type="button" data-
selection-value="color" data-value="${color}" style="--selection-swatch: ${color}" aria-label="Use
${color}"></button>`).join("");
5003:   }
5004:   if (richContextColorSwatches) {
5005:     richContextColorSwatches.innerHTML = commonRichColors.map((color) => `<button type="button" data-
rich-color-value="${color}" style="--selection-swatch: ${color}" aria-label="Use ${color}"></button>`).join("");
5006:   }
5007: }

```

applyStyleToRichSelection - lines 5009-5055

Item	Explanation
Source location	Lines 5009-5055 (47 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function applyStyleToRichSelection(editor, range, command, value = "") {
Touches	browser/Electron DOM, filesystem
Calls detected	textNodesInRichSelection, every, getComputedStyle, includes, forEach, max, min, splitText, createElement, cleanFontFamily, normalizeTextColor, normalizeFontPx
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5009: function applyStyleToRichSelection(editor, range, command, value = "") {
5010:   const nodes = textNodesInRichSelection(range, editor, true);
5011:   if (!nodes.length) return null;
5012:   const makeBold = command === "bold"
5013:     ? !nodes.every((node) => Number.parseInt(getComputedStyle(node.parentElement || editor).fontWeight,
10) >= 600)
5014:     : false;
5015:   const makeItalic = command === "italic"
5016:     ? !nodes.every((node) => getComputedStyle(node.parentElement || editor).fontStyle === "italic")
5017:     : false;
5018:   const makeUnderline = command === "underline"
5019:     ? !nodes.every((node) => getComputedStyle(node.parentElement ||
editor).textDecorationLine.includes("underline"))
5020:     : false;
5021:   const formattedNodes = [];
5022:
5023:   nodes.forEach((node) => {
5024:     const length = node.textContent?.length || 0;
5025:     let start = node === range.startContainer ? range.startOffset : 0;
5026:     let end = node === range.endContainer ? range.endOffset : length;
5027:     start = Math.max(0, Math.min(start, length));
5028:     end = Math.max(start, Math.min(end, length));
5029:     if (start === end) return;
5030:
5031:     if (end < length) node.splitText(end);
5032:     const selectedNode = start > 0 ? node.splitText(start) : node;
5033:     const wrapper = document.createElement("span");
5034:     wrapper.className = "rich-inline-style";
5035:     wrapper.style.display = "inline";
5036:     if (command === "fontName") wrapper.style.fontFamily = cleanFontFamily(value);
5037:     if (command === "foreColor") wrapper.style.color = normalizeTextColor(value);
5038:     if (command === "fontSize") wrapper.style.fontSize = `${normalizeFontPx(value)}px`;
5039:     if (command === "bold") wrapper.style.fontWeight = makeBold ? "700" : "400";
5040:     if (command === "italic") wrapper.style.fontStyle = makeItalic ? "italic" : "normal";
5041:     if (command === "underline") wrapper.style.textDecoration = makeUnderline ? "underline" : "none";
5042:     selectedNode.replaceWith(wrapper);
5043:     wrapper.append(selectedNode);
5044:     formattedNodes.push(selectedNode);
5045:   });
5046:
5047:   if (!formattedNodes.length) return null;
5048:   normalizeInlineStyleSpans(editor);
5049:   const nextRange = document.createRange();
5050:   const firstNode = formattedNodes[0];
5051:   const lastNode = formattedNodes[formattedNodes.length - 1];

```

```

5052:   nextRange.setStart(firstNode, 0);
5053:   nextRange.setEnd(lastNode, lastNode.textContent?.length || 0);
5054:   return nextRange;
5055: }

```

applyRichInlineCommand - lines 5057-5103

Item	Explanation
Source location	Lines 5057-5103 (47 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function applyRichInlineCommand(editor, command, value = "") {
Touches	browser/Electron DOM, filesystem
Calls detected	selectionRangeInsideEditor, cloneRange, rangeBelongsToEditor, applyStyleToRichSelection, showPersistentTextSelection, richBlockFromRange, selectedRichBlock, currentRichBlock, setStatus, selectRichBlock, updateCurrentTextBlock, saveRichEditorToProject
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5057: function applyRichInlineCommand(editor, command, value = "") {
5058:   if (!editor) return;
5059:   const liveRange = selectionRangeInsideEditor(editor);
5060:   const savedRange = liveRange && !liveRange.collapsed
5061:     ? liveRange.cloneRange()
5062:     : rangeBelongsToEditor(activeTextSelectionRange, editor)
5063:       ? activeTextSelectionRange.cloneRange()
5064:       : rangeBelongsToEditor(activeRichSelectionRange, editor)
5065:         ? activeRichSelectionRange.cloneRange()
5066:         : null;
5067:
5068:   let appliedToSelection = false;
5069:   if (savedRange) {
5070:     const nextRange = applyStyleToRichSelection(editor, savedRange, command, value);
5071:     if (nextRange) {
5072:       activeRichSelectionRange = nextRange.cloneRange();
5073:       activeTextSelectionRange = nextRange.cloneRange();
5074:       showPersistentTextSelection(nextRange);
5075:       appliedToSelection = true;
5076:     }
5077:   }
5078:
5079:   if (!appliedToSelection) {
5080:     const block = richBlockFromRange(savedRange, editor) || selectedRichBlock(editor) ||
currentRichBlock(editor);
5081:     if (!block || block.dataset.type !== "paragraph") {
5082:       setStatus("Highlight text or click inside a text line before formatting.");
5083:       return;
5084:     }
5085:     selectRichBlock(block);
5086:     if (command === "fontName") updateCurrentTextBlock(editor, { fontFamily: value });
5087:     if (command === "foreColor") updateCurrentTextBlock(editor, { color: value });
5088:     if (command === "fontSize") updateCurrentTextBlock(editor, { fontPx: value });
5089:     if (command === "bold") updateCurrentTextBlock(editor, { bold: block.dataset.bold !== "true" });
5090:     if (command === "italic") {
5091:       updateCurrentTextBlock(editor, { italic: block.dataset.italic !== "true" });
5092:     }
5093:     if (command === "underline") {
5094:       updateCurrentTextBlock(editor, { underline: block.dataset.underline !== "true" });
5095:     }
5096:     saveRichEditorToProject(editor);
5097:     setStatus("Formatting applied and saved locally for the current text line.");
5098:     return;
5099:   }
5100:
5101:   saveRichEditorToProject(editor);
5102:   setStatus("Unsaved local changes. Click the section save button to keep this formatting.");
5103: }

```

selectRichBlock - lines 5105-5113

Item	Explanation
Source location	Lines 5105-5113 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function selectRichBlock(block) {
Touches	browser/Electron DOM
Calls detected	querySelectorAll, forEach, remove, add, closest, indexOf
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5105: function selectRichBlock(block) {
5106:   document.querySelectorAll(".rich-block.selected").forEach((item) => item.classList.remove("selected"));
5107:   activeSummaryBlock = block || null;
5108:   if (block) {
5109:     block.classList.add("selected");
5110:     const editor = block.closest("[data-rich-editor]");
5111:     if (editor) editor.dataset.activeRichBlockIndex = String([...editor.children].indexOf(block));
5112:   }
5113: }

```

selectedRichBlock - lines 5115-5123

Item	Explanation
Source location	Lines 5115-5123 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function selectedRichBlock(editor) {
Touches	browser/Electron DOM
Calls detected	isInteger, matches, contains, currentRichBlock
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5115: function selectedRichBlock(editor) {
5116:   if (!editor) return null;
5117:   const index = Number(editor.dataset.activeRichBlockIndex);
5118:   if (Number.isInteger(index) && index >= 0 && editor.children[index]?.matches(".rich-block")) {
5119:     return editor.children[index];
5120:   }
5121:   if (activeSummaryBlock && editor.contains(activeSummaryBlock)) return activeSummaryBlock;
5122:   return currentRichBlock(editor);
5123: }

```

syncRichContextMenuControls - lines 5125-5144

Item	Explanation
Source location	Lines 5125-5144 (20 source lines).
Purpose	Keeps two places aligned, such as local data and target files.
Declaration	function syncRichContextMenuControls(block) {
Touches	browser/Electron DOM, filesystem
Calls detected	getComputedStyle, displayRichFontName, round, normalizeTextColor, restoreRichSelection, closest, queryCommandState, some, setAttribute

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5125: function syncRichContextMenuControls(block) {
5126:   if (!block || block.dataset.type !== "paragraph") return;
5127:
5128:   const computed = getComputedStyle(block);
5129:   const family = displayRichFontName(computed.fontFamily) || "Arial";
5130:   const size = String(Math.round(parseFloat(computed.fontSize) || 16));
5131:   const color = normalizeTextColor(block.dataset.color || "") || "#17202a";
5132:   restoreRichSelection(block.closest("[data-rich-editor]"));
5133:   const bold = document.queryCommandState("bold") || block.dataset.bold === "true";
5134:
5135:   if (richFontSelect) {
5136:     richFontSelect.value = [...richFontSelect.options].some((option) => option.value === family ||
option.textContent === family)
5137:       ? family
5138:       : "Arial";
5139:   }
5140:
5141:   if (richFontSize) richFontSize.value = size;
5142:   if (richColorInput) richColorInput.value = color;
5143:   if (richBoldButton) richBoldButton.setAttribute("aria-pressed", bold ? "true" : "false");
5144: }

```

focusTextBlock - lines 5146-5158

Item	Explanation
Source location	Lines 5146-5158 (13 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function focusTextBlock(block) {
Touches	browser/Electron DOM
Calls detected	selectRichBlock, closest, focus, createRange, selectNodeContents, collapse, getSelection, removeAllRanges, addRange, captureRichSelection
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5146: function focusTextBlock(block) {
5147:   if (!block) return;
5148:   selectRichBlock(block);
5149:   const editor = block.closest("[data-rich-editor]");
5150:   editor?.focus();
5151:   const range = document.createRange();
5152:   range.selectNodeContents(block);
5153:   range.collapse(false);
5154:   const selection = window.getSelection();
5155:   selection.removeAllRanges();
5156:   selection.addRange(range);
5157:   captureRichSelection(editor);
5158: }

```

setTextBlockCaretOffset - lines 5160-5174

Item	Explanation
Source location	Lines 5160-5174 (15 source lines).
Purpose	Part of the private builder frontend and editing experience.

Item	Explanation
Declaration	function setTextBlockCaretOffset(block, offset) {
Touches	browser/Electron DOM
Calls detected	selectRichBlock, closest, focus, max, min, createRange, setStart, collapse, getSelection, removeAllRanges, addRange, captureRichSelection
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5160: function setTextBlockCaretOffset(block, offset) {
5161:   if (!block) return;
5162:   selectRichBlock(block);
5163:   const editor = block.closest("[data-rich-editor]");
5164:   editor?.focus();
5165:   const textNode = block.firstChild || block;
5166:   const safeOffset = Math.max(0, Math.min(offset, textNode.textContent?.length || 0));
5167:   const range = document.createRange();
5168:   range.setStart(textNode, safeOffset);
5169:   range.collapse(true);
5170:   const selection = window.getSelection();
5171:   selection.removeAllRanges();
5172:   selection.addRange(range);
5173:   captureRichSelection(editor);
5174: }

```

textBlockCaretOffset - lines 5176-5183

Item	Explanation
Source location	Lines 5176-5183 (8 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function textBlockCaretOffset(block) {
Touches	Mostly local calculations or local UI state.
Calls detected	getSelection, contains, getRangeAt, cloneRange, selectNodeContents, setEnd, toString
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5176: function textBlockCaretOffset(block) {
5177:   const selection = window.getSelection();
5178:   if (!selection?.rangeCount || !block.contains(selection.anchorNode)) return block.textContent.length;
5179:   const range = selection.getRangeAt(0).cloneRange();
5180:   range.selectNodeContents(block);
5181:   range.setEnd(selection.anchorNode, selection.anchorOffset);
5182:   return range.toString().length;
5183: }

```

matchingTextBlock - lines 5185-5198

Item	Explanation
Source location	Lines 5185-5198 (14 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function matchingTextBlock(block, text = "") {
Touches	browser/Electron DOM, filesystem

Item	Explanation
Calls detected	createRichTextBlock
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5185: function matchingTextBlock(block, text = "") {
5186:     return createRichTextBlock(
5187:         text,
5188:         block?.dataset.fontSize || "normal",
5189:         block?.dataset.align || "left",
5190:         block?.dataset.fontFamily || "Arial",
5191:         block?.dataset.fontPx || "",
5192:         block?.dataset.color || "",
5193:         block?.dataset.bold === "true",
5194:         "",
5195:         block?.dataset.italic === "true",
5196:         block?.dataset.underline === "true"
5197:     );
5198: }

```

richInsertionRange - lines 5200-5221

Item	Explanation
Source location	Lines 5200-5221 (22 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function richInsertionRange(editor) {
Touches	browser/Electron DOM
Calls detected	selectionRangeInsideEditor, collapse, restoreRichSelection, querySelectorAll, at, createRange, selectNodeContents
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5200: function richInsertionRange(editor) {
5201:     const currentRange = selectionRangeInsideEditor(editor);
5202:     if (currentRange) {
5203:         currentRange.collapse(true);
5204:         return currentRange;
5205:     }
5206:     if (restoreRichSelection(editor)) {
5207:         const restored = selectionRangeInsideEditor(editor);
5208:         if (restored) {
5209:             restored.collapse(true);
5210:             return restored;
5211:         }
5212:     }
5213:     const lastText = [...editor.querySelectorAll(".rich-text-block")].at(-1);
5214:     if (lastText) {
5215:         const range = document.createRange();
5216:         range.selectNodeContents(lastText);
5217:         range.collapse(false);
5218:         return range;
5219:     }
5220:     return null;
5221: }

```

insertRichBlockAtCursor - lines 5223-5257

Item	Explanation
Source location	Lines 5223-5257 (35 source lines).

Item	Explanation
Purpose	Part of the private builder frontend and editing experience.
Declaration	function insertRichBlockAtCursor(editor, blockElement, insertionRange = null) {
Touches	browser/Electron DOM
Calls detected	cloneRange, richInsertionRange, selectedRichBlock, contains, collapse, createRange, setStart, setEnd, extractContents, after, matchingTextBlock, append
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5223: function insertRichBlockAtCursor(editor, blockElement, insertionRange = null) {
5224:   const range = insertionRange?.cloneRange() || richInsertionRange(editor);
5225:   const rangeNode = range?.startContainer?.nodeType === Node.TEXT_NODE
5226:     ? range.startContainer.parentElement
5227:     : range?.startContainer;
5228:   const current = rangeNode?.closest?(".rich-block") || selectedRichBlock(editor);
5229:   let nextText = null;
5230:
5231:   if (range && current?.dataset.type === "paragraph" && current.contains(range.startContainer)) {
5232:     range.collapse(true);
5233:     const tail = document.createRange();
5234:     tail.setStart(range.startContainer, range.startOffset);
5235:     tail.setEnd(current, current.childNodes.length);
5236:     const trailingContent = tail.extractContents();
5237:     current.after(blockElement);
5238:     nextText = matchingTextBlock(current, "");
5239:     nextText.append(trailingContent);
5240:     blockElement.after(nextText);
5241:   } else if (range?.startContainer === editor) {
5242:     const reference = editor.children[range.startOffset] || null;
5243:     editor.insertBefore(blockElement, reference);
5244:     nextText = createRichTextBlock("");
5245:     blockElement.after(nextText);
5246:   } else if (current) {
5247:     current.after(blockElement);
5248:     nextText = matchingTextBlock(current, "");
5249:     blockElement.after(nextText);
5250:   } else {
5251:     editor.append(blockElement);
5252:     nextText = createRichTextBlock("");
5253:     editor.append(nextText);
5254:   }
5255:
5256:   focusTextBlock(nextText);
5257: }

```

insertRichBlockAfterCursor - lines 5259-5261

Item	Explanation
Source location	Lines 5259-5261 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function insertRichBlockAfterCursor(editor, blockElement) {
Touches	Mostly local calculations or local UI state.
Calls detected	insertRichBlockAtCursor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5259: function insertRichBlockAfterCursor(editor, blockElement) {
5260:   insertRichBlockAtCursor(editor, blockElement);

```

5261: }

cleanupEmptyRichTextBlocks - lines 5263-5275

Item	Explanation
Source location	Lines 5263-5275 (13 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function cleanupEmptyRichTextBlocks(editor) {
Touches	browser/Electron DOM, authentication/security data
Calls detected	querySelectorAll, forEach, richElementPlainText, remove, querySelector, append, createRichTextBlock
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
5263: function cleanupEmptyRichTextBlocks(editor) {
5264:   if (!editor) return;
5265:   [...editor.querySelectorAll(":scope > .rich-text-block")].forEach((block) => {
5266:     const hasVisibleText = richElementPlainText(block);
5267:     if (hasVisibleText) return;
5268:     const previous = block.previousElementSibling;
5269:     const next = block.nextElementSibling;
5270:     const keepTypingBlockAfterImage = previous?.dataset.type === "image" && !next;
5271:     const isOnlyBlock = editor.children.length <= 1;
5272:     if (!isOnlyBlock && !keepTypingBlockAfterImage) block.remove();
5273:   });
5274:   if (!editor.querySelector(":scope > .rich-block")) editor.append(createRichTextBlock(""));
5275: }
```

normalizeRichEditorStructure - lines 5277-5307

Item	Explanation
Source location	Lines 5277-5307 (31 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeRichEditorStructure(editor) {
Touches	browser/Electron DOM, filesystem, authentication/security data
Calls detected	forEach, trim, remove, createRichTextBlock, replaceWith, matches, sanitizeRichInlineHtml, querySelectorAll, add, removeAttribute, applyRichTextBlockStyle, querySelector
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
5277: function normalizeRichEditorStructure(editor) {
5278:   if (!editor) return;
5279:   [...editor.childNodes].forEach((node) => {
5280:     if (node.nodeType === Node.TEXT_NODE) {
5281:       if (!node.textContent.trim()) {
5282:         node.remove();
5283:         return;
5284:       }
5285:       const paragraph = createRichTextBlock(node.textContent);
5286:       node.replaceWith(paragraph);
5287:       return;
5288:     }
5289:
5290:     if (node.nodeType !== Node.ELEMENT_NODE || node.matches(".rich-block")) return;
5291:     const paragraph = createRichTextBlock("");
5292:     paragraph.innerHTML = sanitizeRichInlineHtml(node.innerHTML || node.textContent || "");
5293:     node.replaceWith(paragraph);
5294:   });
5295: }
```



```

5294: });
5295:
5296: editor.querySelectorAll(".rich-text-block").forEach((block) => {
5297:   block.dataset.type = "paragraph";
5298:   block.dataset.fontSize ||= "normal";
5299:   block.dataset.align ||= "left";
5300:   block.dataset.fontFamily ||= "Arial";
5301:   block.classList.add("rich-block");
5302:   block.removeAttribute("contenteditable");
5303:   applyRichTextBlockStyle(block);
5304: });
5305:
5306: if (!editor.querySelector(":scope > .rich-block")) editor.append(createRichTextBlock(""));
5307: }

```

richElementPlainText - lines 5309-5313

Item	Explanation
Source location	Lines 5309-5313 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function richElementPlainText(element) {
Touches	browser/Electron DOM
Calls detected	cloneNode, querySelectorAll, forEach, replaceWith, trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5309: function richElementPlainText(element) {
5310:   const cloneElement = element.cloneNode(true);
5311:   cloneElement.querySelectorAll("br").forEach((lineBreak) => lineBreak.replaceWith("\n"));
5312:   return String(cloneElement.textContent || "").trim();
5313: }

```

updateCurrentTextBlock - lines 5315-5357

Item	Explanation
Source location	Lines 5315-5357 (43 source lines).
Purpose	Updates UI, state, cache, or stored data.
Declaration	function updateCurrentTextBlock(editor, updates) {
Touches	browser/Electron DOM, filesystem
Calls detected	selectedRichBlock, remove, add, call, cleanFontFamily, normalizeFontPx, normalizeTextColor, applyRichTextBlockStyle, saveRichEditorToProject
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5315: function updateCurrentTextBlock(editor, updates) {
5316:   const block = selectedRichBlock(editor);
5317:   if (!block || block.dataset.type !== "paragraph") return;
5318:
5319:   if (updates.fontSize) {
5320:     block.dataset.fontSize = updates.fontSize;
5321:     block.classList.remove("rich-text-small", "rich-text-normal", "rich-text-large");
5322:     block.classList.add(`rich-text-${updates.fontSize}`);
5323:   }
5324:
5325:   if (updates.align) {
5326:     block.dataset.align = updates.align;

```

```

5327:         block.classList.remove("text-left", "text-center", "text-right");
5328:         block.classList.add(`text-${updates.align}`);
5329:     }
5330:
5331:     if (Object.prototype.hasOwnProperty.call(updates, "fontFamily")) {
5332:         block.dataset.fontFamily = cleanFontFamily(updates.fontFamily);
5333:     }
5334:
5335:     if (Object.prototype.hasOwnProperty.call(updates, "fontPx")) {
5336:         block.dataset.fontPx = normalizeFontPx(updates.fontPx);
5337:     }
5338:
5339:     if (Object.prototype.hasOwnProperty.call(updates, "color")) {
5340:         block.dataset.color = normalizeTextColor(updates.color);
5341:     }
5342:
5343:     if (Object.prototype.hasOwnProperty.call(updates, "bold")) {
5344:         block.dataset.bold = updates.bold ? "true" : "false";
5345:     }
5346:
5347:     if (Object.prototype.hasOwnProperty.call(updates, "italic")) {
5348:         block.dataset.italic = updates.italic ? "true" : "false";
5349:     }
5350:
5351:     if (Object.prototype.hasOwnProperty.call(updates, "underline")) {
5352:         block.dataset.underline = updates.underline ? "true" : "false";
5353:     }
5354:
5355:     applyRichTextBlockStyle(block);
5356:     saveRichEditorToProject(editor);
5357: }

```

extractRichSummary - lines 5359-5416

Item	Explanation
Source location	Lines 5359-5416 (58 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function extractRichSummary(editor) {
Touches	browser/Electron DOM, filesystem
Calls detected	normalizeRichEditorStructure, normalizeInlineStyleSpans, querySelectorAll, map, normalizeCropAspect, normalizeCropPosition, normalizeCropZoom, unwrapFormula, normalizeCodeText, querySelector, normalizeCodeLanguage, detectCodeLanguage
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 8 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5359: function extractRichSummary(editor) {
5360:     normalizeRichEditorStructure(editor);
5361:     normalizeInlineStyleSpans(editor);
5362:     const blocks = [...editor.querySelectorAll(".rich-block")].map((element) => {
5363:         const align = element.dataset.align || "left";
5364:         if (element.dataset.type === "image") {
5365:             return {
5366:                 align,
5367:                 caption: element.dataset.caption || "",
5368:                 cropAspect: normalizeCropAspect(element.dataset.cropAspect),
5369:                 cropX: normalizeCropPosition(element.dataset.cropX),
5370:                 cropY: normalizeCropPosition(element.dataset.cropY),
5371:                 cropZoom: normalizeCropZoom(element.dataset.cropZoom),
5372:                 display: element.dataset.display === "download" ? "download" : "show",
5373:                 source: element.dataset.source || "local",
5374:                 title: element.dataset.title || "",
5375:                 type: "image",
5376:                 url: element.dataset.url || ""
5377:             };
5378:         }
5379:         if (element.dataset.type === "formula") {
5380:             return {
5381:                 align,
5382:                 formula: unwrapFormula(element.dataset.formula || element.textContent || ""),
5383:                 type: "formula"
5384:             };
5385:         }
5386:         if (element.dataset.type === "code") {
5387:             const code = normalizeCodeText(element.dataset.code || element.querySelector("code")?.textContent

```

```

|| "", element.dataset.pasteMode || "source");
5388:   if (!code) return null;
5389:   return {
5390:     code,
5391:     language: normalizeCodeLanguage(element.dataset.language || detectCodeLanguage(code)),
5392:     pasteMode: normalizeCodePasteMode(element.dataset.pasteMode),
5393:     type: "code"
5394:   };
5395: }
5396: const text = richElementPlainText(element);
5397: if (!text) return null;
5398: if (isFormulaOnly(text)) {
5399:   return { align: element.dataset.align || "center", formula: unwrapFormula(text), type: "formula" };
5400: }
5401: return {
5402:   align,
5403:   bold: element.dataset.bold === "true",
5404:   color: element.dataset.color || "",
5405:   fontFamily: element.dataset.fontFamily || "",
5406:   fontPx: element.dataset.fontPx || "",
5407:   fontSize: element.dataset.fontSize || "normal",
5408:   html: sanitizeRichInlineHtml(element.innerHTML),
5409:   italic: element.dataset.italic === "true",
5410:   text,
5411:   underline: element.dataset.underline === "true",
5412:   type: "paragraph"
5413: };
5414: }).filter(Boolean);
5415: return { blocks };
5416: }

```

uploadSummaryImageFile - lines 5418-5458

Item	Explanation
Source location	Lines 5418-5458 (41 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	async function uploadSummaryImageFile(file, options = {}) {
Touches	filesystem, network/API requests
Calls detected	selectedProject, isAllowedUpload, setStatus, call, withExtension, readFileAsDataURL, fetch, stringify, json, normalizeCropAspect, normalizeCropPosition, normalizeCropZoom
API endpoints	/api/upload
Storage keys	No local/session storage keys detected.
Async/return clues	3 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5418: async function uploadSummaryImageFile(file, options = {}) {
5419:   const project = selectedProject();
5420:   const projectId = options.projectId || project?.id;
5421:   if (!projectId || !file) return null;
5422:   const validation = isAllowedUpload("media", file.name, file.type);
5423:   if (!validation.ok) {
5424:     setStatus(validation.message);
5425:     return null;
5426:   }
5427:   const displayTitle = Object.prototype.hasOwnProperty.call(options, "title") ? options.title :
file.name;
5428:   const fileName = withExtension(displayTitle || file.name, file.name);
5429:   const data = await readFileAsDataURL(file);
5430:   const response = await fetch("/api/upload", {
5431:     method: "POST",
5432:     headers: { "Content-Type": "application/json" },
5433:     body: JSON.stringify({
5434:       projectId,
5435:       section: options.folder || "summary",
5436:       fileName,
5437:       data
5438:     })
5439:   });
5440:   const result = await response.json();
5441:   if (!response.ok) {
5442:     setStatus(result.error || "Image upload failed.");
5443:     return null;
5444:   }
5445:   return {
5446:     align: options.align || "center",
5447:     caption: options.caption || "",

```

```

5448:     cropAspect: normalizeCropAspect(options.cropAspect),
5449:     cropX: normalizeCropPosition(options.cropX),
5450:     cropY: normalizeCropPosition(options.cropY),
5451:     cropZoom: normalizeCropZoom(options.cropZoom),
5452:     display: options.display === "download" ? "download" : "show",
5453:     source: "local",
5454:     title: displayTitle || "",
5455:     type: "image",
5456:     url: result.url
5457:   });
5458: }

```

updateSummaryImageDialogVisibility - lines 5460-5465

Item	Explanation
Source location	Lines 5460-5465 (6 source lines).
Purpose	Updates UI, state, cache, or stored data.
Declaration	function updateSummaryImageDialogVisibility() {
Touches	browser/Electron DOM
Calls detected	querySelector
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5460: function updateSummaryImageDialogVisibility() {
5461:   const source = summaryImageSource?.value || "local";
5462:   const isLocal = source === "local";
5463:   document.querySelector(".summary-image-local-field").hidden = !isLocal;
5464:   document.querySelector(".summary-image-url-field").hidden = isLocal;
5465: }

```

openSummaryImageDialog - lines 5467-5546

Item	Explanation
Source location	Lines 5467-5546 (80 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	async function openSummaryImageDialog(existingBlock = null, options = {}) {
Touches	browser/Electron DOM, filesystem
Calls detected	querySelector, normalizeCropAspect, normalizeCropZoom, updateSummaryImageDialogVisibility, dialogValue, normalizeLinkTarget, trim, setStatus, urlLooksLikeDirectImage, normalizeCropPosition, displayNameFromUrl, uploadSummaryImageFile
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 7 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5467: async function openSummaryImageDialog(existingBlock = null, options = {}) {
5468:   const heading = summaryImageDialog.querySelector("h2");
5469:   const submitButton = summaryImageDialog.querySelector("button[type='submit']");
5470:   if (heading) heading.textContent = existingBlock ? "Edit image" : "Add image to overview";
5471:   if (submitButton) submitButton.textContent = existingBlock ? "Save image" : "Add image";
5472:   summaryImageSource.value = existingBlock?.dataset.source || "local";
5473:   summaryImageFile.value = "";
5474:   summaryImageUrl.value = existingBlock?.dataset.source && existingBlock.dataset.source !== "local" ?
existingBlock.dataset.url || "" : "";
5475:   summaryImageTitle.value = existingBlock?.dataset.title || "";
5476:   summaryImageCaption.value = existingBlock?.dataset.caption || "";
5477:   summaryImageAlign.value = existingBlock?.dataset.align || "center";
5478:   summaryImageDisplay.value = existingBlock?.dataset.display || "show";

```

```

5479: summaryImageCrop.value = normalizeCropAspect(existingBlock?.dataset.cropAspect);
5480: summaryImageZoom.value = String(normalizeCropZoom(existingBlock?.dataset.cropZoom));
5481: updateSummaryImageDialogVisibility();
5482: const saved = await dialogValue(summaryImageDialog);
5483: if (!saved) return null;
5484: const file = summaryImageFile.files[0];
5485: const source = summaryImageSource.value || "local";
5486: const url = normalizeLinkTarget(summaryImageUrl.value.trim(), { assumeWeb: true });
5487:
5488: if (source !== "local") {
5489:   const existingUrl = existingBlock?.dataset.url || "";
5490:   const nextUrl = url || existingUrl;
5491:   if (!nextUrl) {
5492:     setStatus("Paste an image URL or share link first.");
5493:     return null;
5494:   }
5495:   const display = summaryImageDisplay.value === "download" ? "download" : "show";
5496:   if (display === "show" && !urlLooksLikeDirectImage(nextUrl)) {
5497:     setStatus("Use a direct image URL for visible images, or choose Downloadable file only for share
links and non-image URLs.");
5498:     return null;
5499:   }
5500:   return {
5501:     align: summaryImageAlign.value,
5502:     caption: summaryImageCaption.value.trim(),
5503:     cropAspect: summaryImageCrop.value,
5504:     cropX: normalizeCropPosition(existingBlock?.dataset.cropX),
5505:     cropY: normalizeCropPosition(existingBlock?.dataset.cropY),
5506:     cropZoom: normalizeCropZoom(summaryImageZoom.value),
5507:     display,
5508:     source,
5509:     title: summaryImageTitle.value.trim() || displayNameFromUrl(nextUrl, source === "drive" ? "Google
Drive image" : "Web image"),
5510:     type: "image",
5511:     url: nextUrl
5512:   };
5513: }
5514:
5515: if (!file && !existingBlock) {
5516:   setStatus("Choose an image first.");
5517:   return null;
5518: }
5519: if (!file && existingBlock) {
5520:   return {
5521:     align: summaryImageAlign.value,
5522:     caption: summaryImageCaption.value.trim(),
5523:     cropAspect: summaryImageCrop.value,
5524:     cropX: normalizeCropPosition(existingBlock.dataset.cropX),
5525:     cropY: normalizeCropPosition(existingBlock.dataset.cropY),
5526:     cropZoom: normalizeCropZoom(summaryImageZoom.value),
5527:     display: summaryImageDisplay.value === "download" ? "download" : "show",
5528:     source: existingBlock.dataset.source || "local",
5529:     title: summaryImageTitle.value.trim() || existingBlock.dataset.title || "",
5530:     type: "image",
5531:     url: existingBlock.dataset.url || ""
5532:   };
5533: }
5534: return uploadSummaryImageFile(file, {
5535:   align: summaryImageAlign.value,
5536:   caption: summaryImageCaption.value.trim(),
5537:   cropAspect: summaryImageCrop.value,
5538:   cropX: 50,
5539:   cropY: 50,
5540:   cropZoom: normalizeCropZoom(summaryImageZoom.value),
5541:   display: summaryImageDisplay.value === "download" ? "download" : "show",
5542:   folder: options.folder || "summary",
5543:   projectId: options.projectId || "",
5544:   title: summaryImageTitle.value.trim() || file.name
5545: });
5546: }

```

openSummaryFormulaDialog - lines 5548-5563

Item	Explanation
Source location	Lines 5548-5563 (16 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	async function openSummaryFormulaDialog(existingBlock = null) {
Touches	browser/Electron DOM, filesystem
Calls detected	querySelector, dialogValue, unwrapFormula, trim, setStatus
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	1 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5548: async function openSummaryFormulaDialog(existingBlock = null) {
5549:   const heading = summaryFormulaDialog.querySelector("h2");
5550:   const submitButton = summaryFormulaDialog.querySelector("button[type='submit']");
5551:   if (heading) heading.textContent = existingBlock ? "Edit formula" : "Add formula";
5552:   if (submitButton) submitButton.textContent = existingBlock ? "Save formula" : "Add formula";
5553:   summaryFormulaInput.value = existingBlock?.dataset.formula || "";
5554:   summaryFormulaAlign.value = existingBlock?.dataset.align || "center";
5555:   const saved = await dialogValue(summaryFormulaDialog);
5556:   if (!saved) return null;
5557:   const formula = unwrapFormula(summaryFormulaInput.value.trim());
5558:   if (!formula) {
5559:     setStatus("Paste or type a formula first.");
5560:     return null;
5561:   }
5562:   return { align: summaryFormulaAlign.value, formula, type: "formula" };
5563: }

```

refreshSummaryCodeDialogPreview - lines 5565-5575

Item	Explanation
Source location	Lines 5565-5575 (11 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function refreshSummaryCodeDialogPreview() {
Touches	filesystem
Calls detected	detectCodeLanguage, normalizeCodeLanguage, normalizeCodeText, renderRichCodeBlock
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5565: function refreshSummaryCodeDialogPreview() {
5566:   if (!summaryCodePreview || !summaryCodeInput) return;
5567:   const rawCode = summaryCodeInput.value || "";
5568:   const language = summaryCodeLanguage?.value === "auto"
5569:     ? detectCodeLanguage(rawCode)
5570:     : normalizeCodeLanguage(summaryCodeLanguage?.value || "javascript");
5571:   const code = normalizeCodeText(rawCode, summaryCodePasteMode?.value || "source");
5572:   summaryCodePreview.innerHTML = code
5573:     ? renderRichCodeBlock({ code, language, pasteMode: summaryCodePasteMode?.value || "source" })
5574:     : `<p class="evidence-empty">Code preview appears here.</p>`;
5575: }

```

beautifyCodeClient - lines 5577-5592

Item	Explanation
Source location	Lines 5577-5592 (16 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function beautifyCodeClient(code = "", language = "javascript") {
Touches	filesystem
Calls detected	replace, normalizeCodeLanguage, includes, split, map, trim, max, repeat, match, join, trimEnd
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5577: function beautifyCodeClient(code = "", language = "javascript") {
5578:   const normalized = String(code || "").replace(/\r\n?/g, "\n").replace(/\t/g, " ");
5579:   const lang = normalizeCodeLanguage(language);
5580:   if (["c", "cpp", "java", "javascript", "verilog", "systemverilog"].includes(lang)) {
5581:     let depth = 0;
5582:     return normalized.split("\n").map((rawLine) => {
5583:       const line = rawLine.trim();
5584:       if (!line) return "";
5585:       if (/^[{}]\]/.test(line)) depth = Math.max(0, depth - 1);
5586:       const nextLine = `${" ".repeat(depth)}${line}`;
5587:       const opens = (line.match(/[{}]/g) || []).length;
5588:       const closes = (line.match(/[{}]\]/g) || []).length;
5589:       depth = Math.max(0, depth + opens - closes);
5590:       return nextLine;
5591:     }).join("\n").trimEnd();
5592:   }

```

beautifySummaryCodeDialog - lines 5599-5618

Item	Explanation
Source location	Lines 5599-5618 (20 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	async function beautifySummaryCodeDialog() {
Touches	filesystem, network/API requests
Calls detected	detectCodeLanguage, normalizeCodeLanguage, fetch, now, stringify, json, Error, beautifyCodeClient, refreshSummaryCodeDialogPreview
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5599: async function beautifySummaryCodeDialog() {
5600:   if (!summaryCodeInput) return;
5601:   const language = summaryCodeLanguage?.value === "auto"
5602:     ? detectCodeLanguage(summaryCodeInput.value || "")
5603:     : normalizeCodeLanguage(summaryCodeLanguage?.value || "javascript");
5604:   try {
5605:     const response = await fetch(`/api/code/beautify?t=${Date.now()}`, {
5606:       method: "POST",
5607:       cache: "no-store",
5608:       headers: { "Content-Type": "application/json" },
5609:       body: JSON.stringify({ language, code: summaryCodeInput.value || "" })
5610:     });
5611:     const result = await response.json();
5612:     if (!response.ok || !result.ok) throw new Error(result.error || "Beautifier failed.");
5613:     summaryCodeInput.value = result.code || summaryCodeInput.value;
5614:   } catch {
5615:     summaryCodeInput.value = beautifyCodeClient(summaryCodeInput.value || "", language);
5616:   }
5617:   refreshSummaryCodeDialogPreview();
5618: }

```

openSummaryCodeDialog - lines 5620-5640

Item	Explanation
Source location	Lines 5620-5640 (21 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	async function openSummaryCodeDialog(existingBlock = null, options = {}) {
Touches	browser/Electron DOM, filesystem

Item	Explanation
Calls detected	querySelector, detectCodeLanguage, refreshSummaryCodeDialogPreview, dialogValue, normalizeCodePasteMode, normalizeCodeText, setStatus, normalizeCodeLanguage
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5620: async function openSummaryCodeDialog(existingBlock = null, options = {}) {
5621:   const heading = summaryCodeDialog.querySelector("h2");
5622:   const submitButton = summaryCodeDialog.querySelector("button[type='submit']");
5623:   if (heading) heading.textContent = existingBlock ? "Edit code block" : "Add code block";
5624:   if (submitButton) submitButton.textContent = existingBlock ? "Save code" : "Add code";
5625:   const defaultCode = options.code || "";
5626:   summaryCodeLanguage.value = existingBlock?.dataset.language || (defaultCode ?
detectCodeLanguage(defaultCode) : "auto");
5627:   summaryCodePasteMode.value = existingBlock?.dataset.pasteMode || options.pasteMode || "source";
5628:   summaryCodeInput.value = existingBlock?.dataset.code || defaultCode;
5629:   refreshSummaryCodeDialogPreview();
5630:   const saved = await dialogValue(summaryCodeDialog);
5631:   if (!saved) return null;
5632:   const pasteMode = normalizeCodePasteMode(summaryCodePasteMode.value);
5633:   const code = normalizeCodeText(summaryCodeInput.value, pasteMode);
5634:   if (!code) {
5635:     setStatus("Paste or type code before saving the block.");
5636:     return null;
5637:   }
5638:   const language = summaryCodeLanguage.value === "auto" ? detectCodeLanguage(code) :
normalizeCodeLanguage(summaryCodeLanguage.value);
5639:   return { code, language, pasteMode, type: "code" };
5640: }

```

configureSummaryContextMenu - lines 5642-5665

Item	Explanation
Source location	Lines 5642-5665 (24 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function configureSummaryContextMenu(mode = "rich", options = {}) {
Touches	browser/Electron DOM, filesystem
Calls detected	Set, querySelectorAll, forEach, has
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5642: function configureSummaryContextMenu(mode = "rich", options = {}) {
5643:   const plainActions = new Set([
5644:     "copy-text",
5645:     "paste-text",
5646:     "cut-text",
5647:     "select-all-text",
5648:     "toggle-bold",
5649:     "toggle-italic",
5650:     "toggle-underline"
5651:   ]);
5652:   summaryContextMenu.querySelectorAll("[data-rich-action]").forEach((button) => {
5653:     const action = button.dataset.richAction;
5654:     button.hidden = mode === "plain" ? !plainActions.has(action) : false;
5655:   });
5656:   summaryContextMenu.querySelectorAll(".rich-menu-field").forEach((field) => {
5657:     field.hidden = false;
5658:   });
5659:   if (mode === "rich") {
5660:     const textOnly = Boolean(options.textOnly);
5661:     summaryContextMenu.querySelectorAll("[data-rich-action='add-image'], [data-rich-action='add-
formula'], [data-rich-action='add-code'], [data-rich-action='paste-code']").forEach((button) => {

```



```

5662:         button.hidden = textOnly;
5663:     });
5664: }
5665: }

```

applyPlainTextControlFormat - lines 5667-5701

Item	Explanation
Source location	Lines 5667-5701 (35 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function applyPlainTextControlFormat(control, updates = {}) {
Touches	browser/Electron DOM, filesystem
Calls detected	call, cleanFontFamily, normalizeFontPx, normalizeTextColor, rgbColorToHex, storePlainControlStyle, setStatus
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5667: function applyPlainTextControlFormat(control, updates = {}) {
5668:   if (!control) return;
5669:   if (Object.prototype.hasOwnProperty.call(updates, "fontFamily")) {
5670:     const font = cleanFontFamily(updates.fontFamily);
5671:     control.dataset.builderFontFamily = font;
5672:     control.style.fontFamily = font || "";
5673:   }
5674:   if (Object.prototype.hasOwnProperty.call(updates, "fontPx")) {
5675:     const size = normalizeFontPx(updates.fontPx);
5676:     control.dataset.builderFontPx = size;
5677:     control.style.fontSize = size ? `${size}px` : "";
5678:   }
5679:   if (Object.prototype.hasOwnProperty.call(updates, "color")) {
5680:     const color = normalizeTextColor(updates.color);
5681:     control.dataset.builderColor = color;
5682:     control.style.color = color || "";
5683:     if (richColorInput && color && /^[0-9a-f]{6}$/i.test(rgbColorToHex(color))) {
5684:       richColorInput.value = rgbColorToHex(color);
5685:     }
5686:   }
5687:   if (Object.prototype.hasOwnProperty.call(updates, "bold")) {
5688:     control.dataset.builderBold = updates.bold ? "true" : "false";
5689:     control.style.fontWeight = updates.bold ? "700" : "";
5690:   }
5691:   if (Object.prototype.hasOwnProperty.call(updates, "italic")) {
5692:     control.dataset.builderItalic = updates.italic ? "true" : "false";
5693:     control.style.fontStyle = updates.italic ? "italic" : "";
5694:   }
5695:   if (Object.prototype.hasOwnProperty.call(updates, "underline")) {
5696:     control.dataset.builderUnderline = updates.underline ? "true" : "false";
5697:     control.style.textDecoration = updates.underline ? "underline" : "";
5698:   }
5699:   storePlainControlStyle(control);
5700:   setStatus("Formatting applied and saved locally for this field.");
5701: }

```

syncPlainContextMenuControls - lines 5703-5719

Item	Explanation
Source location	Lines 5703-5719 (17 source lines).
Purpose	Keeps two places aligned, such as local data and target files.
Declaration	function syncPlainContextMenuControls(control) {
Touches	browser/Electron DOM
Calls detected	getComputedStyle, displayRichFontName, round, rgbColorToHex, some, setAttribute
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5703: function syncPlainContextMenuControls(control) {
5704:   if (!control) return;
5705:   const computed = getComputedStyle(control);
5706:   const family = displayRichFontName(control.dataset.builderFontFamily || computed.fontFamily) ||
"Arial";
5707:   const size = String(Math.round(parseFloat(control.dataset.builderFontPx || computed.fontSize) || 16));
5708:   const color = rgbColorToHex(control.dataset.builderColor || computed.color || "#17202a");
5709:   if (richFontSelect) {
5710:     richFontSelect.value = [...richFontSelect.options].some((option) => option.value === family ||
option.textContent === family)
5711:       ? family
5712:       : "Arial";
5713:   }
5714:   if (richFontSize) richFontSize.value = [...richFontSize.options].some((option) => option.value === size
|| option.textContent === size)
5715:     ? size
5716:     : "16";
5717:   if (richColorInput && /^[0-9a-f]{6}$/i.test(color)) richColorInput.value = color;
5718:   if (richBoldButton) richBoldButton.setAttribute("aria-pressed", control.dataset.builderBold === "true"
? "true" : "false");
5719: }

```

plainTextControlFromContextEvent - lines 5721-5728

Item	Explanation
Source location	Lines 5721-5728 (8 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function plainTextControlFromContextEvent(event) {
Touches	Mostly local calculations or local UI state.
Calls detected	closest, toLowerCase, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5721: function plainTextControlFromContextEvent(event) {
5722:   const control = event.target.closest?(".textarea, input");
5723:   if (!control || control.closest("#summary-context-menu")) return null;
5724:   const type = String(control.type || "text").toLowerCase();
5725:   if (control.disabled || control.readOnly) return null;
5726:   if (["button", "checkbox", "color", "file", "hidden", "image", "radio", "range", "reset",
"submit"].includes(type)) return null;
5727:   return control;
5728: }

```

handlePlainTextAction - lines 5730-5778

Item	Explanation
Source location	Lines 5730-5778 (49 source lines).
Purpose	Handles an event, request, command, or user action.
Declaration	async function handlePlainTextAction(action) {
Touches	browser/Electron DOM, filesystem
Calls detected	focus, max, min, slice, writeText, setRangeText, dispatchEvent, Event, readText, select, applyPlainTextControlFormat
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	2 await expression(s), 7 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5730: async function handlePlainTextAction(action) {
5731:   const control = activePlainTextControl;
5732:   if (!control) return;
5733:   control.focus({ preventScroll: true });
5734:   const selection = activePlainTextSelection || {
5735:     start: control.selectionStart ?? 0,
5736:     end: control.selectionEnd ?? control.value.length
5737:   };
5738:   const start = Math.max(0, Math.min(selection.start, control.value.length));
5739:   const end = Math.max(start, Math.min(selection.end, control.value.length));
5740:
5741:   if (action === "copy-text" || action === "cut-text") {
5742:     const text = control.value.slice(start, end) || control.value;
5743:     if (text) await navigator.clipboard.writeText(text);
5744:     if (action === "cut-text" && end > start) {
5745:       control.setRangeText("", start, end, "start");
5746:       control.dispatchEvent(new Event("input", { bubbles: true }));
5747:     }
5748:     return;
5749:   }
5750:
5751:   if (action === "paste-text") {
5752:     const text = await navigator.clipboard.readText();
5753:     if (text) {
5754:       control.setRangeText(text, start, end, "end");
5755:       control.dispatchEvent(new Event("input", { bubbles: true }));
5756:     }
5757:     return;
5758:   }
5759:
5760:   if (action === "select-all-text") {
5761:     control.select();
5762:     activePlainTextSelection = { start: 0, end: control.value.length };
5763:     return;
5764:   }
5765:
5766:   if (action === "toggle-bold") {
5767:     applyPlainTextControlFormat(control, { bold: control.dataset.builderBold !== "true" });
5768:     return;
5769:   }
5770:   if (action === "toggle-italic") {
5771:     applyPlainTextControlFormat(control, { italic: control.dataset.builderItalic !== "true" });
5772:     return;
5773:   }
5774:   if (action === "toggle-underline") {
5775:     applyPlainTextControlFormat(control, { underline: control.dataset.builderUnderline !== "true" });
5776:     return;
5777:   }
5778: }

```

handleRichAction - lines 5780-5878

Item	Explanation
Source location	Lines 5780-5878 (99 source lines).
Purpose	Handles an event, request, command, or user action.
Declaration	async function handleRichAction(action, editor) {
Touches	browser/Electron DOM, filesystem
Calls detected	applyRichInlineCommand, updateCurrentTextBlock, restoreRichSelection, getSelection, toString, selectedRichBlock, writeText, readText, insertPlainTextIntoRichEditor, saveRichEditorToProject, getRangeAt, deleteContents
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	9 await expression(s), 5 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5780: async function handleRichAction(action, editor) {

```

```

5781:   if (!editor) return;
5782:   activeSummaryEditor = editor;
5783:   if (action === "toggle-bold") {
5784:     applyRichInlineCommand(editor, "bold");
5785:   }
5786:   if (action === "toggle-italic") {
5787:     applyRichInlineCommand(editor, "italic");
5788:   }
5789:   if (action === "toggle-underline") {
5790:     applyRichInlineCommand(editor, "underline");
5791:   }
5792:   if (action === "align-left") updateCurrentTextBlock(editor, { align: "left" });
5793:   if (action === "align-center") updateCurrentTextBlock(editor, { align: "center" });
5794:   if (action === "align-right") updateCurrentTextBlock(editor, { align: "right" });
5795:   if (action === "copy-text") {
5796:     restoreRichSelection(editor);
5797:     const selectedText = window.getSelection()?.toString() || "";
5798:     const block = selectedRichBlock(editor);
5799:     const text = selectedText || block?.textContent || "";
5800:     if (text) await navigator.clipboard.writeText(text);
5801:     return;
5802:   }
5803:
5804:   if (action === "paste-text") {
5805:     const text = await navigator.clipboard.readText();
5806:     if (text) {
5807:       insertPlainTextIntoRichEditor(editor, text);
5808:       saveRichEditorToProject(editor);
5809:     }
5810:     return;
5811:   }
5812:   if (action === "cut-text") {
5813:     restoreRichSelection(editor);
5814:     const selection = window.getSelection();
5815:     const selectedText = selection?.toString() || "";
5816:     if (selectedText && selection.rangeCount) {
5817:       await navigator.clipboard.writeText(selectedText);
5818:       const range = selection.getRangeAt(0);
5819:       range.deleteContents();
5820:       captureRichSelection(editor);
5821:       cleanupEmptyRichTextBlocks(editor);
5822:       saveRichEditorToProject(editor);
5823:     }
5824:     return;
5825:   }
5826:   if (action === "select-all-text") {
5827:     const block = selectedRichBlock(editor);
5828:     const target = block?.dataset.type === "paragraph" ? block : editor;
5829:     const range = document.createRange();
5830:     range.selectNodeContents(target);
5831:     const selection = window.getSelection();
5832:     selection.removeAllRanges();
5833:     selection.addRange(range);
5834:     activeRichSelectionRange = range.cloneRange();
5835:     activeTextSelectionRange = range.cloneRange();
5836:     showPersistentTextSelection(range);
5837:     showTextSelectionInspector(editor, range);
5838:     return;
5839:   }
5840:   if (action === "add-image") {
5841:     const imageBlock = await openSummaryImageDialog(null, {
5842:       folder: editor.dataset.richFolder || "summary",
5843:       projectId: editor.dataset.richProjectId || ""
5844:     });
5845:     if (imageBlock) {
5846:       insertRichBlockAfterCursor(editor, createRichImageBlock(imageBlock));
5847:       saveRichEditorToProject(editor);
5848:     }
5849:   }
5850:   if (action === "add-formula") {
5851:     const formulaBlock = await openSummaryFormulaDialog();
5852:     if (formulaBlock) {
5853:       insertRichBlockAfterCursor(editor, createRichFormulaBlock(formulaBlock));
5854:       saveRichEditorToProject(editor);
5855:     }
5856:   }
5857:   if (action === "add-code") {
5858:     const selectedText = window.getSelection()?.toString() || "";
5859:     const codeBlock = await openSummaryCodeDialog(null, { code: selectedText, pasteMode: "source" });
5860:     if (codeBlock) {
5861:       insertRichBlockAfterCursor(editor, createRichCodeBlock(codeBlock));
5862:       saveRichEditorToProject(editor);
5863:     }
5864:   }
5865:   if (action === "paste-code") {
5866:     const text = await navigator.clipboard.readText();
5867:     const codeBlock = await openSummaryCodeDialog(null, { code: text, pasteMode: "source" });
5868:     if (codeBlock) {
5869:       insertRichBlockAfterCursor(editor, createRichCodeBlock(codeBlock));
5870:       saveRichEditorToProject(editor);
5871:     }
5872:   }

```

```

5873:   if (action === "edit-block") {
5874:       await editSelectedRichBlock(editor);
5875:       saveRichEditorToProject(editor);
5876:   }
5877:   if (action === "delete-block") deleteSelectedRichBlock(editor);
5878: }

```

hideSummaryContextMenu - lines 5880-5884

Item	Explanation
Source location	Lines 5880-5884 (5 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function hideSummaryContextMenu() {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5880: function hideSummaryContextMenu() {
5881:     summaryContextMenu.hidden = true;
5882:     activePlainTextControl = null;
5883:     activePlainTextSelection = null;
5884: }

```

editSelectedRichBlock - lines 5886-5909

Item	Explanation
Source location	Lines 5886-5909 (24 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	async function editSelectedRichBlock(editor) {
Touches	browser/Electron DOM, filesystem
Calls detected	selectedRichBlock, selectRichBlock, openSummaryImageDialog, refreshRichImageBlock, openSummaryFormulaDialog, refreshRichFormulaBlock, openSummaryCodeDialog, refreshRichCodeBlock, focusTextBlock, setStatus
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	3 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5886: async function editSelectedRichBlock(editor) {
5887:     const block = selectedRichBlock(editor);
5888:     if (!block) return;
5889:     selectRichBlock(block);
5890:     if (block.dataset.type === "image") {
5891:         const updated = await openSummaryImageDialog(block);
5892:         if (updated) refreshRichImageBlock(block, updated);
5893:         return;
5894:     }
5895:     if (block.dataset.type === "formula") {
5896:         const updated = await openSummaryFormulaDialog(block);
5897:         if (updated) refreshRichFormulaBlock(block, updated);
5898:         return;
5899:     }
5900:     if (block.dataset.type === "code") {
5901:         const updated = await openSummaryCodeDialog(block);
5902:         if (updated) refreshRichCodeBlock(block, updated);
5903:         return;
5904:     }

```

```

5905:   if (block.dataset.type === "paragraph") {
5906:     focusTextBlock(block);
5907:     setStatus("Text block selected. Type directly in the block to edit it.");
5908:   }
5909: }

```

copyRichCodeBlock - lines 5911-5918

Item	Explanation
Source location	Lines 5911-5918 (8 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	async function copyRichCodeBlock(block) {
Touches	browser/Electron DOM
Calls detected	querySelector, writeText, setStatus
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5911: async function copyRichCodeBlock(block) {
5912:   if (!block || block.dataset.type !== "code") return false;
5913:   const code = block.dataset.code || block.querySelector("code")?.textContent || "";
5914:   if (!code) return false;
5915:   await navigator.clipboard.writeText(code);
5916:   setStatus("Code copied to clipboard.");
5917:   return true;
5918: }

```

removeRichBlock - lines 5920-5931

Item	Explanation
Source location	Lines 5920-5931 (12 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function removeRichBlock(block, editor) {
Touches	browser/Electron DOM
Calls detected	matches, remove, querySelector, append, createRichTextBlock, focusTextBlock
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5920: function removeRichBlock(block, editor) {
5921:   if (!block) return;
5922:   const nextFocus = block.previousElementSibling?.matches(".rich-text-block")
5923:     ? block.previousElementSibling
5924:     : block.nextElementSibling?.matches(".rich-text-block")
5925:     ? block.nextElementSibling
5926:     : null;
5927:   block.remove();
5928:   activeSummaryBlock = null;
5929:   if (!editor.querySelector(".rich-block")) editor.append(createRichTextBlock(""));
5930:   focusTextBlock(nextFocus || editor.querySelector(".rich-text-block"));
5931: }

```

deleteSelectedRichBlock - lines 5933-5948

Item	Explanation
Source location	Lines 5933-5948 (16 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function deleteSelectedRichBlock(editor) {
Touches	browser/Electron DOM, filesystem
Calls detected	selectedRichBlock, openDeleteConfirm, removeRichBlock
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5933: function deleteSelectedRichBlock(editor) {
5934:   const block = selectedRichBlock(editor);
5935:   if (!block) return;
5936:   const label = block.dataset.type === "image"
5937:     ? "this overview image"
5938:     : block.dataset.type === "formula"
5939:     ? "this formula"
5940:     : block.dataset.type === "code"
5941:     ? "this code block"
5942:     : "this text block";
5943:   openDeleteConfirm({
5944:     title: "Delete overview block",
5945:     message: `Are you sure you want to delete ${label}?`,
5946:     onConfirm: () => removeRichBlock(block, editor)
5947:   });
5948: }

```

deleteProjectSummary - lines 5950-5966

Item	Explanation
Source location	Lines 5950-5966 (17 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function deleteProjectSummary(project) {
Touches	filesystem
Calls detected	openDeleteConfirm, setStatus, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

5950: function deleteProjectSummary(project) {
5951:   if (!project) return;
5952:   openDeleteConfirm({
5953:     title: "Delete project overview",
5954:     message: "Are you sure you want to delete the current project overview?",
5955:     onConfirm: () => {
5956:       project.summary = "";
5957:       project.summaryRich = null;
5958:       summaryEditorProjectId = "";
5959:       activeSummaryEditor = null;
5960:       activeSummaryBlock = null;
5961:       setStatus("Project overview deleted in the editor. Click Save project to update the portfolio
5962: preview.");
5962:       scheduleAutosave();
5963:       renderAll();
5964:     }
5965:   });
5966: }

```

saveRichSummary - lines 5968-5987

Item	Explanation
Source location	Lines 5968-5987 (20 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	async function saveRichSummary() {
Touches	browser/Electron DOM
Calls detected	selectedProject, querySelector, extractRichSummary, plainTextFromRich, wordCount, setStatus, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
5968: async function saveRichSummary() {
5969:   const project = selectedProject();
5970:   const editor = projectFields.querySelector("[data-rich-editor='summary']");
5971:   if (!project || !editor) return true;
5972:   const rich = extractRichSummary(editor);
5973:   const plain = plainTextFromRich(rich);
5974:   if (wordCount(plain) > 1000) {
5975:     setStatus("Project overview is limited to 1000 words. Shorten the text before saving.");
5976:     return false;
5977:   }
5978:   project.summaryRich = rich;
5979:   project.summary = plain;
5980:   summaryEditorProjectId = "";
5981:   activeSummaryEditor = null;
5982:   activeSummaryBlock = null;
5983:   setStatus("Overview saved in the editor. Click Save project to include this version in the portfolio preview.");
5984:   scheduleAutosave();
5985:   renderAll();
5986:   return true;
5987: }
```

renderFields - lines 5989-6004

Item	Explanation
Source location	Lines 5989-6004 (16 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderFields(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	renderSummaryBuilder, requestAnimationFrame, populateSummaryEditor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
5989: function renderFields(project) {
5990:   if (!project) {
5991:     projectFields.hidden = false;
5992:     projectFields.innerHTML = `<p class="evidence-empty">Add or select a project to begin editing.</p>`;
5993:     return;
5994:   }
5995:
5996:   projectFields.hidden = activeSectionId !== "brief";
5997:   projectWindowTitle.textContent = project.title;
5998:   projectFields.innerHTML = `
5999:     ${renderSummaryBuilder(project)}`
```



```

6000: `;
6001:   if (summaryEditorProjectId === project.id) {
6002:     requestAnimationFrame(() => populateSummaryEditor(project));
6003:   }
6004: }

```

sectionOptions - lines 6006-6016

Item	Explanation
Source location	Lines 6006-6016 (11 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function sectionOptions(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	map, filter, isAnalogProject
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6006: function sectionOptions(project) {
6007:   const custom = (project.sections || []).map((section) => ({
6008:     id: `custom:${section.id}`,
6009:     label: section.title,
6010:     folder: section.id
6011:   }));
6012:   return [
6013:     ...standardSections.filter((section) => !section.analogOnly || isAnalogProject(project)),
6014:     ...custom
6015:   ];
6016: }

```

custom - lines 6007-6011

Item	Explanation
Source location	Lines 6007-6011 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const custom = (project.sections []).map((section) => ({
Touches	Mostly local calculations or local UI state.
Calls detected	map
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6007:   const custom = (project.sections || []).map((section) => ({
6008:     id: `custom:${section.id}`,
6009:     label: section.title,
6010:     folder: section.id
6011:   }));

```

ensureSiteSections - lines 6018-6021

Item	Explanation
Source location	Lines 6018-6021 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.

Item	Explanation
Declaration	function ensureSiteSections() {
Touches	Mostly local calculations or local UI state.
Calls detected	clone
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6018: function ensureSiteSections() {
6019:   catalog.siteSections = catalog.siteSections || [];
6020:   savedPortfolioCatalog.siteSections = savedPortfolioCatalog.siteSections || clone(catalog.siteSections
|| []);
6021: }

```

siteSectionHasContent - lines 6023-6032

Item	Explanation
Source location	Lines 6023-6032 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function siteSectionHasContent(section) {
Touches	Mostly local calculations or local UI state.
Calls detected	richHasContent, some
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6023: function siteSectionHasContent(section) {
6024:   return Boolean(
6025:     section?.description ||
6026:     richHasContent(section?.richDescription) ||
6027:     section?.backgroundImage ||
6028:     (section?.links || []).some((link) => link.label || link.url) ||
6029:     (section?.assets || []).some((asset) => asset.title || asset.url) ||
6030:     (section?.subsections || []).some((item) => item.title || item.description ||
richHasContent(item.richDescription) || (item.links || []).length)
6031:   );
6032: }

```

siteSectionRenderable - lines 6034-6036

Item	Explanation
Source location	Lines 6034-6036 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function siteSectionRenderable(section) {
Touches	Mostly local calculations or local UI state.
Calls detected	siteSectionHasContent
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
6034: function siteSectionRenderable(section) {
6035:   return section?.visible !== false && siteSectionHasContent(section);
6036: }
```

siteSectionLinkCount - lines 6038-6044

Item	Explanation
Source location	Lines 6038-6044 (7 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function siteSectionLinkCount(section) {
Touches	Mostly local calculations or local UI state.
Calls detected	flatMap, filter
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
6038: function siteSectionLinkCount(section) {
6039:   return [
6040:     ...(section.links || []),
6041:     ...(section.assets || []),
6042:     ...(section.subsections || []).flatMap((item) => item.links || [])
6043:   ].filter((item) => item.label || item.url).length;
6044: }
```

renderSiteSectionList - lines 6046-6094

Item	Explanation
Source location	Lines 6046-6094 (49 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderSiteSectionList() {
Touches	Mostly local calculations or local UI state.
Calls detected	ensureSiteSections, map, url, siteSectionLinkCount, richHasContent, renderRichContent, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
6046: function renderSiteSectionList() {
6047:   ensureSiteSections();
6048:   const sections = catalog.siteSections || [];
6049:   siteSectionList.innerHTML = sections.length ? sections.map((section) => `
6050:     <article class="site-section-item ${section.visible === false ? "is-hidden-section" : ""}"
style="${section.backgroundImage ? `--site-section-bg: url('${section.backgroundImage}');` : ""}">
6051:       <div>
6052:         <strong>${section.title || "Untitled section"}</strong>
6053:         <span>${section.visible === false ? "Hidden from website" : "Visible on website"} .
${(section.subsections || []).length} subsection${(section.subsections || []).length === 1 ? "" : "s"} .
${siteSectionLinkCount(section)} link${siteSectionLinkCount(section) === 1 ? "" : "s"}</span>
6054:         ${section.description || richHasContent(section.richDescription) ?
renderRichContent(section.richDescription, section.description || "") : ""}` : ""`>
6055:   `
```

```

6055:         </div>
6056:         ${((section.subsections || []).length ? `
6057:         <div class="site-subsection-list">
6058:             ${((section.subsections || []).map((item, index) => `
6059:             <article class="site-subsection-item">
6060:                 <div>
6061:                     <strong>${item.title || "Untitled subsection"}</strong>
6062:                     ${item.description || richHasContent(item.richDescription) ?
renderRichContent(item.richDescription, item.description || "") : ""}
6063:                 </div>
6064:                 <div class="site-section-actions">
6065:                     <button type="button" data-site-subsection-edit="${section.id}" data-
index="${index}">Edit</button>
6066:                     <button class="danger-icon" type="button" data-site-subsection-delete="${section.id}"
data-index="${index}">Delete</button>
6067:                 </div>
6068:             </article>
6069:             `).join("")}
6070:         </div>
6071:         ` : ""}
6072:         ${((section.links || []).length ? `
6073:         <div class="site-link-list">
6074:             ${((section.links || []).map((link, index) => `
6075:             <span>
6076:                 ${link.label || "Untitled link"}
6077:                 <button type="button" data-site-link-edit="${section.id}" data-
index="${index}">Edit</button>
6078:                 <button type="button" data-site-link-delete="${section.id}" data-
index="${index}">Delete</button>
6079:             </span>
6080:             `).join("")}
6081:         </div>
6082:         ` : ""}
6083:         <div class="site-section-actions">
6084:             <button type="button" data-site-section-toggle="${section.id}">${section.visible === false ?
"Show" : "Hide"}</button>
6085:             <button type="button" data-site-section-edit="${section.id}">Edit</button>
6086:             <button type="button" data-site-subsection-add="${section.id}">Add subsection</button>
6087:             <button type="button" data-site-link-add="${section.id}">Add link</button>
6088:             <button type="button" data-site-section-bg="${section.id}">Background</button>
6089:             ${section.backgroundImage ? `<button type="button" data-site-section-bg-
clear="${section.id}">Clear bg</button>` : ""}
6090:             <button class="danger-icon" type="button" data-site-section-
delete="${section.id}">Delete</button>
6091:         </div>
6092:         </article>
6093:         `).join("") : `<p class="evidence-empty">No extra portfolio areas added yet.</p>`;
6094:     }

```

siteSectionDialogValue - lines 6096-6118

Item	Explanation
Source location	Lines 6096-6118 (23 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	async function siteSectionDialogValue(section = {}) {
Touches	browser/Electron DOM
Calls detected	populateStandaloneRichEditor, dialogValue, trim, setStatus, extractRichSummary, plainTextFromRich, slugify
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6096: async function siteSectionDialogValue(section = {}) {
6097:     sectionDialogTitle.textContent = section.id ? "Edit section" : "Create section";
6098:     sectionTitle.placeholder = "Professional profile, Personal background, Sports...";
6099:     sectionDescription.dataset.placeholder = "Short section overview shown on the website";
6100:     sectionDescription.dataset.richFolder = section.id || "new-portfolio-section";
6101:     sectionDescription.dataset.richProjectId = "_site-sections";
6102:     sectionTitle.value = section.title || "";
6103:     populateStandaloneRichEditor(sectionDescription, section.richDescription, section.description || "");
6104:     const saved = await dialogValue(sectionDialog);
6105:     if (!saved) return null;
6106:     const title = sectionTitle.value.trim();
6107:     if (!title) {
6108:         setStatus("Type a section title before saving.");
6109:         return null;

```

```

6110:   }
6111:   const richDescription = extractRichSummary(sectionDescription);
6112:   return {
6113:     description: plainTextFromRich(richDescription, "\n").trim(),
6114:     id: section.id || slugify(title),
6115:     richDescription,
6116:     title
6117:   };
6118: }

```

siteSubsectionDialogValue - lines 6120-6143

Item	Explanation
Source location	Lines 6120-6143 (24 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	async function siteSubsectionDialogValue(subsection = {}) {
Touches	browser/Electron DOM
Calls detected	populateStandaloneRichEditor, dialogValue, trim, setStatus, extractRichSummary, plainTextFromRich, slugify
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6120: async function siteSubsectionDialogValue(subsection = {}) {
6121:   sectionDialogTitle.textContent = subsection.id ? "Edit subsection" : "Create subsection";
6122:   sectionTitle.placeholder = "Sports, Hobbies, Resume links, Origin...";
6123:   sectionDescription.dataset.placeholder = "Short subsection text shown inside the portfolio area";
6124:   sectionDescription.dataset.richFolder = subsection.id || "new-portfolio-subsection";
6125:   sectionDescription.dataset.richProjectId = "_site-sections";
6126:   sectionTitle.value = subsection.title || "";
6127:   populateStandaloneRichEditor(sectionDescription, subsection.richDescription, subsection.description ||
""");
6128:   const saved = await dialogValue(sectionDialog);
6129:   if (!saved) return null;
6130:   const title = sectionTitle.value.trim();
6131:   if (!title) {
6132:     setStatus("Type a subsection title before saving.");
6133:     return null;
6134:   }
6135:   const richDescription = extractRichSummary(sectionDescription);
6136:   return {
6137:     description: plainTextFromRich(richDescription, "\n").trim(),
6138:     id: subsection.id || slugify(title),
6139:     links: subsection.links || [],
6140:     richDescription,
6141:     title
6142:   };
6143: }

```

dialogLinkValue - lines 6145-6169

Item	Explanation
Source location	Lines 6145-6169 (25 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function dialogLinkValue(link = {}) {
Touches	filesystem
Calls detected	removeEventListener, resolve, trim, setStatus, normalizeLinkTarget, addEventListener, showModal
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
6145: function dialogLinkValue(link = {}) {
6146:   return new Promise((resolve) => {
6147:     siteLinkEyebrow.textContent = link.url ? "Edit portfolio link" : "Add portfolio link";
6148:     siteLinkTitle.textContent = link.url ? "Edit link" : "Add link";
6149:     siteLinkLabel.value = link.label || "";
6150:     siteLinkUrl.value = link.url || "";
6151:     const onClose = () => {
6152:       siteLinkDialog.removeEventListener("close", onClose);
6153:       if (siteLinkDialog.returnValue !== "save") {
6154:         resolve(null);
6155:         return;
6156:       }
6157:       const label = siteLinkLabel.value.trim();
6158:       const url = siteLinkUrl.value.trim();
6159:       if (!label || !url) {
6160:         setStatus("Type both a link label and URL before saving.");
6161:         resolve(null);
6162:         return;
6163:       }
6164:       resolve({ ...link, label, url: normalizeLinkTarget(url, { assumeWeb: true }) });
6165:     };
6166:     siteLinkDialog.addEventListener("close", onClose);
6167:     siteLinkDialog.showModal();
6168:   });
6169: }
```

onClose - lines 6151-6165

Item	Explanation
Source location	Lines 6151-6165 (15 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const onClose = () => {
Touches	filesystem
Calls detected	removeEventListener, resolve, trim, setStatus, normalizeLinkTarget
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
6151:   const onClose = () => {
6152:     siteLinkDialog.removeEventListener("close", onClose);
6153:     if (siteLinkDialog.returnValue !== "save") {
6154:       resolve(null);
6155:       return;
6156:     }
6157:     const label = siteLinkLabel.value.trim();
6158:     const url = siteLinkUrl.value.trim();
6159:     if (!label || !url) {
6160:       setStatus("Type both a link label and URL before saving.");
6161:       resolve(null);
6162:       return;
6163:     }
6164:     resolve({ ...link, label, url: normalizeLinkTarget(url, { assumeWeb: true }) });
6165:   };
```

addSiteSection - lines 6171-6191

Item	Explanation
Source location	Lines 6171-6191 (21 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	async function addSiteSection() {
Touches	Mostly local calculations or local UI state.
Calls detected	ensureSiteSections, siteSectionDialogValue, slugify, push, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6171: async function addSiteSection() {
6172:   ensureSiteSections();
6173:   const input = await siteSectionDialogValue();
6174:   if (!input) return;
6175:   const id = input.id || slugify(input.title);
6176:   catalog.siteSections.push({
6177:     id,
6178:     title: input.title,
6179:     description: input.description,
6180:     richDescription: input.richDescription,
6181:     visible: false,
6182:     backgroundImage: "",
6183:     links: [],
6184:     assets: [],
6185:     subsections: []
6186:   });
6187:   markDraftNeedsSave();
6188:   setStatus("Portfolio area added locally.");
6189:   scheduleAutosave();
6190:   renderAll();
6191: }

```

editSiteSection - lines 6193-6206

Item	Explanation
Source location	Lines 6193-6206 (14 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	async function editSiteSection(sectionId) {
Touches	Mostly local calculations or local UI state.
Calls detected	ensureSiteSections, find, siteSectionDialogValue, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6193: async function editSiteSection(sectionId) {
6194:   ensureSiteSections();
6195:   const section = catalog.siteSections.find((item) => item.id === sectionId);
6196:   if (!section) return;
6197:   const input = await siteSectionDialogValue(section);
6198:   if (!input) return;
6199:   section.title = input.title;
6200:   section.description = input.description;
6201:   section.richDescription = input.richDescription;
6202:   markDraftNeedsSave();
6203:   setStatus("Portfolio area edited locally.");
6204:   scheduleAutosave();
6205:   renderAll();
6206: }

```

addSiteSubsection - lines 6208-6219

Item	Explanation
Source location	Lines 6208-6219 (12 source lines).
Purpose	Part of the private builder frontend and editing experience.

Item	Explanation
Declaration	async function addSiteSubsection(sectionId) {
Touches	Mostly local calculations or local UI state.
Calls detected	find, siteSubsectionDialogValue, push, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6208: async function addSiteSubsection(sectionId) {
6209:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6210:   if (!section) return;
6211:   const input = await siteSubsectionDialogValue();
6212:   if (!input) return;
6213:   section.subsections = section.subsections || [];
6214:   section.subsections.push(input);
6215:   markDraftNeedsSave();
6216:   setStatus("Portfolio subsection added locally.");
6217:   scheduleAutosave();
6218:   renderAll();
6219: }

```

section - lines 6209-6218

Item	Explanation
Source location	Lines 6209-6218 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const section = (catalog.siteSections []).find((item) => item.id === sectionId);
Touches	Mostly local calculations or local UI state.
Calls detected	find, siteSubsectionDialogValue, push, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6209:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6210:   if (!section) return;
6211:   const input = await siteSubsectionDialogValue();
6212:   if (!input) return;
6213:   section.subsections = section.subsections || [];
6214:   section.subsections.push(input);
6215:   markDraftNeedsSave();
6216:   setStatus("Portfolio subsection added locally.");
6217:   scheduleAutosave();
6218:   renderAll();

```

editSiteSubsection - lines 6221-6232

Item	Explanation
Source location	Lines 6221-6232 (12 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	async function editSiteSubsection(sectionId, index) {
Touches	Mostly local calculations or local UI state.
Calls detected	find, siteSubsectionDialogValue, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6221: async function editSiteSubsection(sectionId, index) {
6222:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6223:   const subsection = section?.subsections?.[Number(index)];
6224:   if (!section || !subsection) return;
6225:   const input = await siteSubsectionDialogValue(subsection);
6226:   if (!input) return;
6227:   section.subsections[Number(index)] = input;
6228:   markDraftNeedsSave();
6229:   setStatus("Portfolio subsection edited locally.");
6230:   scheduleAutosave();
6231:   renderAll();
6232: }

```

section - lines 6222-6231

Item	Explanation
Source location	Lines 6222-6231 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const section = (catalog.siteSections []).find((item) => item.id === sectionId);
Touches	Mostly local calculations or local UI state.
Calls detected	find, siteSubsectionDialogValue, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6222:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6223:   const subsection = section?.subsections?.[Number(index)];
6224:   if (!section || !subsection) return;
6225:   const input = await siteSubsectionDialogValue(subsection);
6226:   if (!input) return;
6227:   section.subsections[Number(index)] = input;
6228:   markDraftNeedsSave();
6229:   setStatus("Portfolio subsection edited locally.");
6230:   scheduleAutosave();
6231:   renderAll();

```

deleteSiteSubsection - lines 6234-6248

Item	Explanation
Source location	Lines 6234-6248 (15 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function deleteSiteSubsection(sectionId, index) {
Touches	filesystem
Calls detected	find, openDeleteConfirm, splice, markDraftNeedsSave, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6234: function deleteSiteSubsection(sectionId, index) {
6235:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6236:   const subsection = section?.subsections?.[Number(index)];
6237:   if (!section || !subsection) return;
6238:   openDeleteConfirm({
6239:     title: "Delete subsection",
6240:     message: `Are you sure you want to delete "${subsection.title}"?`,
6241:     onConfirm: () => {
6242:       section.subsections.splice(Number(index), 1);
6243:       markDraftNeedsSave();
6244:       scheduleAutosave();
6245:       renderAll();
6246:     }
6247:   });
6248: }

```

section - lines 6235-6247

Item	Explanation
Source location	Lines 6235-6247 (13 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const section = (catalog.siteSections []).find((item) => item.id === sectionId);
Touches	filesystem
Calls detected	find, openDeleteConfirm, splice, markDraftNeedsSave, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6235:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6236:   const subsection = section?.subsections?.[Number(index)];
6237:   if (!section || !subsection) return;
6238:   openDeleteConfirm({
6239:     title: "Delete subsection",
6240:     message: `Are you sure you want to delete "${subsection.title}"?`,
6241:     onConfirm: () => {
6242:       section.subsections.splice(Number(index), 1);
6243:       markDraftNeedsSave();
6244:       scheduleAutosave();
6245:       renderAll();
6246:     }
6247:   });

```

addSiteSectionLink - lines 6250-6261

Item	Explanation
Source location	Lines 6250-6261 (12 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	async function addSiteSectionLink(sectionId) {
Touches	Mostly local calculations or local UI state.
Calls detected	find, dialogLinkValue, push, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 2 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6250: async function addSiteSectionLink(sectionId) {
6251:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6252:   if (!section) return;
6253:   const input = await dialogLinkValue();
6254:   if (!input) return;
6255:   section.links = section.links || [];
6256:   section.links.push(input);
6257:   markDraftNeedsSave();
6258:   setStatus("Portfolio link added locally.");
6259:   scheduleAutosave();
6260:   renderAll();
6261: }

```

section - lines 6251-6260

Item	Explanation
Source location	Lines 6251-6260 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const section = (catalog.siteSections []).find((item) => item.id === sectionId);
Touches	Mostly local calculations or local UI state.
Calls detected	find, dialogLinkValue, push, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6251:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6252:   if (!section) return;
6253:   const input = await dialogLinkValue();
6254:   if (!input) return;
6255:   section.links = section.links || [];
6256:   section.links.push(input);
6257:   markDraftNeedsSave();
6258:   setStatus("Portfolio link added locally.");
6259:   scheduleAutosave();
6260:   renderAll();

```

editSiteSectionLink - lines 6263-6274

Item	Explanation
Source location	Lines 6263-6274 (12 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	async function editSiteSectionLink(sectionId, index) {
Touches	Mostly local calculations or local UI state.
Calls detected	find, dialogLinkValue, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6263: async function editSiteSectionLink(sectionId, index) {

```

```

6264:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6265:   const link = section?.links?.[Number(index)];
6266:   if (!section || !link) return;
6267:   const input = await dialogLinkValue(link);
6268:   if (!input) return;
6269:   section.links[Number(index)] = input;
6270:   markDraftNeedsSave();
6271:   setStatus("Portfolio link edited locally.");
6272:   scheduleAutosave();
6273:   renderAll();
6274: }

```

section - lines 6264-6273

Item	Explanation
Source location	Lines 6264-6273 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	<code>const section = (catalog.siteSections []).find((item) => item.id === sectionId);</code>
Touches	Mostly local calculations or local UI state.
Calls detected	<code>find</code> , <code>dialogLinkValue</code> , <code>markDraftNeedsSave</code> , <code>setStatus</code> , <code>scheduleAutosave</code> , <code>renderAll</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 <code>await</code> expression(s), 2 <code>return</code> statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6264:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6265:   const link = section?.links?.[Number(index)];
6266:   if (!section || !link) return;
6267:   const input = await dialogLinkValue(link);
6268:   if (!input) return;
6269:   section.links[Number(index)] = input;
6270:   markDraftNeedsSave();
6271:   setStatus("Portfolio link edited locally.");
6272:   scheduleAutosave();
6273:   renderAll();

```

deleteSiteSectionLink - lines 6276-6284

Item	Explanation
Source location	Lines 6276-6284 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	<code>function deleteSiteSectionLink(sectionId, index) {</code>
Touches	Mostly local calculations or local UI state.
Calls detected	<code>find</code> , <code>splice</code> , <code>markDraftNeedsSave</code> , <code>setStatus</code> , <code>scheduleAutosave</code> , <code>renderAll</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 <code>await</code> expression(s), 1 <code>return</code> statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6276: function deleteSiteSectionLink(sectionId, index) {
6277:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6278:   if (!section?.links?.[Number(index)]) return;
6279:   section.links.splice(Number(index), 1);
6280:   markDraftNeedsSave();
6281:   setStatus("Portfolio link deleted locally.");
6282:   scheduleAutosave();
6283:   renderAll();
6284: }

```

section - lines 6277-6283

Item	Explanation
Source location	Lines 6277-6283 (7 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	<code>const section = (catalog.siteSections []).find((item) => item.id === sectionId);</code>
Touches	Mostly local calculations or local UI state.
Calls detected	<code>find</code> , <code>splice</code> , <code>markDraftNeedsSave</code> , <code>setStatus</code> , <code>scheduleAutosave</code> , <code>renderAll</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 <code>await</code> expression(s), 1 <code>return</code> statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
6277: const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6278: if (!section?.links?.[Number(index)]) return;
6279: section.links.splice(Number(index), 1);
6280: markDraftNeedsSave();
6281: setStatus("Portfolio link deleted locally.");
6282: scheduleAutosave();
6283: renderAll();
```

toggleSiteSectionVisibility - lines 6286-6294

Item	Explanation
Source location	Lines 6286-6294 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	<code>function toggleSiteSectionVisibility(sectionId) {</code>
Touches	Mostly local calculations or local UI state.
Calls detected	<code>find</code> , <code>markDraftNeedsSave</code> , <code>setStatus</code> , <code>scheduleAutosave</code> , <code>renderAll</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 <code>await</code> expression(s), 1 <code>return</code> statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
6286: function toggleSiteSectionVisibility(sectionId) {
6287:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6288:   if (!section) return;
6289:   section.visible = section.visible === false;
6290:   markDraftNeedsSave();
6291:   setStatus(section.visible ? "Portfolio area will show on the website preview." : "Portfolio area hidden
from the website preview.");
6292:   scheduleAutosave();
6293:   renderAll();
6294: }
```

section - lines 6287-6293

Item	Explanation
Source location	Lines 6287-6293 (7 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	<code>const section = (catalog.siteSections []).find((item) => item.id === sectionId);</code>
Touches	Mostly local calculations or local UI state.

Item	Explanation
Calls detected	find, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6287:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6288:   if (!section) return;
6289:   section.visible = section.visible === false;
6290:   markDraftNeedsSave();
6291:   setStatus(section.visible ? "Portfolio area will show on the website preview." : "Portfolio area hidden
from the website preview.");
6292:   scheduleAutosave();
6293:   renderAll();

```

deleteSiteSection - lines 6296-6310

Item	Explanation
Source location	Lines 6296-6310 (15 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function deleteSiteSection(sectionId) {
Touches	filesystem
Calls detected	find, openDeleteConfirm, filter, markDraftNeedsSave, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6296: function deleteSiteSection(sectionId) {
6297:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6298:   if (!section) return;
6299:   openDeleteConfirm({
6300:     title: "Delete portfolio area",
6301:     message: `Are you sure you want to delete "${section.title}"?`,
6302:     onConfirm: () => {
6303:       catalog.siteSections = (catalog.siteSections || []).filter((item) => item.id !== sectionId);
6304:       savedPortfolioCatalog.siteSections = (savedPortfolioCatalog.siteSections || []).filter((item) =>
item.id !== sectionId);
6305:       markDraftNeedsSave();
6306:       scheduleAutosave();
6307:       renderAll();
6308:     }
6309:   });
6310: }

```

section - lines 6297-6309

Item	Explanation
Source location	Lines 6297-6309 (13 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const section = (catalog.siteSections []).find((item) => item.id === sectionId);
Touches	filesystem
Calls detected	find, openDeleteConfirm, filter, markDraftNeedsSave, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6297:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6298:   if (!section) return;
6299:   openDeleteConfirm({
6300:     title: "Delete portfolio area",
6301:     message: `Are you sure you want to delete "${section.title}"?`,
6302:     onConfirm: () => {
6303:       catalog.siteSections = (catalog.siteSections || []).filter((item) => item.id !== sectionId);
6304:       savedPortfolioCatalog.siteSections = (savedPortfolioCatalog.siteSections || []).filter((item) =>
item.id !== sectionId);
6305:       markDraftNeedsSave();
6306:       scheduleAutosave();
6307:       renderAll();
6308:     }
6309:   });

```

setSiteSectionBackground - lines 6312-6343

Item	Explanation
Source location	Lines 6312-6343 (32 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	async function setSiteSectionBackground(sectionId) {
Touches	filesystem, network/API requests
Calls detected	find, chooseFile, startsWith, setStatus, readFileAsDataURL, fetch, now, stringify, json, markDraftNeedsSave, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	4 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6312: async function setSiteSectionBackground(sectionId) {
6313:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6314:   if (!section) return;
6315:   const file = await chooseFile();
6316:   if (!file) return;
6317:   if (!file.type.startsWith("image/")) {
6318:     setStatus("Choose an image file for a section background.");
6319:     return;
6320:   }
6321:   const data = await readFileAsDataURL(file);
6322:   const response = await fetch(`/api/upload?t=${Date.now()}`, {
6323:     method: "POST",
6324:     cache: "no-store",
6325:     headers: { "Content-Type": "application/json" },
6326:     body: JSON.stringify({
6327:       projectId: "_site-sections",
6328:       section: section.id,
6329:       fileName: file.name,
6330:       data
6331:     })
6332:   });
6333:   const result = await response.json();
6334:   if (!response.ok) {
6335:     setStatus(result.error || "Background upload failed.");
6336:     return;
6337:   }
6338:   section.backgroundImage = result.url;
6339:   markDraftNeedsSave();
6340:   setStatus("Portfolio area background saved locally.");
6341:   scheduleAutosave();
6342:   renderAll();
6343: }

```

section - lines 6313-6320

Item	Explanation
Source location	Lines 6313-6320 (8 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	<code>const section = (catalog.siteSections []).find((item) => item.id === sectionId);</code>
Touches	Mostly local calculations or local UI state.
Calls detected	<code>find</code> , <code>chooseFile</code> , <code>startsWith</code> , <code>setStatus</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 <code>await</code> expression(s), 3 <code>return</code> statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
6313: const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6314: if (!section) return;
6315: const file = await chooseFile();
6316: if (!file) return;
6317: if (!file.type.startsWith("image/")) {
6318:   setStatus("Choose an image file for a section background.");
6319:   return;
6320: }
```

clearSiteSectionBackground - lines 6345-6353

Item	Explanation
Source location	Lines 6345-6353 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	<code>function clearSiteSectionBackground(sectionId) {</code>
Touches	Mostly local calculations or local UI state.
Calls detected	<code>find</code> , <code>markDraftNeedsSave</code> , <code>setStatus</code> , <code>scheduleAutosave</code> , <code>renderAll</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 <code>await</code> expression(s), 1 <code>return</code> statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
6345: function clearSiteSectionBackground(sectionId) {
6346:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6347:   if (!section) return;
6348:   section.backgroundImage = "";
6349:   markDraftNeedsSave();
6350:   setStatus("Portfolio area background cleared locally.");
6351:   scheduleAutosave();
6352:   renderAll();
6353: }
```

section - lines 6346-6352

Item	Explanation
Source location	Lines 6346-6352 (7 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	<code>const section = (catalog.siteSections []).find((item) => item.id === sectionId);</code>
Touches	Mostly local calculations or local UI state.

Item	Explanation
Calls detected	find, markDraftNeedsSave, setStatus, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6346:   const section = (catalog.siteSections || []).find((item) => item.id === sectionId);
6347:   if (!section) return;
6348:   section.backgroundImage = "";
6349:   markDraftNeedsSave();
6350:   setStatus("Portfolio area background cleared locally.");
6351:   scheduleAutosave();
6352:   renderAll();

```

renderSectionTabs - lines 6355-6369

Item	Explanation
Source location	Lines 6355-6369 (15 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderSectionTabs(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	sectionOptions, map, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6355: function renderSectionTabs(project) {
6356:   if (!project) {
6357:     sectionTabs.innerHTML = "";
6358:     return;
6359:   }
6360:
6361:   sectionTabs.innerHTML = `
6362:     ${sectionOptions(project).map((section) => `
6363:       <button class="${section.id === activeSectionId ? "active" : ""}" type="button" data-section-
id="${section.id}">
6364:         ${section.label}
6365:       </button>
6366:     `).join("")}
6367:     <button type="button" data-add-section="true">Add section</button>
6368:   `;
6369: }

```

renderTextArray - lines 6371-6389

Item	Explanation
Source location	Lines 6371-6389 (19 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderTextArray(project, key, label) {
Touches	Mostly local calculations or local UI state.
Calls detected	map, join, toLowerCase
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6371: function renderTextArray(project, key, label) {
6372:   const items = project[key] || [];
6373:   return
6374:     <div class="section-actions">
6375:       <button class="button primary" type="button" data-open-editor="text-array" data-key="${key}" data-
mode="add">Add ${label}</button>
6376:     </div>
6377:     <div class="builder-items">
6378:       ${items.map((item, index) => `
6379:         <article class="builder-item">
6380:           <div>
6381:             <strong>${typeof item === "string" ? item : item.title || item.name || item.label ||
""}</strong>
6382:           </div>
6383:           <button type="button" data-open-editor="text-array" data-key="${key}" data-mode="edit" data-
index="${index}">Edit</button>
6384:           <button class="danger-icon" type="button" data-delete-array="${key}" data-
index="${index}">Delete</button>
6385:         </article>
6386:       `).join("")} || `<p class="evidence-empty">No ${label.toLowerCase()} added yet.</p>`}
6387:     </div>
6388:   `;
6389: }

```

toolName - lines 6391-6393

Item	Explanation
Source location	Lines 6391-6393 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function toolName(item) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6391: function toolName(item) {
6392:   return typeof item === "string" ? item : item.name || item.title || item.label || "";
6393: }

```

toolDescription - lines 6395-6397

Item	Explanation
Source location	Lines 6395-6397 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function toolDescription(item) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6395: function toolDescription(item) {
6396:   return typeof item === "string" ? "" : item.description || item.usedFor || "";
6397: }

```

renderToolsSection - lines 6399-6420

Item	Explanation
Source location	Lines 6399-6420 (22 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderToolsSection(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	map, toolName, richHasContent, renderRichContent, toolDescription, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6399: function renderToolsSection(project) {
6400:   const items = project.tools || [];
6401:   return
6402:     <div class="section-actions">
6403:       <button class="button primary" type="button" data-open-editor="tool" data-mode="add">Add
tool</button>
6404:     </div>
6405:     <div class="builder-items tool-items">
6406:       ${items.map((item, index) => `
6407:         <article class="builder-item tool-item">
6408:           <div>
6409:             <strong>${toolName(item)} || "Untitled tool"</strong>
6410:           </div>
6411:           ${richHasContent(item?.richDescription)
6412:             ? renderRichContent(item.richDescription, toolDescription(item))
6413:             : `<p>${toolDescription(item)} || "No usage description added yet."</p>`}
6414:           <button type="button" data-open-editor="tool" data-mode="edit" data-
index="${index}">Edit</button>
6415:           <button class="danger-icon" type="button" data-delete-array="tools" data-
index="${index}">Delete</button>
6416:         </article>
6417:       `).join("")} || `<p class="evidence-empty">No tools added yet.</p>`}
6418:     </div>
6419:   `;
6420: }

```

renderObjectSection - lines 6422-6450

Item	Explanation
Source location	Lines 6422-6450 (29 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderObjectSection(project, key, folder, label) {
Touches	filesystem
Calls detected	map, richHasContent, renderRichContent, renderMultilineInlineText, join, toLowerCase
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6422: function renderObjectSection(project, key, folder, label) {
6423:   const items = project[key] || [];
6424:   return `
6425:     <div class="section-actions">
6426:       <button class="button primary" type="button" data-open-editor="object" data-key="${key}" data-
mode="add">Add text item</button>
6427:       ${folder ? `<button class="button secondary" type="button" data-upload="${key}" data-
folder="${folder}">Add file or image</button>` : ""}
6428:     </div>
6429:     <div class="builder-items">
6430:       ${items.map((item, index) => {
6431:         const title = item.title || item.name || item.label || "Untitled item";
6432:         const url = item.url || item.artifact || "";
6433:         const description = item.description || item.caption || item.result || item.method || "";
6434:         return `
6435:           <article class="builder-item">
6436:             <div>
6437:               <strong>${title}</strong>
6438:               <span>${url || "No file attached"}</span>
6439:               ${richHasContent(item.richDescription)
6440:                 ? renderRichContent(item.richDescription, description)
6441:                 : description ? `<p>${renderMultilineInlineText(description)}</p>` : ""}
6442:             </div>
6443:             <button type="button" data-open-editor="object" data-key="${key}" data-mode="edit" data-
index="${index}">Edit</button>
6444:             <button class="danger-icon" type="button" data-delete-item="${key}" data-
index="${index}">Delete</button>
6445:           </article>
6446:         `;
6447:       }).join("") || `<p class="evidence-empty">No ${label.toLowerCase()} added yet.</p>`}
6448:     </div>
6449:   `;
6450: }

```

designArray - lines 6452-6458

Item	Explanation
Source location	Lines 6452-6458 (7 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function designArray(project, pathValue) {
Touches	Mostly local calculations or local UI state.
Calls detected	ensureDesignModel, getByPath, isArray, setByPath
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6452: function designArray(project, pathValue) {
6453:   ensureDesignModel(project);
6454:   const items = getByPath(project, pathValue);
6455:   if (Array.isArray(items)) return items;
6456:   setByPath(project, pathValue, []);
6457:   return getByPath(project, pathValue);
6458: }

```

makeDesignItem - lines 6460-6466

Item	Explanation
Source location	Lines 6460-6466 (7 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function makeDesignItem(kind, title, description = "", url = "") {

Item	Explanation
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 5 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6460: function makeDesignItem(kind, title, description = "", url = "") {
6461:   if (kind === "reference") return { title, description, url, type: "Reference", status: url ? "linked" :
"draft" };
6462:   if (kind === "analysis") return { title, description, type: "Math / Analysis", status: "draft" };
6463:   if (kind === "simulation-file") return { title, description, url, type: "Simulation file", status: url
? "uploaded" : "draft" };
6464:   if (kind === "simulation-result") return { title, description, url, type: "Simulation result", status:
url ? "uploaded" : "draft" };
6465:   return { title, description, url, type: "Document", status: url ? "uploaded" : "draft" };
6466: }

```

renderDesignObjectSection - lines 6468-6496

Item	Explanation
Source location	Lines 6468-6496 (29 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderDesignObjectSection(project, pathValue, folder, label, kind = "document") {
Touches	filesystem
Calls detected	designArray, map, richHasContent, renderRichContent, renderMultilineInlineText, join, toLowerCase
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6468: function renderDesignObjectSection(project, pathValue, folder, label, kind = "document") {
6469:   const items = designArray(project, pathValue);
6470:   return `
6471:     <div class="section-actions">
6472:       <button class="button primary" type="button" data-open-editor="design-object" data-design-
path="${pathValue}" data-design-kind="${kind}" data-mode="add">Add ${label}</button>
6473:       ${folder ? `<button class="button secondary" type="button" data-upload-design="${pathValue}" data-
folder="${folder}" data-design-kind="${kind}">Add file</button>` : ""}
6474:     </div>
6475:     <div class="builder-items">
6476:       ${items.map((item, index) => {
6477:         const title = item.title || item.name || item.label || "Untitled item";
6478:         const url = item.url || item.artifact || "";
6479:         const description = item.description || item.caption || item.result || item.method || "";
6480:         return `
6481:           <article class="builder-item">
6482:             <div>
6483:               <strong>${title}</strong>
6484:               <span>${url || "No file attached"}</span>
6485:               ${richHasContent(item.richDescription)
6486:                 ? renderRichContent(item.richDescription, description)
6487:                 : description ? `<p>${renderMultilineInlineText(description)}</p>` : ""}
6488:             </div>
6489:             <button type="button" data-open-editor="design-object" data-design-path="${pathValue}" data-
design-kind="${kind}" data-mode="edit" data-index="${index}">Edit</button>
6490:             <button class="danger-icon" type="button" data-delete-design-item="${pathValue}" data-
index="${index}">Delete</button>
6491:           </article>
6492:         `;
6493:       }).join("") || `<p class="evidence-empty">No ${label.toLowerCase()} added yet.</p>`}
6494:     </div>
6495:   `;
6496: }

```

renderDesignSummaryField - lines 6498-6530

Item	Explanation
Source location	Lines 6498-6530 (33 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderDesignSummaryField(project, pathValue, label, placeholder) {
Touches	filesystem
Calls detected	getByPath, summaryRichPathFromTextPath, escapeHtml, overviewFolderFromPath
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
6498: function renderDesignSummaryField(project, pathValue, label, placeholder) {
6499:   const textValue = getByPath(project, pathValue) || "";
6500:   const richPath = summaryRichPathFromTextPath(pathValue);
6501:   const richValue = getByPath(project, richPath);
6502:   const hasValue = Boolean(textValue || richValue?.blocks?.length);
6503:
6504:   return `
6505:     <div class="wide-field design-summary-field rich-design-summary-field">
6506:       <div class="summary-builder-heading">
6507:         <span>
6508:           ${label}
6509:           ${hasValue ? `<button class="inline-danger-button" type="button" data-clear-design-
summary="${pathValue}">Clear overview</button>` : ""}
6510:         </span>
6511:       </div>
6512:
6513:       <div class="rich-summary-panel">
6514:         <p class="rich-editor-hint">Right-click to add images, code blocks, apply bold, color, font,
numeric size, alignment, or formulas.</p>
6515:
6516:         <div
6517:           class="rich-summary-editor"
6518:           data-rich-editor="section-overview"
6519:           data-rich-text-path="${escapeHtml(pathValue)}"
6520:           data-rich-path="${escapeHtml(richPath)}"
6521:           data-rich-folder="${escapeHtml(overviewFolderFromPath(pathValue))}"
6522:           data-placeholder="${escapeHtml(placeholder || "")}"
6523:           contenteditable="true"
6524:           spellcheck="true"
6525:           aria-label="${escapeHtml(label)} rich editor"
6526:         ></div>
6527:       </div>
6528:     </div>
6529:   `;
6530: }
```

designPathHasContent - lines 6532-6536

Item	Explanation
Source location	Lines 6532-6536 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function designPathHasContent(project, pathValue) {
Touches	Mostly local calculations or local UI state.
Calls detected	getByPath, isArray, some, itemTitle, richHasContent, summaryRichPathFromTextPath
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
6532: function designPathHasContent(project, pathValue) {
6533:   const value = getByPath(project, pathValue);
6534:   if (Array.isArray(value)) return value.some((item) => itemTitle(item) || item?.description || item?.url
|| item?.artifact || richHasContent(item?.richDescription));
6535:   return Boolean(value || richHasContent(getByPath(project, summaryRichPathFromTextPath(pathValue))));
6536: }
```

renderEmptyDynamicSectionPrompt - lines 6538-6548

Item	Explanation
Source location	Lines 6538-6548 (11 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderEmptyDynamicSectionPrompt(label = "section") {
Touches	Mostly local calculations or local UI state.
Calls detected	escapeHtml
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
6538: function renderEmptyDynamicSectionPrompt(label = "section") {
6539:   return `
6540:     <div class="compile-code-empty">
6541:       <h3>No ${escapeHtml(label)} content added yet</h3>
6542:       <p>Create your own sections and subsections with the Add section button. Empty predefined groups
stay hidden so the project structure is controlled by you.</p>
6543:       <div class="section-actions">
6544:         <button class="button primary" type="button" data-add-section="true">Add section</button>
6545:       </div>
6546:     </div>
6547:   `;
6548: }
```

renderAnalogDesignWorkspace - lines 6550-6593

Item	Explanation
Source location	Lines 6550-6593 (44 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderAnalogDesignWorkspace(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	ensureDesignModel, designPathHasContent, push, renderDesignSummaryField, renderDesignObjectSection, renderObjectSection, join, renderEmptyDynamicSectionPrompt, renderPendingEditor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
6550: function renderAnalogDesignWorkspace(project) {
6551:   ensureDesignModel(project);
6552:   const cards = [];
6553:   if (designPathHasContent(project, "design.brief.summary") || designPathHasContent(project,
"design.brief.files")) {
6554:     cards.push(`
6555:       <section class="design-workspace-card">
6556:         <h3>Design Overview</h3>
6557:         ${renderDesignSummaryField(project, "design.brief.summary", "Design overview", "State the analog
design objective, topology, constraints, and design decisions.")}
6558:         ${renderDesignObjectSection(project, "design.brief.files", "design-brief", "brief file",
```

```

"document"))}
6559:     </section>
6560:   `);
6561: }
6562: if (
6563:   designPathHasContent(project, "design.documentation.summary") ||
6564:   designPathHasContent(project, "design.documentation.files") ||
6565:   designPathHasContent(project, "design.documentation.references") ||
6566:   designPathHasContent(project, "design.documentation.mathAnalysis")
6567: ) {
6568:   cards.push(`
6569:     <section class="design-workspace-card">
6570:       <h3>Documentation</h3>
6571:       ${renderDesignSummaryField(project, "design.documentation.summary", "Documentation overview",
"Summarize schematics, notes, datasheets, calculations, and supporting documents.")}
6572:       ${renderDesignObjectSection(project, "design.documentation.files", "documentation", "document",
"document")}
6573:       ${designPathHasContent(project, "design.documentation.references") ?
`<h4>References</h4>${renderDesignObjectSection(project, "design.documentation.references", "", "reference",
"reference")}` : ""}
6574:       ${designPathHasContent(project, "design.documentation.mathAnalysis") ? `<h4>Math /
Analysis</h4>${renderDesignObjectSection(project, "design.documentation.mathAnalysis", "", "analysis item",
"analysis")}` : ""}
6575:     </section>
6576:   `);
6577: }
6578: if ((project.pcbns || []).length || (project.media || []).length) {
6579:   cards.push(`
6580:     <section class="design-workspace-card">
6581:       <h3>PCB and Visual Evidence</h3>
6582:       ${((project.pcbns || []).length ? renderObjectSection(project, "pcbns", "pcbns", "PCBs") : "")}
6583:       ${((project.media || []).length ? renderObjectSection(project, "media", "images", "Images") : "")}
6584:     </section>
6585:   `);
6586: }
6587: return `
6588:   <div class="section-stack analog-design-workspace">
6589:     ${cards.join("")} || renderEmptyDynamicSectionPrompt("design")}
6590:     ${renderPendingEditor()}
6591:   </div>
6592: `;
6593: }

```

renderAnalogSimulationWorkspace - lines 6595-6628

Item	Explanation
Source location	Lines 6595-6628 (34 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderAnalogSimulationWorkspace(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	ensureDesignModel, designPathHasContent, push, renderDesignSummaryField, renderDesignObjectSection, join, renderEmptyDynamicSectionPrompt, renderPendingEditor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6595: function renderAnalogSimulationWorkspace(project) {
6596:   ensureDesignModel(project);
6597:   const cards = [];
6598:   if (designPathHasContent(project, "design.simulation.summary")) {
6599:     cards.push(`
6600:       <section class="design-workspace-card">
6601:         <h3>Simulation Overview</h3>
6602:         ${renderDesignSummaryField(project, "design.simulation.summary", "Simulation overview", "Explain
what was simulated, why it mattered, and what the results proved.")}
6603:       </section>
6604:     `);
6605:   }
6606:   if (designPathHasContent(project, "design.simulation.files")) {
6607:     cards.push(`
6608:       <section class="design-workspace-card">
6609:         <h3>Simulation Files</h3>
6610:         ${renderDesignObjectSection(project, "design.simulation.files", "simulations", "simulation file",
"simulation-file")}
6611:       </section>

```



```

6612:     `);
6613:   }
6614:   if (designPathHasContent(project, "design.simulation.results")) {
6615:     cards.push(`
6616:       <section class="design-workspace-card">
6617:         <h3>Simulation Results</h3>
6618:         ${renderDesignObjectSection(project, "design.simulation.results", "simulation-results",
"simulation result", "simulation-result")}
6619:       </section>
6620:     `);
6621:   }
6622:   return `
6623:     <div class="section-stack analog-design-workspace">
6624:       ${cards.join("")} || renderEmptyDynamicSectionPrompt("simulation")}
6625:       ${renderPendingEditor()}
6626:     </div>
6627:   `;
6628: }

```

pendingEditorUsesRichDescription - lines 6630-6632

Item	Explanation
Source location	Lines 6630-6632 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function pendingEditorUsesRichDescription(editor = pendingEditor) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6630: function pendingEditorUsesRichDescription(editor = pendingEditor) {
6631:   return Boolean(editor && editor.type !== "text-array");
6632: }

```

pendingEditorFolder - lines 6634-6641

Item	Explanation
Source location	Lines 6634-6641 (8 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function pendingEditorFolder(editor = pendingEditor) {
Touches	Mostly local calculations or local UI state.
Calls detected	slugify
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 6 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6634: function pendingEditorFolder(editor = pendingEditor) {
6635:   if (!editor) return "content";
6636:   if (editor.type === "custom-section") return `custom-${slugify(editor.sectionId ||
"section")}-overview`;
6637:   if (editor.type === "custom") return `custom-${slugify(editor.sectionId || "section")}-subsection`;
6638:   if (editor.type === "design-object") return `design-${slugify(editor.kind || "content")}`;
6639:   if (editor.type === "object") return slugify(editor.key || "content");
6640:   return slugify(editor.type || "content");
6641: }

```

renderPendingRichDescriptionEditor - lines 6643-6666

Item	Explanation
Source location	Lines 6643-6666 (24 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderPendingRichDescriptionEditor(editor = pendingEditor) {
Touches	filesystem
Calls detected	escapeHtml, pendingEditorFolder
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
6643: function renderPendingRichDescriptionEditor(editor = pendingEditor) {
6644:   const label = editor?.type === "tool"
6645:     ? "What it was used for"
6646:     : editor?.type === "custom-section"
6647:       ? "Section content"
6648:       : "Content";
6649:   return `
6650:     <div class="pending-rich-description rich-design-summary-field">
6651:       <div class="summary-builder-heading"><span>${escapeHtml(label)}</span></div>
6652:       <div class="rich-summary-panel">
6653:         <p class="rich-editor-hint">Right-click to add images, code blocks, apply bold, color, font,
numeric size, alignment, or formulas.</p>
6654:         <div
6655:           class="rich-summary-editor"
6656:           data-rich-editor="pending-description"
6657:           data-rich-folder="${escapeHtml(pendingEditorFolder(editor))}"
6658:           data-placeholder="Type or paste the complete content here."
6659:           contenteditable="true"
6660:           spellcheck="true"
6661:           aria-label="${escapeHtml(label)} rich editor"
6662:         ></div>
6663:       </div>
6664:     </div>
6665:   `;
6666: }
```

renderPendingEditor - lines 6668-6692

Item	Explanation
Source location	Lines 6668-6692 (25 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderPendingEditor() {
Touches	filesystem
Calls detected	renderPendingRichDescriptionEditor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
6668: function renderPendingEditor() {
6669:   if (!pendingEditor) return "";
6670:   const title = pendingEditor.title || "";
6671:   const isTool = pendingEditor.type === "tool";
6672:   const isSimpleText = pendingEditor.type === "text-array";
6673:   const isSectionDetails = pendingEditor.type === "custom-section";
6674:   return `
6675:     <form class="pending-editor" id="pending-editor">
```

```

6676:         <div>
6677:             <h3>${pendingEditor.mode === "edit" ? "Edit" : "Add"} ${isTool ? "tool" : isSimpleText ? "item" :
isSectionDetails ? "section" : "subsection"}</h3>
6678:         </div>
6679:         <div class="${isTool ? "tool-editor-grid" : "pending-editor-grid"}">
6680:             <label>
6681:                 <span>${isTool ? "Tool name" : "Title"}</span>
6682:                 <input name="title" type="text" value="${title}" required>
6683:             </label>
6684:             ${isSimpleText ? "" : renderPendingRichDescriptionEditor(pendingEditor)}
6685:         </div>
6686:         <div class="pending-editor-actions">
6687:             <button class="button primary" type="submit">Save</button>
6688:             <button class="button secondary" type="button" data-cancel-pending="true">Cancel</button>
6689:         </div>
6690:     </form>
6691: `;
6692: }

```

openPendingEditor - lines 6694-6701

Item	Explanation
Source location	Lines 6694-6701 (8 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function openPendingEditor(config) {
Touches	browser/Electron DOM
Calls detected	renderSectionContent, selectedProject, requestAnimationFrame, populateRichEditors, querySelector, scrollIntoView
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6694: function openPendingEditor(config) {
6695:     pendingEditor = config;
6696:     renderSectionContent(selectedProject());
6697:     requestAnimationFrame(() => {
6698:         populateRichEditors(sectionContent);
6699:         document.querySelector("#pending-editor")?.scrollIntoView({ behavior: "smooth", block: "nearest" });
6700:     });
6701: }

```

savePendingEditor - lines 6703-6785

Item	Explanation
Source location	Lines 6703-6785 (83 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	function savePendingEditor(form) {
Touches	browser/Electron DOM, filesystem
Calls detected	selectedProject, trim, querySelector, extractRichSummary, plainTextFromRich, setStatus, push, makeItemForSection, designArray, makeDesignItem, find, scheduleAutosave
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6703: function savePendingEditor(form) {
6704:     const project = selectedProject();
6705:     if (!project || !pendingEditor) return;
6706:     const title = form.elements.title.value.trim();
6707:     const richEditor = form.querySelector("[data-rich-editor='pending-description']");

```

```

6708:   const richDescription = richEditor ? extractRichSummary(richEditor) : null;
6709:   const description = richEditor ? plainTextFromRich(richDescription) : "";
6710:   if (!title) {
6711:     setStatus("Type a title before saving.");
6712:     return;
6713:   }
6714:
6715:   if (pendingEditor.type === "tool") {
6716:     project.tools = project.tools || [];
6717:     const nextTool = { name: title, description, richDescription };
6718:     if (pendingEditor.mode === "edit") {
6719:       project.tools[Number(pendingEditor.index)] = nextTool;
6720:     } else {
6721:       project.tools.push(nextTool);
6722:     }
6723:   }
6724:
6725:   if (pendingEditor.type === "object") {
6726:     const key = pendingEditor.key;
6727:     project[key] = project[key] || [];
6728:     if (pendingEditor.mode === "edit") {
6729:       const existing = project[key][Number(pendingEditor.index)] || {};
6730:       const existingUrl = existing.url || existing.artifact || "";
6731:       const nextItem = makeItemForSection(key, title, description, existingUrl);
6732:       project[key][Number(pendingEditor.index)] = { ...existing, ...nextItem, richDescription };
6733:     } else {
6734:       const nextItem = makeItemForSection(key, title, description);
6735:       project[key].push({ ...nextItem, richDescription });
6736:     }
6737:   }
6738:
6739:   if (pendingEditor.type === "design-object") {
6740:     const items = designArray(project, pendingEditor.path);
6741:     const kind = pendingEditor.kind || "document";
6742:     if (pendingEditor.mode === "edit") {
6743:       const existing = items[Number(pendingEditor.index)] || {};
6744:       const existingUrl = existing.url || existing.artifact || "";
6745:       items[Number(pendingEditor.index)] = { ...existing, ...makeDesignItem(kind, title, description,
existingUrl), richDescription };
6746:     } else {
6747:       items.push({ ...makeDesignItem(kind, title, description), richDescription });
6748:     }
6749:   }
6750:
6751:   if (pendingEditor.type === "text-array") {
6752:     const key = pendingEditor.key;
6753:     project[key] = project[key] || [];
6754:     if (pendingEditor.mode === "edit") {
6755:       project[key][Number(pendingEditor.index)] = title;
6756:     } else {
6757:       project[key].push(title);
6758:     }
6759:   }
6760:
6761:   if (pendingEditor.type === "custom") {
6762:     const section = (project.sections || []).find((item) => item.id === pendingEditor.sectionId);
6763:     if (!section) return;
6764:     section.items = section.items || [];
6765:     const nextItem = { title, description, richDescription, type: "Text", status: "draft" };
6766:     if (pendingEditor.mode === "edit") {
6767:       section.items[Number(pendingEditor.index)] = { ...section.items[Number(pendingEditor.index)],
...nextItem };
6768:     } else {
6769:       section.items.push(nextItem);
6770:     }
6771:   }
6772:
6773:   if (pendingEditor.type === "custom-section") {
6774:     const section = (project.sections || []).find((item) => item.id === pendingEditor.sectionId);
6775:     if (!section) return;
6776:     section.title = title;
6777:     section.description = description;
6778:     section.richDescription = richDescription;
6779:   }
6780:
6781:   pendingEditor = null;
6782:   setStatus("Saved in the project editor. Click Save project to include this version in the portfolio
preview.");
6783:   scheduleAutosave();
6784:   renderAll();
6785: }

```

section - lines 6762-6765

Item	Explanation
Source location	Lines 6762-6765 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.

Item	Explanation
Declaration	<code>const section = (project.sections []).find((item) => item.id === pendingEditor.sectionId);</code>
Touches	Mostly local calculations or local UI state.
Calls detected	<code>find</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 <code>await</code> expression(s), 1 <code>return</code> statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6762:     const section = (project.sections || []).find((item) => item.id === pendingEditor.sectionId);
6763:     if (!section) return;
6764:     section.items = section.items || [];
6765:     const nextItem = { title, description, richDescription, type: "Text", status: "draft" };

```

section - lines 6774-6779

Item	Explanation
Source location	Lines 6774-6779 (6 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	<code>const section = (project.sections []).find((item) => item.id === pendingEditor.sectionId);</code>
Touches	Mostly local calculations or local UI state.
Calls detected	<code>find</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 <code>await</code> expression(s), 1 <code>return</code> statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6774:     const section = (project.sections || []).find((item) => item.id === pendingEditor.sectionId);
6775:     if (!section) return;
6776:     section.title = title;
6777:     section.description = description;
6778:     section.richDescription = richDescription;
6779: }

```

openEditorFromButton - lines 6787-6879

Item	Explanation
Source location	Lines 6787-6879 (93 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	<code>function openEditorFromButton(button) {</code>
Touches	<code>browser/Electron DOM</code>
Calls detected	<code>selectedProject</code> , <code>openPendingEditor</code> , <code>toolName</code> , <code>toolDescription</code> , <code>designArray</code> , <code>find</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 <code>await</code> expression(s), 6 <code>return</code> statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6787: function openEditorFromButton(button) {
6788:   const project = selectedProject();
6789:   if (!project) return;
6790:   const type = button.dataset.openEditor;
6791:   const mode = button.dataset.mode || "add";
6792:   const index = button.dataset.index;
6793:
6794:   if (type === "tool") {
6795:     const item = mode === "edit" ? project.tools?.[Number(index)] : {};
6796:     openPendingEditor({
6797:       type,
6798:       mode,
6799:       index,
6800:       title: toolName(item),
6801:       description: toolDescription(item),
6802:       richDescription: item?.richDescription || null
6803:     });
6804:     return;
6805:   }
6806:
6807:   if (type === "object") {
6808:     const key = button.dataset.key;
6809:     const item = mode === "edit" ? project[key]?.[Number(index)] : {};
6810:     openPendingEditor({
6811:       type,
6812:       mode,
6813:       key,
6814:       index,
6815:       title: item?.title || item?.name || item?.label || "",
6816:       description: item?.description || item?.caption || item?.result || item?.method || "",
6817:       richDescription: item?.richDescription || null
6818:     });
6819:     return;
6820:   }
6821:
6822:   if (type === "design-object") {
6823:     const pathValue = button.dataset.designPath;
6824:     const items = designArray(project, pathValue);
6825:     const item = mode === "edit" ? items?.[Number(index)] : {};
6826:     openPendingEditor({
6827:       type,
6828:       mode,
6829:       path: pathValue,
6830:       kind: button.dataset.designKind || "document",
6831:       index,
6832:       title: item?.title || item?.name || item?.label || "",
6833:       description: item?.description || item?.caption || item?.result || item?.method || "",
6834:       richDescription: item?.richDescription || null
6835:     });
6836:     return;
6837:   }
6838:
6839:
6840:   if (type === "custom") {
6841:     const section = (project.sections || []).find((item) => item.id === button.dataset.sectionId);
6842:     const item = mode === "edit" ? section?.items?.[Number(index)] : {};
6843:     openPendingEditor({
6844:       type,
6845:       mode,
6846:       sectionId: button.dataset.sectionId,
6847:       index,
6848:       title: item?.title || "",
6849:       description: item?.description || "",
6850:       richDescription: item?.richDescription || null
6851:     });
6852:     return;
6853:   }
6854:
6855:   if (type === "text-array") {
6856:     const key = button.dataset.key;
6857:     const item = mode === "edit" ? project[key]?.[Number(index)] : "";
6858:     openPendingEditor({
6859:       type,
6860:       mode,
6861:       key,
6862:       index,
6863:       title: typeof item === "string" ? item : item?.title || item?.name || item?.label || ""
6864:     });
6865:     return;
6866:   }
6867:
6868:   if (type === "custom-section") {
6869:     const section = (project.sections || []).find((item) => item.id === button.dataset.sectionId);
6870:     openPendingEditor({
6871:       type,
6872:       mode: "edit",
6873:       sectionId: button.dataset.sectionId,
6874:       title: section?.title || "",
6875:       description: section?.description || "",
6876:       richDescription: section?.richDescription || null
6877:     });
6878:   }

```

6879: }

section - lines 6841-6842

Item	Explanation
Source location	Lines 6841-6842 (2 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	<code>const section = (project.sections []).find((item) => item.id === button.dataset.sectionId);</code>
Touches	browser/Electron DOM
Calls detected	find
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
6841:     const section = (project.sections || []).find((item) => item.id === button.dataset.sectionId);
6842:     const item = mode === "edit" ? section?.items?.[Number(index)] : {};
```

section - lines 6869-6877

Item	Explanation
Source location	Lines 6869-6877 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	<code>const section = (project.sections []).find((item) => item.id === button.dataset.sectionId);</code>
Touches	browser/Electron DOM
Calls detected	find, openPendingEditor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
6869:     const section = (project.sections || []).find((item) => item.id === button.dataset.sectionId);
6870:     openPendingEditor({
6871:       type,
6872:       mode: "edit",
6873:       sectionId: button.dataset.sectionId,
6874:       title: section?.title || "",
6875:       description: section?.description || "",
6876:       richDescription: section?.richDescription || null
6877:     });
```

builderPreviewContent - lines 6881-6888

Item	Explanation
Source location	Lines 6881-6888 (8 source lines).
Purpose	Creates an object, UI structure, process, payload, or generated artifact.
Declaration	<code>function builderPreviewContent(rich, text = "", emptyText = "No content has been added yet.") {</code>
Touches	filesystem
Calls detected	richHasContent, renderRichContent, renderMultilineInlineText, escapeHtml
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6881: function builderPreviewContent(rich, text = "", emptyText = "No content has been added yet.") {
6882:   const content = richHasContent(rich)
6883:   ? renderRichContent(rich, text || "")
6884:   : text
6885:   ? `<p class="rich-paragraph">${renderMultilineInlineText(text)}</p>`
6886:   : `<p class="evidence-empty">${escapeHtml(emptyText)}</p>`;
6887:   return `<div class="builder-item-preview">${content}</div>`;
6888: }

```

renderCustomSection - lines 6890-6933

Item	Explanation
Source location	Lines 6890-6933 (44 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderCustomSection(project, sectionId) {
Touches	Mostly local calculations or local UI state.
Calls detected	find, builderPreviewContent, map, escapeHtml, richHasContent, join, renderPendingEditor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6890: function renderCustomSection(project, sectionId) {
6891:   const section = (project.sections || []).find((item) => item.id === sectionId);
6892:   if (!section) return `<p class="evidence-empty">Section not found.</p>`;
6893:
6894:   return `
6895:     <div class="section-window-heading">
6896:       <h3>${section.title || "Custom section"}</h3>
6897:       <div>
6898:         <button type="button" data-open-editor="custom-section" data-section-id="${section.id}">Edit
6899:       </div>
6900:     </div>
6901:     <div>
6902:       ${builderPreviewContent(section.richDescription, section.description || "", "No section content has
6903:       been added yet.")}
6904:     </div>
6905:     <div class="section-actions">
6906:       <button class="button primary" type="button" data-add-custom-item="${section.id}">Add
6907:     </div>
6908:     <div>
6909:       <button class="button secondary" type="button" data-upload-custom="${section.id}">Add file or
6910:       image</button>
6911:     </div>
6912:     <div class="builder-items">
6913:       ${((section.items || []).map((item, index) => `
6914:         <article class="builder-item">
6915:           <div>
6916:             <strong>${escapeHtml(item.title || (item.url ? "Untitled file" : "Untitled
6917:             subsection"))}</strong>
6918:             ${item.url ? `<span>${escapeHtml(item.url)}</span>` : ""}
6919:             ${richHasContent(item.richDescription) || item.description} ?
6920:             builderPreviewContent(item.richDescription, item.description || "", "") : ""
6921:             ${item.children || []}.length ? `
6922:             <div class="builder-child-files">
6923:               ${item.children || []}.map((child, childIndex) => `
6924:                 <div>
6925:                   <span>${escapeHtml(child.title || "Untitled file")}</span>
6926:                   <button class="danger-icon" type="button" data-delete-custom-
6927:                   child="${escapeHtml(section.id)}" data-index="${index}" data-child-index="${childIndex}">Delete</button>
6928:                 </div>
6929:               `).join("")}
6930:             </div>
6931:           </div>
6932:         </article>
6933:       `)}
6934:     </div>

```



```

6923:         ` : ""}
6924:     </div>
6925:     ${item.url ? "" : `<button type="button" data-upload-custom-
subsection="${escapeHtml(section.id)}" data-index="${index}">Add file</button>`}
6926:     ${item.url ? "" : `<button type="button" data-open-editor="custom" data-mode="edit" data-
section-id="${escapeHtml(section.id)}" data-index="${index}">Edit</button>`}
6927:     <button class="danger-icon" type="button" data-delete-custom-item="${section.id}" data-
index="${index}">Delete</button>
6928: </article>
6929: `).join("") || `<p class="evidence-empty">No subsections added yet.</p>`}
6930: </div>
6931: ${renderPendingEditor()}
6932: `;
6933: }

```

section - lines 6891-6892

Item	Explanation
Source location	Lines 6891-6892 (2 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const section = (project.sections []).find((item) => item.id === sectionId);
Touches	Mostly local calculations or local UI state.
Calls detected	find
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6891:     const section = (project.sections || []).find((item) => item.id === sectionId);
6892:     if (!section) return `<p class="evidence-empty">Section not found.</p>`;

```

compileLanguageOptions - lines 6935-6940

Item	Explanation
Source location	Lines 6935-6940 (6 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function compileLanguageOptions(selected = "javascript") {
Touches	filesystem
Calls detected	normalizeCodeLanguage, map, escapeHtml, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6935: function compileLanguageOptions(selected = "javascript") {
6936:     const normalized = normalizeCodeLanguage(selected);
6937:     return supportedCodeLanguages.map((language) => `
6938:         <option value="${escapeHtml(language.id)}"${language.id === normalized ? " selected" :
""}>${escapeHtml(language.label)}</option>
6939:     `).join("");
6940: }

```

compileRoleOptions - lines 6942-6947

Item	Explanation
Source location	Lines 6942-6947 (6 source lines).

Item	Explanation
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function compileRoleOptions(selected = "design") {
Touches	filesystem
Calls detected	normalizeCompileFileRole, map, escapeHtml, compileFileRoleLabel, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6942: function compileRoleOptions(selected = "design") {
6943:   const normalized = normalizeCompileFileRole(selected, "verilog");
6944:   return ["design", "testbench"].map((role) => `
6945:     <option value="${escapeHtml(role)}"${role === normalized ? " selected" :
""}>${escapeHtml(compileFileRoleLabel(role))}</option>
6946:   `).join("");
6947: }

```

hdlTestbenchTemplate - lines 6949-6973

Item	Explanation
Source location	Lines 6949-6973 (25 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function hdlTestbenchTemplate(language = "systemverilog", designName = "design") {
Touches	filesystem
Calls detected	slugify, replace, dut, clk, reset_n, done, dumpfile, dumpvars, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

6949: function hdlTestbenchTemplate(language = "systemverilog", designName = "design") {
6950:   const tbName = `tb_${slugify(designName || "design").replace(/-/g, "_")} || "design"`;
6951:   const dutName = slugify(designName || "design").replace(/-/g, "_") || "design";
6952:   const commentPrefix = language === "verilog" ? "/*" : "/*";
6953:   return [
6954:     `timescale 1ns/1ps`,
6955:     ``,
6956:     `module ${tbName};`,
6957:     `  ${commentPrefix} Declare testbench signals here.`,
6958:     `  ${commentPrefix} Example: reg clk = 0; reg reset_n = 0; wire done;`,
6959:     ``,
6960:     `  ${commentPrefix} Instantiate the design under test and connect the signals.`,
6961:     `  ${commentPrefix} Example: ${dutName} dut (.clk(clk), .reset_n(reset_n), .done(done));`,
6962:     ``,
6963:     `  initial begin`,
6964:     `    $dumpfile("waveform.vcd");`,
6965:     `    $dumpvars(0, ${tbName});`,
6966:     ``,
6967:     `    ${commentPrefix} Drive stimulus over time here.`,
6968:     `    #100;`,
6969:     `    $finish;`,
6970:     `  end`,
6971:     `endmodule`
6972:   ].join("\n");
6973: }

```

firstHdlModuleName - lines 6975-6978

Item	Explanation
Source location	Lines 6975-6978 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function firstHdlModuleName(code = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	match
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
6975: function firstHdlModuleName(code = "") {
6976:   const match = String(code || "").match(/bmodule\s+([A-Za-z_$][\w$]*)/);
6977:   return match?.[1] || "design";
6978: }
```

compileAppendDestinations - lines 6980-7026

Item	Explanation
Source location	Lines 6980-7026 (47 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function compileAppendDestinations(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	isAnalogProject, ensureDesignModel, forEach, push, summaryRichPathFromTextPath, walkItems
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
6980: function compileAppendDestinations(project) {
6981:   if (!project) return [];
6982:   const destinations = [
6983:     id: "project-overview",
6984:     label: "Project overview",
6985:     richPath: "summaryRich",
6986:     textPath: "summary"
6987:   ];
6988:
6989:   if (isAnalogProject(project)) {
6990:     ensureDesignModel(project);
6991:     [
6992:       ["design-overview", "Design overview", "design.brief.summary"],
6993:       ["documentation-overview", "Documentation overview", "design.documentation.summary"],
6994:       ["simulation-overview", "Simulation overview", "design.simulation.summary"]
6995:     ].forEach(([id, label, textPath]) => {
6996:       destinations.push({ id, label, richPath: summaryRichPathFromTextPath(textPath), textPath });
6997:     });
6998:   }
6999:
7000:   (project.sections || []).forEach((section, index) => {
7001:     destinations.push({
7002:       id: `custom-${section.id || index}`,
7003:       label: `${section.title || `Custom section ${index + 1}`} overview`,
7004:       richPath: `sections.${index}.richDescription`,
7005:       textPath: `sections.${index}.description`
7006:     });
7007:     const walkItems = (items = [], prefix = `sections.${index}.items`, labelPrefix = section.title || `Custom section ${index + 1}`) => {
7008:       (items || []).forEach((item, itemIndex) => {
7009:         const itemPath = `${prefix}.${itemIndex}`;
7010:         const itemTitle = item.title || `Subsection ${itemIndex + 1}`;
```

```

7011:         if (!item.url) {
7012:             destinations.push({
7013:                 id: `custom-${section.id || index}-${itemPath}`,
7014:                 label: `${labelPrefix} / ${itemTitle} overview`,
7015:                 richPath: `${itemPath}.richDescription`,
7016:                 textPath: `${itemPath}.description`
7017:             });
7018:         }
7019:         walkItems(item.children || item.items || [], `${itemPath}.children`, `${labelPrefix} /
${itemTitle}`);
7020:     });
7021:     };
7022:     walkItems(section.items || []);
7023: });
7024:
7025: return destinations;
7026: }

```

compileAppendDestinationOptions - lines 7028-7036

Item	Explanation
Source location	Lines 7028-7036 (9 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function compileAppendDestinationOptions(project, selectedId = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	compileAppendDestinations, some, map, escapeHtml, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7028: function compileAppendDestinationOptions(project, selectedId = "") {
7029:     const destinations = compileAppendDestinations(project);
7030:     const selected = destinations.some((destination) => destination.id === selectedId)
7031:       ? selectedId
7032:       : destinations[0]?.id || "";
7033:     return destinations.map((destination) => `
7034:       <option value="${escapeHtml(destination.id)}"${destination.id === selected ? " selected" :
""}>${escapeHtml(destination.label)}</option>
7035:     `).join("");
7036: }

```

activeCompileFile - lines 7038-7041

Item	Explanation
Source location	Lines 7038-7041 (4 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function activeCompileFile(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	ensureCompileCode, find
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7038: function activeCompileFile(project) {
7039:     const workspace = ensureCompileCode(project);
7040:     return workspace.files.find((file) => file.id === workspace.activeFileId) || workspace.files[0] ||
null;
7041: }

```

compileLogTimestamp - lines 7043-7047

Item	Explanation
Source location	Lines 7043-7047 (5 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function compileLogTimestamp(value = new Date().toISOString()) {
Touches	Mostly local calculations or local UI state.
Calls detected	toISOString, isNaN, getTime, toLocaleTimeString
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
7043: function compileLogTimestamp(value = new Date().toISOString()) {  
7044:   const date = new Date(value);  
7045:   if (Number.isNaN(date.getTime())) return "";  
7046:   return date.toLocaleTimeString([], { hour: "2-digit", minute: "2-digit", second: "2-digit" });  
7047: }
```

addCompileMessage - lines 7049-7062

Item	Explanation
Source location	Lines 7049-7062 (14 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function addCompileMessage(project, text, level = "info") {
Touches	Mostly local calculations or local UI state.
Calls detected	ensureCompileCode, slugify, now, random, toString, slice, toISOString, includes, updateCompileMessagesPanel
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
7049: function addCompileMessage(project, text, level = "info") {  
7050:   const workspace = ensureCompileCode(project);  
7051:   if (!workspace || !text) return;  
7052:   workspace.messages = [  
7053:     {  
7054:       id: slugify(`message-${Date.now()}-${Math.random().toString(16).slice(2)}`),  
7055:       at: new Date().toISOString(),  
7056:       level: ["info", "success", "warning", "error"].includes(level) ? level : "info",  
7057:       text: String(text)  
7058:     },  
7059:     ...(workspace.messages || [])  
7060:   ].slice(0, 80);  
7061:   updateCompileMessagesPanel(project);  
7062: }
```

renderCompileMessages - lines 7064-7073

Item	Explanation
Source location	Lines 7064-7073 (10 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderCompileMessages(workspace) {

Item	Explanation
Touches	Mostly local calculations or local UI state.
Calls detected	isArray, map, escapeHtml, compileLogTimestamp, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7064: function renderCompileMessages(workspace) {
7065:   const messages = Array.isArray(workspace?.messages) ? workspace.messages : [];
7066:   if (!messages.length) return `
```

updateCompileTerminalPanel - lines 7075-7084

Item	Explanation
Source location	Lines 7075-7084 (10 source lines).
Purpose	Updates UI, state, cache, or stored data.
Declaration	function updateCompileTerminalPanel(project, file = activeCompileFile(project)) {
Touches	browser/Electron DOM, filesystem
Calls detected	activeCompileFile, ensureCompileCode, querySelector, requestAnimationFrame
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7075: function updateCompileTerminalPanel(project, file = activeCompileFile(project)) {
7076:   const workspace = ensureCompileCode(project);
7077:   const terminal = file?.lastResult?.terminal || workspace?.terminal || compileTerminalStatus ||
"Compiler output will appear here after you save or run a source file.";
7078:   const terminalElement = sectionContent.querySelector("[data-compile-terminal]");
7079:   if (!terminalElement) return;
7080:   terminalElement.textContent = terminal;
7081:   requestAnimationFrame(() => {
7082:     terminalElement.scrollTop = terminalElement.scrollHeight;
7083:   });
7084: }

```

updateCompileMessagesPanel - lines 7086-7091

Item	Explanation
Source location	Lines 7086-7091 (6 source lines).
Purpose	Updates UI, state, cache, or stored data.
Declaration	function updateCompileMessagesPanel(project) {
Touches	browser/Electron DOM
Calls detected	ensureCompileCode, querySelector, renderCompileMessages
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7086: function updateCompileMessagesPanel(project) {
7087:   const workspace = ensureCompileCode(project);
7088:   const log = sectionContent.querySelector("[data-compile-messages]");
7089:   if (!log) return;
7090:   log.innerHTML = renderCompileMessages(workspace);
7091: }

```

showCompileOutput - lines 7093-7105

Item	Explanation
Source location	Lines 7093-7105 (13 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function showCompileOutput(project, file = activeCompileFile(project)) {
Touches	browser/Electron DOM, filesystem
Calls detected	activeCompileFile, updateCompileTerminalPanel, querySelector, add, scrollIntoView, requestAnimationFrame, remove
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7093: function showCompileOutput(project, file = activeCompileFile(project)) {
7094:   updateCompileTerminalPanel(project, file);
7095:   const panel = sectionContent.querySelector("[data-compile-terminal-panel]");
7096:   const terminal = sectionContent.querySelector("[data-compile-terminal]");
7097:   if (!panel || !terminal) return;
7098:   panel.classList.add("is-focused");
7099:   panel.scrollIntoView({ behavior: "smooth", block: "center" });
7100:   requestAnimationFrame(() => {
7101:     terminal.focus?({ preventScroll: true });
7102:     terminal.scrollTop = terminal.scrollHeight;
7103:   });
7104:   window.setTimeout(() => panel.classList.remove("is-focused"), 1400);
7105: }

```

activeCompileWaveform - lines 7107-7110

Item	Explanation
Source location	Lines 7107-7110 (4 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function activeCompileWaveform(project, file = activeCompileFile(project)) {
Touches	filesystem
Calls detected	activeCompileFile, ensureCompileCode
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7107: function activeCompileWaveform(project, file = activeCompileFile(project)) {
7108:   const workspace = ensureCompileCode(project);

```

```

7109:   return file?.lastResult?.waveform || file?.waveform || workspace?.waveform || null;
7110: }

```

scopeSignalValueAt - lines 7112-7117

Item	Explanation
Source location	Lines 7112-7117 (6 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function scopeSignalValueAt(changes = [], index = 0) {
Touches	authentication/security data
Calls detected	toLowerCase
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7112: function scopeSignalValueAt(changes = [], index = 0) {
7113:   const value = String(changes[index]?.value ?? "x").toLowerCase();
7114:   if (value === "1") return "1";
7115:   if (value === "0") return "0";
7116:   return "x";
7117: }

```

renderScalarWavePath - lines 7119-7140

Item	Explanation
Source location	Lines 7119-7140 (22 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderScalarWavePath(changes = [], maxTime = 1, rowY = 0, left = 150, width = 680) {
Touches	filesystem, authentication/security data
Calls detected	isArray, map, sort, max, min, forEach, xFor, yFor, scopeSignalValueAt, toFixed
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7119: function renderScalarWavePath(changes = [], maxTime = 1, rowY = 0, left = 150, width = 680) {
7120:   const normalized = (Array.isArray(changes) && changes.length ? changes : [{ time: 0, value: "x" }])
7121:     .map((change) => ({ time: Number(change.time) || 0, value: String(change.value ?? "x") }))
7122:     .sort((a, b) => a.time - b.time);
7123:   const top = rowY + 5;
7124:   const bottom = rowY + 23;
7125:   const middle = rowY + 14;
7126:   const yFor = (value) => value === "1" ? top : value === "0" ? bottom : middle;
7127:   const xFor = (time) => left + Math.max(0, Math.min(1, (Number(time) || 0) / Math.max(1, maxTime))) *
width;
7128:   let path = "";
7129:   normalized.forEach((change, index) => {
7130:     const x = xFor(change.time);
7131:     const y = yFor(scopeSignalValueAt(normalized, index));
7132:     if (index === 0) {
7133:       path += `M ${x.toFixed(1)} ${y.toFixed(1)}`;
7134:     } else {
7135:       path += `H ${x.toFixed(1)} V ${y.toFixed(1)}`;
7136:     }
7137:   });
7138:   path += `H ${(left + width).toFixed(1)}`;
7139:   return path;
7140: }

```


yFor - lines 7126-7137

Item	Explanation
Source location	Lines 7126-7137 (12 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const yFor = (value) => value === "1" ? top : value === "0" ? bottom : middle;
Touches	filesystem, authentication/security data
Calls detected	max, min, forEach, xFor, scopeSignalValueAt, toFixed
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
7126: const yFor = (value) => value === "1" ? top : value === "0" ? bottom : middle;
7127: const xFor = (time) => left + Math.max(0, Math.min(1, (Number(time) || 0) / Math.max(1, maxTime))) *
width;
7128: let path = "";
7129: normalized.forEach((change, index) => {
7130:   const x = xFor(change.time);
7131:   const y = yFor(scopeSignalValueAt(normalized, index));
7132:   if (index === 0) {
7133:     path += `M ${x.toFixed(1)} ${y.toFixed(1)}`;
7134:   } else {
7135:     path += `H ${x.toFixed(1)} V ${y.toFixed(1)}`;
7136:   }
7137: });
```

xFor - lines 7127-7137

Item	Explanation
Source location	Lines 7127-7137 (11 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const xFor = (time) => left + Math.max(0, Math.min(1, (Number(time) 0) / Math.max(1, maxTime))) * width;
Touches	filesystem, authentication/security data
Calls detected	max, min, forEach, yFor, scopeSignalValueAt, toFixed
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
7127: const xFor = (time) => left + Math.max(0, Math.min(1, (Number(time) || 0) / Math.max(1, maxTime))) *
width;
7128: let path = "";
7129: normalized.forEach((change, index) => {
7130:   const x = xFor(change.time);
7131:   const y = yFor(scopeSignalValueAt(normalized, index));
7132:   if (index === 0) {
7133:     path += `M ${x.toFixed(1)} ${y.toFixed(1)}`;
7134:   } else {
7135:     path += `H ${x.toFixed(1)} V ${y.toFixed(1)}`;
7136:   }
7137: });
```

renderBusWaveLabels - lines 7142-7152

Item	Explanation
Source location	Lines 7142-7152 (11 source lines).

Item	Explanation
Purpose	Renders UI, markup, or display output.
Declaration	function renderBusWaveLabels(changes = [], maxTime = 1, rowY = 0, left = 150, width = 680) {
Touches	filesystem, authentication/security data
Calls detected	isArray, map, sort, slice, max, min, toFixed, escapeHtml, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7142: function renderBusWaveLabels(changes = [], maxTime = 1, rowY = 0, left = 150, width = 680) {
7143:   const normalized = (Array.isArray(changes) && changes.length ? changes : [{ time: 0, value: "x" }])
7144:   .map((change) => ({ time: Number(change.time) || 0, value: String(change.value ?? "x") }))
7145:   .sort((a, b) => a.time - b.time)
7146:   .slice(0, 12);
7147:   return normalized.map((change, index) => {
7148:     const x = left + Math.max(0, Math.min(1, change.time / Math.max(1, maxTime))) * width;
7149:     const value = change.value.length > 12 ? `${change.value.slice(0, 12)}...` : change.value;
7150:     return `

```

renderHdlScope - lines 7154-7203

Item	Explanation
Source location	Lines 7154-7203 (50 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderHdlScope(waveform) {
Touches	filesystem, authentication/security data
Calls detected	isArray, slice, max, flatMap, map, round, toFixed, join, escapeHtml, renderScalarWavePath, renderBusWaveLabels
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7154: function renderHdlScope(waveform) {
7155:   const signals = Array.isArray(waveform?.signals) ? waveform.signals.slice(0, 16) : [];
7156:   if (!signals.length) {
7157:     return `
7158:       <div class="compile-scope-empty">
7159:         <strong>No waveform scope yet.</strong>
7160:         <span>Run a Verilog/SystemVerilog design with a testbench that calls <code>$dumpfile</code> and
<code>$dumpvars</code>.</span>
7161:       </div>
7162:     `;
7163:   }
7164:   const left = 165;
7165:   const width = 760;
7166:   const rowHeight = 34;
7167:   const top = 28;
7168:   const height = top + signals.length * rowHeight + 28;
7169:   const maxTime = Math.max(1, Number(waveform.maxTime) || Math.max(...signals.flatMap((signal) =>
(signal.changes || []).map((change) => Number(change.time) || 0)), 1));
7170:   const grid = [0, 0.25, 0.5, 0.75, 1].map((ratio) => {
7171:     const x = left + ratio * width;
7172:     const label = Math.round(maxTime * ratio);
7173:     return `

```

```

7176:     const y = top + index * rowHeight;
7177:     const name = String(signal.name || signal.reference || `signal_${index + 1}`);
7178:     const changes = Array.isArray(signal.changes) ? signal.changes : [];
7179:     const scalar = Number(signal.width || 1) <= 1;
7180:     return `
7181:       <g class="scope-row">
7182:         <text x="12" y="${y + 20}" class="scope-signal-label">${escapeHtml(name.length > 24 ?
` ${name.slice(0, 24)}...` : name)}</text>
7183:         <line x1="${left}" x2="${left + width}" y1="${y + 27}" y2="${y + 27}" class="scope-row-line" />
7184:         ${scalar
7185:           ? `<path d="${renderScalarWavePath(changes, maxTime, y, left, width)}" class="scope-wave" />`
7186:           : `<line x1="${left}" x2="${left + width}" y1="${y + 15}" y2="${y + 15}" class="scope-bus"
/>${renderBusWaveLabels(changes, maxTime, y, left, width)}`}
7187:       </g>
7188:     `;
7189:   }).join("");
7190:   return `
7191:     <div class="compile-scope-meta">
7192:       <span>${escapeHtml(waveform.source || "waveform.vcd")}</span>
7193:       <span>${signals.length} signal${signals.length === 1 ? "" : "s"}</span>
7194:       <span>0 to ${escapeHtml(maxTime)} ${escapeHtml(waveform.timeScale || "ticks")}</span>
7195:     </div>
7196:     <div class="compile-scope-scroll">
7197:       <svg class="compile-scope-svg" viewBox="0 0 ${left + width + 20} ${height}" role="img" aria-
label="HDL waveform scope">
7198:         ${grid}
7199:         ${rows}
7200:       </svg>
7201:     </div>
7202:   `;
7203: }

```

renderCompileScopePanel - lines 7205-7217

Item	Explanation
Source location	Lines 7205-7217 (13 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderCompileScopePanel(project, file = activeCompileFile(project)) {
Touches	filesystem, authentication/security data
Calls detected	activeCompileFile, activeCompileWaveform, isHdlLanguage, renderHdlScope
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7205: function renderCompileScopePanel(project, file = activeCompileFile(project)) {
7206:   const waveform = activeCompileWaveform(project, file);
7207:   if (!isHdlLanguage(file?.language) && !waveform) return "";
7208:   return `
7209:     <section class="compile-scope-panel" data-compile-scope-panel aria-label="HDL signal scope">
7210:       <div class="compile-terminal-heading">
7211:         <span class="compile-terminal-title"><span aria-hidden="true" class="scope-dot"></span> Signal
scope</span>
7212:         <button type="button" data-compile-show-scope>Show scope</button>
7213:       </div>
7214:       ${renderHdlScope(waveform)}
7215:     </section>
7216:   `;
7217: }

```

showCompileScope - lines 7219-7228

Item	Explanation
Source location	Lines 7219-7228 (10 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function showCompileScope(project, file = activeCompileFile(project)) {
Touches	browser/Electron DOM, authentication/security data
Calls detected	activeCompileFile, querySelector, addCompileMessage, add, scrollIntoView, remove

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7219: function showCompileScope(project, file = activeCompileFile(project)) {
7220:   const panel = sectionContent.querySelector("[data-compile-scope-panel]");
7221:   if (!panel) {
7222:     addCompileMessage(project, "No HDL scope is available for the active source yet.", "warning");
7223:     return;
7224:   }
7225:   panel.classList.add("is-focused");
7226:   panel.scrollIntoView({ behavior: "smooth", block: "center" });
7227:   window.setTimeout(() => panel.classList.remove("is-focused"), 1400);
7228: }

```

copyCompileOutput - lines 7230-7245

Item	Explanation
Source location	Lines 7230-7245 (16 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	async function copyCompileOutput(project, file = activeCompileFile(project)) {
Touches	filesystem
Calls detected	activeCompileFile, ensureCompileCode, trim, addCompileMessage, writeText, setStatus
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7230: async function copyCompileOutput(project, file = activeCompileFile(project)) {
7231:   const workspace = ensureCompileCode(project);
7232:   const text = file?.lastResult?.terminal || workspace?.terminal || compileTerminalStatus || "";
7233:   if (!text.trim()) {
7234:     addCompileMessage(project, "There is no terminal output to copy yet.", "warning");
7235:     return;
7236:   }
7237:   try {
7238:     await navigator.clipboard.writeText(text);
7239:     addCompileMessage(project, "Terminal output copied to the clipboard.", "success");
7240:     setStatus("Terminal output copied.");
7241:   } catch {
7242:     addCompileMessage(project, "Terminal output could not be copied by this browser.", "warning");
7243:     setStatus("Terminal output could not be copied.");
7244:   }
7245: }

```

clearCompileOutput - lines 7247-7256

Item	Explanation
Source location	Lines 7247-7256 (10 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function clearCompileOutput(project, file = activeCompileFile(project)) {
Touches	filesystem
Calls detected	activeCompileFile, ensureCompileCode, updateCompileTerminalPanel, addCompileMessage, setStatus, scheduleAutosave
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7247: function clearCompileOutput(project, file = activeCompileFile(project)) {
7248:   const workspace = ensureCompileCode(project);
7249:   if (file?.lastResult) file.lastResult.terminal = "";
7250:   if (workspace) workspace.terminal = "";
7251:   compileTerminalStatus = "";
7252:   updateCompileTerminalPanel(project, file);
7253:   addCompileMessage(project, "Terminal output cleared.", "info");
7254:   setStatus("Terminal output cleared.");
7255:   scheduleAutosave();
7256: }

```

richBlocksForCompileAppend - lines 7258-7265

Item	Explanation
Source location	Lines 7258-7265 (8 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function richBlocksForCompileAppend(project, destination) {
Touches	Mostly local calculations or local UI state.
Calls detected	getByPath, richBlocksFromValue, filter, trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7258: function richBlocksForCompileAppend(project, destination) {
7259:   const existingRich = getByPath(project, destination.richPath);
7260:   const existingText = getByPath(project, destination.textPath) || "";
7261:   return richBlocksFromValue(existingRich, existingText).filter((block) => {
7262:     if (block.type !== "paragraph") return true;
7263:     return Boolean(String(block.text || "").trim() || block.html);
7264:   });
7265: }

```

codeBlockForCompileFile - lines 7267-7274

Item	Explanation
Source location	Lines 7267-7274 (8 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function codeBlockForCompileFile(file) {
Touches	filesystem
Calls detected	normalizeCodeText, normalizeCodeLanguage, detectCodeLanguage
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7267: function codeBlockForCompileFile(file) {

```

```

7268:   return {
7269:     code: normalizeCodeText(file?.code || "", "source"),
7270:     language: normalizeCodeLanguage(file?.language || detectCodeLanguage(file?.code || "", file?.fileName)
|| "")),
7271:     pasteMode: "source",
7272:     type: "code"
7273:   };
7274: }

```

compileFileDirtyLabel - lines 7276-7283

Item	Explanation
Source location	Lines 7276-7283 (8 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function compileFileDirtyLabel(file) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 6 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7276: function compileFileDirtyLabel(file) {
7277:   if (!file) return "";
7278:   if (file.dirty) return "Unsaved";
7279:   if (file.lastResult?.ok === true) return "Compiled";
7280:   if (file.lastResult?.ok === false) return "Needs fix";
7281:   if (file.savedAt) return "Saved";
7282:   return "Draft";
7283: }

```

renderCompileCodeSection - lines 7285-7408

Item	Explanation
Source location	Lines 7285-7408 (124 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderCompileCodeSection(project) {
Touches	filesystem, authentication/security data
Calls detected	ensureCompileCode, activeCompileFile, escapeHtml, map, match, replace, compileFileDirtyLabel, join, defaultCodeFileName, compileLanguageOptions, isHdlLanguage, compileRoleOptions
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7285: function renderCompileCodeSection(project) {
7286:   const workspace = ensureCompileCode(project);
7287:   const activeFile = activeCompileFile(project);
7288:   const terminal = activeFile?.lastResult?.terminal || workspace.terminal || compileTerminalStatus ||
"Compiler output will appear here after you save or run a source file.";
7289:   return
7290:     <div class="compile-code-workspace">
7291:       <aside class="compile-code-sidebar" aria-label="Compile code files">
7292:         <div class="compile-code-sidebar-heading">
7293:           <h3>Compile Code</h3>
7294:           <small>${workspace.files.length} source file${workspace.files.length === 1 ? "" : "s"}</small>
7295:         </div>
7296:         <div class="compile-code-workspace-tree" aria-label="Project workspace tree">
7297:           <details open>
7298:             <summary>${escapeHtml(project.title || "Project workspace")}</summary>

```

```

7299:         <div class="compile-code-tree-files">
7300:             ${workspace.files.map((file) => `
7301:                 <button class="compile-code-tree-file${file.id} === workspace.activeFileId ? " is-active"
: "}" type="button" data-compile-select="${escapeHtml(file.id)}">
7302:                     <span class="compile-file-extension">${escapeHtml((file.fileName.match(/\.[^.]?$/)[0]
|| "src").replace(".", ""))}</span>
7303:                     <span>${escapeHtml(file.fileName)}</span>
7304:                     <small>${escapeHtml(compileFileDirtyLabel(file))}</small>
7305:                 </button>
7306:             `).join("") || `<p>No source files yet.</p>`}
7307:         </div>
7308:     </details>
7309: </div>
7310: <div class="compile-code-actions">
7311:     <button type="button" data-compile-add>Add code file</button>
7312:     <button type="button" data-compile-add-testbench>Add testbench</button>
7313:     <button type="button" data-compile-import>Import file</button>
7314:     <button type="button" data-compile-tools>Check compilers</button>
7315: </div>
7316: </aside>
7317:
7318: <section class="compile-code-main" aria-label="Compile code editor">
7319:     ${activeFile ?
7320:     <div class="compile-code-meta-grid">
7321:         <label>
7322:             <span>Title</span>
7323:             <input type="text" data-compile-field="title" value="${escapeHtml(activeFile.title || "")}"
placeholder="Source title" />
7324:         </label>
7325:         <label>
7326:             <span>File name</span>
7327:             <input type="text" data-compile-field="fileName" value="${escapeHtml(activeFile.fileName ||
"" )}" placeholder="${escapeHtml(defaultCodeFileName(activeFile.language))}" />
7328:         </label>
7329:         <label>
7330:             <span>Language</span>
7331:             <select data-compile-
field="language">${compileLanguageOptions(activeFile.language)}</select>
7332:         </label>
7333:         <label>
7334:             <span>${isHdlLanguage(activeFile.language) ? "HDL role" : "Source role"}</span>
7335:             ${isHdlLanguage(activeFile.language)
? `<select data-compile-field="role">${compileRoleOptions(activeFile.role)}</select>`
: `<input type="text" value="Program source" disabled />`}
7336:         </label>
7337:     </div>
7338:     <div class="compile-code-editor-grid compile-code-editor-grid-single">
7339:         <section class="compile-code-preview-panel compile-code-active-panel" aria-label="Active
syntax highlighted code editor">
7340:             <div class="compile-code-preview-heading">
7341:                 <span>${escapeHtml(codeLanguageLabel(activeFile.language))} active editor</span>
7342:             </div>
7343:             <div class="compile-code-active-editor">
7344:                 <pre class="compile-code-preview" aria-hidden="true"><code data-compile-
preview>${tokenizedCodeHtml(activeFile.code || "", activeFile.language)}</code></pre>
7345:                 <textarea data-compile-field="code" data-compile-active-editor spellcheck="false"
wrap="off" rows="18" placeholder="Type or paste code here">${escapeHtml(activeFile.code || "")}</textarea>
7346:             </div>
7347:         </section>
7348:     </div>
7349:     <div class="compile-stdin-field">
7350:         <span>Program input, optional</span>
7351:         <textarea data-compile-field="stdin" rows="4" placeholder="Input passed to stdin when the
code runs">${escapeHtml(activeFile.stdin || "")}</textarea>
7352:     </div>
7353:     <div class="compile-code-command-bar">
7354:         <button type="button" data-compile-save>Save source</button>
7355:         <button type="button" data-compile-beautify>Beautify</button>
7356:         <button type="button" data-compile-compile>Compile</button>
7357:         <button type="button" data-compile-build>Build project</button>
7358:         ${isHdlLanguage(activeFile.language)
? `<button type="button" data-compile-simulate>Simulate</button>`
: `<button type="button" data-compile-run>Run</button>`}
7359:         <button class="compile-output-button" type="button" data-compile-show-output title="Open the
terminal output for this source"><span class="compile-command-icon" aria-hidden="true">&gt;</span><span>Show
output</span></button>
7360:         ${isHdlLanguage(activeFile.language) ? `<button class="compile-output-button" type="button"
data-compile-show-scope title="Open the HDL waveform scope"><span class="compile-command-icon" aria-
hidden="true"></span><span>Show scope</span></button>` : ""}
7361:         <button type="button" data-compile-install>Install tools</button>
7362:         <button class="danger-icon" type="button" data-compile-delete>Delete source</button>
7363:     </div>
7364:     <section class="compile-append-panel" aria-label="Append code to project section">
7365:         <div>
7366:             <strong>Append code to project</strong>
7367:             <span>Insert the active source as a formatted code block into this project.</span>
7368:         </div>
7369:         <label>
7370:             <span>Destination</span>
7371:             <select data-compile-append-target>${compileAppendDestinationOptions(project,
workspace.appendTargetId)}</select>
7372:         </label>

```

```

7377:         <div class="compile-append-actions">
7378:             <button type="button" data-compile-append-code>Append code to project</button>
7379:         </div>
7380:     </section>
7381:     `
7382:     <div class="compile-code-empty">
7383:         <h3>No compile file selected</h3>
7384:         <p>Add a source file for C, C++, SystemVerilog, LTspice, Java, JavaScript, Python, or
HTML.</p>
7385:     </div>
7386: `}
7387: <section class="compile-terminal-panel" data-compile-terminal-panel aria-label="Compiler terminal
output">
7388:     <div class="compile-terminal-heading">
7389:         <span class="compile-terminal-title"><span aria-hidden="true" class="terminal-dot"></span>
OMB Local Terminal</span>
7390:         <span class="compile-terminal-controls">
7391:             <button type="button" data-compile-copy-output>Copy output</button>
7392:             <button type="button" data-compile-clear-output>Clear output</button>
7393:         </span>
7394:     </div>
7395:     <pre class="compile-terminal" data-compile-terminal tabindex="0" role="log" aria-
live="polite">${escapeHtml(terminal)}</pre>
7396:     </section>
7397:     ${renderCompileScopePanel(project, activeFile)}
7398:     <section class="compile-messages-panel" aria-label="Compile messages log">
7399:         <div class="compile-terminal-heading">
7400:             <span>Messages log</span>
7401:             <button type="button" data-compile-clear-messages>Clear log</button>
7402:         </div>
7403:         <div class="compile-messages-log" data-compile-
messages>${renderCompileMessages(workspace)}</div>
7404:     </section>
... excerpt truncated after 120 lines. Open the source file for the rest of this function.

```

renderSectionContent - lines 7410-7472

Item	Explanation
Source location	Lines 7410-7472 (63 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderSectionContent(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	sectionOptions, find, startsWith, renderCustomSection, replace, requestAnimationFrame, populateRichEditors, isAnalogProject, renderAnalogDesignWorkspace, renderObjectSection, filter, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 5 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7410: function renderSectionContent(project) {
7411:     if (!project) {
7412:         sectionContent.hidden = true;
7413:         sectionContent.innerHTML = "";
7414:         return;
7415:     }
7416:
7417:     const section = sectionOptions(project).find((item) => item.id === activeSectionId) ||
standardSections[0];
7418:     const isOverview = section.id === "brief";
7419:     projectFields.hidden = !isOverview;
7420:     sectionContent.hidden = isOverview;
7421:     if (isOverview) {
7422:         sectionContent.innerHTML = "";
7423:         return;
7424:     }
7425:
7426:     if (section.id.startsWith("custom:")) {
7427:         sectionContent.innerHTML = renderCustomSection(project, section.id.replace("custom:", ""));
7428:         requestAnimationFrame(() => populateRichEditors(sectionContent));
7429:         return;
7430:     }
7431:
7432:     const renderers = {
7433:         brief: () => "",
7434:         design: () => {
7435:             if (isAnalogProject(project)) return renderAnalogDesignWorkspace(project);

```



```

7436:         const cards = [
7437:             (project.documents || []).length ? `<section><h3>Documents</h3>${renderObjectSection(project,
"documents", "documents", "Documents")}</section>` : "",
7438:             (project.pcbcs || []).length ? `<section><h3>PCBs Built</h3>${renderObjectSection(project, "pcbcs",
"pcbcs", "PCBs")}</section>` : "",
7439:             (project.media || []).length ? `<section><h3>Images</h3>${renderObjectSection(project, "media",
"images", "Images")}</section>` : ""
7440:         ].filter(Boolean);
7441:         return `
7442:             <div class="section-stack">
7443:                 ${cards.join("") || renderEmptyDynamicSectionPrompt("design")}
7444:                 ${renderPendingEditor()}
7445:             </div>
7446:         `;
7447:     },
7448:     simulation: () => isAnalogProject(project) ? renderAnalogSimulationWorkspace(project) : "",
7449:     "compile-code": () => renderCompileCodeSection(project),
7450:     tests: () => `${renderObjectSection(project, "tests", "tests", "Tests")}${renderPendingEditor()}`,
7451:     tools: () => `
7452:         <div class="section-stack">
7453:             <section>
7454:                 <h3>Tools Used</h3>
7455:                 ${renderToolsSection(project)}
7456:             </section>
7457:             <section>
7458:                 <h3>Languages</h3>
7459:                 ${renderTextArray(project, "languages", "Language")}
7460:             </section>
7461:             <section>
7462:                 <h3>Links</h3>
7463:                 ${renderObjectSection(project, "links", "", "Links")}
7464:             </section>
7465:             ${renderPendingEditor()}
7466:         </div>
7467:     `;
7468: };
7469:
7470: sectionContent.innerHTML = renderers[section.id]();
7471: requestAnimationFrame(() => populateRichEditors(sectionContent));
7472: }

```

fullPortfolioPreviewHtml - lines 7474-7476

Item	Explanation
Source location	Lines 7474-7476 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function fullPortfolioPreviewHtml() {
Touches	Mostly local calculations or local UI state.
Calls detected	fullPortfolioPreviewHtmlExact
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7474: function fullPortfolioPreviewHtml() {
7475:     return fullPortfolioPreviewHtmlExact();
7476: }

```

previewValue - lines 7478-7480

Item	Explanation
Source location	Lines 7478-7480 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function previewValue(source, key, fallback) {
Touches	Mostly local calculations or local UI state.
Calls detected	call

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7478: function previewValue(source, key, fallback) {
7479:   return source && Object.prototype.hasOwnProperty.call(source, key) ? source[key] : fallback;
7480: }

```

fullPortfolioPreviewHtmlExact - lines 7482-7702

Item	Explanation
Source location	Lines 7482-7702 (221 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function fullPortfolioPreviewHtmlExact(dataOverride = null) {
Touches	filesystem, network/API requests, authentication/security data
Calls detected	parsedPortfolioGlobals, normalizeSiteContent, previewValue, normalizeProfile, normalizeFieldStyles, stringify, replaceAll, escapeHtml, plainFieldStyleAttribute, renderRichFieldContent, getFullYear
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7482: function fullPortfolioPreviewHtmlExact(dataOverride = null) {
7483:   const baseHref = `${window.location.origin}/`;
7484:   const parsedGlobals = dataOverride ? null : parsedPortfolioGlobals();
7485:   const siteContent = normalizeSiteContent(previewValue(dataOverride, "siteContent",
parsedGlobals?.siteContent || {}));
7486:   const profile = normalizeProfile(previewValue(dataOverride, "profile", parsedGlobals?.profile || {}));
7487:   const fieldStyles = normalizeFieldStyles(previewValue(dataOverride, "fieldStyles",
parsedGlobals?.fieldStyles || {}));
7488:   const profileRich = previewValue(dataOverride, "profileRich", parsedGlobals?.profileRich || {});
7489:   const siteContentRich = previewValue(dataOverride, "siteContentRich", parsedGlobals?.siteContentRich ||
{});
7490:   const ownerName = profile.displayName || "Portfolio";
7491:   const portfolioLabel = profile.portfolioLabel || "Portfolio";
7492:   const previewDataObject = {
7493:     categories: previewValue(dataOverride, "categories", parsedGlobals?.categories || []),
7494:     fieldStyles,
7495:     funFacts: previewValue(dataOverride, "funFacts", parsedGlobals?.funFacts || []),
7496:     funFactsRich: previewValue(dataOverride, "funFactsRich", parsedGlobals?.funFactsRich || null),
7497:     profile,
7498:     profileRich,
7499:     projects: previewValue(dataOverride, "projects", savedPortfolioCatalog.projects || []),
7500:     siteContent,
7501:     siteContentRich,
7502:     siteSections: previewValue(dataOverride, "siteSections", parsedGlobals?.siteSections || [])
7503:   };
7504:   const previewData = JSON.stringify(previewDataObject).replaceAll("</", "<\\");
7505:
7506:   return `<!DOCTYPE html>
7507: <html lang="en">
7508:   <head>
7509:     <meta charset="UTF-8" />
7510:     <meta name="viewport" content="width=device-width, initial-scale=1.0" />
7511:     <base href="${baseHref}" />
7512:     <meta name="description" content="A portfolio website built with OMB Portfolio Builder for projects,
documents, diagrams, source code, tests, links, and profile information." />
7513:     <meta name="robots" content="index, follow" />
7514:     <meta name="author" content="${escapeHtml(ownerName)}" />
7515:     <meta name="portfolio-ai-endpoint" content="https://omb-portfolio-ai.maurice-baraza-
otieno.workers.dev/api/portfolio-ai" />
7516:     <title>${escapeHtml(ownerName)} | ${escapeHtml(portfolioLabel)}</title>
7517:     <link rel="stylesheet" href="styles.css" />
7518:   </head>
7519:   <body>

```

```

7520:     <header class="site-header" id="top">
7521:         <a class="brand" href="#top" aria-label="Go to top">
7522:             <span class="brand-lockup" aria-hidden="true">
7523:                 <img class="brand-icon" alt="" hidden />
7524:                 <span class="omb-engraving">${escapeHtml(profile.brandText || "Portfolio")}</span>
7525:             </span>
7526:             <span>
7527:                 <strong${plainFieldStyleAttribute(fieldStyles, "profile-display-
name")}>${renderRichFieldContent(profileRich.displayName, ownerName)}</strong>
7528:                 <small${plainFieldStyleAttribute(fieldStyles, "profile-portfolio-
label")}>${renderRichFieldContent(profileRich.portfolioLabel, portfolioLabel)}</small>
7529:             </span>
7530:         </a>
7531:
7532:         <div class="header-right">
7533:             <nav class="site-nav" aria-label="Primary navigation">
7534:                 <a href="#ask-ai">Ask AI</a>
7535:                 <a href="#projects">Projects</a>
7536:                 <a href="#resume">Resume</a>
7537:                 <a href="#process">Process</a>
7538:                 <a href="#contact">Contact</a>
7539:             </nav>
7540:             <a class="header-avatar" href="#contact" aria-label="Go to contact information" hidden>
7541:                 <img alt="" />
7542:             </a>
7543:         </div>
7544:     </header>
7545:
7546:     <main>
7547:         <section class="hero" aria-labelledby="hero-title">
7548:             <img class="hero-image" alt="" hidden />
7549:             <div class="hero-shade" aria-hidden="true"></div>
7550:             <div class="hero-content">
7551:                 <p class="eyebrow">${plainFieldStyleAttribute(fieldStyles, "site-hero-
eyebrow")}>${renderRichFieldContent(siteContentRich.heroEyebrow, siteContent.heroEyebrow)}</p>
7552:                 <h1 id="hero-title">${plainFieldStyleAttribute(fieldStyles, "site-hero-
title")}>${renderRichFieldContent(siteContentRich.heroTitle, siteContent.heroTitle)}</h1>
7553:                 <p class="hero-copy">${plainFieldStyleAttribute(fieldStyles, "site-hero-
copy")}>${renderRichFieldContent(siteContentRich.heroCopy, siteContent.heroCopy)}</p>
7554:                 <div class="hero-actions">
7555:                     <a class="button primary" href="#projects">View projects</a>
7556:                     <a class="button secondary" href="#resume" hidden>View resume</a>
7557:                     <a class="button secondary" href="#" target="_blank" rel="noreferrer" hidden>GitHub
profile</a>
7558:                     <a class="ai-jump-link" href="#ask-ai" aria-label="Jump to the portfolio AI assistant">
7559:                         <svg viewBox="0 0 24 24" aria-hidden="true">
7560:                             <path d="M12 3v3m-5 7H4m16 0h-3M7.5 7.5 5.4 5.4m11.1 2.1 2.1-2.1M8 10h8v7a2 2 0 0 1-2
2h-4a2 2 0 0 1-2-2z" />
7561:                             <path d="M10 13h.01M14 13h.01M10 16h4" />
7562:                         </svg>
7563:                         <span>Ask AI</span>
7564:                     </a>
7565:                 </div>
7566:             </div>
7567:         </section>
7568:
7569:         <section class="fun-facts-section" id="fun-facts-callout" aria-label="Fun facts" hidden></section>
7570:
7571:         <section class="builder-download-section" aria-label="Download portfolio builder">
7572:             <a class="builder-download-link" href="https://github.com/otienomaurice/omb-portfolio-
builder/releases/latest/download/OMB-Portfolio-Builder-Setup-latest-x64.exe" download>
7573:                 <svg viewBox="0 0 24 24" aria-hidden="true">
7574:                     <path d="M12 3v11m0 0 4-4m-4 4-4-4" />
7575:                     <path d="M5 17v2h14v-2" />
7576:                 </svg>
7577:                 <span>Download builder application</span>
7578:             </a>
7579:         </section>
7580:
7581:         <section class="summary-band" aria-label="Portfolio highlights">
7582:             <div class="metric">
7583:                 <strong id="project-count">0</strong>
7584:                 <span>presented projects</span>
7585:             </div>
7586:             <div class="metric">
7587:                 <strong id="project-track-count">0 Tracks</strong>
7588:                 <span id="project-track-labels">project categories</span>
7589:             </div>
7590:             <div class="metric">
7591:                 <strong>Artifacts</strong>
7592:                 <span>source, diagrams, and reports</span>
7593:             </div>
7594:         </section>
7595:
7596:         <section class="section" id="projects" aria-labelledby="projects-title">
7597:             <div class="section-heading">
7598:                 <div>
7599:                     <p class="eyebrow">Selected work</p>
7600:                     <h2 id="projects-title">Engineering Projects</h2>
7601:                 </div>

```

... excerpt truncated after 120 lines. Open the source file for the rest of this function.

renderPreview - lines 7704-7706

Item	Explanation
Source location	Lines 7704-7706 (3 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderPreview() {
Touches	Mostly local calculations or local UI state.
Calls detected	fullPortfolioPreviewHtmlExact
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
7704: function renderPreview() {  
7705:   portfolioPreviewFrame.srcdoc = fullPortfolioPreviewHtmlExact();  
7706: }
```

openPortfolioPreview - lines 7708-7713

Item	Explanation
Source location	Lines 7708-7713 (6 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	async function openPortfolioPreview() {
Touches	browser/Electron DOM
Calls detected	saveAllSections, renderPreview, add, showModal
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
7708: async function openPortfolioPreview() {  
7709:   if (!(await saveAllSections({ refreshOpenPreviews: false }))) return;  
7710:   renderPreview();  
7711:   document.body.classList.add("full-window-open");  
7712:   portfolioPreviewDialog.showModal();  
7713: }
```

renderAll - lines 7715-7732

Item	Explanation
Source location	Lines 7715-7732 (18 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderAll() {
Touches	Mostly local calculations or local UI state.
Calls detected	ensureSiteSections, selectedProject, sectionOptions, some, renderFunFactsEditor, renderSiteContentEditor, renderProfileEditor, renderSiteSectionList, renderTree, renderFields, renderSectionTabs, renderSectionContent
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7715: function renderAll() {
7716:   ensureSiteSections();
7717:   if (!selectedProjectId && catalog.projects.length) selectedProjectId = catalog.projects[0].id;
7718:   const project = selectedProject();
7719:   if (project && !sectionOptions(project).some((section) => section.id === activeSectionId)) {
7720:     activeSectionId = "brief";
7721:   }
7722:   renderFunFactsEditor();
7723:   renderSiteContentEditor();
7724:   renderProfileEditor();
7725:   renderSiteSectionList();
7726:   renderTree();
7727:   renderFields(project);
7728:   renderSectionTabs(project);
7729:   renderSectionContent(project);
7730:   if (portfolioPreviewDialog?.open) renderPreview();
7731:   updateBuilderWorkflow();
7732: }

```

updateProjectField - lines 7734-7768

Item	Explanation
Source location	Lines 7734-7768 (35 source lines).
Purpose	Updates UI, state, cache, or stored data.
Declaration	function updateProjectField(field, value) {
Touches	browser/Electron DOM
Calls detected	selectedProject, split, map, trim, filter, limitWords, slugify, includes, setStatus, scheduleAutosave, scheduleChromeRender, schedulePreviewRender
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7734: function updateProjectField(field, value) {
7735:   const project = selectedProject();
7736:   if (!project) return;
7737:
7738:   let needsChromeRender = false;
7739:   let statusMessage = "Unsaved local changes.";
7740:   if (field === "focus") {
7741:     project.focus = value.split(",").map((item) => item.trim()).filter(Boolean);
7742:   } else if (field === "summary") {
7743:     const limited = limitWords(value, 1000);
7744:     project.summary = limited;
7745:     if (limited !== value && document.activeElement?.dataset?.field === "summary") {
7746:       document.activeElement.value = limited;
7747:       statusMessage = "Project overview is limited to 1000 words.";
7748:     }
7749:   } else if (field === "id") {
7750:     const nextId = slugify(value);
7751:     if (!nextId) return;
7752:     project.id = nextId;
7753:     selectedProjectId = nextId;
7754:     needsChromeRender = true;
7755:   } else {
7756:     project[field] = value;
7757:     needsChromeRender = ["title", "category", "status"].includes(field);
7758:   }
7759:
7760:   if (field === "title") projectWindowTitle.textContent = value || "Untitled project";
7761:   setStatus(statusMessage);
7762:   scheduleAutosave();
7763:   if (needsChromeRender) {
7764:     scheduleChromeRender();
7765:   } else {
7766:     schedulePreviewRender();
7767:   }

```

7768: }

renameSelectedProject - lines 7770-7775

Item	Explanation
Source location	Lines 7770-7775 (6 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function renameSelectedProject() {
Touches	Mostly local calculations or local UI state.
Calls detected	selectedProject, beginTitleRename
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
7770: function renameSelectedProject() {
7771:   const project = selectedProject();
7772:   if (!project) return;
7773:   projectTitleMenu.hidden = true;
7774:   beginTitleRename();
7775: }
```

projectMetaRows - lines 7777-7813

Item	Explanation
Source location	Lines 7777-7813 (37 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function projectMetaRows(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	categoryById, projectTemplateFor, find, flatMap, filter, toLocaleString
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
7777: function projectMetaRows(project) {
7778:   if (!project) return [];
7779:   const category = categoryById(project.category);
7780:   const template = projectTemplateFor(project);
7781:   const savedProject = (savedPortfolioCatalog.projects || []).find((item) => item.id === project.id);
7782:   const design = project.design || {};
7783:   const designFileCount = [
7784:     ...(design.brief?.files || []),
7785:     ...(design.documentation?.files || []),
7786:     ...(design.documentation?.references || []),
7787:     ...(design.documentation?.mathAnalysis || []),
7788:     ...(design.simulation?.files || []),
7789:     ...(design.simulation?.results || [])
7790:   ].length;
7791:   const uploadedFileCount = [
7792:     ...(project.documents || []),
7793:     ...(project.tests || []),
7794:     ...(project.pcbbs || []),
7795:     ...(project.media || []),
7796:     ...(project.links || []),
7797:     ...(project.sections || []).flatMap((section) => section.items || [])
7798:   ].filter((item) => item.url || item.artifact).length + designFileCount;
7799:   return [
7800:     ["Title", project.title || "Untitled project"],
7801:     ["Project ID / folder", project.id || "Not assigned"],
```

```

7802:     ["Category", category.label || project.category || "Not assigned"],
7803:     ["Status", project.status || "Draft"],
7804:     ["Appearance", template.label || project.templateLabel || "No appearance - white project layout"],
7805:     ["Appearance ID", project.templateId || "None"],
7806:     ["Portfolio preview", savedProject ? "Saved into parsed portfolio preview" : "Not yet saved into
portfolio preview"],
7807:     ["Custom sections", String((project.sections || []).length)],
7808:     ["Project files / links", String(uploadedFileCount)],
7809:     ["Tools", String((project.tools || []).length)],
7810:     ["Languages", String((project.languages || []).length)],
7811:     ["Last parsed", savedProject?.portfolioView?.builtAt ? new
Date(savedProject.portfolioView.builtAt).toLocaleString() : "Not parsed yet"]
7812:   ];
7813: }

```

savedProject - lines 7781-7782

Item	Explanation
Source location	Lines 7781-7782 (2 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	<code>const savedProject = (savedPortfolioCatalog.projects []).find((item) => item.id === project.id);</code>
Touches	Mostly local calculations or local UI state.
Calls detected	<code>find</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7781:   const savedProject = (savedPortfolioCatalog.projects || []).find((item) => item.id === project.id);
7782:   const design = project.design || {};

```

openProjectMetaDetails - lines 7815-7827

Item	Explanation
Source location	Lines 7815-7827 (13 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	<code>function openProjectMetaDetails() {</code>
Touches	Mostly local calculations or local UI state.
Calls detected	<code>selectedProject</code> , <code>projectMetaRows</code> , <code>map</code> , <code>join</code> , <code>showModal</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7815: function openProjectMetaDetails() {
7816:   const project = selectedProject();
7817:   if (!project) return;
7818:   projectTitleMenu.hidden = true;
7819:   projectMetaTitle.textContent = project.title || "Project details";
7820:   projectMetaGrid.innerHTML = projectMetaRows(project).map(([label, value]) => `
7821:     <article>
7822:       <span>${label}</span>
7823:       <strong>${value}</strong>
7824:     </article>
7825: `).join("");
7826:   if (!projectMetaDialog.open) projectMetaDialog.showModal();
7827: }

```

selectTitleText - lines 7829-7835

Item	Explanation
Source location	Lines 7829-7835 (7 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function selectTitleText() {
Touches	browser/Electron DOM
Calls detected	getSelection, createRange, selectNodeContents, removeAllRanges, addRange
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
7829: function selectTitleText() {
7830:   const selection = window.getSelection();
7831:   const range = document.createRange();
7832:   range.selectNodeContents(projectWindowTitle);
7833:   selection.removeAllRanges();
7834:   selection.addRange(range);
7835: }
```

beginTitleRename - lines 7837-7846

Item	Explanation
Source location	Lines 7837-7846 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function beginTitleRename() {
Touches	Mostly local calculations or local UI state.
Calls detected	selectedProject, focus, selectTitleText, setStatus
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
7837: function beginTitleRename() {
7838:   const project = selectedProject();
7839:   if (!project) return;
7840:   originalTitleDraft = project.title || "";
7841:   projectWindowTitle.contentEditable = "true";
7842:   projectWindowTitle.focus();
7843:   selectTitleText();
7844:   titleRenameActions.hidden = false;
7845:   setStatus("Edit the title, then click Save title.");
7846: }
```

endTitleRename - lines 7848-7865

Item	Explanation
Source location	Lines 7848-7865 (18 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function endTitleRename(saveChanges) {
Touches	Mostly local calculations or local UI state.

Item	Explanation
Calls detected	selectedProject, trim, getSelection, removeAllRanges, setStatus, scheduleAutosave, scheduleChromeRender
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7848: function endTitleRename(saveChanges) {
7849:   const project = selectedProject();
7850:   if (!project) return;
7851:   const nextTitle = projectWindowTitle.textContent.trim();
7852:   projectWindowTitle.contentEditable = "false";
7853:   titleRenameActions.hidden = true;
7854:   window.getSelection()?.removeAllRanges();
7855:   if (saveChanges && nextTitle) {
7856:     project.title = nextTitle;
7857:     projectWindowTitle.textContent = nextTitle;
7858:     setStatus("Project title saved in the editor. Click Save project to include this version in the
portfolio preview.");
7859:     scheduleAutosave();
7860:     scheduleChromeRender();
7861:   } else {
7862:     projectWindowTitle.textContent = originalTitleDraft || project.title || "Untitled project";
7863:     setStatus("Title edit canceled.");
7864:   }
7865: }

```

makeItemForSection - lines 7867-7874

Item	Explanation
Source location	Lines 7867-7874 (8 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function makeItemForSection(key, title, description = "", url = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 6 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7867: function makeItemForSection(key, title, description = "", url = "") {
7868:   if (key === "documents") return { title, type: "Document", url, status: url ? "uploaded" : "draft" };
7869:   if (key === "tests") return { name: title, method: description || "Validation notes", status: url ?
"uploaded" : "draft", artifact: url, result: description };
7870:   if (key === "pcbs") return { name: title, revision: "Rev A", status: url ? "uploaded" : "draft",
artifact: url };
7871:   if (key === "media") return { title, url, status: url ? "uploaded" : "draft", caption: description };
7872:   if (key === "links") return { label: title, url };
7873:   return { title, description, url, status: url ? "uploaded" : "draft" };
7874: }

```

addTextItem - lines 7876-7891

Item	Explanation
Source location	Lines 7876-7891 (16 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function addTextItem(key) {
Touches	Mostly local calculations or local UI state.

Item	Explanation
Calls detected	selectedProject, prompt, includes, push, makeItemForSection, setStatus, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7876: function addTextItem(key) {
7877:   const project = selectedProject();
7878:   const title = prompt("Type a title for this item.");
7879:   if (!project || !title) return;
7880:
7881:   project[key] = project[key] || [];
7882:   if (["highlights", "tools", "languages"].includes(key)) {
7883:     project[key].push(title);
7884:   } else {
7885:     const description = prompt("Add a short description if you want.") || "";
7886:     project[key].push(makeItemForSection(key, title, description));
7887:   }
7888:   setStatus("Item added in the editor. Click Save project to include this version in the portfolio
preview.");
7889:   scheduleAutosave();
7890:   renderAll();
7891: }

```

editObjectItem - lines 7893-7914

Item	Explanation
Source location	Lines 7893-7914 (22 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function editObjectItem(key, index) {
Touches	Mostly local calculations or local UI state.
Calls detected	selectedProject, prompt, setStatus, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

7893: function editObjectItem(key, index) {
7894:   const project = selectedProject();
7895:   const item = project?.[key]?.[index];
7896:   if (!item) return;
7897:
7898:   const currentTitle = item.title || item.name || item.label || "";
7899:   const nextTitle = prompt("Edit title.", currentTitle);
7900:   if (!nextTitle) return;
7901:   const nextDescription = prompt("Edit description.", item.description || item.caption || item.result ||
item.method || "") || "";
7902:
7903:   if ("title" in item) item.title = nextTitle;
7904:   if ("name" in item) item.name = nextTitle;
7905:   if ("label" in item) item.label = nextTitle;
7906:   if ("caption" in item) item.caption = nextDescription;
7907:   if ("result" in item) item.result = nextDescription;
7908:   if ("description" in item) item.description = nextDescription;
7909:   if (!("caption" in item) && !("result" in item) && !("description" in item)) item.description =
nextDescription;
7910:
7911:   setStatus("Item edited in the editor. Click Save project to include this version in the portfolio
preview.");
7912:   scheduleAutosave();
7913:   renderAll();
7914: }

```

chooseFile - lines 7916-7922

Item	Explanation
Source location	Lines 7916-7922 (7 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function chooseFile() {
Touches	Mostly local calculations or local UI state.
Calls detected	resolve, click
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
7916: function chooseFile() {
7917:   return new Promise((resolve) => {
7918:     filePicker.value = "";
7919:     filePicker.onchange = () => resolve(filePicker.files[0] || null);
7920:     filePicker.click();
7921:   });
7922: }
```

newCompileFile - lines 7924-7942

Item	Explanation
Source location	Lines 7924-7942 (19 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function newCompileFile(language = "javascript", seed = {}) {
Touches	filesystem
Calls detected	normalizeCodeLanguage, safeClientCodeFileName, defaultCodeFileName, normalizeCompileFileRole, inferCompileFileRole, slugify, now, random, toString, slice
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
7924: function newCompileFile(language = "javascript", seed = {}) {
7925:   const normalized = normalizeCodeLanguage(language);
7926:   const fileName = safeClientCodeFileName(seed.fileName || defaultCodeFileName(normalized), normalized);
7927:   const role = normalizeCompileFileRole(seed.role || inferCompileFileRole(fileName, seed.code || "",
normalized), normalized);
7928:   return {
7929:     id: slugify(`${fileName}-${Date.now()}-${Math.random().toString(16).slice(2)}`),
7930:     title: seed.title || fileName,
7931:     fileName,
7932:     language: normalized,
7933:     role,
7934:     code: String(seed.code || ""),
7935:     stdin: "",
7936:     savedPath: "",
7937:     savedAt: "",
7938:     waveform: null,
7939:     lastResult: null,
7940:     dirty: true
7941:   };
7942: }
```

activeCompileWorkspaceAndFile - lines 7944-7947

Item	Explanation
Source location	Lines 7944-7947 (4 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function activeCompileWorkspaceAndFile(project = selectedProject()) {
Touches	Mostly local calculations or local UI state.
Calls detected	selectedProject, ensureCompileCode, activeCompileFile
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
7944: function activeCompileWorkspaceAndFile(project = selectedProject()) {
7945:   const workspace = ensureCompileCode(project);
7946:   return { workspace, file: activeCompileFile(project) };
7947: }
```

updateCompilePreview - lines 7949-7952

Item	Explanation
Source location	Lines 7949-7952 (4 source lines).
Purpose	Updates UI, state, cache, or stored data.
Declaration	function updateCompilePreview(file) {
Touches	browser/Electron DOM, authentication/security data
Calls detected	querySelector, tokenizedCodeHtml
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
7949: function updateCompilePreview(file) {
7950:   const preview = sectionContent.querySelector("[data-compile-preview]");
7951:   if (preview && file) preview.innerHTML = tokenizedCodeHtml(file.code || "", file.language);
7952: }
```

syncCompileEditorScroll - lines 7954-7960

Item	Explanation
Source location	Lines 7954-7960 (7 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function syncCompileEditorScroll(textarea) {
Touches	browser/Electron DOM
Calls detected	closest, querySelector
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
7954: function syncCompileEditorScroll(textarea) {
7955:   const editor = textarea?.closest(".compile-code-active-editor");
7956:   const preview = editor?.querySelector(".compile-code-preview");
7957:   if (!preview || !textarea) return;
7958:   preview.scrollTop = textarea.scrollTop;
7959:   preview.scrollLeft = textarea.scrollLeft;
7960: }
```

compilePayload - lines 7962-7984

Item	Explanation
Source location	Lines 7962-7984 (23 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function compilePayload(project, file, options = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	ensureCompileCode, inferCompileFileRole, map
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
7962: function compilePayload(project, file, options = {}) {
7963:   const workspace = ensureCompileCode(project);
7964:   return {
7965:     projectId: project.id,
7966:     fileId: file.id,
7967:     title: file.title,
7968:     fileName: file.fileName,
7969:     language: file.language,
7970:     role: file.role || inferCompileFileRole(file.fileName, file.code, file.language),
7971:     code: file.code,
7972:     stdin: file.stdin || "",
7973:     workspaceFiles: (workspace.files || []).map((workspaceFile) => ({
7974:       id: workspaceFile.id,
7975:       title: workspaceFile.title,
7976:       fileName: workspaceFile.fileName,
7977:       language: workspaceFile.language,
7978:       role: workspaceFile.role || inferCompileFileRole(workspaceFile.fileName, workspaceFile.code,
workspaceFile.language),
7979:       code: workspaceFile.code || ""
7980:     })),
7981:     action: options.action || "",
7982:     forceRebuild: options.forceRebuild === true
7983:   };
7984: }
```

selectedCompileAppendDestination - lines 7986-7990

Item	Explanation
Source location	Lines 7986-7990 (5 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function selectedCompileAppendDestination(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	ensureCompileCode, compileAppendDestinations, find
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
7986: function selectedCompileAppendDestination(project) {
7987:   const workspace = ensureCompileCode(project);
7988:   const destinations = compileAppendDestinations(project);
7989:   return destinations.find((destination) => destination.id === workspace.appendTargetId) ||
destinations[0] || null;
7990: }
```

appendCompileCodeToProject - lines 7992-8019

Item	Explanation
Source location	Lines 7992-8019 (28 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	function appendCompileCodeToProject(project, file) {
Touches	Mostly local calculations or local UI state.
Calls detected	codeBlockForCompileFile, trim, setStatus, addCompileMessage, selectedCompileAppendDestination, richBlocksForCompileAppend, push, setByPath, plainTextFromRich, toISOString, scheduleAutosave, schedulePreviewRender
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
7992: function appendCompileCodeToProject(project, file) {
7993:   const block = codeBlockForCompileFile(file);
7994:   if (!block.code.trim()) {
7995:     setStatus("Type or import code before appending it to the project.");
7996:     addCompileMessage(project, "Append stopped because the active source is empty.", "warning");
7997:     return false;
7998:   }
7999:
8000:   const destination = selectedCompileAppendDestination(project);
8001:   if (!destination) {
8002:     setStatus("No project destination is available for this code block.");
8003:     addCompileMessage(project, "Append stopped because no destination section is available.", "warning");
8004:     return false;
8005:   }
8006:
8007:   const blocks = richBlocksForCompileAppend(project, destination);
8008:   blocks.push(block);
8009:   setByPath(project, destination.richPath, { blocks });
8010:   setByPath(project, destination.textPath, plainTextFromRich({ blocks }));
8011:
8012:   file.lastAppendedAt = new Date().toISOString();
8013:   addCompileMessage(project, `${file.fileName || file.title || "Source code"} appended to
${destination.label}.`, "success");
8014:   setStatus(`Code appended to ${destination.label}. Save project to include it in the portfolio
preview.`);
8015:   scheduleAutosave();
8016:   schedulePreviewRender();
8017:   updateCompileMessagesPanel(project);
8018:   return true;
8019: }
```

saveCompileFile - lines 8021-8038

Item	Explanation
Source location	Lines 8021-8038 (18 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	async function saveCompileFile(project, file) {
Touches	network/API requests
Calls detected	fetch, now, stringify, compilePayload, json, Error, toISOString, addCompileMessage
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	2 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8021: async function saveCompileFile(project, file) {
8022:   if (!project || !file) return null;
8023:   const response = await fetch(`/api/code/save?t=${Date.now()}`, {
8024:     method: "POST",
8025:     cache: "no-store",
8026:     headers: { "Content-Type": "application/json" },
8027:     body: JSON.stringify(compilePayload(project, file))
8028:   });
8029:   const result = await response.json();
8030:   if (!response.ok || !result.ok) throw new Error(result.error || "Code could not be saved.");
8031:   file.savedPath = result.saved?.sourcePath || "";
8032:   file.savedAt = result.saved?.savedAt || new Date().toISOString();
8033:   file.fileName = result.saved?.fileName || file.fileName;
8034:   file.language = result.saved?.language || file.language;
8035:   file.dirty = false;
8036:   addCompileMessage(project, `Saved ${file.fileName} || "source file"`, "info");
8037:   return result.saved;
8038: }

```

importCompileFile - lines 8040-8063

Item	Explanation
Source location	Lines 8040-8063 (24 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	async function importCompileFile(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	ensureCompileCode, chooseFile, flatMap, split, pop, toLowerCase, includes, setStatus, text, detectCodeLanguage, codeLanguageProfile, newCompileFile
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8040: async function importCompileFile(project) {
8041:   const workspace = ensureCompileCode(project);
8042:   const file = await chooseFile();
8043:   if (!file) return;
8044:   const allowedExtensions = supportedCodeLanguages.flatMap(item => item.extensions || []);
8045:   const extension = `.${String(file.name || "").split(".").pop().toLowerCase()}`;
8046:   if (!allowedExtensions.includes(extension)) {
8047:     setStatus("Choose a supported source file: C, C++, Verilog, SystemVerilog, LTspice, Java, JavaScript,
Python, or HTML.");
8048:     return;
8049:   }
8050:   const code = await file.text();
8051:   const language = detectCodeLanguage(code, file.name);
8052:   const profile = codeLanguageProfile(language);
8053:   const sourceFile = newCompileFile(profile.id, {
8054:     fileName: file.name,
8055:     title: file.name.replace(/\.[^.]*/g, ""),
8056:     code
8057:   });
8058:   workspace.files.push(sourceFile);
8059:   workspace.activeFileId = sourceFile.id;
8060:   setStatus(`Imported ${file.name}. Click Save source before compiling.`);
8061:   scheduleAutosave();
8062:   renderSectionContent(project);
8063: }

```

beautifyCompileFile - lines 8065-8085

Item	Explanation
Source location	Lines 8065-8085 (21 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	async function beautifyCompileFile(project, file) {
Touches	network/API requests
Calls detected	fetch, now, stringify, compilePayload, json, Error, setStatus, scheduleAutosave, renderSectionContent
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8065: async function beautifyCompileFile(project, file) {
8066:   if (!file) return;
8067:   try {
8068:     const response = await fetch(`/api/code/beautify?t=${Date.now()}`, {
8069:       method: "POST",
8070:       cache: "no-store",
8071:       headers: { "Content-Type": "application/json" },
8072:       body: JSON.stringify(compilePayload(project, file))
8073:     });
8074:     const result = await response.json();
8075:     if (!response.ok || !result.ok) throw new Error(result.error || "Beautifier failed.");
8076:     file.code = result.code || file.code;
8077:     file.language = result.language || file.language;
8078:     file.dirty = true;
8079:     setStatus("Code beautified. Review it, then save source.");
8080:     scheduleAutosave();
8081:     renderSectionContent(project);
8082:   } catch (error) {
8083:     setStatus(error.message || "Code beautifier failed.");
8084:   }
8085: }

```

compileActiveFile - lines 8087-8145

Item	Explanation
Source location	Lines 8087-8145 (59 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	async function compileActiveFile(project, file, options = {}) {
Touches	filesystem, network/API requests
Calls detected	ensureCompileCode, isHdlLanguage, addCompileMessage, updateCompileTerminalPanel, updateCompileMessagesPanel, saveCompileFile, fetch, now, stringify, compilePayload, json, toISOString
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	3 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8087: async function compileActiveFile(project, file, options = {}) {
8088:   if (!file) return;
8089:   const workspace = ensureCompileCode(project);
8090:   const action = options.action || (isHdlLanguage(file.language) ? "simulate" : "run");
8091:   const actionLabel = action === "compile" ? "Compile" : action === "build" ? "Build project" : action
=== "simulate" ? "Simulate" : "Run";
8092:   file.lastResult = {
8093:     ok: null,
8094:     terminal: `${actionLabel} started. Please wait...`
8095:   };
8096:   workspace.terminal = file.lastResult.terminal;
8097:   addCompileMessage(project, `${actionLabel} started for ${file.fileName || file.title || "source
file"}`, "info");
8098:   updateCompileTerminalPanel(project, file);

```



```

8099:   updateCompileMessagesPanel(project);
8100:   try {
8101:     await saveCompileFile(project, file);
8102:     const response = await fetch(`/api/code/compile?t=${Date.now()}`, {
8103:       method: "POST",
8104:       cache: "no-store",
8105:       headers: { "Content-Type": "application/json" },
8106:       body: JSON.stringify(compilePayload(project, file, {
8107:         action,
8108:         forceRebuild: options.forceRebuild === true || action === "build"
8109:       })))
8110:   });
8111:   const result = await response.json();
8112:   const compileResult = result.result || {};
8113:   file.lastResult = {
8114:     ok: Boolean(result.ok),
8115:     language: compileResult.language || file.language,
8116:     terminal: compileResult.terminal || result.error || "No compiler output was returned.",
8117:     waveform: compileResult.waveform || null,
8118:     finishedAt: new Date().toISOString()
8119:   };
8120:   file.waveform = compileResult.waveform || null;
8121:   ensureCompileCode(project).waveform = compileResult.waveform || null;
8122:   if (compileResult.saved) {
8123:     file.savedPath = compileResult.saved.sourcePath || file.savedPath;
8124:     file.savedAt = compileResult.saved.savedAt || file.savedAt;
8125:   }
8126:   file.dirty = false;
8127:   ensureCompileCode(project).terminal = file.lastResult.terminal;
8128:   updateCompileTerminalPanel(project, file);
8129:   addCompileMessage(project, result.ok ? `${actionLabel} succeeded for ${file.fileName}.` :
`${actionLabel} completed with errors for ${file.fileName}.`, result.ok ? "success" : "error");
8130:   setStatus(result.ok ? `${actionLabel} succeeded.` : `${actionLabel} completed with errors.`);
8131:   scheduleAutosave();
8132:   renderSectionContent(project);
8133: } catch (error) {
8134:   file.lastResult = {
8135:     ok: false,
8136:     terminal: error.message || "Compile/run failed.",
8137:     finishedAt: new Date().toISOString()
8138:   };
8139:   ensureCompileCode(project).terminal = file.lastResult.terminal;
8140:   updateCompileTerminalPanel(project, file);
8141:   addCompileMessage(project, `${actionLabel} failed for ${file.fileName || "source file"}.`, "error");
8142:   setStatus(error.message || `${actionLabel} failed.`);
8143:   renderSectionContent(project);
8144: }
8145: }

```

checkCompileTools - lines 8147-8162

Item	Explanation
Source location	Lines 8147-8162 (16 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	async function checkCompileTools(project) {
Touches	filesystem, network/API requests
Calls detected	fetch, now, json, Error, entries, map, join, ensureCompileCode, renderSectionContent, setStatus
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8147: async function checkCompileTools(project) {
8148:   try {
8149:     const response = await fetch(`/api/code/tools?t=${Date.now()}`, { cache: "no-store" });
8150:     const result = await response.json();
8151:     if (!response.ok || !result.ok) throw new Error(result.error || "Compiler status failed.");
8152:     const lines = Object.entries(result.tools?.languages || {}).map(([id, status]) => {
8153:       const toolText = Object.entries(status.tools || {}).map(([tool, value]) => `${tool}:
${value}`).join(", ") || "no compiler required";
8154:       return `${status.ready ? "READY" : "MISSING"} ${status.label} (${id}) - ${toolText}`;
8155:     });
8156:     compileTerminalStatus = [`Compile workspace: ${result.tools?.compileRoot || ""}`,
...lines].join("\n");
8157:     ensureCompileCode(project).terminal = compileTerminalStatus;
8158:     renderSectionContent(project);

```

```

8159:   } catch (error) {
8160:     setStatus(error.message || "Compiler status failed.");
8161:   }
8162: }

```

installCompileTools - lines 8164-8187

Item	Explanation
Source location	Lines 8164-8187 (24 source lines).
Purpose	Part of the Compile Code language/tool/build/run flow.
Declaration	async function installCompileTools(project, file) {
Touches	filesystem, network/API requests
Calls detected	codeLanguageLabel, renderSectionContent, fetch, now, stringify, json, toISOString, setStatus
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8164: async function installCompileTools(project, file) {
8165:   if (!file) return;
8166:   file.lastResult = { ok: null, terminal: `Installing tools for ${codeLanguageLabel(file.language)}. This
can take several minutes...` };
8167:   renderSectionContent(project);
8168:   try {
8169:     const response = await fetch(`/api/code/install-tools?t=${Date.now()}`, {
8170:       method: "POST",
8171:       cache: "no-store",
8172:       headers: { "Content-Type": "application/json" },
8173:       body: JSON.stringify({ language: file.language })
8174:     });
8175:     const result = await response.json();
8176:     file.lastResult = {
8177:       ok: Boolean(result.ok),
8178:       terminal: result.terminal || result.error || "Tool installation finished.",
8179:       finishedAt: new Date().toISOString()
8180:     };
8181:     setStatus(result.ok ? "Compiler tool installation finished." : "Compiler tool installation could not
complete.");
8182:     renderSectionContent(project);
8183:   } catch (error) {
8184:     file.lastResult = { ok: false, terminal: error.message || "Tool installation failed." };
8185:     renderSectionContent(project);
8186:   }
8187: }

```

readFileAsDataURL - lines 8189-8196

Item	Explanation
Source location	Lines 8189-8196 (8 source lines).
Purpose	Loads data from disk, network, browser storage, or a service.
Declaration	function readFileAsDataURL(file) {
Touches	filesystem
Calls detected	FileReader, resolve, readAsDataURL
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8189: function readFileAsDataURL(file) {
8190:   return new Promise((resolve, reject) => {

```

```

8191:     const reader = new FileReader();
8192:     reader.onload = () => resolve(reader.result);
8193:     reader.onerror = reject;
8194:     reader.readAsDataURL(file);
8195:   });
8196: }

```

uploadProfileAsset - lines 8198-8242

Item	Explanation
Source location	Lines 8198-8242 (45 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	async function uploadProfileAsset(kind) {
Touches	filesystem, network/API requests
Calls detected	syncProfileFromInputs, chooseFile, Set, has, startsWith, setStatus, includes, extensionFor, toLowerCase, readFileAsDataURL, fetch, now
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	4 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8198: async function uploadProfileAsset(kind) {
8199:   const profile = syncProfileFromInputs();
8200:   const file = await chooseFile();
8201:   if (!file) return;
8202:
8203:   const imageKinds = new Set(["profileImage", "heroImage", "brandImage"]);
8204:   if (imageKinds.has(kind) && !file.type.startsWith("image/")) {
8205:     setStatus("Choose an image file for this profile asset.");
8206:     return;
8207:   }
8208:
8209:   if (kind === "resumeUrl") {
8210:     const allowedResume = [".pdf", ".doc", ".docx"].includes(extensionFor(file.name).toLowerCase());
8211:     if (!allowedResume) {
8212:       setStatus("Choose a PDF, DOC, or DOCX resume file.");
8213:       return;
8214:     }
8215:   }
8216:
8217:   const data = await readFileAsDataURL(file);
8218:   const response = await fetch(`/api/upload?t=${Date.now()}`, {
8219:     method: "POST",
8220:     cache: "no-store",
8221:     headers: { "Content-Type": "application/json" },
8222:     body: JSON.stringify({
8223:       projectId: "_site-profile",
8224:       section: kind,
8225:       fileName: file.name,
8226:       data
8227:     })
8228:   });
8229:   const result = await response.json();
8230:   if (!response.ok) {
8231:     setStatus(result.error || "Profile asset upload failed.");
8232:     return;
8233:   }
8234:
8235:   profile[kind] = result.url;
8236:   catalog.profile = normalizeProfile(profile);
8237:   markDraftNeedsSave();
8238:   setStatus("Profile asset saved locally. Click Save draft before applying to site.");
8239:   scheduleAutosave();
8240:   schedulePreviewRender();
8241:   renderProfileEditor();
8242: }

```

extensionFor - lines 8244-8247

Item	Explanation
Source location	Lines 8244-8247 (4 source lines).

Item	Explanation
Purpose	Part of the private builder frontend and editing experience.
Declaration	function extensionFor(fileName) {
Touches	Mostly local calculations or local UI state.
Calls detected	match
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
8244: function extensionFor(fileName) {
8245:   const match = String(fileName || "").match(/(\.[a-z0-9]+)$/i);
8246:   return match ? match[1] : "";
8247: }
```

withExtension - lines 8249-8253

Item	Explanation
Source location	Lines 8249-8253 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function withExtension(name, fallbackFileName) {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, extensionFor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
8249: function withExtension(name, fallbackFileName) {
8250:   const trimmed = String(name || "").trim();
8251:   if (!trimmed) return fallbackFileName;
8252:   return /\.[a-z0-9]+$/i.test(trimmed) ? trimmed : `${trimmed}${extensionFor(fallbackFileName)}`;
8253: }
```

uploadProfile - lines 8255-8286

Item	Explanation
Source location	Lines 8255-8286 (32 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function uploadProfile(key) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8255: function uploadProfile(key) {
8256:   const anyProjectFile = {
8257:     label: "project file",
8258:     accept: "",
8259:     extensions: [],
8260:     mimeStarts: [],
8261:     allFiles: true
8262:   };
8263:   const profiles = {
8264:     media: {
8265:       ...anyProjectFile,
8266:       label: "image or file"
8267:     },
8268:     pcbs: {
8269:       ...anyProjectFile,
8270:       label: "PCB artifact"
8271:     },
8272:     tests: {
8273:       ...anyProjectFile,
8274:       label: "test artifact"
8275:     },
8276:     documents: {
8277:       ...anyProjectFile,
8278:       label: "document"
8279:     },
8280:     sections: {
8281:       ...anyProjectFile,
8282:       label: "section file"
8283:     }
8284:   };
8285:   return profiles[key] || profiles.sections;
8286: }

```

allowedFileMessage - lines 8288-8291

Item	Explanation
Source location	Lines 8288-8291 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function allowedFileMessage(profile) {
Touches	Mostly local calculations or local UI state.
Calls detected	join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8288: function allowedFileMessage(profile) {
8289:   if (profile.allFiles) return `Add any local file type for this ${profile.label}.`;
8290:   return `Add a ${profile.label}: ${profile.extensions.join(", ")}`;
8291: }

```

isAllowedUpload - lines 8293-8306

Item	Explanation
Source location	Lines 8293-8306 (14 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function isAllowedUpload(key, fileName, mimeType = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	uploadProfile, extensionFor, toLowerCase, includes, startsWith, some, allowedFileMessage
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8293: function isAllowedUpload(key, fileName, mimeType = "") {
8294:   const profile = uploadProfile(key);
8295:   if (profile.allFiles) return { ok: true, message: "" };
8296:   const extension = extensionFor(fileName.toLowerCase());
8297:   if (key === "media" && !profile.extensions.includes(extension) && !mimeType.startsWith("image/")) {
8298:     return { ok: false, message: "Images can only accept image files. Add PDFs under Documents, Tests, or
a custom section instead." };
8299:   }
8300:   const allowedByExtension = profile.extensions.includes(extension);
8301:   const allowedByMime = profile.mimeStarts.some((prefix) => mimeType.startsWith(prefix));
8302:   return {
8303:     ok: allowedByExtension || allowedByMime,
8304:     message: `${fileName} || "This file" is not a supported ${profile.label}.
${allowedFileMessage(profile)}.`
8305:   };
8306: }

```

validateAssetForSection - lines 8308-8317

Item	Explanation
Source location	Lines 8308-8317 (10 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function validateAssetForSection(key, asset) {
Touches	filesystem
Calls detected	isAllowedUpload, normalizeLinkTarget, extensionFromUrl, extensionFor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8308: function validateAssetForSection(key, asset) {
8309:   if (asset.source === "local") {
8310:     return isAllowedUpload(key, asset.fileName || asset.file?.name || "", asset.file?.type || "");
8311:   }
8312:   const url = normalizeLinkTarget(asset.url || "", { assumeWeb: true });
8313:   if (!extensionFromUrl(url) && !extensionFor(asset.title || "")) {
8314:     return { ok: true, message: "" };
8315:   }
8316:   return isAllowedUpload(key, url || asset.title || "", "");
8317: }

```

dialogValue - lines 8319-8328

Item	Explanation
Source location	Lines 8319-8328 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function dialogValue(dialog) {
Touches	Mostly local calculations or local UI state.
Calls detected	removeEventListener, resolve, addEventListener, showModal
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
8319: function dialogValue(dialog) {
8320:   return new Promise((resolve) => {
8321:     const onClose = () => {
8322:       dialog.removeEventListener("close", onClose);
8323:       resolve(dialog.returnValue === "save");
8324:     };
8325:     dialog.addEventListener("close", onClose);
8326:     dialog.showModal();
8327:   });
8328: }
```

onClose - lines 8321-8324

Item	Explanation
Source location	Lines 8321-8324 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const onClose = () => {
Touches	Mostly local calculations or local UI state.
Calls detected	removeEventListener, resolve
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
8321:   const onClose = () => {
8322:     dialog.removeEventListener("close", onClose);
8323:     resolve(dialog.returnValue === "save");
8324:   };
```

openAssetDialog - lines 8330-8392

Item	Explanation
Source location	Lines 8330-8392 (63 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	async function openAssetDialog(key) {
Touches	filesystem
Calls detected	uploadProfile, updateAssetDialogVisibility, dialogValue, trim, setStatus, withExtension, isAllowedUpload, normalizeLinkTarget, displayNameFromUrl, validateAssetForSection, text, inferUrlAssetKind
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 6 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
8330: async function openAssetDialog(key) {
8331:   const profile = uploadProfile(key);
8332:   assetDialogTitle.textContent = `Add ${profile.label}`;
8333:   assetSource.value = "local";
8334:   assetFile.value = "";
8335:   assetFile.accept = profile.accept;
8336:   assetUrl.value = "";
8337:   assetTitle.value = "";
8338:   assetCaption.value = "";
8339:   captionSource.value = "text";
8340:   captionFile.value = "";
8341:   updateAssetDialogVisibility();
8342:
8343:   const saved = await dialogValue(assetDialog);
8344:   if (!saved) return null;
```

```

8345:
8346:   let file = null;
8347:   let url = assetUrl.value.trim();
8348:   let fileName = "";
8349:   let title = assetTitle.value.trim();
8350:
8351:   if (assetSource.value === "local") {
8352:     file = assetFile.files[0];
8353:     if (!file) {
8354:       setStatus("Choose a local file first.");
8355:       return null;
8356:     }
8357:     fileName = withExtension(title, file.name);
8358:     title = title || file.name;
8359:     const validation = isAllowedUpload(key, fileName, file.type);
8360:     if (!validation.ok) {
8361:       setStatus(validation.message);
8362:       return null;
8363:     }
8364:   } else {
8365:     if (!url) {
8366:       setStatus("Paste a web or Google Drive link first.");
8367:       return null;
8368:     }
8369:     url = normalizeLinkTarget(url, { assumeWeb: true });
8370:     title = title || displayNameFromUrl(url, "Linked asset");
8371:     const validation = validateAssetForSection(key, { source: assetSource.value, url, title });
8372:     if (!validation.ok) {
8373:       setStatus(validation.message);
8374:       return null;
8375:     }
8376:   }
8377:
8378:   let caption = assetCaption.value.trim();
8379:   if (captionSource.value === "file" && captionFile.files[0]) {
8380:     caption = await captionFile.files[0].text();
8381:   }
8382:
8383:   return {
8384:     caption,
8385:     file,
8386:     fileName,
8387:     kind: file ? "file" : inferUrlAssetKind(url, assetSource.value),
8388:     source: assetSource.value,
8389:     title,
8390:     url
8391:   };
8392: }

```

uploadToSection - lines 8394-8454

Item	Explanation
Source location	Lines 8394-8454 (61 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	async function uploadToSection(key, folder, customSectionId = "", customItemIndex = null) {
Touches	filesystem, network/API requests
Calls detected	selectedProject, openAssetDialog, validateAssetForSection, setStatus, labelForUrlAssetKind, readFileAsDataURL, fetch, stringify, json, find, push, makeItemForSection
API endpoints	/api/upload
Storage keys	No local/session storage keys detected.
Async/return clues	4 await expression(s), 5 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8394: async function uploadToSection(key, folder, customSectionId = "", customItemIndex = null) {
8395:   const project = selectedProject();
8396:   if (!project) return;
8397:
8398:   const asset = await openAssetDialog(key);
8399:   if (!asset) return;
8400:
8401:   const validation = validateAssetForSection(key, asset);
8402:   if (!validation.ok) {
8403:     setStatus(validation.message);
8404:     return;
8405:   }
8406:

```



```

8407: const sectionFolder = customSectionId || folder || key;
8408: let url = asset.url;
8409: let type = asset.source === "local" ? "File" : labelForUrlAssetKind(asset.kind);
8410:
8411: if (asset.file) {
8412:   const data = await readFileAsDataURL(asset.file);
8413:   const response = await fetch("/api/upload", {
8414:     method: "POST",
8415:     headers: { "Content-Type": "application/json" },
8416:     body: JSON.stringify({
8417:       projectId: project.id,
8418:       section: sectionFolder,
8419:       fileName: asset.fileName,
8420:       data
8421:     })
8422:   });
8423:   const result = await response.json();
8424:   if (!response.ok) {
8425:     setStatus(result.error || "Upload failed.");
8426:     return;
8427:   }
8428:   url = result.url;
8429:   type = asset.file.type || "File";
8430: }
8431:
8432: if (customSectionId) {
8433:   const section = (project.sections || []).find((item) => item.id === customSectionId);
8434:   const uploadedItem = { title: asset.title, description: asset.caption, type, url, status:
asset.source === "local" ? "uploaded" : "linked" };
8435:   if (customItemIndex !== null && customItemIndex !== "") {
8436:     const parent = section?.items?.[Number(customItemIndex)];
8437:     if (!parent || parent.url) return;
8438:     parent.children = parent.children || [];
8439:     parent.children.push(uploadedItem);
8440:   } else {
8441:     section.items = section.items || [];
8442:     section.items.push(uploadedItem);
8443:   }
8444: } else {
8445:   project[key] = project[key] || [];
8446:   project[key].push(makeItemForSection(key, asset.title, asset.caption, url));
8447: }
8448:
8449: setStatus(asset.source === "local"
8450:   ? `Saved ${asset.fileName} to ${url}. Click Save project to include it in the portfolio preview.`
8451:   : `Linked ${asset.title}. Click Save project to include it in the portfolio preview.`);
8452: scheduleAutosave();
8453: renderAll();
8454: }

```

section - lines 8433-8434

Item	Explanation
Source location	Lines 8433-8434 (2 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	<code>const section = (project.sections []).find((item) => item.id === customSectionId);</code>
Touches	Mostly local calculations or local UI state.
Calls detected	find
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8433:   const section = (project.sections || []).find((item) => item.id === customSectionId);
8434:   const uploadedItem = { title: asset.title, description: asset.caption, type, url, status:
asset.source === "local" ? "uploaded" : "linked" };

```

uploadToDesignSection - lines 8456-8502

Item	Explanation
Source location	Lines 8456-8502 (47 source lines).

Item	Explanation
Purpose	Part of the private builder frontend and editing experience.
Declaration	async function uploadToDesignSection(pathValue, folder, kind = "document") {
Touches	filesystem, network/API requests
Calls detected	selectedProject, openAssetDialog, validateAssetForSection, setStatus, labelForUrlAssetKind, readFileAsDataURL, fetch, stringify, json, designArray, push, makeDesignItem
API endpoints	/api/upload
Storage keys	No local/session storage keys detected.
Async/return clues	4 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8456: async function uploadToDesignSection(pathValue, folder, kind = "document") {
8457:   const project = selectedProject();
8458:   if (!project) return;
8459:
8460:   const asset = await openAssetDialog(kind === "simulation-result" ? "tests" : "sections");
8461:   if (!asset) return;
8462:
8463:   const validation = validateAssetForSection(kind === "simulation-result" ? "tests" : "sections", asset);
8464:   if (!validation.ok) {
8465:     setStatus(validation.message);
8466:     return;
8467:   }
8468:
8469:   let url = asset.url;
8470:   let type = asset.source === "local" ? "File" : labelForUrlAssetKind(asset.kind);
8471:   if (asset.file) {
8472:     const data = await readFileAsDataURL(asset.file);
8473:     const response = await fetch("/api/upload", {
8474:       method: "POST",
8475:       headers: { "Content-Type": "application/json" },
8476:       body: JSON.stringify({
8477:         projectId: project.id,
8478:         section: folder || "design",
8479:         fileName: asset.fileName,
8480:         data
8481:       })
8482:     });
8483:     const result = await response.json();
8484:     if (!response.ok) {
8485:       setStatus(result.error || "Upload failed.");
8486:       return;
8487:     }
8488:     url = result.url;
8489:     type = asset.file.type || "File";
8490:   }
8491:
8492:   const items = designArray(project, pathValue);
8493:   items.push({
8494:     ...makeDesignItem(kind, asset.title, asset.caption, url),
8495:     type
8496:   });
8497:   setStatus(asset.source === "local"
8498:     ? `Saved ${asset.fileName} to ${url}. Click Save project to include it in the portfolio preview.`
8499:     : `Linked ${asset.title}. Click Save project to include it in the portfolio preview.`);
8500:   scheduleAutosave();
8501:   renderAll();
8502: }

```

openSectionDialog - lines 8504-8523

Item	Explanation
Source location	Lines 8504-8523 (20 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	async function openSectionDialog() {
Touches	filesystem
Calls detected	dialogValue, trim, setStatus
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8504: async function openSectionDialog() {
8505:   sectionDialogTitle.textContent = "Create a section";
8506:   sectionTitle.placeholder = "Design, Measurements, Firmware...";
8507:   sectionDescription.placeholder = "Optional description shown to recruiters";
8508:   sectionTitle.value = "";
8509:   sectionDescription.value = "";
8510:   const saved = await dialogValue(sectionDialog);
8511:   if (!saved) return null;
8512:
8513:   const title = sectionTitle.value.trim();
8514:   if (!title) {
8515:     setStatus("Type a section title before saving.");
8516:     return null;
8517:   }
8518:
8519:   return {
8520:     description: sectionDescription.value.trim(),
8521:     title
8522:   };
8523: }

```

openCategoryDialog - lines 8525-8531

Item	Explanation
Source location	Lines 8525-8531 (7 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function openCategoryDialog() {
Touches	Mostly local calculations or local UI state.
Calls detected	showModal, focus
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8525: function openCategoryDialog() {
8526:   categoryTitle.value = "";
8527:   categoryDescription.value = "";
8528:   categoryAccent.value = "#1677a8";
8529:   categoryDialog.showModal();
8530:   setTimeout(() => categoryTitle.focus(), 0);
8531: }

```

saveCategoryFromDialog - lines 8533-8559

Item	Explanation
Source location	Lines 8533-8559 (27 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	function saveCategoryFromDialog() {
Touches	filesystem
Calls detected	trim, setStatus, slugify, Set, map, has, normalizeCategory, add, refreshCategoryControls, renderAll, scheduleAutosave
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8533: function saveCategoryFromDialog() {
8534:   const label = categoryTitle.value.trim();
8535:   if (!label) {
8536:     setStatus("Type a category name before saving.");
8537:     return;
8538:   }
8539:   const baseId = slugify(label);
8540:   const ids = new Set((catalog.categories || []).map((category) => category.id));
8541:   let id = baseId;
8542:   let suffix = 2;
8543:   while (ids.has(id)) {
8544:     id = `${baseId}-${suffix}`;
8545:     suffix += 1;
8546:   }
8547:   const category = normalizeCategory({
8548:     accent: categoryAccent.value,
8549:     description: categoryDescription.value,
8550:     id,
8551:     label
8552:   });
8553:   catalog.categories = [...(catalog.categories || []), category];
8554:   expandedCategories.add(category.id);
8555:   refreshCategoryControls();
8556:   renderAll();
8557:   scheduleAutosave();
8558:   setStatus(`Category "${category.label}" added locally. It is now available under Add new project.`);
8559: }

```

addCustomSection - lines 8561-8582

Item	Explanation
Source location	Lines 8561-8582 (22 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	async function addCustomSection() {
Touches	Mostly local calculations or local UI state.
Calls detected	selectedProject, openSectionDialog, slugify, Set, map, has, push, setStatus, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8561: async function addCustomSection() {
8562:   const project = selectedProject();
8563:   if (!project) return;
8564:
8565:   const sectionInput = await openSectionDialog();
8566:   if (!sectionInput) return;
8567:
8568:   let id = slugify(sectionInput.title);
8569:   const existingIds = new Set((project.sections || []).map((section) => section.id));
8570:   let suffix = 2;
8571:   while (existingIds.has(id)) {
8572:     id = `${slugify(sectionInput.title)}-${suffix}`;
8573:     suffix += 1;
8574:   }
8575:
8576:   project.sections = project.sections || [];
8577:   project.sections.push({ id, title: sectionInput.title, description: sectionInput.description, items: [] });
8578:   activeSectionId = `custom:${id}`;
8579:   setStatus("New section added in the editor. Click Save project to include this version in the portfolio preview.");
8580:   scheduleAutosave();
8581:   renderAll();
8582: }

```

addCustomItem - lines 8584-8596

Item	Explanation
Source location	Lines 8584-8596 (13 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function addCustomItem(sectionId) {
Touches	Mostly local calculations or local UI state.
Calls detected	selectedProject, find, prompt, push, setStatus, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
8584: function addCustomItem(sectionId) {
8585:   const project = selectedProject();
8586:   const section = (project.sections || []).find((item) => item.id === sectionId);
8587:   const title = prompt("Type a subsection title.");
8588:   if (!section || !title) return;
8589:   const description = prompt("Paste or type a text description if you want.") || "";
8590:
8591:   section.items = section.items || [];
8592:   section.items.push({ title, description, type: "Text", status: "draft" });
8593:   setStatus("Subsection added locally.");
8594:   scheduleAutosave();
8595:   renderAll();
8596: }
```

section - lines 8586-8589

Item	Explanation
Source location	Lines 8586-8589 (4 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const section = (project.sections []).find((item) => item.id === sectionId);
Touches	Mostly local calculations or local UI state.
Calls detected	find, prompt
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
8586:   const section = (project.sections || []).find((item) => item.id === sectionId);
8587:   const title = prompt("Type a subsection title.");
8588:   if (!section || !title) return;
8589:   const description = prompt("Paste or type a text description if you want.") || "";
```

editCustomItem - lines 8598-8609

Item	Explanation
Source location	Lines 8598-8609 (12 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function editCustomItem(sectionId, index) {
Touches	Mostly local calculations or local UI state.

Item	Explanation
Calls detected	selectedProject, find, prompt, setStatus, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8598: function editCustomItem(sectionId, index) {
8599:   const project = selectedProject();
8600:   const item = (project.sections || []).find((section) => section.id === sectionId)?.items?.[index];
8601:   if (!item) return;
8602:   const title = prompt("Edit subsection title.", item.title || "");
8603:   if (!title) return;
8604:   item.title = title;
8605:   item.description = prompt("Edit description.", item.description || "") || "";
8606:   setStatus("Subsection edited locally.");
8607:   scheduleAutosave();
8608:   renderAll();
8609: }

```

item - lines 8600-8608

Item	Explanation
Source location	Lines 8600-8608 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const item = (project.sections []).find((section) => section.id === sectionId)?.items?.[index];
Touches	Mostly local calculations or local UI state.
Calls detected	find, prompt, setStatus, scheduleAutosave, renderAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8600:   const item = (project.sections || []).find((section) => section.id === sectionId)?.items?.[index];
8601:   if (!item) return;
8602:   const title = prompt("Edit subsection title.", item.title || "");
8603:   if (!title) return;
8604:   item.title = title;
8605:   item.description = prompt("Edit description.", item.description || "") || "";
8606:   setStatus("Subsection edited locally.");
8607:   scheduleAutosave();
8608:   renderAll();

```

catalogForSave - lines 8611-8633

Item	Explanation
Source location	Lines 8611-8633 (23 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function catalogForSave(endpoint) {
Touches	network/API requests
Calls detected	parsedPortfolioGlobals, clone
API endpoints	/api/apply-catalog
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8611: function catalogForSave(endpoint) {
8612:   const parsedGlobals = parsedPortfolioGlobals();
8613:   if (endpoint === "/api/apply-catalog") {
8614:     return {
8615:       ...savedPortfolioCatalog,
8616:       ...parsedGlobals,
8617:       projects: clone(savedPortfolioCatalog.projects || [])
8618:     };
8619:   }
8620:   return {
8621:     ...catalog,
8622:     categories: parsedGlobals.categories,
8623:     fieldStyles: parsedGlobals.fieldStyles,
8624:     funFacts: parsedGlobals.funFacts,
8625:     funFactsRich: parsedGlobals.funFactsRich,
8626:     profile: parsedGlobals.profile,
8627:     profileRich: parsedGlobals.profileRich,
8628:     siteContent: parsedGlobals.siteContent,
8629:     siteContentRich: parsedGlobals.siteContentRich,
8630:     siteSections: clone(catalog.siteSections || []),
8631:     projects: clone(catalog.projects || [])
8632:   };
8633: }

```

saveCatalog - lines 8635-8697

Item	Explanation
Source location	Lines 8635-8697 (63 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	async function saveCatalog(endpoint, message) {
Touches	network/API requests
Calls detected	catalogForSave, showBuilderError, setStatus, setSaveState, fetch, now, stringify, json, showPublishResult
API endpoints	/api/apply-catalog, /api/save-draft
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8635: async function saveCatalog(endpoint, message) {
8636:   const targetCatalog = catalogForSave(endpoint);
8637:   if (endpoint === "/api/apply-catalog" && !draftSavedSinceChanges) {
8638:     showBuilderError(
8639:       "Save draft required",
8640:       "Please click Save draft before applying this portfolio to the live site.",
8641:       "Apply to site was stopped before commit and push. Save the draft, then click Apply to site again."
8642:     );
8643:     setStatus("Apply to site stopped. Save draft first.");
8644:     setSaveState("dirty", "Save draft needed");
8645:     return;
8646:   }
8647:   if (endpoint === "/api/apply-catalog" && !(targetCatalog.projects || []).length) {
8648:     showBuilderError(
8649:       "No saved project preview",
8650:       "Save a project or use Save all sections before applying to the live site.",
8651:       "Apply to site was stopped before commit and push."
8652:     );
8653:     setStatus("Apply to site stopped. Save a project or use Save all sections first.");
8654:     setSaveState("dirty", "Preview needs projects");
8655:     return;
8656:   }
8657:   try {
8658:     clearTimeout(autosaveTimer);
8659:     saveDraftButton.disabled = true;
8660:     applyCatalogButton.disabled = true;
8661:     const applying = endpoint === "/api/apply-catalog";
8662:     setStatus(applying ? "Applying site changes..." : "Saving draft...");
8663:     setSaveState(applying ? "publishing" : "saving", applying ? "Publishing..." : "Saving...");

```

```

8665:     const response = await fetch(`${endpoint}?t=${Date.now()}`, {
8666:       method: "POST",
8667:       cache: "no-store",
8668:       headers: { "Content-Type": "application/json" },
8669:       body: JSON.stringify({ catalog: targetCatalog })
8670:     });
8671:     const result = await response.json();
8672:     if (!response.ok) {
8673:       setStatus(result.error || "Save failed.");
8674:       setSaveState("error", endpoint === "/api/apply-catalog" ? "Publish failed" : "Save failed");
8675:       return;
8676:     }
8677:     if (result.publish) {
8678:       setStatus(result.publish.pushed
8679:         ? `${message}: ${result.file}. Pushed to ${result.publish.branch}.`
8680:         : `${message}: ${result.file}. Git push failed: ${result.publish.error}`);
8681:       setSaveState(result.publish.pushed ? "published" : "error", result.publish.pushed ? "Site applied"
: "Push failed");
8682:       showPublishResult(result);
8683:       return;
8684:     }
8685:     if (endpoint === "/api/save-draft") {
8686:       draftSavedSinceChanges = true;
8687:       setSaveState("saved", "Draft saved");
8688:     }
8689:     setStatus(`${message}: ${result.file}`);
8690:   } catch {
8691:     setStatus("Save failed. Make sure the local server window is still running.");
8692:     setSaveState("error", "Save failed");
8693:   } finally {
8694:     saveDraftButton.disabled = false;
8695:     applyCatalogButton.disabled = false;
8696:   }
8697: }

```

loadData - lines 8699-8745

Item	Explanation
Source location	Lines 8699-8745 (47 source lines).
Purpose	Loads data from disk, network, browser storage, or a service.
Declaration	async function loadData() {
Touches	filesystem, network/API requests
Calls detected	all, fetch, now, json, call, normalizeFunFacts, clone, normalizeFieldStyles, normalizeProfile, normalizeSiteContent, filter, has
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	3 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8699: async function loadData() {
8700:   const [catalogResponse, templateResponse] = await Promise.all([
8701:     fetch(`/api/catalog?t=${Date.now()}`, { cache: "no-store" }),
8702:     fetch(`/api/templates?t=${Date.now()}`, { cache: "no-store" })
8703:   ]);
8704:   const catalogData = await catalogResponse.json();
8705:   const templateData = await templateResponse.json();
8706:
8707:   catalog = catalogData.catalog;
8708:   catalog.funFacts = Object.prototype.hasOwnProperty.call(catalog, "funFacts")
8709:     ? normalizeFunFacts(catalog.funFacts)
8710:     : clone(defaultFunFacts);
8711:   catalog.funFactsRich = catalog.funFactsRich || null;
8712:   catalog.fieldStyles = normalizeFieldStyles(catalog.fieldStyles || {});
8713:   catalog.siteContentRich = catalog.siteContentRich || {};
8714:   catalog.profileRich = catalog.profileRich || {};
8715:   catalog.profile = normalizeProfile(catalog.profile || {});
8716:   catalog.siteContent = normalizeSiteContent(catalog.siteContent || {});
8717:   catalog.siteSections = catalog.siteSections || [];
8718:   catalog.projects = (catalog.projects || []).filter((project) => !starterProjectIds.has(project.id));
8719:   catalog.projects.forEach((project) => {
8720:     if (isAnalogProject(project)) ensureDesignModel(project);
8721:     ensureCompileCode(project);
8722:   });
8723:   savedPortfolioCatalog = {
8724:     categories: clone(catalog.categories || []),
8725:     fieldStyles: clone(catalog.fieldStyles || {}),

```



```

8726:     funFacts: clone(catalog.funFacts || []),
8727:     funFactsRich: clone(catalog.funFactsRich || null),
8728:     profile: clone(normalizeProfile(catalog.profile || {})),
8729:     profileRich: clone(catalog.profileRich || {}),
8730:     siteContent: clone(catalog.siteContent || defaultSiteContent),
8731:     siteContentRich: clone(catalog.siteContentRich || {}),
8732:     projects: [],
8733:     siteSections: clone(catalog.siteSections || [])
8734: };
8735: templates = templateData.templates || [];
8736: setStatus(`Loaded ${catalogData.source} catalog. Builder controls are local-only.`);
8737: draftSavedSinceChanges = true;
8738: setSaveState("loaded", "Loaded");
8739:
8740: templateSelect.innerHTML = groupedTemplateOptions();
8741: refreshCategoryControls();
8742: selectedProjectId = catalog.projects[0]?.id || "";
8743: expandedCategories = new Set(catalog.categories.map((category) => category.id));
8744: renderAll();
8745: }

```

prepareBuilderGuideTopicLinks - lines 8758-8768

Item	Explanation
Source location	Lines 8758-8768 (11 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function prepareBuilderGuideTopicLinks() {
Touches	browser/Electron DOM
Calls detected	querySelectorAll, forEach, querySelector, setAttribute, trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8758: function prepareBuilderGuideTopicLinks() {
8759:   builderGuideSections?.querySelectorAll("[data-guide-section]").forEach((section) => {
8760:     const summary = section.querySelector("summary");
8761:     if (!summary) return;
8762:     summary.setAttribute("role", "link");
8763:     summary.setAttribute("tabindex", "0");
8764:     summary.setAttribute("aria-label", `Open ${summary.textContent.trim()}`);
8765:     summary.title = "Open this guide topic";
8766:     section.open = false;
8767:   });
8768: }

```

updateBuilderGuideStats - lines 8770-8849

Item	Explanation
Source location	Lines 8770-8849 (80 source lines).
Purpose	Updates UI, state, cache, or stored data.
Declaration	function updateBuilderGuideStats() {
Touches	browser/Electron DOM
Calls detected	replace, entries, forEach, querySelector, guideSectionText, join, toLowerCase, filterBuilderGuide, trim, querySelectorAll, split, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8770: function updateBuilderGuideStats() {
8771:   if (!builderGuideDialog) return;
8772:   const projectCount = catalog.projects?.length || 0;
8773:   const categoryCount = catalog.categories?.length || 0;
8774:   const parsedCount = savedPortfolioCatalog.projects?.length || 0;
8775:   const targetLabel = currentPublishTarget?.remote
8776:     ? `Target: ${currentPublishTarget.remote.replace(/^https:\/\/github\.com\/\//i, "")}`
8777:     : "Target: not connected";
8778:   const values = {
8779:     projects: `Projects: ${projectCount} (${parsedCount} parsed)`,
8780:     categories: `Categories: ${categoryCount}`,
8781:     draft: draftSavedSinceChanges ? "Draft: saved" : "Draft: unsaved changes",
8782:     target: targetLabel
8783:   };
8784:   Object.entries(values).forEach(([key, value]) => {
8785:     const node = builderGuideDialog.querySelector(`[data-guide-stat='${key}']`);
8786:     if (node) node.textContent = value;
8787:   });
8788: }
8789:
8790: function guideSectionText(section) {
8791:   return [
8792:     section.querySelector("summary")?.textContent || "",
8793:     section.textContent || "",
8794:     section.dataset.guideSection || ""
8795:   ].join(" ").toLowerCase();
8796: }
8797:
8798: function filterBuilderGuide() {
8799:   if (!builderGuideSections) return;
8800:   const query = String(builderGuideSearch?.value || "").trim().toLowerCase();
8801:   const sections = [...builderGuideSections.querySelectorAll("[data-guide-section]")];
8802:   let visibleCount = 0;
8803:   sections.forEach((section) => {
8804:     const topicMatch = activeBuilderGuideTopic === "all" || (section.dataset.guideSection ||
8805:       "").split(/\s+/).includes(activeBuilderGuideTopic);
8806:     const queryMatch = !query || guideSectionText(section).includes(query);
8807:     const visible = topicMatch && queryMatch;
8808:     section.hidden = !visible;
8809:     if (visible) {
8810:       visibleCount += 1;
8811:       section.open = false;
8812:     }
8813:   });
8814:   if (builderGuideResults) {
8815:     const topicText = activeBuilderGuideTopic === "all" ? "all topics" : activeBuilderGuideTopic;
8816:     builderGuideResults.textContent = visibleCount
8817:       ? `Showing ${visibleCount} guide topic${visibleCount === 1 ? "" : "s"} for ${topicText}${query ? " matching "${query}"` : ""}`
8818:       : `No guide topics match "${query}"`;
8819:   }
8820: }
8821: function setBuilderGuideTopic(topic = "all") {
8822:   activeBuilderGuideTopic = topic || "all";
8823:   builderGuideDialog?.querySelectorAll("[data-guide-topic]").forEach((button) => {
8824:     button.setAttribute("aria-pressed", button.dataset.guideTopic === activeBuilderGuideTopic ? "true" :
8825:       "false");
8826:   });
8827:   filterBuilderGuide();
8828: }
8829: function closeGuideAndFocus(target) {
8830:   closeDialogElement(builderGuideDialog);
8831:   requestAnimationFrame(() => {
8832:     target?.scrollIntoView?({ behavior: "smooth", block: "center" });
8833:     if (target?.focus) target.focus({ preventScroll: true });
8834:   });
8835: }
8836:
8837: function handleBuilderGuideAction(action = "") {
8838:   if (action === "profile") closeGuideAndFocus(profileDisplayNameInput);
8839:   if (action === "site-content") closeGuideAndFocus(siteHeroTitleInput);
8840:   if (action === "fun-facts") closeGuideAndFocus(funFactsInput);
8841:   if (action === "projects") closeGuideAndFocus(projectTree);
8842:   if (action === "preview") {
8843:     closeDialogElement(builderGuideDialog);
8844:     openPortfolioPreviewButton?.click();
8845:   }
8846:   if (action === "target") {
8847:     closeDialogElement(builderGuideDialog);
8848:     publishTargetOpen?.click();
8849:   }

```

guideSectionText - lines 8790-8796

Item	Explanation
Source location	Lines 8790-8796 (7 source lines).

Item	Explanation
Purpose	Part of the private builder frontend and editing experience.
Declaration	function guideSectionText(section) {
Touches	browser/Electron DOM
Calls detected	querySelector, join, toLowerCase
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8790: function guideSectionText(section) {
8791:   return [
8792:     section.querySelector("summary")?.textContent || "",
8793:     section.textContent || "",
8794:     section.dataset.guideSection || ""
8795:   ].join(" ").toLowerCase();
8796: }

```

filterBuilderGuide - lines 8798-8819

Item	Explanation
Source location	Lines 8798-8819 (22 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function filterBuilderGuide() {
Touches	browser/Electron DOM
Calls detected	trim, toLowerCase, querySelectorAll, forEach, split, includes, guideSectionText
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8798: function filterBuilderGuide() {
8799:   if (!builderGuideSections) return;
8800:   const query = String(builderGuideSearch?.value || "").trim().toLowerCase();
8801:   const sections = [...builderGuideSections.querySelectorAll("[data-guide-section]")];
8802:   let visibleCount = 0;
8803:   sections.forEach((section) => {
8804:     const topicMatch = activeBuilderGuideTopic === "all" || (section.dataset.guideSection ||
8805:       "").split(/\s+/).includes(activeBuilderGuideTopic);
8806:     const queryMatch = !query || guideSectionText(section).includes(query);
8807:     const visible = topicMatch && queryMatch;
8808:     section.hidden = !visible;
8809:     if (visible) {
8810:       visibleCount += 1;
8811:       section.open = false;
8812:     }
8813:   });
8814:   if (builderGuideResults) {
8815:     const topicText = activeBuilderGuideTopic === "all" ? "all topics" : activeBuilderGuideTopic;
8816:     builderGuideResults.textContent = visibleCount
8817:       ? `Showing ${visibleCount} guide topic${visibleCount === 1 ? "" : "s"} for ${topicText}${query ? `
8818:         matching "${query}"` : ""}.`
8819:       : `No guide topics match "${query}"`;

```

setBuilderGuideTopic - lines 8821-8827

Item	Explanation
Source location	Lines 8821-8827 (7 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function setBuilderGuideTopic(topic = "all") {
Touches	browser/Electron DOM
Calls detected	querySelectorAll, forEach, setAttribute, filterBuilderGuide
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8821: function setBuilderGuideTopic(topic = "all") {
8822:   activeBuilderGuideTopic = topic || "all";
8823:   builderGuideDialog?.querySelectorAll("[data-guide-topic]").forEach((button) => {
8824:     button.setAttribute("aria-pressed", button.dataset.guideTopic === activeBuilderGuideTopic ? "true" :
"false");
8825:   });
8826:   filterBuilderGuide();
8827: }

```

closeGuideAndFocus - lines 8829-8835

Item	Explanation
Source location	Lines 8829-8835 (7 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function closeGuideAndFocus(target) {
Touches	Mostly local calculations or local UI state.
Calls detected	closeDialogElement, requestAnimationFrame, focus
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8829: function closeGuideAndFocus(target) {
8830:   closeDialogElement(builderGuideDialog);
8831:   requestAnimationFrame(() => {
8832:     target?.scrollIntoView?({ behavior: "smooth", block: "center" });
8833:     if (target?.focus) target.focus({ preventScroll: true });
8834:   });
8835: }

```

handleBuilderGuideAction - lines 8837-8854

Item	Explanation
Source location	Lines 8837-8854 (18 source lines).
Purpose	Handles an event, request, command, or user action.
Declaration	function handleBuilderGuideAction(action = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	closeGuideAndFocus, closeDialogElement, click
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8837: function handleBuilderGuideAction(action = "") {
8838:   if (action === "profile") closeGuideAndFocus(profileDisplayNameInput);
8839:   if (action === "site-content") closeGuideAndFocus(siteHeroTitleInput);
8840:   if (action === "fun-facts") closeGuideAndFocus(funFactsInput);
8841:   if (action === "projects") closeGuideAndFocus(projectTree);
8842:   if (action === "preview") {
8843:     closeDialogElement(builderGuideDialog);
8844:     openPortfolioPreviewButton?.click();
8845:   }
8846:   if (action === "target") {
8847:     closeDialogElement(builderGuideDialog);
8848:     publishTargetOpen?.click();
8849:   }
8850:   if (action === "updates") {
8851:     closeDialogElement(builderGuideDialog);
8852:     appUpdateCheckButton?.click();
8853:   }
8854: }

```

openBuilderGuideTopic - lines 8856-8878

Item	Explanation
Source location	Lines 8856-8878 (23 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function openBuilderGuideTopic(section) {
Touches	browser/Electron DOM, authentication/security data
Calls detected	querySelector, trim, replaceChildren, cloneNode, add, append, createElement, showModal, requestAnimationFrame, focus, updateDialogWindowButtons
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

8856: function openBuilderGuideTopic(section) {
8857:   if (!section || !builderGuideTopicDialog || !builderGuideTopicTitle || !builderGuideTopicBody) return;
8858:   const summary = section.querySelector("summary");
8859:   const source = section.querySelector(":scope > div");
8860:   const title = summary?.textContent?.trim() || "Builder guide topic";
8861:   builderGuideTopicTitle.textContent = title;
8862:   builderGuideTopicBody.replaceChildren();
8863:   if (source) {
8864:     const copy = source.cloneNode(true);
8865:     copy.classList.add("builder-guide-topic-copy");
8866:     builderGuideTopicBody.append(copy);
8867:   } else {
8868:     const empty = document.createElement("p");
8869:     empty.textContent = "This guide topic does not have written instructions yet.";
8870:     builderGuideTopicBody.append(empty);
8871:   }
8872:   if (!builderGuideTopicDialog.open) builderGuideTopicDialog.showModal();
8873:   requestAnimationFrame(() => {
8874:     builderGuideTopicBody.scrollTop = 0;
8875:     builderGuideTopicBody.focus({ preventScroll: true });
8876:     updateDialogWindowButtons(builderGuideTopicDialog);
8877:   });
8878: }

```

handleRichEditorContextMenu - lines 9312-9372

Item	Explanation
Source location	Lines 9312-9372 (61 source lines).
Purpose	Handles an event, request, command, or user action.
Declaration	function handleRichEditorContextMenu(event) {
Touches	browser/Electron DOM
Calls detected	richEditorFromContextEvent, preventDefault, stopPropagation, closest, clearPersistentTextSelection, hideTextSelectionInspector, configureSummaryContextMenu, selectionRangeInsideEditor, cloneRange, contains, restoreRichSelection, captureRichSelection
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9312: function handleRichEditorContextMenu(event) {
9313:     const editor = richEditorFromContextEvent(event);
9314:     const textBlock = event.target.closest?(".rich-text-block");
9315:
9316:     if (!editor && !textBlock) return;
9317:
9318:     event.preventDefault();
9319:     event.stopPropagation();
9320:     const nextEditor = editor || textBlock.closest("[data-rich-editor]");
9321:     if (activeSummaryEditor && nextEditor && activeSummaryEditor !== nextEditor) {
9322:         activeRichSelectionRange = null;
9323:         activeTextSelectionRange = null;
9324:         clearPersistentTextSelection(true);
9325:         summaryContextMenu.hidden = true;
9326:     }
9327:     hideTextSelectionInspector();
9328:
9329:     activePlainTextControl = null;
9330:     activePlainTextSelection = null;
9331:     activeSummaryEditor = nextEditor;
9332:     configureSummaryContextMenu("rich", { textOnly: activeSummaryEditor.dataset.richTextOnly === "true"
});
9333:     const liveRange = selectionRangeInsideEditor(activeSummaryEditor);
9334:     const savedRange = activeTextSelectionRange && !activeTextSelectionRange.collapsed
9335:         ? activeTextSelectionRange.cloneRange()
9336:         : null;
9337:     const savedContainer = savedRange?.commonAncestorContainer?.nodeType === Node.TEXT_NODE
9338:         ? savedRange.commonAncestorContainer.parentElement
9339:         : savedRange?.commonAncestorContainer;
9340:     if (liveRange && !liveRange.collapsed) {
9341:         activeRichSelectionRange = liveRange.cloneRange();
9342:         activeTextSelectionRange = liveRange.cloneRange();
9343:     } else if (savedRange && savedContainer && activeSummaryEditor.contains(savedContainer)) {
9344:         activeRichSelectionRange = savedRange.cloneRange();
9345:         restoreRichSelection(activeSummaryEditor);
9346:     } else {
9347:         captureRichSelection(activeSummaryEditor);
9348:     }
9349:     selectRichBlock(event.target.closest?(".rich-block") || currentRichBlock(activeSummaryEditor));
9350:     syncRichContextMenuControls(activeSummaryBlock);
9351:
9352:     const menuHost = activeSummaryEditor.closest("dialog") || document.body;
9353:     if (summaryContextMenu.parentElement !== menuHost) menuHost.append(summaryContextMenu);
9354:
9355:     summaryContextMenu.style.left = `${event.clientX}px`;
9356:     summaryContextMenu.style.top = `${event.clientY}px`;
9357:     summaryContextMenu.style.maxHeight = "";
9358:     summaryContextMenu.hidden = false;
9359:     const hostRect = menuHost.getBoundingClientRect();
9360:     const bounds = {
9361:         bottom: Math.min(window.innerHeight - 8, hostRect.bottom - 8),
9362:         left: Math.max(8, hostRect.left + 8),
9363:         right: Math.min(window.innerWidth - 8, hostRect.right - 8),
9364:         top: Math.max(8, hostRect.top + 8)
9365:     };
9366:     summaryContextMenu.style.maxHeight = `${Math.max(180, bounds.bottom - bounds.top)}px`;
9367:     const menuRect = summaryContextMenu.getBoundingClientRect();
9368:     const left = Math.max(bounds.left, Math.min(event.clientX, bounds.right - menuRect.width));
9369:     const top = Math.max(bounds.top, Math.min(event.clientY, bounds.bottom - menuRect.height));
9370:     summaryContextMenu.style.left = `${left}px`;
9371:     summaryContextMenu.style.top = `${top}px`;
9372: }

```

handlePlainTextContextMenu - lines 9398-9433

Item	Explanation
Source location	Lines 9398-9433 (36 source lines).
Purpose	Handles an event, request, command, or user action.
Declaration	function handlePlainTextContextMenu(event) {
Touches	browser/Electron DOM
Calls detected	plainTextControlFromContextEvent, preventDefault, hideTextSelectionInspector, configureSummaryContextMenu, syncPlainContextMenuControls, closest, append, getBoundingClientRect, min, max
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
9398: function handlePlainTextContextMenu(event) {
9399:   const control = plainTextControlFromContextEvent(event);
9400:   if (!control) return;
9401:
9402:   event.preventDefault();
9403:   hideTextSelectionInspector();
9404:   activeSummaryEditor = null;
9405:   activeSummaryBlock = null;
9406:   activePlainTextControl = control;
9407:   activePlainTextSelection = {
9408:     start: control.selectionStart ?? 0,
9409:     end: control.selectionEnd ?? control.value.length
9410:   };
9411:   configureSummaryContextMenu("plain");
9412:   syncPlainContextMenuControls(control);
9413:
9414:   const menuHost = control.closest("dialog") || document.body;
9415:   if (summaryContextMenu.parentElement !== menuHost) menuHost.append(summaryContextMenu);
9416:   summaryContextMenu.style.left = `${event.clientX}px`;
9417:   summaryContextMenu.style.top = `${event.clientY}px`;
9418:   summaryContextMenu.style.maxHeight = "";
9419:   summaryContextMenu.hidden = false;
9420:   const hostRect = menuHost.getBoundingClientRect();
9421:   const bounds = {
9422:     bottom: Math.min(window.innerHeight - 8, hostRect.bottom - 8),
9423:     left: Math.max(8, hostRect.left + 8),
9424:     right: Math.min(window.innerWidth - 8, hostRect.right - 8),
9425:     top: Math.max(8, hostRect.top + 8)
9426:   };
9427:   summaryContextMenu.style.maxHeight = `${Math.max(140, bounds.bottom - bounds.top)}px`;
9428:   const menuRect = summaryContextMenu.getBoundingClientRect();
9429:   const left = Math.max(bounds.left, Math.min(event.clientX, bounds.right - menuRect.width));
9430:   const top = Math.max(bounds.top, Math.min(event.clientY, bounds.bottom - menuRect.height));
9431:   summaryContextMenu.style.left = `${left}px`;
9432:   summaryContextMenu.style.top = `${top}px`;
9433: }
```

handleRichEditorPaste - lines 9437-9480

Item	Explanation
Source location	Lines 9437-9480 (44 source lines).
Purpose	Handles an event, request, command, or user action.
Declaration	async function handleRichEditorPaste(event) {
Touches	browser/Electron DOM
Calls detected	closest, find, startsWith, preventDefault, setStatus, uploadSummaryImageFile, getAsFile, insertRichBlockAtCursor, createRichImageBlock, saveRichEditorToProject, getData, insertHtmlIntoRichEditor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 5 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9437: async function handleRichEditorPaste(event) {
9438:   const editor = event.target.closest("[data-rich-editor]");
9439:   if (!editor) return;
9440:
9441:   const imageItem = [...(event.clipboardData?.items || [])].find((item) =>
item.type.startsWith("image/"));
9442:
9443:   if (imageItem) {
9444:     event.preventDefault();
9445:     if (editor.dataset.richTextOnly === "true") {
9446:       setStatus("Images can only be pasted into overview editors, not title, profile, or contact
fields.");
9447:       return;
9448:     }
9449:
9450:     const imageBlock = await uploadSummaryImageFile(imageItem.getAsFile(), {
9451:       align: "center",
9452:       display: "show",
9453:       folder: editor.dataset.richFolder || "summary",
9454:       projectId: editor.dataset.richProjectId || "",
9455:       title: ""
9456:     });
9457:
9458:     if (imageBlock) {
9459:       insertRichBlockAtCursor(editor, createRichImageBlock(imageBlock));
9460:       saveRichEditorToProject(editor);
9461:     }
9462:
9463:     return;
9464:   }
9465:
9466:   const pastedHtml = event.clipboardData?.getData("text/html") || "";
9467:   if (pastedHtml) {
9468:     event.preventDefault();
9469:     insertHtmlIntoRichEditor(editor, pastedHtml);
9470:     saveRichEditorToProject(editor);
9471:     return;
9472:   }
9473:
9474:   const pastedText = event.clipboardData?.getData("text/plain") || "";
9475:   if (!pastedText) return;
9476:
9477:   event.preventDefault();
9478:   insertPlainTextIntoRichEditor(editor, pastedText);
9479:   saveRichEditorToProject(editor);
9480: }

```

restoreRichInsertionPoint - lines 9494-9503

Item	Explanation
Source location	Lines 9494-9503 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function restoreRichInsertionPoint(editor) {
Touches	browser/Electron DOM
Calls detected	restoreRichSelection, focusTextBlock, selectedRichBlock, querySelector, focus, selectionRangeInsideEditor, deleteContents
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9494: function restoreRichInsertionPoint(editor) {
9495:   if (!restoreRichSelection(editor)) focusTextBlock(selectedRichBlock(editor) ||
editor.querySelector(".rich-text-block"));
9496:   editor.focus({ preventScroll: true });
9497:
9498:   const range = selectionRangeInsideEditor(editor);

```



```

9499:     if (!range) return null;
9500:
9501:     range.deleteContents();
9502:     return range;
9503: }

```

insertFragmentIntoRichEditor - lines 9505-9521

Item	Explanation
Source location	Lines 9505-9521 (17 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function insertFragmentIntoRichEditor(editor, fragment) {
Touches	browser/Electron DOM
Calls detected	restoreRichInsertionPoint, insertNode, createRange, setStartAfter, setStart, collapse, getSelection, removeAllRanges, addRange, captureRichSelection
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9505: function insertFragmentIntoRichEditor(editor, fragment) {
9506:     const range = restoreRichInsertionPoint(editor);
9507:     if (!range || !fragment) return;
9508:
9509:     const lastInsertedNode = fragment.lastChild;
9510:     range.insertNode(fragment);
9511:
9512:     const nextRange = document.createRange();
9513:     if (lastInsertedNode) nextRange.setStartAfter(lastInsertedNode);
9514:     else nextRange.setStart(range.startContainer, range.startOffset);
9515:     nextRange.collapse(true);
9516:
9517:     const selection = window.getSelection();
9518:     selection.removeAllRanges();
9519:     selection.addRange(nextRange);
9520:     captureRichSelection(editor);
9521: }

```

insertPlainTextIntoRichEditor - lines 9523-9533

Item	Explanation
Source location	Lines 9523-9533 (11 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function insertPlainTextIntoRichEditor(editor, pastedText) {
Touches	browser/Electron DOM
Calls detected	replace, createDocumentFragment, split, forEach, append, createElement, createTextNode, insertFragmentIntoRichEditor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9523: function insertPlainTextIntoRichEditor(editor, pastedText) {
9524:     const textToInsert = String(pastedText || "").replace(/\r\n?/g, "\n");
9525:     if (!textToInsert) return;
9526:
9527:     const fragment = document.createDocumentFragment();
9528:     textToInsert.split("\n").forEach((line, index) => {
9529:         if (index > 0) fragment.append(document.createElement("br"));
9530:         if (line) fragment.append(document.createTextNode(line));

```

```

9531:     });
9532:     insertFragmentIntoRichEditor(editor, fragment);
9533: }

```

insertHtmlIntoRichEditor - lines 9535-9543

Item	Explanation
Source location	Lines 9535-9543 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function insertHtmlIntoRichEditor(editor, pastedHtml) {
Touches	browser/Electron DOM, filesystem
Calls detected	replace, createElement, sanitizeRichInlineHtml, insertFragmentIntoRichEditor, cloneNode
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9535: function insertHtmlIntoRichEditor(editor, pastedHtml) {
9536:     const normalizedHtml = String(pastedHtml || "")
9537:     .replace(/<\/(?<:p|div|li|h[1-6]|tr)>/gi, "<br>")
9538:     .replace(/</(?<:p|div|li|h[1-6]|tr|td|th)[^>]*>/gi, "")
9539:     .replace(/<br\s*\/?>\s*(<br\s*\/?>\s*){2,}/gi, "<br><br>");
9540:     const template = document.createElement("template");
9541:     template.innerHTML = sanitizeRichInlineHtml(normalizedHtml);
9542:     insertFragmentIntoRichEditor(editor, template.content.cloneNode(true));
9543: }

```

handleRichEditorKeydown - lines 9565-9610

Item	Explanation
Source location	Lines 9565-9610 (46 source lines).
Purpose	Handles an event, request, command, or user action.
Declaration	function handleRichEditorKeydown(event) {
Touches	browser/Electron DOM
Calls detected	closest, includes, toLowerCase, preventDefault, applyRichInlineCommand, execCommand, captureRichSelection, selectionRangeInsideEditor, contains, deleteContents, createRange, setStart
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 6 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9565: function handleRichEditorKeydown(event) {
9566:     const editor = event.target.closest("[data-rich-editor]");
9567:     if (!editor) return;
9568:
9569:     if ((event.ctrlKey || event.metaKey) && ["b", "i", "u"].includes(event.key.toLowerCase())) {
9570:         event.preventDefault();
9571:         const command = event.key.toLowerCase() === "b"
9572:             ? "bold"
9573:             : event.key.toLowerCase() === "i"
9574:                 ? "italic"
9575:                 : "underline";
9576:         applyRichInlineCommand(editor, command);
9577:         return;
9578:     }
9579:
9580:     if (event.key !== "Enter") return;
9581:
9582:     const block = event.target.closest(".rich-text-block");
9583:     if (!block) return;

```

```

9584:
9585:     event.preventDefault();
9586:     if (event.shiftKey) {
9587:         document.execCommand("insertLineBreak", false);
9588:         captureRichSelection(editor);
9589:         return;
9590:     }
9591:
9592:     const range = selectionRangeInsideEditor(editor);
9593:     if (!range || !block.contains(range.startContainer)) return;
9594:     range.deleteContents();
9595:     const tail = document.createRange();
9596:     tail.setStart(range.startContainer, range.startOffset);
9597:     tail.setEnd(block, block.childNodes.length);
9598:     const trailingContent = tail.extractContents();
9599:     const nextBlock = matchingTextBlock(block, "");
9600:     nextBlock.append(trailingContent);
9601:     block.after(nextBlock);
9602:     const nextRange = document.createRange();
9603:     nextRange.selectNodeContents(nextBlock);
9604:     nextRange.collapse(true);
9605:     const selection = window.getSelection();
9606:     selection.removeAllRanges();
9607:     selection.addRange(nextRange);
9608:     captureRichSelection(editor);
9609:     setStatus("Unsaved local changes.");
9610: }

```

handleRichEditorFocus - lines 9632-9637

Item	Explanation
Source location	Lines 9632-9637 (6 source lines).
Purpose	Handles an event, request, command, or user action.
Declaration	function handleRichEditorFocus(event) {
Touches	browser/Electron DOM
Calls detected	closest, execCommand, captureRichSelection
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9632: function handleRichEditorFocus(event) {
9633:     const editor = event.target.closest("[data-rich-editor]");
9634:     if (!editor) return;
9635:     document.execCommand("defaultParagraphSeparator", false, "p");
9636:     captureRichSelection(editor);
9637: }

```

handleRichEditorInput - lines 9639-9644

Item	Explanation
Source location	Lines 9639-9644 (6 source lines).
Purpose	Handles an event, request, command, or user action.
Declaration	function handleRichEditorInput(event) {
Touches	Mostly local calculations or local UI state.
Calls detected	closest, captureRichSelection, setStatus
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9639: function handleRichEditorInput(event) {
9640:   const editor = event.target.closest("[data-rich-editor]");
9641:   if (!editor) return;
9642:   captureRichSelection(editor);
9643:   setStatus("Unsaved local changes.");
9644: }

```

handleRichEditorBeforeInput - lines 9646-9653

Item	Explanation
Source location	Lines 9646-9653 (8 source lines).
Purpose	Handles an event, request, command, or user action.
Declaration	function handleRichEditorBeforeInput(event) {
Touches	Mostly local calculations or local UI state.
Calls detected	closest, clearPersistentTextSelection
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9646: function handleRichEditorBeforeInput(event) {
9647:   const editor = event.target.closest("[data-rich-editor]");
9648:   if (!editor) return;
9649:   const inputType = event.inputType || "";
9650:   if (!/insertFromPaste|insertFromDrop/i.test(inputType) && /^(insert|delete|history)/i.test(inputType))
{
9651:     clearPersistentTextSelection(true);
9652:   }
9653: }

```

finishTextSelectionGesture - lines 9728-9753

Item	Explanation
Source location	Lines 9728-9753 (26 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function finishTextSelectionGesture(event) {
Touches	Mostly local calculations or local UI state.
Calls detected	getSelection, selectionRangeInsideEditor, cloneRange, showPersistentTextSelection, rangeBelongsToEditor, clearPersistentTextSelection, refreshTextSelectionInspector
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9728: function finishTextSelectionGesture(event) {
9729:   if (event?.button === 2 || event?.target?.closest?("#text-selection-inspector, #summary-context-
menu")) return;
9730:   selectionGestureActive = false;
9731:   clearTimeout(selectionGestureFinishTimer);
9732:   selectionGestureFinishTimer = setTimeout(() => {
9733:     const selection = window.getSelection();
9734:     let node = selection?.anchorNode;
9735:     if (node?.nodeType === Node.TEXT_NODE) node = node.parentElement;
9736:     const editor = node?.closest?("[data-rich-editor]");
9737:     const range = editor ? selectionRangeInsideEditor(editor) : null;
9738:     if (editor && range && !range.collapsed) {
9739:       activeSummaryEditor = editor;
9740:       activeRichSelectionRange = range.cloneRange();
9741:       activeTextSelectionRange = range.cloneRange();
9742:       selectionGestureProducedRange = true;

```

```

9743:     showPersistentTextSelection(range);
9744:   } else if (!selectionGestureProducedRange) {
9745:     if (editor && rangeBelongsToEditor(activeTextSelectionRange, editor)) {
9746:       showPersistentTextSelection(activeTextSelectionRange);
9747:     } else {
9748:       clearPersistentTextSelection(true);
9749:     }
9750:   }
9751:   refreshTextSelectionInspector();
9752: }, 40);
9753: }

```

endSelectionInspectorDrag - lines 9814-9819

Item	Explanation
Source location	Lines 9814-9819 (6 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function endSelectionInspectorDrag() {
Touches	browser/Electron DOM
Calls detected	remove
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9814: function endSelectionInspectorDrag() {
9815:   if (!activeSelectionInspectorDrag) return;
9816:   activeSelectionInspectorDrag.target?.releasePointerCapture?.(activeSelectionInspectorDrag.pointerId);
9817:   textSelectionInspector?.classList.remove("is-dragging");
9818:   activeSelectionInspectorDrag = null;
9819: }

```

handleSelectionInspectorTextField - lines 9843-9849

Item	Explanation
Source location	Lines 9843-9849 (7 source lines).
Purpose	Handles an event, request, command, or user action.
Declaration	function handleSelectionInspectorTextField(event) {
Touches	filesystem
Calls detected	preventDefault, applySelectionInspectorFormat
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9843: function handleSelectionInspectorTextField(event) {
9844:   if (event.type === "keydown" && event.key !== "Enter") return;
9845:   if (event.type === "keydown") event.preventDefault();
9846:   if (event.target === selectionFontInput) applySelectionInspectorFormat("font", event.target.value);
9847:   if (event.target === selectionColorInput) applySelectionInspectorFormat("color", event.target.value);
9848:   if (event.target === selectionSizeInput) applySelectionInspectorFormat("size", event.target.value);
9849: }

```

scheduleSelectionInspectorTextField - lines 9851-9859

Item	Explanation
Source location	Lines 9851-9859 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function scheduleSelectionInspectorTextField(event) {
Touches	filesystem
Calls detected	cleanFontFamily, applySelectionInspectorFormat, normalizeTextColor, normalizeFontPx
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9851: function scheduleSelectionInspectorTextField(event) {
9852:   clearTimeout(selectionInspectorInputTimer);
9853:   const target = event.target;
9854:   selectionInspectorInputTimer = setTimeout(() => {
9855:     if (target === selectionFontInput && cleanFontFamily(target.value))
applySelectionInspectorFormat("font", target.value);
9856:     if (target === selectionColorInput && normalizeTextColor(target.value))
applySelectionInspectorFormat("color", target.value);
9857:     if (target === selectionSizeInput && normalizeFontPx(target.value))
applySelectionInspectorFormat("size", target.value);
9858:   }, 350);
9859: }

```

focusAdjacentImageText - lines 9871-9882

Item	Explanation
Source location	Lines 9871-9882 (12 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function focusAdjacentImageText(block, direction) {
Touches	Mostly local calculations or local UI state.
Calls detected	closest, matches, matchingTextBlock, before, after, focusTextBlock, saveRichEditorToProject
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9871: function focusAdjacentImageText(block, direction) {
9872:   const editor = block?.closest("[data-rich-editor]");
9873:   if (!editor) return;
9874:   const sibling = direction === "before" ? block.previousElementSibling : block.nextElementSibling;
9875:   const textBlock = sibling?.matches(".rich-text-block") ? sibling : matchingTextBlock(block, "");
9876:   if (textBlock !== sibling) {
9877:     if (direction === "before") block.before(textBlock);
9878:     else block.after(textBlock);
9879:   }
9880:   focusTextBlock(textBlock);
9881:   saveRichEditorToProject(editor);
9882: }

```

handleRichCaretClick - lines 9884-9889

Item	Explanation
Source location	Lines 9884-9889 (6 source lines).
Purpose	Handles an event, request, command, or user action.

Item	Explanation
Declaration	function handleRichCaretClick(event) {
Touches	browser/Electron DOM
Calls detected	closest, focusAdjacentImageText
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9884: function handleRichCaretClick(event) {
9885:   const caretButton = event.target.closest("[data-rich-caret]");
9886:   if (!caretButton) return false;
9887:   focusAdjacentImageText(caretButton.closest(".rich-image-block"), caretButton.dataset.richCaret);
9888:   return true;
9889: }

```

handleRichDragStart - lines 9891-9910

Item	Explanation
Source location	Lines 9891-9910 (20 source lines).
Purpose	Handles an event, request, command, or user action.
Declaration	function handleRichDragStart(event) {
Touches	browser/Electron DOM
Calls detected	closest, preventDefault, selectRichBlock, add, clearData, setData
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9891: function handleRichDragStart(event) {
9892:   const handle = event.target.closest("[data-rich-drag-handle]");
9893:   const imageBlock = event.target.closest(".rich-image-block");
9894:   const isControl = event.target.closest("button, input, select, textarea, a, .rich-block-actions, [data-rich-caret]");
9895:   const block = handle?.closest(".rich-block") || (!isControl ? imageBlock : null);
9896:   if (!block) {
9897:     if (imageBlock) event.preventDefault();
9898:     return;
9899:   }
9900:   activeRichDragBlock = block;
9901:   selectRichBlock(activeRichDragBlock);
9902:   activeRichDragBlock.classList.add("is-dragging");
9903:   if (event.dataTransfer) {
9904:     event.dataTransfer.clearData();
9905:     event.dataTransfer.effectAllowed = "move";
9906:     event.dataTransfer.dropEffect = "move";
9907:     event.dataTransfer.setData("text/x-rich-block", "move");
9908:     event.dataTransfer.setData("text/plain", activeRichDragBlock.dataset.title ||
activeRichDragBlock.dataset.type || "image");
9909:   }
9910: }

```

handleRichDragOver - lines 9912-9919

Item	Explanation
Source location	Lines 9912-9919 (8 source lines).
Purpose	Handles an event, request, command, or user action.

Item	Explanation
Declaration	function handleRichDragOver(event) {
Touches	Mostly local calculations or local UI state.
Calls detected	closest, preventDefault, updateRichDropMarker
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9912: function handleRichDragOver(event) {
9913:   if (!activeRichDragBlock) return;
9914:   const editor = event.target.closest("[data-rich-editor]");
9915:   if (!editor || editor !== activeRichDragBlock.closest("[data-rich-editor]")) return;
9916:   event.preventDefault();
9917:   if (event.dataTransfer) event.dataTransfer.dropEffect = "move";
9918:   updateRichDropMarker(editor, activeRichDragBlock, event.clientX, event.clientY, event.target);
9919: }

```

richRangeFromPoint - lines 9921-9933

Item	Explanation
Source location	Lines 9921-9933 (13 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function richRangeFromPoint(x, y) {
Touches	browser/Electron DOM
Calls detected	caretRangeFromPoint, caretPositionFromPoint, createRange, setStart, collapse
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9921: function richRangeFromPoint(x, y) {
9922:   if (document.caretRangeFromPoint) return document.caretRangeFromPoint(x, y);
9923:   if (document.caretPositionFromPoint) {
9924:     const position = document.caretPositionFromPoint(x, y);
9925:     if (position) {
9926:       const range = document.createRange();
9927:       range.setStart(position.offsetNode, position.offset);
9928:       range.collapse(true);
9929:       return range;
9930:     }
9931:   }
9932:   return null;
9933: }

```

richDropLocation - lines 9935-9950

Item	Explanation
Source location	Lines 9935-9950 (16 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function richDropLocation(editor, movingBlock, x, y, eventTarget = null) {
Touches	browser/Electron DOM
Calls detected	validDropRangeForEditor, richRangeFromPoint, contains, elementFromPoint
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9935: function richDropLocation(editor, movingBlock, x, y, eventTarget = null) {
9936:   const previousPointerEvents = movingBlock?.style.pointerEvents || "";
9937:   if (movingBlock) movingBlock.style.pointerEvents = "none";
9938:   try {
9939:     const dropRange = validDropRangeForEditor(editor, movingBlock, richRangeFromPoint(x, y));
9940:     const pointTarget = eventTarget && !movingBlock?.contains(eventTarget)
9941:       ? eventTarget
9942:       : document.elementFromPoint(x, y);
9943:     return {
9944:       dropRange,
9945:       target: pointTarget?.closest?(".rich-block") || null
9946:     };
9947:   } finally {
9948:     if (movingBlock) movingBlock.style.pointerEvents = previousPointerEvents;
9949:   }
9950: }

```

validDropRangeForEditor - lines 9952-9960

Item	Explanation
Source location	Lines 9952-9960 (9 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function validDropRangeForEditor(editor, movingBlock, range) {
Touches	Mostly local calculations or local UI state.
Calls detected	contains, closest
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9952: function validDropRangeForEditor(editor, movingBlock, range) {
9953:   const rangeNode = range?.startContainer?.nodeType === Node.TEXT_NODE
9954:     ? range.startContainer.parentElement
9955:     : range?.startContainer;
9956:   if (!range || !rangeNode || !editor.contains(rangeNode)) return null;
9957:   if (movingBlock?.contains(rangeNode)) return null;
9958:   if (rangeNode.closest("[contenteditable='false']")) return null;
9959:   return range;
9960: }

```

ensureRichDropMarker - lines 9962-9971

Item	Explanation
Source location	Lines 9962-9971 (10 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function ensureRichDropMarker() {
Touches	browser/Electron DOM
Calls detected	createElement, setAttribute, append
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9962: function ensureRichDropMarker() {
9963:   if (!activeRichDropMarker) {
9964:     activeRichDropMarker = document.createElement("div");
9965:     activeRichDropMarker.className = "rich-drop-marker";
9966:     activeRichDropMarker.setAttribute("aria-hidden", "true");
9967:     document.body.append(activeRichDropMarker);
9968:   }
9969:   activeRichDropMarker.hidden = false;
9970:   return activeRichDropMarker;
9971: }

```

hideRichDropMarker - lines 9973-9975

Item	Explanation
Source location	Lines 9973-9975 (3 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function hideRichDropMarker() {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9973: function hideRichDropMarker() {
9974:   if (activeRichDropMarker) activeRichDropMarker.hidden = true;
9975: }

```

updateRichDropMarker - lines 9977-10006

Item	Explanation
Source location	Lines 9977-10006 (30 source lines).
Purpose	Updates UI, state, cache, or stored data.
Declaration	function updateRichDropMarker(editor, movingBlock, x, y, eventTarget = null) {
Touches	Mostly local calculations or local UI state.
Calls detected	ensureRichDropMarker, richDropLocation, getClientRects, getBoundingClientRect, max, min, contains
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

9977: function updateRichDropMarker(editor, movingBlock, x, y, eventTarget = null) {
9978:   if (!editor || !movingBlock) return;
9979:   const marker = ensureRichDropMarker();
9980:   const { dropRange, target } = richDropLocation(editor, movingBlock, x, y, eventTarget);
9981:   if (dropRange) {
9982:     const rangeRect = dropRange.getClientRects()[0];
9983:     const anchor = dropRange.startContainer.nodeType === Node.TEXT_NODE
9984:       ? dropRange.startContainer.parentElement
9985:       : dropRange.startContainer;
9986:     const anchorRect = anchor?.getBoundingClientRect?().();

```

```

9987:     const rect = rangeRect || anchorRect || editor.getBoundingClientRect();
9988:     marker.className = "rich-drop-marker caret";
9989:     marker.style.left = `${Math.max(rect.left, Math.min(x, rect.right || rect.left))}px`;
9990:     marker.style.top = `${rect.top}px`;
9991:     marker.style.width = "2px";
9992:     marker.style.height = `${Math.max(22, rect.height || 22)}px`;
9993:     return;
9994: }
9995:
9996: const targetRect = (target && editor.contains(target) && target !== movingBlock)
9997:   ? target.getBoundingClientRect()
9998:   : editor.getBoundingClientRect();
9999: const imageRect = movingBlock.getBoundingClientRect();
10000: const beforeTarget = !target || y < targetRect.top + targetRect.height / 2;
10001: marker.className = "rich-drop-marker block";
10002: marker.style.left = `${targetRect.left}px`;
10003: marker.style.top = `${beforeTarget ? targetRect.top : targetRect.bottom}px`;
10004: marker.style.width = `${Math.min(targetRect.width, Math.max(160, imageRect.width ||
targetRect.width))}px`;
10005: marker.style.height = "3px";
10006: }

```

moveRichBlockToPoint - lines 10008-10029

Item	Explanation
Source location	Lines 10008-10029 (22 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function moveRichBlockToPoint(editor, movingBlock, x, y, eventTarget = null, options = {}) {
Touches	browser/Electron DOM
Calls detected	richDropLocation, remove, insertRichBlockAtCursor, contains, append, getBoundingClientRect, before, after, cleanupEmptyRichTextBlocks, focusAdjacentImageText, saveRichEditorToProject, selectRichBlock
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

10008: function moveRichBlockToPoint(editor, movingBlock, x, y, eventTarget = null, options = {}) {
10009:   if (!editor || !movingBlock) return;
10010:   const { dropRange, target } = richDropLocation(editor, movingBlock, x, y, eventTarget);
10011:
10012:   movingBlock.classList.remove("is-dragging");
10013:   if (target === movingBlock) return;
10014:
10015:   if (dropRange) {
10016:     insertRichBlockAtCursor(editor, movingBlock, dropRange);
10017:   } else if (!target || !editor.contains(target)) {
10018:     editor.append(movingBlock);
10019:   } else if (y < target.getBoundingClientRect().top + target.getBoundingClientRect().height / 2) {
10020:     target.before(movingBlock);
10021:   } else {
10022:     target.after(movingBlock);
10023:   }
10024:
10025:   cleanupEmptyRichTextBlocks(editor);
10026:   if (options.focusAfter !== false) focusAdjacentImageText(movingBlock, "after");
10027:   if (options.commit !== false) saveRichEditorToProject(editor);
10028:   if (options.focusAfter === false) selectRichBlock(movingBlock);
10029: }

```

handleRichDrop - lines 10031-10039

Item	Explanation
Source location	Lines 10031-10039 (9 source lines).
Purpose	Handles an event, request, command, or user action.
Declaration	function handleRichDrop(event) {
Touches	Mostly local calculations or local UI state.
Calls detected	closest, preventDefault, moveRichBlockToPoint, hideRichDropMarker

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

10031: function handleRichDrop(event) {
10032:   if (!activeRichDragBlock) return;
10033:   const editor = event.target.closest("[data-rich-editor]");
10034:   if (!editor || editor !== activeRichDragBlock.closest("[data-rich-editor]")) return;
10035:   event.preventDefault();
10036:   moveRichBlockToPoint(editor, activeRichDragBlock, event.clientX, event.clientY, event.target);
10037:   hideRichDropMarker();
10038:   activeRichDragBlock = null;
10039: }

```

imageMoveDragControl - lines 10041-10043

Item	Explanation
Source location	Lines 10041-10043 (3 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function imageMoveDragControl(target) {
Touches	Mostly local calculations or local UI state.
Calls detected	closest
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

10041: function imageMoveDragControl(target) {
10042:   return target.closest("button, input, select, textarea, a, .rich-block-actions, [data-rich-caret]");
10043: }

```

beginRichImageMoveDrag - lines 10045-10061

Item	Explanation
Source location	Lines 10045-10061 (17 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function beginRichImageMoveDrag(event) {
Touches	Mostly local calculations or local UI state.
Calls detected	closest, imageMoveDragControl
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

10045: function beginRichImageMoveDrag(event) {
10046:   if (event.button !== 0) return;
10047:   const block = event.target.closest(".rich-image-block");
10048:   if (!block || imageMoveDragControl(event.target)) return;
10049:   if (event.target.closest(".rich-image-viewport.crop-active")) return;

```

```

10050:   const editor = block.closest("[data-rich-editor]");
10051:   if (!editor) return;
10052:   activeRichImageMoveDrag = {
10053:     block,
10054:     editor,
10055:     moved: false,
10056:     pointerId: event.pointerId,
10057:     startX: event.clientX,
10058:     startY: event.clientY
10059:   };
10060:   block.setPointerCapture?.(event.pointerId);
10061: }

```

moveRichImageMoveDrag - lines 10063-10077

Item	Explanation
Source location	Lines 10063-10077 (15 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function moveRichImageMoveDrag(event) {
Touches	browser/Electron DOM
Calls detected	hypot, selectRichBlock, add, updateRichDropMarker, preventDefault
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

10063: function moveRichImageMoveDrag(event) {
10064:   if (!activeRichImageMoveDrag) return;
10065:   const drag = activeRichImageMoveDrag;
10066:   const distance = Math.hypot(event.clientX - drag.startX, event.clientY - drag.startY);
10067:   if (!drag.moved && distance < 6) return;
10068:   if (!drag.moved) {
10069:     drag.moved = true;
10070:     activeRichDragBlock = drag.block;
10071:     selectRichBlock(drag.block);
10072:     drag.block.classList.add("is-dragging");
10073:     document.body.classList.add("rich-image-dragging");
10074:   }
10075:   updateRichDropMarker(drag.editor, drag.block, event.clientX, event.clientY, event.target);
10076:   event.preventDefault();
10077: }

```

endRichImageMoveDrag - lines 10079-10092

Item	Explanation
Source location	Lines 10079-10092 (14 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function endRichImageMoveDrag(event) {
Touches	browser/Electron DOM
Calls detected	moveRichBlockToPoint, hideRichDropMarker, preventDefault, remove
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

10079: function endRichImageMoveDrag(event) {
10080:   if (!activeRichImageMoveDrag) return;
10081:   const drag = activeRichImageMoveDrag;
10082:   drag.block.releasePointerCapture?.(drag.pointerId);
10083:   if (drag.moved) {

```

```

10084:     moveRichBlockToPoint(drag.editor, drag.block, event.clientX, event.clientY);
10085:     hideRichDropMarker();
10086:     event.preventDefault();
10087: }
10088: drag.block.classList.remove("is-dragging");
10089: document.body.classList.remove("rich-image-dragging");
10090: activeRichImageMoveDrag = null;
10091: activeRichDragBlock = null;
10092: }

```

beginImageCropDrag - lines 10094-10109

Item	Explanation
Source location	Lines 10094-10109 (16 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function beginImageCropDrag(event) {
Touches	browser/Electron DOM, filesystem
Calls detected	closest, normalizeCropPosition, max, getBoundingClientRect, preventDefault
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

10094: function beginImageCropDrag(event) {
10095:   const viewport = event.target.closest(".rich-image-viewport.crop-active");
10096:   if (!viewport || event.button !== 0) return;
10097:   const block = viewport.closest(".rich-image-block");
10098:   activeImageCropDrag = {
10099:     block,
10100:     editor: block.closest("[data-rich-editor]"),
10101:     startX: event.clientX,
10102:     startY: event.clientY,
10103:     cropX: normalizeCropPosition(block.dataset.cropX),
10104:     cropY: normalizeCropPosition(block.dataset.cropY),
10105:     width: Math.max(1, viewport.getBoundingClientRect().width),
10106:     height: Math.max(1, viewport.getBoundingClientRect().height)
10107:   };
10108:   event.preventDefault();
10109: }

```

moveImageCropDrag - lines 10111-10121

Item	Explanation
Source location	Lines 10111-10121 (11 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function moveImageCropDrag(event) {
Touches	browser/Electron DOM, filesystem
Calls detected	normalizeCropPosition, querySelector, setProperty
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

10111: function moveImageCropDrag(event) {
10112:   if (!activeImageCropDrag) return;
10113:   const { block, startX, startY, cropX, cropY, width, height } = activeImageCropDrag;
10114:   const nextX = normalizeCropPosition(cropX - ((event.clientX - startX) / width) * 100);
10115:   const nextY = normalizeCropPosition(cropY - ((event.clientY - startY) / height) * 100);
10116:   block.dataset.cropX = String(nextX);
10117:   block.dataset.cropY = String(nextY);

```

```

10118:   const viewport = block.querySelector(".rich-image-viewport");
10119:   viewport?.style.setProperty("--crop-x", `${nextX}%`);
10120:   viewport?.style.setProperty("--crop-y", `${nextY}%`);
10121: }

```

endImageCropDrag - lines 10123-10127

Item	Explanation
Source location	Lines 10123-10127 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	function endImageCropDrag() {
Touches	Mostly local calculations or local UI state.
Calls detected	saveRichEditorToProject
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

10123: function endImageCropDrag() {
10124:   if (!activeImageCropDrag) return;
10125:   saveRichEditorToProject(activeImageCropDrag.editor);
10126:   activeImageCropDrag = null;
10127: }

```

handleStandaloneRichClick - lines 10211-10226

Item	Explanation
Source location	Lines 10211-10226 (16 source lines).
Purpose	Handles an event, request, command, or user action.
Declaration	async function handleStandaloneRichClick(event) {
Touches	browser/Electron DOM
Calls detected	handleRichCaretClick, closest, selectRichBlock, copyRichCodeBlock, editSelectedRichBlock, deleteSelectedRichBlock, saveRichEditorToProject
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

10211: async function handleStandaloneRichClick(event) {
10212:   if (handleRichCaretClick(event)) return;
10213:   const blockAction = event.target.closest("[data-rich-block-action]");
10214:   const richBlock = event.target.closest(".rich-block");
10215:   if (richBlock) selectRichBlock(richBlock);
10216:   if (!blockAction) return;
10217:   const editor = blockAction.closest("[data-rich-editor]");
10218:   selectRichBlock(blockAction.closest(".rich-block"));
10219:   if (blockAction.dataset.richBlockAction === "copy-code") {
10220:     await copyRichCodeBlock(blockAction.closest(".rich-block"));
10221:     return;
10222:   }
10223:   if (blockAction.dataset.richBlockAction === "edit") await editSelectedRichBlock(editor);
10224:   if (blockAction.dataset.richBlockAction === "delete") deleteSelectedRichBlock(editor);
10225:   saveRichEditorToProject(editor);
10226: }

```

hasDataset - lines 10323-10327

Item	Explanation
Source location	Lines 10323-10327 (5 source lines).
Purpose	Part of the private builder frontend and editing experience.
Declaration	const hasDataset = (key) => Object.prototype.hasOwnProperty.call(button.dataset, key);
Touches	browser/Electron DOM
Calls detected	call, addCustomSection
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

10323:   const hasDataset = (key) => Object.prototype.hasOwnProperty.call(button.dataset, key);
10324:   if (button.dataset.addSection) {
10325:     await addCustomSection();
10326:     return;
10327:   }

```

Representative opening snippet

```

1: const builderGuideOpen = document.querySelector("#builder-guide-open");
2: const builderGuideDialog = document.querySelector("#builder-guide-dialog");
3: const builderGuideClose = document.querySelector("#builder-guide-close");
4: const builderGuideSearch = document.querySelector("#builder-guide-search");
5: const builderGuideResults = document.querySelector("#builder-guide-results");
6: const builderGuideSections = document.querySelector("#builder-guide-sections");
7: const builderGuideTopicDialog = document.querySelector("#builder-guide-topic-dialog");
8: const builderGuideTopicTitle = document.querySelector("#builder-guide-topic-title");
9: const builderGuideTopicBody = document.querySelector("#builder-guide-topic-body");
10: const builderGuideTopicClose = document.querySelector("#builder-guide-topic-close");
11: const templateSelect = document.querySelector("#template-select");
12: const newCategorySelect = document.querySelector("#new-category-select");
13: const placementSelect = document.querySelector("#placement-select");
14: const newProjectTitle = document.querySelector("#new-project-title");
15: const addProjectButton = document.querySelector("#add-project");
16: const addCategoryButton = document.querySelector("#add-category");
17: const addSiteSectionButton = document.querySelector("#add-site-section");
18: const funFactsInput = document.querySelector("#fun-facts-input");
19: const funFactsCount = document.querySelector("#fun-facts-count");
20: const saveFunFactsButton = document.querySelector("#save-fun-facts");
21: const clearFunFactsButton = document.querySelector("#clear-fun-facts");
22: const siteHeroEyebrowInput = document.querySelector("#site-hero-eyebrow");
23: const siteHeroTitleInput = document.querySelector("#site-hero-title");
24: const siteHeroCopyInput = document.querySelector("#site-hero-copy");
25: const saveSiteContentButton = document.querySelector("#save-site-content");
26: const profileDisplayNameInput = document.querySelector("#profile-display-name");
27: const profilePortfolioLabelInput = document.querySelector("#profile-portfolio-label");
28: const profileContactIntroInput = document.querySelector("#profile-contact-intro");
29: const profileEmailInput = document.querySelector("#profile-email");
30: const profilePhoneInput = document.querySelector("#profile-phone");
31: const profileGithubInput = document.querySelector("#profile-github");
32: const profileLinkedinInput = document.querySelector("#profile-linkedin");
33: const profileWebsiteInput = document.querySelector("#profile-website");
34: const saveProfileContentButton = document.querySelector("#save-profile-content");
35: const profileImageUploadButton = document.querySelector("#profile-image-upload");
36: const profileHeroUploadButton = document.querySelector("#profile-hero-upload");
37: const profileResumeUploadButton = document.querySelector("#profile-resume-upload");
38: const profileBrandUploadButton = document.querySelector("#profile-brand-upload");
39: const profileAssetStatus = document.querySelector("#profile-asset-status");
40: const categoryDropdown = document.querySelector("#category-dropdown");
41: const siteSectionList = document.querySelector("#site-section-list");
42: const projectCreateDialog = document.querySelector("#project-create-dialog");
43: const projectCreateForm = document.querySelector("#project-create-form");
44: const projectCreateCategory = document.querySelector("#project-create-category");
45: const projectCreateCancel = document.querySelector("#project-create-cancel");
46: const categoryDialog = document.querySelector("#category-dialog");
47: const categoryForm = document.querySelector("#category-form");
48: const categoryTitle = document.querySelector("#category-title");
49: const categoryDescription = document.querySelector("#category-description");
50: const categoryAccent = document.querySelector("#category-accent");
51: const categoryCancel = document.querySelector("#category-cancel");
52: const projectTree = document.querySelector("#project-tree");
53: const projectSearchInput = document.querySelector("#project-search");
54: const projectSearchClear = document.querySelector("#project-search-clear");
55: const projectSearchStatus = document.querySelector("#project-search-status");
56: const projectDialog = document.querySelector("#project-dialog");

```



```
57: const projectWindowTitle = document.querySelector("#project-window-title");
58: const projectWindowClose = document.querySelector("#project-window-close");
59: const projectWindowDelete = document.querySelector("#project-window-delete");
60: const viewProjectPreviewButton = document.querySelector("#view-project-preview");
61: const saveProjectButton = document.querySelector("#save-project");
62: const saveProjectCloseButton = document.querySelector("#save-project-close");
63: const projectTitleMenu = document.querySelector("#project-title-menu");
64: const titleRenameActions = document.querySelector("#title-rename-actions");
65: const titleRenameSave = document.querySelector("#title-rename-save");
66: const titleRenameCancel = document.querySelector("#title-rename-cancel");
67: const projectMetaDialog = document.querySelector("#project-meta-dialog");
68: const projectMetaTitle = document.querySelector("#project-meta-title");
69: const projectMetaGrid = document.querySelector("#project-meta-grid");
70: const projectMetaClose = document.querySelector("#project-meta-close");
71: const projectFields = document.querySelector("#project-fields");
72: const sectionTabs = document.querySelector("#section-tabs");
```

Safe change checklist

- Confirm which app, website, installer, workflow, or document consumes this file.
- Change it on a feature branch from development.
- Regenerate docs if the file purpose, API surface, or behavior changes.
- Run the smallest relevant smoke test before merging into development.

Important Code Reference: template-preview.html

Item	Explanation
File	template-preview.html
Category	Website/builder UI source
Folder role	Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
Size	92,731 bytes
Extension	.html
MIME type	text/html
Text lines	1,349

Why this file exists

- Builder HTML shell loaded inside the Electron app and local preview runtime.

How it is used

Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.

How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

Important implementation signals

Item	Explanation
Installer at line 11	Windows setup, update, shortcut, or installation behavior. Evidence: <link rel="shortcut icon" href="assets/favicon.ico" />
Rich text at line 64	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"
Rich text at line 79	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"
Rich text at line 94	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"
Rich text at line 118	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"
Rich text at line 133	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"
Rich text at line 148	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"
Rich text at line 163	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"
Rich text at line 178	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"
Network request at line 192	Frontend-backend communication or public web/API calls. Evidence: data-placeholder="https://github.com/username"
Rich text at line 193	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"
Network request at line 207	Frontend-backend communication or public web/API calls. Evidence: data-placeholder="https://linkedin.com/in/username"
Rich text at line 208	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"
Network request at line 222	Frontend-backend communication or public web/API calls. Evidence: data-placeholder="https://example.com"
Rich text at line 223	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"
Rich text at line 259	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: contenteditable="true"
Browser DOM at line 391	Builder or website interface behavior in the browser/Electron renderer. Evidence: Resume should be a public-safe PDF, not a private editable document.

Item	Explanation
Installer at line 420	Windows setup, update, shortcut, or installation behavior. Evidence: All project creation starts from the main Add project control. This keeps project creation consistent and prevents hidden category-specific add buttons from creating projects with incomplete metadata.
Security or auth at line 432	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: Only categories that contain projects show project rows. Empty categories stay as clean category headers until a project is added.
Rich text at line 462	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <p>Rich text fields behave like document editors. Click where you want to type, highlight the exact words you want to edit, then apply formatting to that selection. Formatting should not change an entire paragraph unless
Rich text at line 463	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <p>The builder stores the text as structured content so the portfolio can render it cleanly later. That means the editor may standardize pasted formatting, but it should preserve intentional choices such as bold text, fo
Rich text at line 464	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <p>Front-page, profile, contact, fun facts, and project overview fields now use the same selection-level text behavior. Highlight the exact words you want to edit, then use the right-click menu or floating selection insp
Rich text at line 466	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: Right-click highlighted text in any builder text field to copy, cut, paste, select all, change font, change size, change color, and apply bold, italic, or underline.
Rich text at line 467	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: The default typing font is Arial.
Rich text at line 468	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: If selected text has mixed font, size, or color, the inspector fields are blank so you know the selection is mixed.
Rich text at line 469	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: Use Enter for a new paragraph. Pasted line breaks remain line breaks.
Rich text at line 470	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: Typing and wrapping should not create fake new paragraphs; only hard enters and pasted line breaks do.
Rich text at line 471	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: Use the Code block command or Paste as code for source snippets so they render differently from general text.
Rich text at line 477	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: Keep color changes readable on the selected project appearance.
Rich text at line 478	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: If a pasted paragraph looks strange, clear the formatting and reapply only the formatting you need.
Rich text at line 480	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <p>If the selection menu disappears, click back inside the rich editor and highlight the text again. Before saving, read the section in the project preview to make sure line breaks, lists, and emphasized words still look
Rich text at line 484	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <summary>6. Links, formulas, code, and pasted content</summary>
Cloudflare or AI at line 487	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <p>Use this section when pasting content from ChatGPT, documentation, GitHub, LinkedIn, LTspice notes, equations, source code, or lab references. The builder tries to recognize the content, but you should still check the
Compiler or simulator at line 491	Part of the Compile Code workspace or language/tool detection. Evidence: Use Code block or Paste as code for C, C++, SystemVerilog, LTspice, Java, JavaScript, Python, and HTML snippets.
Rich text at line 492	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: Choose Basic paste when you want the builder to clean extra trailing whitespace, or Keep source spacing when indentation is important.
Compiler or simulator at line 495	Part of the Compile Code workspace or language/tool detection. Evidence: For Verilog and SystemVerilog, use Simulate to run the simulator. Simulation requires a Testbench file; plain Compile can still check or build HDL design files.
Security or auth at line 496	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: Use Add testbench to create a starter testbench with \$dumpfile and \$dumpvars. Those waveform calls feed the signal scope, where HDL signals can be viewed over time.
Network request at line 503	Frontend-backend communication or public web/API calls. Evidence: Use full links when possible: https://github.com/user/repo is clearer than only github.com.
Compiler or simulator at line 507	Part of the Compile Code workspace or language/tool detection. Evidence: Use Compile Code to test C, C++, Java, JavaScript, Python, HTML checks, and any installed Verilog, SystemVerilog, or LTspice command-line tools before deciding what to publish later.
Security or auth at line 508	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: Use Show output for terminal logs and Show scope for HDL timing waveforms after a successful testbench simulation.

Item	Explanation
Rich text at line 510	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: If a link is accidentally treated as a formula, remove the formula block and paste it again as normal text or link content.
Cloudflare or AI at line 512	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <p>When pasting from ChatGPT, review formulas and code before saving. If the content is a link, it should render as a clickable link, not as a formula. If the content is source code, it should render inside a dark code b
Security or auth at line 519	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <p>Images are best for visual evidence: schematics, PCB layouts, test benches, oscilloscope captures, simulation plots, block diagrams, timing diagrams, microscope photos, assembled boards, or setup photos. Files are bes
Rich text at line 521	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: Paste an image into a rich editor to place it below the cursor line.
Rich text at line 522	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: Use Add image when you need source selection, rename, caption, alignment, crop, or visible/download-only control.
Compiler or simulator at line 535	Part of the Compile Code workspace or language/tool detection. Evidence: <p>Project evidence is saved under docs/project-folder/section-or-subsection/filename. For example, a SystemVerilog design file could be saved as docs/laser-link/design/filter.sv. The upload status shows the generated pa
Security or auth at line 581	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <summary>10. Publishing target and authentication</summary>
Security or auth at line 583	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <p>Publishing target controls where Apply to site can push. A target is saved only after write access is verified, because the builder should never publish to a website unless the current user has permission to update th
Security or auth at line 584	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <p>This protects your portfolio from accidental pushes and also lets another user install the builder for their own site without getting access to your website. The target describes where the public portfolio will live;
Security or auth at line 588	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: Click Authenticate target to verify write access for that repository and branch.
Git command at line 589	Repository state, publishing, branch checks, or release automation. Evidence: Complete GitHub/Git Credential Manager sign-in only when prompted.
Security or auth at line 590	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: Use Load from target only after authentication succeeds and only when you want to import that website repository into this builder.
Security or auth at line 595	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: Authenticate target does not import content; Load from target does not authenticate a new target.
Git command at line 596	Repository state, publishing, branch checks, or release automation. Evidence: If authentication fails, read the message before retrying; it usually explains whether Git, credentials, branch state, or repository access caused the failure.
Security or auth at line 598	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <p>GitHub authorization is cached per repository and branch. Daily reuse avoids repeated prompts, and repeated successful authorization can extend trust for that exact target. If the target changes, authenticate again be
Git command at line 605	Repository state, publishing, branch checks, or release automation. Evidence: The builder can be used offline for most editing work. Publishing is the only part that needs the network and a Git target with write access.
Git command at line 611	Repository state, publishing, branch checks, or release automation. Evidence: If offline, keep editing and saving drafts. Publish when the network and Git target are available.
Security or auth at line 617	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: Use Apply to site only after the preview looks correct and the target is authenticated.
Installer at line 625	Windows setup, update, shortcut, or installation behavior. Evidence: <p>Use Check updates to manually look for a newer builder. Updates come from GitHub Releases after changes are merged into the release branch and a new installer is built. Use the Preferences menu to switch between light
Installer at line 629	Windows setup, update, shortcut, or installation behavior. Evidence: The app downloads the installer, closes the current builder, updates the existing install, and reopens the app.
Installer at line 631	Windows setup, update, shortcut, or installation behavior. Evidence: If an old uninstaller fails, setup stops the builder process and continues updating the same folder.
Installer at line 636	Windows setup, update, shortcut, or installation behavior. Evidence: If a release is paused, wait for the next version instead of forcing the old installer.
Installer at line 637	Windows setup, update, shortcut, or installation behavior. Evidence: If Windows blocks a process from closing, use Task Manager as Administrator to end the stuck installer or old builder process.
Installer at line 638	Windows setup, update, shortcut, or installation behavior. Evidence: If the app opens but looks stale, close it fully and reopen it from the desktop shortcut.

Item	Explanation
Installer at line 640	Windows setup, update, shortcut, or installation behavior. Evidence: <p>If the app cannot update, close any old OMB Builder process in Task Manager and reopen the app from the desktop shortcut. Avoid installing another copy in a different folder; duplicate copies can make Windows shortcut
Browser DOM at line 646	Builder or website interface behavior in the browser/Electron renderer. Evidence: <p>Anything pushed to the portfolio repository can become public. Treat the portfolio as a public engineering document. If a file would be unsafe in a public GitHub repository, it should not be uploaded or published thro
Security or auth at line 647	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <p>Good portfolio evidence is public-safe, clearly labeled, and useful to a visitor. Private credentials, private lab data, personal identity documents, and unpublished confidential work should stay out of the portfolio.
Security or auth at line 649	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: Do not upload private keys, passwords, private tokens, or private lab data.
Security or auth at line 652	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: Keep authentication local. The builder should not commit credentials.
Security or auth at line 657	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: Check code files for tokens, API keys, Wi-Fi credentials, private comments, or machine-specific paths.
Cloudflare or AI at line 660	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: Use Security report to review app download counts, current publishing authorization status, and whether visitor logging requires Cloudflare analytics.
Security or auth at line 662	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <p>If another user installs the builder, they should connect their own target unless you explicitly authorize access to your target. The builder should help people create portfolios, but it should not give anyone silent
Git command at line 669	Repository state, publishing, branch checks, or release automation. Evidence: The routine is intentionally repetitive because publishing is safer when every project has been parsed and every section has been previewed before Git pushes anything.
Security or auth at line 684	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: For publishing target changes, authenticate before loading or applying site data.
Rich text at line 755	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: Accent color
Rich text at line 756	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <input id="category-accent" type="color" value="#1677a8" />
Rich text at line 780	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <button type="button" data-title-action="paste">Paste as title</button>
Git command at line 868	Repository state, publishing, branch checks, or release automation. Evidence: <input id="publish-target-repository" type="text" placeholder="https://github.com/USERNAME/USERNAME.github.io.git" />
Security or auth at line 871	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: Authenticate target verifies that this computer can write to the selected GitHub Pages repository. Load from target imports the latest compatible website data from that authenticated repository into this builder. Authent
Security or auth at line 874	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <button class="button primary" id="publish-target-auth" type="submit">Authenticate target</button>

Representative opening snippet

```

1: <!DOCTYPE html>
2: <html lang="en">
3:   <head>
4:     <meta charset="UTF-8" />
5:     <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6:     <meta name="robots" content="noindex, nofollow" />
7:     <title>Local Portfolio Builder</title>
8:     <link rel="icon" type="image/png" sizes="16x16" href="assets/favicon-16.png" />
9:     <link rel="icon" type="image/png" sizes="32x32" href="assets/favicon-32.png" />
10:    <link rel="icon" type="image/png" sizes="192x192" href="assets/favicon-192.png" />
11:    <link rel="shortcut icon" href="assets/favicon.ico" />
12:    <link rel="apple-touch-icon" sizes="180x180" href="assets/apple-touch-icon.png" />
13:    <link rel="stylesheet" href="styles.css" />
14:    <link rel="stylesheet" href="template-preview.css" />
15:  </head>
16:  <body class="template-preview-page">
17:    <header class="template-header">
18:      <a class="brand" href="index.html#top" aria-label="Back to portfolio">
19:        <span class="brand-lockup" aria-hidden="true">
20:          
21:          <span class="omb-engraving">OMB</span>
22:        </span>
23:      </span>
24:      <strong>Local Portfolio Builder</strong>
25:      <small>Build, preview, save locally, then publish when ready</small>

```

```

26:         </span>
27:     </a>
28:     <div class="builder-header-actions">
29:         <span class="builder-save-state" id="builder-save-state" data-state="loaded">Loaded</span>
30:         <a class="button secondary" href="index.html#projects" target="_blank"
rel="noreferrer">Portfolio</a>
31:         <button class="button secondary" id="publish-target-open" type="button">Publishing
target</button>
32:         <button class="button secondary" id="security-report-open" type="button">Security
report</button>
33:         <button class="button secondary" id="app-update-check" type="button">Check updates</button>
34:         <button class="button primary" id="save-draft" type="button">Save draft</button>
35:         <button class="button secondary" id="apply-catalog" type="button">Apply to site</button>
36:     </div>
37: </header>
38:
39: <main class="builder-workspace">
40:     <aside class="builder-sidebar" aria-labelledby="library-title">
41:         <button class="builder-guide-panel" id="builder-guide-open" type="button" aria-label="Open
builder guide">
42:             <span class="builder-guide-icon" aria-hidden="true"></span>
43:             <span>
44:                 <strong id="library-title">Portfolio Builder</strong>
45:                 <small>Click for a quick build guide</small>
46:             </span>
47:         </button>
48:
49:         <div class="builder-panel site-section-panel">
50:             <div class="panel-title-row">
51:                 <h2>Front page text</h2>
52:                 <button class="button secondary compact-button" id="save-site-content"
type="button">Save</button>
53:             </div>
54:             <div class="front-page-text-fields">
55:                 <label>
56:                     <span>Small label</span>
57:                     <div
58:                         class="rich-summary-editor front-rich-field"
59:                         id="site-hero-eyebrow"
60:                         data-rich-editor="site-field"
61:                         data-site-field="heroEyebrow"
62:                         data-rich-text-only="true"
63:                         data-placeholder="Engineering portfolio"
64:                         contenteditable="true"
65:                         spellcheck="true"
66:                         role="textbox"
67:                         aria-label="Small front page label"
68:                     ></div>
69:                 </label>
70:                 <label>
71:                     <span>Main headline</span>
72:                     <div

```

Safe change checklist

- Confirm which app, website, installer, workflow, or document consumes this file.
- Change it on a feature branch from development.
- Regenerate docs if the file purpose, API surface, or behavior changes.
- Run the smallest relevant smoke test before merging into development.

Important Code Reference: template-preview.css

Item	Explanation
File	template-preview.css
Category	Website/builder UI source
Folder role	Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
Size	131,453 bytes
Extension	.css
MIME type	text/css
Text lines	5,749

Why this file exists

- Builder application CSS for dialogs, windows, rich editors, compile code, dark mode, guide windows, and local previews.

How it is used

Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.

How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

Important implementation signals

Item	Explanation
Rich text at line 34	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #16384a;
Rich text at line 36	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.78rem;
Rich text at line 37	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 900;
Rich text at line 43	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: #f5c56d;
Rich text at line 44	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #5f4207;
Rich text at line 50	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: #93c5fd;
Rich text at line 51	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #164780;
Rich text at line 57	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: #8eefac;
Rich text at line 58	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #14532d;
Rich text at line 63	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: #fca5a5;
Rich text at line 64	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #7f1d1d;
Rich text at line 70	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: #d6effa;
Rich text at line 71	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #16384a;
Rich text at line 74	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color 140ms ease,
Rich text at line 75	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: background-color 140ms ease,
Rich text at line 76	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color 140ms ease,
Rich text at line 83	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: #b8e6f7;
Rich text at line 84	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #16384a;
Rich text at line 91	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: #ffffff;
Rich text at line 92	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #16384a;

Item	Explanation
Rich text at line 148	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--ink);
Rich text at line 149	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 1.18rem;
Rich text at line 173	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--muted);
Rich text at line 174	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.72rem;
Rich text at line 175	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 900;
Rich text at line 184	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--ink);
Rich text at line 186	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font: inherit;
Rich text at line 187	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 700;
Rich text at line 191	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: #4b91db;
Rich text at line 198	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--muted);
Rich text at line 199	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.78rem;
Rich text at line 205	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: inherit;
Rich text at line 222	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--ink);
Rich text at line 223	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.96rem;
Rich text at line 241	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #16384a;
Rich text at line 243	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.75rem;
Rich text at line 244	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 900;
Rich text at line 266	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #16384a;
Rich text at line 278	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--ink);
Rich text at line 279	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 1rem;
Rich text at line 283	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--muted);
Rich text at line 284	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.78rem;
Rich text at line 292	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border: 2px solid currentColor;
Rich text at line 304	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: background: currentColor;
Rich text at line 309	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: box-shadow: 0 7px 0 currentColor;
Rich text at line 338	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #16384a;
Rich text at line 340	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.78rem;
Rich text at line 341	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 900;
Rich text at line 356	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--muted);
Rich text at line 357	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.78rem;
Rich text at line 358	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 900;
Rich text at line 366	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--ink);
Rich text at line 368	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font: inherit;
Rich text at line 369	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.9rem;
Rich text at line 378	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #16384a;
Rich text at line 380	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font: inherit;
Rich text at line 381	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.8rem;
Rich text at line 382	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 900;
Rich text at line 388	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #ffffff;
Rich text at line 390	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: #1f6ed4;
Rich text at line 395	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--muted);

Item	Explanation
Rich text at line 396	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.82rem;
Rich text at line 433	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--ink);
Rich text at line 435	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.92rem;
Rich text at line 436	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 950;
Rich text at line 456	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #087f9b;
Rich text at line 457	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 950;
Rich text at line 471	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #16384a;
Rich text at line 473	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.7rem;
Rich text at line 474	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 950;
Rich text at line 490	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #ffffff;
Rich text at line 496	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #ffffff;
Rich text at line 500	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: rgba(255, 255, 255, 0.7);
Rich text at line 542	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--muted);
Rich text at line 543	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.86rem;
Rich text at line 555	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #2b5064;
Rich text at line 556	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.86rem;
Rich text at line 582	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #fafdff;
Rich text at line 583	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.78rem;
Rich text at line 601	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--muted);

CSS selectors and visual hooks

Item	Explanation
Line 1	.template-preview-page
Line 5	.template-header
Line 20	.section-actions
Line 27	.builder-save-state
Line 42	.builder-save-state[data-state="dirty"]
Line 49	.builder-save-state[data-state="publishing"]
Line 56	.builder-save-state[data-state="published"]
Line 62	.builder-save-state[data-state="error"]
Line 69	.template-preview-page button
Line 82	.template-preview-page button:hover
Line 90	.template-preview-page button:active
Line 98	.builder-workspace
Line 107	.builder-inspector
Line 116	.template-preview
Line 123	.builder-panel
Line 129	.project-library-heading
Line 137	.project-library-heading > div
Line 143	.project-library-heading h2

Item	Explanation
Line 147	.project-library-heading h2
Line 153	.project-library-tools
Line 165	.project-search-field
Line 172	.workflow-card span
Line 179	.project-search-field input
Line 190	.project-search-field input:focus
Line 196	#project-search-status
Line 203	.tree-category-toggle mark
Line 210	.workflow-card
Line 219	.workflow-card strong
Line 228	.workflow-actions
Line 234	.workflow-metrics span
Line 247	.workflow-actions .button
Line 253	.template-preview > .section-heading
Line 257	.builder-guide-panel
Line 273	.builder-guide-panel small
Line 277	.builder-guide-panel strong
Line 282	.builder-guide-panel small
Line 287	.builder-guide-icon
Line 297	.builder-guide-icon::after
Line 307	.builder-guide-icon::before
Line 312	.builder-guide-icon::after
Line 318	.builder-guide-dialog-content
Line 328	.builder-guide-actions
Line 334	.builder-guide-live span
Line 344	.builder-guide-tools
Line 353	.builder-guide-tools label
Line 361	.builder-guide-tools input
Line 373	.builder-guide-actions button
Line 387	.builder-guide-actions button:hover
Line 393	.builder-guide-results
Line 399	.builder-guide-sections
Line 416	.builder-guide-section
Line 424	.builder-guide-section[hidden]
Line 428	.builder-guide-section summary
Line 444	.builder-guide-section summary::-webkit-details-marker
Line 448	.builder-guide-section summary::before
Line 462	.builder-guide-section summary::after
Line 480	.builder-guide-section[open] summary
Line 485	.builder-guide-section[open] summary::before
Line 489	.builder-guide-section summary:hover

Item	Explanation
Line 495	.builder-guide-section summary: hover::after
Line 499	.builder-guide-section summary: hover::after
Line 504	.builder-guide-section > div
Line 513	.builder-guide-topic-dialog
Line 517	.builder-guide-topic-content
Line 522	.builder-guide-topic-body
Line 534	.builder-guide-topic-copy
Line 540	.builder-guide-topic-body p
Line 550	.builder-guide-topic-body ul
Line 561	.builder-guide-topic-body li
Line 565	.publish-result-box
Line 570	.publish-result-box p
Line 574	.publish-result-box pre
Line 588	.publish-target-form
Line 595	.publish-target-form label
Line 600	.publish-target-field-note
Line 606	.publish-target-form input
Line 616	.publish-target-current
Line 627	.publish-target-current div
Line 633	.publish-target-current strong
Line 637	.publish-target-current span
Line 641	.publish-target-readiness
Line 648	.publish-target-readiness strong
Line 653	.publish-target-readiness ul
Line 661	.publish-target-readiness li.ready
Line 665	.publish-target-readiness li.blocked
Line 669	.publish-target-tool-actions
Line 675	.publish-target-note
Line 682	.profile-setup-fields
Line 687	.profile-setup-fields label: first-child
Line 691	.profile-asset-actions
Line 697	.profile-asset-status
Line 704	.app-update-dialog
Line 708	.security-report-dialog
Line 712	.app-update-content
Line 717	.security-report-body
Line 727	.security-report-body p
Line 734	.app-update-body p: first-child
Line 739	.security-report-grid
Line 745	.security-report-card
Line 756	.security-report-card h3

Item	Explanation
Line 763	.security-report-card strong
Line 769	.security-report-card ul
Line 779	.security-report-assets
Line 787	.security-report-assets li
Line 797	.security-report-assets li:last-child
Line 801	.security-report-assets span:first-child
Line 808	.app-update-release-meta
Line 816	.app-update-dialog-actions
Line 820	.app-update-dialog-actions [hidden]
Line 824	.portfolio-preview-button
Line 834	.portfolio-preview-icon
Line 842	.portfolio-preview-icon::before
Line 854	.portfolio-preview-icon::after
Line 867	.save-all-icon
Line 876	.save-all-icon::after
Line 881	.save-all-icon::before
Line 890	.save-all-icon::after
Line 902	.builder-panel p
Line 906	.builder-panel h1
Line 911	.builder-panel h2

Representative opening snippet

```

1: .template-preview-page {
2:   background: #eef8fc;
3: }
4:
5: .template-header {
6:   position: sticky;
7:   top: 0;
8:   z-index: 20;
9:   display: flex;
10:  align-items: center;
11:  justify-content: space-between;
12:  gap: 18px;
13:  padding: 12px clamp(18px, 4vw, 56px);
14:  border-bottom: 1px solid rgba(201, 222, 234, 0.84);
15:  background: rgba(214, 239, 250, 0.96);
16:  backdrop-filter: blur(16px);
17: }
18:
19: .builder-header-actions,
20: .section-actions {
21:   display: flex;
22:   flex-wrap: wrap;
23:   gap: 10px;
24:   align-items: center;
25: }
26:
27: .builder-save-state {
28:   display: inline-flex;
29:   align-items: center;
30:   min-height: 34px;
31:   padding: 0 12px;
32:   border: 1px solid #9bd4ea;
33:   border-radius: 999px;
34:   color: #16384a;
35:   background: #f5fbfe;
36:   font-size: 0.78rem;
37:   font-weight: 900;
38:   letter-spacing: 0.01em;
39:   white-space: nowrap;
40: }
41:

```

```

42: .builder-save-state[data-state="dirty"] {
43:   border-color: #f5c56d;
44:   color: #5f4207;
45:   background: #fff7dc;
46: }
47:
48: .builder-save-state[data-state="saving"],
49: .builder-save-state[data-state="publishing"] {
50:   border-color: #93c5fd;
51:   color: #164780;
52:   background: #dbeafe;
53: }
54:
55: .builder-save-state[data-state="saved"],
56: .builder-save-state[data-state="published"] {
57:   border-color: #86efac;
58:   color: #14532d;
59:   background: #dcfce7;
60: }
61:
62: .builder-save-state[data-state="error"] {
63:   border-color: #fca5a5;
64:   color: #7f1d1d;
65:   background: #fee2e2;
66: }
67:
68: .template-preview-page .button,
69: .template-preview-page button {
70:   border-color: #d6effa;
71:   color: #16384a;
72:   background: #d6effa;

```

Safe change checklist

- Confirm which app, website, installer, workflow, or document consumes this file.
- Change it on a feature branch from development.
- Regenerate docs if the file purpose, API surface, or behavior changes.
- Run the smallest relevant smoke test before merging into development.

Important Code Reference: script.js

Item	Explanation
File	script.js
Category	JavaScript source
Folder role	Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
Size	158,556 bytes
Extension	.js
MIME type	application/javascript
Text lines	3,710

Why this file exists

- Public website JavaScript for navigation, portfolio rendering, search, AI UI, and interactive project windows.

How it is used

Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.

How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

API endpoints mentioned or called

Item	Explanation
/api/portfolio-ai	Line 1492. Handles AI assistant questions.
/api/portfolio-ai	Line 1493. Handles AI assistant questions.

Important implementation signals

Item	Explanation
Browser DOM at line 1	Builder or website interface behavior in the browser/Electron renderer. Evidence: const grid = document.querySelector("#project-grid");
Browser DOM at line 2	Builder or website interface behavior in the browser/Electron renderer. Evidence: const searchInput = document.querySelector("#project-search");
Browser DOM at line 3	Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectFilters = document.querySelector("#project-filters");
Browser DOM at line 4	Builder or website interface behavior in the browser/Electron renderer. Evidence: let filterButtons = [...document.querySelectorAll(".filter-button")];
Browser DOM at line 5	Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectCount = document.querySelector("#project-count");
Browser DOM at line 6	Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectTrackCount = document.querySelector("#project-track-count");
Browser DOM at line 7	Builder or website interface behavior in the browser/Electron renderer. Evidence: const projectTrackLabels = document.querySelector("#project-track-labels");
Browser DOM at line 8	Builder or website interface behavior in the browser/Electron renderer. Evidence: const year = document.querySelector("#year");
Browser DOM at line 9	Builder or website interface behavior in the browser/Electron renderer. Evidence: const funFactsCallout = document.querySelector("#fun-facts-callout");
Browser DOM at line 10	Builder or website interface behavior in the browser/Electron renderer. Evidence: const heroEyebrow = document.querySelector(".hero-content .eyebrow");

Item	Explanation
Browser DOM at line 11	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const heroTitle = document.querySelector("#hero-title");</code>
Browser DOM at line 12	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const heroCopy = document.querySelector(".hero-copy");</code>
Browser DOM at line 13	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const brandName = document.querySelector(".brand strong");</code>
Browser DOM at line 14	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const brandSubtitle = document.querySelector(".brand small");</code>
Browser DOM at line 15	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const brandIcon = document.querySelector(".brand-icon");</code>
Browser DOM at line 16	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const brandText = document.querySelector(".omb-engraving");</code>
Browser DOM at line 17	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const headerAvatar = document.querySelector(".header-avatar");</code>
Browser DOM at line 18	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const headerAvatarImage = document.querySelector(".header-avatar img");</code>
Browser DOM at line 19	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const heroImage = document.querySelector(".hero-image");</code>
Browser DOM at line 20	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const resumeSection = document.querySelector("#resume");</code>
Browser DOM at line 21	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const contactBand = document.querySelector("#contact");</code>
Browser DOM at line 22	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const profilePhoto = document.querySelector(".profile-photo");</code>
Browser DOM at line 23	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const contactTitle = document.querySelector("#contact-title");</code>
Browser DOM at line 24	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const contactIntro = document.querySelector(".contact-intro");</code>
Browser DOM at line 25	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const contactDetails = document.querySelector(".contact-details");</code>
Browser DOM at line 26	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const contactLinks = document.querySelector(".contact-links");</code>
Browser DOM at line 27	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const footerOwner = document.querySelector(".site-footer span");</code>
Browser DOM at line 29	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const aiAssistantForm = document.querySelector("#ai-assistant-form");</code>
Browser DOM at line 30	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const aiAssistantPanel = document.querySelector("#ask-ai");</code>
Browser DOM at line 31	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const aiAssistantInput = document.querySelector("#ai-assistant-input");</code>
Browser DOM at line 32	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const aiAssistantLog = document.querySelector("#ai-assistant-log");</code>
Browser DOM at line 33	Builder or website interface behavior in the browser/Electron renderer. Evidence: <code>const aiAssistantStatus = document.querySelector("#ai-assistant-status");</code>
Cloudflare or AI at line 34	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>const aiClearChatButton = document.querySelector("#ai-clear-chat");</code>
Cloudflare or AI at line 57	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <code>let assistantChatHistory = [];</code>
Compiler or simulator at line 73	Part of the Compile Code workspace or language/tool detection. Evidence: <code>{ id: "verilog", label: "Verilog", aliases: ["verilog", "v"] },</code>
Compiler or simulator at line 74	Part of the Compile Code workspace or language/tool detection. Evidence: <code>{ id: "systemverilog", label: "SystemVerilog", aliases: ["systemverilog", "system verilog", "sv"] },</code>
Compiler or simulator at line 76	Part of the Compile Code workspace or language/tool detection. Evidence: <code>{ id: "java", label: "Java", aliases: ["java"] },</code>

Item	Explanation
Compiler or simulator at line 78	Part of the Compile Code workspace or language/tool detection. Evidence: { id: "python", label: "Python", aliases: ["python", "py"] },
Compiler or simulator at line 104	Part of the Compile Code workspace or language/tool detection. Evidence: "embedded-wireless-node": "appearance-deep-navy-cyan-click",
Browser DOM at line 175	Builder or website interface behavior in the browser/Electron renderer. Evidence: dialog.classList.add("is-draggable-dialog");
Browser DOM at line 184	Builder or website interface behavior in the browser/Electron renderer. Evidence: if (dialog.querySelector("[data-section-dialog-resize-handle]")) return;
Browser DOM at line 186	Builder or website interface behavior in the browser/Electron renderer. Evidence: const handle = document.createElement("div");
Browser DOM at line 188	Builder or website interface behavior in the browser/Electron renderer. Evidence: handle.dataset.sectionDialogResizeHandle = direction;
Browser DOM at line 197	Builder or website interface behavior in the browser/Electron renderer. Evidence: dialog.classList.add("is-resized-dialog");
Browser DOM at line 255	Builder or website interface behavior in the browser/Electron renderer. Evidence: dialog.classList.add("is-dragging-dialog");
Browser DOM at line 273	Builder or website interface behavior in the browser/Electron renderer. Evidence: activeSectionDialogDrag.dialog.classList.remove("is-dragging-dialog");
Browser DOM at line 281	Builder or website interface behavior in the browser/Electron renderer. Evidence: if (!handle !dialog?.open dialog.classList.contains("is-minimized-dialog")) return;
Browser DOM at line 285	Builder or website interface behavior in the browser/Electron renderer. Evidence: direction: handle.dataset.sectionDialogResizeHandle,
Browser DOM at line 293	Builder or website interface behavior in the browser/Electron renderer. Evidence: dialog.classList.add("is-resizing-dialog");
Browser DOM at line 309	Builder or website interface behavior in the browser/Electron renderer. Evidence: activeSectionDialogResize.dialog.classList.remove("is-resizing-dialog");
Browser DOM at line 314	Builder or website interface behavior in the browser/Electron renderer. Evidence: const minimized = dialog.classList.contains("is-minimized-dialog");
Browser DOM at line 315	Builder or website interface behavior in the browser/Electron renderer. Evidence: const button = dialog.querySelector(".section-view-minimize");
Browser DOM at line 324	Builder or website interface behavior in the browser/Electron renderer. Evidence: dialog.classList.toggle("is-minimized-dialog");
Browser DOM at line 331	Builder or website interface behavior in the browser/Electron renderer. Evidence: document.addEventListener("pointerdown", beginSectionDialogResize);
Browser DOM at line 332	Builder or website interface behavior in the browser/Electron renderer. Evidence: document.addEventListener("pointerdown", beginSectionDialogDrag);
Browser DOM at line 333	Builder or website interface behavior in the browser/Electron renderer. Evidence: document.addEventListener("pointermove", moveSectionDialogResize);
Browser DOM at line 334	Builder or website interface behavior in the browser/Electron renderer. Evidence: document.addEventListener("pointermove", moveSectionDialogDrag);
Browser DOM at line 335	Builder or website interface behavior in the browser/Electron renderer. Evidence: document.addEventListener("pointerup", endSectionDialogResize);
Browser DOM at line 336	Builder or website interface behavior in the browser/Electron renderer. Evidence: document.addEventListener("pointerup", endSectionDialogDrag);
Browser DOM at line 337	Builder or website interface behavior in the browser/Electron renderer. Evidence: document.addEventListener("pointercancel", endSectionDialogResize);
Browser DOM at line 338	Builder or website interface behavior in the browser/Electron renderer. Evidence: document.addEventListener("pointercancel", endSectionDialogDrag);
Browser DOM at line 339	Builder or website interface behavior in the browser/Electron renderer. Evidence: window.addEventListener("resize", () => {
Browser DOM at line 340	Builder or website interface behavior in the browser/Electron renderer. Evidence: document.querySelectorAll(".section-view-dialog.is-draggable-dialog[open]").forEach((dialog) => {
Rich text at line 370	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: function normalizeCodePasteMode(value = "") {

Item	Explanation
Rich text at line 374	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: function normalizeCodeText(value = "", pasteMode = "source") {
Rich text at line 376	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: if (normalizeCodePasteMode(pasteMode) === "source") return normalized.replace(/t/g, " ").replace(/s+\$g, "");
Compiler or simulator at line 400	Part of the Compile Code workspace or language/tool detection. Evidence: if (/b(always_ff always_comb always_latch logic interface covergroup assert s+property typedef s+enum class s+w+)\b/.test(code)) return "systemverilog";
Compiler or simulator at line 401	Part of the Compile Code workspace or language/tool detection. Evidence: if (/b(module endmodule always assign reg wire initial posedge negedge)\b/.test(code)) return "verilog";
Cloudflare or AI at line 402	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: if (/^s*\.(tran ac dc op model subckt ends param)\b/im.test(code) /bVw+ s+ w+ s+ w+ s+(?DC SIN PULSE)?/i.test(code)) return "Itspice";
Compiler or simulator at line 403	Part of the Compile Code workspace or language/tool detection. Evidence: if (/b(def elif import s+w+ from s+w+ s+import self None True False)\b/.test(code)) return "python";
Compiler or simulator at line 404	Part of the Compile Code workspace or language/tool detection. Evidence: if (/b(public s+class private s+ protected s+ static s+void s+main System\.out)\b/.test(code)) return "java";
Security or auth at line 412	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: function tokenizedCodeHtml(code = "", language = "javascript") {
Security or auth at line 414	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: const tokens = [];
Security or auth at line 417	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: const token = `@@CODE_TOKEN_\${tokens.length}@`;
Security or auth at line 418	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: tokens.push(`<span class="\${className}">\${match}</span>`);
Security or auth at line 419	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: return token;
Security or auth at line 424	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: protect(/<!--[\s\S]*?-->/g, "code-token-comment");
Security or auth at line 425	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: protect(/(<V?)([a-zA-Z0-9-:~+])([\\s\S]?)(V?>)/g, "code-token-tag");
Compiler or simulator at line 427	Part of the Compile Code workspace or language/tool detection. Evidence: const commentPattern = ["python", "Itspice"].includes(normalizeCodeLanguage(language))
Security or auth at line 430	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: protect(commentPattern, "code-token-comment");

Important constants and variables

Item	Explanation
grid	Line 1: const grid = document.querySelector("#project-grid");
searchInput	Line 2: const searchInput = document.querySelector("#project-search");
projectFilters	Line 3: const projectFilters = document.querySelector("#project-filters");
filterButtons	Line 4: let filterButtons = [...document.querySelectorAll(".filter-button")];
projectCount	Line 5: const projectCount = document.querySelector("#project-count");
projectTrackCount	Line 6: const projectTrackCount = document.querySelector("#project-track-count");
projectTrackLabels	Line 7: const projectTrackLabels = document.querySelector("#project-track-labels");
year	Line 8: const year = document.querySelector("#year");
funFactsCallout	Line 9: const funFactsCallout = document.querySelector("#fun-facts-callout");
heroEyebrow	Line 10: const heroEyebrow = document.querySelector(".hero-content .eyebrow");
heroTitle	Line 11: const heroTitle = document.querySelector("#hero-title");
heroCopy	Line 12: const heroCopy = document.querySelector(".hero-copy");
brandName	Line 13: const brandName = document.querySelector(".brand strong");
brandSubtitle	Line 14: const brandSubtitle = document.querySelector(".brand small");

Item	Explanation
brandIcon	Line 15: const brandIcon = document.querySelector(".brand-icon");
brandText	Line 16: const brandText = document.querySelector(".omb-engraving");
headerAvatar	Line 17: const headerAvatar = document.querySelector(".header-avatar");
headerAvatarImage	Line 18: const headerAvatarImage = document.querySelector(".header-avatar img");
heroImage	Line 19: const heroImage = document.querySelector(".hero-image");
resumeSection	Line 20: const resumeSection = document.querySelector("#resume");
contactBand	Line 21: const contactBand = document.querySelector("#contact");
profilePhoto	Line 22: const profilePhoto = document.querySelector(".profile-photo");
contactTitle	Line 23: const contactTitle = document.querySelector("#contact-title");
contactIntro	Line 24: const contactIntro = document.querySelector(".contact-intro");
contactDetails	Line 25: const contactDetails = document.querySelector(".contact-details");
contactLinks	Line 26: const contactLinks = document.querySelector(".contact-links");
footerOwner	Line 27: const footerOwner = document.querySelector(".site-footer span");
searchWrap	Line 28: const searchWrap = searchInput?.closest(".search-wrap");
aiAssistantForm	Line 29: const aiAssistantForm = document.querySelector("#ai-assistant-form");
aiAssistantPanel	Line 30: const aiAssistantPanel = document.querySelector("#ask-ai");
aiAssistantInput	Line 31: const aiAssistantInput = document.querySelector("#ai-assistant-input");
aiAssistantLog	Line 32: const aiAssistantLog = document.querySelector("#ai-assistant-log");
aiAssistantStatus	Line 33: const aiAssistantStatus = document.querySelector("#ai-assistant-status");
aiClearChatButton	Line 34: const aiClearChatButton = document.querySelector("#ai-clear-chat");
categories	Line 36: let categories = [];
projects	Line 37: let projects = [];
siteSections	Line 38: let siteSections = [];
funFacts	Line 39: let funFacts = [];
funFactsRich	Line 40: let funFactsRich = null;
fieldStyles	Line 41: let fieldStyles = {};
siteContent	Line 42: let siteContent = null;
siteContentRich	Line 43: let siteContentRich = {};
profile	Line 44: let profile = null;
profileRich	Line 45: let profileRich = {};
activeFilter	Line 46: let activeFilter = "all";
activeSectionDialogDrag	Line 47: let activeSectionDialogDrag = null;
activeSectionDialogResize	Line 48: let activeSectionDialogResize = null;
sectionDialogDragEnabled	Line 49: let sectionDialogDragEnabled = false;
isApplyingSectionRoute	Line 50: let isApplyingSectionRoute = false;
searchPanel	Line 51: let searchPanel = null;
searchStatus	Line 52: let searchStatus = null;
searchLimitSelect	Line 53: let searchLimitSelect = null;
currentSearchResults	Line 54: let currentSearchResults = [];
assistantSourceCounter	Line 55: let assistantSourceCounter = 0;
assistantSourceMap	Line 56: let assistantSourceMap = new Map();

Item	Explanation
assistantChatHistory	Line 57: let assistantChatHistory = [];
searchableEntries	Line 58: let searchableEntries = [];
searchResultLimit	Line 59: let searchResultLimit = 10;
searchHighlightTimer	Line 60: let searchHighlightTimer = 0;
searchHighlightName	Line 61: const searchHighlightName = "portfolio-search-highlight";
sectionRouteKey	Line 63: const sectionRouteKey = "portfolio-section";
sectionRouteParams	Line 64: const sectionRouteParams = {
supportedCodeLanguages	Line 70: const supportedCodeLanguages = [
legacyTemplateSkins	Line 82: const legacyTemplateSkins = {
defaultSiteContent	Line 108: const defaultSiteContent = {
defaultProfile	Line 114: const defaultProfile = {
items	Line 137: const items = Array.isArray(value) ? value : String(value "").split(/\r?\n/);
facts	Line 146: const facts = normalizeFunFacts(funFacts);
hasRichFacts	Line 147: const hasRichFacts = Boolean(funFactsRich?.blocks?.length);
rect	Line 162: const rect = dialog.getBoundingClientRect();
margin	Line 163: const margin = 12;
maxLeft	Line 164: const maxLeft = Math.max(margin, window.innerWidth - rect.width - margin);
maxTop	Line 165: const maxTop = Math.max(margin, window.innerHeight - rect.height - margin);
rect	Line 173: const rect = dialog.getBoundingClientRect();
position	Line 174: const position = clampSectionDialogPosition(rect.left, rect.top, dialog);
handle	Line 186: const handle = document.createElement("div");
rect	Line 196: const rect = dialog.getBoundingClientRect();
margin	Line 204: const margin = 12;
minWidth	Line 205: const minWidth = Math.min(320, Math.max(180, window.innerWidth - margin * 2));
minHeight	Line 206: const minHeight = Math.min(170, Math.max(120, window.innerHeight - margin * 2));

JSON/YAML-style keys detected

Item	Explanation
skin-light-blue	Line 83: "skin-light-blue": "appearance-light-blue-red-click",
skin-clean-white	Line 84: "skin-clean-white": "appearance-white-blue-click",
skin-deep-navy	Line 85: "skin-deep-navy": "appearance-deep-navy-cyan-click",
skin-red-warm	Line 86: "skin-red-warm": "appearance-warm-red-pale-click",
skin-emerald-instrument	Line 87: "skin-emerald-instrument": "appearance-emerald-mint-click",
skin-amber-power	Line 88: "skin-amber-power": "appearance-amber-cream-click",
skin-violet-mixed	Line 89: "skin-violet-mixed": "appearance-violet-lilac-click",
skin-graphite-asic	Line 90: "skin-graphite-asic": "appearance-graphite-white-click",
analog-opamp-topology	Line 91: "analog-opamp-topology": "appearance-light-blue-red-click",
analog-power-charger	Line 92: "analog-power-charger": "appearance-amber-cream-click",
analog-filter-front-end	Line 93: "analog-filter-front-end": "appearance-light-blue-red-click",
analog-sensor-interface	Line 94: "analog-sensor-interface": "appearance-emerald-mint-click",
analog-mixed-signal-timing	Line 95: "analog-mixed-signal-timing": "appearance-violet-lilac-click",

Item	Explanation
digital-fpga-pipeline	Line 96: "digital-fpga-pipeline": "appearance-deep-navy-cyan-click",
digital-asic-block	Line 97: "digital-asic-block": "appearance-graphite-white-click",
digital-verification-suite	Line 98: "digital-verification-suite": "appearance-graphite-white-click",
digital-interface-controller	Line 99: "digital-interface-controller": "appearance-deep-navy-cyan-click",
digital-signal-processing	Line 100: "digital-signal-processing": "appearance-violet-lilac-click",
embedded-stm32-sensor	Line 101: "embedded-stm32-sensor": "appearance-emerald-mint-click",
embedded-rtos-control	Line 102: "embedded-rtos-control": "appearance-graphite-white-click",
embedded-power-monitor	Line 103: "embedded-power-monitor": "appearance-amber-cream-click",
embedded-wireless-node	Line 104: "embedded-wireless-node": "appearance-deep-navy-cyan-click",
embedded-motor-control	Line 105: "embedded-motor-control": "appearance-violet-lilac-click"
Brief	Line 792: "Brief": "Overview",
Project Brief	Line 793: "Project Brief": "Overview",
Design Brief	Line 794: "Design Brief": "Design Overview",
Simulation Brief	Line 795: "Simulation Brief": "Simulation Overview"
--responsive-section-card-min	Line 1029: "--responsive-section-card-min": profile.sectionCardMin,
--responsive-content-max	Line 1030: "--responsive-content-max": profile.contentMax,
--responsive-media-max	Line 1031: "--responsive-media-max": profile.mediaMax,
--responsive-touch-target	Line 1032: "--responsive-touch-target": profile.touchTarget
--accent	Line 1047: "--accent": accent,
--template-bg	Line 1048: "--template-bg": visual.background,
--template-panel	Line 1049: "--template-panel": visual.panel,
--template-accent	Line 1050: "--template-accent": visual.accent,
--template-hover	Line 1051: "--template-hover": visual.hover,
--template-click	Line 1052: "--template-click": visual.click,
--template-click-text	Line 1053: "--template-click-text": visual.clickText,
--template-line	Line 1054: "--template-line": visual.line,
--template-text	Line 1055: "--template-text": visual.text,
--accent	Line 1080: "--accent": accent,
--template-bg	Line 1081: "--template-bg": visual.background,
--template-panel	Line 1082: "--template-panel": visual.panel,
--template-accent	Line 1083: "--template-accent": visual.accent,
--template-hover	Line 1084: "--template-hover": visual.hover,
--template-click	Line 1085: "--template-click": visual.click,
--template-click-text	Line 1086: "--template-click-text": visual.clickText,
--template-line	Line 1087: "--template-line": visual.line,
--template-text	Line 1088: "--template-text": visual.text,

Event handlers and user interactions

Item	Explanation
Line 331	document.addEventListener("pointerdown", beginSectionDialogResize);
Line 332	document.addEventListener("pointerdown", beginSectionDialogDrag);

Item	Explanation
Line 333	<code>document.addEventListener("pointermove", moveSectionDialogResize);</code>
Line 334	<code>document.addEventListener("pointermove", moveSectionDialogDrag);</code>
Line 335	<code>document.addEventListener("pointerup", endSectionDialogResize);</code>
Line 336	<code>document.addEventListener("pointerup", endSectionDialogDrag);</code>
Line 337	<code>document.addEventListener("pointercancel", endSectionDialogResize);</code>
Line 338	<code>document.addEventListener("pointercancel", endSectionDialogDrag);</code>
Line 339	<code>window.addEventListener("resize", () => {</code>
Line 1214	<code>button.addEventListener("click", () => {</code>
Line 3272	<code>searchLimitSelect.addEventListener("change", () => {</code>
Line 3276	<code>searchPanel.addEventListener("mousedown", (event) => event.preventDefault());</code>
Line 3277	<code>searchPanel.addEventListener("click", (event) => {</code>
Line 3434	<code>dialog.querySelector(".section-view-close").addEventListener("click", () => closeSectionDialog(dialog));</code>
Line 3435	<code>dialog.addEventListener("cancel", () => clearSectionRouteInHistory());</code>
Line 3436	<code>dialog.addEventListener("close", () => {</code>
Line 3445	<code>dialog.querySelector(".section-view-back").addEventListener("click", () => {</code>
Line 3457	<code>dialog.querySelector(".section-view-forward").addEventListener("click", () => {</code>
Line 3464	<code>dialog.querySelector("#section-view-content").addEventListener("click", (event) => {</code>
Line 3562	<code>window.addEventListener("popstate", (event) => {</code>
Line 3567	<code>searchInput.addEventListener("input", handleSearchInput);</code>
Line 3568	<code>searchInput.addEventListener("focus", showSearchPanelIfNeeded);</code>
Line 3569	<code>searchInput.addEventListener("keydown", (event) => {</code>
Line 3580	<code>document.addEventListener("click", (event) => {</code>
Line 3586	<code>document.addEventListener("click", (event) => {</code>
Line 3599	<code>aiAssistantForm?.addEventListener("submit", (event) => {</code>
Line 3606	<code>aiClearChatButton?.addEventListener("click", clearAssistantChat);</code>
Line 3607	<code>window.addEventListener("resize", updateAssistantPanelGrowth);</code>
Line 3609	<code>aiAssistantLog?.addEventListener("click", (event) => {</code>
Line 3616	<code>grid.addEventListener("click", (event) => {</code>

Function-by-function detail

normalize - lines 132-134

Item	Explanation
Source location	Lines 132-134 (3 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	<code>function normalize(value) {</code>
Touches	<code>filesystem</code>
Calls detected	<code>toLowerCase</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

132: function normalize(value) {
133:   return String(value || "").toLowerCase();
134: }

```

normalizeFunFacts - lines 136-142

Item	Explanation
Source location	Lines 136-142 (7 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeFunFacts(value) {
Touches	filesystem
Calls detected	isArray, split, map, replace, trim, filter, slice
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

136: function normalizeFunFacts(value) {
137:   const items = Array.isArray(value) ? value : String(value || "").split(/\r?\n/);
138:   return items
139:     .map((item) => String(item || "").replace(/\s+/g, " ").trim())
140:     .filter(Boolean)
141:     .slice(0, 10);
142: }

```

renderFunFacts - lines 144-159

Item	Explanation
Source location	Lines 144-159 (16 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderFunFacts() {
Touches	filesystem
Calls detected	normalizeFunFacts, renderRichContent, slice, join, map, renderInlineMath
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

144: function renderFunFacts() {
145:   if (!funFactsCallout) return;
146:   const facts = normalizeFunFacts(funFacts);
147:   const hasRichFacts = Boolean(funFactsRich?.blocks?.length);
148:   funFactsCallout.hidden = !facts.length && !hasRichFacts;
149:   funFactsCallout.innerHTML = facts.length || hasRichFacts ? `
150:     <div class="fun-facts-window">
151:       <span class="fun-facts-label">Fun fact:</span>
152:       <div class="fun-facts-lines">
153:         ${hasRichFacts
154:           ? renderRichContent({ blocks: funFactsRich.blocks.slice(0, 3) }, facts.join("\n"))
155:           : facts.map((fact) => `<p class="fun-fact-line">${renderInlineMath(fact)}</p>`).join("")}
156:       </div>

```

```

157:     </div>
158:     :   "";
159: }

```

clampSectionDialogPosition - lines 161-170

Item	Explanation
Source location	Lines 161-170 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function clampSectionDialogPosition(left, top, dialog) {
Touches	Mostly local calculations or local UI state.
Calls detected	getBoundingClientRect, max, min
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

161: function clampSectionDialogPosition(left, top, dialog) {
162:   const rect = dialog.getBoundingClientRect();
163:   const margin = 12;
164:   const maxLeft = Math.max(margin, window.innerWidth - rect.width - margin);
165:   const maxTop = Math.max(margin, window.innerHeight - rect.height - margin);
166:   return {
167:     left: Math.min(Math.max(left, margin), maxLeft),
168:     top: Math.min(Math.max(top, margin), maxTop)
169:   };
170: }

```

anchorSectionDialog - lines 172-181

Item	Explanation
Source location	Lines 172-181 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function anchorSectionDialog(dialog) {
Touches	browser/Electron DOM
Calls detected	getBoundingClientRect, clampSectionDialogPosition, add
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

172: function anchorSectionDialog(dialog) {
173:   const rect = dialog.getBoundingClientRect();
174:   const position = clampSectionDialogPosition(rect.left, rect.top, dialog);
175:   dialog.classList.add("is-draggable-dialog");
176:   dialog.style.left = `${position.left}px`;
177:   dialog.style.top = `${position.top}px`;
178:   dialog.style.right = "auto";
179:   dialog.style.bottom = "auto";
180:   dialog.style.margin = "0";
181: }

```

ensureSectionResizeHandles - lines 183-192

Item	Explanation
Source location	Lines 183-192 (10 source lines).

Item	Explanation
Purpose	Part of the public website runtime.
Declaration	function ensureSectionResizeHandles(dialog) {
Touches	browser/Electron DOM
Calls detected	querySelector, forEach, createElement, setAttribute, append
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

183: function ensureSectionResizeHandles(dialog) {
184:   if (dialog.querySelector("[data-section-dialog-resize-handle]")) return;
185:   ["n", "e", "s", "w", "ne", "nw", "se", "sw"].forEach((direction) => {
186:     const handle = document.createElement("div");
187:     handle.className = `dialog-resize-handle resize-${direction}`;
188:     handle.dataset.sectionDialogResizeHandle = direction;
189:     handle.setAttribute("aria-hidden", "true");
190:     dialog.append(handle);
191:   });
192: }

```

anchorSectionDialogForResize - lines 194-201

Item	Explanation
Source location	Lines 194-201 (8 source lines).
Purpose	Part of the public website runtime.
Declaration	function anchorSectionDialogForResize(dialog) {
Touches	browser/Electron DOM
Calls detected	anchorSectionDialog, getBoundingClientRect, add
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

194: function anchorSectionDialogForResize(dialog) {
195:   anchorSectionDialog(dialog);
196:   const rect = dialog.getBoundingClientRect();
197:   dialog.classList.add("is-resized-dialog");
198:   dialog.style.width = `${rect.width}px`;
199:   dialog.style.height = `${rect.height}px`;
200:   dialog.style.maxHeight = "none";
201: }

```

sectionDialogResizeBounds - lines 203-230

Item	Explanation
Source location	Lines 203-230 (28 source lines).
Purpose	Part of the public website runtime.
Declaration	function sectionDialogResizeBounds(state, event) {
Touches	Mostly local calculations or local UI state.
Calls detected	min, max, includes
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

203: function sectionDialogResizeBounds(state, event) {
204:   const margin = 12;
205:   const minWidth = Math.min(320, Math.max(180, window.innerWidth - margin * 2));
206:   const minHeight = Math.min(170, Math.max(120, window.innerHeight - margin * 2));
207:   const direction = state.direction;
208:   let { left, top, width, height } = state.rect;
209:   const dx = event.clientX - state.startX;
210:   const dy = event.clientY - state.startY;
211:
212:   if (direction.includes("e")) {
213:     width = Math.min(Math.max(state.rect.width + dx, minWidth), window.innerWidth - left - margin);
214:   }
215:   if (direction.includes("s")) {
216:     height = Math.min(Math.max(state.rect.height + dy, minHeight), window.innerHeight - top - margin);
217:   }
218:   if (direction.includes("w")) {
219:     const right = state.rect.left + state.rect.width;
220:     left = Math.min(Math.max(state.rect.left + dx, margin), right - minWidth);
221:     width = right - left;
222:   }
223:   if (direction.includes("n")) {
224:     const bottom = state.rect.top + state.rect.height;
225:     top = Math.min(Math.max(state.rect.top + dy, margin), bottom - minHeight);
226:     height = bottom - top;
227:   }
228:
229:   return { height, left, top, width };
230: }

```

sectionDialogFromDragEvent - lines 232-239

Item	Explanation
Source location	Lines 232-239 (8 source lines).
Purpose	Part of the public website runtime.
Declaration	function sectionDialogFromDragEvent(event) {
Touches	Mostly local calculations or local UI state.
Calls detected	closest
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

232: function sectionDialogFromDragEvent(event) {
233:   if (event.target.closest("[data-section-dialog-resize-handle]")) return null;
234:   const handle = event.target.closest("[data-section-dialog-drag='true']");
235:   if (!handle) return null;
236:   if (event.target.closest("button, a, input, textarea, select, label, summary")) return null;
237:   const dialog = handle.closest("dialog");
238:   return dialog?.open ? dialog : null;
239: }

```

beginSectionDialogDrag - lines 241-257

Item	Explanation
Source location	Lines 241-257 (17 source lines).
Purpose	Part of the public website runtime.
Declaration	function beginSectionDialogDrag(event) {

Item	Explanation
Touches	browser/Electron DOM
Calls detected	sectionDialogFromDragEvent, anchorSectionDialog, getBoundingClientRect, add, preventDefault
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

241: function beginSectionDialogDrag(event) {
242:   if (event.button !== 0) return;
243:   const dialog = sectionDialogFromDragEvent(event);
244:   if (!dialog) return;
245:   anchorSectionDialog(dialog);
246:   const rect = dialog.getBoundingClientRect();
247:   activeSectionDialogDrag = {
248:     dialog,
249:     offsetX: event.clientX - rect.left,
250:     offsetY: event.clientY - rect.top,
251:     pointerId: event.pointerId,
252:     pointerTarget: event.target
253:   };
254:   event.target.setPointerCapture?.(event.pointerId);
255:   dialog.classList.add("is-dragging-dialog");
256:   event.preventDefault();
257: }

```

moveSectionDialogDrag - lines 259-268

Item	Explanation
Source location	Lines 259-268 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function moveSectionDialogDrag(event) {
Touches	Mostly local calculations or local UI state.
Calls detected	clampSectionDialogPosition
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

259: function moveSectionDialogDrag(event) {
260:   if (!activeSectionDialogDrag) return;
261:   const next = clampSectionDialogPosition(
262:     event.clientX - activeSectionDialogDrag.offsetX,
263:     event.clientY - activeSectionDialogDrag.offsetY,
264:     activeSectionDialogDrag.dialog
265:   );
266:   activeSectionDialogDrag.dialog.style.left = `${next.left}px`;
267:   activeSectionDialogDrag.dialog.style.top = `${next.top}px`;
268: }

```

endSectionDialogDrag - lines 270-275

Item	Explanation
Source location	Lines 270-275 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function endSectionDialogDrag() {
Touches	browser/Electron DOM

Item	Explanation
Calls detected	remove
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

270: function endSectionDialogDrag() {
271:   if (!activeSectionDialogDrag) return;
272:   activeSectionDialogDrag.pointerTarget?.releasePointerCapture?.(activeSectionDialogDrag.pointerId);
273:   activeSectionDialogDrag.dialog.classList.remove("is-dragging-dialog");
274:   activeSectionDialogDrag = null;
275: }

```

beginSectionDialogResize - lines 277-295

Item	Explanation
Source location	Lines 277-295 (19 source lines).
Purpose	Part of the public website runtime.
Declaration	function beginSectionDialogResize(event) {
Touches	browser/Electron DOM
Calls detected	closest, contains, anchorSectionDialogForResize, getBoundingClientRect, add, preventDefault
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

277: function beginSectionDialogResize(event) {
278:   if (event.button !== 0) return;
279:   const handle = event.target.closest("[data-section-dialog-resize-handle]");
280:   const dialog = handle?.closest("dialog");
281:   if (!handle || !dialog?.open || dialog.classList.contains("is-minimized-dialog")) return;
282:   anchorSectionDialogForResize(dialog);
283:   activeSectionDialogResize = {
284:     dialog,
285:     direction: handle.dataset.sectionDialogResizeHandle,
286:     pointerId: event.pointerId,
287:     pointerTarget: event.target,
288:     rect: dialog.getBoundingClientRect(),
289:     startX: event.clientX,
290:     startY: event.clientY
291:   };
292:   event.target.setPointerCapture?.(event.pointerId);
293:   dialog.classList.add("is-resizing-dialog");
294:   event.preventDefault();
295: }

```

moveSectionDialogResize - lines 297-304

Item	Explanation
Source location	Lines 297-304 (8 source lines).
Purpose	Part of the public website runtime.
Declaration	function moveSectionDialogResize(event) {
Touches	Mostly local calculations or local UI state.
Calls detected	sectionDialogResizeBounds
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

297: function moveSectionDialogResize(event) {
298:   if (!activeSectionDialogResize) return;
299:   const next = sectionDialogResizeBounds(activeSectionDialogResize, event);
300:   activeSectionDialogResize.dialog.style.left = `${next.left}px`;
301:   activeSectionDialogResize.dialog.style.top = `${next.top}px`;
302:   activeSectionDialogResize.dialog.style.width = `${next.width}px`;
303:   activeSectionDialogResize.dialog.style.height = `${next.height}px`;
304: }

```

endSectionDialogResize - lines 306-311

Item	Explanation
Source location	Lines 306-311 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function endSectionDialogResize() {
Touches	browser/Electron DOM
Calls detected	remove
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

306: function endSectionDialogResize() {
307:   if (!activeSectionDialogResize) return;
308:   activeSectionDialogResize.pointerTarget?.releasePointerCapture?.(activeSectionDialogResize.pointerId);
309:   activeSectionDialogResize.dialog.classList.remove("is-resizing-dialog");
310:   activeSectionDialogResize = null;
311: }

```

updateSectionDialogMinimize - lines 313-320

Item	Explanation
Source location	Lines 313-320 (8 source lines).
Purpose	Updates UI, state, cache, or stored data.
Declaration	function updateSectionDialogMinimize(dialog) {
Touches	browser/Electron DOM
Calls detected	contains, querySelector, setAttribute
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

313: function updateSectionDialogMinimize(dialog) {
314:   const minimized = dialog.classList.contains("is-minimized-dialog");
315:   const button = dialog.querySelector(".section-view-minimize");
316:   if (!button) return;
317:   button.textContent = minimized ? "+" : "-";

```

```

318:   button.title = minimized ? "Restore window" : "Minimize window";
319:   button.setAttribute("aria-label", minimized ? "Restore window" : "Minimize window");
320: }

```

toggleSectionDialogMinimized - lines 322-326

Item	Explanation
Source location	Lines 322-326 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function toggleSectionDialogMinimized(dialog) {
Touches	browser/Electron DOM
Calls detected	anchorSectionDialog, toggle, updateSectionDialogMinimize
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

322: function toggleSectionDialogMinimized(dialog) {
323:   anchorSectionDialog(dialog);
324:   dialog.classList.toggle("is-minimized-dialog");
325:   updateSectionDialogMinimize(dialog);
326: }

```

enableSectionDialogDrag - lines 328-347

Item	Explanation
Source location	Lines 328-347 (20 source lines).
Purpose	Part of the public website runtime.
Declaration	function enableSectionDialogDrag() {
Touches	browser/Electron DOM, filesystem
Calls detected	addEventListener, querySelectorAll, forEach, getBoundingClientRect, clampSectionDialogPosition
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

328: function enableSectionDialogDrag() {
329:   if (sectionDialogDragEnabled) return;
330:   sectionDialogDragEnabled = true;
331:   document.addEventListener("pointerdown", beginSectionDialogResize);
332:   document.addEventListener("pointerdown", beginSectionDialogDrag);
333:   document.addEventListener("pointermove", moveSectionDialogResize);
334:   document.addEventListener("pointermove", moveSectionDialogDrag);
335:   document.addEventListener("pointerup", endSectionDialogResize);
336:   document.addEventListener("pointerup", endSectionDialogDrag);
337:   document.addEventListener("pointercancel", endSectionDialogResize);
338:   document.addEventListener("pointercancel", endSectionDialogDrag);
339:   window.addEventListener("resize", () => {
340:     document.querySelectorAll(".section-view-dialog.is-draggable-dialog[open]").forEach((dialog) => {
341:       const rect = dialog.getBoundingClientRect();
342:       const next = clampSectionDialogPosition(rect.left, rect.top, dialog);
343:       dialog.style.left = `${next.left}px`;
344:       dialog.style.top = `${next.top}px`;
345:     });
346:   });
347: }

```

escapeHtml - lines 349-356

Item	Explanation
Source location	Lines 349-356 (8 source lines).
Purpose	Part of the public website runtime.
Declaration	function escapeHtml(value) {
Touches	Mostly local calculations or local UI state.
Calls detected	replaceAll
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

349: function escapeHtml(value) {
350:   return String(value || "")
351:     .replaceAll("&", "&amp;")
352:     .replaceAll("<", "&lt;")
353:     .replaceAll(">", "&gt;")
354:     .replaceAll("'", "&quot;")
355:     .replaceAll('"', "&#039;");
356: }

```

normalizeCodeLanguage - lines 358-364

Item	Explanation
Source location	Lines 358-364 (7 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeCodeLanguage(value = "") {
Touches	filesystem
Calls detected	trim, toLowerCase, replace, find, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

358: function normalizeCodeLanguage(value = "") {
359:   const clean = String(value || "").trim().toLowerCase().replace(/[_-]+/g, " ");
360:   const match = supportedCodeLanguages.find((language) =>
361:     language.id === clean || language.aliases.includes(clean)
362:   );
363:   return match?.id || "javascript";
364: }

```

codeLanguageLabel - lines 366-368

Item	Explanation
Source location	Lines 366-368 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function codeLanguageLabel(value = "") {
Touches	filesystem
Calls detected	find, normalizeCodeLanguage
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

366: function codeLanguageLabel(value = "") {
367:   return supportedCodeLanguages.find((language) => language.id === normalizeCodeLanguage(value))?.label
|| "Code";
368: }

```

normalizeCodePasteMode - lines 370-372

Item	Explanation
Source location	Lines 370-372 (3 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeCodePasteMode(value = "") {
Touches	filesystem
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

370: function normalizeCodePasteMode(value = "") {
371:   return value === "source" ? "source" : "basic";
372: }

```

normalizeCodeText - lines 374-383

Item	Explanation
Source location	Lines 374-383 (10 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeCodeText(value = "", pasteMode = "source") {
Touches	filesystem
Calls detected	replace, normalizeCodePasteMode, split, map, trimEnd, join, trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

374: function normalizeCodeText(value = "", pasteMode = "source") {
375:   const normalized = String(value || "").replace(/\r\n?/g, "\n").replace(/\u00a0/g, " ");
376:   if (normalizeCodePasteMode(pasteMode) === "source") return normalized.replace(/\t/g, "
").replace(/\s+$/g, "");
377:   return normalized
378:     .split("\n")
379:     .map((line) => line.trimEnd())
380:     .join("\n")
381:     .replace(/\n{4,}/g, "\n\n\n")
382:     .trim();
383: }

```

sourceLooksCpp - lines 385-389

Item	Explanation
Source location	Lines 385-389 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function sourceLooksCpp(source = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
385: function sourceLooksCpp(source = "") {
386:   const text = String(source || "");
387:   return
#include\s*<(?:iostream|string|vector|array|map|unordered_map|memory|algorithm|optional|variant)>/.test(text)
||
388:   /\b(?:std::|cout|cin|cerr|namespace|template\s*<|class\s+\w+|new\s+\w+|delete\s+|constexpr|nullptr|using
\s+namespace)\b/.test(text);
389: }
```

sourceLooksC - lines 391-395

Item	Explanation
Source location	Lines 391-395 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function sourceLooksC(source = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
391: function sourceLooksC(source = "") {
392:   const text = String(source || "");
393:   return #include\s*<(?:stdio|stdlib|string|stdint|stdbool|math)\.h>/.test(text) ||
394:   /\b(?:printf|scanf|malloc|calloc|realloc|free|struct\s+\w+|typedef\s+struct)\b/.test(text);
395: }
```

detectCodeLanguage - lines 397-410

Item	Explanation
Source location	Lines 397-410 (14 source lines).
Purpose	Part of the public website runtime.
Declaration	function detectCodeLanguage(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	b, sourceLooksCpp, sourceLooksC
API endpoints	None detected inside this function body.


```

ster|restrict|return|short|signed|sizeof|static|struct|switch|typedef|union|unsigned|void|volatile|while",
441:   cpp: "alignas|alignof|and|and_eq|asm|auto|bitand|bitor|bool|break|case|catch|char|char8_t|char16_t|ch
ar32_t|class|compl|concept|const|constexpr|constexpr|constinit|const_cast|continue|co_await|co_return|co_yield|d
ecltype|default|delete|do|double|dynamic_cast|else|enum|explicit|export|extern|false|float|for|friend|if|inline|
int|long|mutable|namespace|new|noexcept|not|not_eq|nullptr|operator|or|or_eq|private|protected|public|register|r
einterpret_cast|requires|return|short|signed|sizeof|static|static_assert|static_cast|struct|switch|template|this
|thread_local|throw|true|try|typedef|typeid|typename|union|unsigned|using|virtual|void|volatile|wchar_t|while|x
or|xor_eq",
442:   verilog: "always|and|assign|begin|buf|case|casex|casez|deassign|default|defparam|disable|edge|else|en
d|endcase|endfunction|endmodule|endprimitive|endspecify|endtable|endtask|event|for|force|forever|fork|function|g
enerate|genvar|if|initial|inout|input|integer|join|module|nand|negedge|nor|not|or|output|parameter|posedge|primi
tive|reg|release|repeat|signed|specify|supply0|supply1|table|task|tri|wand|while|wire|wor|xnor|xor",
443:   systemverilog: "always|always_comb|always_ff|always_latch|assign|automatic|begin|bit|case|class|clock
ing|covergroup|default|disable|do|else|end|endcase|endclass|endclocking|endfunction|endmodule|endpackage|endtask
|enum|for|forever|function|generate|genvar|if|initial|input|int|interface|logic|module|negedge|output|package|pa
rameter|posedge|reg|return|signed|task|typedef|wire",
444:   ltspice: "ac|dc|end|ends|four|func|global|ic|include|lib|meas|model|nodeset|op|options|param|plot|pro
be|save|step|subckt|temp|tran",
445:   java: "abstract|assert|boolean|break|byte|case|catch|char|class|const|continue|default|do|double|else
|enum|extends|final|finally|float|for|if|implements|import|instanceof|int|interface|long|native|new|null|package
|private|protected|public|return|short|static|strictfp|super|switch|synchronized|this|throw|throws|transient|try
|void|volatile|while",
446:   javascript: "async|await|break|case|catch|class|const|continue|debugger|default|delete|do|else|export
|extends|false|finally|for|from|function|if|import|in|instanceof|let|new|null|of|return|static|super|switch|this
|throw|true|try|typeof|undefined|var|void|while|yield",
447:   python: "and|as|assert|async|await|break|class|continue|def|del|elif|else|except|False|finally|for|fr
om|global|if|import|in|is|lambda|None|nonlocal|not|or|pass|raise|return|True|try|while|with|yield",
448:   html: "html|head|body|main|section|article|div|span|script|style|link|meta|title|header|footer|nav|bu
tton|input|form|label|img|a|p|pre|code"
449:   };
450:   const keywords = keywordMap[normalizeCodeLanguage(language)] || keywordMap.javascript;
451:   html = html.replace(new RegExp(`\\b(${keywords})\\b`, "g"), `

```

protect - lines 415-421

Item	Explanation
Source location	Lines 415-421 (7 source lines).
Purpose	Part of the public website runtime.
Declaration	const protect = (pattern, className) => {
Touches	authentication/security data
Calls detected	replace, push
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

415:   const protect = (pattern, className) => {
416:     html = html.replace(pattern, (match) => {
417:       const token = `@@CODE_TOKEN_${tokens.length}@@`;
418:       tokens.push(`

```

renderRichCodeBlock - lines 456-465

Item	Explanation
Source location	Lines 456-465 (10 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderRichCodeBlock(block = {}) {
Touches	filesystem, authentication/security data
Calls detected	normalizeCodeLanguage, detectCodeLanguage, normalizeCodeText, escapeHtml, codeLanguageLabel, tokenizedCodeHtml

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

456: function renderRichCodeBlock(block = {}) {
457:   const language = normalizeCodeLanguage(block.language || detectCodeLanguage(block.code || ""));
458:   const code = normalizeCodeText(block.code || "", block.pasteMode || "source");
459:   return `
460:     <figure class="rich-code-block rich-code-language-${language}">
461:       <figcaption><span>&lt;&gt;</span> ${escapeHtml(codeLanguageLabel(language))}</figcaption>
462:       <pre><code>${tokenizedCodeHtml(code, language)}</code></pre>
463:     </figure>
464:   `;
465: }

```

renderPlainMultiline - lines 467-469

Item	Explanation
Source location	Lines 467-469 (3 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderPlainMultiline(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	escapeHtml, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

467: function renderPlainMultiline(value = "") {
468:   return escapeHtml(value).replace(/\r\n?|\n/g, "<br>");
469: }

```

normalizeSiteContent - lines 471-477

Item	Explanation
Source location	Lines 471-477 (7 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeSiteContent(content = {}) {
Touches	filesystem
Calls detected	trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

471: function normalizeSiteContent(content = {}) {
472:   return {
473:     heroCopy: String(content.heroCopy || "").trim() || defaultSiteContent.heroCopy,
474:     heroEyebrow: String(content.heroEyebrow || "").trim() || defaultSiteContent.heroEyebrow,

```

```

475:     heroTitle: String(content.heroTitle || "").trim() || defaultSiteContent.heroTitle
476:   };
477: }

```

normalizeProfile - lines 479-495

Item	Explanation
Source location	Lines 479-495 (17 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeProfile(profileValue = {}) {
Touches	filesystem
Calls detected	trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

479: function normalizeProfile(profileValue = {}) {
480:   return {
481:     brandImage: String(profileValue.brandImage || "").trim(),
482:     brandText: String(profileValue.brandText || "").trim() || String(profileValue.displayName ||
483:   "").trim() || defaultProfile.brandText,
484:     contactIntro: String(profileValue.contactIntro || "").trim(),
485:     displayName: String(profileValue.displayName || "").trim(),
486:     email: String(profileValue.email || "").trim(),
487:     githubUrl: String(profileValue.githubUrl || "").trim(),
488:     heroImage: String(profileValue.heroImage || "").trim(),
489:     linkedinUrl: String(profileValue.linkedinUrl || "").trim(),
490:     phone: String(profileValue.phone || "").trim(),
491:     portfolioLabel: String(profileValue.portfolioLabel || "").trim() || defaultProfile.portfolioLabel,
492:     profileImage: String(profileValue.profileImage || "").trim(),
493:     resumeUrl: String(profileValue.resumeUrl || "").trim(),
494:     websiteUrl: String(profileValue.websiteUrl || "").trim()
495:   };
496: }

```

mailComposeLink - lines 497-500

Item	Explanation
Source location	Lines 497-500 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function mailComposeLink(email = "") {
Touches	network/API requests
Calls detected	trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

497: function mailComposeLink(email = "") {
498:   const clean = String(email || "").trim();
499:   return clean ? `https://mail.google.com/mail/?view=cm&fs=1&to=${encodeURIComponent(clean)}` : "";
500: }

```

phoneLink - lines 502-505

Item	Explanation
Source location	Lines 502-505 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function phoneLink(phone = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
502: function phoneLink(phone = "") {
503:   const clean = String(phone || "").trim();
504:   return clean ? `tel:${clean.replace(/^[^d+]/g, "")}` : "";
505: }
```

renderSiteContent - lines 507-515

Item	Explanation
Source location	Lines 507-515 (9 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderSiteContent() {
Touches	filesystem
Calls detected	normalizeSiteContent, renderRichFieldContent, applyPlainFieldStyle
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
507: function renderSiteContent() {
508:   const content = normalizeSiteContent(siteContent || {});
509:   if (heroEyebrow) heroEyebrow.innerHTML = renderRichFieldContent(siteContentRich.heroEyebrow,
content.heroEyebrow);
510:   if (heroTitle) heroTitle.innerHTML = renderRichFieldContent(siteContentRich.heroTitle,
content.heroTitle);
511:   if (heroCopy) heroCopy.innerHTML = renderRichFieldContent(siteContentRich.heroCopy, content.heroCopy);
512:   applyPlainFieldStyle(heroEyebrow, "site-hero-eyebrow");
513:   applyPlainFieldStyle(heroTitle, "site-hero-title");
514:   applyPlainFieldStyle(heroCopy, "site-hero-copy");
515: }
```

renderProfile - lines 517-627

Item	Explanation
Source location	Lines 517-627 (111 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderProfile() {
Touches	browser/Electron DOM, filesystem
Calls detected	normalizeProfile, mailComposeLink, phoneLink, normalizeLinkTarget, filter, renderRichFieldContent, applyPlainFieldStyle, slice, toUpperCase, setAttribute, querySelectorAll, forEach
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

517: function renderProfile() {
518:   const current = normalizeProfile(profile || {});
519:   const displayName = current.displayName || "Portfolio";
520:   const label = current.portfolioLabel || "Portfolio";
521:   const emailLink = mailComposeLink(current.email);
522:   const callLink = phoneLink(current.phone);
523:   const links = [
524:     current.email ? { label: "Email", url: emailLink, external: true } : null,
525:     current.phone ? { label: "Call", url: callLink } : null,
526:     current.githubUrl ? { label: "GitHub", url: normalizeLinkTarget(current.githubUrl, { assumeWeb: true
527:   })), external: true } : null,
528:     current.linkedinUrl ? { label: "LinkedIn", url: normalizeLinkTarget(current.linkedinUrl, { assumeWeb:
529:   true })), external: true } : null,
530:     current.websiteUrl ? { label: "Website", url: normalizeLinkTarget(current.websiteUrl, { assumeWeb:
531:   true })), external: true } : null,
532:     current.resumeUrl ? { label: "Resume", url: current.resumeUrl, external: false } : null
533:   ].filter(Boolean);
534:   document.title = `${displayName} | ${label}`;
535:   if (brandName) brandName.innerHTML = renderRichFieldContent(profileRich.displayName, displayName);
536:   if (brandSubtitle) brandSubtitle.innerHTML = renderRichFieldContent(profileRich.portfolioLabel, label);
537:   applyPlainFieldStyle(brandName, "profile-display-name");
538:   applyPlainFieldStyle(brandSubtitle, "profile-portfolio-label");
539:   if (brandText) brandText.textContent = current.brandText || displayName.slice(0, 3).toUpperCase() ||
"PORT";
540:   if (brandIcon) {
541:     if (current.brandImage) {
542:       brandIcon.src = current.brandImage;
543:       brandIcon.hidden = false;
544:     } else {
545:       brandIcon.hidden = true;
546:     }
547:   }
548:   if (headerAvatar && headerAvatarImage) {
549:     headerAvatar.hidden = !current.profileImage;
550:     if (current.profileImage) {
551:       headerAvatarImage.src = current.profileImage;
552:       headerAvatarImage.alt = displayName;
553:       headerAvatar.setAttribute("aria-label", `Go to ${displayName} contact information`);
554:     }
555:   }
556:   if (heroImage) {
557:     if (current.heroImage) {
558:       heroImage.hidden = false;
559:       heroImage.src = current.heroImage;
560:       heroImage.alt = `${displayName} portfolio background`;
561:     } else {
562:       heroImage.hidden = true;
563:     }
564:   }
565:   document.querySelectorAll('a[href="#resume"]').forEach((link) => {
566:     link.hidden = !current.resumeUrl;
567:   });
568:   const heroGithubLink = [...document.querySelectorAll(".hero-actions a")]
569:     .find((link) => /github/i.test(link.textContent || ""));
570:   if (heroGithubLink) {
571:     heroGithubLink.hidden = !current.githubUrl;
572:     if (current.githubUrl) heroGithubLink.href = normalizeLinkTarget(current.githubUrl, { assumeWeb: true
573:   });
574:   }
575:   if (resumeSection) {
576:     resumeSection.hidden = !current.resumeUrl;
577:     resumeSection.querySelectorAll("a").forEach((link) => {
578:       link.href = current.resumeUrl || "#";
579:     });
580:     const resumeObject = resumeSection.querySelector("object");
581:     if (resumeObject) resumeObject.data = current.resumeUrl || "";
582:     const fallbackLink = resumeSection.querySelector("object a");
583:     if (fallbackLink) fallbackLink.href = current.resumeUrl || "#";
584:   }
585:   if (profilePhoto) {
586:     profilePhoto.hidden = !current.profileImage;
587:     if (current.profileImage) {
588:       profilePhoto.src = current.profileImage;
589:     }
590:   }

```

```

591:     profilePhoto.alt = `Portrait of ${displayName}`;
592:   }
593: }
594: if (contactTitle) contactTitle.innerHTML = renderRichFieldContent(profileRich.displayName,
displayName);
595: applyPlainFieldStyle(contactTitle, "profile-display-name");
596: if (contactIntro) {
597:   contactIntro.innerHTML = renderRichFieldContent(
598:     profileRich.contactIntro,
599:     current.contactIntro || (current.displayName ? `${displayName}'s contact information and public
links.` : "Add contact details in the builder.")
600:   );
601: }
602: applyPlainFieldStyle(contactIntro, "profile-contact-intro");
603: if (contactDetails) {
604:   contactDetails.innerHTML = [
605:     current.email ? `` : "",
606:     current.phone ? `` : ""
607:   \\].filter\\(Boolean\\).join\\(""\\);
608: }
609: if \\(contactLinks\\) {
610:   contactLinks.innerHTML = links.map\\(\\(link\\) => {
611:     const styleId = link.label === "GitHub"
612:       ? "profile-github"
613:       : link.label === "LinkedIn"
614:       ? "profile-linkedin"
615:       : link.label === "Website"
616:       ? "profile-website"
617:       : link.label === "Email"
618:       ? "profile-email"
619:       : link.label === "Call"
620:       ? "profile-phone"
621:       : "";
622:     return `

```

textBlocksFromPlainText - lines 629-637

Item	Explanation
Source location	Lines 629-637 (9 source lines).
Purpose	Part of the public website runtime.
Declaration	function textBlocksFromPlainText(text) {
Touches	filesystem
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

629: function textBlocksFromPlainText(text) {
630:   return [{
631:     align: "left",
632:     fontFamily: "Arial",
633:     fontSize: "normal",
634:     text: String(text || ""),
635:     type: "paragraph"
636:   }];
637: }

```

unwrapFormula - lines 639-652

Item	Explanation
Source location	Lines 639-652 (14 source lines).
Purpose	Part of the public website runtime.
Declaration	function unwrapFormula(value) {
Touches	filesystem
Calls detected	trim, match
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

639: function unwrapFormula(value) {
640:   const text = String(value || "").trim();
641:   const wrappers = [
642:     [/^\$\$([\s\S]+\)\$\$\$/, 1],
643:     [/^\$\$([\s\S]+\)\$\$/, 1],
644:     [/^\$\$([\s\S]+\)\$/, 1],
645:     [/^\$([\^$]+\)\$\$/, 1]
646:   ];
647:   for (const [pattern, group] of wrappers) {
648:     const match = text.match(pattern);
649:     if (match) return match[group].trim();
650:   }
651:   return text;
652: }

```

renderInlineMath - lines 654-663

Item	Explanation
Source location	Lines 654-663 (10 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderInlineMath(text) {
Touches	filesystem
Calls detected	escapeHtml, replace, b, match, slice, normalizeLinkTarget
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

654: function renderInlineMath(text) {
655:   const escaped = escapeHtml(text);
656:   const withMath = escaped.replace(/\$([\^$]+\)\$/g, '<span class="rich-inline-formula">$1</span>');
657:   return withMath.replace(/\b(https?:\/\/[\^s<]+|www\.[\^s<]+|(?:(?:github|linkedin|gitlab|bitbucket|drive|docs)\.com\/[\^s<]+\))/gi, (match) => {
658:     const trailing = match.match(/[.,;:!?]+\$/)?.[0] || "";
659:     const clean = trailing ? match.slice(0, -trailing.length) : match;
660:     const href = normalizeLinkTarget(clean, { assumeWeb: true });
661:     return `<a href="${href}" target="_blank" rel="noreferrer">${clean}</a>${trailing}`;
662:   });
663: }

```

renderMultilineInlineText - lines 665-667

Item	Explanation
Source location	Lines 665-667 (3 source lines).
Purpose	Renders UI, markup, or display output.

Item	Explanation
Declaration	function renderMultilineInlineText(text = "") {
Touches	filesystem
Calls detected	renderInlineMath, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

665: function renderMultilineInlineText(text = "") {
666:   return renderInlineMath(text).replace(/\r\n?|\n/g, "<br>");
667: }

```

looksLikeBareWebAddress - lines 669-676

Item	Explanation
Source location	Lines 669-676 (8 source lines).
Purpose	Part of the public website runtime.
Declaration	function looksLikeBareWebAddress(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, split, toLowerCase
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

669: function looksLikeBareWebAddress(value = "") {
670:   const clean = String(value || "").trim();
671:   if (!clean || /\s/.test(clean) || /^(\.?\.\?\/|#|mailto:|tel:)/i.test(clean)) return false;
672:   if (/^[a-z][a-z0-9+.-]*$/i.test(clean)) return false;
673:   const host = clean.split(/[/?#]/)[0].toLowerCase();
674:   if (!/^[a-z0-9-]+\.(?![a-z0-9-]+)$/.test(host)) return false;
675:   return !/\.(pdf|docx?|xlsx?|pptx?|zip|7z|rar|png|jpe?g|gif|svg|webp|txt|md|csv|json|xml|log|c|h|cpp|hpp|py|js|mjs|ts|v|sv|vhd|l?spice|cir|net|asc|sch|kicad_sch|kicad_pcb)$/i.test(host);
676: }

```

looksLikeWebOrContactText - lines 678-686

Item	Explanation
Source location	Lines 678-686 (9 source lines).
Purpose	Part of the public website runtime.
Declaration	function looksLikeWebOrContactText(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, b, looksLikeBareWebAddress, split, some, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 6 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
678: function looksLikeWebOrContactText(value = "") {
679:   const clean = String(value || "").trim();
680:   if (!clean) return false;
681:   if (/^b(?:https?:\/\/|www\.)[^s<>'"]+/i.test(clean)) return true;
682:   if (/^b[a-z0-9._%+~]@[a-z0-9.-]+\.[a-z]{2,}\b/i.test(clean)) return true;
683:   if (/^b(?:github|linkedin|gitlab|bitbucket|drive|docs)\.com\/[^s<>'"]+/i.test(clean)) return true;
684:   if (looksLikeBareWebAddress(clean)) return true;
685:   return clean.split(/\s+/).some((word) => looksLikeBareWebAddress(word.replace(/^[('"]+|[)]'";:;!]?$/g,
  "")));
686: }
```

normalizeLinkTarget - lines 688-695

Item	Explanation
Source location	Lines 688-695 (8 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeLinkTarget(target = "", options = {}) {
Touches	filesystem, network/API requests
Calls detected	trim, looksLikeBareWebAddress
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 5 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
688: function normalizeLinkTarget(target = "", options = {}) {
689:   const clean = String(target || "").trim();
690:   if (!clean) return "";
691:   if (/^\/\//.test(clean)) return `https:${clean}`;
692:   if (/^www\./i.test(clean)) return `https://${clean}`;
693:   if (options.assumeWeb && looksLikeBareWebAddress(clean)) return `https://${clean}`;
694:   return clean;
695: }
```

isWebsiteLinkItem - lines 697-705

Item	Explanation
Source location	Lines 697-705 (9 source lines).
Purpose	Part of the public website runtime.
Declaration	function isWebsiteLinkItem(item = {}, target = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, filter, join, toLowerCase, b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
697: function isWebsiteLinkItem(item = {}, target = "") {
698:   const clean = String(target || "").trim();
699:   if (/^(https?:)?\/\//i.test(clean) || /^www\./i.test(clean)) return true;
700:   const typeText = [item.kind, item.type, item.status, item.meta, item.label, item.title]
701:     .filter(Boolean)
702:     .join(" ")
703:     .toLowerCase();
704:   return /^b(link|website|web link|url|external|github|linkedin|drive|social|profile)\b/i.test(typeText);
705: }
```

linkifyRichTextNodes - lines 707-742

Item	Explanation
Source location	Lines 707-742 (36 source lines).
Purpose	Part of the public website runtime.
Declaration	function linkifyRichTextNodes(root) {
Touches	browser/Electron DOM, filesystem
Calls detected	createTreeWalker, acceptNode, closest, nextNode, push, forEach, createDocumentFragment, replace, b, append, createTextNode, slice
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
707: function linkifyRichTextNodes(root) {
708:   const walker = document.createTreeWalker(root, NodeFilter.SHOW_TEXT, {
709:     acceptNode(node) {
710:       if (!node.textContent ||
!/(https?:\/\/|www\.|(?:(?:github|linkedin|gitlab|bitbucket|drive|docs)\.com\/)/i.test(node.textContent)) return
NodeFilter.FILTER_REJECT;
711:       return node.parentElement?.closest("a") ? NodeFilter.FILTER_REJECT : NodeFilter.FILTER_ACCEPT;
712:     }
713:   });
714:   const nodes = [];
715:   let node = walker.nextNode();
716:   while (node) {
717:     nodes.push(node);
718:     node = walker.nextNode();
719:   }
720:
721:   nodes.forEach((textNode) => {
722:     const text = textNode.textContent || "";
723:     const fragment = document.createDocumentFragment();
724:     let cursor = 0;
725:     text.replace(/(https?:\/\/|www\.|(?:(?:github|linkedin|gitlab|bitbucket|drive|docs)\.com
\/|\/|s<]+)gi, (match, _url, offset) => {
726:       if (offset > cursor) fragment.append(document.createTextNode(text.slice(cursor, offset)));
727:       const trailing = match.match(/[,.;:!?]+$/)?.[0] || "";
728:       const clean = trailing ? match.slice(0, -trailing.length) : match;
729:       const link = document.createElement("a");
730:       link.href = normalizeLinkTarget(clean, { assumeWeb: true });
731:       link.target = "_blank";
732:       link.rel = "noreferrer";
733:       link.textContent = clean;
734:       fragment.append(link);
735:       if (trailing) fragment.append(document.createTextNode(trailing));
736:       cursor = offset + match.length;
737:       return match;
738:     });
739:     if (cursor < text.length) fragment.append(document.createTextNode(text.slice(cursor)));
740:     textNode.replaceWith(fragment);
741:   });
742: }
```

sanitizeRichInlineHtml - lines 744-787

Item	Explanation
Source location	Lines 744-787 (44 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function sanitizeRichInlineHtml(value = "") {
Touches	browser/Electron DOM, filesystem
Calls detected	createElement, Set, querySelectorAll, forEach, has, replaceWith, getAttribute, trim, normalizeLinkTarget, cleanFontFamily, normalizeFontPx, normalizeTextColor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

744: function sanitizeRichInlineHtml(value = "") {
745:   const template = document.createElement("template");
746:   template.innerHTML = String(value || "");
747:   const allowedTags = new Set(["A", "B", "BR", "EM", "I", "SPAN", "STRONG", "U"]);
748:
749:   [...template.content.querySelectorAll("*")].forEach((node) => {
750:     if (!allowedTags.has(node.tagName)) {
751:       node.replaceWith(...node.childNodes);
752:       return;
753:     }
754:
755:     const href = node.tagName === "A" ? String(node.getAttribute("href") || "").trim() : "";
756:     const normalizedHref = normalizeLinkTarget(href, { assumeWeb: true });
757:     const safeHref = /^(https?:|mailto:|tel:|#|\/)/i.test(normalizedHref) ? normalizedHref : "";
758:     const fontFamily = cleanFontFamily(node.style.fontFamily || "");
759:     const fontPx = normalizeFontPx(parseFloat(node.style.fontSize || ""));
760:     const color = normalizeTextColor(node.style.color || "");
761:     const rawFontWeight = node.style.fontWeight || "";
762:     const fontWeight = /^(bold|[6-9]00)$/i.test(rawFontWeight)
763:       ? "700"
764:       : /^(normal|[1-5]00)$/i.test(rawFontWeight)
765:       ? "400"
766:       : "";
767:     const fontStyle = node.style.fontStyle === "italic" ? "italic" : "";
768:     const textDecoration = node.style.textDecoration.includes("underline") ? "underline" : "";
769:
770:     [...node.attributes].forEach((attribute) => node.removeAttribute(attribute.name));
771:
772:     if (node.tagName === "A" && safeHref) {
773:       node.setAttribute("href", safeHref);
774:       node.setAttribute("target", "_blank");
775:       node.setAttribute("rel", "noreferrer");
776:     }
777:     if (fontFamily) node.style.fontFamily = fontFamily;
778:     if (fontPx) node.style.fontSize = `${fontPx}px`;
779:     if (color) node.style.color = color;
780:     if (fontWeight) node.style.fontWeight = fontWeight;
781:     if (fontStyle) node.style.fontStyle = fontStyle;
782:     if (textDecoration) node.style.textDecoration = textDecoration;
783:   });
784:
785:   linkifyRichTextNodes(template.content);
786:   return template.innerHTML;
787: }

```

displayTitle - lines 789-798

Item	Explanation
Source location	Lines 789-798 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function displayTitle(value, fallback = "Untitled item") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

789: function displayTitle(value, fallback = "Untitled item") {
790:   const title = String(value || fallback).trim();
791:   const replacements = {
792:     "Brief": "Overview",
793:     "Project Brief": "Overview",
794:     "Design Brief": "Design Overview",
795:     "Simulation Brief": "Simulation Overview"

```

```

796:   };
797:   return replacements[title] || title;
798: }

```

cleanFontFamily - lines 800-807

Item	Explanation
Source location	Lines 800-807 (8 source lines).
Purpose	Part of the public website runtime.
Declaration	function cleanFontFamily(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	split, map, replace, trim, filter, slice, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

800: function cleanFontFamily(value = "") {
801:   return String(value)
802:     .split(",")
803:     .map((item) => item.replace(/^[^\w\s-]/g, "").trim())
804:     .filter(Boolean)
805:     .slice(0, 3)
806:     .join(", ");
807: }

```

normalizeFontPx - lines 809-812

Item	Explanation
Source location	Lines 809-812 (4 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeFontPx(value) {
Touches	filesystem
Calls detected	isFinite
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

809: function normalizeFontPx(value) {
810:   const size = Number(value);
811:   return Number.isFinite(size) && size >= 8 && size <= 24 ? String(size) : "";
812: }

```

normalizeTextColor - lines 813-820

Item	Explanation
Source location	Lines 813-820 (8 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeTextColor(value = "") {
Touches	filesystem
Calls detected	trim

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

813: function normalizeTextColor(value = "") {
814:     const color = String(value || "").trim();
815:     return /^(#[0-9a-f]{3}|[0-9a-f]{6})$/i.test(color) ||
816:         /^rgba?\(/i.test(color) ||
817:         /^hsla?\(/i.test(color)
818:         ? color
819:         : "";
820: }

```

rgbColorToHex - lines 822-830

Item	Explanation
Source location	Lines 822-830 (9 source lines).
Purpose	Part of the public website runtime.
Declaration	function rgbColorToHex(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, toLowerCase, match, map, max, min, toString, padStart, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

822: function rgbColorToHex(value = "") {
823:     const color = String(value || "").trim();
824:     if (/^[0-9a-f]{6}$/i.test(color)) return color.toLowerCase();
825:     const shortHex = color.match(/^[0-9a-f]([0-9a-f])([0-9a-f])$/i);
826:     if (shortHex) return
`#${shortHex[1]}${shortHex[1]}${shortHex[2]}${shortHex[2]}${shortHex[3]}${shortHex[3]}`.toLowerCase();
827:     const rgb = color.match(/^rgba?\((\d+),\s*(\d+),\s*(\d+)/i);
828:     if (!rgb) return color;
829:     return `#${[rgb[1], rgb[2], rgb[3]].map((part) => Math.max(0, Math.min(255,
Number(part))).toString(16).padStart(2, "0")).join("")}`;
830: }

```

normalizePlainFieldStyle - lines 832-844

Item	Explanation
Source location	Lines 832-844 (13 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizePlainFieldStyle(style = {}) {
Touches	filesystem
Calls detected	cleanFontFamily, normalizeFontPx, normalizeTextColor, rgbColorToHex
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
832: function normalizePlainFieldStyle(style = {}) {
833:   const normalized = {};
834:   const fontFamily = cleanFontFamily(style.fontFamily || "");
835:   const fontPx = normalizeFontPx(style.fontPx || style.fontSize || "");
836:   const color = normalizeTextColor(style.color || "");
837:   if (fontFamily) normalized.fontFamily = fontFamily;
838:   if (fontPx) normalized.fontPx = fontPx;
839:   if (color) normalized.color = rgbColorToHex(color);
840:   if (style.bold === true || style.bold === "true") normalized.bold = true;
841:   if (style.italic === true || style.italic === "true") normalized.italic = true;
842:   if (style.underline === true || style.underline === "true") normalized.underline = true;
843:   return normalized;
844: }
```

normalizeFieldStyles - lines 846-852

Item	Explanation
Source location	Lines 846-852 (7 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeFieldStyles(styles = {}) {
Touches	filesystem
Calls detected	fromEntries, entries, map, normalizePlainFieldStyle, filter, keys
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
846: function normalizeFieldStyles(styles = {}) {
847:   return Object.fromEntries(
848:     Object.entries(styles || {})
849:       .map(([fieldId, style]) => [fieldId, normalizePlainFieldStyle(style)])
850:       .filter(([ , style]) => Object.keys(style).length)
851:   );
852: }
```

plainFieldStyleToCss - lines 854-864

Item	Explanation
Source location	Lines 854-864 (11 source lines).
Purpose	Part of the public website runtime.
Declaration	function plainFieldStyleToCss(style = {}) {
Touches	filesystem
Calls detected	normalizePlainFieldStyle, push, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
854: function plainFieldStyleToCss(style = {}) {
855:   const normalized = normalizePlainFieldStyle(style);
856:   const rules = [];
857:   if (normalized.fontFamily) rules.push(`font-family: ${normalized.fontFamily}`);
858:   if (normalized.fontPx) rules.push(`font-size: ${normalized.fontPx}px`);
859:   if (normalized.color) rules.push(`color: ${normalized.color}`);
860:   if (normalized.bold) rules.push("font-weight: 700");
861:   if (normalized.italic) rules.push("font-style: italic");
862:   if (normalized.underline) rules.push("text-decoration: underline");
863:   return rules.join(" ");
}
```

864: }

applyPlainFieldStyle - lines 866-875

Item	Explanation
Source location	Lines 866-875 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function applyPlainFieldStyle(element, fieldId = "") {
Touches	filesystem
Calls detected	normalizePlainFieldStyle
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
866: function applyPlainFieldStyle(element, fieldId = "") {
867:   if (!element) return;
868:   const style = normalizePlainFieldStyle(fieldStyles[fieldId] || {});
869:   element.style.fontFamily = style.fontFamily || "";
870:   element.style.fontSize = style.fontPx ? `${style.fontPx}px` : "";
871:   element.style.color = style.color || "";
872:   element.style.fontWeight = style.bold ? "700" : "";
873:   element.style.fontStyle = style.italic ? "italic" : "";
874:   element.style.textDecoration = style.underline ? "underline" : "";
875: }
```

plainFieldStyleAttribute - lines 877-880

Item	Explanation
Source location	Lines 877-880 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function plainFieldStyleAttribute(fieldId = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	plainFieldStyleToCss, escapeHtml
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
877: function plainFieldStyleAttribute(fieldId = "") {
878:   const css = plainFieldStyleToCss(fieldStyles[fieldId] || {});
879:   return css ? `style="${escapeHtml(css)}"` : "";
880: }
```

richTextStyle - lines 882-896

Item	Explanation
Source location	Lines 882-896 (15 source lines).
Purpose	Part of the public website runtime.
Declaration	function richTextStyle(block = {}) {
Touches	filesystem
Calls detected	cleanFontFamily, normalizeFontPx, normalizeTextColor, push, escapeHtml, join

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

882: function richTextStyle(block = {}) {
883:   const styles = [];
884:   const fontFamily = cleanFontFamily(block.fontFamily || "Arial") || "Arial";
885:   const fontPx = normalizeFontPx(block.fontPx);
886:   const color = normalizeTextColor(block.color || "");
887:
888:   if (fontFamily) styles.push(`font-family: ${fontFamily}`);
889:   if (fontPx) styles.push(`font-size: ${fontPx}px`);
890:   if (color) styles.push(`color: ${color}`);
891:   if (block.bold) styles.push("font-weight: 700");
892:   if (block.italic) styles.push("font-style: italic");
893:   if (block.underline) styles.push("text-decoration: underline");
894:
895:   return styles.length ? ` style="${escapeHtml(styles.join("; "))}"` : "";
896: }

```

normalizeCropAspect - lines 898-900

Item	Explanation
Source location	Lines 898-900 (3 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeCropAspect(value = "") {
Touches	filesystem
Calls detected	includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

898: function normalizeCropAspect(value = "") {
899:   return ["1 / 1", "4 / 3", "16 / 9", "3 / 4"].includes(value) ? value : "original";
900: }

```

normalizeCropZoom - lines 902-905

Item	Explanation
Source location	Lines 902-905 (4 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeCropZoom(value = 1) {
Touches	filesystem
Calls detected	isFinite, min, max
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

902: function normalizeCropZoom(value = 1) {
903:     const zoom = Number(value);
904:     return Number.isFinite(zoom) ? Math.min(3, Math.max(1, zoom)) : 1;
905: }

```

normalizeCropPosition - lines 907-910

Item	Explanation
Source location	Lines 907-910 (4 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeCropPosition(value = 50) {
Touches	filesystem
Calls detected	isFinite, min, max
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

907: function normalizeCropPosition(value = 50) {
908:     const position = Number(value);
909:     return Number.isFinite(position) ? Math.min(100, Math.max(0, position)) : 50;
910: }

```

richImageCropStyle - lines 912-921

Item	Explanation
Source location	Lines 912-921 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function richImageCropStyle(block = {}) {
Touches	filesystem
Calls detected	normalizeCropAspect, normalizeCropZoom, normalizeCropPosition, push, escapeHtml, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

912: function richImageCropStyle(block = {}) {
913:     const aspect = normalizeCropAspect(block.cropAspect);
914:     const styles = [
915:         `--crop-zoom: ${normalizeCropZoom(block.cropZoom)}`,
916:         `--crop-x: ${normalizeCropPosition(block.cropX)}%`,
917:         `--crop-y: ${normalizeCropPosition(block.cropY)}%`
918:     ];
919:     if (aspect !== "original") styles.push(`--crop-aspect: ${aspect}`);
920:     return ` style="${escapeHtml(styles.join("; "))}"`;
921: }

```

richImageDownloadLink - lines 923-940

Item	Explanation
Source location	Lines 923-940 (18 source lines).
Purpose	Part of the public website runtime.
Declaration	function richImageDownloadLink(block = {}) {

Item	Explanation
Touches	filesystem
Calls detected	escapeHtml, cleanRichImageTitle, normalizeLinkTarget, isWebsiteLinkItem, linkAttributes, downloadAttribute
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

923: function richImageDownloadLink(block = {}) {
924:   const label = escapeHtml(cleanRichImageTitle(block) || block.caption || "Image file");
925:   const url = block.url || "#";
926:   const target = normalizeLinkTarget(url, { assumeWeb: isWebsiteLinkItem(block, url) });
927:
928:   return `
929:   <p class="rich-download-only">
930:     <a
931:       class="resource-link"
932:       href="${escapeHtml(target)}"
933:       ${linkAttributes(target, block)}
934:       ${downloadAttribute(target, block)}
935:     >
936:       ${label}
937:     </a>
938:   </p>
939:   `;
940: }

```

cleanRichImageTitle - lines 942-945

Item	Explanation
Source location	Lines 942-945 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function cleanRichImageTitle(block = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

942: function cleanRichImageTitle(block = {}) {
943:   const title = String(block.title || "").trim();
944:   return /^pasted image\b/i.test(title) ? "" : title;
945: }

```

renderRichContent - lines 947-982

Item	Explanation
Source location	Lines 947-982 (36 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderRichContent(rich, fallbackText = "") {
Touches	filesystem
Calls detected	textBlocksFromPlainText, map, includes, richImageDownloadLink, cleanRichImageTitle, normalizeLinkTarget, normalizeCropAspect, richImageCropStyle, escapeHtml, unwrapFormula, looksLikeWebOrContactText, renderInlineMath

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 7 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

947: function renderRichContent(rich, fallbackText = "") {
948:   const blocks = rich?.blocks?.length ? rich.blocks : textBlocksFromPlainText(fallbackText);
949:   return `
950:     <div class="rich-content">
951:       ${blocks.map((block) => {
952:         const align = ["left", "center", "right"].includes(block.align) ? block.align : "left";
953:         if (block.type === "image") {
954:           if (block.display === "download") return richImageDownloadLink(block);
955:           const title = cleanRichImageTitle(block);
956:           const imageSrc = normalizeLinkTarget(block.url, { assumeWeb: true });
957:           return `
958:             <figure class="rich-image justify-${align}">
959:               <span class="rich-image-viewport crop-${normalizeCropAspect(block.cropAspect) ===
"original" ? "original" : "active"}"${richImageCropStyle(block)}>
960:                 
961:               </span>
962:               ${title || block.caption} ? `<figcaption>${title} ?
`<strong>${escapeHtml(title)}</strong>` : ""}${block.caption ? `<span>${escapeHtml(block.caption)}</span>` :
""}</figcaption>` : ""}
963:             </figure>
964:           `;
965:         }
966:         if (block.type === "formula") {
967:           const formula = unwrapFormula(block.formula);
968:           if (looksLikeWebOrContactText(formula)) {
969:             return `<p class="rich-paragraph rich-text-normal
text-${align}">${renderInlineMath(formula)}</p>`;
970:           }
971:           return `<div class="rich-formula justify-${align}">${escapeHtml(formula)}</div>`;
972:         }
973:         if (block.type === "code") return renderRichCodeBlock(block);
974:         const size = ["small", "normal", "large"].includes(block.fontSize) ? block.fontSize : "normal";
975:         const content = block.html
976:           ? sanitizeRichInlineHtml(block.html)
977:           : renderInlineMath(block.text || "");
978:         return `<p class="rich-paragraph rich-text-${size}
text-${align}"${richTextStyle(block)}>${content}</p>`;
979:       }).join("")}
980:     </div>
981:   `;
982: }

```

renderRichFieldContent - lines 984-994

Item	Explanation
Source location	Lines 984-994 (11 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderRichFieldContent(rich, fallbackText = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	textBlocksFromPlainText, map, includes, sanitizeRichInlineHtml, renderInlineMath, richTextStyle, filter, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

984: function renderRichFieldContent(rich, fallbackText = "") {
985:   const blocks = rich?.blocks?.length ? rich.blocks : textBlocksFromPlainText(fallbackText);
986:   return blocks.map((block) => {
987:     if (block.type !== "paragraph") return "";
988:     const align = ["left", "center", "right"].includes(block.align) ? block.align : "left";
989:     const content = block.html

```

```

990:         ? sanitizeRichInlineHtml(block.html)
991:         : renderInlineMath(block.text || "");
992:     return `<span class="rich-field-line text-${align}"${richTextStyle(block)}>${content}</span>`;
993:   }).filter(Boolean).join("");
994: }

```

slugLabel - lines 996-999

Item	Explanation
Source location	Lines 996-999 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function slugLabel(value) {
Touches	Mostly local calculations or local UI state.
Calls detected	find
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

996: function slugLabel(value) {
997:   const category = categories.find((item) => item.id === value);
998:   return category ? category.label : value;
999: }

```

canonicalTemplateId - lines 1001-1003

Item	Explanation
Source location	Lines 1001-1003 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function canonicalTemplateId(id) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1001: function canonicalTemplateId(id) {
1002:   return legacyTemplateSkins[id] || id || "";
1003: }

```

projectTemplateId - lines 1005-1007

Item	Explanation
Source location	Lines 1005-1007 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function projectTemplateId(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	canonicalTemplateId
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1005: function projectTemplateId(project) {
1006:   return canonicalTemplateId(project?.portfolioView?.template?.id || project?.templateId || "");
1007: }
```

projectTemplateClass - lines 1009-1012

Item	Explanation
Source location	Lines 1009-1012 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function projectTemplateClass(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	projectTemplateId
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1009: function projectTemplateClass(project) {
1010:   const templateId = projectTemplateId(project);
1011:   return templateId ? `project-template project-template-${templateId}` : "project-template-white";
1012: }
```

responsiveClassName - lines 1014-1019

Item	Explanation
Source location	Lines 1014-1019 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function responsiveClassName(value, fallback) {
Touches	Mostly local calculations or local UI state.
Calls detected	toLowerCase, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1014: function responsiveClassName(value, fallback) {
1015:   return String(value || fallback || "")
1016:     .toLowerCase()
1017:     .replace(/[a-z0-9]+/g, "-")
1018:     .replace(/^|-$/g, "") || fallback;
1019: }
```

projectResponsiveClass - lines 1021-1024

Item	Explanation
Source location	Lines 1021-1024 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function projectResponsiveClass(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	projectResponsiveProfile, responsiveClassName
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1021: function projectResponsiveClass(project) {
1022:   const profile = projectResponsiveProfile(project);
1023:   return `responsive-project responsive-density-${responsiveClassName(profile.density, "balanced")}
responsive-card-${responsiveClassName(profile.cardLayout, "single")}`;
1024: }

```

responsiveStyleValues - lines 1026-1034

Item	Explanation
Source location	Lines 1026-1034 (9 source lines).
Purpose	Part of the public website runtime.
Declaration	function responsiveStyleValues(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	projectResponsiveProfile
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1026: function responsiveStyleValues(project) {
1027:   const profile = projectResponsiveProfile(project);
1028:   return {
1029:     "--responsive-section-card-min": profile.sectionCardMin,
1030:     "--responsive-content-max": profile.contentMax,
1031:     "--responsive-media-max": profile.mediaMax,
1032:     "--responsive-touch-target": profile.touchTarget
1033:   };
1034: }

```

hasPublicTemplate - lines 1036-1038

Item	Explanation
Source location	Lines 1036-1038 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function hasPublicTemplate(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	projectTemplateId
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1036: function hasPublicTemplate(project) {
1037:   return Boolean(projectTemplateId(project));
1038: }

```

projectTemplateVisual - lines 1040-1042

Item	Explanation
Source location	Lines 1040-1042 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function projectTemplateVisual(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1040: function projectTemplateVisual(project) {
1041:   return project?.portfolioView?.template?.visual || project?.templateVisual || null;
1042: }

```

projectTemplateStyle - lines 1044-1062

Item	Explanation
Source location	Lines 1044-1062 (19 source lines).
Purpose	Part of the public website runtime.
Declaration	function projectTemplateStyle(project, accent) {
Touches	Mostly local calculations or local UI state.
Calls detected	projectTemplateVisual, responsiveStyleValues, entries, filter, map, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1044: function projectTemplateStyle(project, accent) {
1045:   const visual = projectTemplateVisual(project) || {};
1046:   const values = {
1047:     "--accent": accent,
1048:     "--template-bg": visual.background,
1049:     "--template-panel": visual.panel,
1050:     "--template-accent": visual.accent,
1051:     "--template-hover": visual.hover,
1052:     "--template-click": visual.click,
1053:     "--template-click-text": visual.clickText,
1054:     "--template-line": visual.line,
1055:     "--template-text": visual.text,
1056:     ...responsiveStyleValues(project)
1057:   };
1058:   return Object.entries(values)

```



```

1059:     .filter(([, value]) => value)
1060:     .map(([key, value]) => `${key}: ${value}`)
1061:     .join("; ");
1062: }

```

applyProjectTemplateToElement - lines 1064-1098

Item	Explanation
Source location	Lines 1064-1098 (35 source lines).
Purpose	Part of the public website runtime.
Declaration	function applyProjectTemplateToElement(element, project, accent) {
Touches	browser/Electron DOM
Calls detected	filter, startsWith, forEach, remove, projectTemplateClass, split, add, projectResponsiveClass, projectTemplateVisual, responsiveStyleValues, entries, setProperty
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1064: function applyProjectTemplateToElement(element, project, accent) {
1065:   if (!element) return;
1066:   [...element.classList]
1067:   .filter((className) =>
1068:     className === "project-template" ||
1069:     className === "project-template-white" ||
1070:     className === "responsive-project" ||
1071:     className.startsWith("project-template-") ||
1072:     className.startsWith("responsive-density-") ||
1073:     className.startsWith("responsive-card-")
1074:   )
1075:   .forEach((className) => element.classList.remove(className));
1076:   projectTemplateClass(project).split(" ").forEach((className) => element.classList.add(className));
1077:   projectResponsiveClass(project).split(" ").forEach((className) => element.classList.add(className));
1078:   const visual = projectTemplateVisual(project) || {};
1079:   const values = {
1080:     "--accent": accent,
1081:     "--template-bg": visual.background,
1082:     "--template-panel": visual.panel,
1083:     "--template-accent": visual.accent,
1084:     "--template-hover": visual.hover,
1085:     "--template-click": visual.click,
1086:     "--template-click-text": visual.clickText,
1087:     "--template-line": visual.line,
1088:     "--template-text": visual.text,
1089:     ...responsiveStyleValues(project)
1090:   };
1091:   Object.entries(values).forEach(([key, value]) => {
1092:     if (value) {
1093:       element.style.setProperty(key, value);
1094:     } else {
1095:       element.style.removeProperty(key);
1096:     }
1097:   });
1098: }

```

parsedItemTerms - lines 1100-1110

Item	Explanation
Source location	Lines 1100-1110 (11 source lines).
Purpose	Part of the public website runtime.
Declaration	function parsedItemTerms(item) {
Touches	filesystem
Calls detected	flatMap
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1100: function parsedItemTerms(item) {
1101:   return [
1102:     item?.title,
1103:     item?.description,
1104:     item?.meta,
1105:     item?.url,
1106:     item?.kind,
1107:     ...(item?.rich?.blocks || []).flatMap((block) => [block.text, block.formula, block.title,
block.caption]),
1108:     ...(item?.children || []).flatMap(parsedItemTerms)
1109:   ];
1110: }

```

flattenProject - lines 1112-1165

Item	Explanation
Source location	Lines 1112-1165 (54 source lines).
Purpose	Part of the public website runtime.
Declaration	function flattenProject(project) {
Touches	filesystem
Calls detected	slugLabel, electronicsSearchKeywords, flatMap, map, filter, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1112: function flattenProject(project) {
1113:   const categoryLabel = slugLabel(project.category);
1114:   const electronicsKeywords = window.electronicsSearchKeywords
1115:     ? window.electronicsSearchKeywords(project, categoryLabel)
1116:     : [];
1117:   const richSummaryTerms = (project.summaryRich?.blocks || []).flatMap((block) => [
1118:     block.text,
1119:     block.formula,
1120:     block.title,
1121:     block.caption
1122:   ]);
1123:   const parsedChildTerms = (project.portfolioView?.sections || []).flatMap((section) =>
1124:     (section.items || []).flatMap(parsedItemTerms)
1125:   );
1126:   const design = project.design || {};
1127:   const designTerms = [
1128:     design.brief?.summary,
1129:     ...(design.brief?.files || []).flatMap((item) => [item.title, item.description, item.url]),
1130:     design.documentation?.summary,
1131:     ...(design.documentation?.files || []).flatMap((item) => [item.title, item.description, item.url]),
1132:     ...(design.documentation?.references || []).flatMap((item) => [item.title, item.description,
item.url]),
1133:     ...(design.documentation?.mathAnalysis || []).flatMap((item) => [item.title, item.description]),
1134:     design.simulation?.summary,
1135:     ...(design.simulation?.files || []).flatMap((item) => [item.title, item.description, item.url]),
1136:     ...(design.simulation?.results || []).flatMap((item) => [item.title, item.description, item.url])
1137:   ];
1138:   return [
1139:     project.id,
1140:     project.title,
1141:     categoryLabel,
1142:     project.status,
1143:     project.summary,
1144:     ...richSummaryTerms,
1145:     ...parsedChildTerms,
1146:     ...designTerms,
1147:     ...(project.focus || []),
1148:     ...(project.highlights || []),
1149:     ...(project.tools || []).map((item) => typeof item === "string" ? item : [item.name, item.title,

```

```

item.label, item.description].filter(Boolean).join(" "),
1150:   ...(project.languages || []),
1151:   ...(project.documents || []).flatMap((item) => [item.title, item.type, item.status, item.url]),
1152:   ...(project.tests || []).flatMap((item) => [item.name, item.method, item.status, item.result,
item.artifact]),
1153:   ...(project.pcbbs || []).flatMap((item) => [item.name, item.revision, item.status, item.artifact]),
1154:   ...(project.media || []).flatMap((item) => [item.title, item.caption, item.url]),
1155:   ...(project.sections || []).flatMap((section) => [
1156:     section.title,
1157:     section.description,
1158:     ...(section.items || []).flatMap((item) => [item.title, item.description, item.type, item.status,
item.url])
1159:   ]),
1160:   ...(project.links || []).flatMap((item) => [item.label, item.url]),
1161:   ...electronicsKeywords
1162: ]
1163:   .map(normalize)
1164:   .join(" ");
1165: }

```

richSummaryTerms - lines 1117-1126

Item	Explanation
Source location	Lines 1117-1126 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	const richSummaryTerms = (project.summaryRich?.blocks []).flatMap((block) => [
Touches	filesystem
Calls detected	flatMap
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1117:   const richSummaryTerms = (project.summaryRich?.blocks || []).flatMap((block) => [
1118:     block.text,
1119:     block.formula,
1120:     block.title,
1121:     block.caption
1122:   ]);
1123:   const parsedChildTerms = (project.portfolioView?.sections || []).flatMap((section) =>
1124:     (section.items || []).flatMap(parsedItemTerms)
1125:   );
1126:   const design = project.design || {};

```

parsedChildTerms - lines 1123-1126

Item	Explanation
Source location	Lines 1123-1126 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	const parsedChildTerms = (project.portfolioView?.sections []).flatMap((section) =>
Touches	filesystem
Calls detected	flatMap
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1123:   const parsedChildTerms = (project.portfolioView?.sections || []).flatMap((section) =>
1124:     (section.items || []).flatMap(parsedItemTerms)
1125:   );

```

```
1126:   const design = project.design || {};
```

projectMatches - lines 1167-1169

Item	Explanation
Source location	Lines 1167-1169 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function projectMatches(project, query) {
Touches	Mostly local calculations or local UI state.
Calls detected	flattenProject, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1167: function projectMatches(project, query) {  
1168:   return !query || flattenProject(project).includes(query);  
1169: }
```

projectVisible - lines 1171-1174

Item	Explanation
Source location	Lines 1171-1174 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function projectVisible(project, query) {
Touches	Mostly local calculations or local UI state.
Calls detected	projectMatches
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1171: function projectVisible(project, query) {  
1172:   const categoryMatch = activeFilter === "all" || project.category === activeFilter;  
1173:   return categoryMatch && projectMatches(project, query);  
1174: }
```

normalizeCategory - lines 1176-1184

Item	Explanation
Source location	Lines 1176-1184 (9 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeCategory(category = {}) {
Touches	filesystem
Calls detected	trim, normalizeTextColor, toLowerCase, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1176: function normalizeCategory(category = {}) {
1177:   const label = String(category.label || category.title || category.id || "Project category").trim() ||
"Project category";
1178:   return {
1179:     accent: normalizeTextColor(category.accent || "" || "#1677a8",
1180:     description: String(category.description || "").trim(),
1181:     id: String(category.id || label.toLowerCase().replace(/^[a-z0-9]+/g, "-").replace(/^-|-$/g, "")) ||
"category"),
1182:     label
1183:   };
1184: }

```

hydrateProjectCategories - lines 1186-1201

Item	Explanation
Source location	Lines 1186-1201 (16 source lines).
Purpose	Part of the public website runtime.
Declaration	function hydrateProjectCategories(rawCategories = [], rawProjects = []) {
Touches	filesystem
Calls detected	map, Set, forEach, trim, has, split, filter, charAt, toUpperCase, slice, join, push
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1186: function hydrateProjectCategories(rawCategories = [], rawProjects = []) {
1187:   const normalized = rawCategories.map(normalizeCategory);
1188:   const knownIds = new Set(normalized.map((category) => category.id));
1189:   rawProjects.forEach((project) => {
1190:     const id = String(project?.category || "").trim();
1191:     if (!id || knownIds.has(id)) return;
1192:     const label = id
1193:       .split(/[-_\s]+/)
1194:       .filter(Boolean)
1195:       .map((part) => part.charAt(0).toUpperCase() + part.slice(1))
1196:       .join(" ") || "Project category";
1197:     normalized.push(normalizeCategory({ id, label, description: "Project category" }));
1198:     knownIds.add(id);
1199:   });
1200:   return normalized;
1201: }

```

setActiveFilter - lines 1203-1209

Item	Explanation
Source location	Lines 1203-1209 (7 source lines).
Purpose	Part of the public website runtime.
Declaration	function setActiveFilter(filter = "all") {
Touches	browser/Electron DOM
Calls detected	some, forEach, toggle
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1203: function setActiveFilter(filter = "all") {
1204:   activeFilter = filter;
1205:   if (activeFilter !== "all" && !categories.some((category) => category.id === activeFilter)) {
1206:     activeFilter = "all";
1207:   }
1208:   filterButtons.forEach((item) => item.classList.toggle("active", item.dataset.filter === activeFilter));
1209: }

```

bindFilterButtons - lines 1211-1220

Item	Explanation
Source location	Lines 1211-1220 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function bindFilterButtons() {
Touches	browser/Electron DOM
Calls detected	querySelectorAll, forEach, addEventListener, setActiveFilter, renderProjects, updateSearchDropdown
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1211: function bindFilterButtons() {
1212:   filterButtons = [...document.querySelectorAll(".filter-button")];
1213:   filterButtons.forEach((button) => {
1214:     button.addEventListener("click", () => {
1215:       setActiveFilter(button.dataset.filter || "all");
1216:       renderProjects();
1217:       updateSearchDropdown();
1218:     });
1219:   });
1220: }

```

renderCategoryFilters - lines 1222-1241

Item	Explanation
Source location	Lines 1222-1241 (20 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderCategoryFilters() {
Touches	Mostly local calculations or local UI state.
Calls detected	map, join, escapeHtml, bindFilterButtons, setActiveFilter
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1222: function renderCategoryFilters() {
1223:   if (projectTrackCount) {
1224:     projectTrackCount.textContent = `${categories.length} Track${categories.length === 1 ? "" : "s"}`;
1225:   }
1226:   if (projectTrackLabels) {
1227:     projectTrackLabels.textContent = categories.length

```

```

1228:         ? categories.map((category) => category.label).join(", ")
1229:         : "project categories";
1230:     }
1231:     if (projectFilters) {
1232:         projectFilters.innerHTML = `
1233:         <button class="filter-button active" type="button" data-filter="all">All</button>
1234:         ${categories.map((category) => `
1235:         <button class="filter-button" type="button" data-
filter="${escapeHtml(category.id)}">${escapeHtml(category.label)}</button>
1236:         `).join("")}
1237:         `;
1238:     }
1239:     bindFilterButtons();
1240:     setActiveFilter(activeFilter);
1241: }

```

flattenSearchText - lines 1243-1251

Item	Explanation
Source location	Lines 1243-1251 (9 source lines).
Purpose	Part of the public website runtime.
Declaration	function flattenSearchText(values = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	flat, filter, map, stringify, join, replace, trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1243: function flattenSearchText(values = []) {
1244:     return values
1245:         .flat(Infinity)
1246:         .filter((value) => value !== null && value !== undefined)
1247:         .map((value) => typeof value === "object" ? JSON.stringify(value) : String(value))
1248:         .join(" ")
1249:         .replace(/\s+/g, " ")
1250:         .trim();
1251: }

```

richTextTerms - lines 1253-1264

Item	Explanation
Source location	Lines 1253-1264 (12 source lines).
Purpose	Part of the public website runtime.
Declaration	function richTextTerms(rich) {
Touches	filesystem
Calls detected	flatMap
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1253: function richTextTerms(rich) {
1254:     return (rich?.blocks || []).flatMap((block) => [
1255:         block.text,
1256:         block.html,
1257:         block.formula,
1258:         block.code,
1259:         block.language,
1260:         block.title,

```

```

1261:     block.caption,
1262:     block.url
1263:   });
1264: }

```

addSearchEntry - lines 1266-1276

Item	Explanation
Source location	Lines 1266-1276 (11 source lines).
Purpose	Part of the public website runtime.
Declaration	function addSearchEntry(entries, entry) {
Touches	filesystem
Calls detected	flattenSearchText, normalize, push
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1266: function addSearchEntry(entries, entry) {
1267:   const text = flattenSearchText([entry.title, entry.type, entry.context, entry.text, entry.url]);
1268:   if (!entry.title && !text) return null;
1269:   const storedEntry = {
1270:     ...entry,
1271:     normalizedText: normalize(text),
1272:     text
1273:   };
1274:   entries.push(storedEntry);
1275:   return storedEntry;
1276: }

```

searchSnippet - lines 1278-1286

Item	Explanation
Source location	Lines 1278-1286 (9 source lines).
Purpose	Part of the public website runtime.
Declaration	function searchSnippet(text = "", query = "") {
Touches	filesystem
Calls detected	replace, trim, normalize, indexOf, slice, max, min
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1278: function searchSnippet(text = "", query = "") {
1279:   const clean = String(text || "").replace(/\s+/g, " ").trim();
1280:   if (!clean) return "";
1281:   const index = normalize(clean).indexOf(normalize(query));
1282:   if (index < 0) return clean.slice(0, 150);
1283:   const start = Math.max(0, index - 54);
1284:   const end = Math.min(clean.length, index + query.length + 80);
1285:   return `${start ? "... " : ""}${clean.slice(start, end)}${end < clean.length ? "... " : ""}`;
1286: }

```

entryScore - lines 1288-1314

Item	Explanation
Source location	Lines 1288-1314 (27 source lines).
Purpose	Part of the public website runtime.
Declaration	function entryScore(entry, query) {
Touches	filesystem
Calls detected	normalize, trim, split, filter, startsWith, includes, forEach
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1288: function entryScore(entry, query) {
1289:   const normalizedQuery = normalize(query).trim();
1290:   if (!normalizedQuery) return 0;
1291:   const title = normalize(entry.title);
1292:   const type = normalize(entry.type);
1293:   const context = normalize(entry.context);
1294:   const text = entry.normalizedText || "";
1295:   const words = normalizedQuery.split(/\s+/).filter(Boolean);
1296:   let score = 0;
1297:
1298:   if (title === normalizedQuery) score += 160;
1299:   if (title.startsWith(normalizedQuery)) score += 120;
1300:   if (title.includes(normalizedQuery)) score += 90;
1301:   if (context.includes(normalizedQuery)) score += 55;
1302:   if (type.includes(normalizedQuery)) score += 35;
1303:   if (text.includes(normalizedQuery)) score += 45;
1304:   words.forEach((word) => {
1305:     if (title.includes(word)) score += 16;
1306:     if (context.includes(word)) score += 10;
1307:     if (text.includes(word)) score += 6;
1308:   });
1309:
1310:   if (score > 0 && entry.kind === "file") score += 18;
1311:   if (score > 0 && entry.kind === "section") score += 14;
1312:   if (score > 0 && entry.kind === "project") score += 12;
1313:   return score;
1314: }

```

entryMatches - lines 1316-1318

Item	Explanation
Source location	Lines 1316-1318 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function entryMatches(entry, query) {
Touches	Mostly local calculations or local UI state.
Calls detected	entryScore
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1316: function entryMatches(entry, query) {
1317:   return entryScore(entry, query) > 0;
1318: }

```

fileIsBrowserSearchable - lines 1320-1323

Item	Explanation
Source location	Lines 1320-1323 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function fileIsBrowserSearchable(url = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	split, toLowerCase
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1320: function fileIsBrowserSearchable(url = "") {
1321:   const clean = String(url || "").split(/[?#]/)[0].toLowerCase();
1322:   return /\.(txt|md|markdown|csv|json|xml|log|c|h|cpp|hpp|py|js|mjs|ts|v|sv|vhd|?|spice|cir|net|asc|sch|k
icad_sch|kicad_pcb)$/i.test(clean);
1323: }

```

indexSearchableFileText - lines 1325-1338

Item	Explanation
Source location	Lines 1325-1338 (14 source lines).
Purpose	Part of the public website runtime.
Declaration	async function indexSearchableFileText(entry) {
Touches	filesystem, network/API requests
Calls detected	fileIsBrowserSearchable, startsWith, fetch, text, flattenSearchText, slice, normalize, updateSearchDropdown
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1325: async function indexSearchableFileText(entry) {
1326:   if (!entry.url || !fileIsBrowserSearchable(entry.url)) return;
1327:   if (/^(https?:)?\/\//i.test(entry.url) && !entry.url.startsWith(window.location.origin)) return;
1328:   try {
1329:     const response = await fetch(entry.url);
1330:     if (!response.ok) return;
1331:     const text = await response.text();
1332:     entry.text = flattenSearchText([entry.text, text.slice(0, 60000)]);
1333:     entry.normalizedText = normalize(entry.text);
1334:     updateSearchDropdown();
1335:   } catch {
1336:     // Files remain searchable by title, path, caption, and catalog text if direct text fetch is
unavailable.
1337:   }
1338: }

```

addProjectSearchEntries - lines 1340-1414

Item	Explanation
Source location	Lines 1340-1414 (75 source lines).
Purpose	Part of the public website runtime.
Declaration	function addProjectSearchEntries(entries, project) {
Touches	filesystem

Item	Explanation
Calls detected	slugLabel, richTextTerms, electronicsSearchKeywords, addSearchEntry, filter, sectionHasRenderableContent, forEach, displayTitle, parsedItemTerms, fileNameFromUrl, itemUrl, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1340: function addProjectSearchEntries(entries, project) {
1341:   const categoryLabel = slugLabel(project.category);
1342:   const baseContext = `${categoryLabel} / ${project.title}`;
1343:   const projectTerms = [
1344:     project.id,
1345:     project.summary,
1346:     richTextTerms(project.summaryRich),
1347:     project.focus,
1348:     project.highlights,
1349:     project.tools,
1350:     project.languages,
1351:     project.links,
1352:     window.electronicsSearchKeywords ? window.electronicsSearchKeywords(project, categoryLabel) : []
1353:   ];
1354:
1355:   addSearchEntry(entries, {
1356:     context: categoryLabel,
1357:     kind: "project",
1358:     projectId: project.id,
1359:     title: project.title,
1360:     type: "Project",
1361:     text: projectTerms
1362:   });
1363:
1364:   const sections = (project.portfolioView?.sections || []).filter((section) => section.id !== "brief" &&
sectionHasRenderableContent(section));
1365:   sections.forEach((section, sectionIndex) => {
1366:     addSearchEntry(entries, {
1367:       context: baseContext,
1368:       kind: "section",
1369:       projectId: project.id,
1370:       sectionIndex: String(sectionIndex),
1371:       title: displayTitle(section.title, "Section"),
1372:       type: "Project section",
1373:       text: [section.description, richTextTerms(section.rich), parsedItemTerms(section)]
1374:     });
1375:
1376:     const walkItems = (items = [], path = [], parentTitles = []) => {
1377:       items.forEach((item, index) => {
1378:         const nextPath = [...path, index];
1379:         const title = displayTitle(item.title || item.name || item.label || fileNameFromUrl(item.url ||
""), "Project item");
1380:         const url = itemUrl(item);
1381:         const context = [baseContext, displayTitle(section.title, "Section"),
...parentTitles].filter(Boolean).join(" / ");
1382:         const entry = {
1383:           context,
1384:           kind: url ? "file" : "subsection",
1385:           projectId: project.id,
1386:           resourcePath: pathToString(nextPath),
1387:           sectionIndex: String(sectionIndex),
1388:           title,
1389:           type: url ? "File or uploaded asset" : "Subsection",
1390:           url,
1391:           text: [item.description, item.meta, item.kind, item.type, url, richTextTerms(item.rich),
parsedItemTerms(item)]
1392:         };
1393:         const storedEntry = addSearchEntry(entries, entry);
1394:         if (url && storedEntry) indexSearchableFileText(storedEntry);
1395:         walkItems(nodeChildren(item), nextPath, [...parentTitles, title]);
1396:       });
1397:     };
1398:     walkItems(section.items || []);
1399:   });
1400:
1401:   uniqueDownloads(collectDownloadsFromValue(project, "Project file", [])).forEach((download) => {
1402:     const entry = {
1403:       context: baseContext,
1404:       kind: "file",
1405:       projectId: project.id,
1406:       title: download.title || fileNameFromUrl(download.url),
1407:       type: download.type || "Uploaded file",
1408:       url: download.url,
1409:       text: [download.description, download.status, download.url]

```

```

1410:     };
1411:     const storedEntry = addSearchEntry(entries, entry);
1412:     if (storedEntry) indexSearchableFileText(storedEntry);
1413:   });
1414: }

```

sections - lines 1364-1399

Item	Explanation
Source location	Lines 1364-1399 (36 source lines).
Purpose	Part of the public website runtime.
Declaration	const sections = (project.portfolioView?.sections []).filter((section) => section.id !== "brief" && sectionHasRenderableContent(section));
Touches	filesystem
Calls detected	filter, sectionHasRenderableContent, forEach, addSearchEntry, displayTitle, richTextTerms, parsedItemTerms, fileNameFromUrl, itemUrl, join, pathToString, indexSearchableFileText
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1364:   const sections = (project.portfolioView?.sections || []).filter((section) => section.id !== "brief" &&
sectionHasRenderableContent(section));
1365:   sections.forEach((section, sectionIndex) => {
1366:     addSearchEntry(entries, {
1367:       context: baseContext,
1368:       kind: "section",
1369:       projectId: project.id,
1370:       sectionIndex: String(sectionIndex),
1371:       title: displayTitle(section.title, "Section"),
1372:       type: "Project section",
1373:       text: [section.description, richTextTerms(section.rich), parsedItemTerms(section)]
1374:     });
1375:
1376:     const walkItems = (items = [], path = [], parentTitles = []) => {
1377:       items.forEach((item, index) => {
1378:         const nextPath = [...path, index];
1379:         const title = displayTitle(item.title || item.name || item.label || fileNameFromUrl(item.url ||
""), "Project item");
1380:         const url = itemUrl(item);
1381:         const context = [baseContext, displayTitle(section.title, "Section"),
...parentTitles].filter(Boolean).join(" / ");
1382:         const entry = {
1383:           context,
1384:           kind: url ? "file" : "subsection",
1385:           projectId: project.id,
1386:           resourcePath: pathToString(nextPath),
1387:           sectionIndex: String(sectionIndex),
1388:           title,
1389:           type: url ? "File or uploaded asset" : "Subsection",
1390:           url,
1391:           text: [item.description, item.meta, item.kind, item.type, url, richTextTerms(item.rich),
parsedItemTerms(item)]
1392:         };
1393:         const storedEntry = addSearchEntry(entries, entry);
1394:         if (url && storedEntry) indexSearchableFileText(storedEntry);
1395:         walkItems(nodeChildren(item), nextPath, [...parentTitles, title]);
1396:       });
1397:     };
1398:     walkItems(section.items || []);
1399:   });

```

addSiteSectionSearchEntries - lines 1416-1463

Item	Explanation
Source location	Lines 1416-1463 (48 source lines).
Purpose	Part of the public website runtime.
Declaration	function addSiteSectionSearchEntries(entries) {

Item	Explanation
Touches	filesystem
Calls detected	normalizeProfile, splice, forEach, addSearchEntry, indexSearchableFileText, filter, richTextTerms
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1416: function addSiteSectionSearchEntries(entries) {
1417:   const current = normalizeProfile(profile || {});
1418:   const pageSections = [
1419:     { id: "top", title: heroTitle?.textContent || "Front page", type: "Page section", text:
[heroEyebrow?.textContent, heroCopy?.textContent] },
1420:     { id: "projects", title: "Engineering Projects", type: "Directory", text: "project directory
categories files sections" },
1421:     { id: "process", title: "Process", type: "Page section", text: "engineering process design testing
documentation" },
1422:     { id: "contact", title: "Contact", type: "Page section", text: [current.displayName, current.email,
current.phone, current.githubUrl, current.linkedinUrl, current.websiteUrl, current.contactIntro] }
1423:   ];
1424:   if (current.resumeUrl) {
1425:     pageSections.splice(2, 0, { id: "resume", title: "Resume", type: "Page section", text: "resume
professional profile document" });
1426:   }
1427:   pageSections.forEach((section) => addSearchEntry(entries, {
1428:     domTarget: section.id,
1429:     kind: "page",
1430:     title: section.title,
1431:     type: section.type,
1432:     text: section.text
1433:   }));
1434:
1435:   if (current.resumeUrl) {
1436:     const resumeTextEntry = addSearchEntry(entries, {
1437:       context: "Professional Profile / Resume",
1438:       kind: "file",
1439:       title: "Resume",
1440:       type: "Resume",
1441:       url: current.resumeUrl,
1442:       text: [current.displayName, current.portfolioLabel, current.contactIntro, current.resumeUrl]
1443:     });
1444:     if (resumeTextEntry) indexSearchableFileText(resumeTextEntry);
1445:   }
1446:
1447:   (siteSections || []).filter(siteSectionRenderable).forEach((section) => {
1448:     addSearchEntry(entries, {
1449:       domTarget: section.id,
1450:       kind: "page",
1451:       title: section.title || "Portfolio section",
1452:       type: "Portfolio section",
1453:       text: [section.description, richTextTerms(section.richDescription), section.links, section.assets]
1454:     });
1455:     (section.subsections || []).forEach((item) => addSearchEntry(entries, {
1456:       domTarget: section.id,
1457:       kind: "page",
1458:       title: item.title || "Portfolio subsection",
1459:       type: "Portfolio subsection",
1460:       text: [item.description, richTextTerms(item.richDescription), item.links]
1461:     }));
1462:   });
1463: }

```

rebuildSearchIndex - lines 1465-1470

Item	Explanation
Source location	Lines 1465-1470 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function rebuildSearchIndex() {
Touches	Mostly local calculations or local UI state.
Calls detected	addSiteSectionSearchEntries, forEach, addProjectSearchEntries
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1465: function rebuildSearchIndex() {
1466:   const entries = [];
1467:   addSiteSectionSearchEntries(entries);
1468:   projects.forEach((project) => addProjectSearchEntries(entries, project));
1469:   searchableEntries = entries;
1470: }

```

searchResultsFor - lines 1472-1486

Item	Explanation
Source location	Lines 1472-1486 (15 source lines).
Purpose	Part of the public website runtime.
Declaration	function searchResultsFor(query) {
Touches	filesystem
Calls detected	normalize, trim, Set, map, entryScore, filter, sort, localeCompare, join, has, add
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1472: function searchResultsFor(query) {
1473:   const normalizedQuery = normalize(query).trim();
1474:   if (!normalizedQuery) return [];
1475:   const seen = new Set();
1476:   return searchableEntries
1477:     .map((entry) => ({ ...entry, score: entryScore(entry, normalizedQuery) }))
1478:     .filter((entry) => entry.score > 0)
1479:     .sort((a, b) => b.score - a.score || a.title.localeCompare(b.title))
1480:     .filter((entry) => {
1481:       const key = [entry.kind, entry.projectId, entry.sectionIndex, entry.resourcePath, entry.domTarget,
entry.url, entry.title].join("|");
1482:       if (seen.has(key)) return false;
1483:       seen.add(key);
1484:       return true;
1485:     });
1486: }

```

assistantEndpoint - lines 1488-1494

Item	Explanation
Source location	Lines 1488-1494 (7 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantEndpoint() {
Touches	browser/Electron DOM, network/API requests
Calls detected	trim, querySelector, includes
API endpoints	/api/portfolio-ai
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1488: function assistantEndpoint() {
1489:   if (typeof window.OMB_AI_ENDPOINT === "string") return window.OMB_AI_ENDPOINT.trim();
1490:   const metaEndpoint = document.querySelector('meta[name="portfolio-ai-endpoint"]')?.content || "";
1491:   if (metaEndpoint.trim()) return String(metaEndpoint).trim();
1492:   if (["localhost", "127.0.0.1"].includes(window.location.hostname)) return "/api/portfolio-ai";
1493:   return "/api/portfolio-ai";
1494: }
```

assistantSourceLabel - lines 1496-1505

Item	Explanation
Source location	Lines 1496-1505 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantSourceLabel(result = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	split, map, trim, filter, slice, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1496: function assistantSourceLabel(result = {}) {
1497:   const title = result.title || "Portfolio item";
1498:   if (result.kind === "project" || result.kind === "page") return title;
1499:   const contextParts = String(result.context || "")
1500:     .split("/")
1501:     .map((item) => item.trim())
1502:     .filter(Boolean);
1503:   const specificContext = contextParts.slice(-2).join(" / ");
1504:   return [title, specificContext].filter(Boolean).join(" - ");
1505: }
```

assistantProjectSummary - lines 1507-1522

Item	Explanation
Source location	Lines 1507-1522 (16 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantProjectSummary(project = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	slugLabel, filter, sectionHasRenderableContent, map, displayTitle
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1507: function assistantProjectSummary(project = {}) {
1508:   const categoryLabel = slugLabel(project.category);
1509:   const sections = (project.portfolioView?.sections || [])
1510:     .filter((section) => section.id !== "brief" && sectionHasRenderableContent(section))
1511:     .map((section) => displayTitle(section.title, "Section"));
1512:   return {
1513:     category: categoryLabel,
1514:     focus: project.focus || [],
1515:     id: project.id,
1516:     links: project.links || [],
1517:     sections,
1518:     summary: project.summary || "",
1519:     title: project.portfolioView?.title || project.title,
```

```

1520:     tools: project.tools || []
1521:   };
1522: }

```

assistantProjectTitleAcronyms - lines 1524-1535

Item	Explanation
Source location	Lines 1524-1535 (12 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantProjectTitleAcronyms(title = "") {
Touches	filesystem
Calls detected	normalize, split, filter, Set, min, add, slice, map, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1524: function assistantProjectTitleAcronyms(title = "") {
1525:   const words = normalize(title)
1526:     .split(/\s+/)
1527:     .filter((word) => word.length > 2);
1528:   const acronyms = new Set();
1529:   for (let start = 0; start < words.length; start += 1) {
1530:     for (let length = 2; length <= Math.min(5, words.length - start); length += 1) {
1531:       acronyms.add(words.slice(start, start + length).map((word) => word[0]).join(""));
1532:     }
1533:   }
1534:   return [...acronyms].filter((item) => item.length >= 2);
1535: }

```

assistantProjectIsNamedInQuestion - lines 1537-1553

Item	Explanation
Source location	Lines 1537-1553 (17 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantProjectIsNamedInQuestion(question = "", project = {}) {
Touches	filesystem
Calls detected	normalize, Set, includes, split, assistantProjectTitleAcronyms, some, has, filter, min
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 6 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1537: function assistantProjectIsNamedInQuestion(question = "", project = {}) {
1538:   const clean = normalize(question);
1539:   if (!clean) return false;
1540:   const fillerWords = new Set(["with", "from", "into", "using", "project", "design", "system",
"systems"]);
1541:   const rawTitle = project.portfolioView?.title || project.title || project.id;
1542:   const title = normalize(rawTitle);
1543:   if (!title) return false;
1544:   if (clean.includes(title)) return true;
1545:   const cleanWords = new Set(clean.split(/\s+/));
1546:   const acronymMatch = assistantProjectTitleAcronyms(rawTitle)
1547:     .some((acronym) => cleanWords.has(acronym));
1548:   if (acronymMatch) return true;
1549:   const words = title.split(/\s+/).filter((word) => word.length > 3 && !fillerWords.has(word));
1550:   if (!words.length) return false;
1551:   const matches = words.filter((word) => clean.includes(word)).length;
1552:   return words.length <= 2 ? matches === words.length : matches >= Math.min(3, words.length);

```



```
1553: }
```

assistantNamedProjectIds - lines 1555-1559

Item	Explanation
Source location	Lines 1555-1559 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantNamedProjectIds(question = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	filter, assistantProjectIsNamedInQuestion, map
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1555: function assistantNamedProjectIds(question = "") {
1556:   return projects
1557:     .filter((project) => assistantProjectIsNamedInQuestion(question, project))
1558:     .map((project) => project.id);
1559: }
```

assistantProjectTitleMatches - lines 1561-1563

Item	Explanation
Source location	Lines 1561-1563 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantProjectTitleMatches(question = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	assistantNamedProjectIds
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1561: function assistantProjectTitleMatches(question = "") {
1562:   return assistantNamedProjectIds(question).length > 0;
1563: }
```

assistantUrlKind - lines 1565-1577

Item	Explanation
Source location	Lines 1565-1577 (13 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantUrlKind(url = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	split, toLowerCase, fileIsBrowserSearchable
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 10 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1565: function assistantUrlKind(url = "") {
1566:   const clean = String(url || "").split(/[?#]/)[0].toLowerCase();
1567:   if (!clean) return "unknown";
1568:   if (/github\.com|raw\.githubusercontent\.com|gist\.githubusercontent\.com/i.test(clean)) return
"github";
1569:   if (/linkedin\.com/i.test(clean)) return "linkedin";
1570:   if (/\. (png|jpe?g|gif|webp|svg)$/i.test(clean)) return "image";
1571:   if (/\.pdf$/i.test(clean)) return "pdf";
1572:   if (/\. (zip|7z|rar)$/i.test(clean)) return "archive";
1573:   if (/\. (xlsx?|csv)$/i.test(clean)) return "spreadsheet";
1574:   if (fileIsBrowserSearchable(clean)) return "text";
1575:   if (/^https?:\/\/\//i.test(clean)) return "webpage";
1576:   return "file";
1577: }

```

assistantAbsoluteUrl - lines 1579-1658

Item	Explanation
Source location	Lines 1579-1658 (80 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantAbsoluteUrl(url = "") {
Touches	filesystem
Calls detected	normalizeLinkTarget, looksLikeBareWebAddress, URL, assistantLinkRecordsFromValue, uniqueDownloads, collectDownloadsFromValue, forEach, push, assistantUrlKind, fileNameFromUrl, assistantUrlsFromTextValue, flattenSearchText
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 9 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1579: function assistantAbsoluteUrl(url = "") {
1580:   const target = normalizeLinkTarget(url, { assumeWeb: looksLikeBareWebAddress(url) ||
/^https?:\/\/\//i.test(url) });
1581:   if (!target) return "";
1582:   try {
1583:     return new URL(target, window.location.href).href;
1584:   } catch {
1585:     return target;
1586:   }
1587: }
1588:
1589: function assistantLinkRecordsFromValue(value, context = "") {
1590:   const records = [];
1591:   const downloads = uniqueDownloads(collectDownloadsFromValue(value, "Portfolio asset", []));
1592:   downloads.forEach((download) => {
1593:     const rawUrl = download.url || download.href || download.path || "";
1594:     if (!rawUrl) return;
1595:     records.push({
1596:       context,
1597:       description: download.description || "",
1598:       kind: download.kind || assistantUrlKind(rawUrl),
1599:       label: download.title || download.label || fileNameFromUrl(rawUrl),
1600:       url: assistantAbsoluteUrl(rawUrl)
1601:     });
1602:   });
1603:   return records;
1604: }
1605:
1606: function assistantUrlsFromTextValue(value, context = "") {
1607:   const text = flattenSearchText([value]);
1608:   const matches = text.match(/\b(?:https?:\/\/\|www\.)[^s<>"])+/gi) || [];
1609:   return [...new Set(matches.map((url) => url.replace(/[(),.;:!?]+$/g, "")))]
1610:     .slice(0, 16)
1611:     .map((url) => ({
1612:       context,

```

```

1613:         description: "",
1614:         kind: assistantUrlKind(url),
1615:         label: /github\.com/i.test(url) ? "GitHub repository or profile" : fileNameFromUrl(url),
1616:         type: "Public link",
1617:         url: assistantAbsoluteUrl(url)
1618:     }));
1619: }
1620:
1621: function assistantPublicProfileLinks() {
1622:     const records = [];
1623:     const add = (item = {}, context = "") => {
1624:         const rawUrl = item.url || item.href || item.path || "";
1625:         if (!rawUrl) return;
1626:         const label = item.label || item.title || fileNameFromUrl(rawUrl);
1627:         records.push({
1628:             context,
1629:             kind: item.kind || assistantUrlKind(rawUrl),
1630:             label,
1631:             text: item.description || "",
1632:             url: assistantAbsoluteUrl(rawUrl)
1633:         });
1634:     };
1635:
1636:     const current = normalizeProfile(profile || {});
1637:     if (current.resumeUrl) add({ label: "Resume", kind: "resume", url: current.resumeUrl }, "Professional
Profile");
1638:     if (current.githubUrl) add({ label: "GitHub", kind: "github", url: current.githubUrl }, "Professional
Profile");
1639:     if (current.linkedinUrl) add({ label: "LinkedIn", kind: "linkedin", url: current.linkedinUrl },
"Professional Profile");
1640:     if (current.websiteUrl) add({ label: "Website", kind: "webpage", url: current.websiteUrl },
"Professional Profile");
1641:
1642:     (siteSections || []).forEach((section) => {
1643:         (section.links || []).forEach((link) => add(link, section.title || "Portfolio section"));
1644:     });
1645:
1646:     return records.filter((record, index, array) => (
1647:         record.url && array.findIndex((item) => item.url === record.url && item.label === record.label) ===
index
1648:     ));
1649: }
1650:
1651: function assistantProjectEvidence(project = {}) {
1652:     const downloads = uniqueDownloads(collectDownloadsFromValue(project, "Project asset", []));
1653:     const downloadRecords = downloads.slice(0, 26).map((download) => {
1654:         const rawUrl = download.url || "";
1655:         return {
1656:             description: download.description || "",
1657:             kind: assistantUrlKind(rawUrl),
1658:             label: download.title || download.label || fileNameFromUrl(rawUrl),

```

assistantLinkRecordsFromValue - lines 1589-1604

Item	Explanation
Source location	Lines 1589-1604 (16 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantLinkRecordsFromValue(value, context = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	uniqueDownloads, collectDownloadsFromValue, forEach, push, assistantUrlKind, fileNameFromUrl, assistantAbsoluteUrl
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1589: function assistantLinkRecordsFromValue(value, context = "") {
1590:     const records = [];
1591:     const downloads = uniqueDownloads(collectDownloadsFromValue(value, "Portfolio asset", []));
1592:     downloads.forEach((download) => {
1593:         const rawUrl = download.url || download.href || download.path || "";
1594:         if (!rawUrl) return;
1595:         records.push({
1596:             context,
1597:             description: download.description || "",
1598:             kind: download.kind || assistantUrlKind(rawUrl),

```

```

1599:         label: download.title || download.label || fileNameFromUrl(rawUrl),
1600:         url: assistantAbsoluteUrl(rawUrl)
1601:     });
1602: });
1603: return records;
1604: }

```

assistantUrlsFromTextValue - lines 1606-1619

Item	Explanation
Source location	Lines 1606-1619 (14 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantUrlsFromTextValue(value, context = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	flattenSearchText, match, b, Set, map, replace, slice, assistantUrlKind, fileNameFromUrl, assistantAbsoluteUrl
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1606: function assistantUrlsFromTextValue(value, context = "") {
1607:   const text = flattenSearchText([value]);
1608:   const matches = text.match(/\b(?:https?:\/\/|www\.)[^s<>"])+/gi) || [];
1609:   return [...new Set(matches.map((url) => url.replace(/[.,:!?]+$/g, "")))]
1610:     .slice(0, 16)
1611:     .map((url) => ({
1612:       context,
1613:       description: "",
1614:       kind: assistantUrlKind(url),
1615:       label: /github\.com/i.test(url) ? "GitHub repository or profile" : fileNameFromUrl(url),
1616:       type: "Public link",
1617:       url: assistantAbsoluteUrl(url)
1618:     }));
1619: }

```

assistantPublicProfileLinks - lines 1621-1649

Item	Explanation
Source location	Lines 1621-1649 (29 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantPublicProfileLinks() {
Touches	filesystem
Calls detected	fileNameFromUrl, push, assistantUrlKind, assistantAbsoluteUrl, normalizeProfile, add, forEach, filter, findIndex
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1621: function assistantPublicProfileLinks() {
1622:   const records = [];
1623:   const add = (item = {}, context = "") => {
1624:     const rawUrl = item.url || item.href || item.path || "";
1625:     if (!rawUrl) return;
1626:     const label = item.label || item.title || fileNameFromUrl(rawUrl);
1627:     records.push({
1628:       context,
1629:       kind: item.kind || assistantUrlKind(rawUrl),
1630:       label,
1631:       text: item.description || "",
1632:       url: assistantAbsoluteUrl(rawUrl)

```

```

1633:     });
1634:   });
1635:
1636:   const current = normalizeProfile(profile || {});
1637:   if (current.resumeUrl) add({ label: "Resume", kind: "resume", url: current.resumeUrl }, "Professional
Profile");
1638:   if (current.githubUrl) add({ label: "GitHub", kind: "github", url: current.githubUrl }, "Professional
Profile");
1639:   if (current.linkedinUrl) add({ label: "LinkedIn", kind: "linkedin", url: current.linkedinUrl },
"Professional Profile");
1640:   if (current.websiteUrl) add({ label: "Website", kind: "webpage", url: current.websiteUrl },
"Professional Profile");
1641:
1642:   (siteSections || []).forEach((section) => {
1643:     (section.links || []).forEach((link) => add(link, section.title || "Portfolio section"));
1644:   });
1645:
1646:   return records.filter((record, index, array) => (
1647:     record.url && array.findIndex((item) => item.url === record.url && item.label === record.label) ===
index
1648:   ));
1649: }

```

assistantProjectEvidence - lines 1651-1667

Item	Explanation
Source location	Lines 1651-1667 (17 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantProjectEvidence(project = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	uniqueDownloads, collectDownloadsFromValue, slice, map, assistantUrlKind, fileNameFromUrl, assistantAbsoluteUrl, assistantUrlsFromTextValue, filter, findIndex
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1651: function assistantProjectEvidence(project = {}) {
1652:   const downloads = uniqueDownloads(collectDownloadsFromValue(project, "Project asset", []));
1653:   const downloadRecords = downloads.slice(0, 26).map((download) => {
1654:     const rawUrl = download.url || "";
1655:     return {
1656:       description: download.description || "",
1657:       kind: assistantUrlKind(rawUrl),
1658:       label: download.title || download.label || fileNameFromUrl(rawUrl),
1659:       type: download.type || "",
1660:       url: assistantAbsoluteUrl(rawUrl)
1661:     };
1662:   });
1663:   const textLinkRecords = assistantUrlsFromTextValue([project.summary, project.links,
project.portfolioView, project.portfolioView?.title || project.title];
1664:   return [...downloadRecords, ...textLinkRecords].filter((record, index, array) => (
1665:     record.url && array.findIndex((item) => item.url === record.url) === index
1666:   ));
1667: }

```

assistantQuestionIsCasual - lines 1669-1675

Item	Explanation
Source location	Lines 1669-1675 (7 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantQuestionIsCasual(question = "") {
Touches	filesystem
Calls detected	normalize
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1669: function assistantQuestionIsCasual(question = "") {
1670:   const clean = normalize(question);
1671:   return
1672:   /^(hi|hello|hey|good\s+(morning|afternoon|evening)|yo|sup|thanks|thank\s+you|ok|okay)\b/.test(clean)
1673:   || /\^(how\s+are\s+you|what\s+can\s+you\s+help.*|what\s+do\s+you\s+do|who\s+are\s+you|help|help\s+me|c
1674:   an\s+you\s+help|nice\s+to\s+meet\s+you)\??$/ .test(clean)
1675:   || /\^(what'\s+what\s+is|do\s+you\s+know)\s+my\s+name\??$/ .test(clean)
1676:   || /\^who\s+am\s+i\??$/ .test(clean);
1677: }
```

assistantQuestionHasPortfolioIntent - lines 1677-1682

Item	Explanation
Source location	Lines 1677-1682 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantQuestionHasPortfolioIntent(question = "") {
Touches	filesystem
Calls detected	normalize, assistantProjectTitleMatches, b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1677: function assistantQuestionHasPortfolioIntent(question = "") {
1678:   const clean = normalize(question);
1679:   return assistantProjectTitleMatches(question)
1680:   || /\b(maurice|otieno|portfolio|resume|github|linkedin|contact|email|phone|project|projects|repo|repo
1681:   sitory|repositories|file|files|document|documents|artifact|artifacts|link|links|download|open|show|where|built|b
1682:   uild|created|designed|implemented)\b/.test(clean)
1683:   || /\b(your|his|maurice's)\s+(project|projects|resume|github|linkedin|portfolio|work|email|phone|cont
1684:   act|repo|repository|files?|documents?|links?)\b/.test(clean);
1685: }
```

assistantConversationSuggestsPortfolioContext - lines 1684-1688

Item	Explanation
Source location	Lines 1684-1688 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantConversationSuggestsPortfolioContext(history = []) {
Touches	filesystem
Calls detected	slice, some, b, normalize
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1684: function assistantConversationSuggestsPortfolioContext(history = []) {
```

```

1685:   return history
1686:     .slice(-4)
1687:     .some((item) => /\b(maurice|otieno|portfolio|project|resume|github|linkedin|electronics|engineering)\
b/.test(normalize(item?.content)));
1688: }

```

assistantQuestionAllowsPublicLookup - lines 1690-1695

Item	Explanation
Source location	Lines 1690-1695 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantQuestionAllowsPublicLookup(question = "", intent = assistantQuestionIntent(question)) {
Touches	filesystem
Calls detected	assistantQuestionIntent, normalize, b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1690: function assistantQuestionAllowsPublicLookup(question = "", intent = assistantQuestionIntent(question)) {
1691:   const clean = normalize(question);
1692:   return intent === "general_engineering"
1693:     || intent === "general_knowledge"
1694:     || /\b(github|linkedin|repo|repository|repositories|public\s+link|website|web\s+page|profile|resume|u
ploadeds+file|document|source\s+code)\b/.test(clean);
1695: }

```

assistantQuestionLooksConceptual - lines 1697-1702

Item	Explanation
Source location	Lines 1697-1702 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantQuestionLooksConceptual(question = "") {
Touches	filesystem
Calls detected	normalize, assistantQuestionIsEngineeringRelated, b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1697: function assistantQuestionLooksConceptual(question = "") {
1698:   const clean = normalize(question);
1699:   if (!assistantQuestionIsEngineeringRelated(question)) return false;
1700:   return
/^(what|what\s+is|what\s+are|what's|define|explain|describe|how|why|compare|differentiate)\b/.test(clean)
1701:     || /\b(definition|meaning|basics|overview|introduction|difference\s+between|how\s+does|how\s+do|why\s
+does|why\s+do)\b/.test(clean);
1702: }

```

assistantQuestionIntent - lines 1704-1717

Item	Explanation
Source location	Lines 1704-1717 (14 source lines).
Purpose	Part of the public website runtime.

Item	Explanation
Declaration	function assistantQuestionIntent(question = "", history = assistantChatHistory) {
Touches	filesystem
Calls detected	assistantQuestionIsCasual, normalize, assistantQuestionHasPortfolioIntent, assistantConversationSuggestsPortfolioContext, b, assistantQuestionIsEngineeringRelated, assistantQuestionLooksConceptual
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 7 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1704: function assistantQuestionIntent(question = "", history = assistantChatHistory) {
1705:   if (assistantQuestionIsCasual(question)) return "general_conversation";
1706:   if (/^(what'?s|what\s+is|do\s+you\s+know)\s+my\s+name\??$/i.test(normalize(question))) ||
/who\s+am\s+i\??$/i.test(normalize(question))) {
1707:     return "general_conversation";
1708:   }
1709:   const hasPortfolioIntent = assistantQuestionHasPortfolioIntent(question);
1710:   if (!hasPortfolioIntent && assistantConversationSuggestsPortfolioContext(history) &&
/b(name|you|your|he|his|hlm|that|this|it)\b/i.test(normalize(question))) {
1711:     return "general_conversation";
1712:   }
1713:   if (!hasPortfolioIntent && assistantQuestionIsEngineeringRelated(question)) return
"general_engineering";
1714:   if (!hasPortfolioIntent && assistantQuestionLooksConceptual(question)) return "general_engineering";
1715:   if (hasPortfolioIntent) return "portfolio_specific";
1716:   return "general_knowledge";
1717: }

```

assistantQuestionTokens - lines 1719-1730

Item	Explanation
Source location	Lines 1719-1730 (12 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantQuestionTokens(question = "") {
Touches	filesystem, authentication/security data
Calls detected	Set, normalize, split, map, trim, filter, has
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1719: function assistantQuestionTokens(question = "") {
1720:   const stopWords = new Set([
1721:     "about", "again", "also", "and", "are", "can", "could", "does", "for", "from", "give", "have", "his",
1722:     "into", "link", "links", "maurice", "me", "my", "open", "otieno", "please", "portfolio", "project",
1723:     "projects", "show", "tell", "that", "the", "their", "this", "what", "where", "which", "who", "with",
1724:     "work", "your"
1725:   ]);
1726:   return normalize(question)
1727:     .split(/[\a-z0-9+#+.]+/i)
1728:     .map((token) => token.trim())
1729:     .filter((token) => token.length > 2 && !stopWords.has(token));
1730: }

```

assistantSpecificSourceTokens - lines 1732-1739

Item	Explanation
Source location	Lines 1732-1739 (8 source lines).

Item	Explanation
Purpose	Part of the public website runtime.
Declaration	function assistantSpecificSourceTokens(question = "") {
Touches	authentication/security data
Calls detected	Set, assistantQuestionTokens, filter, has
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1732: function assistantSpecificSourceTokens(question = "") {
1733:   const genericTokens = new Set([
1734:     "built", "build", "created", "designed", "did", "done", "engineering", "file", "files", "hardware",
1735:     "item", "items", "section", "sections", "system", "systems", "thing", "things", "tool", "tools",
1736:     "use", "used", "using", "website"
1737:   ]);
1738:   return assistantQuestionTokens(question).filter((token) => !genericTokens.has(token));
1739: }

```

assistantPromptTargetsProjectLanding - lines 1741-1746

Item	Explanation
Source location	Lines 1741-1746 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantPromptTargetsProjectLanding(question = "") {
Touches	filesystem
Calls detected	normalize, b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1741: function assistantPromptTargetsProjectLanding(question = "") {
1742:   const clean = normalize(question);
1743:   return /\b(open|show|view|go\s+to|take\s+me\s+to)\b/.test(clean)
1744:     && /\b(project|overview|page)\b/.test(clean)
1745:     && /\b(code|source|repo|repository|github|file|files|document|documents|test|tests|result|results|pc
b|schematic|simulation|tool|tools|resume|linkedin)\b/.test(clean);
1746: }

```

assistantEntryRelationScore - lines 1748-1762

Item	Explanation
Source location	Lines 1748-1762 (15 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantEntryRelationScore(result = {}, question = "") {
Touches	filesystem, authentication/security data
Calls detected	assistantSpecificSourceTokens, normalize, filter, join, includes, b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1748: function assistantEntryRelationScore(result = {}, question = "") {
1749:   const tokens = assistantSpecificSourceTokens(question);
1750:   const haystack = normalize([result.title, result.context, result.type, result.text,
result.url].filter(Boolean).join(" "));
1751:   const tokenHits = tokens.filter((token) => haystack.includes(token)).length;
1752:   let relation = tokenHits * 30;
1753:
1754:   if (tokens.length && tokenHits === tokens.length) relation += 25;
1755:   if (Number(result.score || 0) >= 120) relation += 15;
1756:   if (Number(result.score || 0) >= 80) relation += 8;
1757:   if (result.kind === "project") relation += 12;
1758:   if (result.kind === "file" && /\b(file|files|download|document|documents|code|source|github|repo|repository|test|tests|result|results|pcb|schematic|simulation)\b/.test(normalize(question))) {
1759:     relation += 16;
1760:   }
1761:   return relation;
1762: }

```

assistantNamedProjectTopicTokens - lines 1764-1776

Item	Explanation
Source location	Lines 1764-1776 (13 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantNamedProjectTopicTokens(question = "", namedIds = assistantNamedProjectIds(question)) {
Touches	filesystem, authentication/security data
Calls detected	assistantNamedProjectIds, Set, map, find, filter, forEach, normalize, split, add, assistantProjectTitleAcronyms, assistantSpecificSourceTokens, has
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1764: function assistantNamedProjectTopicTokens(question = "", namedIds = assistantNamedProjectIds(question)) {
1765:   const projectVocabulary = new Set();
1766:   namedIds
1767:     .map((id) => projects.find((project) => project.id === id))
1768:     .filter(Boolean)
1769:     .forEach((project) => {
1770:       const title = project.portfolioView?.title || project.title || project.id;
1771:       normalize(title).split(/\s+/).filter((word) => word.length > 2).forEach((word) =>
projectVocabulary.add(word));
1772:       assistantProjectTitleAcronyms(title).forEach((acronym) => projectVocabulary.add(acronym));
1773:       projectVocabulary.add(normalize(project.id));
1774:     });
1775:   return assistantSpecificSourceTokens(question).filter((token) => !projectVocabulary.has(token));
1776: }

```

assistantSourcesForDisplay - lines 1778-1813

Item	Explanation
Source location	Lines 1778-1813 (36 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantSourcesForDisplay(question = "", results = [], intent = assistantQuestionIntent(question)) {
Touches	filesystem, authentication/security data
Calls detected	assistantQuestionIntent, assistantSpecificSourceTokens, assistantNamedProjectIds, Set, assistantNamedProjectTopicTokens, b, normalize, assistantPromptTargetsProjectLanding, map, assistantEntryRelationScore, filter, has

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 14 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1778: function assistantSourcesForDisplay(question = "", results = [], intent =
assistantQuestionIntent(question)) {
1779:   if (intent !== "portfolio_specific") return [];
1780:   const tokens = assistantSpecificSourceTokens(question);
1781:   const namedProjectIdList = assistantNamedProjectIds(question);
1782:   const namedProjectIds = new Set(namedProjectIdList);
1783:   const namedProjectTopicTokens = assistantNamedProjectTopicTokens(question, namedProjectIdList);
1784:   const asksForLinks = /\b(open|show|where|link|links|github|linkedin|resume|download|file|files|repo|rep
ository|source\s+code)\b/.test(normalize(question));
1785:   const asksForProjectLanding = assistantPromptTargetsProjectLanding(question);
1786:   const pageLinkIntent = /\b(resume|github|linkedin|contact|email|phone)\b/.test(normalize(question));
1787:   const asksForSpecificArtifact = /\b(code|source|repo|repository|github|file|files|document|documents|te
st|tests|result|results|pcb|schematic|simulation|tool|tools)\b/.test(normalize(question));
1788:
1789:   if (!namedProjectIds.size && !pageLinkIntent && tokens.length === 0) return [];
1790:
1791:   const filtered = results
1792:     .map((result) => ({ ...result, relation: assistantEntryRelationScore(result, question) })))
1793:     .filter((result) => {
1794:       if (!result) return false;
1795:       if (namedProjectIds.size && !namedProjectIds.has(result.projectId)) return false;
1796:       if (asksForProjectLanding) return result.kind === "project";
1797:       const haystack = normalize([result.title, result.context, result.type, result.text,
result.url].filter(Boolean).join(" "));
1798:       const tokenHits = tokens.filter((token) => haystack.includes(token)).length;
1799:       const topicHits = namedProjectTopicTokens.filter((token) => haystack.includes(token)).length;
1800:       if (namedProjectIds.size && namedProjectTopicTokens.length && result.kind !== "project" &&
topicHits === 0) return false;
1801:       if (namedProjectIds.size && namedProjectTopicTokens.length && result.kind === "project" &&
asksForSpecificArtifact) return false;
1802:       if (namedProjectIds.size && result.kind === "project") return true;
1803:       if (pageLinkIntent && tokenHits >= 1) return true;
1804:       if (!tokens.length) return false;
1805:       if (tokenHits >= Math.min(2, tokens.length)) return true;
1806:       if (asksForLinks && tokenHits >= 1 && result.relation >= 35) return true;
1807:       return result.relation >= 55;
1808:     });
1809:
1810:   return filtered
1811:     .sort((a, b) => b.relation - a.relation || Number(b.score || 0) - Number(a.score || 0))
1812:     .slice(0, asksForLinks ? 4 : 2);
1813: }

```

assistantContextForQuestion - lines 1815-1890

Item	Explanation
Source location	Lines 1815-1890 (76 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantContextForQuestion(question = "", intent = assistantQuestionIntent(question), knownResults = null) {
Touches	filesystem
Calls detected	assistantQuestionIntent, includes, searchResultsFor, slice, Set, map, filter, assistantNamedProjectIds, find, b, normalize, assistantPublicProfileLinks
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1815: function assistantContextForQuestion(question = "", intent = assistantQuestionIntent(question),
knownResults = null) {
1816:   const directResults = ["general_engineering", "general_knowledge",
"general_conversation"].includes(intent)

```

```

1817:     ? []
1818:     : (knownResults || searchResultsFor(question)).slice(0, 8);
1819: const projectIds = [...new Set([
1820:   ...directResults.map((item) => item.projectId).filter(Boolean),
1821:   ...assistantNamedProjectIds(question)
1822: ])]);
1823: const matchedProjects = projectIds
1824:   .map((id) => projects.find((project) => project.id === id))
1825:   .filter(Boolean)
1826:   .slice(0, 5);
1827: const includePortfolioLinkContext = intent === "portfolio_specific";
1828: const wantsPublicCode = /\b(github|repo|repository|repositories|source\s|code|code|snippet|firmware|ver
ilog|systemverilog|python|javascript)\b/.test(normalize(question));
1829: const publicProfileFetches = includePortfolioLinkContext
1830:   ? assistantPublicProfileLinks()
1831:   .map((item) => ({ ...item, question }));
1832:   .sort((a, b) => {
1833:     if (!wantsPublicCode) return 0;
1834:     const aGithub = /github/i.test(`${a.kind || ""} ${a.url || ""}`) ? 1 : 0;
1835:     const bGithub = /github/i.test(`${b.kind || ""} ${b.url || ""}`) ? 1 : 0;
1836:     return bGithub - aGithub;
1837:   })
1838:   : [];
1839: const resultFetches = directResults.map((result) => ({
1840:   context: result.context,
1841:   kind: assistantUrlKind(result.url),
1842:   title: result.title,
1843:   type: result.type,
1844:   url: assistantAbsoluteUrl(result.url || ""),
1845:   question
1846: }));
1847: const projectFetches = matchedProjects.flatMap((project) => assistantProjectEvidence(project)
1848:   .filter((item) => ["text", "webpage", "github", "linkedin", "resume_text"].includes(item.kind))
1849:   .slice(0, 6)
1850:   .map((item) => ({ ...item, question })));
1851: const sourceFetches = [
1852:   ...(wantsPublicCode ? publicProfileFetches : []),
1853:   ...resultFetches,
1854:   ...(wantsPublicCode ? [] : publicProfileFetches),
1855:   ...projectFetches
1856: ].filter((item) => item.url);
1857: const uniqueSourceFetches = sourceFetches.filter((item, index, array) => (
1858:   array.findIndex((candidate) => candidate.url === item.url) === index
1859: )).slice(0, 14);
1860: return {
1861:   categories: categories.map((category) => ({ id: category.id, label: category.label })),
1862:   intent,
1863:   profile: normalizeProfile(profile || {}),
1864:   knowledgeManifest: {
1865:     publicProfiles: includePortfolioLinkContext ? assistantPublicProfileLinks() : [],
1866:     publicSourcePolicy: "Only public profile and project links shown in the portfolio are used. GitHub
repository links can be expanded into repository metadata, README text, and selected public source files when
the visitor asks for code.",
1867:     matchedProjectEvidence: matchedProjects.map((project) => ({
1868:       evidence: assistantProjectEvidence(project),
1869:       projectId: project.id,
1870:       title: project.portfolioView?.title || project.title
1871:     })),
1872:     note: "Image records include filenames, captions, descriptions, and neighboring portfolio text. Use
a vision-capable backend before claiming visual details not present in captions or text."
1873:   },
1874:   portfolioContextPolicy: intent !== "portfolio_specific"
1875:     ? "Answer from general knowledge first. Do not lead with the portfolio owner's project context
unless the visitor explicitly asks to connect the concept to the portfolio."
1876:     : "Answer from the configured portfolio context first. Do not invent project details outside the
supplied context.",
1877:   projects: matchedProjects.map(assistantProjectSummary),
1878:   question,
1879:   results: directResults.map((result) => ({
1880:     context: result.context,
1881:     kind: result.kind,
1882:     text: searchSnippet(result.text, question),
1883:     title: result.title,
1884:     type: result.type,
1885:     url: result.url
1886: })),
1887:   sourceFetches: uniqueSourceFetches,
1888:   siteSections: siteSections.filter(siteSectionRenderable).map((section) => ({ id: section.id, title:
section.title })))
1889:   };
1890: }

```

assistantRemember - lines 1892-1901

Item	Explanation
Source location	Lines 1892-1901 (10 source lines).
Purpose	Part of the public website runtime.

Item	Explanation
Declaration	function assistantRemember(role, content = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, push, slice
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1892: function assistantRemember(role, content = "") {
1893:   const cleanRole = role === "assistant" ? "assistant" : "user";
1894:   const cleanContent = String(content || "").trim();
1895:   if (!cleanContent) return;
1896:   assistantChatHistory.push({
1897:     role: cleanRole,
1898:     content: cleanContent.slice(0, 1400)
1899:   });
1900:   assistantChatHistory = assistantChatHistory.slice(-10);
1901: }

```

assistantConversationForRequest - lines 1903-1912

Item	Explanation
Source location	Lines 1903-1912 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantConversationForRequest(currentQuestion = "", priorConversation = assistantChatHistory) {
Touches	Mostly local calculations or local UI state.
Calls detected	slice, map, push
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1903: function assistantConversationForRequest(currentQuestion = "", priorConversation = assistantChatHistory)
{
1904:   const history = priorConversation.slice(-8).map((item) => ({
1905:     role: item.role === "assistant" ? "assistant" : "user",
1906:     content: String(item.content || "").slice(0, 1200)
1907:   }));
1908:   if (currentQuestion) {
1909:     history.push({ role: "user", content: String(currentQuestion).slice(0, 1200) });
1910:   }
1911:   return history;
1912: }

```

assistantFallbackResults - lines 1914-1972

Item	Explanation
Source location	Lines 1914-1972 (59 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantFallbackResults(question = "", intent = assistantQuestionIntent(question)) {
Touches	filesystem
Calls detected	assistantQuestionIntent, normalize, trim, assistantNamedProjectIds, searchResultsFor, filter, includes, slice, map, find, slugLabel, flattenSearchText

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 9 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1914: function assistantFallbackResults(question = "", intent = assistantQuestionIntent(question)) {
1915:   if (intent !== "portfolio_specific") return [];
1916:   const clean = normalize(question).trim();
1917:   if (!clean) return [];
1918:   const namedProjectIds = assistantNamedProjectIds(question);
1919:   let results = searchResultsFor(clean);
1920:   if (namedProjectIds.length && results.length) {
1921:     const focusedResults = results.filter((result) => namedProjectIds.includes(result.projectId));
1922:     if (focusedResults.length) return focusedResults.slice(0, 6);
1923:   }
1924:   if (results.length) return results.slice(0, 6);
1925:
1926:   if (namedProjectIds.length) {
1927:     return namedProjectIds
1928:       .map((id) => projects.find((project) => project.id === id))
1929:       .filter(Boolean)
1930:       .map((project) => ({
1931:         context: slugLabel(project.category),
1932:         kind: "project",
1933:         projectId: project.id,
1934:         text: flattenSearchText([project.summary, project.focus, project.tools, project.languages]),
1935:         title: project.portfolioView?.title || project.title,
1936:         type: "Project"
1937:       })))
1938:     .slice(0, 6);
1939:   }
1940:
1941:   const category = categories.find((item) => clean.includes(normalize(item.label)) ||
clean.includes(normalize(item.id)));
1942:   if (category) {
1943:     return projects
1944:       .filter((project) => project.category === category.id)
1945:       .map((project) => ({
1946:         context: category.label,
1947:         kind: "project",
1948:         projectId: project.id,
1949:         text: flattenSearchText([project.summary, project.focus, project.tools, project.languages]),
1950:         title: project.portfolioView?.title || project.title,
1951:         type: "Project"
1952:       })))
1953:     .slice(0, 6);
1954:   }
1955:
1956:   if (/resume|professional|contact|email|phone|github|linkedin|profile/.test(clean)) {
1957:     return searchResultsFor("resume contact github professional profile").slice(0, 6);
1958:   }
1959:
1960:   if (/project|work|portfolio|hardware|design/.test(clean)) {
1961:     return projects.slice(0, 6).map((project) => ({
1962:       context: slugLabel(project.category),
1963:       kind: "project",
1964:       projectId: project.id,
1965:       text: project.summary || "",
1966:       title: project.portfolioView?.title || project.title,
1967:       type: "Project"
1968:     }));
1969:   }
1970:
1971:   return [];
1972: }

```

assistantQuestionIsEngineeringRelated - lines 1974-1977

Item	Explanation
Source location	Lines 1974-1977 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantQuestionIsEngineeringRelated(question = "") {
Touches	filesystem
Calls detected	normalize, b

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1974: function assistantQuestionIsEngineeringRelated(question = "") {
1975:   const clean = normalize(question);
1976:   return /\b(op\s*amp|amplifier|analog|mixed\s*signal|adc|dac|filter|vco|oscillator|pwm|charger|rectifier
|regulator|buck|boost|ldo|mosfet|bjt|transistor|diode|pcb|schematic|layout|ground|noise|frequency|gain|phase|lts
pice|kicad|vivado|verilog|vhdl|fpga|asic|stm32|mcu|microcontroller|embedded|firmware|rtos|i2c|spi|uart|sensor|co
ntrol|signal|circuit|electronics|hardware|power)\b/.test(clean);
1977: }

```

assistantGeneralEngineeringAnswer - lines 1979-2091

Item	Explanation
Source location	Lines 1979-2091 (113 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantGeneralEngineeringAnswer(question = "") {
Touches	filesystem
Calls detected	normalize, assistantQuestionIsEngineeringRelated, join, b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 9 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1979: function assistantGeneralEngineeringAnswer(question = "") {
1980:   const clean = normalize(question);
1981:   if (!assistantQuestionIsEngineeringRelated(question)) return "";
1982:
1983:   if (!\bembedded\s+systems?\b/.test(clean)) {
1984:     return [
1985:       "Embedded systems is a field of engineering focused on computers that are built into larger
products, machines, instruments, or control systems.",
1986:       "",
1987:       "Unlike a laptop or desktop computer, an embedded system is usually designed for a specific job. It
combines hardware, firmware, sensors, actuators, communication interfaces, and timing constraints so the
product can make measurements, control outputs, or respond to events reliably.",
1988:       "",
1989:       "Common examples include:",
1990:       "- A microcontroller controlling a motor drive.",
1991:       "- A sensor node collecting temperature, pressure, or motion data.",
1992:       "- A medical device running real-time measurement firmware.",
1993:       "- An automotive controller reading sensors and driving actuators.",
1994:       "",
1995:       "The important engineering ideas are timing, interrupts, power use, peripheral interfaces,
reliability, and how the firmware proves the hardware is behaving correctly."
1996:     ].join("\n");
1997:   }
1998:
1999:   if (!\b(op\s*amp|amplifier|fully differential|differential)\b/.test(clean)) {
2000:     return [
2001:       "A useful way to think about an op amp is as a high-gain error-correcting block. With negative
feedback, the circuit around it sets the useful behavior: gain, filtering, buffering, summing, integration, or
differential conversion.",
2002:       "",
2003:       "For a fully differential op amp:",
2004:       "- The signal is carried on two opposite-phase outputs instead of one output referenced to
ground.",
2005:       "- The differential output improves noise immunity and dynamic range.",
2006:       "- A common-mode feedback loop is normally needed to hold the average output voltage at the desired
common-mode level.",
2007:       "",
2008:       "In portfolio terms, this connects most naturally to analog and mixed-signal design, especially
filters, VCO control paths, ADC front ends, and precision measurement circuits."
2009:     ].join("\n");

```

```

2010: }
2011:
2012: if (/b(pwm|vco|oscillator|frequency)\b/.test(clean)) {
2013:     return [
2014:         "PWM-driven VCO work is about converting a digital timing signal into an analog control quantity
and then into a frequency.",
2015:         "",
2016:         "The usual signal chain is:",
2017:         "- PWM duty cycle encodes the control value.",
2018:         "- A low-pass filter turns the PWM waveform into an average voltage.",
2019:         "- The VCO maps that voltage to oscillation frequency.",
2020:         "- Tests compare duty cycle, control voltage, frequency range, ripple, and stability.",
2021:         "",
2022:         "That is a mixed-signal problem because digital timing, analog filtering, oscillator behavior, and
measurement all interact."
2023:     ].join("\n");
2024: }
2025:
2026: if (/b(fpga|asic|verilog|vhdl|vivado|digital)\b/.test(clean)) {
2027:     return [
2028:         "FPGA and ASIC work both implement digital hardware, but the engineering tradeoffs are different.",
2029:         "",
2030:         "- FPGA design is reconfigurable, fast to test, and useful for prototyping or deployment when
flexibility matters.",
2031:         "- ASIC design targets a fixed silicon implementation, so verification, timing closure, power,
area, and design-for-test become much more permanent decisions.",
2032:         "- Verilog or VHDL describes hardware structure and behavior; it is not software running line by
line.",
2033:         "",
2034:         "For a recruiter, strong evidence includes block diagrams, RTL, simulation waveforms, timing
reports, verification plans, and notes explaining design tradeoffs."
2035:     ].join("\n");
2036: }
2037:
2038: if (/b(stm32|mcu|microcontroller|embedded|firmware|rtos|i2c|spi|uart)\b/.test(clean)) {
2039:     return [
2040:         "Embedded work is strongest when the firmware is tied clearly to hardware behavior.",
2041:         "",
2042:         "A good explanation usually covers:",
2043:         "- The MCU role in the system.",
2044:         "- The peripherals used, such as ADC, PWM, timers, I2C, SPI, UART, or GPIO.",
2045:         "- How timing, interrupts, sampling, power, and fault handling were managed.",
2046:         "- What was measured to prove the system worked.",
2047:         "",
2048:         "STM32CubeIDE is useful because it helps configure peripherals, generate startup code, debug
firmware, and manage the build flow around STM32 devices."
2049:     ].join("\n");
2050: }
2051:
2052: if (/b(pcb|layout|kicad|schematic|ground|noise)\b/.test(clean)) {
2053:     return [
2054:         "PCB design is where the schematic becomes a physical electrical system.",
2055:         "",
2056:         "The important ideas are:",
2057:         "- Current returns matter as much as current paths.",
2058:         "- Grounding, decoupling, trace width, component placement, and loop area affect noise and
stability.",
2059:         "- Analog, power, and switching sections should be arranged so noisy currents do not corrupt
sensitive measurements.",
2060:         "- A good portfolio PCB section should show schematic intent, layout decisions, board files, bring-
up notes, and test results.",
2061:         "",
2062:         "KiCad evidence is strongest when the design files are paired with clear measurements or bring-up
photos."
2063:     ].join("\n");
2064: }
2065:
2066: if (/b(charger|rectifier|ac\s*\to\s*\dc|regulator|buck|boost|ldo|power)\b/.test(clean)) {
2067:     return [
2068:         "An AC-to-DC charger or power supply usually has a few major stages.",
2069:         "",
2070:         "- Input protection and filtering handle transients and noise.",
2071:         "- Rectification converts AC into pulsating DC.",
2072:         "- Bulk capacitance smooths the rectified waveform.",
2073:         "- Regulation sets the desired output voltage or current.",
2074:         "- Load testing checks ripple, thermal behavior, regulation, and fault response.",
2075:         "",
2076:         "For portfolio presentation, the best proof is a clear schematic, expected waveforms, measured
output behavior, and a short explanation of component choices."
2077:     ].join("\n");
2078: }
2079:
2080: return [
2081:     "I can answer this as a general electronics or hardware-engineering question, even if it is not
directly tied to one saved project yet.",
2082:     "",
2083:     "The best way to approach it is:",
2084:     "- Define the signal or energy flow.",
2085:     "- Identify the active devices, passive networks, and control points.",
2086:     "- State what should be measured to prove the design works.",
2087:     "- Connect the explanation back to a project artifact: schematic, code, test data, PCB, or
simulation."

```



```

2088:     "",
2089:     "If you ask about a specific circuit, tool, waveform, or failure mode, I can make the explanation
more concrete."
2090:   ].join("\n");
2091: }

```

assistantLocalAnswer - lines 2093-2188

Item	Explanation
Source location	Lines 2093-2188 (96 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantLocalAnswer(question = "", results = [], intent = assistantQuestionIntent(question)) {
Touches	filesystem
Calls detected	assistantQuestionIntent, normalizeProfile, normalize, b, join, assistantGeneralEngineeringAnswer, replace, Set, map, filter, find, slice
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 11 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2093: function assistantLocalAnswer(question = "", results = [], intent = assistantQuestionIntent(question)) {
2094:   const ownerName = normalizeProfile(profile || {}).displayName || "the portfolio owner";
2095:   if (intent === "general_conversation") {
2096:     const clean = normalize(question);
2097:     if (/^(hi|hello|hey|yo|sup)\b/.test(clean)) {
2098:       return "Hi, what can I do for you?";
2099:     }
2100:     if (/^(what'?s|what\s+is|do\s+you\s+know)\s+my\s+name\??$/i.test(clean) ||
/who\s+am\s+i\??$/i.test(clean)) {
2101:       return `I am an AI agent for ${ownerName}'s portfolio. I know the portfolio owner is ${ownerName},
but I do not know a visitor's personal name unless they tell me.`;
2102:     }
2103:     if (/^(who are you|what are you)\b/.test(clean)) {
2104:       return `I am an AI agent for ${ownerName}'s portfolio. I can help with projects, resume links,
public links, files, and related engineering questions.`;
2105:     }
2106:     return [
2107:       `I am here with you. I can help explore ${ownerName}'s portfolio, explain projects, summarize
files, open relevant sections, or answer related engineering questions.`,
2108:       "",
2109:       "You can ask things like:",
2110:       "- What is embedded systems?",
2111:       "- Show me a project.",
2112:       "- What tools were used?",
2113:       "- Explain the difference between FPGA and ASIC design."
2114:     ].join("\n");
2115:   }
2116:   if (intent === "general_engineering") {
2117:     const generalAnswer = assistantGeneralEngineeringAnswer(question);
2118:     if (generalAnswer) return generalAnswer;
2119:     return [
2120:       "I can answer that as a general engineering question, but I need a little more detail to make the
answer useful.",
2121:       "",
2122:       "Try asking about a specific circuit, device, tool, protocol, waveform, or failure mode."
2123:     ].join("\n");
2124:   }
2125:   if (intent === "general_knowledge") {
2126:     const generalAnswer = assistantGeneralEngineeringAnswer(question);
2127:     if (generalAnswer) return generalAnswer;
2128:     return [
2129:       "I can answer that as a general question from the local fallback, but the live model will usually
give a deeper answer when it is reachable.",
2130:       "",
2131:       "Short answer:",
2132:       `~ ${question.replace(/\?+$/, "")} is a topic I can discuss by breaking it into definition,
purpose, main parts, and practical examples.`
2133:     ].join("\n");
2134:     if (question.toLowerCase().includes(ownerName.toLowerCase())) {
2135:       return `If the question connects to Maurice's portfolio, I will use the saved project context and links
as evidence.`;
2136:     }
2137:     return `If it is broader than the portfolio, I will answer it as a general engineering or technology
concept first.`;
2138:   }
2139: }

```

```

2140:   if (!projects.length) return "The project catalog is still loading. Try again in a moment.";
2141:   if (!results.length) {
2142:     return [
2143:       `I could not find a strong portfolio match for "${question}".`,
2144:       "",
2145:       "Try asking about:",
2146:       "- A project name or category.",
2147:       "- A tool such as LTspice, KiCad, Vivado, STM32CubeIDE, or GitHub.",
2148:       "- A hardware topic such as op amps, PCB testing, VCOs, embedded firmware, FPGA, ASIC, or power
circuits."
2149:     ].join("\n");
2150:   }
2151:
2152:   const projectIds = [...new Set(results.map((item) => item.projectId).filter(Boolean))];
2153:   const matchedProjects = projectIds
2154:     .map((id) => projects.find((project) => project.id === id))
2155:     .filter(Boolean);
2156:   const projectNames = matchedProjects.map((project) => project.portfolioView?.title ||
project.title).slice(0, 3);
2157:   const fileCount = results.filter((item) => item.kind === "file").length;
2158:   const sectionCount = results.filter((item) => item.kind === "section" || item.kind ===
"subsection").length;
2159:   const pageCount = results.filter((item) => item.kind === "page").length;
2160:   const parts = [];
2161:
2162:   if (projectNames.length) {
2163:     parts.push(`The strongest portfolio match is ${projectNames.join(projectNames.length > 2 ? ", " : "
and ")}.`);
2164:   } else if (pageCount) {
2165:     parts.push("The strongest matches are in the profile, resume, contact, or portfolio information
sections.");
2166:   } else {
2167:     parts.push(`I found ${results.length} relevant portfolio item${results.length === 1 ? "" : "s"}.`);
2168:   }
2169:
2170:   parts.push("");
2171:   parts.push("What I found:");
2172:   if (sectionCount || fileCount) {
2173:     parts.push(`- ${sectionCount} section match${sectionCount === 1 ? "" : "es"}.`);
2174:     if (fileCount) parts.push(`- ${fileCount} file or link match${fileCount === 1 ? "" : "es"}.`);
2175:   }
2176:   if (pageCount) parts.push(`- ${pageCount} page or profile match${pageCount === 1 ? "" : "es"}.`);
2177:   if (projectNames.length) {
2178:     parts.push(`- Best project candidate${projectNames.length === 1 ? "" : "s"}: ${projectNames.join(",
")}.`);
2179:   }
2180:
2181:   const firstSnippet = searchSnippet(results[0].text, question);
2182:   if (firstSnippet) {
2183:     parts.push("");
2184:     parts.push("Why it matched:");
2185:     parts.push(firstSnippet);
2186:   }
2187:   return parts.join("\n");
2188: }

```

assistantSourcesMarkup - lines 2190-2208

Item	Explanation
Source location	Lines 2190-2208 (19 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantSourcesMarkup(results = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	slice, map, set, escapeHtml, assistantSourceLabel, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2190: function assistantSourcesMarkup(results = []) {
2191:   const sources = results.slice(0, 5);
2192:   if (!sources.length) return "";
2193:   const sourceButtons = sources.map((source) => {
2194:     const id = String(assistantSourceCounter++);
2195:     assistantSourceMap.set(id, source);
2196:     return `

```

```

2197:         <button type="button" data-ai-source-
id="${id}">${escapeHtml(assistantSourceLabel(source))}</button>
2198:     `;
2199:   }).join("");
2200:   return `
2201:     <div class="ai-source-panel">
2202:       <p class="ai-source-heading">Relevant portfolio links</p>
2203:       <div class="ai-source-list" aria-label="Assistant sources">
2204:         ${sourceButtons}
2205:       </div>
2206:     </div>
2207:   `;
2208: }

```

setAssistantStatus - lines 2210-2214

Item	Explanation
Source location	Lines 2210-2214 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function setAssistantStatus(message, state = "ready") {
Touches	browser/Electron DOM
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2210: function setAssistantStatus(message, state = "ready") {
2211:   if (!aiAssistantStatus) return;
2212:   aiAssistantStatus.textContent = message;
2213:   aiAssistantStatus.dataset.state = state;
2214: }

```

assistantDesktopDockedHeight - lines 2216-2221

Item	Explanation
Source location	Lines 2216-2221 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function assistantDesktopDockedHeight() {
Touches	Mostly local calculations or local UI state.
Calls detected	getComputedStyle, getPropertyValue, trim, endsWith, min, max
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2216: function assistantDesktopDockedHeight() {
2217:   if (!aiAssistantPanel) return 360;
2218:   const value = getComputedStyle(aiAssistantPanel).getPropertyValue("--ai-console-rest-height").trim();
2219:   if (value.endsWith("px")) return Number.parseFloat(value) || 360;
2220:   return Math.min(Math.max(window.innerHeight * 0.38, 340), 390);
2221: }

```

updateAssistantPanelGrowth - lines 2223-2246

Item	Explanation
Source location	Lines 2223-2246 (24 source lines).
Purpose	Updates UI, state, cache, or stored data.
Declaration	function updateAssistantPanelGrowth() {
Touches	browser/Electron DOM, filesystem
Calls detected	matchMedia, removeProperty, remove, assistantDesktopDockedHeight, getBoundingClientRect, querySelector, querySelectorAll, reduce, min, round, max, setProperty
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2223: function updateAssistantPanelGrowth() {
2224:   if (!aiAssistantPanel || !aiAssistantLog || !aiAssistantForm) return;
2225:   const isDesktop = window.matchMedia("(min-width: 901px)").matches;
2226:   if (!isDesktop) {
2227:     aiAssistantPanel.style.removeProperty("--ai-console-live-height");
2228:     aiAssistantPanel.classList.remove("ai-assistant-panel-expanded");
2229:     return;
2230:   }
2231:
2232:   const dockedHeight = assistantDesktopDockedHeight();
2233:   const formHeight = aiAssistantForm.getBoundingClientRect().height || 46;
2234:   const clearHeight = aiClearChatButton?.getBoundingClientRect().height || 38;
2235:   const titleHeight = document.querySelector("#ai-assistant-title")?.getBoundingClientRect().height ||
34;
2236:   const chromeHeight = titleHeight + formHeight + clearHeight + 92;
2237:   const messages = [...aiAssistantLog.querySelectorAll(".ai-message")];
2238:   const messageGap = messages.length > 1 ? (messages.length - 1) * 10 : 0;
2239:   const messageHeight = messages.reduce((sum, message) => sum + message.getBoundingClientRect().height,
0) + messageGap;
2240:   const contentHeight = messageHeight + chromeHeight;
2241:   const maxHeight = Math.min(window.innerHeight * 0.76, 720);
2242:   const nextHeight = Math.round(Math.max(dockedHeight, Math.min(contentHeight, maxHeight)));
2243:
2244:   aiAssistantPanel.style.setProperty("--ai-console-live-height", `${nextHeight}px`);
2245:   aiAssistantPanel.classList.toggle("ai-assistant-panel-expanded", nextHeight > dockedHeight + 12);
2246: }

```

renderAssistantInline - lines 2248-2250

Item	Explanation
Source location	Lines 2248-2250 (3 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderAssistantInline(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	renderInlineMath, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2248: function renderAssistantInline(value = "") {
2249:   return renderInlineMath(value).replace(/\^*([\^*]+)\^*/g, "<strong>$1</strong>");
2250: }

```

renderAssistantAnswerContent - lines 2252-2304

Item	Explanation
Source location	Lines 2252-2304 (53 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderAssistantAnswerContent(content = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	replace, split, push, map, renderAssistantInline, join, escapeHtml, forEach, trim, match, flushCode, flushBullets
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 7 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2252: function renderAssistantAnswerContent(content = "") {
2253:   const lines = String(content || "").replace(/\r\n?/g, "\n").split("\n");
2254:   const html = [];
2255:   let bulletItems = [];
2256:   let codeBlock = null;
2257:   const flushBullets = () => {
2258:     if (!bulletItems.length) return;
2259:     html.push(`<ul>${bulletItems.map((item) =>
`<li>${renderAssistantInline(item)}</li>`).join("")}</ul>`);
2260:     bulletItems = [];
2261:   };
2262:   const flushCode = () => {
2263:     if (!codeBlock) return;
2264:     const language = codeBlock.language ? ` data-language="${escapeHtml(codeBlock.language)}" ` : "";
2265:     html.push(`<pre class="ai-code-
block"${language}><code>${escapeHtml(codeBlock.lines.join("\n"))}</code></pre>`);
2266:     codeBlock = null;
2267:   };
2268:
2269:   lines.forEach((line) => {
2270:     const trimmed = line.trim();
2271:     const fence = trimmed.match(/^````([a-z0-9_+.#-]*)\s*$/i);
2272:     if (fence) {
2273:       if (codeBlock) {
2274:         flushCode();
2275:       } else {
2276:         flushBullets();
2277:         codeBlock = { language: fence[1] || "", lines: [] };
2278:       }
2279:       return;
2280:     }
2281:     if (codeBlock) {
2282:       codeBlock.lines.push(line);
2283:       return;
2284:     }
2285:     if (!trimmed) {
2286:       flushBullets();
2287:       return;
2288:     }
2289:     const bullet = trimmed.match(/^[-*\u2022]\s+(.+)/u);
2290:     if (bullet) {
2291:       bulletItems.push(bullet[1]);
2292:       return;
2293:     }
2294:     flushBullets();
2295:     if (/^[^!?]{2,64}:$/.test(trimmed)) {
2296:       html.push(`<p><strong>${renderAssistantInline(trimmed.slice(0, -1))}</strong></p>`);
2297:     } else {
2298:       html.push(`<p>${renderAssistantInline(trimmed)}</p>`);
2299:     }
2300:   });
2301:   flushCode();
2302:   flushBullets();
2303:   return `<div class="ai-answer-content">${html.join("")} || "<p>I am ready.</p>"</div>`;
2304: }

```

flushBullets - lines 2257-2261

Item	Explanation
Source location	Lines 2257-2261 (5 source lines).
Purpose	Part of the public website runtime.

Item	Explanation
Declaration	const flushBullets = () => {
Touches	Mostly local calculations or local UI state.
Calls detected	push, map, renderAssistantInline, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2257:   const flushBullets = () => {
2258:     if (!bulletItems.length) return;
2259:     html.push(`<ul>${bulletItems.map((item) =>
`<li>${renderAssistantInline(item)}</li>`).join("")}</ul>`);
2260:     bulletItems = [];
2261:   };

```

flushCode - lines 2262-2267

Item	Explanation
Source location	Lines 2262-2267 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	const flushCode = () => {
Touches	Mostly local calculations or local UI state.
Calls detected	escapeHtml, push, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2262:   const flushCode = () => {
2263:     if (!codeBlock) return;
2264:     const language = codeBlock.language ? ` data-language="${escapeHtml(codeBlock.language)}"` : "";
2265:     html.push(`<pre class="ai-code-
block"${language}><code>${escapeHtml(codeBlock.lines.join("\n"))}</code></pre>`);
2266:     codeBlock = null;
2267:   };

```

appendAssistantMessage - lines 2306-2317

Item	Explanation
Source location	Lines 2306-2317 (12 source lines).
Purpose	Part of the public website runtime.
Declaration	function appendAssistantMessage(role, content, sources = []) {
Touches	browser/Electron DOM, filesystem
Calls detected	createElement, renderAssistantAnswerContent, assistantSourcesMarkup, renderMultilineInlineText, append, updateAssistantPanelGrowth
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2306: function appendAssistantMessage(role, content, sources = []) {
2307:   if (!aiAssistantLog) return;
2308:   const article = document.createElement("article");
2309:   article.className = `ai-message ai-message-${role}`;
2310:   article.innerHTML = role === "assistant"
2311:     ? `${renderAssistantAnswerContent(content)}${assistantSourcesMarkup(sources)}`
2312:     : `<p>${renderMultilineInlineText(content)}</p>`;
2313:   aiAssistantLog.append(article);
2314:   updateAssistantPanelGrowth();
2315:   aiAssistantLog.scrollTop = aiAssistantLog.scrollHeight;
2316:   return article;
2317: }

```

scrollAssistantAnswerToStart - lines 2319-2329

Item	Explanation
Source location	Lines 2319-2329 (11 source lines).
Purpose	Part of the public website runtime.
Declaration	function scrollAssistantAnswerToStart(article) {
Touches	Mostly local calculations or local UI state.
Calls detected	max, requestAnimationFrame
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2319: function scrollAssistantAnswerToStart(article) {
2320:   if (!aiAssistantLog || !article) return;
2321:   const scrollToAnswer = () => {
2322:     aiAssistantLog.scrollTop = Math.max(0, article.offsetTop - aiAssistantLog.offsetTop - 4);
2323:   };
2324:   if (typeof window.requestAnimationFrame === "function") {
2325:     window.requestAnimationFrame(scrollToAnswer);
2326:   } else {
2327:     window.setTimeout(scrollToAnswer, 0);
2328:   }
2329: }

```

scrollToAnswer - lines 2321-2323

Item	Explanation
Source location	Lines 2321-2323 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	const scrollToAnswer = () => {
Touches	Mostly local calculations or local UI state.
Calls detected	max
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2321:   const scrollToAnswer = () => {
2322:     aiAssistantLog.scrollTop = Math.max(0, article.offsetTop - aiAssistantLog.offsetTop - 4);
2323:   };

```

appendAssistantPendingMessage - lines 2331-2335

Item	Explanation
Source location	Lines 2331-2335 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function appendAssistantPendingMessage() {
Touches	browser/Electron DOM
Calls detected	appendAssistantMessage, add
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2331: function appendAssistantPendingMessage() {
2332:   const article = appendAssistantMessage("assistant", "Thinking", []);
2333:   article?.classList.add("ai-message-pending");
2334:   return article;
2335: }

```

replaceAssistantMessage - lines 2337-2346

Item	Explanation
Source location	Lines 2337-2346 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function replaceAssistantMessage(article, content, sources = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	appendAssistantMessage, renderAssistantAnswerContent, assistantSourcesMarkup, updateAssistantPanelGrowth, scrollAssistantAnswerToStart
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2337: function replaceAssistantMessage(article, content, sources = []) {
2338:   if (!article) {
2339:     appendAssistantMessage("assistant", content, sources);
2340:     return;
2341:   }
2342:   article.className = "ai-message ai-message-assistant";
2343:   article.innerHTML = `${renderAssistantAnswerContent(content)}${assistantSourcesMarkup(sources)}`;
2344:   updateAssistantPanelGrowth();
2345:   scrollAssistantAnswerToStart(article);
2346: }

```

setAssistantBusy - lines 2348-2352

Item	Explanation
Source location	Lines 2348-2352 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function setAssistantBusy(isBusy) {

Item	Explanation
Touches	browser/Electron DOM, filesystem
Calls detected	querySelector
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2348: function setAssistantBusy(isBusy) {
2349:   const submitButton = aiAssistantForm?.querySelector('button[type="submit"]');
2350:   if (aiAssistantInput) aiAssistantInput.disabled = isBusy;
2351:   if (submitButton) submitButton.disabled = isBusy;
2352: }

```

clearAssistantChat - lines 2354-2363

Item	Explanation
Source location	Lines 2354-2363 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function clearAssistantChat() {
Touches	Mostly local calculations or local UI state.
Calls detected	Map, setAssistantStatus, updateAssistantPanelGrowth, focus
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2354: function clearAssistantChat() {
2355:   if (!aiAssistantLog) return;
2356:   assistantSourceCounter = 0;
2357:   assistantSourceMap = new Map();
2358:   assistantChatHistory = [];
2359:   aiAssistantLog.innerHTML = "";
2360:   setAssistantStatus("Chat cleared");
2361:   updateAssistantPanelGrowth();
2362:   aiAssistantInput?.focus();
2363: }

```

assistantRemoteTimeoutMs - lines 2365-2374

Item	Explanation
Source location	Lines 2365-2374 (10 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	function assistantRemoteTimeoutMs(question = "", context = {}, options = {}) {
Touches	filesystem
Calls detected	normalize, isArray, some, toLowerCase, b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2365: function assistantRemoteTimeoutMs(question = "", context = {}, options = {}) {
2366:   const clean = normalize(question);
2367:   const sourceFetches = Array.isArray(context.sourceFetches) ? context.sourceFetches : [];
2368:   const usesGitHub = sourceFetches.some((source) => String(source.kind || "").toLowerCase() === "github"
|| /github\.com/i.test(source.url || ""));
2369:   if (usesGitHub || /\b(github|repo|repository|repositories|source\s+code|code|snippet|firmware|verilog|s
ystemverilog|python|javascript|circuit\s+file)\b/.test(clean)) {
2370:     return 45000;
2371:   }
2372:   if (options.allowWebSearch) return 30000;
2373:   return 18000;
2374: }
```

askRemoteAssistant - lines 2376-2411

Item	Explanation
Source location	Lines 2376-2411 (36 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	async function askRemoteAssistant(question, context, options = {}) {
Touches	network/API requests
Calls detected	assistantEndpoint, AbortController, abort, assistantRemoteTimeoutMs, fetch, stringify, Error, json, trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	2 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2376: async function askRemoteAssistant(question, context, options = {}) {
2377:   const endpoint = assistantEndpoint();
2378:   if (!endpoint) return null;
2379:   const controller = new AbortController();
2380:   let didTimeout = false;
2381:   const timer = setTimeout(() => {
2382:     didTimeout = true;
2383:     controller.abort();
2384:   }, assistantRemoteTimeoutMs(question, context, options));
2385:   let response;
2386:   try {
2387:     response = await fetch(endpoint, {
2388:       method: "POST",
2389:       headers: { "Content-Type": "application/json" },
2390:       signal: controller.signal,
2391:       body: JSON.stringify({
2392:         context,
2393:         conversation: options.conversation || [],
2394:         intent: options.intent || context.intent || "portfolio_specific",
2395:         allowWebSearch: Boolean(options.allowWebSearch),
2396:         question,
2397:         mode: "portfolio-plus-general-knowledge",
2398:         responseType: "chatgpt-like-answer-first-links-second"
2399:       })
2400:     });
2401:   } catch (error) {
2402:     if (didTimeout) throw new Error("AI request timed out while reading public sources.");
2403:     throw error;
2404:   } finally {
2405:     clearTimeout(timer);
2406:   }
2407:   if (!response.ok) throw new Error("AI service unavailable");
2408:   const data = await response.json();
2409:   const answer = String(data.answer || data.message || "").trim();
2410:   return answer ? { answer, model: data.model || "", usedWebSearch: Boolean(data.usedWebSearch) } : null;
2411: }
```

answerAssistantQuestion - lines 2413-2458

Item	Explanation
Source location	Lines 2413-2458 (46 source lines).
Purpose	Part of the public website runtime.

Item	Explanation
Declaration	async function answerAssistantQuestion(question = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, slice, appendAssistantMessage, assistantRemember, appendAssistantPendingMessage, setAssistantBusy, setAssistantStatus, assistantEndpoint, assistantQuestionIntent, assistantFallbackResults, assistantSourcesForDisplay, assistantContextForQuestion
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2413: async function answerAssistantQuestion(question = "") {
2414:   const cleanQuestion = String(question || "").trim();
2415:   if (!cleanQuestion) return;
2416:   const priorConversation = assistantChatHistory.slice();
2417:   appendAssistantMessage("user", cleanQuestion);
2418:   assistantRemember("user", cleanQuestion);
2419:   const pendingMessage = appendAssistantPendingMessage();
2420:   setAssistantBusy(true);
2421:   try {
2422:     setAssistantStatus(assistantEndpoint() ? "Composing AI answer" : "Composing local answer",
"working");
2423:
2424:     const intent = assistantQuestionIntent(cleanQuestion, priorConversation);
2425:     const rawSources = assistantFallbackResults(cleanQuestion, intent);
2426:     const sources = assistantSourcesForDisplay(cleanQuestion, rawSources, intent);
2427:     const context = assistantContextForQuestion(cleanQuestion, intent, rawSources);
2428:     const conversation = assistantConversationForRequest(cleanQuestion, priorConversation);
2429:     try {
2430:       const remoteAnswer = await askRemoteAssistant(cleanQuestion, context, {
2431:         allowWebSearch: assistantQuestionAllowsPublicLookup(cleanQuestion, intent),
2432:         conversation,
2433:         intent
2434:       });
2435:       if (remoteAnswer?.answer) {
2436:         replaceAssistantMessage(pendingMessage, remoteAnswer.answer, sources);
2437:         assistantRemember("assistant", remoteAnswer.answer);
2438:         setAssistantStatus(remoteAnswer.model ? `AI answer ready: ${remoteAnswer.model}` : "AI answer
ready");
2439:         return;
2440:       }
2441:     } catch (error) {
2442:       setAssistantStatus(error?.message || "Local answer used", "error");
2443:     }
2444:
2445:     const fallbackAnswer = assistantLocalAnswer(cleanQuestion, sources, intent);
2446:     replaceAssistantMessage(pendingMessage, fallbackAnswer, sources);
2447:     assistantRemember("assistant", fallbackAnswer);
2448:     if (!assistantEndpoint()) setAssistantStatus("Local portfolio intelligence");
2449:   } catch {
2450:     const errorAnswer = "I had trouble processing that question. Try rephrasing it with a project name,
tool name, or electronics topic.";
2451:     replaceAssistantMessage(pendingMessage, errorAnswer, []);
2452:     assistantRemember("assistant", errorAnswer);
2453:     setAssistantStatus("Assistant error", "error");
2454:   } finally {
2455:     setAssistantBusy(false);
2456:     aiAssistantInput?.focus();
2457:   }
2458: }

```

linkAttributes - lines 2459-2462

Item	Explanation
Source location	Lines 2459-2462 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function linkAttributes(url, item = {}) {
Touches	filesystem
Calls detected	normalizeLinkTarget, isWebsiteLinkItem
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2459: function linkAttributes(url, item = {}) {
2460:   const target = normalizeLinkTarget(url, { assumeWeb: isWebsiteLinkItem(item, url) });
2461:   return /^https?:\/\//i.test(target) ? ' target="_blank" rel="noreferrer"' : "";
2462: }

```

isLocalDownloadTarget - lines 2464-2467

Item	Explanation
Source location	Lines 2464-2467 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function isLocalDownloadTarget(target = "") {
Touches	filesystem
Calls detected	normalizeLinkTarget
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2464: function isLocalDownloadTarget(target = "") {
2465:   const value = normalizeLinkTarget(target, { assumeWeb: true });
2466:   return Boolean(value) && !/^(https?:)?\/\//i.test(value) && !/^(mailto:|tel:|#)/i.test(value);
2467: }

```

downloadAttribute - lines 2469-2472

Item	Explanation
Source location	Lines 2469-2472 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function downloadAttribute(target = "", item = {}) {
Touches	filesystem
Calls detected	normalizeLinkTarget, isWebsiteLinkItem, isLocalDownloadTarget
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2469: function downloadAttribute(target = "", item = {}) {
2470:   const value = normalizeLinkTarget(target, { assumeWeb: isWebsiteLinkItem(item, target) });
2471:   return isLocalDownloadTarget(value) ? " download" : "";
2472: }

```

fileNameFromUrl - lines 2474-2481

Item	Explanation
Source location	Lines 2474-2481 (8 source lines).
Purpose	Part of the public website runtime.
Declaration	function fileNameFromUrl(url = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	split, pop
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2474: function fileNameFromUrl(url = "") {
2475:     const clean = String(url || "").split(/[?#]/)[0].split("/").pop() || "Download file";
2476:     try {
2477:         return decodeURIComponent(clean);
2478:     } catch {
2479:         return clean;
2480:     }
2481: }

```

resourceLink - lines 2483-2501

Item	Explanation
Source location	Lines 2483-2501 (19 source lines).
Purpose	Part of the public website runtime.
Declaration	function resourceLink(item = {}, label = item.label item.title item.name fileNameFromUrl(item.url item.artifact)) {
Touches	filesystem
Calls detected	fileNameFromUrl, normalizeLinkTarget, isWebsiteLinkItem, escapeHtml, linkAttributes, downloadAttribute
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2483: function resourceLink(item = {}, label = item.label || item.title || item.name ||
fileNameFromUrl(item.url || item.artifact)) {
2484:     const rawTarget = item.url || item.artifact || item.href || item.file || item.path || item.src || "";
2485:     const target = normalizeLinkTarget(rawTarget, { assumeWeb: isWebsiteLinkItem(item, rawTarget) });
2486:
2487:     if (!target || item.status === "planned") {
2488:         return `<span class="resource-link muted">${escapeHtml(label || "Planned file")}</span>`;
2489:     }
2490:
2491:     return `
2492:     <a
2493:         class="resource-link"
2494:         href="${escapeHtml(target)}"
2495:         ${linkAttributes(target, item)}
2496:         ${downloadAttribute(target, item)}
2497:     >
2498:         ${escapeHtml(label || fileNameFromUrl(target))}
2499:     </a>
2500: `;
2501: }

```

pillList - lines 2503-2510

Item	Explanation
Source location	Lines 2503-2510 (8 source lines).
Purpose	Part of the public website runtime.
Declaration	function pillList(items, className = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	map, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2503: function pillList(items, className = "") {
2504:   return (items || [])
2505:     .map((item) => {
2506:       const label = typeof item === "string" ? item : item.name || item.title || item.label || "";
2507:       return label ? `

```

evidenceList - lines 2512-2518

Item	Explanation
Source location	Lines 2512-2518 (7 source lines).
Purpose	Part of the public website runtime.
Declaration	function evidenceList(items, renderItem, emptyMessage) {
Touches	Mostly local calculations or local UI state.
Calls detected	map, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2512: function evidenceList(items, renderItem, emptyMessage) {
2513:   if (!items || !items.length) {
2514:     return `
```

detailBlock - lines 2520-2527

Item	Explanation
Source location	Lines 2520-2527 (8 source lines).
Purpose	Part of the public website runtime.
Declaration	function detailBlock(title, className, content) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2520: function detailBlock(title, className, content) {
2521:   return `
2522:     <details class="${className}">
2523:       <summary>${title}</summary>
2524:       ${content}
2525:     </details>
2526:   `;
2527: }

```

mediaGrid - lines 2529-2549

Item	Explanation
Source location	Lines 2529-2549 (21 source lines).
Purpose	Part of the public website runtime.
Declaration	function mediaGrid(items) {
Touches	filesystem
Calls detected	map, escapeHtml, normalizeLinkTarget, isWebsiteLinkItem, linkAttributes, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2529: function mediaGrid(items) {
2530:   if (!items || !items.length) {
2531:     return `<p class="evidence-empty">No project images have been added yet.</p>`;
2532:   }
2533: }
2534: return `
2535:   <div class="media-grid">
2536:     ${items.map((item) => `
2537:       <figure>
2538:         <a href="${escapeHtml(normalizeLinkTarget(item.url, { assumeweb: isWebsiteLinkItem(item,
item.url) })))}"${linkAttributes(item.url, item)}>
2539:         
2540:         </a>
2541:         <figcaption>
2542:           <strong>${escapeHtml(item.title)}</strong>
2543:           <span>${escapeHtml(item.caption || "")}</span>
2544:         </figcaption>
2545:       </figure>
2546:     `).join("")}
2547:   </div>
2548: `;
2549: }

```

siteSectionHasContent - lines 2551-2560

Item	Explanation
Source location	Lines 2551-2560 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function siteSectionHasContent(section) {
Touches	Mostly local calculations or local UI state.
Calls detected	some

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2551: function siteSectionHasContent(section) {
2552:   return Boolean(
2553:     section?.description ||
2554:     section?.richDescription?.blocks?.length ||
2555:     section?.backgroundImage ||
2556:     (section?.links || []).some((link) => link.label || link.url) ||
2557:     (section?.assets || []).some((asset) => asset.title || asset.url) ||
2558:     (section?.subsections || []).some((item) => item.title || item.description ||
item.richDescription?.blocks?.length || (item.links || []).length)
2559:   );
2560: }

```

siteSectionRenderable - lines 2562-2564

Item	Explanation
Source location	Lines 2562-2564 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function siteSectionRenderable(section) {
Touches	Mostly local calculations or local UI state.
Calls detected	siteSectionHasContent
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2562: function siteSectionRenderable(section) {
2563:   return section?.visible !== false && siteSectionHasContent(section);
2564: }

```

renderDynamicLinks - lines 2566-2570

Item	Explanation
Source location	Lines 2566-2570 (5 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderDynamicLinks(items = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	filter, map, resourceLink, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2566: function renderDynamicLinks(items = []) {
2567:   const links = items.filter((item) => item?.url && (item.label || item.title));
2568:   if (!links.length) return "";

```



```

2569:   return `<div class="resource-list dynamic-link-list">${links.map((item) => resourceLink(item,
item.label || item.title || "Open")).join("")}</div>`;
2570: }

```

renderSiteSections - lines 2572-2609

Item	Explanation
Source location	Lines 2572-2609 (38 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderSiteSections() {
Touches	browser/Electron DOM
Calls detected	querySelector, filter, map, url, escapeHtml, renderRichContent, renderDynamicLinks, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2572: function renderSiteSections() {
2573:   const mount = document.querySelector("#dynamic-sections");
2574:   if (!mount) return;
2575:   const visibleSections = (siteSections || []).filter(siteSectionRenderable);
2576:   mount.innerHTML = visibleSections.map((section) => {
2577:     const style = section.backgroundImage ? ` style="--section-bg:
url('${escapeHtml(section.backgroundImage)}')` : "";
2578:     const links = [...(section.links || []), ...(section.assets || [])];
2579:     const subsections = (section.subsections || []).filter((item) => item.title || item.description ||
item.richDescription?.blocks?.length || (item.links || []).length);
2580:     return `
2581:       <section class="section dynamic-section" id="${escapeHtml(section.id || "")}"${style}>
2582:         <div class="dynamic-section-surface">
2583:           <div class="section-heading">
2584:             <div>
2585:               <h2>${escapeHtml(section.title || "Untitled section")}</h2>
2586:             </div>
2587:           </div>
2588:           ${section.description || section.richDescription?.blocks?.length
? `<div class="dynamic-section-copy">${renderRichContent(section.richDescription,
section.description || "")}</div>`
2590:             : ""}
2591:           ${subsections.length ? `
2592:             <div class="dynamic-section-grid">
2593:               ${subsections.map((item) => `
2594:                 <article class="dynamic-section-card">
2595:                   <h3>${escapeHtml(item.title || "Untitled")}</h3>
2596:                   ${item.description || item.richDescription?.blocks?.length
? renderRichContent(item.richDescription, item.description || "")
2598:                     : ""}
2599:                   ${renderDynamicLinks(item.links || [])}
2600:                 </article>
2601:               `).join("")}
2602:             </div>
2603:             : ""}
2604:             ${renderDynamicLinks(links)}
2605:           </div>
2606:         </section>
2607:       `;
2608:   }).join("");
2609: }

```

subsections - lines 2579-2581

Item	Explanation
Source location	Lines 2579-2581 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	const subsections = (section.subsections []).filter((item) => item.title item.description item.richDescription?.blocks?.length (item.links []).length);
Touches	Mostly local calculations or local UI state.

Item	Explanation
Calls detected	filter, escapeHtml
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2579:     const subsections = (section.subsections || []).filter((item) => item.title || item.description ||
item.richDescription?.blocks?.length || (item.links || []).length);
2580:     return `
2581:       <section class="section dynamic-section" id="${escapeHtml(section.id || "")}"${style}>

```

customSectionBlocks - lines 2611-2622

Item	Explanation
Source location	Lines 2611-2622 (12 source lines).
Purpose	Part of the public website runtime.
Declaration	function customSectionBlocks(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	map, detailBlock, evidenceList, resourceLink, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2611: function customSectionBlocks(project) {
2612:   return (project.sections || []).map((section) => detailBlock(section.title, "evidence-block evidence-
wide", `
2613:     ${section.description ? `<p class="evidence-empty">${section.description}</p>` : ""}
2614:     ${evidenceList(section.items || [], (item) => `
2615:       <li>
2616:         ${item.url ? resourceLink(item, item.title) : `<strong>${item.title}</strong>`}
2617:         <span>${item.type || "Section item"} &middot; ${item.status || "tracked"}</span>
2618:         ${item.description ? `<p>${item.description}</p>` : ""}
2619:       </li>
2620:     `, "No content has been added yet.")]
2621:   `)).join("");
2622: }

```

itemUrl - lines 2623-2625

Item	Explanation
Source location	Lines 2623-2625 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function itemUrl(item = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2623: function itemUrl(item = {}) {  
2624:     return item.url || item.artifact || item.href || item.file || item.path || item.src || "";  
2625: }
```

itemLabel - lines 2627-2630

Item	Explanation
Source location	Lines 2627-2630 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function itemLabel(item = {}, fallback = "Download file") {
Touches	Mostly local calculations or local UI state.
Calls detected	itemUrl, fileNameFromUrl
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2627: function itemLabel(item = {}, fallback = "Download file") {  
2628:     const url = itemUrl(item);  
2629:     return item.title || item.label || item.name || item.caption || fileNameFromUrl(url) || fallback;  
2630: }
```

normalizeDownloadAsset - lines 2632-2643

Item	Explanation
Source location	Lines 2632-2643 (12 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeDownloadAsset(item = {}, fallbackType = "File") {
Touches	filesystem
Calls detected	itemUrl, itemLabel
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2632: function normalizeDownloadAsset(item = {}, fallbackType = "File") {  
2633:     const url = itemUrl(item);  
2634:     if (!url || item.status === "planned") return null;  
2635:     return {  
2636:         title: itemLabel(item),  
2637:         url,  
2638:         status: item.status || "uploaded",  
2639:         type: item.type || item.kind || item.meta || fallbackType,  
2640:         description: item.description || item.caption || item.summary || ""  
2641:     };  
2642: }  
2643: }
```

collectRichDownloads - lines 2645-2659

Item	Explanation
Source location	Lines 2645-2659 (15 source lines).
Purpose	Part of the public website runtime.

Item	Explanation
Declaration	function collectRichDownloads(rich, fallbackType = "File", output = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	forEach, push, cleanRichImageTitle, fileNameFromUrl
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2645: function collectRichDownloads(rich, fallbackType = "File", output = []) {
2646:   (rich?.blocks || []).forEach((block) => {
2647:     if (!block?.url) return;
2648:
2649:     output.push({
2650:       title: cleanRichImageTitle(block) || block.title || block.caption ||
fileNameFromUrl(block.url),
2651:       url: block.url,
2652:       status: "uploaded",
2653:       type: block.type === "image" ? "Image" : fallbackType,
2654:       description: block.caption || ""
2655:     });
2656:   });
2657:
2658:   return output;
2659: }

```

collectDownloadsFromValue - lines 2661-2689

Item	Explanation
Source location	Lines 2661-2689 (29 source lines).
Purpose	Part of the public website runtime.
Declaration	function collectDownloadsFromValue(value, fallbackType = "File", output = []) {
Touches	filesystem
Calls detected	isArray, forEach, normalizeDownloadAsset, push, collectRichDownloads
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2661: function collectDownloadsFromValue(value, fallbackType = "File", output = []) {
2662:   if (!value) return output;
2663:
2664:   if (Array.isArray(value)) {
2665:     value.forEach((item) => collectDownloadsFromValue(item, fallbackType, output));
2666:     return output;
2667:   }
2668:
2669:   if (typeof value !== "object") return output;
2670:
2671:   const asset = normalizeDownloadAsset(value, fallbackType);
2672:   if (asset) output.push(asset);
2673:
2674:   collectRichDownloads(value.rich, fallbackType, output);
2675:   collectRichDownloads(value.summaryRich, fallbackType, output);
2676:
2677:   [
2678:     "files",
2679:     "assets",
2680:     "items",
2681:     "children",
2682:     "links",
2683:     "results",
2684:     "references",

```

```

2685:         "mathAnalysis"
2686:     ].forEach((key) => collectDownloadsFromValue(value[key], fallbackType, output));
2687:
2688:     return output;
2689: }

```

uniqueDownloads - lines 2691-2700

Item	Explanation
Source location	Lines 2691-2700 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function uniqueDownloads(items = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	Set, filter, has, add
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2691: function uniqueDownloads(items = []) {
2692:     const seen = new Set();
2693:
2694:     return items.filter((item) => {
2695:         const key = `${item.url}:${item.title}`;
2696:         if (!item.url || seen.has(key)) return false;
2697:         seen.add(key);
2698:         return true;
2699:     });
2700: }

```

renderDownloadBlock - lines 2702-2717

Item	Explanation
Source location	Lines 2702-2717 (16 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderDownloadBlock(title, items = []) {
Touches	filesystem
Calls detected	uniqueDownloads, detailBlock, escapeHtml, map, resourceLink, renderMultilineInlineText, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2702: function renderDownloadBlock(title, items = []) {
2703:     const downloads = uniqueDownloads(items);
2704:     if (!downloads.length) return "";
2705:
2706:     return detailBlock(escapeHtml(title), "evidence-block evidence-wide download-evidence-block", `
2707:     <ul class="download-list">
2708:         ${downloads.map((item) => `
2709:             <li>
2710:                 ${resourceLink(item, item.title)}
2711:                 <span>${escapeHtml(item.type || "File")} &middot; ${escapeHtml(item.status ||
"uploaded")}</span>
2712:                 ${item.description ? `<p>${renderMultilineInlineText(item.description)}</p>` : ""}
2713:             </li>
2714:         `).join("")}
2715:     </ul>
2716:     `);
2717: }

```

renderCollectedDownloadSections - lines 2719-2760

Item	Explanation
Source location	Lines 2719-2760 (42 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderCollectedDownloadSections(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	uniqueDownloads, collectDownloadsFromValue, push, renderDownloadBlock, addBlock, forEach, displayTitle, filter, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2719: function renderCollectedDownloadSections(project) {
2720:   const blocks = [];
2721:
2722:   const addBlock = (title, source, fallbackType = "File") => {
2723:     const downloads = uniqueDownloads(collectDownloadsFromValue(source, fallbackType, []));
2724:     if (downloads.length) blocks.push(renderDownloadBlock(title, downloads));
2725:   };
2726:
2727:   addBlock("Overview downloads", [
2728:     { summaryRich: project.summaryRich },
2729:     project.design?.brief?.files,
2730:     project.design?.brief?.summaryRich
2731:   ], "Overview file");
2732:
2733:   addBlock("Design downloads", [
2734:     project.design?.documentation?.files,
2735:     project.design?.documentation?.references,
2736:     project.design?.documentation?.mathAnalysis
2737:   ], "Design file");
2738:
2739:   addBlock("Simulation downloads", [
2740:     project.design?.simulation?.files,
2741:     project.design?.simulation?.results
2742:   ], "Simulation file");
2743:
2744:   addBlock("Project documents", project.documents, "Document");
2745:   addBlock("Test artifacts", project.tests, "Test artifact");
2746:   addBlock("PCB artifacts", project.pcb, "PCB artifact");
2747:   addBlock("Images and media", project.media, "Image");
2748:
2749:   (project.sections || []).forEach((section) => {
2750:     addBlock(`${displayTitle(section.title, "Custom section")} downloads`, section.items ||
section.files || [], section.title || "Section file");
2751:   });
2752:
2753:   (project.portfolioView?.sections || []).
2754:     .filter((section) => section.id !== "brief")
2755:     .forEach((section) => {
2756:       addBlock(`${displayTitle(section.title, "Section")} downloads`, section.items || [],
section.title || "Section file");
2757:     });
2758:
2759:   return blocks.join("");
2760: }
```

renderParsedBriefBlock - lines 2761-2771

Item	Explanation
Source location	Lines 2761-2771 (11 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderParsedBriefBlock(section, fallbackSummary = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	renderRichContent

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2761: function renderParsedBriefBlock(section, fallbackSummary = "") {
2762:   const briefItem = section?.items?.[0] || {};
2763:   const briefText = briefItem.description || fallbackSummary || "";
2764:   if (!briefItem.rich?.blocks?.length && !briefText) return "";
2765:   return
2766:     <details class="project-brief-default parsed-summary" open>
2767:       <summary>Overview</summary>
2768:       ${renderRichContent(briefItem.rich, briefText)}
2769:     </details>
2770:   `;
2771: }

```

pathToString - lines 2773-2775

Item	Explanation
Source location	Lines 2773-2775 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function pathToString(path = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2773: function pathToString(path = []) {
2774:   return path.join(".");
2775: }

```

pathFromString - lines 2777-2780

Item	Explanation
Source location	Lines 2777-2780 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function pathFromString(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	split, map, filter, isInteger
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2777: function pathFromString(value = "") {
2778:   if (!value) return [];
2779:   return value.split(".").map((item) => Number(item)).filter((item) => Number.isInteger(item));

```

2780: }

sectionRouteState - lines 2782-2789

Item	Explanation
Source location	Lines 2782-2789 (8 source lines).
Purpose	Part of the public website runtime.
Declaration	function sectionRouteState(projectId, sectionIndex, resourcePath = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2782: function sectionRouteState(projectId, sectionIndex, resourcePath = "") {
2783:   return {
2784:     appwindow: sectionRouteKey,
2785:     projectId: String(projectId || ""),
2786:     resourcePath: String(resourcePath || ""),
2787:     sectionIndex: String(sectionIndex ?? "")
2788:   };
2789: }
```

canUseSectionHistory - lines 2791-2793

Item	Explanation
Source location	Lines 2791-2793 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function canUseSectionHistory() {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2791: function canUseSectionHistory() {
2792:   return Boolean(window.history && typeof window.history.pushState === "function" && typeof
window.history.replaceState === "function");
2793: }
```

sectionBaseUrl - lines 2795-2799

Item	Explanation
Source location	Lines 2795-2799 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function sectionBaseUrl() {
Touches	Mostly local calculations or local UI state.
Calls detected	URL, values, forEach, delete

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2795: function sectionBaseUrl() {
2796:   const url = new URL(window.location.href);
2797:   Object.values(sectionRouteParams).forEach((param) => url.searchParams.delete(param));
2798:   return url;
2799: }

```

sectionRouteUrl - lines 2801-2809

Item	Explanation
Source location	Lines 2801-2809 (9 source lines).
Purpose	Part of the public website runtime.
Declaration	function sectionRouteUrl(projectId, sectionIndex, resourcePath = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	sectionBaseUrl, set, delete
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2801: function sectionRouteUrl(projectId, sectionIndex, resourcePath = "") {
2802:   const url = sectionBaseUrl();
2803:   url.searchParams.set(sectionRouteParams.project, String(projectId || ""));
2804:   url.searchParams.set(sectionRouteParams.section, String(sectionIndex ?? ""));
2805:   if (resourcePath) url.searchParams.set(sectionRouteParams.path, String(resourcePath));
2806:   else url.searchParams.delete(sectionRouteParams.path);
2807:   url.hash = "projects";
2808:   return url;
2809: }

```

sectionRouteFromState - lines 2811-2819

Item	Explanation
Source location	Lines 2811-2819 (9 source lines).
Purpose	Part of the public website runtime.
Declaration	function sectionRouteFromState(state) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2811: function sectionRouteFromState(state) {
2812:   if (!state || state.appWindow !== sectionRouteKey) return null;
2813:   if (!state.projectId || state.sectionIndex === "") return null;

```

```

2814:   return {
2815:     projectId: String(state.projectId),
2816:     resourcePath: String(state.resourcePath || ""),
2817:     sectionIndex: String(state.sectionIndex)
2818:   };
2819: }

```

sectionRouteFromLocation - lines 2821-2831

Item	Explanation
Source location	Lines 2821-2831 (11 source lines).
Purpose	Part of the public website runtime.
Declaration	function sectionRouteFromLocation() {
Touches	Mostly local calculations or local UI state.
Calls detected	URLSearchParams, get
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2821: function sectionRouteFromLocation() {
2822:   const params = new URLSearchParams(window.location.search);
2823:   const projectId = params.get(sectionRouteParams.project);
2824:   const sectionIndex = params.get(sectionRouteParams.section);
2825:   if (!projectId || sectionIndex === null) return null;
2826:   return {
2827:     projectId,
2828:     resourcePath: params.get(sectionRouteParams.path) || "",
2829:     sectionIndex
2830:   };
2831: }

```

sameSectionRoute - lines 2833-2838

Item	Explanation
Source location	Lines 2833-2838 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function sameSectionRoute(left, right) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2833: function sameSectionRoute(left, right) {
2834:   return Boolean(left && right &&
2835:     left.projectId === right.projectId &&
2836:     String(left.sectionIndex) === String(right.sectionIndex) &&
2837:     String(left.resourcePath || "") === String(right.resourcePath || ""));
2838: }

```

syncSectionRouteToHistory - lines 2840-2851

Item	Explanation
Source location	Lines 2840-2851 (12 source lines).

Item	Explanation
Purpose	Keeps two places aligned, such as local data and target files.
Declaration	function syncSectionRouteToHistory(projectId, sectionIndex, resourcePath = "", mode = "push") {
Touches	Mostly local calculations or local UI state.
Calls detected	canUseSectionHistory, sectionRouteState, sectionRouteUrl, sectionRouteFromState, sectionRouteFromLocation, sameSectionRoute
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2840: function syncSectionRouteToHistory(projectId, sectionIndex, resourcePath = "", mode = "push") {
2841:   if (isApplyingSectionRoute || mode === "none" || !canUseSectionHistory()) return;
2842:   const route = sectionRouteState(projectId, sectionIndex, resourcePath);
2843:   const url = sectionRouteUrl(projectId, sectionIndex, resourcePath);
2844:   const currentRoute = sectionRouteFromState(history.state) || sectionRouteFromLocation();
2845:   const method = mode === "replace" || sameSectionRoute(currentRoute, route) ? "replaceState" :
"pushState";
2846:   try {
2847:     history[method](route, "", url);
2848:   } catch {
2849:     // Preview iframes and restricted browser contexts may reject history writes.
2850:   }
2851: }

```

clearSectionRouteInHistory - lines 2853-2860

Item	Explanation
Source location	Lines 2853-2860 (8 source lines).
Purpose	Part of the public website runtime.
Declaration	function clearSectionRouteInHistory() {
Touches	Mostly local calculations or local UI state.
Calls detected	canUseSectionHistory, replaceState, sectionBaseUrl
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2853: function clearSectionRouteInHistory() {
2854:   if (isApplyingSectionRoute || !canUseSectionHistory()) return;
2855:   try {
2856:     history.replaceState({ appWindow: "portfolio-base" }, "", sectionBaseUrl());
2857:   } catch {
2858:     // The visible dialog is already closed; URL cleanup is best effort.
2859:   }
2860: }

```

initializeSectionHistoryState - lines 2862-2871

Item	Explanation
Source location	Lines 2862-2871 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function initializeSectionHistoryState() {
Touches	Mostly local calculations or local UI state.

Item	Explanation
Calls detected	canUseSectionHistory, sectionRouteFromLocation, sectionRouteFromState, replaceState, sectionRouteState, sectionRouteUrl
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2862: function initializeSectionHistoryState() {
2863:   if (!canUseSectionHistory()) return;
2864:   const route = sectionRouteFromLocation() || sectionRouteFromState(history.state);
2865:   try {
2866:     if (route) history.replaceState(sectionRouteState(route.projectId, route.sectionIndex,
route.resourcePath), "", sectionRouteUrl(route.projectId, route.sectionIndex, route.resourcePath));
2867:     else if (!history.state?.appwindow) history.replaceState({ appwindow: "portfolio-base" }, "",
window.location.href);
2868:   } catch {
2869:     // Browser history state is progressive enhancement for the portfolio windows.
2870:   }
2871: }

```

nodeAtPath - lines 2873-2882

Item	Explanation
Source location	Lines 2873-2882 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function nodeAtPath(section, path = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2873: function nodeAtPath(section, path = []) {
2874:   let node = section;
2875:   let children = section?.items || [];
2876:   for (const index of path) {
2877:     node = children[index];
2878:     if (!node) return null;
2879:     children = node.children || [];
2880:   }
2881:   return node;
2882: }

```

nodeChildren - lines 2884-2886

Item	Explanation
Source location	Lines 2884-2886 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function nodeChildren(node) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2884: function nodeChildren(node) {
2885:   return node?.items || node?.children || [];
2886: }

```

richHasRenderableContent - lines 2888-2897

Item	Explanation
Source location	Lines 2888-2897 (10 source lines).
Purpose	Part of the public website runtime.
Declaration	function richHasRenderableContent(rich) {
Touches	filesystem
Calls detected	some
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2888: function richHasRenderableContent(rich) {
2889:   return Boolean(rich?.blocks?.some((block) =>
2890:     block.type === "image" && block.url ||
2891:     block.type === "formula" && block.formula ||
2892:     block.type === "code" && block.code ||
2893:     block.text ||
2894:     block.title ||
2895:     block.caption
2896:   ));
2897: }

```

nodeHasRenderableContent - lines 2899-2910

Item	Explanation
Source location	Lines 2899-2910 (12 source lines).
Purpose	Part of the public website runtime.
Declaration	function nodeHasRenderableContent(node) {
Touches	Mostly local calculations or local UI state.
Calls detected	richHasRenderableContent, nodeChildren, some, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 5 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2899: function nodeHasRenderableContent(node) {
2900:   if (!node) return false;
2901:   if (node.kind === "summary") return Boolean(node.description || richHasRenderableContent(node.rich));
2902:   const children = nodeChildren(node);
2903:   if (node.kind === "subsection") {
2904:     return Boolean(node.description || node.url || richHasRenderableContent(node.rich) ||
children.some(nodeHasRenderableContent));
2905:   }

```

```

2906:   if (["tool", "language"].includes(node.kind)) {
2907:     return Boolean(node.title || node.description || node.url || richHasRenderableContent(node.rich) ||
children.some(nodeHasRenderableContent));
2908:   }
2909:   return Boolean(node.description || node.url || richHasRenderableContent(node.rich) ||
children.some(nodeHasRenderableContent));
2910: }

```

sectionHasRenderableContent - lines 2912-2914

Item	Explanation
Source location	Lines 2912-2914 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function sectionHasRenderableContent(section) {
Touches	Mostly local calculations or local UI state.
Calls detected	richHasRenderableContent, some
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2912: function sectionHasRenderableContent(section) {
2913:   return Boolean(section?.description || richHasRenderableContent(section?.rich) || (section?.items ||
[])).some(nodeHasRenderableContent));
2914: }

```

responsiveChildren - lines 2916-2918

Item	Explanation
Source location	Lines 2916-2918 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function responsiveChildren(node) {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2916: function responsiveChildren(node) {
2917:   return node?.items || node?.children || [];
2918: }

```

richContainsMedia - lines 2920-2922

Item	Explanation
Source location	Lines 2920-2922 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function richContainsMedia(rich) {
Touches	Mostly local calculations or local UI state.
Calls detected	some

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2920: function richContainsMedia(rich) {
2921:   return Boolean(rich?.blocks?.some((block) => block.type === "image" && block.url));
2922: }

```

responsiveNodeHasMedia - lines 2924-2930

Item	Explanation
Source location	Lines 2924-2930 (7 source lines).
Purpose	Part of the public website runtime.
Declaration	function responsiveNodeHasMedia(node) {
Touches	Mostly local calculations or local UI state.
Calls detected	richContainsMedia, responsiveChildren, some
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2924: function responsiveNodeHasMedia(node) {
2925:   if (!node) return false;
2926:   if (node.kind === "image" || richContainsMedia(node.rich)) return true;
2927:   const url = String(node.url || "");
2928:   if (/\. (apng|avif|gif|jpe?g|png|svg|webp)(\?|#|$)/i.test(url)) return true;
2929:   return responsiveChildren(node).some(responsiveNodeHasMedia);
2930: }

```

responsiveNodeCount - lines 2932-2935

Item	Explanation
Source location	Lines 2932-2935 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function responsiveNodeCount(node) {
Touches	Mostly local calculations or local UI state.
Calls detected	nodeHasRenderableContent, responsiveChildren, reduce
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2932: function responsiveNodeCount(node) {
2933:   if (!node || !nodeHasRenderableContent(node)) return 0;
2934:   return 1 + responsiveChildren(node).reduce((count, child) => count + responsiveNodeCount(child), 0);
2935: }

```

responsiveNodeDepth - lines 2937-2942

Item	Explanation
Source location	Lines 2937-2942 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function responsiveNodeDepth(node) {
Touches	Mostly local calculations or local UI state.
Calls detected	nodeHasRenderableContent, responsiveChildren, filter, max, map
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2937: function responsiveNodeDepth(node) {
2938:   if (!node || !nodeHasRenderableContent(node)) return 0;
2939:   const children = responsiveChildren(node).filter(nodeHasRenderableContent);
2940:   if (!children.length) return 1;
2941:   return 1 + Math.max(...children.map(responsiveNodeDepth));
2942: }

```

inferResponsiveProfile - lines 2944-2976

Item	Explanation
Source location	Lines 2944-2976 (33 source lines).
Purpose	Part of the public website runtime.
Declaration	function inferResponsiveProfile(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	filter, sectionHasRenderableContent, reduce, responsiveNodeCount, max, map, some, responsiveNodeHasMedia, hasPublicTemplate
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2944: function inferResponsiveProfile(project) {
2945:   const sections = project?.portfolioView?.sections || [];
2946:   const visibleSections = sections.filter((section) => section?.id !== "brief" &&
sectionHasRenderableContent(section));
2947:   const itemCount = visibleSections.reduce(
2948:     (count, section) => count + (section.items || []).reduce((sum, item) => sum +
responsiveNodeCount(item), 0),
2949:     0
2950:   );
2951:   const maxDepth = visibleSections.reduce(
2952:     (depth, section) => Math.max(depth, ...(section.items || []).map(responsiveNodeDepth), 1),
2953:     1
2954:   );
2955:   const hasMedia = visibleSections.some((section) => responsiveNodeHasMedia(section));
2956:   const density = itemCount >= 18 || visibleSections.length >= 6 || maxDepth >= 4
2957:     ? "dense"
2958:     : itemCount <= 6 && visibleSections.length <= 2
2959:     ? "simple"
2960:     : "balanced";
2961:   return {
2962:     version: 1,
2963:     breakpoints: {
2964:       mobile: 640,
2965:       tablet: 900,
2966:       desktop: 1180
2967:     },
2968:     cardLayout: hasPublicTemplate(project) ? "single" : "split",
2969:     contentMax: hasMedia || maxDepth >= 3 ? "1180px" : "980px",
2970:     density,

```



```

2971:     mediaMax: hasMedia ? "860px" : "720px",
2972:     sectionCardMin: density === "dense" ? "210px" : density === "simple" ? "280px" : "240px",
2973:     touchTarget: "44px",
2974:     windowMode: "full-screen"
2975:   };
2976: }

```

projectResponsiveProfile - lines 2978-2982

Item	Explanation
Source location	Lines 2978-2982 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function projectResponsiveProfile(project) {
Touches	Mostly local calculations or local UI state.
Calls detected	inferResponsiveProfile
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2978: function projectResponsiveProfile(project) {
2979:   const savedProfile = project?.portfolioView?.responsive;
2980:   if (savedProfile?.version) return { ...inferResponsiveProfile(project), ...savedProfile };
2981:   return inferResponsiveProfile(project);
2982: }

```

nodeSummary - lines 2984-2992

Item	Explanation
Source location	Lines 2984-2992 (9 source lines).
Purpose	Part of the public website runtime.
Declaration	function nodeSummary(title, rich, text, emptyMessage = "No summary has been added yet.") {
Touches	Mostly local calculations or local UI state.
Calls detected	escapeHtml, renderRichContent
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2984: function nodeSummary(title, rich, text, emptyMessage = "No summary has been added yet.") {
2985:   if (!rich?.blocks?.length && !text) return "";
2986:   return `
2987:     <details class="parsed-summary" open>
2988:       <summary>${escapeHtml(title)}</summary>
2989:       ${renderRichContent(rich, text || "")}
2990:     </details>
2991:   `;
2992: }

```

nodeSummary - lines 2994-2997

Item	Explanation
Source location	Lines 2994-2997 (4 source lines).
Purpose	Part of the public website runtime.

Item	Explanation
Declaration	function nodeIsOverview(item = {}) {
Touches	filesystem
Calls detected	normalize, displayTitle, endsWith
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2994: function nodeIsOverview(item = {}) {
2995:   const title = normalize(displayTitle(item.title || item.label || ""));
2996:   return item.kind === "summary" || title === "overview" || title.endsWith(" overview");
2997: }

```

nodeOverviewDetails - lines 2999-3017

Item	Explanation
Source location	Lines 2999-3017 (19 source lines).
Purpose	Part of the public website runtime.
Declaration	function nodeOverviewDetails(node, children = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	filter, push, renderRichContent, forEach, renderInlineSectionFiles, nodeChildren, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2999: function nodeOverviewDetails(node, children = []) {
3000:   const overviewItems = children.filter(nodeIsOverview);
3001:   const blocks = [];
3002:   if (node?.rich?.blocks?.length || node?.description) {
3003:     blocks.push(renderRichContent(node.rich, node.description || ""));
3004:   }
3005:   overviewItems.forEach((item) => {
3006:     if (item.rich?.blocks?.length || item.description) blocks.push(renderRichContent(item.rich,
item.description || ""));
3007:     const itemFiles = renderInlineSectionFiles(nodeChildren(item));
3008:     if (itemFiles) blocks.push(itemFiles);
3009:   });
3010:   if (!blocks.length) return "";
3011:   return `
3012:     <details class="parsed-summary section-overview-default" open>
3013:       <summary>Overview</summary>
3014:       ${blocks.join("")}
3015:     </details>
3016:   `;
3017: }

```

parsedNodeCard - lines 3019-3032

Item	Explanation
Source location	Lines 3019-3032 (14 source lines).
Purpose	Part of the public website runtime.
Declaration	function parsedNodeCard(node, projectId, sectionIndex, path) {
Touches	Mostly local calculations or local UI state.

Item	Explanation
Calls detected	displayTitle, pathToString, escapeHtml
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3019: function parsedNodeCard(node, projectId, sectionIndex, path) {
3020:   const title = displayTitle(node.title);
3021:   return `
3022:     <button
3023:       class="section-open-card subsection-open-card"
3024:       type="button"
3025:       data-section-project="${projectId}"
3026:       data-section-index="${sectionIndex}"
3027:       data-resource-path="${pathToString(path)}"
3028:     >
3029:       <span>${escapeHtml(title)}</span>
3030:     </button>
3031:   `;
3032: }

```

parsedChildCards - lines 3034-3042

Item	Explanation
Source location	Lines 3034-3042 (9 source lines).
Purpose	Part of the public website runtime.
Declaration	function parsedChildCards(children, projectId, sectionIndex, basePath = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	filter, nodeHasRenderableContent, nodeIsOverview, map, parsedNodeCard, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3034: function parsedChildCards(children, projectId, sectionIndex, basePath = []) {
3035:   const visibleChildren = children.filter((child) => nodeHasRenderableContent(child) && !child.url &&
!nodeIsOverview(child));
3036:   if (!visibleChildren.length) return "";
3037:   return `
3038:     <div class="subsection-grid section-content-grid">
3039:       ${children.map((child, index) => nodeHasRenderableContent(child) && !child.url &&
!nodeIsOverview(child) ? parsedNodeCard(child, projectId, sectionIndex, [...basePath, index]) : "").join("")}
3040:     </div>
3041:   `;
3042: }

```

parsedNodeContent - lines 3044-3058

Item	Explanation
Source location	Lines 3044-3058 (15 source lines).
Purpose	Part of the public website runtime.
Declaration	function parsedNodeContent(node, projectId, sectionIndex, path = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	renderInlineSectionFiles, nodeChildren, filter, nodeIsOverview, nodeOverviewDetails, parsedChildCards
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3044: function parsedNodeContent(node, projectId, sectionIndex, path = []) {
3045:   const isContainer = !node.kind || node.kind === "subsection";
3046:   if (!isContainer && node.url) return renderInlineSectionFiles([node]);
3047:
3048:   const children = nodeChildren(node);
3049:   const contentChildren = children.filter((child) => !nodeIsOverview(child));
3050:
3051:   return `
3052:   <article class="parsed-window-panel section-directory">
3053:     ${nodeOverviewDetails(node, children)}
3054:     ${parsedChildCards(children, projectId, sectionIndex, path)}
3055:     ${renderInlineSectionFiles(contentChildren)}
3056:   </article>
3057: `;
3058: }

```

parsedSectionContent - lines 3060-3064

Item	Explanation
Source location	Lines 3060-3064 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function parsedSectionContent(section, projectId, sectionIndex, path = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	nodeAtPath, parsedNodeContent
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3060: function parsedSectionContent(section, projectId, sectionIndex, path = []) {
3061:   const node = path.length ? nodeAtPath(section, path) : section;
3062:   if (!node) return `<p class="evidence-empty">This section could not be found.</p>`;
3063:   return parsedNodeContent(node, projectId, sectionIndex, path);
3064: }

```

parsedSection - lines 3066-3080

Item	Explanation
Source location	Lines 3066-3080 (15 source lines).
Purpose	Part of the public website runtime.
Declaration	function parsedSection(section, index, project) {
Touches	Mostly local calculations or local UI state.
Calls detected	filter, some, escapeHtml, displayTitle
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3066: function parsedSection(section, index, project) {
3067:   const visibleItems = (section.items || []).filter(nodeHasRenderableContent);
3068:   const hasImages = visibleItems.some((item) => item.kind === "image");
3069:
3070:   return `
3071:     <button
3072:       class="section-open-card evidence-block ${hasImages ? "evidence-wide" : ""}"
3073:       type="button"
3074:       data-section-project="${project.id}"
3075:       data-section-index="${index}"
3076:     >
3077:       <span>${escapeHtml(displayTitle(section.title, "Section"))}</span>
3078:     </button>
3079:   `;
3080: }

```

projectCard - lines 3082-3164

Item	Explanation
Source location	Lines 3082-3164 (83 source lines).
Purpose	Part of the public website runtime.
Declaration	function projectCard(project) {
Touches	filesystem
Calls detected	find, filter, sectionHasRenderableContent, projectTemplateClass, projectResponsiveClass, projectTemplateStyle, renderParsedBriefBlock, map, parsedSection, join, renderMultilineInlineText, detailBlock
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3082: function projectCard(project) {
3083:   if (project.portfolioView) {
3084:     const category = categories.find((item) => item.id === project.category) || {};
3085:     const accent = category.accent || "#117c7a";
3086:     const sections = project.portfolioView.sections || [];
3087:     const briefSection = sections.find((section) => section.id === "brief");
3088:     const otherSections = sections.filter((section) => section.id !== "brief" &&
sectionHasRenderableContent(section));
3089:
3090:     return `
3091:       <article class="project-card catalog-card ${projectTemplateClass(project)}
${projectResponsiveClass(project)}" id="${project.id}" style="${projectTemplateStyle(project, accent)}">
3092:         <div class="project-body">
3093:           <h3>${project.portfolioView.title || project.title}</h3>
3094:           ${renderParsedBriefBlock(briefSection, project.summary)}
3095:           <div class="evidence-grid" aria-label="${project.title} parsed project content">
3096:             ${otherSections.map((section, index) => parsedSection(section, index, project)).join("")}
3097:           </div>
3098:         </div>
3099:       </article>
3100:     `;
3101:   }
3102:
3103:   const category = categories.find((item) => item.id === project.category) || {};
3104:   const accent = category.accent || "#117c7a";
3105:
3106:   return `
3107:     <article class="project-card catalog-card ${projectTemplateClass(project)}
${projectResponsiveClass(project)}" id="${project.id}" style="${projectTemplateStyle(project, accent)}">
3108:       <div class="project-body">
3109:         <h3>${project.title}</h3>
3110:         <p class="rich-paragraph">${renderMultilineInlineText(project.summary)}</p>
3111:
3112:         ${project.highlights && project.highlights.length ? detailBlock("Project highlights", "project-
drawer", `
3113:           <ul class="highlight-list">
3114:             ${project.highlights.map((item) => `<li>${item}</li>`).join("")}
3115:           </ul>
3116:         `) : ""}
3117:
3118:         <div class="evidence-grid" aria-label="${project.title} evidence blocks">
3119:           ${detailBlock("Documents", "evidence-block", evidenceList(project.documents, (item) => `
3120:             <li>
3121:               ${resourceLink(item, item.title)}
3122:               <span>${item.type || "Document"} &middot; ${item.status || "tracked"}</span>
3123:             </li>

```

```

3124:         ` , "No document artifact has been added yet."))}
3125:
3126:     ${detailBlock("Tests and results", "evidence-block", evidenceList(project.tests, (item) => `
3127:         <li>
3128:             ${resourceLink({ url: item.artifact, status: item.status }, item.name)}
3129:             <span>${item.method || "Validation"} &middot; ${item.status || "tracked"}</span>
3130:             ${item.result ? `<p>${item.result}</p>` : ""}
3131:         </li>
3132:     ` , "No test artifact has been added yet."))}
3133:
3134:     ${detailBlock("PCBs built", "evidence-block", evidenceList(project.pcb, (item) => `
3135:         <li>
3136:             ${resourceLink({ url: item.artifact, status: item.status }, item.name)}
3137:             <span>${item.revision || "Revision"} &middot; ${item.status || "tracked"}</span>
3138:         </li>
3139:     ` , "No PCB build has been added yet."))}
3140:
3141:     ${detailBlock("Images", "evidence-block evidence-wide", mediaGrid(project.media))}
3142:     ${customSectionBlocks(project)}
3143: </div>
3144:
3145:     ${detailBlock("Tools and implementation files", "project-drawer", `
3146:         <div class="project-tooling">
3147:             <div>
3148:                 <h4>Tools Used</h4>
3149:                 <div class="project-meta">${pillList(project.tools, "tool-tag")}</div>
3150:             </div>
3151:             <div>
3152:                 <h4>Languages</h4>
3153:                 <div class="project-meta">${pillList(project.languages, "language-tag")}</div>
3154:             </div>
3155:         </div>
3156:     `)}
3157:
3158:     <div class="resource-list">
3159:         ${project.links || []}.map((item) => resourceLink(item)).join("")
3160:     </div>
3161: </div>
3162: </article>
3163: `;
3164: }

```

categorySection - lines 3166-3181

Item	Explanation
Source location	Lines 3166-3181 (16 source lines).
Purpose	Part of the public website runtime.
Declaration	function categorySection(category, visibleProjects) {
Touches	Mostly local calculations or local UI state.
Calls detected	escapeHtml, map, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3166: function categorySection(category, visibleProjects) {
3167:     return `
3168:         <section class="category-section ${visibleProjects.length ? "" : "empty-category-section"}" aria-
3169:             labelledby="${escapeHtml(category.id)}-title">
3170:             <div class="category-heading">
3171:                 <div>
3172:                     <h3 id="${escapeHtml(category.id)}-title">${escapeHtml(category.label)}</h3>
3173:                 </div>
3174:                 ${visibleProjects.length ? `<span>${visibleProjects.length} project${visibleProjects.length === 1
3175: ? "" : "s"}</span>` : ""}
3176:             </div>
3177:             ${visibleProjects.length && category.description ? `<p class="category-
3178: description">${escapeHtml(category.description)}</p>` : ""}
3179:             <div class="category-projects">
3180:                 ${visibleProjects.map(projectCard).join("")}
3181:             </div>
3182:         </section>
3183:     `;
3184: }

```

escapeRegExp - lines 3183-3185

Item	Explanation
Source location	Lines 3183-3185 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function escapeRegExp(value = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
3183: function escapeRegExp(value = "") {
3184:   return String(value).replace(/[*+?^${}()|[\]\\]/g, "\\$&");
3185: }
```

highlightQueryHtml - lines 3187-3193

Item	Explanation
Source location	Lines 3187-3193 (7 source lines).
Purpose	Part of the public website runtime.
Declaration	function highlightQueryHtml(value = "", query = "") {
Touches	Mostly local calculations or local UI state.
Calls detected	escapeHtml, trim, escapeRegExp, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
3187: function highlightQueryHtml(value = "", query = "") {
3188:   const escaped = escapeHtml(value);
3189:   const cleanQuery = String(query || "").trim();
3190:   if (!cleanQuery) return escaped;
3191:   const pattern = new RegExp(`${escapeRegExp(escapeHtml(cleanQuery))}`, "ig");
3192:   return escaped.replace(pattern, '<mark class="search-result-mark">$1</mark>');
3193: }
```

clearSearchHighlights - lines 3195-3197

Item	Explanation
Source location	Lines 3195-3197 (3 source lines).
Purpose	Part of the public website runtime.
Declaration	function clearSearchHighlights() {
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3195: function clearSearchHighlights() {
3196:   window.CSS?.highlights?.delete?.(searchHighlightName);
3197: }

```

textNodeSearchTargets - lines 3199-3204

Item	Explanation
Source location	Lines 3199-3204 (6 source lines).
Purpose	Part of the public website runtime.
Declaration	function textNodeSearchTargets() {
Touches	browser/Electron DOM
Calls detected	querySelector, filter
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3199: function textNodeSearchTargets() {
3200:   return [
3201:     document.querySelector("main"),
3202:     document.querySelector("#section-view-dialog[open] .section-view-content")
3203:   ].filter(Boolean);
3204: }

```

applySearchHighlights - lines 3206-3238

Item	Explanation
Source location	Lines 3206-3238 (33 source lines).
Purpose	Part of the public website runtime.
Declaration	function applySearchHighlights(query) {
Touches	browser/Electron DOM, filesystem
Calls detected	clearSearchHighlights, trim, normalize, textNodeSearchTargets, forEach, createTreeWalker, acceptNode, includes, closest, nextNode, indexOf, createRange
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 5 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3206: function applySearchHighlights(query) {
3207:   clearSearchHighlights();
3208:   const cleanQuery = String(query || "").trim();
3209:   if (!cleanQuery || !window.CSS?.highlights || typeof window.Highlight !== "function") return 0;
3210:   const ranges = [];
3211:   const needle = normalize(cleanQuery);
3212:
3213:   textNodeSearchTargets().forEach((root) => {
3214:     const walker = document.createTreeWalker(root, NodeFilter.SHOW_TEXT, {
3215:       acceptNode(node) {
3216:         if (!node.textContent || !normalize(node.textContent).includes(needle)) return
NodeFilter.FILTER_REJECT;
3217:         if (node.parentElement?.closest("script, style, input, textarea, select, button, .search-engine,

```



```

.site-header")) return NodeFilter.FILTER_REJECT;
3218:     return NodeFilter.FILTER_ACCEPT;
3219:   }
3220: });
3221: let node = walker.nextNode();
3222: while (node) {
3223:   const haystack = normalize(node.textContent);
3224:   let index = haystack.indexOf(needle);
3225:   while (index >= 0) {
3226:     const range = document.createRange();
3227:     range.setStart(node, index);
3228:     range.setEnd(node, index + cleanQuery.length);
3229:     ranges.push(range);
3230:     index = haystack.indexOf(needle, index + Math.max(1, cleanQuery.length));
3231:   }
3232:   node = walker.nextNode();
3233: }
3234: });
3235:
3236: if (ranges.length) window.CSS.highlights.set(searchHighlightName, new window.Highlight(...ranges));
3237: return ranges.length;
3238: }

```

queueSearchHighlights - lines 3240-3243

Item	Explanation
Source location	Lines 3240-3243 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function queueSearchHighlights() {
Touches	Mostly local calculations or local UI state.
Calls detected	applySearchHighlights
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3240: function queueSearchHighlights() {
3241:   clearTimeout(searchHighlightTimer);
3242:   searchHighlightTimer = setTimeout(() => applySearchHighlights(searchInput?.value || ""), 30);
3243: }

```

ensureSearchPanel - lines 3245-3283

Item	Explanation
Source location	Lines 3245-3283 (39 source lines).
Purpose	Part of the public website runtime.
Declaration	function ensureSearchPanel() {
Touches	browser/Electron DOM
Calls detected	createElement, replaceWith, append, querySelector, addEventListener, updateSearchDropdown, preventDefault, closest, goToSearchResult
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3245: function ensureSearchPanel() {
3246:   if (!searchInput || !searchWrap || searchPanel) return;
3247:   const shell = document.createElement("div");
3248:   shell.className = "search-engine";
3249:   searchWrap.replaceWith(shell);

```

```

3250: shell.append(searchWrap);
3251: searchPanel = document.createElement("div");
3252: searchPanel.className = "search-results-panel";
3253: searchPanel.hidden = true;
3254: searchPanel.innerHTML = `
3255:   <div class="search-results-tools">
3256:     <span id="search-status">Type to search projects, files, sections, and phrases.</span>
3257:     <label>
3258:       <span>Show</span>
3259:       <select id="search-result-limit" aria-label="Number of search results">
3260:         <option value="5">5</option>
3261:         <option value="10" selected>10</option>
3262:         <option value="15">15</option>
3263:         <option value="20">20</option>
3264:       </select>
3265:     </label>
3266:   </div>
3267:   <div class="search-results-list" role="listbox" aria-label="Search results"></div>
3268: `;
3269: shell.append(searchPanel);
3270: searchStatus = searchPanel.querySelector("#search-status");
3271: searchLimitSelect = searchPanel.querySelector("#search-result-limit");
3272: searchLimitSelect.addEventListener("change", () => {
3273:   searchResultLimit = Number(searchLimitSelect.value) || 10;
3274:   updateSearchDropdown();
3275: });
3276: searchPanel.addEventListener("mousedown", (event) => event.preventDefault());
3277: searchPanel.addEventListener("click", (event) => {
3278:   const button = event.target.closest("[data-search-result-index]");
3279:   if (!button) return;
3280:   const result = currentSearchResults[Number(button.dataset.searchResultIndex)];
3281:   if (result) goToSearchResult(result);
3282: });
3283: }

```

setSearchMessage - lines 3285-3289

Item	Explanation
Source location	Lines 3285-3289 (5 source lines).
Purpose	Part of the public website runtime.
Declaration	function setSearchMessage(message, state = "neutral") {
Touches	browser/Electron DOM
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3285: function setSearchMessage(message, state = "neutral") {
3286:   if (!searchStatus) return;
3287:   searchStatus.textContent = message;
3288:   searchStatus.dataset.state = state;
3289: }

```

renderSearchResults - lines 3291-3327

Item	Explanation
Source location	Lines 3291-3327 (37 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderSearchResults(query, visibleCount) {
Touches	browser/Electron DOM
Calls detected	ensureSearchPanel, querySelector, trim, searchResultsFor, slice, setAttribute, setSearchMessage, removeAttribute, map, highlightQueryHtml, escapeHtml, filter
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3291: function renderSearchResults(query, visibleCount) {
3292:   ensureSearchPanel();
3293:   if (!searchPanel) return;
3294:   const list = searchPanel.querySelector(".search-results-list");
3295:   const cleanQuery = String(query || "").trim();
3296:   currentSearchResults = searchResultsFor(cleanQuery);
3297:   const limitedResults = currentSearchResults.slice(0, searchResultLimit);
3298:   searchPanel.hidden = !cleanQuery;
3299:   searchInput.setAttribute("aria-expanded", String(Boolean(cleanQuery)));
3300:
3301:   if (!cleanQuery) {
3302:     list.innerHTML = "";
3303:     setSearchMessage("Type to search projects, files, sections, and phrases.");
3304:     searchInput.removeAttribute("aria-invalid");
3305:     return;
3306:   }
3307:
3308:   if (!currentSearchResults.length) {
3309:     list.innerHTML = "";
3310:     setSearchMessage(`"${cleanQuery}" was not found on this portfolio.`, "not-found");
3311:     searchInput.setAttribute("aria-invalid", "true");
3312:     return;
3313:   }
3314:
3315:   searchInput.removeAttribute("aria-invalid");
3316:   const visibleText = visibleCount
3317:     ? `${visibleCount} visible match${visibleCount === 1 ? "" : "es"} highlighted.`
3318:     : "Matches found elsewhere. Choose a result below.";
3319:   setSearchMessage(`${visibleText} Showing ${limitedResults.length} of ${currentSearchResults.length}.`,
3320: "found");
3321:   list.innerHTML = limitedResults.map((result, index) => `
3322:     <button class="search-result-item" type="button" role="option" data-search-result-index="${index}">
3323:       <span class="search-result-title">${highlightQueryHtml(result.title || "Untitled result",
3324: cleanQuery)}</span>
3325:       <span class="search-result-meta">${escapeHtml([result.type, result.context].filter(Boolean).join("
3326: / ")).join("<br>")}</span>
3327:       ${searchSnippet(result.text, cleanQuery)} ? <span class="search-result-
3328: snippet">${highlightQueryHtml(searchSnippet(result.text, cleanQuery), cleanQuery)}</span>` : ""}
3329:     </button>
3330:   `).join("");
3331: }
```

updateSearchDropdown - lines 3329-3333

Item	Explanation
Source location	Lines 3329-3333 (5 source lines).
Purpose	Updates UI, state, cache, or stored data.
Declaration	function updateSearchDropdown() {
Touches	Mostly local calculations or local UI state.
Calls detected	applySearchHighlights, renderSearchResults
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3329: function updateSearchDropdown() {
3330:   const query = searchInput?.value || "";
3331:   const visibleCount = applySearchHighlights(query);
3332:   renderSearchResults(query, visibleCount);
3333: }
```

showSearchPanelIfNeeded - lines 3335-3338

Item	Explanation
Source location	Lines 3335-3338 (4 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function showSearchPanelIfNeeded() {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, updateSearchDropdown
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3335: function showSearchPanelIfNeeded() {
3336:   if (!searchInput?.value?.trim()) return;
3337:   updateSearchDropdown();
3338: }

```

scrollToDomTarget - lines 3340-3352

Item	Explanation
Source location	Lines 3340-3352 (13 source lines).
Purpose	Part of the public website runtime.
Declaration	function scrollToDomTarget(id) {
Touches	browser/Electron DOM
Calls detected	getElementById, closeSectionDialogForRoute, scrollIntoView, replaceState, queueSearchHighlights
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3340: function scrollToDomTarget(id) {
3341:   const target = id ? document.getElementById(id) : null;
3342:   if (!target) return false;
3343:   closeSectionDialogForRoute();
3344:   target.scrollIntoView({ behavior: "smooth", block: "start" });
3345:   try {
3346:     history.replaceState(history.state || { appWindow: "portfolio-base" }, "", `#${id}`);
3347:   } catch {
3348:     // Scrolling is still the important behavior.
3349:   }
3350:   queueSearchHighlights();
3351:   return true;
3352: }

```

goToSearchResult - lines 3354-3377

Item	Explanation
Source location	Lines 3354-3377 (24 source lines).
Purpose	Part of the public website runtime.
Declaration	function goToSearchResult(result) {
Touches	browser/Electron DOM
Calls detected	scrollToDomTarget, openParsedSection, queueSearchHighlights, getElementById, setActiveFilter, renderProjects, scrollIntoView, open

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3354: function goToSearchResult(result) {
3355:   if (!result) return;
3356:   searchPanel.hidden = true;
3357:   if (result.domTarget && scrollToDomTarget(result.domTarget)) return;
3358:
3359:   if (result.projectId && result.sectionIndex !== undefined) {
3360:     openParsedSection(result.projectId, result.sectionIndex, result.resourcePath || "");
3361:     queueSearchHighlights();
3362:     return;
3363:   }
3364:
3365:   if (result.projectId) {
3366:     if (!document.getElementById(result.projectId)) {
3367:       setActiveFilter("all");
3368:       renderProjects();
3369:     }
3370:     const target = document.getElementById(result.projectId);
3371:     target?.scrollIntoView({ behavior: "smooth", block: "center" });
3372:     queueSearchHighlights();
3373:     return;
3374:   }
3375:
3376:   if (result.url) window.open(result.url, "_blank", "noopener");
3377: }

```

handleSearchInput - lines 3379-3382

Item	Explanation
Source location	Lines 3379-3382 (4 source lines).
Purpose	Handles an event, request, command, or user action.
Declaration	function handleSearchInput() {
Touches	Mostly local calculations or local UI state.
Calls detected	renderProjects, updateSearchDropdown
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3379: function handleSearchInput() {
3380:   renderProjects();
3381:   updateSearchDropdown();
3382: }

```

renderProjects - lines 3384-3409

Item	Explanation
Source location	Lines 3384-3409 (26 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderProjects() {
Touches	filesystem
Calls detected	normalize, trim, filter, projectVisible, queueSearchHighlights, map, categorySection, join
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3384: function renderProjects() {
3385:   const query = normalize(searchInput.value).trim();
3386:   const visible = projects.filter((project) => projectVisible(project, query));
3387:
3388:   projectCount.textContent = projects.length;
3389:
3390:   if (!visible.length) {
3391:     grid.innerHTML = `
3392:       <div class="empty-state">
3393:         <h3>No projects match that view.</h3>
3394:         <p>Search covers categories, project names, documents, tests, PCB builds, tools, languages, and
links.</p>
3395:       </div>
3396:     `;
3397:     queueSearchHighlights();
3398:     return;
3399:   }
3400:
3401:   grid.innerHTML = categories
3402:     .map((category) => {
3403:       const visibleProjects = visible.filter((project) => project.category === category.id);
3404:       const shouldShowEmptyCategory = !query && (activeFilter === "all" || activeFilter === category.id);
3405:       return visibleProjects.length || shouldShowEmptyCategory ? categorySection(category,
visibleProjects) : "";
3406:     })
3407:     .join("");
3408:   queueSearchHighlights();
3409: }

```

ensureSectionDialog - lines 3411-3470

Item	Explanation
Source location	Lines 3411-3470 (60 source lines).
Purpose	Part of the public website runtime.
Declaration	function ensureSectionDialog() {
Touches	browser/Electron DOM
Calls detected	querySelector, createElement, append, addEventListener, closeSectionDialog, clearSectionRouteInHistory, remove, updateSectionDialogMinimize, focus, pathFromString, sectionForwardStack, push
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 5 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3411: function ensureSectionDialog() {
3412:   let dialog = document.querySelector("#section-view-dialog");
3413:   if (dialog) return dialog;
3414:
3415:   dialog = document.createElement("dialog");
3416:   dialog.id = "section-view-dialog";
3417:   dialog.className = "section-view-dialog";
3418:   dialog.innerHTML = `
3419:     <div class="section-view-shell">
3420:       <div class="section-view-heading">
3421:         <div class="section-view-titlebar">
3422:           <button class="section-view-back" type="button" title="Previous view" aria-label="Previous
view" hidden>&larr;</button>
3423:           <button class="section-view-forward" type="button" title="Next view" aria-label="Next view"
hidden>&rarr;</button>
3424:           <h2 id="section-view-title">Section</h2>
3425:         </div>
3426:         <div class="section-view-actions">
3427:           <button class="section-view-close" type="button" aria-label="Close section">&times;</button>
3428:         </div>

```

```

3429:         </div>
3430:         <div class="section-view-content" id="section-view-content"></div>
3431:     </div>
3432: `;
3433:     document.body.append(dialog);
3434:     dialog.querySelector(".section-view-close").addEventListener("click", () =>
closeSectionDialog(dialog));
3435:     dialog.addEventListener("cancel", () => clearSectionRouteInHistory());
3436:     dialog.addEventListener("close", () => {
3437:         document.body.classList.remove("full-window-open");
3438:         dialog.classList.remove("is-minimized-dialog");
3439:         updateSectionDialogMinimize(dialog);
3440:         const returnTarget = document.querySelector(
3441:             `[data-section-project="${dialog.dataset.projectId}"][data-section-
index="${dialog.dataset.sectionIndex}"]`
3442:         );
3443:         returnTarget?.focus({ preventScroll: true });
3444:     });
3445:     dialog.querySelector(".section-view-back").addEventListener("click", () => {
3446:         const path = pathFromString(dialog.dataset.resourcePath || "");
3447:         if (!path.length) {
3448:             closeSectionDialog(dialog);
3449:             return;
3450:         }
3451:         const stack = sectionForwardStack(dialog);
3452:         stack.push(pathToString(path));
3453:         dialog.dataset.forwardStack = JSON.stringify(stack);
3454:         path.pop();
3455:         openParsedSection(dialog.dataset.projectId, dialog.dataset.sectionIndex, pathToString(path), {
historyMode: "replace", preserveForward: true });
3456:     });
3457:     dialog.querySelector(".section-view-forward").addEventListener("click", () => {
3458:         const stack = sectionForwardStack(dialog);
3459:         const nextPath = stack.pop();
3460:         if (!nextPath) return;
3461:         dialog.dataset.forwardStack = JSON.stringify(stack);
3462:         openParsedSection(dialog.dataset.projectId, dialog.dataset.sectionIndex, nextPath, { historyMode:
"replace", preserveForward: true });
3463:     });
3464:     dialog.querySelector("#section-view-content").addEventListener("click", (event) => {
3465:         const nodeButton = event.target.closest("[data-resource-path]");
3466:         if (!nodeButton) return;
3467:         openParsedSection(nodeButton.dataset.sectionProject, nodeButton.dataset.sectionIndex,
nodeButton.dataset.resourcePath);
3468:     });
3469:     return dialog;
3470: }

```

sectionForwardStack - lines 3472-3478

Item	Explanation
Source location	Lines 3472-3478 (7 source lines).
Purpose	Part of the public website runtime.
Declaration	function sectionForwardStack(dialog) {
Touches	browser/Electron DOM
Calls detected	parse
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3472: function sectionForwardStack(dialog) {
3473:     try {
3474:         return JSON.parse(dialog.dataset.forwardStack || "[]");
3475:     } catch {
3476:         return [];
3477:     }
3478: }

```

updateSectionNavigation - lines 3480-3486

Item	Explanation
Source location	Lines 3480-3486 (7 source lines).
Purpose	Updates UI, state, cache, or stored data.
Declaration	function updateSectionNavigation(dialog, path) {
Touches	browser/Electron DOM
Calls detected	querySelector, setAttribute, sectionForwardStack
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3480: function updateSectionNavigation(dialog, path) {
3481:   const back = dialog.querySelector(".section-view-back");
3482:   back.hidden = false;
3483:   back.title = path.length ? "Previous view" : "Back to projects";
3484:   back.setAttribute("aria-label", path.length ? "Previous view" : "Back to projects");
3485:   dialog.querySelector(".section-view-forward").hidden = !sectionForwardStack(dialog).length;
3486: }

```

closeSectionDialog - lines 3488-3491

Item	Explanation
Source location	Lines 3488-3491 (4 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function closeSectionDialog(dialog) {
Touches	Mostly local calculations or local UI state.
Calls detected	close, clearSectionRouteInHistory
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3488: function closeSectionDialog(dialog) {
3489:   dialog.close();
3490:   clearSectionRouteInHistory();
3491: }

```

closeOrStepBackSectionDialog - lines 3493-3504

Item	Explanation
Source location	Lines 3493-3504 (12 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function closeOrStepBackSectionDialog(dialog) {
Touches	browser/Electron DOM
Calls detected	pathFromString, sectionForwardStack, push, pathToString, stringify, pop, openParsedSection, closeSectionDialog
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3493: function closeOrStepBackSectionDialog(dialog) {
3494:   const path = pathFromString(dialog.dataset.resourcePath || "");
3495:   if (path.length) {
3496:     const stack = sectionForwardStack(dialog);
3497:     stack.push(pathToString(path));
3498:     dialog.dataset.forwardStack = JSON.stringify(stack);
3499:     path.pop();
3500:     openParsedSection(dialog.dataset.projectId, dialog.dataset.sectionIndex, pathToString(path), {
historyMode: "replace", preserveForward: true });
3501:     return;
3502:   }
3503:   closeSectionDialog(dialog);
3504: }

```

openParsedSection - lines 3506-3532

Item	Explanation
Source location	Lines 3506-3532 (27 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	function openParsedSection(projectId, sectionIndex, resourcePath = "", options = {}) {
Touches	browser/Electron DOM
Calls detected	find, filter, sectionHasRenderableContent, pathFromString, nodeAtPath, nodeHasRenderableContent, ensureSectionDialog, applyProjectTemplateToElement, pathToString, syncSectionRouteToHistory, querySelector, displayTitle
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3506: function openParsedSection(projectId, sectionIndex, resourcePath = "", options = {}) {
3507:   const project = projects.find((item) => item.id === projectId);
3508:   const sections = (project?.portfolioView?.sections || []).filter((section) => section.id !== "brief" &&
sectionHasRenderableContent(section));
3509:   const section = sections[Number(sectionIndex)];
3510:   if (!section) return false;
3511:   const path = pathFromString(resourcePath);
3512:   const node = path.length ? nodeAtPath(section, path) : section;
3513:   if (!node) return false;
3514:   if (path.length ? !nodeHasRenderableContent(node) : !sectionHasRenderableContent(section)) return
false;
3515:
3516:   const dialog = ensureSectionDialog();
3517:   const category = categories.find((item) => item.id === project.category) || {};
3518:   applyProjectTemplateToElement(dialog, project, category.accent || "#117c7a");
3519:   dialog.dataset.projectId = projectId;
3520:   dialog.dataset.sectionIndex = String(sectionIndex);
3521:   dialog.dataset.resourcePath = pathToString(path);
3522:   if (!options.preserveForward) dialog.dataset.forwardStack = "[]";
3523:   syncSectionRouteToHistory(projectId, sectionIndex, pathToString(path), options.historyMode || "push");
3524:   dialog.querySelector("#section-view-title").textContent = displayTitle(node.title || section.title,
"Section");
3525:   updateSectionNavigation(dialog, path);
3526:   dialog.querySelector("#section-view-content").innerHTML = parsedSectionContent(section, projectId,
Number(sectionIndex), path);
3527:   document.body.classList.add("full-window-open");
3528:   if (!dialog.open) dialog.showModal();
3529:   dialog.scrollTop = 0;
3530:   queueSearchHighlights();
3531:   return true;
3532: }

```

sections - lines 3508-3514

Item	Explanation
Source location	Lines 3508-3514 (7 source lines).
Purpose	Part of the public website runtime.
Declaration	<code>const sections = (project?.portfolioView?.sections []).filter((section) => section.id !== "brief" && sectionHasRenderableContent(section));</code>
Touches	Mostly local calculations or local UI state.
Calls detected	<code>filter</code> , <code>sectionHasRenderableContent</code> , <code>pathFromString</code> , <code>nodeAtPath</code> , <code>nodeHasRenderableContent</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 <code>await</code> expression(s), 3 <code>return</code> statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3508:   const sections = (project?.portfolioView?.sections || []).filter((section) => section.id !== "brief" &&
sectionHasRenderableContent(section));
3509:   const section = sections[Number(sectionIndex)];
3510:   if (!section) return false;
3511:   const path = pathFromString(resourcePath);
3512:   const node = path.length ? nodeAtPath(section, path) : section;
3513:   if (!node) return false;
3514:   if (path.length ? !nodeHasRenderableContent(node) : !sectionHasRenderableContent(section)) return
false;

```

closeSectionDialogForRoute - lines 3534-3537

Item	Explanation
Source location	Lines 3534-3537 (4 source lines).
Purpose	Controls a window, dialog, panel, menu, or visible state.
Declaration	<code>function closeSectionDialogForRoute() {</code>
Touches	browser/Electron DOM
Calls detected	<code>querySelector</code> , <code>close</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 <code>await</code> expression(s), 0 <code>return</code> statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3534: function closeSectionDialogForRoute() {
3535:   const dialog = document.querySelector("#section-view-dialog");
3536:   if (dialog?.open) dialog.close();
3537: }

```

applySectionRoute - lines 3539-3555

Item	Explanation
Source location	Lines 3539-3555 (17 source lines).
Purpose	Part of the public website runtime.
Declaration	<code>function applySectionRoute(route) {</code>
Touches	Mostly local calculations or local UI state.
Calls detected	<code>closeSectionDialogForRoute</code> , <code>openParsedSection</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3539: function applySectionRoute(route) {
3540:   isApplyingSectionRoute = true;
3541:   try {
3542:     if (!route) {
3543:       closeSectionDialogForRoute();
3544:       return;
3545:     }
3546:
3547:     const opened = openParsedSection(route.projectId, route.sectionIndex, route.resourcePath, {
3548:       historyMode: "none",
3549:       preserveForward: true
3550:     });
3551:     if (!opened) closeSectionDialogForRoute();
3552:   } finally {
3553:     isApplyingSectionRoute = false;
3554:   }
3555: }

```

restoreSectionRouteAfterCatalogLoad - lines 3557-3560

Item	Explanation
Source location	Lines 3557-3560 (4 source lines).
Purpose	Part of the public website runtime.
Declaration	function restoreSectionRouteAfterCatalogLoad() {
Touches	Mostly local calculations or local UI state.
Calls detected	initializeSectionHistoryState, applySectionRoute, sectionRouteFromLocation, sectionRouteFromState
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3557: function restoreSectionRouteAfterCatalogLoad() {
3558:   initializeSectionHistoryState();
3559:   applySectionRoute(sectionRouteFromLocation() || sectionRouteFromState(history.state));
3560: }

```

renderInlineSectionFiles - lines 3621-3639

Item	Explanation
Source location	Lines 3621-3639 (19 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderInlineSectionFiles(items = []) {
Touches	filesystem
Calls detected	uniqueDownloads, map, normalizeDownloadAsset, filter, escapeHtml, normalizeLinkTarget, isWebsiteLinkItem, linkAttributes, downloadAttribute, fileNameFromUrl, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3621: function renderInlineSectionFiles(items = []) {
3622:   const downloads = uniqueDownloads(items.map((item) => normalizeDownloadAsset(item,
"File")).filter(Boolean));
3623:   if (!downloads.length) return "";
3624:
3625:   return `
3626:     <div class="section-file-list">
3627:       ${downloads.map((item) => `
3628:         <a
3629:           class="section-file-link"
3630:           href="${escapeHtml(normalizeLinkTarget(item.url, { assumeWeb: isWebsiteLinkItem(item, item.url)
3631:         })}"}"
3632:           ${linkAttributes(item.url, item)}
3633:           ${downloadAttribute(item.url, item)}
3634:         >
3635:           ${escapeHtml(item.title || fileNameFromUrl(item.url))}
3636:         </a>
3637:       `).join("")}
3638:     </div>
3639:   `;
3640: }

```

normalizeTextColor - lines 3640-3647

Item	Explanation
Source location	Lines 3640-3647 (8 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalizeTextColor(value = "") {
Touches	filesystem
Calls detected	trim
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3640: function normalizeTextColor(value = "") {
3641:   const color = String(value || "").trim();
3642:   return /^[0-9a-f]{3}|[0-9a-f]{6}$/i.test(color) ||
3643:     /^rgba?\(/i.test(color) ||
3644:     /^hsla?\(/i.test(color)
3645:     ? color
3646:     : "";
3647: }

```

loadProjectCatalog - lines 3648-3708

Item	Explanation
Source location	Lines 3648-3708 (61 source lines).
Purpose	Loads data from disk, network, browser storage, or a service.
Declaration	function loadProjectCatalog() {
Touches	filesystem, network/API requests
Calls detected	hydrateProjectCategories, normalizeFieldStyles, normalizeSiteContent, normalizeProfile, normalizeFunFacts, renderProfile, renderSiteContent, renderFunFacts, renderSiteSections, renderCategoryFilters, renderProjects, rebuildSearchIndex
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

3648: function loadProjectCatalog() {

```

```

3649:   if (window.__PORTFOLIO_CATALOG__) {
3650:     projects = window.__PORTFOLIO_CATALOG__.projects || [];
3651:     categories = hydrateProjectCategories(window.__PORTFOLIO_CATALOG__.categories || [], projects);
3652:     siteSections = window.__PORTFOLIO_CATALOG__.siteSections || [];
3653:     fieldStyles = normalizeFieldStyles(window.__PORTFOLIO_CATALOG__.fieldStyles || {});
3654:     siteContent = normalizeSiteContent(window.__PORTFOLIO_CATALOG__.siteContent || {});
3655:     siteContentRich = window.__PORTFOLIO_CATALOG__.siteContentRich || {};
3656:     profile = normalizeProfile(window.__PORTFOLIO_CATALOG__.profile || {});
3657:     profileRich = window.__PORTFOLIO_CATALOG__.profileRich || {};
3658:     funFacts = normalizeFunFacts(window.__PORTFOLIO_CATALOG__.funFacts || []);
3659:     funFactsRich = window.__PORTFOLIO_CATALOG__.funFactsRich || null;
3660:     renderProfile();
3661:     renderSiteContent();
3662:     renderFunFacts();
3663:     renderSiteSections();
3664:     renderCategoryFilters();
3665:     renderProjects();
3666:     rebuildSearchIndex();
3667:     updateSearchDropdown();
3668:     restoreSectionRouteAfterCatalogLoad();
3669:     return;
3670:   }
3671:
3672:   const catalogUrl = new URL("projects.json", window.location.href);
3673:   catalogUrl.searchParams.set("v", String(Date.now()));
3674:   fetch(catalogUrl.href, { cache: "no-store" })
3675:     .then((response) => {
3676:       if (!response.ok) throw new Error("Could not load projects.json");
3677:       return response.json();
3678:     })
3679:     .then((data) => {
3680:       projects = data.projects || [];
3681:       categories = hydrateProjectCategories(data.categories || [], projects);
3682:       siteSections = data.siteSections || [];
3683:       fieldStyles = normalizeFieldStyles(data.fieldStyles || {});
3684:       siteContent = normalizeSiteContent(data.siteContent || {});
3685:       siteContentRich = data.siteContentRich || {};
3686:       profile = normalizeProfile(data.profile || {});
3687:       profileRich = data.profileRich || {};
3688:       funFacts = normalizeFunFacts(data.funFacts || []);
3689:       funFactsRich = data.funFactsRich || null;
3690:       renderProfile();
3691:       renderSiteContent();
3692:       renderFunFacts();
3693:       renderSiteSections();
3694:       renderCategoryFilters();
3695:       renderProjects();
3696:       rebuildSearchIndex();
3697:       updateSearchDropdown();
3698:       restoreSectionRouteAfterCatalogLoad();
3699:     })
3700:     .catch(() => {
3701:       grid.innerHTML = `
3702:         <div class="empty-state">
3703:           <h3>Project data did not load.</h3>
3704:           <p>The project catalog is temporarily unavailable.</p>
3705:         </div>
3706:       `;
3707:     });
3708: }

```

Representative opening snippet

```

1: const grid = document.querySelector("#project-grid");
2: const searchInput = document.querySelector("#project-search");
3: const projectFilters = document.querySelector("#project-filters");
4: let filterButtons = [...document.querySelectorAll(".filter-button")];
5: const projectCount = document.querySelector("#project-count");
6: const projectTrackCount = document.querySelector("#project-track-count");
7: const projectTrackLabels = document.querySelector("#project-track-labels");
8: const year = document.querySelector("#year");
9: const funFactsCallout = document.querySelector("#fun-facts-callout");
10: const heroEyebrow = document.querySelector(".hero-content .eyebrow");
11: const heroTitle = document.querySelector("#hero-title");
12: const heroCopy = document.querySelector(".hero-copy");
13: const brandName = document.querySelector(".brand strong");
14: const brandSubtitle = document.querySelector(".brand small");
15: const brandIcon = document.querySelector(".brand-icon");
16: const brandText = document.querySelector(".omb-engraving");
17: const headerAvatar = document.querySelector(".header-avatar");
18: const headerAvatarImage = document.querySelector(".header-avatar img");
19: const heroImage = document.querySelector(".hero-image");
20: const resumeSection = document.querySelector("#resume");
21: const contactBand = document.querySelector("#contact");
22: const profilePhoto = document.querySelector(".profile-photo");
23: const contactTitle = document.querySelector("#contact-title");
24: const contactIntro = document.querySelector(".contact-intro");
25: const contactDetails = document.querySelector(".contact-details");
26: const contactLinks = document.querySelector(".contact-links");
27: const footerOwner = document.querySelector(".site-footer span");

```

```

28: const searchWrap = searchInput?.closest(".search-wrap");
29: const aiAssistantForm = document.querySelector("#ai-assistant-form");
30: const aiAssistantPanel = document.querySelector("#ask-ai");
31: const aiAssistantInput = document.querySelector("#ai-assistant-input");
32: const aiAssistantLog = document.querySelector("#ai-assistant-log");
33: const aiAssistantStatus = document.querySelector("#ai-assistant-status");
34: const aiClearChatButton = document.querySelector("#ai-clear-chat");
35:
36: let categories = [];
37: let projects = [];
38: let siteSections = [];
39: let funFacts = [];
40: let funFactsRich = null;
41: let fieldStyles = {};
42: let siteContent = null;
43: let siteContentRich = {};
44: let profile = null;
45: let profileRich = {};
46: let activeFilter = "all";
47: let activeSectionDialogDrag = null;
48: let activeSectionDialogResize = null;
49: let sectionDialogDragEnabled = false;
50: let isApplyingSectionRoute = false;
51: let searchPanel = null;
52: let searchStatus = null;
53: let searchLimitSelect = null;
54: let currentSearchResults = [];
55: let assistantSourceCounter = 0;
56: let assistantSourceMap = new Map();
57: let assistantChatHistory = [];
58: let searchableEntries = [];
59: let searchResultLimit = 10;
60: let searchHighlightTimer = 0;
61: const searchHighlightName = "portfolio-search-highlight";
62:
63: const sectionRouteKey = "portfolio-section";
64: const sectionRouteParams = {
65:   path: "portfolioPath",
66:   project: "portfolioProject",
67:   section: "portfolioSection"
68: };
69:
70: const supportedCodeLanguages = [
71:   { id: "c", label: "C", aliases: ["c"] },
72:   { id: "cpp", label: "C++", aliases: ["cpp", "c++", "cplusplus"] },

```

Safe change checklist

- Confirm which app, website, installer, workflow, or document consumes this file.
- Change it on a feature branch from development.
- Regenerate docs if the file purpose, API surface, or behavior changes.
- Run the smallest relevant smoke test before merging into development.

Important Code Reference: index.html

Item	Explanation
File	index.html
Category	Website/builder UI source
Folder role	Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
Size	10,799 bytes
Extension	.html
MIME type	text/html
Text lines	268

Why this file exists

- Public website HTML shell that recruiters and visitors load in the browser.

How it is used

Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.

How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

Important implementation signals

Item	Explanation
Security or auth at line 11	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: <meta name="author" content="Portfolio Owner" />
Cloudflare or AI at line 12	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <meta name="portfolio-ai-endpoint" content="https://omb-portfolio-ai.maurice-baraza-otieno.workers.dev/api/portfolio-ai" />
Installer at line 17	Windows setup, update, shortcut, or installation behavior. Evidence: <link rel="shortcut icon" href="assets/favicon.ico" />
Network request at line 38	Frontend-backend communication or public web/API calls. Evidence: "@context": "https://schema.org",
Compiler or simulator at line 55	Part of the Compile Code workspace or language/tool detection. Evidence: "Python",
Network request at line 126	Frontend-backend communication or public web/API calls. Evidence: href="https://github.com/otienomaurice/omb-portfolio-builder/releases/latest/download/OMB-Portfolio-Builder-Setup-latest-x64.exe"
Compiler or simulator at line 160	Part of the Compile Code workspace or language/tool detection. Evidence: <input id="project-search" type="search" placeholder="LTspice, PCB, frequency response, Verilog..." />
Cloudflare or AI at line 242	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <div class="ai-assistant-log" id="ai-assistant-log" aria-live="polite" aria-label="AI chat messages"></div>
Cloudflare or AI at line 253	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: <button class="ai-clear-chat" id="ai-clear-chat" type="button">Clear chat</button>

JSON/YAML-style keys detected

Item	Explanation
@context	Line 38: "@context": "https://schema.org",
@type	Line 39: "@type": "ProfilePage",
dateModified	Line 40: "dateModified": "2026-06-04",
mainEntity	Line 41: "mainEntity": {

Item	Explanation
@type	Line 42: "@type": "Person",
name	Line 43: "name": "Portfolio Owner",
url	Line 44: "url": "",
image	Line 45: "image": "",
jobTitle	Line 46: "jobTitle": "",
description	Line 47: "description": "Portfolio owner profile.",
sameAs	Line 48: "sameAs": [],
knowsAbout	Line 49: "knowsAbout": [

Representative opening snippet

```

1: <!DOCTYPE html>
2: <html lang="en">
3:   <head>
4:     <meta charset="UTF-8" />
5:     <meta name="viewport" content="width=device-width, initial-scale=1.0" />
6:     <meta
7:       name="description"
8:       content="A portfolio website built with OMB Portfolio Builder for projects, documents, diagrams,
source code, tests, links, and profile information."
9:     />
10:    <meta name="robots" content="index, follow" />
11:    <meta name="author" content="Portfolio Owner" />
12:    <meta name="portfolio-ai-endpoint" content="https://omb-portfolio-ai.maurice-baraza-
otieno.workers.dev/api/portfolio-ai" />
13:    <link rel="canonical" href="" />
14:    <link rel="icon" type="image/png" sizes="16x16" href="assets/favicon-16.png" />
15:    <link rel="icon" type="image/png" sizes="32x32" href="assets/favicon-32.png" />
16:    <link rel="icon" type="image/png" sizes="192x192" href="assets/favicon-192.png" />
17:    <link rel="shortcut icon" href="assets/favicon.ico" />
18:    <link rel="apple-touch-icon" sizes="180x180" href="assets/apple-touch-icon.png" />
19:    <meta property="og:type" content="profile" />
20:    <meta property="og:title" content="Portfolio" />
21:    <meta
22:      property="og:description"
23:      content="Explore projects, documents, tests, PCBs, tools, links, and source code."
24:    />
25:    <meta property="og:url" content="" />
26:    <meta property="og:image" content="" />
27:    <meta name="twitter:card" content="summary_large_image" />
28:    <meta name="twitter:title" content="Portfolio" />
29:    <meta
30:      name="twitter:description"
31:      content="A project portfolio built with OMB Portfolio Builder."
32:    />
33:    <meta name="twitter:image" content="" />
34:    <title>Portfolio</title>
35:    <link rel="stylesheet" href="styles.css?v=2026062203" />
36:    <script type="application/ld+json">
37:      {
38:        "@context": "https://schema.org",
39:        "@type": "ProfilePage",
40:        "dateModified": "2026-06-04",
41:        "mainEntity": {
42:          "@type": "Person",
43:          "name": "Portfolio Owner",
44:          "url": "",
45:          "image": "",
46:          "jobTitle": "",
47:          "description": "Portfolio owner profile.",
48:          "sameAs": [],
49:          "knowsAbout": [
50:            "Embedded systems",
51:            "ASIC design",
52:            "Digital hardware",
53:            "Software engineering",
54:            "Microcontrollers",
55:            "Python",
56:            "C++"
57:          ]
58:        }
59:      }
60:    </script>
61:  </head>
62:  <body>
63:    <header class="site-header" id="top">
64:      <a class="brand" href="#top" aria-label="Go to top">
65:        <span class="brand-lockup" aria-hidden="true">

```



```
66:         
67:         <span class="omb-engraving">OMB</span>
68:     </span>
69: <span>
70:     <strong>Portfolio</strong>
71:     <small>Portfolio</small>
72: </span>
```

Safe change checklist

- Confirm which app, website, installer, workflow, or document consumes this file.
- Change it on a feature branch from development.
- Regenerate docs if the file purpose, API surface, or behavior changes.
- Run the smallest relevant smoke test before merging into development.

Important Code Reference: styles.css

Item	Explanation
File	styles.css
Category	Website/builder UI source
Folder role	Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
Size	53,685 bytes
Extension	.css
MIME type	text/css
Text lines	2,735

Why this file exists

- Public website CSS for layout, contact area, projects, responsive behavior, AI panel, and mobile/desktop rendering.

How it is used

Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.

How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

Important implementation signals

Item	Explanation
Rich text at line 2	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color-scheme: light;
Rich text at line 27	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--ink);
Rich text at line 29	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-family: Inter, ui-sans-serif, system-ui, -apple-system, BlinkMacSystemFont, "Segoe UI", sans-serif;
Rich text at line 38	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: inherit;
Rich text at line 99	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #fff;
Rich text at line 104	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.64rem;
Rich text at line 105	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 900;
Rich text at line 116	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--muted);
Rich text at line 117	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.78rem;
Rich text at line 132	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--muted);
Rich text at line 133	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.95rem;
Rich text at line 134	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 700;
Rich text at line 138	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--teal);
Rich text at line 196	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--teal);
Rich text at line 197	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.78rem;
Rich text at line 198	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 800;
Rich text at line 213	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: clamp(2.45rem, 6vw, 5.75rem);
Rich text at line 220	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: clamp(1.85rem, 4vw, 3.2rem);

Item	Explanation
Rich text at line 227	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 1.1rem;
Rich text at line 233	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #2e4d62;
Rich text at line 234	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: clamp(1rem, 2vw, 1.18rem);
Rich text at line 252	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 800;
Rich text at line 260	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #fff;
Rich text at line 277	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #083c4c;
Rich text at line 279	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 850;
Rich text at line 282	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: transition: background 160ms ease, border-color 160ms ease, color 160ms ease, transform 160ms ease;
Rich text at line 289	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: stroke: currentColor;
Rich text at line 297	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: rgba(11, 114, 136, 0.72);
Rich text at line 299	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #062f3b;
Rich text at line 334	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--muted);
Rich text at line 356	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--ink);
Rich text at line 357	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: clamp(0.98rem, 1.4vw, 1.12rem);
Rich text at line 373	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border: 1px solid color-mix(in srgb, var(--line), #ffff 18%);
Rich text at line 385	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--ink);
Rich text at line 386	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 1rem;
Rich text at line 390	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--muted);
Rich text at line 391	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.92rem;
Rich text at line 434	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #086a7c;
Rich text at line 436	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.78rem;
Rich text at line 437	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 950;
Rich text at line 448	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: var(--ink);
Rich text at line 449	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: clamp(0.95rem, 1.3vw, 1.08rem);
Rich text at line 450	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 720;
Rich text at line 468	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #06384f;
Rich text at line 470	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.94rem;
Rich text at line 471	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 900;
Rich text at line 478	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: transition: background 140ms ease, border-color 140ms ease, color 140ms ease, transform 140ms ease;
Rich text at line 483	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #041c2a;
Rich text at line 484	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: #47b8df;
Rich text at line 498	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: stroke: currentColor;
Rich text at line 547	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #162636;
Rich text at line 548	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: clamp(1.45rem, 2vw, 1.85rem);
Rich text at line 553	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #0b778d;
Rich text at line 554	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.78rem;
Rich text at line 555	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 900;

Item	Explanation
Cloudflare or AI at line 559	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: .ai-clear-chat {
Rich text at line 568	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #9f2222;
Rich text at line 570	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font: inherit;
Rich text at line 571	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.82rem;
Rich text at line 572	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 850;
Rich text at line 574	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: transition: background 150ms ease, border-color 150ms ease, color 150ms ease, transform 150ms ease;
Cloudflare or AI at line 577	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: .ai-clear-chat:hover,
Cloudflare or AI at line 578	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: .ai-clear-chat:focus-visible {
Rich text at line 579	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: #b42222;
Rich text at line 580	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #ffffff;
Cloudflare or AI at line 585	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: .ai-clear-chat:active {
Rich text at line 586	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #9f2222;
Rich text at line 592	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #8a5a00;
Rich text at line 596	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #a62929;
Rich text at line 625	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #23384b;
Rich text at line 639	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #23384b;
Rich text at line 644	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #102b3b;
Rich text at line 655	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #eaf6ff;
Rich text at line 656	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font: 0.82rem/1.5 Consolas, "Liberation Mono", "Courier New", monospace;
Rich text at line 661	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: inherit;
Rich text at line 666	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: border-color: rgba(16, 121, 144, 0.36);
Rich text at line 706	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #526d82;
Rich text at line 707	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-size: 0.78rem;
Rich text at line 708	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: font-weight: 900;
Rich text at line 716	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: color: #102b3b;

CSS selectors and visual hooks

Item	Explanation
Line 1	:root
Line 21	html
Line 25	body
Line 33	body.full-window-open
Line 37	a
Line 44	input:focus-visible
Line 49	.site-header
Line 64	.brand
Line 71	.brand-lockup
Line 79	.brand-icon
Line 89	.omb-engraving

Item	Explanation
Line 111	.brand small
Line 115	.brand small
Line 120	.header-right
Line 128	.site-nav
Line 137	.site-nav a:hover
Line 141	.header-avatar
Line 152	.header-avatar img
Line 160	.hero
Line 170	.hero-content
Line 174	.hero-image
Line 180	.hero-shade
Line 186	.hero-content
Line 194	.eyebrow
Line 206	p
Line 210	h1
Line 218	h2
Line 225	h3
Line 230	.hero-copy
Line 237	.hero-actions
Line 246	.resource-link
Line 255	.button
Line 259	.button.primary
Line 264	.button.secondary
Line 269	.ai-jump-link
Line 285	.ai-jump-link svg
Line 296	.ai-jump-link:focus-visible
Line 303	.resume-section
Line 308	.resume-actions
Line 314	.resume-viewer
Line 324	.resume-viewer object
Line 331	.resume-viewer p
Line 337	.dynamic-section
Line 347	.dynamic-section-surface
Line 354	.dynamic-section-copy
Line 362	.dynamic-section-grid
Line 368	.dynamic-section-card
Line 380	.dynamic-section-card p
Line 384	.dynamic-section-card h3
Line 389	.dynamic-section-card p
Line 396	.dynamic-link-list
Line 400	.fun-facts-section

Item	Explanation
Line 409	.fun-facts-section[hidden]
Line 413	.fun-facts-window
Line 427	.fun-facts-label
Line 441	.fun-facts-lines
Line 446	.fun-fact-line
Line 454	.builder-download-section
Line 462	.builder-download-link
Line 482	.builder-download-link:focus-visible
Line 489	.builder-download-link:active
Line 494	.builder-download-link svg
Line 505	.sr-only
Line 517	.ai-assistant-panel
Line 528	.ai-assistant-console
Line 545	.ai-assistant-console h2
Line 552	.ai-assistant-status
Line 559	.ai-clear-chat
Line 578	.ai-clear-chat:focus-visible
Line 585	.ai-clear-chat:active
Line 591	.ai-assistant-status[data-state="working"]
Line 595	.ai-assistant-status[data-state="error"]
Line 599	.ai-assistant-log
Line 613	.ai-message
Line 623	.ai-message p
Line 629	.ai-answer-content
Line 634	.ai-answer-content ul
Line 643	.ai-answer-content strong
Line 647	.ai-code-block
Line 660	.ai-code-block code
Line 664	.ai-message-user
Line 670	.ai-message-assistant
Line 674	.ai-message-pending p::after
Line 691	.ai-source-list
Line 698	.ai-source-panel
Line 704	.ai-source-heading
Line 713	.ai-assistant-form button
Line 724	.ai-source-list button
Line 732	.ai-source-list button:focus-visible
Line 738	.ai-assistant-form
Line 745	.ai-assistant-form input
Line 757	.ai-assistant-form input:focus-visible
Line 762	.ai-assistant-form button

Item	Explanation
Line 772	.ai-assistant-form button:focus-visible
Line 776	.summary-band
Line 784	.metric
Line 790	.metric strong
Line 796	.metric span
Line 803	.section
Line 807	.section-heading
Line 816	.search-wrap
Line 825	.search-engine
Line 830	.search-engine .search-wrap
Line 834	.search-wrap input
Line 845	::highlight(portfolio-search-highlight)
Line 850	.search-results-panel
Line 867	.search-results-panel[hidden]
Line 871	.search-results-tools
Line 883	.search-results-tools [data-state="not-found"]
Line 887	.search-results-tools [data-state="found"]
Line 891	.search-results-tools label
Line 898	.search-results-tools select
Line 909	.search-results-list
Line 914	.search-result-item
Line 928	.search-result-item:focus-visible
Line 934	.search-result-title
Line 940	.search-result-snippet
Line 947	.search-result-snippet
Line 951	.search-result-mark
Line 958	.filters

Representative opening snippet

```

1: :root {
2:   color-scheme: light;
3:   --ink: #17202a;
4:   --muted: #526d82;
5:   --line: #c9deea;
6:   --paper: #eef8fc;
7:   --panel: #ffffff;
8:   --teal: #0d7f91;
9:   --amber: #d18b21;
10:  --coral: #bc5648;
11:  --graphite: #223244;
12:  --omb-copper: #c7772d;
13:  --focus: #0b6fce;
14:  --shadow: 0 18px 48px rgba(23, 32, 42, 0.14);
15: }
16:
17: * {
18:   box-sizing: border-box;
19: }
20:
21: html {
22:   scroll-behavior: smooth;
23: }
24:
25: body {
26:   margin: 0;

```

```

27:   color: var(--ink);
28:   background: var(--paper);
29:   font-family: Inter, ui-sans-serif, system-ui, -apple-system, BlinkMacSystemFont, "Segoe UI", sans-
serif;
30:   line-height: 1.5;
31: }
32:
33: body.full-window-open {
34:   overflow: hidden;
35: }
36:
37: a {
38:   color: inherit;
39:   text-decoration: none;
40: }
41:
42: a:focus-visible,
43: button:focus-visible,
44: input:focus-visible {
45:   outline: 3px solid var(--focus);
46:   outline-offset: 3px;
47: }
48:
49: .site-header {
50:   position: sticky;
51:   top: 0;
52:   z-index: 20;
53:   display: flex;
54:   align-items: center;
55:   justify-content: space-between;
56:   gap: 24px;
57:   min-height: 72px;
58:   padding: 12px clamp(18px, 4vw, 56px);
59:   border-bottom: 1px solid rgba(201, 222, 234, 0.84);
60:   background: rgba(238, 248, 252, 0.92);
61:   backdrop-filter: blur(16px);
62: }
63:
64: .brand {
65:   display: inline-flex;
66:   align-items: center;
67:   gap: 12px;
68:   min-width: 0;
69: }
70:
71: .brand-lockup {
72:   position: relative;

```

Safe change checklist

- Confirm which app, website, installer, workflow, or document consumes this file.
- Change it on a feature branch from development.
- Regenerate docs if the file purpose, API surface, or behavior changes.
- Run the smallest relevant smoke test before merging into development.

Important Code Reference: cloudflare/portfolio-ai-worker.js

Item	Explanation
File	cloudflare/portfolio-ai-worker.js
Category	JavaScript source
Folder role	Cloudflare Worker/deployment area. Used for public AI backend and Cloudflare deployment notes.
Size	10,435 bytes
Extension	.js
MIME type	application/javascript
Text lines	276

Why this file exists

- Cloudflare Worker backend for the public portfolio AI assistant and fallback intelligence path.

How it is used

Cloudflare Worker/deployment area. Used for public AI backend and Cloudflare deployment notes.

How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

API endpoints mentioned or called

Item	Explanation
/api/portfolio-ai	Line 228. Handles AI assistant questions.

Important implementation signals

Item	Explanation
Network request at line 2	Frontend-backend communication or public web/API calls. Evidence: "https://mauriceotieno.com",
Network request at line 3	Frontend-backend communication or public web/API calls. Evidence: "https://www.mauriceotieno.com",
Network request at line 4	Frontend-backend communication or public web/API calls. Evidence: "https://otienomaurice.github.io",
Network request at line 5	Frontend-backend communication or public web/API calls. Evidence: "http://localhost:8080",
Network request at line 6	Frontend-backend communication or public web/API calls. Evidence: "http://localhost:8081",
Network request at line 7	Frontend-backend communication or public web/API calls. Evidence: "http://127.0.0.1:8080",
Network request at line 8	Frontend-backend communication or public web/API calls. Evidence: "http://127.0.0.1:8081"
Security or auth at line 11	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: function json(data, status = 200, headers = {}) {
Security or auth at line 14	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: headers: {
Local storage at line 16	Local persistence, draft state, auth cache, update snooze, or browser-side caching. Evidence: "Cache-Control": "no-store",
Security or auth at line 17	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: ...headers
Security or auth at line 30	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: function corsHeaders(request, env) {
Security or auth at line 31	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: const origin = request.headers.get("Origin") "";

Item	Explanation
Security or auth at line 36	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "Access-Control-Allow-Headers": "Content-Type",
Cloudflare or AI at line 54	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: function extractOpenAiText(data = {}) {
Security or auth at line 89	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "Never invent portfolio projects, credentials, files, test results, or private repository access.",
Cloudflare or AI at line 121	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: async function callOpenAi(body, env) {
Cloudflare or AI at line 122	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: const model = env.OPENAI_MODEL "gpt-4.1-mini";
Cloudflare or AI at line 124	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: model,
Cloudflare or AI at line 135	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: max_output_tokens: Number(env.OPENAI_MAX_OUTPUT_TOKENS 1000)
Cloudflare or AI at line 138	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: const response = await fetch("https://api.openai.com/v1/responses", {
Security or auth at line 140	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: headers: {
Cloudflare or AI at line 141	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "Authorization": `Bearer \${env.OPENAI_API_KEY}`,
Cloudflare or AI at line 151	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: error: data?.error?.message "OpenAI request failed."
Cloudflare or AI at line 156	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: answer: extractOpenAiText(data),
Cloudflare or AI at line 157	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: model
Cloudflare or AI at line 166	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: error: "Cloudflare Workers AI binding is not available for this Worker."
Cloudflare or AI at line 170	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: const model = env.WORKERS_AI_MODEL "@cf/meta/llama-3.2-3b-instruct";
Cloudflare or AI at line 171	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: const data = await env.AI.run(model, {
Cloudflare or AI at line 173	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: max_tokens: Number(env.OPENAI_MAX_OUTPUT_TOKENS 1000)
Cloudflare or AI at line 178	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: model
Cloudflare or AI at line 210	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "The full AI model is not reachable from this Worker right now, so this is a fallback answer from the available context."
Cloudflare or AI at line 214	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "I can answer that in a general way, but the full AI model is not reachable from this Worker right now.",
Network request at line 223	Frontend-backend communication or public web/API calls. Evidence: async fetch(request, env) {
Security or auth at line 224	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: const cors = corsHeaders(request, env);
Security or auth at line 225	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: if (request.method === "OPTIONS") return new Response(null, { status: 204, headers: cors });
Cloudflare or AI at line 228	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: if (request.method !== "POST" !url.pathname.endsWith("/api/portfolio-ai")) {
Cloudflare or AI at line 243	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: let provider = env.OPENAI_API_KEY ? "openai" : "cloudflare-workers-ai";
Cloudflare or AI at line 244	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: let result = env.OPENAI_API_KEY
Cloudflare or AI at line 245	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: ? await callOpenAi({ ...body, question }, env)
Cloudflare or AI at line 247	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: if (!result.ok && env.OPENAI_API_KEY && env.AI) {
Cloudflare or AI at line 248	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: provider = "cloudflare-workers-ai";

Item	Explanation
Cloudflare or AI at line 254	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: model: "worker-fallback",
Cloudflare or AI at line 262	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: model: result.model,
Cloudflare or AI at line 269	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: model: "worker-fallback",

Important constants and variables

Item	Explanation
defaultAllowedOrigins	Line 1: const defaultAllowedOrigins = [
origin	Line 31: const origin = request.headers.get("Origin") "";
allowOrigin	Line 32: const allowOrigin = allowedOrigins(env).includes(origin) ? origin : defaultAllowedOrigins[0];
parts	Line 56: const parts = [];
messages	Line 96: const messages = [{ role: "system", content: buildInstructions() }];
model	Line 122: const model = env.OPENAI_MODEL "gpt-4.1-mini";
payload	Line 123: const payload = {
response	Line 138: const response = await fetch("https://api.openai.com/v1/responses", {
data	Line 146: const data = await response.json().catch(() => ({}));
model	Line 170: const model = env.WORKERS_AI_MODEL "@cf/meta/llama-3.2-3b-instruct";
data	Line 171: const data = await env.AI.run(model, {
question	Line 183: const question = String(body.question "").trim();
clean	Line 184: const clean = question.toLowerCase();
context	Line 185: const context = body.context {};
projects	Line 186: const projects = Array.isArray(context.projects) ? context.projects : [];
owner	Line 187: const owner = context.owner context.profile?.displayName "Maurice Otieno";
cors	Line 224: const cors = corsHeaders(request, env);
url	Line 227: const url = new URL(request.url);
body	Line 232: let body = {};
question	Line 239: const question = clamp(body.question, 1200).trim();
provider	Line 243: let provider = env.OPENAI_API_KEY ? "openai" : "cloudflare-workers-ai";
result	Line 244: let result = env.OPENAI_API_KEY

JSON/YAML-style keys detected

Item	Explanation
Content-Type	Line 15: "Content-Type": "application/json; charset=utf-8",
Cache-Control	Line 16: "Cache-Control": "no-store",
Access-Control-Allow-Origin	Line 34: "Access-Control-Allow-Origin": allowOrigin,
Access-Control-Allow-Methods	Line 35: "Access-Control-Allow-Methods": "POST, OPTIONS",
Access-Control-Allow-Headers	Line 36: "Access-Control-Allow-Headers": "Content-Type",
Access-Control-Max-Age	Line 37: "Access-Control-Max-Age": "86400",
Vary	Line 38: "Vary": "Origin"
Authorization	Line 141: "Authorization": `Bearer \${env.OPENAI_API_KEY}`,
Content-Type	Line 142: "Content-Type": "application/json"

Function-by-function detail

json - lines 11-20

Item	Explanation
Source location	Lines 11-20 (10 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	function json(data, status = 200, headers = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	Response, stringify
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
11: function json(data, status = 200, headers = {}) {
12:   return new Response(JSON.stringify(data, null, 2), {
13:     status,
14:     headers: {
15:       "Content-Type": "application/json; charset=utf-8",
16:       "Cache-Control": "no-store",
17:       ...headers
18:     }
19:   });
20: }
```

allowedOrigins - lines 22-28

Item	Explanation
Source location	Lines 22-28 (7 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	function allowedOrigins(env) {
Touches	Mostly local calculations or local UI state.
Calls detected	split, map, trim, filter, concat
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
22: function allowedOrigins(env) {
23:   return String(env.ALLOWED_ORIGINS || "")
24:     .split(",")
25:     .map((origin) => origin.trim())
26:     .filter(Boolean)
27:     .concat(defaultAllowedOrigins);
28: }
```

corsHeaders - lines 30-40

Item	Explanation
Source location	Lines 30-40 (11 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	function corsHeaders(request, env) {

Item	Explanation
Touches	Mostly local calculations or local UI state.
Calls detected	get, allowedOrigins, includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

30: function corsHeaders(request, env) {
31:   const origin = request.headers.get("Origin") || "";
32:   const allowOrigin = allowedOrigins(env).includes(origin) ? origin : defaultAllowedOrigins[0];
33:   return {
34:     "Access-Control-Allow-Origin": allowOrigin,
35:     "Access-Control-Allow-Methods": "POST, OPTIONS",
36:     "Access-Control-Allow-Headers": "Content-Type",
37:     "Access-Control-Max-Age": "86400",
38:     "Vary": "Origin"
39:   };
40: }

```

clamp - lines 42-44

Item	Explanation
Source location	Lines 42-44 (3 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	function clamp(value, maxLength) {
Touches	Mostly local calculations or local UI state.
Calls detected	slice
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

42: function clamp(value, maxLength) {
43:   return String(value || "").slice(0, maxLength);
44: }

```

cleanConversation - lines 46-52

Item	Explanation
Source location	Lines 46-52 (7 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	function cleanConversation(conversation = []) {
Touches	Mostly local calculations or local UI state.
Calls detected	isArray, slice, map, clamp, filter
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
46: function cleanConversation(conversation = []) {
47:   if (!Array.isArray(conversation)) return [];
48:   return conversation.slice(-8).map((item) => ({
49:     role: item?.role === "assistant" ? "assistant" : "user",
50:     content: clamp(item?.content || item?.text || "", 1400)
51:   })).filter((item) => item.content);
52: }
```

extractOpenAiText - lines 54-64

Item	Explanation
Source location	Lines 54-64 (11 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	function extractOpenAiText(data = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, push, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
54: function extractOpenAiText(data = {}) {
55:   if (typeof data.output_text === "string" && data.output_text.trim()) return data.output_text.trim();
56:   const parts = [];
57:   for (const item of data.output || []) {
58:     for (const content of item.content || []) {
59:       if (typeof content.text === "string") parts.push(content.text);
60:       if (typeof content.output_text === "string") parts.push(content.output_text);
61:     }
62:   }
63:   return parts.join("\n").trim();
64: }
```

extractWorkersAiText - lines 66-79

Item	Explanation
Source location	Lines 66-79 (14 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	function extractWorkersAiText(data = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	trim, isArray, map, filter, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 9 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
66: function extractWorkersAiText(data = {}) {
67:   if (typeof data.response === "string" && data.response.trim()) return data.response.trim();
68:   if (typeof data.result?.response === "string" && data.result.response.trim()) return
data.result.response.trim();
69:   if (typeof data.output_text === "string" && data.output_text.trim()) return data.output_text.trim();
70:   if (Array.isArray(data.output)) {
71:     return data.output.map((item) => {
72:       if (typeof item === "string") return item;
73:       if (typeof item?.text === "string") return item.text;
74:       if (typeof item?.content === "string") return item.content;
75:       return "";
76:     }).filter(Boolean).join("\n").trim();
```

```

77:   }
78:   return "";
79: }

```

buildInstructions - lines 81-93

Item	Explanation
Source location	Lines 81-93 (13 source lines).
Purpose	Creates an object, UI structure, process, payload, or generated artifact.
Declaration	function buildInstructions() {
Touches	authentication/security data
Calls detected	join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

81: function buildInstructions() {
82:   return [
83:     "You are the AI assistant for a public engineering portfolio website.",
84:     "Answer naturally like a helpful technical mentor and recruiter-facing portfolio guide.",
85:     "If the question is general, answer the concept directly before mentioning portfolio context.",
86:     "If the question is about the portfolio owner, projects, files, resume, tools, links, or sections,
answer from the supplied portfolio context.",
87:     "Use recent conversation to keep a conversational flow. Greetings should receive short greetings.",
88:     "Only include portfolio links when they are directly relevant to the visitor's question.",
89:     "Never invent portfolio projects, credentials, files, test results, or private repository access.",
90:     "If a requested detail is not in the context, say what is missing and give a useful next step.",
91:     "Keep answers concise, readable, and specific. Use bullets when they make the answer easier to scan."
92:   ].join("\n");
93: }

```

buildMessages - lines 95-105

Item	Explanation
Source location	Lines 95-105 (11 source lines).
Purpose	Creates an object, UI structure, process, payload, or generated artifact.
Declaration	function buildMessages(body = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	buildInstructions, cleanConversation, push, buildUserPrompt
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

95: function buildMessages(body = {}) {
96:   const messages = [{ role: "system", content: buildInstructions() }];
97:   for (const item of cleanConversation(body.conversation)) {
98:     messages.push({
99:       role: item.role === "assistant" ? "assistant" : "user",
100:      content: item.content
101:     });
102:   }
103:   messages.push({ role: "user", content: buildUserPrompt(body) });
104:   return messages;
105: }

```

buildUserPrompt - lines 107-119

Item	Explanation
Source location	Lines 107-119 (13 source lines).
Purpose	Creates an object, UI structure, process, payload, or generated artifact.
Declaration	function buildUserPrompt(body = {}) {
Touches	Mostly local calculations or local UI state.
Calls detected	clamp, stringify, cleanConversation, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

107: function buildUserPrompt(body = {}) {
108:   return [
109:     `Visitor question: ${clamp(body.question, 1200)}`,
110:     `Question intent: ${clamp(body.intent || "portfolio_specific", 80)}`,
111:     `Web/public lookup requested by browser: ${body.allowWebSearch === true ? "yes" : "no"}`,
112:     "",
113:     "Recent conversation JSON:",
114:     clamp(JSON.stringify(cleanConversation(body.conversation), null, 2), 6000),
115:     "",
116:     "Portfolio context JSON:",
117:     clamp(JSON.stringify(body.context || {}, null, 2), 20000)
118:   ].join("\n");
119: }

```

callOpenAi - lines 121-145

Item	Explanation
Source location	Lines 121-145 (25 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	async function callOpenAi(body, env) {
Touches	network/API requests, authentication/security data
Calls detected	buildInstructions, buildUserPrompt, fetch, stringify
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

121: async function callOpenAi(body, env) {
122:   const model = env.OPENAI_MODEL || "gpt-4.1-mini";
123:   const payload = {
124:     model,
125:     input: [
126:       {
127:         role: "developer",
128:         content: [{ type: "input_text", text: buildInstructions() }]
129:       },
130:       {
131:         role: "user",
132:         content: [{ type: "input_text", text: buildUserPrompt(body) }]
133:       }
134:     ],
135:     max_output_tokens: Number(env.OPENAI_MAX_OUTPUT_TOKENS || 1000)
136:   };
137:
138:   const response = await fetch("https://api.openai.com/v1/responses", {
139:     method: "POST",
140:     headers: {
141:       "Authorization": `Bearer ${env.OPENAI_API_KEY}`,
142:       "Content-Type": "application/json"
143:     },

```



```

144:     body: JSON.stringify(payload)
145:   });

```

callWorkersAi - lines 161-180

Item	Explanation
Source location	Lines 161-180 (20 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	async function callWorkersAi(body, env) {
Touches	authentication/security data
Calls detected	run, buildMessages, extractWorkersAiText
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

161: async function callWorkersAi(body, env) {
162:   if (!env.AI || typeof env.AI.run !== "function") {
163:     return {
164:       ok: false,
165:       status: 503,
166:       error: "Cloudflare Workers AI binding is not available for this Worker."
167:     };
168:   }
169:
170:   const model = env.WORKERS_AI_MODEL || "@cf/meta/llama-3.2-3b-instruct";
171:   const data = await env.AI.run(model, {
172:     messages: buildMessages(body),
173:     max_tokens: Number(env.OPENAI_MAX_OUTPUT_TOKENS || 1000)
174:   });
175:   return {
176:     ok: true,
177:     answer: extractWorkersAiText(data),
178:     model
179:   };
180: }

```

fallbackWorkerAnswer - lines 182-220

Item	Explanation
Source location	Lines 182-220 (39 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	function fallbackWorkerAnswer(body = {}) {
Touches	filesystem
Calls detected	trim, toLowerCase, isArray, slice, map, filter, join, b
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

182: function fallbackWorkerAnswer(body = {}) {
183:   const question = String(body.question || "").trim();
184:   const clean = question.toLowerCase();
185:   const context = body.context || {};
186:   const projects = Array.isArray(context.projects) ? context.projects : [];
187:   const owner = context.owner || context.profile?.displayName || "Maurice Otieno";
188:
189:   if (/^(hi|hello|hey|yo)\b/.test(clean)) {
190:     return `Hi, what can I do for you? I can help with ${owner}'s projects, files, resume links, or
related electronics concepts.`;

```

```

191:   }
192:   if (/\\bembedded systems?\\b/.test(clean)) {
193:     return [
194:       "Embedded systems is a field of engineering where software is built into dedicated hardware to
control a device, measure signals, communicate with peripherals, or respond to real-time events.",
195:       "",
196:       "Typical embedded work combines microcontrollers or processors, firmware, timing, interrupts,
communication buses, sensors, actuators, and hardware validation.",
197:       "",
198:       projects.length
199:       ? `In this portfolio, I would connect that explanation back to saved projects such as
${projects.slice(0, 3).map((project) => project.title || project.name).filter(Boolean).join(", ")} when those
projects are relevant.`
200:       : "When project context is available, I connect the explanation to the saved project evidence."
201:     ].join("\\n");
202:   }
203:   if (projects.length &&
/b(project|portfolio|maurice|file|github|resume|tool|design|test)\\b/.test(clean)) {
204:     return [
205:       "I can use the portfolio context for ${owner}.",
206:       "",
207:       "Relevant saved project areas include:",
208:       ...projects.slice(0, 6).map((project) => `- ${project.title || project.name || "Untitled
project"}`),
209:       "",
210:       "The full AI model is not reachable from this Worker right now, so this is a fallback answer from
the available context."
211:     ].join("\\n");
212:   }
213:   return [
214:     "I can answer that in a general way, but the full AI model is not reachable from this Worker right
now.",
215:     "",
216:     "A useful answer should define the concept, explain the main parts, give a practical example, and
connect it to a project artifact when portfolio context is relevant.",
217:     "",
218:     `Question received: ${question}`
219:   ].join("\\n");
220: }

```

Representative opening snippet

```

1: const defaultAllowedOrigins = [
2:   "https://mauriceotieno.com",
3:   "https://www.mauriceotieno.com",
4:   "https://otienomaurice.github.io",
5:   "http://localhost:8080",
6:   "http://localhost:8081",
7:   "http://127.0.0.1:8080",
8:   "http://127.0.0.1:8081"
9: ];
10:
11: function json(data, status = 200, headers = {}) {
12:   return new Response(JSON.stringify(data, null, 2), {
13:     status,
14:     headers: {
15:       "Content-Type": "application/json; charset=utf-8",
16:       "Cache-Control": "no-store",
17:       ...headers
18:     }
19:   });
20: }
21:
22: function allowedOrigins(env) {
23:   return String(env.ALLOWED_ORIGINS || "").
24:     .split(",")
25:     .map((origin) => origin.trim())
26:     .filter(Boolean)
27:     .concat(defaultAllowedOrigins);
28: }
29:
30: function corsHeaders(request, env) {
31:   const origin = request.headers.get("Origin") || "";
32:   const allowOrigin = allowedOrigins(env).includes(origin) ? origin : defaultAllowedOrigins[0];
33:   return {
34:     "Access-Control-Allow-Origin": allowOrigin,
35:     "Access-Control-Allow-Methods": "POST, OPTIONS",
36:     "Access-Control-Allow-Headers": "Content-Type",
37:     "Access-Control-Max-Age": "86400",
38:     "Vary": "Origin"
39:   };
40: }
41:
42: function clamp(value, maxLength) {
43:   return String(value || "").slice(0, maxLength);
44: }
45:
46: function cleanConversation(conversation = []) {
47:   if (!Array.isArray(conversation)) return [];
48:   return conversation.slice(-8).map((item) => ({

```

```

49:     role: item?.role === "assistant" ? "assistant" : "user",
50:     content: clamp(item?.content || item?.text || "", 1400)
51:   })).filter((item) => item.content);
52: }
53:
54: function extractOpenAiText(data = {}) {
55:   if (typeof data.output_text === "string" && data.output_text.trim()) return data.output_text.trim();
56:   const parts = [];
57:   for (const item of data.output || []) {
58:     for (const content of item.content || []) {
59:       if (typeof content.text === "string") parts.push(content.text);
60:       if (typeof content.output_text === "string") parts.push(content.output_text);
61:     }
62:   }
63:   return parts.join("\n").trim();
64: }
65:
66: function extractWorkersAiText(data = {}) {
67:   if (typeof data.response === "string" && data.response.trim()) return data.response.trim();
68:   if (typeof data.result?.response === "string" && data.result.response.trim()) return
data.result.response.trim();
69:   if (typeof data.output_text === "string" && data.output_text.trim()) return data.output_text.trim();
70:   if (Array.isArray(data.output)) {
71:     return data.output.map((item) => {
72:       if (typeof item === "string") return item;

```

Safe change checklist

- Confirm which app, website, installer, workflow, or document consumes this file.
- Change it on a feature branch from development.
- Regenerate docs if the file purpose, API surface, or behavior changes.
- Run the smallest relevant smoke test before merging into development.

Important Code Reference: cloudflare/wrangler.toml

Item	Explanation
File	cloudflare/wrangler.toml
Category	Configuration or structured data
Folder role	Cloudflare Worker/deployment area. Used for public AI backend and Cloudflare deployment notes.
Size	573 bytes
Extension	.toml
MIME type	unknown
Text lines	18

Why this file exists

- Wrangler configuration used to deploy the Cloudflare Worker.

How it is used

Cloudflare Worker/deployment area. Used for public AI backend and Cloudflare deployment notes.

How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

Important implementation signals

Item	Explanation
Cloudflare or AI at line 1	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: name = "omb-portfolio-ai"
Cloudflare or AI at line 2	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: main = "portfolio-ai-worker.js"
Cloudflare or AI at line 7	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: { pattern = "mauriceotieno.com/api/portfolio-ai", zone_name = "mauriceotieno.com" },
Cloudflare or AI at line 8	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: { pattern = "www.mauriceotieno.com/api/portfolio-ai", zone_name = "mauriceotieno.com" }
Cloudflare or AI at line 12	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: OPENAI_MODEL = "gpt-4.1-mini"
Cloudflare or AI at line 13	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: OPENAI_MAX_OUTPUT_TOKENS = "1000"
Cloudflare or AI at line 14	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: WORKERS_AI_MODEL = "@cf/meta/llama-3.2-3b-instruct"
Network request at line 15	Frontend-backend communication or public web/API calls. Evidence: ALLOWED_ORIGINS = "https://mauriceotieno.com,https://www.mauriceotieno.com,https://otienomaurice.github.io"

Representative opening snippet

```
1: name = "omb-portfolio-ai"
2: main = "portfolio-ai-worker.js"
3: compatibility_date = "2026-07-01"
4: workers_dev = true
5:
6: routes = [
7:   { pattern = "mauriceotieno.com/api/portfolio-ai", zone_name = "mauriceotieno.com" },
8:   { pattern = "www.mauriceotieno.com/api/portfolio-ai", zone_name = "mauriceotieno.com" }
9: ]
10:
11: [vars]
12: OPENAI_MODEL = "gpt-4.1-mini"
13: OPENAI_MAX_OUTPUT_TOKENS = "1000"
14: WORKERS_AI_MODEL = "@cf/meta/llama-3.2-3b-instruct"
15: ALLOWED_ORIGINS =
```

```
"https://mauriceotieno.com,https://www.mauriceotieno.com,https://otienomaurice.github.io"  
16:  
17: [ai]  
18: binding = "AI"
```

Safe change checklist

- Confirm which app, website, installer, workflow, or document consumes this file.
- Change it on a feature branch from development.
- Regenerate docs if the file purpose, API surface, or behavior changes.
- Run the smallest relevant smoke test before merging into development.

Important Code Reference: build/installer.nsh

Item	Explanation
File	build/installer.nsh
Category	NSIS installer script
Folder role	Packaging and installer customization. Used while creating the Windows app installer.
Size	29,416 bytes
Extension	.nsh
MIME type	unknown
Text lines	570

Why this file exists

- NSIS installer customization script for update behavior, shortcut placement, old install detection, and setup details.

How it is used

Packaging and installer customization. Used while creating the Windows app installer.

How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

Important implementation signals

Item	Explanation
Installer at line 6	Windows setup, update, shortcut, or installation behavior. Evidence: <code>!ifndef BUILD_UNINSTALLER</code>
Installer at line 8	Windows setup, update, shortcut, or installation behavior. Evidence: <code>Var OMB_ShortcutCheckbox</code>
Installer at line 10	Windows setup, update, shortcut, or installation behavior. Evidence: <code>Var OMB_CreateDesktopShortcutState</code>
Installer at line 24	Windows setup, update, shortcut, or installation behavior. Evidence: <code>\${NSD_CreateLabel} 0 0 100% 12u "Tools and shortcuts"</code>
Git command at line 27	Repository state, publishing, branch checks, or release automation. Evidence: <code>\${NSD_CreateLabel} 0 18u 100% 36u "The desktop app includes its own runtime. Users do not need to install Node.js or npm.\$\r\$\nGit for Windows with Git Credential Manager is only needed when publishing to GitHub."</code>
Installer at line 30	Windows setup, update, shortcut, or installation behavior. Evidence: <code>\${NSD_CreateCheckbox} 0 62u 100% 12u "Create a desktop shortcut"</code>
Installer at line 31	Windows setup, update, shortcut, or installation behavior. Evidence: <code>Pop \$OMB_ShortcutCheckbox</code>
Installer at line 32	Windows setup, update, shortcut, or installation behavior. Evidence: <code>\${If} \$OMB_CreateDesktopShortcutState == ""</code>
Installer at line 33	Windows setup, update, shortcut, or installation behavior. Evidence: <code>StrCpy \$OMB_CreateDesktopShortcutState \${BST_CHECKED}</code>
Installer at line 35	Windows setup, update, shortcut, or installation behavior. Evidence: <code>\${If} \$OMB_CreateDesktopShortcutState == \${BST_CHECKED}</code>
Installer at line 36	Windows setup, update, shortcut, or installation behavior. Evidence: <code>\${NSD_Check} \$OMB_ShortcutCheckbox</code>
Git command at line 39	Repository state, publishing, branch checks, or release automation. Evidence: <code>\${NSD_CreateCheckbox} 0 82u 100% 24u "Check publishing tools and install Git for Windows if Git/Git Credential Manager is missing"</code>
Git command at line 48	Repository state, publishing, branch checks, or release automation. Evidence: <code>\${NSD_CreateLabel} 0 114u 100% 38u "Existing shared tools are not uninstalled silently. If a compatible tool is already available, setup skips it. If Git is missing or unusable, setup will try to install Git for Windows"</code>
Installer at line 55	Windows setup, update, shortcut, or installation behavior. Evidence: <code>\${NSD_GetState} \$OMB_ShortcutCheckbox \$OMB_CreateDesktopShortcutState</code>

Item	Explanation
Git command at line 66	Repository state, publishing, branch checks, or release automation. Evidence: IfFileExists "\$PROGRAMFILES64\Git\cmd\git.exe" 0 +2
Git command at line 68	Repository state, publishing, branch checks, or release automation. Evidence: IfFileExists "\$PROGRAMFILES\Git\cmd\git.exe" 0 +2
Git command at line 70	Repository state, publishing, branch checks, or release automation. Evidence: IfFileExists "\$PROGRAMFILES32\Git\cmd\git.exe" 0 +2
Git command at line 72	Repository state, publishing, branch checks, or release automation. Evidence: IfFileExists "\$LOCALAPPDATA\Programs\Git\cmd\git.exe" 0 +2
Git command at line 76	Repository state, publishing, branch checks, or release automation. Evidence: nsExec::ExecToStack 'cmd /c git --version'
Security or auth at line 86	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: Function OMBCheckGitCredentialManagerAvailable
Git command at line 87	Repository state, publishing, branch checks, or release automation. Evidence: nsExec::ExecToStack 'cmd /c git credential-manager --version'
Security or auth at line 100	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: Call OMBCheckGitCredentialManagerAvailable
Git command at line 105	Repository state, publishing, branch checks, or release automation. Evidence: DetailPrint "Git for Windows and Git Credential Manager were found. Skipping publishing tool installation."
Git command at line 114	Repository state, publishing, branch checks, or release automation. Evidence: MessageBox MB_OK MB_ICONINFORMATION "OMB Portfolio Builder was installed, but Git for Windows was not found and Windows Package Manager is unavailable. Open Publishing target > Install Git inside the app, or install Git
Git command at line 118	Repository state, publishing, branch checks, or release automation. Evidence: DetailPrint "Installing or repairing Git for Windows. This can take several minutes."
Git command at line 119	Repository state, publishing, branch checks, or release automation. Evidence: ExecWait 'winget install --id Git.Git -e --source winget --accept-source-agreements --accept-package-agreements' \$0
Git command at line 121	Repository state, publishing, branch checks, or release automation. Evidence: MessageBox MB_OK MB_ICONINFORMATION "OMB Portfolio Builder was installed, but Git for Windows setup did not complete. Open Publishing target > Install Git inside the app when you are ready to publish."
Git command at line 123	Repository state, publishing, branch checks, or release automation. Evidence: DetailPrint "Git for Windows setup completed."
Installer at line 149	Windows setup, update, shortcut, or installation behavior. Evidence: MessageBox MB_OK MB_ICONINFORMATION "OMB Portfolio Builder \$OMB_ExistingInstallVersion is already installed.\$r\$n\$r\$nThis installer is also version \${VERSION}, so setup will stop without changing the app."
Installer at line 157	Windows setup, update, shortcut, or installation behavior. Evidence: MessageBox MB_OK MB_ICONINFORMATION "OMB Portfolio Builder \$OMB_ExistingInstallVersion is already installed.\$r\$n\$r\$nThis installer is older (\${VERSION}), so setup will stop without changing the app."
Installer at line 161	Windows setup, update, shortcut, or installation behavior. Evidence: DetailPrint "OMB Portfolio Builder \$OMB_ExistingInstallVersion is newer than installer \${VERSION}. Setup stopped."
Installer at line 168	Windows setup, update, shortcut, or installation behavior. Evidence: MessageBox MB_OK MB_ICONSTOP "Setup found an existing local OMB Portfolio Builder workspace:\$r\$n\$r\$n\$PROFILE\OMB\builder\desktop-builder\$r\$n\$r\$nRemove, uninstall, or rename that copy before installing this releas
Installer at line 172	Windows setup, update, shortcut, or installation behavior. Evidence: MessageBox MB_OK MB_ICONSTOP "Setup found an existing local OMB Portfolio Builder workspace:\$r\$n\$r\$n\$DOCUMENTS\OMB\desktop-builder\$r\$n\$r\$nRemove, uninstall, or rename that copy before installing this release. Thi
Installer at line 176	Windows setup, update, shortcut, or installation behavior. Evidence: MessageBox MB_OK MB_ICONSTOP "Setup found an existing local OMB Portfolio Builder workspace:\$r\$n\$r\$n\$PROFILE\OneDrive\Documents\OMB\desktop-builder\$r\$n\$r\$nRemove, uninstall, or rename that copy before installing
Installer at line 180	Windows setup, update, shortcut, or installation behavior. Evidence: MessageBox MB_OK MB_ICONSTOP "Setup found an existing local OMB Portfolio Builder workspace:\$r\$n\$r\$n\$PROFILE\Documents\OMB\desktop-builder\$r\$n\$r\$nRemove, uninstall, or rename that copy before installing this rele
Installer at line 184	Windows setup, update, shortcut, or installation behavior. Evidence: MessageBox MB_OK MB_ICONSTOP "Setup found an existing OMB Portfolio Builder development copy:\$r\$n\$r\$n\$DOCUMENTS\omb-portfolio-builder\$r\$n\$r\$nRemove, uninstall, or rename that copy before installing this release."

Item	Explanation
Installer at line 188	Windows setup, update, shortcut, or installation behavior. Evidence: MessageBox MB_OK MB_ICONSTOP "Setup found an existing OMB Portfolio Builder development copy.\$\r\$\n\$\r\$\n\$PROFILE\OneDrive\Documents\omb-portfolio-builder\$\r\$\n\$\r\$\nRemove, uninstall, or rename that copy before installi
File read at line 208	Reads local builder state, project files, website assets, or configuration. Evidence: FileWrite \$0 " try { \$\$packageText = Get-Content -LiteralPath (Join-Path \$\$Directory 'package.json') -Raw -ErrorAction Stop } catch { return \$\$false }\$\r\$\n"
File read at line 215	Reads local builder state, project files, website assets, or configuration. Evidence: FileWrite \$0 " try { \$\$packageText = Get-Content -LiteralPath (Join-Path \$\$Directory 'package.json') -Raw -ErrorAction Stop } catch { return \$\$false }\$\r\$\n"
Installer at line 220	Windows setup, update, shortcut, or installation behavior. Evidence: FileWrite \$0 " \$\$managedRoot = Join-Path \$\$env:LOCALAPPDATA 'OMB Portfolio Builder'\$\r\$\n"
Git command at line 227	Repository state, publishing, branch checks, or release automation. Evidence: FileWrite \$0 " if (\$\$name -in @('node_modules','git','dist','.vs',\$\$Recycle.Bin','System Volume Information','WinSxS')) { return \$\$true }\$\r\$\n"
Installer at line 251	Windows setup, update, shortcut, or installation behavior. Evidence: FileWrite \$0 "\$\$registryKeys = @(HKLM:\Software\Microsoft\Windows\CurrentVersion\Uninstall*,HKCU:\Software\Microsoft\Windows\CurrentVersion\Uninstall*)\$\r\$\n"
Installer at line 261	Windows setup, update, shortcut, or installation behavior. Evidence: FileWrite \$0 "foreach (\$\$common in @(\$\$env:ProgramFiles, \$\$programFilesX86, (Join-Path \$\$env:LOCALAPPDATA 'Programs'), \$\$env:USERPROFILE, (Join-Path \$\$env:PUBLIC 'Desktop')) { if (\$\$common -and -not \$\$roots.Contains(\$\$c
Installer at line 282	Windows setup, update, shortcut, or installation behavior. Evidence: MessageBox MB_OK MB_ICONSTOP "Setup found another OMB Portfolio Builder copy on this machine.\$\r\$\n\$\r\$\n\$R3\$\r\$\n\$\r\$\nTo prevent duplicate installations and stale builder features, remove, uninstall, or rename the exis
Installer at line 285	Windows setup, update, shortcut, or installation behavior. Evidence: MessageBox MB_OK MB_ICONSTOP "Setup could not complete the duplicate-installation scan. To avoid duplicate builder copies, setup will stop. Run the installer again as Administrator, or remove old OMB Portfolio Builder co
Installer at line 298	Windows setup, update, shortcut, or installation behavior. Evidence: Function OMBDisableBuiltInOldUninstallerForInPlaceUpdate
Installer at line 299	Windows setup, update, shortcut, or installation behavior. Evidence: DetailPrint "Preparing in-place update. The old uninstaller handoff will be skipped."
Installer at line 302	Windows setup, update, shortcut, or installation behavior. Evidence: DeleteRegValue HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" "UninstallString"
Installer at line 303	Windows setup, update, shortcut, or installation behavior. Evidence: DeleteRegValue HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" "QuietUninstallString"
Installer at line 304	Windows setup, update, shortcut, or installation behavior. Evidence: DeleteRegValue HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" "UninstallString"
Installer at line 305	Windows setup, update, shortcut, or installation behavior. Evidence: DeleteRegValue HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" "QuietUninstallString"
Installer at line 308	Windows setup, update, shortcut, or installation behavior. Evidence: DeleteRegValue HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" "UninstallString"
Installer at line 309	Windows setup, update, shortcut, or installation behavior. Evidence: DeleteRegValue HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" "QuietUninstallString"
Installer at line 310	Windows setup, update, shortcut, or installation behavior. Evidence: DeleteRegValue HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" "UninstallString"
Installer at line 311	Windows setup, update, shortcut, or installation behavior. Evidence: DeleteRegValue HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" "QuietUninstallString"
Installer at line 315	Windows setup, update, shortcut, or installation behavior. Evidence: Function OMBUninstallExistingInstallIfPresent
Installer at line 326	Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R6 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" DisplayVersion
Installer at line 327	Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R5 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" QuietUninstallString
Installer at line 329	Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R5 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" UninstallString
Installer at line 332	Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R7 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" InstallLocation
Installer at line 337	Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R6 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" DisplayVersion

Item	Explanation
Installer at line 338	Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R5 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" QuietUninstallString
Installer at line 340	Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R5 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" UninstallString
Installer at line 343	Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R7 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" InstallLocation
Installer at line 350	Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R6 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" DisplayVersion
Installer at line 351	Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R5 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" QuietUninstallString
Installer at line 353	Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R5 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" UninstallString
Installer at line 356	Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R7 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" InstallLocation
Installer at line 360	Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R6 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" DisplayVersion
Installer at line 361	Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R5 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" QuietUninstallString
Installer at line 363	Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R5 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" UninstallString
Installer at line 366	Windows setup, update, shortcut, or installation behavior. Evidence: ReadRegStr \$R7 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\\${UNINSTALL_APP_KEY}" InstallLocation
Installer at line 373	Windows setup, update, shortcut, or installation behavior. Evidence: IfFileExists "\$LOCALAPPDATA\Programs\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe" 0 +3
Installer at line 374	Windows setup, update, shortcut, or installation behavior. Evidence: StrCpy \$R7 "\$LOCALAPPDATA\Programs\OMB Portfolio Builder"
Installer at line 375	Windows setup, update, shortcut, or installation behavior. Evidence: StrCpy \$R5 ""\$LOCALAPPDATA\Programs\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe" /S'
Installer at line 378	Windows setup, update, shortcut, or installation behavior. Evidence: IfFileExists "\$PROGRAMFILES64\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe" 0 +3
Installer at line 380	Windows setup, update, shortcut, or installation behavior. Evidence: StrCpy \$R5 ""\$PROGRAMFILES64\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe" /S'
Installer at line 383	Windows setup, update, shortcut, or installation behavior. Evidence: IfFileExists "\$PROGRAMFILES\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe" 0 +3
Installer at line 385	Windows setup, update, shortcut, or installation behavior. Evidence: StrCpy \$R5 ""\$PROGRAMFILES\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe" /S'
Installer at line 440	Windows setup, update, shortcut, or installation behavior. Evidence: DetailPrint "Migrating legacy OMB application install to the AppData per-user install location."

Function-by-function detail

OMBToolsPageCreate - lines 17-52

Item	Explanation
Source location	Lines 17-52 (36 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	Function OMBToolsPageCreate
Touches	authentication/security data
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

17: Function OMBToolsPageCreate
18:   nsDialogs::Create 1018
19:   Pop $OMB_ToolsDialog
20:   ${If} $OMB_ToolsDialog == error
21:     Abort
22:   ${EndIf}
23:
24:   ${NSD_CreateLabel} 0 0 100% 12u "Tools and shortcuts"
25:   Pop $0
26:
27:   ${NSD_CreateLabel} 0 18u 100% 36u "The desktop app includes its own runtime. Users do not need to
install Node.js or npm. $r$\nGit for Windows with Git Credential Manager is only needed when publishing to
GitHub."
28:   Pop $0
29:
30:   ${NSD_CreateCheckbox} 0 62u 100% 12u "Create a desktop shortcut"
31:   Pop $OMB_ShortcutCheckbox
32:   ${If} $OMB_CreateDesktopShortcutState == ""
33:     StrCpy $OMB_CreateDesktopShortcutState ${BST_CHECKED}
34:   ${EndIf}
35:   ${If} $OMB_CreateDesktopShortcutState == ${BST_CHECKED}
36:     ${NSD_Check} $OMB_ShortcutCheckbox
37:   ${EndIf}
38:
39:   ${NSD_CreateCheckbox} 0 82u 100% 24u "Check publishing tools and install Git for Windows if Git/Git
Credential Manager is missing"
40:   Pop $OMB_PublishingToolsCheckbox
41:   ${If} $OMB_InstallPublishingToolsState == ""
42:     StrCpy $OMB_InstallPublishingToolsState ${BST_CHECKED}
43:   ${EndIf}
44:   ${If} $OMB_InstallPublishingToolsState == ${BST_CHECKED}
45:     ${NSD_Check} $OMB_PublishingToolsCheckbox
46:   ${EndIf}
47:
48:   ${NSD_CreateLabel} 0 114u 100% 38u "Existing shared tools are not uninstalled silently. If a compatible
tool is already available, setup skips it. If Git is missing or unusable, setup will try to install Git for
Windows side-by-side using Windows Package Manager."
49:   Pop $0
50:
51:   nsDialogs::Show
52: FunctionEnd

```

OMBToolsPageLeave - lines 54-62

Item	Explanation
Source location	Lines 54-62 (9 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	Function OMBToolsPageLeave
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

54: Function OMBToolsPageLeave
55:   ${NSD_GetState} $OMB_ShortcutCheckbox $OMB_CreateDesktopShortcutState
56:   ${NSD_GetState} $OMB_PublishingToolsCheckbox $OMB_InstallPublishingToolsState
57:   ${If} $OMB_IsUpdateInstall == "1"
58:     ${AndIf} $INSTRDIR != $OMB_ExistingInstallLocation
59:       MessageBox MB_OK|MB_ICONSTOP "OMB Portfolio Builder is already installed on this
machine.$r$\n$r$\nUpdates must use the current installed
location:$r$\n$OMB_ExistingInstallLocation$r$\n$r$\nSetup will not create another installed copy in a
different directory."
60:       Abort

```

```

61:  ${EndIf}
62: FunctionEnd

```

OMBCheckGitAvailable - lines 64-84

Item	Explanation
Source location	Lines 64-84 (21 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	Function OMBCheckGitAvailable
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

64: Function OMBCheckGitAvailable
65:   StrCpy $R9 "0"
66:   IfFileExists "$PROGRAMFILES64\Git\cmd\git.exe" 0 +2
67:   StrCpy $R9 "1"
68:   IfFileExists "$PROGRAMFILES\Git\cmd\git.exe" 0 +2
69:   StrCpy $R9 "1"
70:   IfFileExists "$PROGRAMFILES32\Git\cmd\git.exe" 0 +2
71:   StrCpy $R9 "1"
72:   IfFileExists "$LOCALAPPDATA\Programs\Git\cmd\git.exe" 0 +2
73:   StrCpy $R9 "1"
74:
75:   ${If} $R9 == "0"
76:     nsExec::ExecToStack 'cmd /c git --version'
77:     Pop $0
78:     Pop $1
79:     ${If} $0 == 0
80:       StrCpy $R9 "1"
81:     ${EndIf}
82:   ${EndIf}
83:   Push $R9
84: FunctionEnd

```

OMBCheckGitCredentialManagerAvailable - lines 86-95

Item	Explanation
Source location	Lines 86-95 (10 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	Function OMBCheckGitCredentialManagerAvailable
Touches	authentication/security data
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

86: Function OMBCheckGitCredentialManagerAvailable
87:   nsExec::ExecToStack 'cmd /c git credential-manager --version'
88:   Pop $0
89:   Pop $1
90:   ${If} $0 == 0
91:     Push "1"
92:   ${Else}
93:     Push "0"
94:   ${EndIf}

```

OMBInstallGitIfNeeded - lines 97-125

Item	Explanation
Source location	Lines 97-125 (29 source lines).
Purpose	Part of publishing, Git, target repository, or authorization behavior.
Declaration	Function OMBInstallGitIfNeeded
Touches	filesystem, network/API requests, authentication/security data
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

97: Function OMBInstallGitIfNeeded
98:   Call OMBCheckGitAvailable
99:   Pop $R9
100:   Call OMBCheckGitCredentialManagerAvailable
101:   Pop $R8
102:
103:   ${If} $R9 == "1"
104:   ${AndIf} $R8 == "1"
105:     DetailPrint "Git for Windows and Git Credential Manager were found. Skipping publishing tool
installation."
106:     Return
107:   ${EndIf}
108:
109:   DetailPrint "Publishing tools are missing or incomplete. Checking Windows Package Manager."
110:   nsExec::ExecToStack 'where.exe winget'
111:   Pop $0
112:   Pop $1
113:   ${If} $0 != 0
114:     MessageBox MB_OK|MB_ICONINFORMATION "OMB Portfolio Builder was installed, but Git for Windows was not
found and Windows Package Manager is unavailable. Open Publishing target > Install Git inside the app, or
install Git for Windows with Git Credential Manager from https://git-scm.com/download/win."
115:     Return
116:   ${EndIf}
117:
118:   DetailPrint "Installing or repairing Git for Windows. This can take several minutes."
119:   ExecWait 'winget install --id Git.Git -e --source winget --accept-source-agreements --accept-package-
agreements' $0
120:   ${If} $0 != 0
121:     MessageBox MB_OK|MB_ICONINFORMATION "OMB Portfolio Builder was installed, but Git for Windows setup
did not complete. Open Publishing target > Install Git inside the app when you are ready to publish."
122:   ${Else}
123:     DetailPrint "Git for Windows setup completed."
124:   ${EndIf}
125: FunctionEnd

```

OMBReadExistingAppVersion - lines 127-139

Item	Explanation
Source location	Lines 127-139 (13 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	Function OMBReadExistingAppVersion
Touches	Mostly local calculations or local UI state.
Calls detected	Write
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
127: Function OMBReadExistingAppVersion
128:   StrCpy $OMB_ExistingInstallVersion ""
129:   IfFileExists "$R7\OMB Portfolio Builder.exe" omb_read_existing_app_version 0
130:   Return
131:
132:   omb_read_existing_app_version:
133:   nsExec::ExecToStack 'powershell.exe -NoProfile -ExecutionPolicy Bypass -Command "$$p = '$R7\OMB
Portfolio Builder.exe'; try { $$v = (Get-Item -LiteralPath $$p).VersionInfo.FileVersion; if (-not $$v) { $$v =
(Get-Item -LiteralPath $$p).VersionInfo.ProductVersion }; [Console]::Write($$v) } catch { exit 1 }'"
134:   Pop $R4
135:   Pop $R3
136:   ${If} $R4 == 0
137:   StrCpy $OMB_ExistingInstallVersion $R3
138:   ${EndIf}
139: FunctionEnd
```

OMBStopIfExistingInstallsCurrent - lines 141-164

Item	Explanation
Source location	Lines 141-164 (24 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	Function OMBStopIfExistingInstallsCurrent
Touches	filesystem
Calls detected	older
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
141: Function OMBStopIfExistingInstallsCurrent
142:   ${If} $OMB_ExistingInstallVersion == ""
143:   Return
144:   ${EndIf}
145:
146:   ${VersionCompare} "$OMB_ExistingInstallVersion" "${VERSION}" $R4
147:   ${If} $R4 == 0
148:   IfSilent omb_existing_install_current_silent
149:   MessageBox MB_OK|MB_ICONINFORMATION "OMB Portfolio Builder $OMB_ExistingInstallVersion is already
installed.$\r$\n\rThis installer is also version ${VERSION}, so setup will stop without changing the app."
150:   Quit
151:
152:   omb_existing_install_current_silent:
153:   DetailPrint "OMB Portfolio Builder $OMB_ExistingInstallVersion is already installed. Setup stopped
because the installed version is current."
154:   Quit
155:   ${ElseIf} $R4 == 1
156:   IfSilent omb_existing_install_newer_silent
157:   MessageBox MB_OK|MB_ICONINFORMATION "OMB Portfolio Builder $OMB_ExistingInstallVersion is already
installed.$\r$\n\rThis installer is older (${VERSION}), so setup will stop without changing the app."
158:   Quit
159:
160:   omb_existing_install_newer_silent:
161:   DetailPrint "OMB Portfolio Builder $OMB_ExistingInstallVersion is newer than installer ${VERSION}.
Setup stopped."
162:   Quit
163:   ${EndIf}
164: FunctionEnd
```

OMBCheckKnownBuilderWorkspace - lines 166-190

Item	Explanation
Source location	Lines 166-190 (25 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	Function OMBCheckKnownBuilderWorkspace
Touches	Mostly local calculations or local UI state.

Item	Explanation
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

166: Function OMBCheckKnownBuilderWorkspace
167:   IfFileExists "$PROFILE\OMB\builder\desktop-builder\package.json" 0 +3
168:   MessageBox MB_OK|MB_ICONSTOP "Setup found an existing local OMB Portfolio Builder
workspace:$r$\n$r$\n$PROFILE\OMB\builder\desktop-builder$r$\n$r$\nRemove, uninstall, or rename that copy
before installing this release. This prevents an older working builder from staying on the machine beside the
installed app."
169:   Quit
170:
171:   IfFileExists "$DOCUMENTS\OMB\desktop-builder\package.json" 0 +3
172:   MessageBox MB_OK|MB_ICONSTOP "Setup found an existing local OMB Portfolio Builder
workspace:$r$\n$r$\n$DOCUMENTS\OMB\desktop-builder$r$\n$r$\nRemove, uninstall, or rename that copy before
installing this release. This prevents an older working builder from staying on the machine beside the installed
app."
173:   Quit
174:
175:   IfFileExists "$PROFILE\OneDrive\Documents\OMB\desktop-builder\package.json" 0 +3
176:   MessageBox MB_OK|MB_ICONSTOP "Setup found an existing local OMB Portfolio Builder
workspace:$r$\n$r$\n$PROFILE\OneDrive\Documents\OMB\desktop-builder$r$\n$r$\nRemove, uninstall, or rename
that copy before installing this release. This prevents an older working builder from staying on the machine
beside the installed app."
177:   Quit
178:
179:   IfFileExists "$PROFILE\Documents\OMB\desktop-builder\package.json" 0 +3
180:   MessageBox MB_OK|MB_ICONSTOP "Setup found an existing local OMB Portfolio Builder
workspace:$r$\n$r$\n$PROFILE\Documents\OMB\desktop-builder$r$\n$r$\nRemove, uninstall, or rename that copy
before installing this release. This prevents an older working builder from staying on the machine beside the
installed app."
181:   Quit
182:
183:   IfFileExists "$DOCUMENTS\omb-portfolio-builder\package.json" 0 +3
184:   MessageBox MB_OK|MB_ICONSTOP "Setup found an existing OMB Portfolio Builder development
copy:$r$\n$r$\n$DOCUMENTS\omb-portfolio-builder$r$\n$r$\nRemove, uninstall, or rename that copy before
installing this release."
185:   Quit
186:
187:   IfFileExists "$PROFILE\OneDrive\Documents\omb-portfolio-builder\package.json" 0 +3
188:   MessageBox MB_OK|MB_ICONSTOP "Setup found an existing OMB Portfolio Builder development
copy:$r$\n$r$\n$PROFILE\OneDrive\Documents\omb-portfolio-builder$r$\n$r$\nRemove, uninstall, or rename that
copy before installing this release."
189:   Quit
190: FunctionEnd

```

OMBWriteDuplicateScanScript - lines 192-272

Item	Explanation
Source location	Lines 192-272 (81 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	Function OMBWriteDuplicateScanScript
Touches	filesystem
Calls detected	Hit, IsNullOrWhiteSpace, Contains, Add, DesktopBuilder, not, LiteralPath, BuilderWorkspace, SkipDirectory, Windows, Tree, Push
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 17 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

192: Function OMBWriteDuplicateScanScript
193:   InitPluginsDir
194:   FileOpen $0 "$PLUGINSDIR\omb-duplicate-scan.ps1" w

```

```

195: FileWrite $0 "$ErrorActionPreference = 'SilentlyContinue'\r\n"
196: FileWrite $0 "$$hits = New-Object System.Collections.Generic.List[string]\r\n"
197: FileWrite $0 "function Add-Hit([string]$$Kind, [string]$$Candidate) {$\r\n"
198: FileWrite $0 "    if ([string]::IsNullOrEmpty($$Candidate)) { return }\r\n"
199: FileWrite $0 "    try { $$resolved = (Resolve-Path -LiteralPath $$Candidate -ErrorAction Stop).Path }
catch { $$resolved = $$Candidate }\r\n"
200: FileWrite $0 "    if ($$resolved -match '\\\\OMB-Portfolio-Builder-Setup-[^\\]+\\.exe$$') { return
}\r\n"
201: FileWrite $0 "    $$line = $$Kind + ' - ' + $$resolved\r\n"
202: FileWrite $0 "    if (-not $$hits.Contains($$line)) { [void]$$hits.Add($$line) }\r\n"
203: FileWrite $0 "}"
204: FileWrite $0 "function Test-DesktopBuilder([string]$$Directory) {$\r\n"
205: FileWrite $0 "    if (-not (Test-Path -LiteralPath (Join-Path $$Directory 'package.json'))) { return
$$false }\r\n"
206: FileWrite $0 "    if (-not (Test-Path -LiteralPath (Join-Path $$Directory 'main.cjs'))) { return $$false
}\r\n"
207: FileWrite $0 "    if (-not (Test-Path -LiteralPath (Join-Path $$Directory 'builder-site'))) { return
$$false }\r\n"
208: FileWrite $0 "    try { $$packageText = Get-Content -LiteralPath (Join-Path $$Directory 'package.json')
-Raw -ErrorAction Stop } catch { return $$false }\r\n"
209: FileWrite $0 "    return $$packageText -match 'OMB Portfolio Builder|omb-portfolio-
builder|portfoliobuilder'\r\n"
210: FileWrite $0 "}"
211: FileWrite $0 "function Test-BuilderWorkspace([string]$$Directory) {$\r\n"
212: FileWrite $0 "    if (-not (Test-Path -LiteralPath (Join-Path $$Directory 'package.json'))) { return
$$false }\r\n"
213: FileWrite $0 "    if (-not (Test-Path -LiteralPath (Join-Path $$Directory 'main.cjs'))) { return $$false
}\r\n"
214: FileWrite $0 "    if (-not (Test-Path -LiteralPath (Join-Path $$Directory 'server.mjs'))) { return
$$false }\r\n"
215: FileWrite $0 "    try { $$packageText = Get-Content -LiteralPath (Join-Path $$Directory 'package.json')
-Raw -ErrorAction Stop } catch { return $$false }\r\n"
216: FileWrite $0 "    return $$packageText -match 'OMB Portfolio Builder|omb-portfolio-
builder|portfoliobuilder'\r\n"
217: FileWrite $0 "}"
218: FileWrite $0 "function Should-SkipDirectory([string]$$Directory) {$\r\n"
219: FileWrite $0 "    $$name = Split-Path -Leaf $$Directory\r\n"
220: FileWrite $0 "    $$managedRoot = Join-Path $env:LOCALAPPDATA 'OMB Portfolio Builder'\r\n"
221: FileWrite $0 "    $$legacyRoot = Join-Path $env:USERPROFILE 'OMB'\r\n"
222: FileWrite $0 "    $$managedBuilder = Join-Path $$managedRoot 'builder'\r\n"
223: FileWrite $0 "    $$managedPortfolio = Join-Path $$managedRoot 'portfolio'\r\n"
224: FileWrite $0 "    $$legacyBuilder = Join-Path $$legacyRoot 'builder'\r\n"
225: FileWrite $0 "    $$legacyPortfolio = Join-Path $$legacyRoot 'portfolio'\r\n"
226: FileWrite $0 "    if ($$Directory -ieq $$managedBuilder -or $$Directory -ieq $$managedPortfolio -or
$$Directory -ieq $$legacyBuilder -or $$Directory -ieq $$legacyPortfolio) { return $$true }\r\n"
227: FileWrite $0 "    if ($$name -in @('node_modules', '.git', 'dist', '.vs', '$$Recycle.Bin', 'System Volume
Information', 'WinSxS')) { return $$true }\r\n"
228: FileWrite $0 "    if ($$Directory -match '\\\\Windows\\\\($$)') { return $$true }\r\n"
229: FileWrite $0 "    return $$false\r\n"
230: FileWrite $0 "}"
231: FileWrite $0 "function Scan-Tree([string]$$Root) {$\r\n"
232: FileWrite $0 "    if (-not (Test-Path -LiteralPath $$Root)) { return }\r\n"
233: FileWrite $0 "    $$stack = New-Object System.Collections.Generic.Stack[string>\r\n"
234: FileWrite $0 "    $$stack.Push($$Root)\r\n"
235: FileWrite $0 "    while ($$stack.Count -gt 0) {$\r\n"
236: FileWrite $0 "        $$dir = $$stack.Pop()\r\n"
237: FileWrite $0 "        if (Should-SkipDirectory $$dir) { continue }\r\n"
238: FileWrite $0 "        if ((Split-Path -Leaf $$dir) -ieq 'desktop-builder' -and (Test-DesktopBuilder $$dir))
{ Add-Hit 'Local desktop-builder workspace' $$dir }\r\n"
239: FileWrite $0 "        if ((Split-Path -Leaf $$dir) -ieq 'builder' -and (Test-BuilderWorkspace $$dir)) {
Add-Hit 'Local builder workspace' $$dir }\r\n"
240: FileWrite $0 "        try { $$children = Get-ChildItem -LiteralPath $$dir -Force -ErrorAction Stop } catch
{ continue }\r\n"
241: FileWrite $0 "        foreach ($$child in $$children) {$\r\n"
242: FileWrite $0 "            if ($$child.PSIsContainer) {$\r\n"
243: FileWrite $0 "                if (-not (Should-SkipDirectory $$child.FullName)) {
$$stack.Push($$child.FullName) }\r\n"
244: FileWrite $0 "            } else {$\r\n"
245: FileWrite $0 "                if ($$child.Name -ieq 'OMB Portfolio Builder.exe') { Add-Hit 'Installed app
executable' $$child.FullName }\r\n"
246: FileWrite $0 "                if ($$child.Name -like 'OMB-Portfolio-Builder-Portable-*.exe') { Add-Hit
'Portable app executable' $$child.FullName }\r\n"
247: FileWrite $0 "            }\r\n"
248: FileWrite $0 "        }\r\n"
249: FileWrite $0 "    }\r\n"
250: FileWrite $0 "}"
251: FileWrite $0 "```$registryKeys = @('HKLM:\Software\Microsoft\Windows\CurrentVersion\Uninstall\*', 'HKCU:\S
oftware\Microsoft\Windows\CurrentVersion\Uninstall\*')\r\n"
252: FileWrite $0 "foreach ($$key in $$registryKeys) {$\r\n"
253: FileWrite $0 "    Get-ItemProperty $$key -ErrorAction SilentlyContinue | Where-Object { $_.DisplayName
-like 'OMB Portfolio Builder*' } | ForEach-Object {$\r\n"
254: FileWrite $0 "        if ($_.InstallLocation) { Add-Hit 'Registered Windows installation'
$_InstallLocation }\r\n"
255: FileWrite $0 "        elseif ($_.DisplayIcon) { Add-Hit 'Registered Windows installation' $_DisplayIcon
}\r\n"
256: FileWrite $0 "    }\r\n"
257: FileWrite $0 "}"
258: FileWrite $0 "```$roots = New-Object System.Collections.Generic.List[string]\r\n"
259: FileWrite $0 "Get-CimInstance Win32_LogicalDisk -Filter 'DriveType=3' -ErrorAction SilentlyContinue |
ForEach-Object { [void]$$roots.Add(($_.DeviceID + '\')) }\r\n"
260: FileWrite $0 "```$programFilesX86 = [Environment]::GetEnvironmentVariable('ProgramFiles(x86)')\r\n"
261: FileWrite $0 "foreach ($$common in @($env:ProgramFiles, $programFilesX86, (Join-Path

```

```

$Env:LOCALAPPDATA 'Programs'), $Env:USERPROFILE, (Join-Path $Env:PUBLIC 'Desktop')) { if ($$common -and -not
$$roots.Contains($$common)) { [void]$$roots.Add($$common) } }$r$n"
262: FileWrite $0 "foreach ($$root in $$roots) { Scan-Tree $$root }$r$n"
263: FileWrite $0 "$$Unique = $$hits | Sort-Object -Unique$r$n"
264: FileWrite $0 "if ($$Unique.Count -gt 0) {$r$n"
265: FileWrite $0 "  $$preview = $$Unique | Select-Object -First 10$r$n"
266: FileWrite $0 "  if ($$Unique.Count -gt 10) { $$preview += ('... and ' + ($$Unique.Count - 10) + '
more') }$r$n"
267: FileWrite $0 "  Write-Output ($$preview -join [Environment]::NewLine)$r$n"
268: FileWrite $0 "  exit 42$r$n"
269: FileWrite $0 "}$r$n"
270: FileWrite $0 "exit 0$r$n"
271: FileClose $0
272: FunctionEnd

```

OMBScanForDuplicateBuilderCopies - lines 274-288

Item	Explanation
Source location	Lines 274-288 (15 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	Function OMBScanForDuplicateBuilderCopies
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

274: Function OMBScanForDuplicateBuilderCopies
275:   DetailPrint "Scanning this machine for existing OMB Portfolio Builder copies."
276:   Call OMBWriteDuplicateScanScript
277:   nsExec::ExecToStack 'powershell.exe -NoProfile -ExecutionPolicy Bypass -File "$PLUGINDIR\omb-
duplicate-scan.ps1"'
278:   Pop $R4
279:   Pop $R3
280:
281:   ${If} $R4 == 42
282:     MessageBox MB_OK|MB_ICONSTOP "Setup found another OMB Portfolio Builder copy on this
machine.$r$n$r$n$r$n$R3$r$n$r$nTo prevent duplicate installations and stale builder features, remove,
uninstall, or rename the existing copy first, then run this installer again."
283:     Quit
284:   ${ElseIf} $R4 != 0
285:     MessageBox MB_OK|MB_ICONSTOP "Setup could not complete the duplicate-installation scan. To avoid
duplicate builder copies, setup will stop. Run the installer again as Administrator, or remove old OMB Portfolio
Builder copies manually before installing."
286:     Quit
287:   ${EndIf}
288: FunctionEnd

```

OMBStopBuilderProcessesForUpdate - lines 290-296

Item	Explanation
Source location	Lines 290-296 (7 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	Function OMBStopBuilderProcessesForUpdate
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
290: Function OMBStopBuilderProcessesForUpdate
291:     DetailPrint "Stopping running OMB Portfolio Builder processes before update."
292:     nsExec::ExecToStack 'cmd /c taskkill /IM "OMB Portfolio Builder.exe" /F /T'
293:     Pop $R4
294:     Pop $R3
295:     Sleep 1500
296: FunctionEnd
```

OMBDisableBuiltInOldUninstallerForInPlaceUpdate - lines 298-313

Item	Explanation
Source location	Lines 298-313 (16 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	Function OMBDisableBuiltInOldUninstallerForInPlaceUpdate
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
298: Function OMBDisableBuiltInOldUninstallerForInPlaceUpdate
299:     DetailPrint "Preparing in-place update. The old uninstaller handoff will be skipped."
300:
301:     SetRegView 64
302:     DeleteRegValue HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
"UninstallString"
303:     DeleteRegValue HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
"QuietUninstallString"
304:     DeleteRegValue HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
"UninstallString"
305:     DeleteRegValue HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
"QuietUninstallString"
306:
307:     SetRegView 32
308:     DeleteRegValue HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
"UninstallString"
309:     DeleteRegValue HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
"QuietUninstallString"
310:     DeleteRegValue HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
"UninstallString"
311:     DeleteRegValue HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
"QuietUninstallString"
312:     SetRegView lastused
313: FunctionEnd
```

OMBUninstallExistingInstallIfPresent - lines 315-497

Item	Explanation
Source location	Lines 315-497 (183 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	Function OMBUninstallExistingInstallIfPresent
Touches	filesystem
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
315: Function OMBUninstallExistingInstallIfPresent
316:   StrCpy $OMB_ExistingInstallLocation ""
317:   StrCpy $OMB_ExistingInstallVersion ""
318:   StrCpy $OMB_IsUpdateInstall "0"
319:   StrCpy $OMB_ManualApplicationInstall "0"
320:   StrCpy $R7 ""
321:   StrCpy $R6 ""
322:   StrCpy $R5 ""
323:
324:   SetRegView 64
325:   ReadRegStr $R7 HKCU "Software\${APP_GUID}" InstallLocation
326:   ReadRegStr $R6 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
DisplayVersion
327:   ReadRegStr $R5 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
QuietUninstallString
328:   ${If} $R5 == ""
329:     ReadRegStr $R5 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
UninstallString
330:   ${EndIf}
331:   ${If} $R7 == ""
332:     ReadRegStr $R7 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
InstallLocation
333:   ${EndIf}
334:
335:   ${If} $R7 == ""
336:     ReadRegStr $R7 HKLM "Software\${APP_GUID}" InstallLocation
337:     ReadRegStr $R6 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
DisplayVersion
338:     ReadRegStr $R5 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
QuietUninstallString
339:     ${If} $R5 == ""
340:       ReadRegStr $R5 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
UninstallString
341:     ${EndIf}
342:     ${If} $R7 == ""
343:       ReadRegStr $R7 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
InstallLocation
344:     ${EndIf}
345:     ${EndIf}
346:
347:     ${If} $R7 == ""
348:       SetRegView 32
349:       ReadRegStr $R7 HKCU "Software\${APP_GUID}" InstallLocation
350:       ReadRegStr $R6 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
DisplayVersion
351:       ReadRegStr $R5 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
QuietUninstallString
352:       ${If} $R5 == ""
353:         ReadRegStr $R5 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
UninstallString
354:       ${EndIf}
355:       ${If} $R7 == ""
356:         ReadRegStr $R7 HKCU "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
InstallLocation
357:       ${EndIf}
358:       ${If} $R7 == ""
359:         ReadRegStr $R7 HKLM "Software\${APP_GUID}" InstallLocation
360:         ReadRegStr $R6 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
DisplayVersion
361:         ReadRegStr $R5 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
QuietUninstallString
362:         ${If} $R5 == ""
363:           ReadRegStr $R5 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
UninstallString
364:         ${EndIf}
365:         ${If} $R7 == ""
366:           ReadRegStr $R7 HKLM "Software\Microsoft\Windows\CurrentVersion\Uninstall\${UNINSTALL_APP_KEY}"
InstallLocation
367:         ${EndIf}
368:         ${EndIf}
369:         SetRegView lastused
370:         ${EndIf}
371:
372:         ${If} $R7 == ""
373:           IfFileExists "$LOCALAPPDATA\Programs\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe" 0 +3
374:           StrCpy $R7 "$LOCALAPPDATA\Programs\OMB Portfolio Builder"
375:           StrCpy $R5 "'$LOCALAPPDATA\Programs\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe' /S'
376:         ${EndIf}
377:         ${If} $R7 == ""
378:           IfFileExists "$PROGRAMFILES64\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe" 0 +3
379:           StrCpy $R7 "$PROGRAMFILES64\OMB Portfolio Builder"
380:           StrCpy $R5 "'$PROGRAMFILES64\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe' /S'
381:         ${EndIf}
382:         ${If} $R7 == ""
383:           IfFileExists "$PROGRAMFILES\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe" 0 +3
384:           StrCpy $R7 "$PROGRAMFILES\OMB Portfolio Builder"
385:           StrCpy $R5 "'$PROGRAMFILES\OMB Portfolio Builder\Uninstall OMB Portfolio Builder.exe' /S'
386:         ${EndIf}
387:         ${If} $R7 == ""
388:           IfFileExists "$PROFILE\OMB\application\OMB Portfolio Builder\OMB Portfolio Builder.exe" 0 +4
```

```

389:         StrCpy $R7 "$PROFILE\OMB\application\OMB Portfolio Builder"
390:         StrCpy $R6 "a legacy app-folder copy"
391:         StrCpy $OMB_ManualApplicationInstall "1"
392:     ${EndIf}
393:     ${If} $R7 == ""
394:         IfFileExists "$PROFILE\OMB\application\OMB Portfolio Builder.exe" 0 +4
395:         StrCpy $R7 "$PROFILE\OMB\application"
396:         StrCpy $R6 "a standalone app-folder copy"
397:         StrCpy $OMB_ManualApplicationInstall "1"
398:     ${EndIf}
399:     ${If} $R7 == ""
400:         IfFileExists "$PROFILE\OneDrive\Documents\OMB\application\OMB Portfolio Builder\OMB Portfolio
Builder.exe" 0 +4
401:         StrCpy $R7 "$PROFILE\OneDrive\Documents\OMB\application\OMB Portfolio Builder"
402:         StrCpy $R6 "a legacy app-folder copy"
403:         StrCpy $OMB_ManualApplicationInstall "1"
404:     ${EndIf}
405:     ${If} $R7 == ""
406:         IfFileExists "$PROFILE\OneDrive\Documents\OMB\application\OMB Portfolio Builder.exe" 0 +4
407:         StrCpy $R7 "$PROFILE\OneDrive\Documents\OMB\application"
408:         StrCpy $R6 "a standalone app-folder copy"
409:         StrCpy $OMB_ManualApplicationInstall "1"
410:     ${EndIf}
411:     ${If} $R7 == ""
412:         IfFileExists "$PROFILE\OneDrive\Desktop\OMB\application\OMB Portfolio Builder.exe" 0 +4
413:         StrCpy $R7 "$PROFILE\OneDrive\Desktop\OMB\application"
414:         StrCpy $R6 "a standalone app-folder copy"
415:         StrCpy $OMB_ManualApplicationInstall "1"
416:     ${EndIf}
417:     ${If} $R7 == ""
418:         IfFileExists "$DOCUMENTS\OMB\application\OMB Portfolio Builder.exe" 0 +4
419:         StrCpy $R7 "$DOCUMENTS\OMB\application"
420:         StrCpy $R6 "a standalone app-folder copy"
421:         StrCpy $OMB_ManualApplicationInstall "1"
422:     ${EndIf}
423:
424:     ${If} $OMB_ManualApplicationInstall == "1"
425:         StrCpy $R5 ""
426:     ${EndIf}
427:
428:     ${If} $R7 != ""
429:         ${If} $OMB_ManualApplicationInstall == "1"
430:             Call OMBReadExistingAppVersion
431:         ${Else}
432:             StrCpy $OMB_ExistingInstallVersion $R6
433:         ${EndIf}
434:     ${If} $R7 == "$PROFILE\OMB\application\OMB Portfolio Builder"
... excerpt truncated after 120 lines. Open the source file for the rest of this function.

```

OMBUninstallPageCreate - lines 546-560

Item	Explanation
Source location	Lines 546-560 (15 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	Function OMBUninstallPageCreate
Touches	Mostly local calculations or local UI state.
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

546: Function OMBUninstallPageCreate
547:     nsDialogs::Create 1018
548:     Pop $OMB_UninstallDialog
549:     ${If} $OMB_UninstallDialog == error
550:         Abort
551:     ${EndIf}
552:
553:     ${NSD_CreateLabel} 0 0 100% 12u "Uninstall OMB Portfolio Builder"
554:     Pop $0
555:
556:     ${NSD_CreateLabel} 0 20u 100% 90u "Uninstall will remove:$\r$\n$\r$\n- OMB Portfolio Builder
application files$\r$\n- OMB Portfolio Builder shortcuts$\r$\n- Local app data created by this application when
Windows requests app-data cleanup$\r$\n$\r$\nUninstall will not silently remove shared tools such as Git for
Windows, Node.js, pnpm, browsers, or your Git repositories."

```

```

557:   Pop $0
558:
559:   nsDialogs::Show
560: FunctionEnd

```

Representative opening snippet

```

1: !include LogicLib.nsh
2: !include nsDialogs.nsh
3: !include WinMessages.nsh
4: !include WordFunc.nsh
5:
6: !ifndef BUILD_UNINSTALLER
7: Var OMB_ToolsDialog
8: Var OMB_ShortcutCheckbox
9: Var OMB_PublishingToolsCheckbox
10: Var OMB_CreateDesktopShortcutState
11: Var OMB_InstallPublishingToolsState
12: Var OMB_ExistingInstallLocation
13: Var OMB_ExistingInstallVersion
14: Var OMB_IsUpdateInstall
15: Var OMB_ManualApplicationInstall
16:
17: Function OMBToolsPageCreate
18:   nsDialogs::Create 1018
19:   Pop $OMB_ToolsDialog
20:   ${If} $OMB_ToolsDialog == error
21:     Abort
22:   ${EndIf}
23:
24:   ${NSD_CreateLabel} 0 0 100% 12u "Tools and shortcuts"
25:   Pop $0
26:
27:   ${NSD_CreateLabel} 0 18u 100% 36u "The desktop app includes its own runtime. Users do not need to
install Mode.js or pnpm.$r$\nGit for Windows with Git Credential Manager is only needed when publishing to
GitHub."
28:   Pop $0
29:
30:   ${NSD_CreateCheckbox} 0 62u 100% 12u "Create a desktop shortcut"
31:   Pop $OMB_ShortcutCheckbox
32:   ${If} $OMB_CreateDesktopShortcutState == ""
33:     StrCpy $OMB_CreateDesktopShortcutState ${BST_CHECKED}
34:   ${EndIf}
35:   ${If} $OMB_CreateDesktopShortcutState == ${BST_CHECKED}
36:     ${NSD_Check} $OMB_ShortcutCheckbox
37:   ${EndIf}
38:
39:   ${NSD_CreateCheckbox} 0 82u 100% 24u "Check publishing tools and install Git for Windows if Git/Git
Credential Manager is missing"
40:   Pop $OMB_PublishingToolsCheckbox
41:   ${If} $OMB_InstallPublishingToolsState == ""
42:     StrCpy $OMB_InstallPublishingToolsState ${BST_CHECKED}
43:   ${EndIf}
44:   ${If} $OMB_InstallPublishingToolsState == ${BST_CHECKED}
45:     ${NSD_Check} $OMB_PublishingToolsCheckbox
46:   ${EndIf}
47:
48:   ${NSD_CreateLabel} 0 114u 100% 38u "Existing shared tools are not uninstalled silently. If a compatible
tool is already available, setup skips it. If Git is missing or unusable, setup will try to install Git for
Windows side-by-side using Windows Package Manager."
49:   Pop $0
50:
51:   nsDialogs::Show
52: FunctionEnd
53:
54: Function OMBToolsPageLeave
55:   ${NSD_GetState} $OMB_ShortcutCheckbox $OMB_CreateDesktopShortcutState
56:   ${NSD_GetState} $OMB_PublishingToolsCheckbox $OMB_InstallPublishingToolsState
57:   ${If} $OMB_IsUpdateInstall == "1"
58:     ${AndIf} $INSTDIR != $OMB_ExistingInstallLocation
59:     MessageBox MB_OK|MB_ICONSTOP "OMB Portfolio Builder is already installed on this
machine.$r$\n$r$\nUpdates must use the current installed
location:$r$\n$OMB_ExistingInstallLocation$r$\n$r$\nSetup will not create another installed copy in a
different directory."
60:     Abort
61:   ${EndIf}
62: FunctionEnd
63:
64: Function OMBCheckGitAvailable
65:   StrCpy $R9 "0"
66:   IfFileExists "$PROGRAMFILES64\Git\cmd\git.exe" 0 +2
67:   StrCpy $R9 "1"
68:   IfFileExists "$PROGRAMFILES\Git\cmd\git.exe" 0 +2
69:   StrCpy $R9 "1"
70:   IfFileExists "$PROGRAMFILES32\Git\cmd\git.exe" 0 +2
71:   StrCpy $R9 "1"
72:   IfFileExists "$LOCALAPPDATA\Programs\Git\cmd\git.exe" 0 +2

```

Safe change checklist

- Confirm which app, website, installer, workflow, or document consumes this file.
- Change it on a feature branch from development.
- Regenerate docs if the file purpose, API surface, or behavior changes.
- Run the smallest relevant smoke test before merging into development.

Important Code Reference: docs/build_complete_guide.py

Item	Explanation
File	docs/build_complete_guide.py
Category	Python tooling
Folder role	Documentation and generated reference area. Used to explain the app and source structure.
Size	185,742 bytes
Extension	.py
MIME type	text/x-python
Text lines	2,821

Why this file exists

- Tracked repository file used by the builder, website, installer, documentation, or release process.

How it is used

Documentation and generated reference area. Used to explain the app and source structure.

How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

Imports, requires, and external dependencies

Item	Explanation
Line 3	import html
Line 4	import json
Line 5	import math
Line 6	import mimetypes
Line 7	import re
Line 8	import shutil
Line 9	import subprocess
Line 10	import textwrap

API endpoints mentioned or called

Item	Explanation
/api/...	Line 99. Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact behavior.
/api/code/save	Line 455. Saves source code into the local compile workspace.
/api/code/compile	Line 648. Compiles or runs a source file and returns terminal output.
/api/code/compile	Line 660. Compiles or runs a source file and returns terminal output.
/api/portfolio-ai	Line 672. Handles AI assistant questions.
/api/portfolio-ai	Line 703. Handles AI assistant questions.

Important implementation signals

Item	Explanation
Rich text at line 14	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: from PIL import Image, ImageDraw, ImageFont
Rich text at line 20	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: from docx.shared import Inches, Pt, RGBColor
Rich text at line 23	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: from reportlab.lib import colors
Installer at line 50	Windows setup, update, shortcut, or installation behavior. Evidence: ".github/workflows/build-windows-builder.yml": "GitHub Actions workflow that builds the Windows installer and portable application, checks the stable download link, enforces release version bumps, uploads artifacts, and
Git command at line 52	Repository state, publishing, branch checks, or release automation. Evidence: ".gitignore": "Git ignore rules for build outputs, temporary files, and local-only data that should not be committed.",
Cloudflare or AI at line 55	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "BRANCHING.md": "Human-readable branch model explaining development, main, feature branches, and release flow.",
Git command at line 56	Repository state, publishing, branch checks, or release automation. Evidence: "Backgrounds/.gitkeep": "Keeps the Backgrounds folder in Git even when no background image files are present.",
Cloudflare or AI at line 57	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "README.md": "Main README for the standalone builder application: install, updates, workspaces, text editing, publishing, build commands, branch model, and uninstall.",
Cloudflare or AI at line 58	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "_headers": "Cloudflare Pages style HTTP headers for security and caching behavior on the public website.",
Installer at line 67	Windows setup, update, shortcut, or installation behavior. Evidence: "assets/omb-app-icon.ico": "Windows ICO application icon used by Electron Builder and shortcuts.",
Installer at line 69	Windows setup, update, shortcut, or installation behavior. Evidence: "build/installer.nsh": "NSIS installer customization script for update behavior, shortcut placement, old install detection, and setup details.",
Cloudflare or AI at line 71	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "cloudflare/README.md": "Cloudflare setup notes for deploying the AI Worker endpoint and wiring the public site to the backend.",
Cloudflare or AI at line 72	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "cloudflare/portfolio-ai-worker.js": "Cloudflare Worker backend for the public portfolio AI assistant and fallback intelligence path.",
Cloudflare or AI at line 73	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "cloudflare/wrangler.toml": "Wrangler configuration used to deploy the Cloudflare Worker.",
Git command at line 74	Repository state, publishing, branch checks, or release automation. Evidence: "docs/.gitkeep": "Keeps the docs folder in Git even before generated documents are added.",
Installer at line 77	Windows setup, update, shortcut, or installation behavior. Evidence: "installer-notes.txt": "Installer license/notes text shown during Windows setup.",
Compiler or simulator at line 79	Part of the Compile Code workspace or language/tool detection. Evidence: "package.json": "Node/Electron package manifest with scripts, dependencies, version, installer settings, and extra resources.",
Security or auth at line 87	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "server.mjs": "Local backend server used by the builder for drafts, parsing, publishing, compiler execution, authentication checks, and file operations.",
Git command at line 96	Repository state, publishing, branch checks, or release automation. Evidence: ("High-Level System Map", "The system has two personalities: a private builder app and a public website. The builder is where content is created and verified. The website is what recruiters see after publishing.", "Owner
Security or auth at line 98	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: ("Private Builder Versus Public Website", "The builder includes editing controls, local draft storage, authentication checks, compiler tools, and publishing actions. The public website removes those controls and only dis
Git command at line 99	Repository state, publishing, branch checks, or release automation. Evidence: ("Local Backend Role", "The frontend cannot safely do everything by itself. The local backend handles file writes, repository operations, compiler execution, publishing checks, and generated site output.", "template-prev
Git command at line 101	Repository state, publishing, branch checks, or release automation. Evidence: ("GitHub Pages Publishing", "The target repository receives static files. GitHub Pages then serves those files over HTTPS. The builder does not run on the public site.", "Local publish mirror -> git commit -> git push ->
Cloudflare or AI at line 102	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: ("Cloudflare Role", "Cloudflare can sit in front of the domain and can also host Worker endpoints such as the portfolio AI backend. DNS points the custom domain to the website host.", "Visitor DNS lookup -> Cloudflare DN
Cloudflare or AI at line 103	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: ("AI Backend Role", "The website AI should answer from portfolio context when the question is about Maurice's work and use general knowledge when the question is general. The Worker endpoint keeps API keys off the public

Item	Explanation
Installer at line 105	Windows setup, update, shortcut, or installation behavior. Evidence: ("Installer And Update Role", "The installer places the app in the per-user AppData Programs folder. Updates should replace that install instead of creating duplicate copies.", "GitHub Release -> installer download -> up
Cloudflare or AI at line 106	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: ("Branch Model", "Development collects work. Main is the release branch. Feature branches should merge into development first. Main should receive only development.", "feature branch -> development -> main -> release wor
Security or auth at line 107	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: ("Security Boundary", "Editing is local and authenticated. Public visitors should not get editing tools. Publishing requires GitHub write access to the target repository.", "Visitor can read public site\nOwner can edit b
Cloudflare or AI at line 115	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "Cloudflare Role": "cloudflare-ai-flow",
Cloudflare or AI at line 116	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "AI Backend Role": "cloudflare-ai-flow",
Installer at line 118	Windows setup, update, shortcut, or installation behavior. Evidence: "Installer And Update Role": "release-pipeline",
Cloudflare or AI at line 119	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "Branch Model": "release-pipeline",
Git command at line 124	Repository state, publishing, branch checks, or release automation. Evidence: ("git status", "Shows what branch you are on and whether files are changed, staged, or clean.", "Run it before edits, before commits, and before merges."),
Git command at line 125	Repository state, publishing, branch checks, or release automation. Evidence: ("git switch development", "Moves your working copy to the development branch.", "Use development as the integration branch before anything goes to main."),
Git command at line 126	Repository state, publishing, branch checks, or release automation. Evidence: ("git switch -c codex/example-change", "Creates a new feature branch from the branch you are currently on.", "Use this before changing the builder so development stays stable."),
Git command at line 127	Repository state, publishing, branch checks, or release automation. Evidence: ("git add <file>", "Stages a changed file for the next commit.", "Only stage files that belong to the current change."),
Git command at line 128	Repository state, publishing, branch checks, or release automation. Evidence: ("git commit -m 'message'", "Records staged changes as a named checkpoint.", "Write messages that explain why the change exists."),
Git command at line 129	Repository state, publishing, branch checks, or release automation. Evidence: ("git push origin <branch>", "Uploads your branch to GitHub.", "This lets GitHub run workflows and enables pull requests."),
Git command at line 130	Repository state, publishing, branch checks, or release automation. Evidence: ("git pull --ff-only", "Updates the current branch only if Git can fast-forward safely.", "Avoids surprise merge commits."),
Git command at line 131	Repository state, publishing, branch checks, or release automation. Evidence: ("git fetch --tags --force", "Downloads branch and tag metadata from GitHub.", "Used before checking whether a release tag already exists."),
Compiler or simulator at line 132	Part of the Compile Code workspace or language/tool detection. Evidence: ("pnpm install --frozen-lockfile", "Installs Node dependencies exactly as pinned in pnpm-lock.yaml.", "Makes release builds repeatable."),
Installer at line 135	Windows setup, update, shortcut, or installation behavior. Evidence: ("pnpm run installer", "Builds the NSIS Windows installer.", "Use this when testing installation and update behavior."),
Cloudflare or AI at line 138	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: ("wrangler deploy", "Deploys the Cloudflare Worker backend.", "Use it when the AI endpoint code changes."),
Compiler or simulator at line 139	Part of the Compile Code workspace or language/tool detection. Evidence: ("gcc file.c -o app.exe", "Compiles C source into a Windows executable.", "The builder uses similar logic when C tools are available."),
Compiler or simulator at line 141	Part of the Compile Code workspace or language/tool detection. Evidence: ("javac Main.java", "Compiles Java source into .class bytecode.", "Java has a compile step before running."),
Compiler or simulator at line 142	Part of the Compile Code workspace or language/tool detection. Evidence: ("java Main", "Runs compiled Java bytecode.", "The builder shows this output in the terminal panel."),
Compiler or simulator at line 143	Part of the Compile Code workspace or language/tool detection. Evidence: ("python script.py", "Runs Python source directly.", "Python is interpreted by default."),
Compiler or simulator at line 144	Part of the Compile Code workspace or language/tool detection. Evidence: ("node script.js", "Runs JavaScript through Node.js.", "Useful for builder scripts and JavaScript examples."),
Compiler or simulator at line 145	Part of the Compile Code workspace or language/tool detection. Evidence: ("iverilog -g2012 design.sv -o sim.vvp", "Compiles Verilog/SystemVerilog into an Icarus simulation file.", "Prepares HDL for simulation."),
Compiler or simulator at line 146	Part of the Compile Code workspace or language/tool detection. Evidence: ("vvp sim.vvp", "Runs Icarus Verilog simulation output.", "This is how HDL terminal output appears."),
Git command at line 147	Repository state, publishing, branch checks, or release automation. Evidence: ("git push --dry-run origin main", "Tests whether GitHub would accept a push without changing the remote.", "Used during authorization checks."),

Item	Explanation
Git command at line 148	Repository state, publishing, branch checks, or release automation. Evidence: ("git remote -v", "Shows which GitHub repository the local folder talks to.", "Important for publishing targets."),
Git command at line 149	Repository state, publishing, branch checks, or release automation. Evidence: ("git branch --show-current", "Prints the active branch name.", "The builder must know whether it is pushing the right branch."),
Git command at line 150	Repository state, publishing, branch checks, or release automation. Evidence: ("git tag --list builder-v*", "Lists builder release tags.", "Release workflows use this to prevent duplicate releases."),
Installer at line 151	Windows setup, update, shortcut, or installation behavior. Evidence: ("Get-ChildItem", "PowerShell command that lists files and folders.", "Used to inspect workspaces and installer output."),
Process execution at line 153	Runs Git, compilers, installers, or helper tools outside the JavaScript runtime. Evidence: ("Start-Process", "PowerShell command that starts another program.", "The updater uses detached process launching."),
Cloudflare or AI at line 158	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "Profile identity and contact details", "Front-page hero text", "Fun facts block", "Portfolio areas", "Project categories", "Project creation flow", "Project overview", "Design sections", "Simulation sections", "Files an
Installer at line 163	Windows setup, update, shortcut, or installation behavior. Evidence: ("The app does not open", "Check whether the installed executable exists under AppData Local Programs. If it does, start it directly. If it fails, reinstall from the latest GitHub Release."),
Installer at line 164	Windows setup, update, shortcut, or installation behavior. Evidence: ("Update does not restart", "The update handoff must close Electron, run the installer, and then launch the new executable. Check the updater log and whether another builder process is still running."),
Security or auth at line 165	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: ("GitHub authentication succeeds but builder cannot push", "Authentication and repository freshness are different. Pull or load from target first if the remote branch is ahead."),
Local storage at line 167	Local persistence, draft state, auth cache, update snooze, or browser-side caching. Evidence: ("Website changes show on desktop but not phone", "Check cache, confirm the published files changed, and verify the mobile renderer is using the same projects.json and CSS."),
Cloudflare or AI at line 168	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: ("AI endpoint unavailable", "Verify the Worker route, environment secrets, Cloudflare deployment, and browser network errors."),
Local storage at line 169	Local persistence, draft state, auth cache, update snooze, or browser-side caching. Evidence: ("Compiler takes too long", "Check whether the first run is installing or detecting tools. Cached runs should be faster after the first successful compile."),
Compiler or simulator at line 170	Part of the Compile Code workspace or language/tool detection. Evidence: ("SystemVerilog output missing", "Check that Icarus Verilog and vvp are available and that the file is a simulation testbench or has a runnable top module."),
Rich text at line 171	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: ("Right-click formatting does not apply", "Confirm the active editor saved its selection before the menu changed font, size, or color."),
Rich text at line 172	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: ("Links are treated as formulas", "URL detection should win before formula detection. Pasted http, https, and www links must become normal links."),
Installer at line 174	Windows setup, update, shortcut, or installation behavior. Evidence: ("Installer reports an old installation", "The installer searches per-user and legacy locations. Remove stale install records only after confirming the active app path."),
Git command at line 182	Repository state, publishing, branch checks, or release automation. Evidence: "What HTML does", "What CSS does", "What JavaScript does", "What JSON does", "What Markdown does", "What YAML does", "What TOML does", "What an executable is", "What an installer is", "What a portable app is", "What a lo
Git command at line 188	Repository state, publishing, branch checks, or release automation. Evidence: ("Electron vs pure web builder", "Electron gives desktop file access and packaging. A pure web builder would be easier to access anywhere, but browser security would block many local file, compiler, and Git operations.")
Installer at line 190	Windows setup, update, shortcut, or installation behavior. Evidence: ("Per-user install vs Program Files", "Per-user installs update without admin prompts. Program Files feels traditional but causes UAC friction and update failures."),
Git command at line 193	Repository state, publishing, branch checks, or release automation. Evidence: ("Git authentication cache vs always prompt", "Caching reduces annoyance. Always prompting is stricter but makes normal publishing painful."),
Cloudflare or AI at line 194	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: ("Cloudflare Worker AI vs browser-only AI", "Worker AI hides secrets and can access server-side providers. Browser-only AI would expose keys or be limited to local browser models."),
Git command at line 209	Repository state, publishing, branch checks, or release automation. Evidence: ("git", "Git target repo\ncommit + push", 1040, 140, 250, 135, "#FEF3C7"),
Cloudflare or AI at line 212	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: ("ai", "Cloudflare Worker\nportfolio AI endpoint", 1040, 565, 250, 130, "#CCFBF1"),
Git command at line 218	Repository state, publishing, branch checks, or release automation. Evidence: ("backend", "git", "publishes"),

Item	Explanation
Git command at line 219	Repository state, publishing, branch checks, or release automation. Evidence: ("git", "site", "hosts files"),
Cloudflare or AI at line 221	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: ("site", "ai", "chat API"),
Git command at line 233	Repository state, publishing, branch checks, or release automation. Evidence: ("system", "Filesystem, Git,\ncompilers, updater", 1045, 250, 300, 140, "#DCFCE7"),
Installer at line 275	Windows setup, update, shortcut, or installation behavior. Evidence: ("release", "GitHub Release\ninstaller + portable", 1015, 370, 300, 130, "#EDE9FE"),
Compiler or simulator at line 293	Part of the Compile Code workspace or language/tool detection. Evidence: ("lang", "Language profile\nC, C++, Java,\nPython, JS, HDL", 705, 150, 260, 145, "#FEF3C7"),
Security or auth at line 295	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: ("scope", "HDL testbench\n+ signal scope", 1030, 315, 250, 125, "#DCFCE7"),

JSON/YAML-style keys detected

Item	Explanation
.github/workflows/build-windows-builder.yml	Line 50: ".github/workflows/build-windows-builder.yml": "GitHub Actions workflow that builds the Windows installer and portable application, checks the stable download link, enforces releases",
.github/workflows/main-branch-gate.yml	Line 51: ".github/workflows/main-branch-gate.yml": "GitHub Actions guard that rejects direct pushes to main and rejects pull requests into main unless the source branch is development.",
.gitignore	Line 52: ".gitignore": "Git ignore rules for build outputs, temporary files, and local-only data that should not be committed.",
.nojekyll	Line 53: ".nojekyll": "Tells GitHub Pages not to process the site with Jekyll, so folders and static files publish exactly as generated.",
.well-known/security.txt	Line 54: ".well-known/security.txt": "Public security contact file used by browsers, researchers, and scanners to find responsible disclosure information.",
BRANCHING.md	Line 55: "BRANCHING.md": "Human-readable branch model explaining development, main, feature branches, and release flow.",
Backgrounds/.gitkeep	Line 56: "Backgrounds/.gitkeep": "Keeps the Backgrounds folder in Git even when no background image files are present.",
README.md	Line 57: "README.md": "Main README for the standalone builder application: install, updates, workspaces, text editing, publishing, build commands, branch model, and uninstall.",
_headers	Line 58: "_headers": "Cloudflare Pages style HTTP headers for security and caching behavior on the public website.",
assets/apple-touch-icon.png	Line 59: "assets/apple-touch-icon.png": "Apple touch icon shown when the public site is saved on iOS or compatible launch surfaces.",
assets/favicon-16.png	Line 60: "assets/favicon-16.png": "Small browser favicon used by tabs and browser UI.",
assets/favicon-192.png	Line 61: "assets/favicon-192.png": "Large web app icon for Android and installable browser surfaces.",
assets/favicon-32.png	Line 62: "assets/favicon-32.png": "Standard browser favicon size.",
assets/favicon-48.png	Line 63: "assets/favicon-48.png": "Windows/browser icon size used by some desktop surfaces.",
assets/favicon.ico	Line 64: "assets/favicon.ico": "ICO favicon bundle used by browsers that request favicon.ico.",
assets/omb-app-icon-256.png	Line 65: "assets/omb-app-icon-256.png": "Builder application icon source at 256 pixels.",
assets/omb-app-icon-512.png	Line 66: "assets/omb-app-icon-512.png": "Builder application icon source at 512 pixels.",
assets/omb-app-icon.ico	Line 67: "assets/omb-app-icon.ico": "Windows ICO application icon used by Electron Builder and shortcuts.",
assets/omb-mark.png	Line 68: "assets/omb-mark.png": "OMB snake/mark image used for branding in the site and builder.",
build/installer.nsh	Line 69: "build/installer.nsh": "NSIS installer customization script for update behavior, shortcut placement, old install detection, and setup details.",
builder-rich-future-sections.js	Line 70: "builder-rich-future-sections.js": "Builder-side helper code for richer future portfolio sections and section defaults.",
cloudflare/README.md	Line 71: "cloudflare/README.md": "Cloudflare setup notes for deploying the AI Worker endpoint and wiring the public site to the backend.",
cloudflare/portfolio-ai-worker.js	Line 72: "cloudflare/portfolio-ai-worker.js": "Cloudflare Worker backend for the public portfolio AI assistant and fallback intelligence path.",

Item	Explanation
cloudflare/wrangler.toml	Line 73: "cloudflare/wrangler.toml": "Wrangler configuration used to deploy the Cloudflare Worker.",
docs/.gitkeep	Line 74: "docs/.gitkeep": "Keeps the docs folder in Git even before generated documents are added.",
electronics-search.js	Line 75: "electronics-search.js": "Electronics vocabulary and search support used by the website search behavior.",
index.html	Line 76: "index.html": "Public website HTML shell that recruiters and visitors load in the browser.",
installer-notes.txt	Line 77: "installer-notes.txt": "Installer license/notes text shown during Windows setup.",
main.cjs	Line 78: "main.cjs": "Electron main process: creates the desktop window, starts the local backend, handles menus, update handoff, and application lifecycle.",
package.json	Line 79: "package.json": "Node/Electron package manifest with scripts, dependencies, version, installer settings, and extra resources.",
pnpm-lock.yaml	Line 80: "pnpm-lock.yaml": "Pinned dependency lockfile so installs and GitHub Actions use the same dependency versions.",
pnpm-workspace.yaml	Line 81: "pnpm-workspace.yaml": "pnpm workspace declaration for the repository.",
portfolio-README.md	Line 82: "portfolio-README.md": "README for the generated public portfolio website rather than the builder application.",
project-templates.json	Line 83: "project-templates.json": "Appearance template definitions used by projects to control visual feel rather than content.",
projects.json	Line 84: "projects.json": "Public project catalog consumed by the website after parsing builder data.",
robots.txt	Line 85: "robots.txt": "Crawler instruction file for search engines.",
script.js	Line 86: "script.js": "Public website JavaScript for navigation, portfolio rendering, search, AI UI, and interactive project windows.",
server.mjs	Line 87: "server.mjs": "Local backend server used by the builder for drafts, parsing, publishing, compiler execution, authentication checks, and file operations.",
styles.css	Line 88: "styles.css": "Public website CSS for layout, contact area, projects, responsive behavior, AI panel, and mobile/desktop rendering.",
template-preview.css	Line 89: "template-preview.css": "Builder application CSS for dialogs, windows, rich editors, compile code, dark mode, guide windows, and local previews.",
template-preview.html	Line 90: "template-preview.html": "Builder HTML shell loaded inside the Electron app and local preview runtime.",
template-preview.js	Line 91: "template-preview.js": "Builder frontend brain: state handling, rich editors, project builder, previews, parser triggers, publishing UI, compile workspace, and guide windows.",
High-Level System Map	Line 111: "High-Level System Map": "system-overview",
Why Electron Was Chosen	Line 112: "Why Electron Was Chosen": "electron-runtime",
Portfolio Parser Role	Line 113: "Portfolio Parser Role": "builder-to-site-files",
GitHub Pages Publishing	Line 114: "GitHub Pages Publishing": "builder-to-site-files",
Cloudflare Role	Line 115: "Cloudflare Role": "cloudflare-ai-flow",
AI Backend Role	Line 116: "AI Backend Role": "cloudflare-ai-flow",
Compiler Workspace Role	Line 117: "Compiler Workspace Role": "compile-code-flow",
Installer And Update Role	Line 118: "Installer And Update Role": "release-pipeline",
Branch Model	Line 119: "Branch Model": "release-pipeline",
name	Line 202: "name": "system-overview",
title	Line 203: "title": "Private Builder To Public Portfolio",
nodes	Line 204: "nodes": [
edges	Line 214: "edges": [
name	Line 225: "name": "electron-runtime",
title	Line 226: "title": "Electron Runtime Inside The Desktop App",
nodes	Line 227: "nodes": [
edges	Line 235: "edges": [

Item	Explanation
name	Line 245: "name": "builder-to-site-files",
title	Line 246: "title": "How Builder Edits Become Website Files",
nodes	Line 247: "nodes": [
edges	Line 256: "edges": [
name	Line 267: "name": "release-pipeline",
title	Line 268: "title": "Builder Release Pipeline",
nodes	Line 269: "nodes": [
edges	Line 278: "edges": [
name	Line 288: "name": "compile-code-flow",
title	Line 289: "title": "Compile Code Workspace",
nodes	Line 290: "nodes": [
edges	Line 300: "edges": [
name	Line 311: "name": "tool-build-map",
title	Line 312: "title": "What Tool Builds What",
nodes	Line 313: "nodes": [
edges	Line 322: "edges": [
name	Line 333: "name": "file-communication-map",
title	Line 334: "title": "How Key Files Communicate",
nodes	Line 335: "nodes": [
edges	Line 346: "edges": [
name	Line 359: "name": "cloudflare-ai-flow",

Event handlers and user interactions

Item	Explanation
Line 460	"sectionContent.addEventListener('click', async (event) => {\n const button = event.target.closest('[data-compile-run]');\n if (button) await compileActiveFile(project, file);\n})
Line 1596	if ".addEventListener(" in line:

Function-by-function detail

tracked_files - lines 897-901

Item	Explanation
Source location	Lines 897-901 (5 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def tracked_files() -> list[str]:
Touches	Mostly local calculations or local UI state.
Calls detected	run, strip, splitlines
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
897: def tracked_files() -> list[str]:
898:     result = subprocess.run(["git", "ls-files"], cwd=REPO, text=True, capture_output=True, check=True)
899:     return [line.strip() for line in result.stdout.splitlines() if line.strip()]
900:
901:
```

is_generated_reference_file - lines 925-928

Item	Explanation
Source location	Lines 925-928 (4 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def is_generated_reference_file(repo_file: str) -> bool:
Touches	Mostly local calculations or local UI state.
Calls detected	startswith
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
925: def is_generated_reference_file(repo_file: str) -> bool:
926:     return repo_file in GENERATED_REFERENCE_FILES or repo_file.startswith(GENERATED_REFERENCE_PREFIXES)
927:
928:
```

primary_tracked_files - lines 929-932

Item	Explanation
Source location	Lines 929-932 (4 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def primary_tracked_files(files: list[str]) -> list[str]:
Touches	Mostly local calculations or local UI state.
Calls detected	is_generated_reference_file
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
929: def primary_tracked_files(files: list[str]) -> list[str]:
930:     return [file for file in files if not is_generated_reference_file(file)]
931:
932:
```

important_code_files - lines 956-960

Item	Explanation
Source location	Lines 956-960 (5 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def important_code_files(files: list[str]) -> list[str]:
Touches	Mostly local calculations or local UI state.
Calls detected	set, is_generated_reference_file

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

956: def important_code_files(files: list[str]) -> list[str]:
957:     available = set(files)
958:     return [file for file in IMPORTANT_CODE_FILES if file in available and not
is_generated_reference_file(file)]
959:
960:

```

load_package - lines 961-967

Item	Explanation
Source location	Lines 961-967 (7 source lines).
Purpose	Loads data from disk, network, browser storage, or a service.
Declaration	def load_package() -> dict:
Touches	Mostly local calculations or local UI state.
Calls detected	loads, read_text
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

961: def load_package() -> dict:
962:     try:
963:         return json.loads((REPO / "package.json").read_text(encoding="utf-8"))
964:     except Exception:
965:         return {}
966:
967:

```

load_font - lines 968-975

Item	Explanation
Source location	Lines 968-975 (8 source lines).
Purpose	Loads data from disk, network, browser storage, or a service.
Declaration	def load_font(size: int, bold: bool = False) -> ImageFont.ImageFont:
Touches	Mostly local calculations or local UI state.
Calls detected	Path, exists, truetype, str, load_default
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

968: def load_font(size: int, bold: bool = False) -> ImageFont.ImageFont:
969:     font_name = "arialbd.ttf" if bold else "arial.ttf"
970:     font_path = Path(r"C:\Windows\Fonts") / font_name
971:     if font_path.exists():

```

```

972:         return ImageFont.truetype(str(font_path), size=size)
973:     return ImageFont.load_default()
974:
975:

```

wrap_for_box - lines 976-995

Item	Explanation
Source location	Lines 976-995 (20 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def wrap_for_box(label: str, font: ImageFont.ImageFont, max_width: int) -> list[str]:
Touches	Mostly local calculations or local UI state.
Calls detected	str, splitlines, split, append, getbbox
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

976: def wrap_for_box(label: str, font: ImageFont.ImageFont, max_width: int) -> list[str]:
977:     lines: list[str] = []
978:     for part in str(label).splitlines():
979:         words = part.split()
980:         if not words:
981:             lines.append("")
982:             continue
983:         current = words[0]
984:         for word in words[1:]:
985:             candidate = f"{current} {word}"
986:             bbox = font.getbbox(candidate)
987:             if bbox[2] - bbox[0] <= max_width:
988:                 current = candidate
989:             else:
990:                 lines.append(current)
991:                 current = word
992:         lines.append(current)
993:     return lines
994:
995:

```

draw_arrow - lines 996-1008

Item	Explanation
Source location	Lines 996-1008 (13 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def draw_arrow(draw: ImageDraw.ImageDraw, start: tuple[int, int], end: tuple[int, int], color: str, width: int = 4) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	line, atan2, int, cos, sin, polygon
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

996: def draw_arrow(draw: ImageDraw.ImageDraw, start: tuple[int, int], end: tuple[int, int], color: str,
width: int = 4) -> None:
997:     draw.line([start, end], fill=color, width=width)
998:     angle = math.atan2(end[1] - start[1], end[0] - start[0])
999:     length = 18
1000:     spread = 0.55
1001:     points = [
1002:         end,

```

```

1003:         (int(end[0] - length * math.cos(angle - spread)), int(end[1] - length * math.sin(angle -
spread))),
1004:         (int(end[0] - length * math.cos(angle + spread)), int(end[1] - length * math.sin(angle +
spread))),
1005:     ]
1006:     draw.polygon(points, fill=color)
1007:
1008:

```

edge_points - lines 1009-1024

Item	Explanation
Source location	Lines 1009-1024 (16 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def edge_points(source: tuple[int, int, int, int], target: tuple[int, int, int, int]) -> tuple[tuple[int, int], tuple[int, int]]:
Touches	Mostly local calculations or local UI state.
Calls detected	abs
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1009: def edge_points(source: tuple[int, int, int, int], target: tuple[int, int, int, int]) -> tuple[tuple[int,
int], tuple[int, int]]:
1010:     sx1, sy1, sx2, sy2 = source
1011:     tx1, ty1, tx2, ty2 = target
1012:     sc = ((sx1 + sx2) // 2, (sy1 + sy2) // 2)
1013:     tc = ((tx1 + tx2) // 2, (ty1 + ty2) // 2)
1014:     dx = tc[0] - sc[0]
1015:     dy = tc[1] - sc[1]
1016:     if abs(dx) >= abs(dy):
1017:         start = (sx2 if dx >= 0 else sx1, sc[1])
1018:         end = (tx1 if dx >= 0 else tx2, tc[1])
1019:     else:
1020:         start = (sc[0], sy2 if dy >= 0 else sy1)
1021:         end = (tc[0], ty1 if dy >= 0 else ty2)
1022:     return start, end
1023:
1024:

```

make_diagrams - lines 1025-1074

Item	Explanation
Source location	Lines 1025-1074 (50 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def make_diagrams() -> dict[str, Path]:
Touches	filesystem
Calls detected	makedirs, load_font, Draw, rounded_rectangle, text, line, edge_points, draw_arrow, getbbox, hypot, max, wrap_for_box
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1025: def make_diagrams() -> dict[str, Path]:
1026:     DIAGRAM_DIR.mkdir(parents=True, exist_ok=True)
1027:     title_font = load_font(38, bold=True)
1028:     box_font = load_font(24, bold=True)
1029:     label_font = load_font(18, bold=False)
1030:     diagram_paths: dict[str, Path] = {}
1031:

```



```

1032:     for spec in DIAGRAM_SPECS:
1033:         image = Image.new("RGB", (1600, 900), "#F8F8FD")
1034:         draw = ImageDraw.Draw(image)
1035:         draw.rounded_rectangle((24, 24, 1576, 876), radius=26, outline="#B7D7EA", width=3,
fill="#FFFFFF")
1036:         draw.text((60, 48), spec["title"], fill="#0B2545", font=title_font)
1037:         draw.line((60, 105, 1540, 105), fill="#B7D7EA", width=3)
1038:
1039:         boxes: dict[str, tuple[int, int, int, int]] = {}
1040:         for node_id, label, x, y, w, h, fill in spec["nodes"]:
1041:             boxes[node_id] = (x, y, x + w, y + h)
1042:
1043:         for source_id, target_id, *maybe_label in spec["edges"]:
1044:             source = boxes[source_id]
1045:             target = boxes[target_id]
1046:             start, end = edge_points(source, target)
1047:             draw_arrow(draw, start, end, "#2563EB", width=4)
1048:             if maybe_label:
1049:                 label = maybe_label[0]
1050:                 mid = ((start[0] + end[0]) // 2, (start[1] + end[1]) // 2)
1051:                 text_box = label_font.getbbox(label)
1052:                 tw = text_box[2] - text_box[0] + 16
1053:                 th = text_box[3] - text_box[1] + 10
1054:                 if math.hypot(end[0] - start[0], end[1] - start[1]) > max(tw + 28, 95):
1055:                     draw.rounded_rectangle((mid[0] - tw // 2, mid[1] - th // 2, mid[0] + tw // 2, mid[1]
+ th // 2), radius=9, fill="#FFF6FF", outline="#BFD8FE", width=1)
1056:                     draw.text((mid[0] - tw // 2 + 8, mid[1] - th // 2 + 4), label, fill="#1E3A8A",
font=label_font)
1057:
1058:         for node_id, label, x, y, w, h, fill in spec["nodes"]:
1059:             draw.rounded_rectangle((x, y, x + w, y + h), radius=20, fill=fill, outline="#668EA8",
width=3)
1060:             lines = wrap_for_box(label, box_font, w - 28)
1061:             line_height = 30
1062:             total_h = len(lines) * line_height
1063:             start_y = y + (h - total_h) // 2
1064:             for index, line in enumerate(lines):
1065:                 bbox = box_font.getbbox(line)
1066:                 tw = bbox[2] - bbox[0]
1067:                 draw.text((x + (w - tw) // 2, start_y + index * line_height), line, fill="#102033",
font=box_font)
1068:
1069:         path = DIAGRAM_DIR / f"{spec['name']}.png"
1070:         image.save(path)
1071:         diagram_paths[spec["name"]] = path
1072:     return diagram_paths
1073:
1074:

```

add_diagram - lines 1075-1088

Item	Explanation
Source location	Lines 1075-1088 (14 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_diagram(doc: Document, path: Path, caption: str) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	exists, add_paragraph, add_run, add_picture, str, Inches, Pt, RGBColor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1075: def add_diagram(doc: Document, path: Path, caption: str) -> None:
1076:     if not path.exists():
1077:         return
1078:     paragraph = doc.add_paragraph()
1079:     paragraph.alignment = WD_ALIGN_PARAGRAPH.CENTER
1080:     run = paragraph.add_run()
1081:     run.add_picture(str(path), width=Inches(6.35))
1082:     cap = doc.add_paragraph(caption)
1083:     cap.alignment = WD_ALIGN_PARAGRAPH.CENTER
1084:     for run in cap.runs:
1085:         run.font.size = Pt(9)
1086:         run.font.color.rgb = RGBColor(85, 85, 85)
1087:
1088:

```

shade - lines 1089-1095

Item	Explanation
Source location	Lines 1089-1095 (7 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def shade(cell, fill: str) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	get_or_add_tcPr, OxmlElement, set, qn, append
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1089: def shade(cell, fill: str) -> None:
1090:     tc_pr = cell._tc.get_or_add_tcPr()
1091:     shd = OxmlElement("w:shd")
1092:     shd.set(qn("w:fill"), fill)
1093:     tc_pr.append(shd)
1094:
1095:
```

cell_text - lines 1096-1104

Item	Explanation
Source location	Lines 1096-1104 (9 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def cell_text(cell, text: str, bold: bool = False) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	add_run, str, Pt
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1096: def cell_text(cell, text: str, bold: bool = False) -> None:
1097:     cell.text = ""
1098:     run = cell.paragraphs[0].add_run(str(text))
1099:     run.bold = bold
1100:     run.font.name = "Calibri"
1101:     run.font.size = Pt(9)
1102:     cell.vertical_alignment = WD_CELL_VERTICAL_ALIGNMENT.CENTER
1103:
1104:
```

add_table - lines 1105-1121

Item	Explanation
Source location	Lines 1105-1121 (17 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_table(doc: Document, rows: list[tuple[str, str]]) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	Inches, cell_text, shade, add_row

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1105: def add_table(doc: Document, rows: list[tuple[str, str]]) -> None:
1106:     table = doc.add_table(rows=1, cols=2)
1107:     table.alignment = WD_TABLE_ALIGNMENT.CENTER
1108:     table.autofit = False
1109:     table.columns[0].width = Inches(1.55)
1110:     table.columns[1].width = Inches(4.95)
1111:     head = table.rows[0].cells
1112:     cell_text(head[0], "Item", True)
1113:     cell_text(head[1], "Explanation", True)
1114:     shade(head[0], "E8EEF5")
1115:     shade(head[1], "E8EEF5")
1116:     for label, value in rows:
1117:         cells = table.add_row().cells
1118:         cell_text(cells[0], label, True)
1119:         cell_text(cells[1], value, False)
1120:
1121:

```

add_code - lines 1122-1129

Item	Explanation
Source location	Lines 1122-1129 (8 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_code(doc: Document, text: str) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	add_paragraph, add_run, rstrip, set, qn
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1122: def add_code(doc: Document, text: str) -> None:
1123:     p = doc.add_paragraph()
1124:     p.style = doc.styles["CodeBlock"]
1125:     run = p.add_run((text or "").rstrip())
1126:     run.font.name = "Consolas"
1127:     run._element.rPr.rFonts.set(qn("w: eastAsia"), "Consolas")
1128:
1129:

```

add_note - lines 1130-1144

Item	Explanation
Source location	Lines 1130-1144 (15 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_note(doc: Document, title: str, body: str) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	add_table, Inches, cell, shade, add_run, RGBColor, Pt, add_paragraph
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1130: def add_note(doc: Document, title: str, body: str) -> None:
1131:     table = doc.add_table(rows=1, cols=1)
1132:     table.alignment = WD_TABLE_ALIGNMENT.CENTER
1133:     table.autofit = False
1134:     table.columns[0].width = Inches(6.35)
1135:     cell = table.cell(0, 0)
1136:     shade(cell, "F4F8FC")
1137:     title_run = cell.paragraphs[0].add_run(title)
1138:     title_run.bold = True
1139:     title_run.font.color.rgb = RGBColor(31, 77, 120)
1140:     title_run.font.size = Pt(10)
1141:     paragraph = cell.add_paragraph(body)
1142:     paragraph.style = "Body Text"
1143:
1144:

```

bullets - lines 1145-1149

Item	Explanation
Source location	Lines 1145-1149 (5 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def bullets(doc: Document, items: list[str]) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	add_paragraph
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1145: def bullets(doc: Document, items: list[str]) -> None:
1146:     for item in items:
1147:         doc.add_paragraph(item, style="List Bullet")
1148:
1149:

```

numbers - lines 1150-1154

Item	Explanation
Source location	Lines 1150-1154 (5 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def numbers(doc: Document, items: list[str]) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	add_paragraph
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1150: def numbers(doc: Document, items: list[str]) -> None:

```

```

1151:         for item in items:
1152:             doc.add_paragraph(item, style="List Number")
1153:
1154:

```

setup_document - lines 1155-1204

Item	Explanation
Source location	Lines 1155-1204 (50 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def setup_document() -> Document:
Touches	filesystem
Calls detected	Document, Inches, set, qn, Pt, from_string, add_style, RGBColor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1155: def setup_document() -> Document:
1156:     doc = Document()
1157:     section = doc.sections[0]
1158:     section.page_height = Inches(11)
1159:     section.page_width = Inches(8.5)
1160:     for section in doc.sections:
1161:         section.top_margin = Inches(0.65)
1162:         section.bottom_margin = Inches(0.6)
1163:         section.left_margin = Inches(0.65)
1164:         section.right_margin = Inches(0.65)
1165:         section.header_distance = Inches(0.28)
1166:         section.footer_distance = Inches(0.28)
1167:
1168:     normal = doc.styles["Normal"]
1169:     normal.font.name = "Calibri"
1170:     normal._element.rPr.rFonts.set(qn("w:eastAsia"), "Calibri")
1171:     normal.font.size = Pt(10)
1172:     normal.paragraph_format.space_after = Pt(3)
1173:     normal.paragraph_format.line_spacing = 1.08
1174:
1175:     for name, size, color, before, after in [
1176:         ("Heading 1", 14, "2E74B5", 9, 4),
1177:         ("Heading 2", 12, "2E74B5", 7, 3),
1178:         ("Heading 3", 10.5, "1F4D78", 5, 2),
1179:     ]:
1180:         style = doc.styles[name]
1181:         style.font.name = "Calibri"
1182:         style._element.rPr.rFonts.set(qn("w:eastAsia"), "Calibri")
1183:         style.font.size = Pt(size)
1184:         style.font.color.rgb = RGBColor.from_string(color)
1185:         style.paragraph_format.space_before = Pt(before)
1186:         style.paragraph_format.space_after = Pt(after)
1187:
1188:     code = doc.styles.add_style("CodeBlock", 1)
1189:     code.font.name = "Consolas"
1190:     code._element.rPr.rFonts.set(qn("w:eastAsia"), "Consolas")
1191:     code.font.size = Pt(7.2)
1192:     code.paragraph_format.space_before = Pt(2)
1193:     code.paragraph_format.space_after = Pt(3)
1194:     code.paragraph_format.line_spacing = 1.0
1195:
1196:     header = doc.sections[0].header.paragraphs[0]
1197:     header.text = "OMB Portfolio Builder and Website Complete Guide"
1198:     header.alignment = WD_ALIGN_PARAGRAPH.RIGHT
1199:     for run in header.runs:
1200:         run.font.size = Pt(8)
1201:         run.font.color.rgb = RGBColor(85, 85, 85)
1202:     return doc
1203:
1204:

```

add_cover - lines 1205-1222

Item	Explanation
Source location	Lines 1205-1222 (18 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_cover(doc: Document, package: dict) -> None:
Touches	filesystem
Calls detected	add_paragraph, add_run, Pt, RGBColor, get, add_note, add_page_break
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1205: def add_cover(doc: Document, package: dict) -> None:
1206:     title = doc.add_paragraph()
1207:     title.alignment = WD_ALIGN_PARAGRAPH.CENTER
1208:     run = title.add_run("OMB Portfolio Builder and Website\nComplete Creation Guide")
1209:     run.bold = True
1210:     run.font.size = Pt(26)
1211:     run.font.color.rgb = RGBColor(11, 37, 69)
1212:     subtitle = doc.add_paragraph()
1213:     subtitle.alignment = WD_ALIGN_PARAGRAPH.CENTER
1214:     r = subtitle.add_run("A beginner-friendly manual explaining the desktop app, public website, GitHub
workflows, installer, parser, compiler tools, AI backend, and publishing system.")
1215:     r.font.size = Pt(12)
1216:     meta = doc.add_paragraph()
1217:     meta.alignment = WD_ALIGN_PARAGRAPH.CENTER
1218:     meta.add_run(f"Repository: otienomaurice/omb-portfolio-builder\nVersion documented:
{package.get('version', 'unknown')}\nGenerated: July 4, 2026")
1219:     add_note(doc, "How this guide is written", "This guide starts from the highest-level idea and then
drills down. It intentionally explains terms that engineers usually skip: what Electron is, what a workflow is,
what a local backend is, why Git branches exist, and why the builder and website are separated.")
1220:     doc.add_page_break()
1221:
1222:

```

add_reading_map - lines 1223-1236

Item	Explanation
Source location	Lines 1223-1236 (14 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_reading_map(doc: Document) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	add_heading, add_paragraph, add_table, add_page_break
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1223: def add_reading_map(doc: Document) -> None:
1224:     doc.add_heading("Reading Map", level=1)
1225:     doc.add_paragraph("Use this manual in layers. First read the high-level design pages. Then read the
app flow pages. After that, use the file-by-file and command pages as a reference when you need to understand a
specific file or command.")
1226:     add_table(doc, [
1227:         ("Part 1", "High-level architecture and big-picture mental model."),
1228:         ("Part 2", "Builder app internals: Electron, local backend, parser, rich editors, previews,
compile code, and publishing."),
1229:         ("Part 3", "Public website internals: HTML, CSS, JavaScript, projects.json, assets, search, AI,
and mobile behavior."),
1230:         ("Part 4", "GitHub, branches, workflows, releases, installers, and updates."),
1231:         ("Part 5", "Every tracked file explained in plain English."),
1232:         ("Part 6", "Commands, tradeoffs, troubleshooting, and safe operating procedures."),

```

```

1233:     })
1234:     doc.add_page_break()
1235:
1236:

```

add_entry_points - lines 1237-1253

Item	Explanation
Source location	Lines 1237-1253 (17 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_entry_points(doc: Document) -> None:
Touches	filesystem
Calls detected	add_heading, add_paragraph, add_table, numbers, add_page_break
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1237: def add_entry_points(doc: Document) -> None:
1238:     doc.add_heading("Where Someone Starts: Entry Points", level=1)
1239:     doc.add_paragraph("An entry point is the first file, executable, or function that starts a flow. This
project has several entry points because normal users, developers, GitHub Actions, Cloudflare, and installers
all start in different places.")
1240:     add_table(doc, [(name, f"{entry}. {meaning}") for name, entry, meaning in ENTRY_POINTS])
1241:     doc.add_heading("Fastest way to orient yourself", level=2)
1242:     numbers(doc, [
1243:         "If you are using the installed app, start with OMB Portfolio Builder.exe.",
1244:         "If you are debugging desktop startup, start with package.json and main.cjs.",
1245:         "If you are debugging local APIs, start with server.mjs.",
1246:         "If you are debugging builder UI, start with template-preview.html, template-preview.js, and
template-preview.css.",
1247:         "If you are debugging the public website, start with index.html, script.js, styles.css, and
projects.json.",
1248:         "If you are debugging installer behavior, start with package.json build.nsis and
build/installer.nsh.",
1249:         "If you are debugging releases, start with .github/workflows/build-windows-builder.yml.",
1250:     ])
1251:     doc.add_page_break()
1252:
1253:

```

add_flow_walkthroughs - lines 1254-1264

Item	Explanation
Source location	Lines 1254-1264 (11 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_flow_walkthroughs(doc: Document, diagrams: dict[str, Path]) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	add_heading, add_diagram, add_paragraph, add_table, add_page_break
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1254: def add_flow_walkthroughs(doc: Document, diagrams: dict[str, Path]) -> None:
1255:     for title, diagram_name, steps, debug_note in FLOW_WALKTHROUGHS:
1256:         doc.add_heading(f"Detailed Walkthrough: {title}", level=1)
1257:         add_diagram(doc, diagrams[diagram_name], f"Context diagram for {title}.")
1258:         doc.add_paragraph("This walkthrough follows the real direction of work through the system. Read
it slowly: each line is one handoff from one layer to another.")

```

```

1259:         add_table(doc, steps)
1260:         doc.add_heading("Debug note", level=2)
1261:         doc.add_paragraph(debug_note)
1262:         doc.add_page_break()
1263:
1264:

```

add_data_contracts - lines 1265-1277

Item	Explanation
Source location	Lines 1265-1277 (13 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_data_contracts(doc: Document) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	add_heading, add_paragraph, add_code, add_page_break
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1265: def add_data_contracts(doc: Document) -> None:
1266:     for title, purpose, example, note in DATA_CONTRACTS:
1267:         doc.add_heading(f"Data Contract: {title}", level=1)
1268:         doc.add_paragraph(purpose)
1269:         doc.add_heading("Example shape", level=2)
1270:         add_code(doc, example)
1271:         doc.add_heading("How to read this", level=2)
1272:         doc.add_paragraph(note)
1273:         doc.add_heading("Why contracts matter", level=2)
1274:         doc.add_paragraph("A data contract is an agreement between two pieces of software. If the
frontend sends one shape and the backend expects another shape, the feature breaks even if both files are
individually valid code.")
1275:         doc.add_page_break()
1276:
1277:

```

endpoint_purpose - lines 1278-1307

Item	Explanation
Source location	Lines 1278-1307 (30 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def endpoint_purpose(endpoint: str) -> str:
Touches	filesystem, authentication/security data
Calls detected	No obvious named calls detected by the generator.
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 14 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1278: def endpoint_purpose(endpoint: str) -> str:
1279:     if "code/compile" in endpoint:
1280:         return "Compiles or runs a source file and returns terminal output."
1281:     if "code/save" in endpoint:
1282:         return "Saves source code into the local compile workspace."
1283:     if "code/beautify" in endpoint:
1284:         return "Formats source code for cleaner display and editing."
1285:     if "code/tools" in endpoint:
1286:         return "Checks compiler/runtime availability."
1287:     if "code/install" in endpoint:
1288:         return "Attempts to install missing compiler tools."

```



```

1289:     if "publish-target" in endpoint:
1290:         return "Reads or writes the Git publishing target and authorization state."
1291:     if "publish" in endpoint:
1292:         return "Publishes generated website files to the target repository."
1293:     if "catalog" in endpoint:
1294:         return "Loads project/catalog data for the builder."
1295:     if "templates" in endpoint:
1296:         return "Loads appearance template definitions."
1297:     if "app-update" in endpoint:
1298:         return "Checks or starts builder app update behavior."
1299:     if "security-report" in endpoint:
1300:         return "Returns security/download/auth reporting for the builder."
1301:     if "portfolio-ai" in endpoint:
1302:         return "Handles AI assistant questions."
1303:     if "system-check" in endpoint:
1304:         return "Checks local tool/system readiness."
1305:     return "Project-specific local API endpoint. Inspect server.mjs and the fetch caller for exact
behavior."
1306:
1307:

```

extract_api_endpoints - lines 1308-1320

Item	Explanation
Source location	Lines 1308-1320 (13 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def extract_api_endpoints() -> list[tuple[str, str, str]]:
Touches	network/API requests
Calls detected	exists, enumerate, read_text, splitlines, findall, rstrip, min, get, str, sorted, items
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1308: def extract_api_endpoints() -> list[tuple[str, str, str]]:
1309:     endpoints: dict[tuple[str, str], int] = {}
1310:     for file_name in ["template-preview.js", "script.js", "server.mjs", "index.html"]:
1311:         path = REPO / file_name
1312:         if not path.exists():
1313:             continue
1314:         for line_no, line in enumerate(path.read_text(encoding="utf-8", errors="replace").splitlines(),
start=1):
1315:             for match in re.findall(r"['\"](/api/[A-Za-z0-9_./-]+)", line):
1316:                 clean = match.rstrip("/")
1317:                 endpoints[(clean, file_name)] = min(line_no, endpoints.get((clean, file_name), line_no))
1318:     return [(endpoint, file_name, str(line_no)) for (endpoint, file_name), line_no in
sorted(endpoints.items())]
1319:
1320:

```

add_api_endpoint_catalog - lines 1321-1343

Item	Explanation
Source location	Lines 1321-1343 (23 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_api_endpoint_catalog(doc: Document) -> None:
Touches	filesystem
Calls detected	extract_api_endpoints, add_heading, add_paragraph, add_page_break, endpoint_purpose, add_table
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1321: def add_api_endpoint_catalog(doc: Document) -> None:
1322:     endpoints = extract_api_endpoints()
1323:     doc.add_heading("API Endpoint Catalog", level=1)
1324:     doc.add_paragraph("This catalog is generated from strings found in the source files. It tells you
which /api paths are used and where to start reading when a request fails.")
1325:     if not endpoints:
1326:         doc.add_paragraph("No /api endpoints were found by the scanner.")
1327:         doc.add_page_break()
1328:         return
1329:     rows = [(endpoint, f"Seen in {file_name} near line {line_no}. Purpose: {endpoint_purpose(endpoint)}")
for endpoint, file_name, line_no in endpoints]
1330:     add_table(doc, rows)
1331:     doc.add_page_break()
1332:     for endpoint, file_name, line_no in endpoints:
1333:         doc.add_heading(f"API Detail: {endpoint}", level=1)
1334:         add_table(doc, [
1335:             ("Where seen", f"{file_name} near line {line_no}"),
1336:             ("Purpose", endpoint_purpose(endpoint)),
1337:             ("Frontend question", "Which button, form, or page action sends this request?"),
1338:             ("Backend question", "Which handler in server.mjs receives this path and what files/tools
does it touch?"),
1339:             ("Debugging", "Check browser network status, response JSON, server logs, and whether stale
cache was avoided with no-store or a timestamp."),
1340:         ])
1341:         doc.add_page_break()
1342:
1343:
```

function_purpose - lines 1344-1382

Item	Explanation
Source location	Lines 1344-1382 (39 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def function_purpose(file_name: str, function_name: str) -> str:
Touches	filesystem, authentication/security data
Calls detected	lower, startswith
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 18 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1344: def function_purpose(file_name: str, function_name: str) -> str:
1345:     name = function_name.lower()
1346:     if name.startswith("render"):
1347:         return "Renders UI, markup, or display output."
1348:     if name.startswith("handle"):
1349:         return "Handles an event, request, command, or user action."
1350:     if name.startswith("update"):
1351:         return "Updates UI, state, cache, or stored data."
1352:     if name.startswith("save") or name.startswith("write") or name.startswith("store"):
1353:         return "Persists data to local state, disk, credentials, or browser storage."
1354:     if name.startswith("read") or name.startswith("load") or name.startswith("fetch"):
1355:         return "Loads data from disk, network, browser storage, or a service."
1356:     if "compile" in name or "compiler" in name:
1357:         return "Part of the Compile Code language/tool/build/run flow."
1358:     if "publish" in name or "git" in name or "remote" in name:
1359:         return "Part of publishing, Git, target repository, or authorization behavior."
1360:     if "auth" in name or "credential" in name or "access" in name:
1361:         return "Part of authentication, authorization, or credential checking."
1362:     if "cache" in name:
1363:         return "Reads, writes, or validates cached state."
1364:     if name.startswith("validate") or name.startswith("normalize") or name.startswith("safe") or
name.startswith("sanitize"):
1365:         return "Validates or normalizes input so later code receives safe predictable values."
1366:     if name.startswith("create") or name.startswith("new") or name.startswith("build"):
1367:         return "Creates an object, UI structure, process, payload, or generated artifact."
1368:     if name.startswith("open") or name.startswith("close") or name.startswith("show") or
name.startswith("hide"):
1369:         return "Controls a window, dialog, panel, menu, or visible state."
1370:     if name.startswith("sync"):
1371:         return "Keeps two places aligned, such as local data and target files."
1372:     if file_name == "main.cjs":
1373:         return "Part of Electron desktop startup, window control, workspace setup, menu behavior, or
update handoff."
```

```

1374:     if file_name == "server.mjs":
1375:         return "Part of the local backend API, filesystem, compiler, Git, AI, update, or publish
workflow."
1376:     if file_name == "template-preview.js":
1377:         return "Part of the private builder frontend and editing experience."
1378:     if file_name == "script.js":
1379:         return "Part of the public website runtime."
1380:     return "Named function in the repository. Inspect the surrounding lines to learn its exact inputs and
outputs."
1381:
1382:

```

line_excerpt - lines 1417-1426

Item	Explanation
Source location	Lines 1417-1426 (10 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def line_excerpt(lines: list[str], start: int, end: int, max_lines: int = 120) -> str:
Touches	Mostly local calculations or local UI state.
Calls detected	max, min, len, enumerate, append, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1417: def line_excerpt(lines: list[str], start: int, end: int, max_lines: int = 120) -> str:
1418:     selected = lines[max(start - 1, 0): min(end, len(lines))]
1419:     truncated = len(selected) > max_lines
1420:     selected = selected[:max_lines]
1421:     rendered = [f"{start + index:>5}: {line}" for index, line in enumerate(selected)]
1422:     if truncated:
1423:         rendered.append(f"    ... excerpt truncated after {max_lines} lines. Open the source file for
the rest of this function.")
1424:     return "\n".join(rendered)
1425:
1426:

```

find_block_end - lines 1427-1442

Item	Explanation
Source location	Lines 1427-1442 (16 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def find_block_end(lines: list[str], start_index: int) -> int:
Touches	Mostly local calculations or local UI state.
Calls detected	range, min, len, sub, count, startswith
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1427: def find_block_end(lines: list[str], start_index: int) -> int:
1428:     brace_balance = 0
1429:     saw_brace = False
1430:     for index in range(start_index, min(start_index + 500, len(lines))):
1431:         line = re.sub(r"//.*$", "", lines[index])
1432:         brace_balance += line.count("{")
1433:         brace_balance -= line.count("}")
1434:         if "{" in line:
1435:             saw_brace = True
1436:         if saw_brace and brace_balance <= 0 and index > start_index:

```

```

1437:         return index + 1
1438:         if not saw_brace and index > start_index and not lines[index].startswith((" ", "\t")):
1439:             return index
1440:     return min(start_index + 80, len(lines))
1441:
1442:

```

extract_function_blocks - lines 1443-1488

Item	Explanation
Source location	Lines 1443-1488 (46 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def extract_function_blocks(repo_file: str, lines: list[str]) -> list[dict]:
Touches	browser/Electron DOM, filesystem, network/API requests, authentication/security data
Calls detected	enumerate, strip, search, group, lstrip, startswith, next, range, len, min, find_block_end, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1443: def extract_function_blocks(repo_file: str, lines: list[str]) -> list[dict]:
1444:     functions = []
1445:     for index, line in enumerate(lines):
1446:         stripped = line.strip()
1447:         for pattern in FUNCTION_PATTERNS:
1448:             match = pattern.search(line)
1449:             if not match:
1450:                 continue
1451:             name = match.group(1)
1452:             if line.lstrip().startswith("Function "):
1453:                 end = next((i + 1 for i in range(index + 1, len(lines)) if
lines[i].lstrip().startswith("FunctionEnd")), min(index + 80, len(lines)))
1454:             elif line.lstrip().startswith("def "):
1455:                 end = next((i for i in range(index + 1, len(lines)) if lines[i] and not
lines[i].startswith((" ", "\t", "@"))), min(index + 120, len(lines)))
1456:             else:
1457:                 end = find_block_end(lines, index)
1458:             body = "\n".join(lines[index:end])
1459:             call_candidates = re.findall(r"\b([A-Za-z_$][\w$]*)\s*\(", body)
1460:             calls = []
1461:             for candidate in call_candidates:
1462:                 if candidate not in COMMON_CALL_WORDS and candidate != name and candidate not in calls:
1463:                     calls.append(candidate)
1464:             endpoint_hits = sorted(set(match.rstrip("/") for match in
re.findall(r"['\"](/api/[A-Za-z0-9_-]+)", body)))
1465:             local_storage_keys = sorted(set(match for match in
re.findall(r"(?:localStorage|sessionStorage)\.(?:getItem|setItem|removeItem)\(['\"](?:[^\"]+)", body)))
1466:             functions.append({
1467:                 "line": index + 1,
1468:                 "end_line": end,
1469:                 "name": name,
1470:                 "purpose": function_purpose(Path(repo_file).name, name),
1471:                 "signature": stripped[:240],
1472:                 "excerpt": line_excerpt(lines, index + 1, end, max_lines=120),
1473:                 "line_count": max(end - index, 1),
1474:                 "calls": calls[:30],
1475:                 "endpoints": endpoint_hits[:30],
1476:                 "storage_keys": local_storage_keys[:30],
1477:                 "returns": len(re.findall(r"\breturn\b", body)),
1478:                 "awaits": len(re.findall(r"\bawait\b", body)),
1479:                 "touches_dom":
bool(re.search(r"document\.querySelector|getElementById|classList|dataset", body)),
1480:                 "touches_files":
bool(re.search(r"readFile|writeFile|appendFile|copyFile|mknod|rm|unlink|fs\.", body, re.IGNORECASE)),
1481:                 "runs_process": bool(re.search(r"\bspawn\b|bexecFile\b|bexec\b", body, re.IGNORECASE)),
1482:                 "uses_network": bool(re.search(r"\bfetch\s*\(|https?:\/\/|api/", body, re.IGNORECASE)),
1483:                 "uses_security":
bool(re.search(r"auth|token|secret|credential|permission|authorize|scope", body, re.IGNORECASE)),
1484:             })
1485:             break
1486:     return functions
1487:
1488:

```

collect_signal_hits - lines 1489-1506

Item	Explanation
Source location	Lines 1489-1506 (18 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def collect_signal_hits(lines: list[str]) -> list[dict]:
Touches	Mostly local calculations or local UI state.
Calls detected	enumerate, strip, search, append
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1489: def collect_signal_hits(lines: list[str]) -> list[dict]:
1490:     hits = []
1491:     for line_no, line in enumerate(lines, start=1):
1492:         stripped = line.strip()
1493:         if not stripped:
1494:             continue
1495:         for name, pattern, explanation in SOURCE_SIGNAL_RULES:
1496:             if pattern.search(line):
1497:                 hits.append({
1498:                     "line": line_no,
1499:                     "type": name,
1500:                     "evidence": stripped[:220],
1501:                     "meaning": explanation,
1502:                 })
1503:                 break
1504:     return hits
1505:
1506:
```

function_detail_rows - lines 1507-1531

Item	Explanation
Source location	Lines 1507-1531 (25 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def function_detail_rows(item: dict) -> list[tuple[str, str]]:
Touches	authentication/security data
Calls detected	append, join, expression, statement
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	1 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1507: def function_detail_rows(item: dict) -> list[tuple[str, str]]:
1508:     touches = []
1509:     if item["touches_dom"]:
1510:         touches.append("browser/Electron DOM")
1511:     if item["touches_files"]:
1512:         touches.append("filesystem")
1513:     if item["runs_process"]:
1514:         touches.append("external processes")
1515:     if item["uses_network"]:
1516:         touches.append("network/API requests")
1517:     if item["uses_security"]:
1518:         touches.append("authentication/security data")
1519:     return [
1520:         ("Source location", f"Lines {item['line']}-{item['end_line']} ({item['line_count']} source
lines)."),
```

```

1521:         ("Purpose", item["purpose"]),
1522:         ("Declaration", item["signature"]),
1523:         ("Touches", " ", ".join(touches) if touches else "Mostly local calculations or local UI state."),
1524:         ("Calls detected", " ", ".join(item["calls"][:12]) if item["calls"] else "No obvious named calls
detected by the generator."),
1525:         ("API endpoints", " ", ".join(item["endpoints"]) if item["endpoints"] else "None detected inside
this function body."),
1526:         ("Storage keys", " ", ".join(item["storage_keys"]) if item["storage_keys"] else "No local/session
storage keys detected."),
1527:         ("Async/return clues", f"{item['awaits']} await expression(s), {item['returns']} return
statement(s)."),
1528:         ("Debug approach", "Trigger the user action that reaches this function, inspect inputs/state
before the function, then inspect the returned value, DOM update, file write, process output, or API response
after it runs."),
1529:     ]
1530:
1531:

```

extract_functions - lines 1532-1547

Item	Explanation
Source location	Lines 1532-1547 (16 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def extract_functions() -> list[tuple[str, int, str, str]]:
Touches	Mostly local calculations or local UI state.
Calls detected	exists, enumerate, read_text, splitlines, search, group, append, function_purpose
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1532: def extract_functions() -> list[tuple[str, int, str, str]]:
1533:     function_rows: list[tuple[str, int, str, str]] = []
1534:     for file_name in ["main.cjs", "server.mjs", "template-preview.js", "script.js",
"cloudflare/portfolio-ai-worker.js", "docs/build_complete_guide.py", "build/installer.nsh"]:
1535:         path = REPO / file_name
1536:         if not path.exists():
1537:             continue
1538:         for line_no, line in enumerate(path.read_text(encoding="utf-8", errors="replace").splitlines(),
start=1):
1539:             for pattern in FUNCTION_PATTERNS:
1540:                 match = pattern.search(line)
1541:                 if match:
1542:                     name = match.group(1)
1543:                     function_rows.append((file_name, line_no, name, function_purpose(file_name, name)))
1544:                     break
1545:     return function_rows
1546:
1547:

```

is_text_code_file - lines 1553-1561

Item	Explanation
Source location	Lines 1553-1561 (9 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def is_text_code_file(repo_file: str) -> bool:
Touches	Mostly local calculations or local UI state.
Calls detected	is_generated_reference_file, endswith, is_file, lower
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1553: def is_text_code_file(repo_file: str) -> bool:
1554:     path = REPO / repo_file
1555:     if is_generated_reference_file(repo_file):
1556:         return False
1557:     if repo_file.endswith((".docx", ".pdf", ".png", ".ico")):
1558:         return False
1559:     return path.is_file() and path.suffix.lower() in TEXT_CODE_EXTENSIONS
1560:
1561:
```

source_lines - lines 1562-1568

Item	Explanation
Source location	Lines 1562-1568 (7 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def source_lines(repo_file: str) -> list[str]:
Touches	Mostly local calculations or local UI state.
Calls detected	exists, is_file, read_text, splitlines
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1562: def source_lines(repo_file: str) -> list[str]:
1563:     path = REPO / repo_file
1564:     if not path.exists() or not path.is_file():
1565:         return []
1566:     return path.read_text(encoding="utf-8", errors="replace").splitlines()
1567:
1568:
```

extract_source_facts - lines 1569-1628

Item	Explanation
Source location	Lines 1569-1628 (60 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def extract_source_facts(repo_file: str) -> dict:
Touches	network/API requests
Calls detected	source_lines, join, extract_function_blocks, enumerate, strip, match, len, append, group, findall, rstrip, search
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1569: def extract_source_facts(repo_file: str) -> dict:
1570:     lines = source_lines(repo_file)
1571:     text = "\n".join(lines)
1572:     functions = extract_function_blocks(repo_file, lines)
1573:     variables = []
1574:     imports = []
1575:     endpoints = []
1576:     selectors = []
1577:     events = []
1578:     routes = []
1579:     json_keys = []
1580:     for line_no, line in enumerate(lines, start=1):
1581:         stripped = line.strip()
1582:         var_match = re.match(r"^\s*(?:const|let|var)\s+([A-Za-z_$][\w$]*)\s*=", line)
```

```

1583:         if var_match and len(variables) < 140:
1584:             variables.append({
1585:                 "line": line_no,
1586:                 "name": var_match.group(1),
1587:                 "statement": stripped[:180],
1588:             })
1589:         if re.match(r"^\s*import\s+", line) or re.match(r"^\s*(?:const|let|var)\s+\S+\s*=\s*require\(",
line):
1590:             imports.append({"line": line_no, "statement": stripped[:220]})
1591:         for match in re.findall(r"['\"](/api/[A-Za-z0-9_-./-]+)", line):
1592:             endpoints.append({"line": line_no, "endpoint": match.rstrip("/")})
1593:         route_match = re.search(r"^\s*bapp\.(get|post|put|delete|patch)\(['\"]([^\"]+)\)", line)
1594:         if route_match:
1595:             routes.append({"line": line_no, "method": route_match.group(1).upper(), "path":
route_match.group(2)})
1596:         if ".addEventListener(" in line:
1597:             events.append({"line": line_no, "statement": stripped[:180]})
1598:         json_match = re.match(r"^\s*"([^\"]+)\s*:", line)
1599:         if json_match and len(json_keys) < 120:
1600:             json_keys.append({"line": line_no, "key": json_match.group(1), "statement": stripped[:180]})
1601:         if repo_file.endswith(".css"):
1602:             selector_match = re.match(r"^\s*"([.#A-Za-z0-9_\-:\[\]\(\)\s>+.=\\"]+)\s*\{", line)
1603:             if selector_match and len(selectors) < 140:
1604:                 selectors.append({"line": line_no, "selector": selector_match.group(1).strip()[:160]})
1605:         return {
1606:             "repo_file": repo_file,
1607:             "line_count": len(lines),
1608:             "size_bytes": (REPO / repo_file).stat().st_size if (REPO / repo_file).exists() else 0,
1609:             "description": FILE_DESCRIPTIONS.get(repo_file, "Tracked source/configuration file in the OMB
Portfolio Builder repository."),
1610:             "functions": functions,
1611:             "variables": variables,
1612:             "imports": imports,
1613:             "endpoints": endpoints,
1614:             "routes": routes,
1615:             "selectors": selectors,
1616:             "events": events,
1617:             "json_keys": json_keys,
1618:             "signals": collect_signal_hits(lines),
1619:             "mentions_server": "server.mjs" in text,
1620:             "mentions_template_preview": "template-preview" in text,
1621:             "mentions_script": "script.js" in text,
1622:             "mentions_projects_json": "projects.json" in text,
1623:             "mentions_github": "github" in text.lower(),
1624:             "mentions_cloudflare": "cloudflare" in text.lower() or "worker" in text.lower(),
1625:             "snippet": line_excerpt(lines, 1, min(72, len(lines)), max_lines=72) if lines else "",
1626:         }
1627:
1628:

```

fact_relationship_summary - lines 1629-1645

Item	Explanation
Source location	Lines 1629-1645 (17 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def fact_relationship_summary(facts: dict) -> str:
Touches	Mostly local calculations or local UI state.
Calls detected	get, append, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1629: def fact_relationship_summary(facts: dict) -> str:
1630:     related = []
1631:     if facts.get("mentions_template_preview"):
1632:         related.append("builder frontend")
1633:     if facts.get("mentions_server"):
1634:         related.append("local backend")
1635:     if facts.get("mentions_script"):
1636:         related.append("public website runtime")
1637:     if facts.get("mentions_projects_json"):
1638:         related.append("portfolio catalog")
1639:     if facts.get("mentions_cloudflare"):
1640:         related.append("Cloudflare/AI layer")
1641:     if facts.get("mentions_github"):

```



```

1642:         related.append("GitHub/release layer")
1643:     return ", ".join(related) if related else "Mostly local to its own feature area."
1644:
1645:

```

folder_role - lines 1646-1663

Item	Explanation
Source location	Lines 1646-1663 (18 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def folder_role(repo_file: str) -> str:
Touches	Mostly local calculations or local UI state.
Calls detected	startswith
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 8 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1646: def folder_role(repo_file: str) -> str:
1647:     if repo_file.startswith(".github/workflows/"):
1648:         return "GitHub automation. GitHub Actions reads this during CI/release/branch-gate runs."
1649:     if repo_file.startswith(".well-known/"):
1650:         return "Public web metadata. Browsers, security tools, and crawlers may request this path."
1651:     if repo_file.startswith("assets/"):
1652:         return "Public and desktop visual asset. Used for favicons, app icons, branding, or install
surfaces."
1653:     if repo_file.startswith("build/"):
1654:         return "Packaging and installer customization. Used while creating the Windows app installer."
1655:     if repo_file.startswith("cloudflare/"):
1656:         return "Cloudflare Worker/deployment area. Used for public AI backend and Cloudflare deployment
notes."
1657:     if repo_file.startswith("docs/"):
1658:         return "Documentation and generated reference area. Used to explain the app and source
structure."
1659:     if repo_file.startswith("Backgrounds/"):
1660:         return "Portfolio background asset folder placeholder."
1661:     return "Repository root. Usually an entry file, app config, public website file, package manifest, or
top-level documentation."
1662:
1663:

```

file_category - lines 1664-1688

Item	Explanation
Source location	Lines 1664-1688 (25 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def file_category(repo_file: str, path: Path) -> str:
Touches	Mostly local calculations or local UI state.
Calls detected	lower, endswith, startswith
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 11 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1664: def file_category(repo_file: str, path: Path) -> str:
1665:     suffix = path.suffix.lower()
1666:     if repo_file.endswith(".gitkeep"):
1667:         return "Folder marker"
1668:     if repo_file.startswith(".github/workflows/"):
1669:         return "GitHub workflow"

```

```

1670:     if suffix in {".png", ".ico", ".jpg", ".jpeg", ".webp", ".svg"}:
1671:         return "Image/icon asset"
1672:     if suffix in {".docx", ".pdf"}:
1673:         return "Generated document"
1674:     if suffix in {".js", ".mjs", ".cjs"}:
1675:         return "JavaScript source"
1676:     if suffix in {".html", ".css"}:
1677:         return "Website/builder UI source"
1678:     if suffix in {".json", ".toml", ".yaml", ".yml"}:
1679:         return "Configuration or structured data"
1680:     if suffix in {".md", ".txt", ""}:
1681:         return "Documentation or text control file"
1682:     if suffix == ".nsh":
1683:         return "NSIS installer script"
1684:     if suffix == ".py":
1685:         return "Python tooling"
1686:     return "Repository file"
1687:
1688:

```

image_metadata - lines 1689-1704

Item	Explanation
Source location	Lines 1689-1704 (16 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def image_metadata(path: Path) -> dict:
Touches	filesystem
Calls detected	exists, lower, open, lstrip, upper, getattr, str
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1689: def image_metadata(path: Path) -> dict:
1690:     if not path.exists() or path.suffix.lower() not in {".png", ".jpg", ".jpeg", ".webp", ".ico"}:
1691:         return {}
1692:     try:
1693:         with Image.open(path) as image:
1694:             return {
1695:                 "format": image.format or path.suffix.lower().lstrip(".").upper(),
1696:                 "width": image.size[0],
1697:                 "height": image.size[1],
1698:                 "mode": image.mode,
1699:                 "frames": getattr(image, "n_frames", 1),
1700:             }
1701:     except Exception as exc:
1702:         return {"error": str(exc)}
1703:
1704:

```

file_specific_notes - lines 1705-1733

Item	Explanation
Source location	Lines 1705-1733 (29 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def file_specific_notes(repo_file: str, category: str) -> list[str]:
Touches	Mostly local calculations or local UI state.
Calls detected	append, startswith, endswith
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1705: def file_specific_notes(repo_file: str, category: str) -> list[str]:
1706:     notes = []
1707:     if repo_file in FILE_DESCRIPTIONS:
1708:         notes.append(FILE_DESCRIPTIONS[repo_file])
1709:     if repo_file.startswith("assets/favicon"):
1710:         notes.append("This favicon participates in browser tab identity, mobile install surfaces, and
public website branding.")
1711:     if repo_file.startswith("assets/omb-app-icon"):
1712:         notes.append("This app icon is used by the Windows desktop application, installer metadata, and
shortcuts.")
1713:     if repo_file == "assets/omb-mark.png":
1714:         notes.append("This is the OMB visual mark used by the website and builder branding.")
1715:     if repo_file == "_headers":
1716:         notes.append("This file describes security and caching headers for hosts that support static
_headers files, such as Cloudflare Pages style deployments.")
1717:     if repo_file == ".nojekyll":
1718:         notes.append("This makes GitHub Pages serve static files directly without Jekyll filtering.")
1719:     if repo_file == ".gitattributes":
1720:         notes.append("This protects binary artifacts such as DOCX, PDF, PNG, ICO, and installers from
line-ending conversion.")
1721:     if repo_file == ".gitignore":
1722:         notes.append("This keeps temporary build output, local-only data, and generated noise out of
version control.")
1723:     if repo_file.endswith(".gitkeep"):
1724:         notes.append("This empty marker exists so Git keeps an otherwise empty folder.")
1725:     if category == "Image/icon asset":
1726:         notes.append("Do not edit binary assets with a text editor. Replace them with properly exported
image/icon files.")
1727:     if category == "GitHub workflow":
1728:         notes.append("Workflow changes should be tested on a feature branch because mistakes can block
merges or releases.")
1729:     if not notes:
1730:         notes.append("Tracked repository file used by the builder, website, installer, documentation, or
release process.")
1731:     return notes
1732:
1733:
```

extract_file_facts - lines 1734-1763

Item	Explanation
Source location	Lines 1734-1763 (30 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def extract_file_facts(repo_file: str) -> dict:
Touches	Mostly local calculations or local UI state.
Calls detected	lower, file_category, is_text_code_file, source_lines, exists, is_file, guess_type, str, stat, folder_role, get, file_specific_notes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1734: def extract_file_facts(repo_file: str) -> dict:
1735:     path = REPO / repo_file
1736:     suffix = path.suffix.lower()
1737:     category = file_category(repo_file, path)
1738:     text_file = is_text_code_file(repo_file) or suffix in {".gitignore", ".gitattributes", ".nojekyll",
"", ".txt", ".md"}
1739:     lines = source_lines(repo_file) if text_file and path.exists() and path.is_file() else []
1740:     mime_type, _encoding = mimetypes.guess_type(str(path))
1741:     facts = {
1742:         "repo_file": repo_file,
1743:         "path": path,
1744:         "exists": path.exists(),
1745:         "size_bytes": path.stat().st_size if path.exists() and path.is_file() else 0,
1746:         "suffix": suffix or "(none)",
1747:         "category": category,
1748:         "mime_type": mime_type or "unknown",
1749:         "folder_role": folder_role(repo_file),
1750:         "description": FILE_DESCRIPTIONS.get(repo_file, "Tracked repository file used by the builder or
public website."),
1751:         "notes": file_specific_notes(repo_file, category),
1752:         "is_text": bool(lines or text_file),
1753:         "line_count": len(lines),
```

```

1754:         "snippet": line_excerpt(lines, 1, min(72, len(lines)), max_lines=72) if lines else "",
1755:         "image": image_metadata(path),
1756:     }
1757: if is_text_code_file(repo_file):
1758:     facts["source"] = extract_source_facts(repo_file)
1759: else:
1760:     facts["source"] = None
1761: return facts
1762:
1763:

```

add_file_fact_summary - lines 1764-1782

Item	Explanation
Source location	Lines 1764-1782 (19 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_file_fact_summary(doc: Document, facts: dict) -> None:
Touches	filesystem
Calls detected	append, add_table
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1764: def add_file_fact_summary(doc: Document, facts: dict) -> None:
1765:     rows = [
1766:         ("File", facts["repo_file"]),
1767:         ("Category", facts["category"]),
1768:         ("Folder role", facts["folder_role"]),
1769:         ("Size", f"{facts['size_bytes']:,} bytes"),
1770:         ("Extension", facts["suffix"]),
1771:         ("MIME type", facts["mime_type"]),
1772:         ("Text lines", f"{facts['line_count']:,}" if facts["is_text"] else "Not a readable text file"),
1773:     ]
1774:     if facts["image"]:
1775:         image = facts["image"]
1776:         if "error" in image:
1777:             rows.append(("Image metadata", f"Could not inspect image: {image['error']}"))
1778:         else:
1779:             rows.append(("Image metadata", f"{image['format']} {image['width']}x{image['height']}, mode {image['mode']}, frames {image['frames']}"))
1780:     add_table(doc, rows)
1781:
1782:

```

add_file_fact_sections - lines 1783-1808

Item	Explanation
Source location	Lines 1783-1808 (26 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_file_fact_sections(doc: Document, facts: dict) -> None:
Touches	filesystem
Calls detected	add_heading, add_paragraph, add_source_fact_sections, add_code, numbers
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1783: def add_file_fact_sections(doc: Document, facts: dict) -> None:
1784:     doc.add_heading("why this file exists", level=2)

```

```

1785:     for note in facts["notes"]:
1786:         doc.add_paragraph(note, style="List Bullet")
1787:     doc.add_heading("How it is used", level=2)
1788:     doc.add_paragraph(facts["folder_role"])
1789:     if facts["source"]:
1790:         add_source_fact_sections(doc, facts["source"], include_excerpts=True)
1791:     elif facts["snippet"]:
1792:         doc.add_heading("Readable content snippet", level=2)
1793:         add_code(doc, facts["snippet"])
1794:     elif facts["image"]:
1795:         doc.add_heading("Asset handling note", level=2)
1796:         doc.add_paragraph("This file is documented by metadata instead of raw bytes. Binary image/icon
bytes are not useful to print in a Word/PDF reference. Use an image editor or icon tool when changing it.")
1797:     else:
1798:         doc.add_heading("Binary or marker note", level=2)
1799:         doc.add_paragraph("This file is either binary, empty, or a marker/config file whose value comes
from its presence and path rather than readable content.")
1800:     doc.add_heading("Safe change checklist", level=2)
1801:     numbers(doc, [
1802:         "Confirm which app, website, installer, workflow, or document consumes this file.",
1803:         "Change it on a feature branch from development.",
1804:         "Regenerate docs if the file purpose, API surface, or behavior changes.",
1805:         "Run the smallest relevant smoke test before merging into development.",
1806:     ])
1807:
1808:

```

code_reference_name - lines 1809-1812

Item	Explanation
Source location	Lines 1809-1812 (4 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def code_reference_name(repo_file: str, suffix: str) -> str:
Touches	Mostly local calculations or local UI state.
Calls detected	sub, strip
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1809: def code_reference_name(repo_file: str, suffix: str) -> str:
1810:     return re.sub(r"^[A-Za-z0-9_.-]+", "_", repo_file).strip("_") + suffix
1811:
1812:

```

remove_directory - lines 1813-1826

Item	Explanation
Source location	Lines 1813-1826 (14 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def remove_directory(path: Path) -> None:
Touches	filesystem
Calls detected	exists, remove_readonly, Path, chmod, rmtree
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1813: def remove_directory(path: Path) -> None:
1814:     if not path.exists():

```

```

1815:         return
1816:
1817:     def remove_readonly(function, item_path, excinfo):
1818:         try:
1819:             Path(item_path).chmod(0o700)
1820:             function(item_path)
1821:         except Exception:
1822:             raise excinfo
1823:
1824:     shutil.rmtree(path, onexc=remove_readonly)
1825:
1826:

```

remove_readonly - lines 1817-1826

Item	Explanation
Source location	Lines 1817-1826 (10 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def remove_readonly(function, item_path, excinfo):
Touches	filesystem
Calls detected	Path, chmod, rmtree
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1817:     def remove_readonly(function, item_path, excinfo):
1818:         try:
1819:             Path(item_path).chmod(0o700)
1820:             function(item_path)
1821:         except Exception:
1822:             raise excinfo
1823:
1824:     shutil.rmtree(path, onexc=remove_readonly)
1825:
1826:

```

add_source_fact_summary - lines 1827-1840

Item	Explanation
Source location	Lines 1827-1840 (14 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_source_fact_summary(doc: Document, facts: dict) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	add_table, fact_relationship_summary, str, len
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1827: def add_source_fact_summary(doc: Document, facts: dict) -> None:
1828:     add_table(doc, [
1829:         ("File", facts["repo_file"]),
1830:         ("Purpose", facts["description"]),
1831:         ("Lines", f"{facts['line_count']:,}"),
1832:         ("Size", f"{facts['size_bytes']:,} bytes"),
1833:         ("Talks to", fact_relationship_summary(facts)),
1834:         ("Functions", str(len(facts["functions"]))),
1835:         ("Variables/constants", str(len(facts["variables"]))),
1836:         ("API mentions", str(len(facts["endpoints"]))),

```

```

1837:         ("Signals", str(len(facts["signals"]))),
1838:     ])
1839:
1840:

```

add_source_fact_sections - lines 1841-1887

Item	Explanation
Source location	Lines 1841-1887 (47 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_source_fact_sections(doc: Document, facts: dict, include_excerpts: bool = True) -> None:
Touches	filesystem
Calls detected	add_heading, add_paragraph, add_table, endpoint_purpose, function_detail_rows, add_code, numbers
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1841: def add_source_fact_sections(doc: Document, facts: dict, include_excerpts: bool = True) -> None:
1842:     doc.add_heading("How this file participates in the system", level=2)
1843:     doc.add_paragraph("This generated section reads the actual file and points out how it communicates
with other pieces of the app. It is intentionally concrete: line numbers, declarations, endpoints, events,
source excerpts, and debugging cues.")
1844:     if facts["imports"]:
1845:         doc.add_heading("Imports, requires, and external dependencies", level=2)
1846:         add_table(doc, [(f"Line {item['line']}", item["statement"]) for item in facts["imports"][:60]])
1847:     if facts["routes"]:
1848:         doc.add_heading("Backend routes implemented here", level=2)
1849:         add_table(doc, [(f"{item['method']} {item['path']}", f"Line {item['line']}".
endpoint_purpose(item['path'])) for item in facts["routes"][:80]])
1850:     if facts["endpoints"]:
1851:         doc.add_heading("API endpoints mentioned or called", level=2)
1852:         add_table(doc, [(item["endpoint"], f"Line {item['line']}". endpoint_purpose(item['endpoint'])) for item in facts["endpoints"][:80]])
1853:     if facts["signals"]:
1854:         doc.add_heading("Important implementation signals", level=2)
1855:         add_table(doc, [(f"{item['type']} at line {item['line']}", f"{item['meaning']} Evidence:
{item['evidence']}") for item in facts["signals"][:80]])
1856:     if facts["variables"]:
1857:         doc.add_heading("Important constants and variables", level=2)
1858:         add_table(doc, [(item["name"], f"Line {item['line']}: {item['statement']}") for item in
facts["variables"][:80]])
1859:     if facts["json_keys"]:
1860:         doc.add_heading("JSON/YAML-style keys detected", level=2)
1861:         add_table(doc, [(item["key"], f"Line {item['line']}: {item['statement']}") for item in
facts["json_keys"][:80]])
1862:     if facts["events"]:
1863:         doc.add_heading("Event handlers and user interactions", level=2)
1864:         add_table(doc, [(f"Line {item['line']}", item["statement"]) for item in facts["events"][:80]])
1865:     if facts["selectors"]:
1866:         doc.add_heading("CSS selectors and visual hooks", level=2)
1867:         add_table(doc, [(f"Line {item['line']}", item["selector"]) for item in facts["selectors"][:120]])
1868:     if facts["functions"]:
1869:         doc.add_heading("Function-by-function detail", level=2)
1870:         for item in facts["functions"]:
1871:             doc.add_heading(f"{item['name']} - lines {item['line']}-{item['end_line']}", level=3)
1872:             add_table(doc, function_detail_rows(item))
1873:             if include_excerpts:
1874:                 doc.add_paragraph("Source excerpt:")
1875:                 add_code(doc, item["excerpt"])
1876:             doc.add_heading("Representative opening snippet", level=2)
1877:             add_code(doc, facts["snippet"] or "(empty file)")
1878:             doc.add_heading("Beginner debugging checklist for this file", level=2)
1879:             numbers(doc, [
1880:                 "Name the user action, command, or workflow that reaches this file.",
1881:                 "Find the exact function or route that owns the behavior.",
1882:                 "Check the inputs: DOM state, request body, file path, local storage value, compiler source, or
Git branch.",
1883:                 "Check the outputs: returned JSON, updated DOM, written file, terminal output, generated project
catalog, or published website asset.",
1884:                 "If the issue crosses a boundary, test the boundary directly: browser console for frontend,
endpoint call for backend, command line for Git/compiler, Cloudflare logs for Worker behavior.",
1885:             ])
1886:
1887:

```

setup_code_reference_document - lines 1888-1901

Item	Explanation
Source location	Lines 1888-1901 (14 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def setup_code_reference_document(title: str, subtitle: str) -> Document:
Touches	Mostly local calculations or local UI state.
Calls detected	setup_document, add_paragraph, add_run, Pt, RGBColor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1888: def setup_code_reference_document(title: str, subtitle: str) -> Document:
1889:     doc = setup_document()
1890:     title_p = doc.add_paragraph()
1891:     title_p.alignment = WD_ALIGN_PARAGRAPH.LEFT
1892:     run = title_p.add_run(title)
1893:     run.bold = True
1894:     run.font.size = Pt(22)
1895:     run.font.color.rgb = RGBColor(11, 37, 69)
1896:     doc.add_paragraph(subtitle)
1897:     doc.add_paragraph(f"Generated from repository state at {REPO}.")
1898:     doc.add_paragraph("")
1899:     return doc
1900:
1901:
```

compact_generated_docx - lines 1902-1922

Item	Explanation
Source location	Lines 1902-1922 (21 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def compact_generated_docx(doc: Document, keep_first_page_break: bool = True) -> int:
Touches	Mostly local calculations or local UI state.
Calls detected	list, any, qn, get, iter, getparent, remove
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1902: def compact_generated_docx(doc: Document, keep_first_page_break: bool = True) -> int:
1903:     """Remove generator-inserted page breaks so pages fill naturally."""
1904:     kept = 0
1905:     removed = 0
1906:     for paragraph in list(doc.paragraphs):
1907:         has_page_break = any(
1908:             element.tag == qn("w:br") and element.get(qn("w:type")) == "page"
1909:             for element in paragraph._element.iter()
1910:         )
1911:         if not has_page_break:
1912:             continue
1913:         if keep_first_page_break and kept == 0:
1914:             kept += 1
1915:             continue
1916:         parent = paragraph._element.getparent()
1917:         if parent is not None:
1918:             parent.remove(paragraph._element)
1919:             removed += 1
1920:     return removed
```


1921:
1922:

write_single_code_reference_docx - lines 1923-1933

Item	Explanation
Source location	Lines 1923-1933 (11 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	def write_single_code_reference_docx(facts: dict, output_path: Path) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	setup_code_reference_document, add_source_fact_summary, add_source_fact_sections, compact_generated_docx, save
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1923: def write_single_code_reference_docx(facts: dict, output_path: Path) -> None:
1924:     doc = setup_code_reference_document(
1925:         f"Code Reference: {facts['repo_file']}",
1926:         "A source-level explanation with function details, file communication, variables, endpoints, and
debugging notes.",
1927:     )
1928:     add_source_fact_summary(doc, facts)
1929:     add_source_fact_sections(doc, facts, include_excerpts=True)
1930:     compact_generated_docx(doc, keep_first_page_break=False)
1931:     doc.save(output_path)
1932:
1933:
```

pdf_styles - lines 1934-1963

Item	Explanation
Source location	Lines 1934-1963 (30 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def pdf_styles():
Touches	Mostly local calculations or local UI state.
Calls detected	getSampleStyleSheet, add, ParagraphStyle, HexColor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
1934: def pdf_styles():
1935:     styles = getSampleStyleSheet()
1936:     styles.add(ParagraphStyle(
1937:         name="CodeTiny",
1938:         parent=styles["Code"],
1939:         fontName="Courier",
1940:         fontSize=6.4,
1941:         leading=7.6,
1942:         backColor=colors.HexColor("#F5F8FB"),
1943:         borderColor=colors.HexColor("#D8E6F0"),
1944:         borderWidth=0.25,
1945:         borderPadding=4,
1946:         spaceAfter=7,
1947:     ))
1948:     styles.add(ParagraphStyle(
1949:         name="SmallBody",
1950:         parent=styles["BodyText"],
```

```

1951:         fontSize=8.5,
1952:         leading=11.0,
1953:         spaceAfter=5,
1954:     ))
1955:     styles.add(ParagraphStyle(
1956:         name="TinyTable",
1957:         parent=styles["BodyText"],
1958:         fontSize=7.0,
1959:         leading=8.5,
1960:     ))
1961:     return styles
1962:
1963:

```

pdf_paragraph - lines 1964-1967

Item	Explanation
Source location	Lines 1964-1967 (4 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def pdf_paragraph(text: str, style) -> Paragraph:
Touches	Mostly local calculations or local UI state.
Calls detected	Paragraph, escape, str, replace
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1964: def pdf_paragraph(text: str, style) -> Paragraph:
1965:     return Paragraph(html.escape(str(text)).replace("\n", "<br/>"), style)
1966:
1967:

```

pdf_code - lines 1968-1976

Item	Explanation
Source location	Lines 1968-1976 (9 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def pdf_code(text: str, styles) -> Preformatted:
Touches	filesystem
Calls detected	in, splitlines, replace, wrap, extend, Preformatted, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1968: def pdf_code(text: str, styles) -> Preformatted:
1969:     wrapped_lines = []
1970:     for raw_line in (text or "").splitlines():
1971:         expanded = raw_line.replace("\t", " ")
1972:         chunks = textwrap.wrap(expanded, width=112, replace_whitespace=False, drop_whitespace=False) or
1973:         [""]
1974:         wrapped_lines.extend(chunks)
1975:     return Preformatted("\n".join(wrapped_lines), styles["CodeTiny"])
1976:

```

pdf_table - lines 1977-1992

Item	Explanation
Source location	Lines 1977-1992 (16 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def pdf_table(rows: list[tuple[str, str]], styles):
Touches	Mostly local calculations or local UI state.
Calls detected	pdf_paragraph, extend, Table, setStyle, TableStyle, HexColor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1977: def pdf_table(rows: list[tuple[str, str]], styles):
1978:     data = [[pdf_paragraph("Item", styles["TinyTable"]), pdf_paragraph("Explanation",
styles["TinyTable"])]
1979:     data.extend([[pdf_paragraph(label, styles["TinyTable"]), pdf_paragraph(value, styles["TinyTable"])]
for label, value in rows])
1980:     table = Table(data, colWidths=[1.45 * inch, 5.25 * inch], repeatRows=1)
1981:     table.setStyle(TableStyle([
1982:         ("BACKGROUND", (0, 0), (-1, 0), colors.HexColor("#E8EEF5")),
1983:         ("GRID", (0, 0), (-1, -1), 0.25, colors.HexColor("#B7CFE0")),
1984:         ("VALIGN", (0, 0), (-1, -1), "TOP"),
1985:         ("LEFTPADDING", (0, 0), (-1, -1), 5),
1986:         ("RIGHTPADDING", (0, 0), (-1, -1), 5),
1987:         ("TOPPADDING", (0, 0), (-1, -1), 4),
1988:         ("BOTTOMPADDING", (0, 0), (-1, -1), 4),
1989:     ]))
1990:     return table
1991:
1992:

```

add_pdf_source_fact_sections - lines 1993-2031

Item	Explanation
Source location	Lines 1993-2031 (39 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_pdf_source_fact_sections(story: list, facts: dict, styles, include_excerpts: bool = True) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	append, Paragraph, pdf_paragraph, pdf_table, endpoint_purpose, function_detail_rows, pdf_code
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

1993: def add_pdf_source_fact_sections(story: list, facts: dict, styles, include_excerpts: bool = True) ->
None:
1994:     story.append(Paragraph("How this file participates in the system", styles["Heading2"]))
1995:     story.append(pdf_paragraph("This generated section reads the actual file and points out line numbers,
declarations, endpoints, events, source excerpts, and debugging cues.", styles["SmallBody"]))
1996:     if facts["imports"]:
1997:         story.append(Paragraph("Imports, requires, and external dependencies", styles["Heading2"]))
1998:         story.append(pdf_table([(f"Line {item['line']}", item["statement"]) for item in
facts["imports"][:60]], styles))
1999:         if facts["routes"]:
2000:             story.append(Paragraph("Backend routes implemented here", styles["Heading2"]))
2001:             story.append(pdf_table([(f"{item['method']} {item['path']}", f"Line {item['line']}.
{endpoint_purpose(item['path'])}") for item in facts["routes"][:80]], styles))
2002:             if facts["endpoints"]:
2003:                 story.append(Paragraph("API endpoints mentioned or called", styles["Heading2"]))
2004:                 story.append(pdf_table([(item["endpoint"], f"Line {item['line']}.
{endpoint_purpose(item['endpoint'])}") for item in facts["endpoints"][:80]], styles))
2005:                 if facts["signals"]:

```

```

2006:         story.append(Paragraph("Important implementation signals", styles["Heading2"]))
2007:         story.append(pdf_table([(f"{item['type']} at line {item['line']}", f"{item['meaning']} Evidence:
{item['evidence']}") for item in facts["signals"][:80]], styles))
2008:         if facts["variables"]:
2009:             story.append(Paragraph("Important constants and variables", styles["Heading2"]))
2010:             story.append(pdf_table([(item["name"], f"Line {item['line']}: {item['statement']}") for item in
facts["variables"][:80]], styles))
2011:         if facts["json_keys"]:
2012:             story.append(Paragraph("JSON/YAML-style keys detected", styles["Heading2"]))
2013:             story.append(pdf_table([(item["key"], f"Line {item['line']}: {item['statement']}") for item in
facts["json_keys"][:80]], styles))
2014:         if facts["events"]:
2015:             story.append(Paragraph("Event handlers and user interactions", styles["Heading2"]))
2016:             story.append(pdf_table([(f"Line {item['line']}", item["statement"]) for item in
facts["events"][:80]], styles))
2017:         if facts["selectors"]:
2018:             story.append(Paragraph("CSS selectors and visual hooks", styles["Heading2"]))
2019:             story.append(pdf_table([(f"Line {item['line']}", item["selector"]) for item in
facts["selectors"][:120]], styles))
2020:         if facts["functions"]:
2021:             story.append(Paragraph("Function-by-function detail", styles["Heading2"]))
2022:             for item in facts["functions"]:
2023:                 story.append(Paragraph(f"{item['name']} - lines {item['line']}-{item['end_line']}",
styles["Heading3"]))
2024:                 story.append(pdf_table(function_detail_rows(item), styles))
2025:                 if include_excerpts:
2026:                     story.append(pdf_paragraph("Source excerpt:", styles["SmallBody"]))
2027:                     story.append(pdf_code(item["excerpt"], styles))
2028:                 story.append(Paragraph("Representative opening snippet", styles["Heading2"]))
2029:                 story.append(pdf_code(facts["snippet"] or "(empty file)", styles))
2030:
2031:

```

write_single_code_reference_pdf - lines 2032-2055

Item	Explanation
Source location	Lines 2032-2055 (24 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	def write_single_code_reference_pdf(facts: dict, output_path: Path) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	pdf_styles, Paragraph, pdf_paragraph, pdf_table, fact_relationship_summary, str, len, Spacer, add_pdf_source_fact_sections, SimpleDocTemplate, build
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2032: def write_single_code_reference_pdf(facts: dict, output_path: Path) -> None:
2033:     if not REPORTLAB_AVAILABLE:
2034:         return
2035:     styles = pdf_styles()
2036:     story = [
2037:         Paragraph(f"Code Reference: {facts['repo_file']}", styles["Title"]),
2038:         pdf_paragraph("A source-level explanation with function details, file communication, variables,
endpoints, and debugging notes.", styles["SmallBody"]),
2039:         pdf_table([
2040:             ("File", facts["repo_file"]),
2041:             ("Purpose", facts["description"]),
2042:             ("Lines", f"{facts['line_count']:,}"),
2043:             ("Size", f"{facts['size_bytes']:,} bytes"),
2044:             ("Talks to", fact_relationship_summary(facts)),
2045:             ("Functions", str(len(facts["functions"]))),
2046:             ("Variables/constants", str(len(facts["variables"]))),
2047:             ("API mentions", str(len(facts["endpoints"]))),
2048:             ("Signals", str(len(facts["signals"]))),
2049:         ], styles),
2050:         Spacer(1, 0.12 * inch),
2051:     ]
2052:     add_pdf_source_fact_sections(story, facts, styles, include_excerpts=True)
2053:     SimpleDocTemplate(str(output_path), pagesize=LETTER, rightMargin=0.55 * inch, leftMargin=0.55 * inch,
topMargin=0.55 * inch, bottomMargin=0.55 * inch).build(story)
2054:
2055:

```

write_master_code_reference_docx - lines 2056-2084

Item	Explanation
Source location	Lines 2056-2084 (29 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	def write_master_code_reference_docx(code_files: list[str], facts_by_file: dict[str, dict], diagrams: dict[str, Path]) -> None:
Touches	network/API requests, authentication/security data
Calls detected	setup_code_reference_document, add_diagram, add_heading, numbers, add_table, len, signal, add_page_break, add_source_fact_summary, add_source_fact_sections, compact_generated_docx, save
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2056: def write_master_code_reference_docx(code_files: list[str], facts_by_file: dict[str, dict], diagrams:
dict[str, Path]) -> None:
2057:     doc = setup_code_reference_document(
2058:         "OMB Portfolio Builder Code Reference",
2059:         "A generated Word reference that explains the important source files, all discovered functions,
communication paths, variables, endpoints, installers, AI paths, and compiler paths.",
2060:     )
2061:     add_diagram(doc, diagrams["file-communication-map"], "Main source file communication map.")
2062:     add_diagram(doc, diagrams["frontend-backend-cloudflare"], "Frontend, backend, Cloudflare, and AI
boundary.")
2063:     add_diagram(doc, diagrams["compile-code-flow"], "Compile Code workspace and simulator flow.")
2064:     doc.add_heading("How to use this document", level=1)
2065:     numbers(doc, [
2066:         "Start with the file you want to understand.",
2067:         "Read the summary table to understand its role.",
2068:         "Read implementation signals to see whether the file touches Git, files, compilers, Cloudflare,
authentication, DOM, storage, or network calls.",
2069:         "Read function details to understand line ranges, side effects, calls, endpoints, and source
excerpts.",
2070:         "Use the debugging checklist to trace a behavior from click to function to file/API/output.",
2071:     ])
2072:     doc.add_heading("Generated file index", level=1)
2073:     add_table(doc, [(repo_file, f"{facts_by_file[repo_file]['line_count']},") lines,
{len(facts_by_file[repo_file]['functions'])} function(s), {len(facts_by_file[repo_file]['signals'])}
signal(s).") for repo_file in code_files])
2074:     doc.add_page_break()
2075:     for repo_file in code_files:
2076:         facts = facts_by_file[repo_file]
2077:         doc.add_heading(f"Source File: {repo_file}", level=1)
2078:         add_source_fact_summary(doc, facts)
2079:         add_source_fact_sections(doc, facts, include_excerpts=True)
2080:         doc.add_page_break()
2081:     compact_generated_docx(doc, keep_first_page_break=True)
2082:     doc.save(CODE_REFERENCE_MASTER_DOCX)
2083:
2084:
```

write_master_code_reference_pdf - lines 2085-2123

Item	Explanation
Source location	Lines 2085-2123 (39 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	def write_master_code_reference_pdf(code_files: list[str], facts_by_file: dict[str, dict], diagrams: dict[str, Path]) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	pdf_styles, Paragraph, pdf_paragraph, get, exists, append, PdfImage, str, pdf_table, len, signal, PageBreak
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2085: def write_master_code_reference_pdf(code_files: list[str], facts_by_file: dict[str, dict], diagrams:
dict[str, Path]) -> None:
2086:     if not REPORTLAB_AVAILABLE:
2087:         return
2088:     styles = pdf_styles()
2089:     story = [
2090:         Paragraph("OMB Portfolio Builder Code Reference", styles["Title"]),
2091:         pdf_paragraph("A generated PDF reference that explains the important source files, discovered
functions, communication paths, variables, endpoints, installers, AI paths, and compiler paths.",
styles["SmallBody"]),
2092:     ]
2093:     for diagram_name, caption in [
2094:         ("file-communication-map", "Main source file communication map."),
2095:         ("frontend-backend-cloudflare", "Frontend, backend, Cloudflare, and AI boundary."),
2096:         ("compile-code-flow", "Compile Code workspace and simulator flow."),
2097:     ]:
2098:         path = diagrams.get(diagram_name)
2099:         if path and path.exists():
2100:             story.append(PdfImage(str(path), width=6.6 * inch, height=3.72 * inch))
2101:             story.append(pdf_paragraph(caption, styles["SmallBody"]))
2102:             story.append(Paragraph("Generated file index", styles["Heading1"]))
2103:             story.append(pdf_table([(repo_file, f"{facts_by_file[repo_file]['line_count']:,} lines,
{len(facts_by_file[repo_file]['functions'])} function(s), {len(facts_by_file[repo_file]['signals'])}
signal(s).") for repo_file in code_files], styles))
2104:             story.append(PageBreak())
2105:             for repo_file in code_files:
2106:                 facts = facts_by_file[repo_file]
2107:                 story.append(Paragraph(f"Source File: {repo_file}", styles["Heading1"]))
2108:                 story.append(pdf_table([
2109:                     ("File", facts["repo_file"]),
2110:                     ("Purpose", facts["description"]),
2111:                     ("Lines", f"{facts['line_count']:,}"),
2112:                     ("Size", f"{facts['size_bytes']:,} bytes"),
2113:                     ("Talks to", fact_relationship_summary(facts)),
2114:                     ("Functions", str(len(facts["functions"]))),
2115:                     ("Variables/constants", str(len(facts["variables"]))),
2116:                     ("API mentions", str(len(facts["endpoints"]))),
2117:                     ("Signals", str(len(facts["signals"]))),
2118:                 ], styles))
2119:                 add_pdf_source_fact_sections(story, facts, styles, include_excerpts=True)
2120:                 story.append(PageBreak())
2121:             SimpleDocTemplate(str(CODE_REFERENCE_MASTER_PDF), pagesize=LETTER, rightMargin=0.55 * inch,
leftMargin=0.55 * inch, topMargin=0.55 * inch, bottomMargin=0.55 * inch).build(story)
2122:
2123:

```

write_single_file_reference_docx - lines 2124-2134

Item	Explanation
Source location	Lines 2124-2134 (11 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	def write_single_file_reference_docx(facts: dict, output_path: Path) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	setup_code_reference_document, add_file_fact_summary, add_file_fact_sections, compact_generated_docx, save
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2124: def write_single_file_reference_docx(facts: dict, output_path: Path) -> None:
2125:     doc = setup_code_reference_document(
2126:         f"Important Code Reference: {facts['repo_file']}",
2127:         "A focused code/control-file explanation covering purpose, owner layer, APIs, variables,
implementation signals, source excerpts, and debugging rules.",
2128:     )
2129:     add_file_fact_summary(doc, facts)
2130:     add_file_fact_sections(doc, facts)

```

```

2131: compact_generated_docx(doc, keep_first_page_break=False)
2132: doc.save(output_path)
2133:
2134:

```

add_pdf_file_fact_sections - lines 2135-2161

Item	Explanation
Source location	Lines 2135-2161 (27 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_pdf_file_fact_sections(story: list, facts: dict, styles) -> None:
Touches	browser/Electron DOM, filesystem
Calls detected	append, Paragraph, pdf_paragraph, add_pdf_source_fact_sections, pdf_code
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2135: def add_pdf_file_fact_sections(story: list, facts: dict, styles) -> None:
2136:     story.append(Paragraph("Why this file exists", styles["Heading2"]))
2137:     for note in facts["notes"]:
2138:         story.append(pdf_paragraph(f"- {note}", styles["SmallBody"]))
2139:     story.append(Paragraph("How it is used", styles["Heading2"]))
2140:     story.append(pdf_paragraph(facts["folder_role"], styles["SmallBody"]))
2141:     if facts["source"]:
2142:         add_pdf_source_fact_sections(story, facts["source"], styles, include_excerpts=True)
2143:     elif facts["snippet"]:
2144:         story.append(Paragraph("Readable content snippet", styles["Heading2"]))
2145:         story.append(pdf_code(facts["snippet"], styles))
2146:     elif facts["image"]:
2147:         story.append(Paragraph("Asset handling note", styles["Heading2"]))
2148:         story.append(pdf_paragraph("This file is documented by metadata instead of raw bytes. Binary
image/icon bytes are not useful to print in a reference document.", styles["SmallBody"]))
2149:     else:
2150:         story.append(Paragraph("Binary or marker note", styles["Heading2"]))
2151:         story.append(pdf_paragraph("This file is either binary, empty, or a marker/config file whose
value comes from its presence and path rather than readable content.", styles["SmallBody"]))
2152:     story.append(Paragraph("Safe change checklist", styles["Heading2"]))
2153:     for item in [
2154:         "Confirm which app, website, installer, workflow, or document consumes this file.",
2155:         "Change it on a feature branch from development.",
2156:         "Regenerate docs if the file purpose, API surface, or behavior changes.",
2157:         "Run the smallest relevant smoke test before merging into development.",
2158:     ]:
2159:         story.append(pdf_paragraph(f"- {item}", styles["SmallBody"]))
2160:
2161:

```

write_single_file_reference_pdf - lines 2162-2190

Item	Explanation
Source location	Lines 2162-2190 (29 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	def write_single_file_reference_pdf(facts: dict, output_path: Path) -> None:
Touches	filesystem
Calls detected	pdf_styles, append, Paragraph, pdf_paragraph, pdf_table, Spacer, add_pdf_file_fact_sections, SimpleDocTemplate, str, build
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2162: def write_single_file_reference_pdf(facts: dict, output_path: Path) -> None:
2163:     if not REPORTLAB_AVAILABLE:
2164:         return
2165:     styles = pdf_styles()
2166:     rows = [
2167:         ("File", facts["repo_file"]),
2168:         ("Category", facts["category"]),
2169:         ("Folder role", facts["folder_role"]),
2170:         ("Size", f"{facts['size_bytes']:,} bytes"),
2171:         ("Extension", facts["suffix"]),
2172:         ("MIME type", facts["mime_type"]),
2173:         ("Text lines", f"{facts['line_count']:,}" if facts["is_text"] else "Not a readable text file"),
2174:     ]
2175:     if facts["image"]:
2176:         image = facts["image"]
2177:         if "error" in image:
2178:             rows.append(("Image metadata", f"Could not inspect image: {image['error']}"))
2179:         else:
2180:             rows.append(("Image metadata", f"{image['format']} {image['width']}x{image['height']}, mode {image['mode']}, frames {image['frames']}"))
2181:     story = [
2182:         Paragraph(f"Important Code Reference: {facts['repo_file']}", styles["Title"]),
2183:         pdf_paragraph("A focused code/control-file explanation covering purpose, owner layer, APIs, variables, implementation signals, source excerpts, and debugging rules.", styles["SmallBody"]),
2184:         pdf_table(rows, styles),
2185:         Spacer(1, 0.12 * inch),
2186:     ]
2187:     add_pdf_file_fact_sections(story, facts, styles)
2188:     SimpleDocTemplate(str(output_path), pagesize=LETTER, rightMargin=0.55 * inch, leftMargin=0.55 * inch, topMargin=0.55 * inch, bottomMargin=0.55 * inch).build(story)
2189:
2190:
```

write_master_file_reference_docx - lines 2191-2215

Item	Explanation
Source location	Lines 2191-2215 (25 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	def write_master_file_reference_docx(file_facts: list[dict], diagrams: dict[str, Path]) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	setup_code_reference_document, add_diagram, Counter, add_heading, add_table, str, len, join, sorted, items, add_page_break, add_file_fact_summary
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2191: def write_master_file_reference_docx(file_facts: list[dict], diagrams: dict[str, Path]) -> None:
2192:     doc = setup_code_reference_document(
2193:         "OMB Portfolio Builder Important Code Reference",
2194:         "A generated Word reference for the code and control files that drive the builder, website, installer, Cloudflare AI worker, workflows, parser, compile workspace, and generated documentation.",
2195:     )
2196:     add_diagram(doc, diagrams["file-communication-map"], "Main file communication map for the builder and public website.")
2197:     category_counts = Counter(facts["category"] for facts in file_facts)
2198:     doc.add_heading("Coverage summary", level=1)
2199:     add_table(doc, [
2200:         ("Important files documented", str(len(file_facts))),
2201:         ("Selection rule", "Only behavior-driving code/control files are included. Favicons, binary assets, and passive marker files are intentionally skipped here."),
2202:         ("Categories", ", ".join(f"{name}: {count}" for name, count in sorted(category_counts.items()))),
2203:     ])
2204:     doc.add_heading("File index", level=1)
2205:     add_table(doc, [(facts["repo_file"], f"{facts['category']}. {facts['folder_role']}") for facts in file_facts])
2206:     doc.add_page_break()
2207:     for facts in file_facts:
2208:         doc.add_heading(f"Important Code Reference: {facts['repo_file']}", level=1)
2209:         add_file_fact_summary(doc, facts)
2210:         add_file_fact_sections(doc, facts)
2211:         doc.add_page_break()
2212:     compact_generated_docx(doc, keep_first_page_break=True)
2213:     doc.save(FILE_REFERENCE_MASTER_DOCX)
```


2214:
2215:

write_master_file_reference_pdf - lines 2216-2256

Item	Explanation
Source location	Lines 2216-2256 (41 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	def write_master_file_reference_pdf(file_facts: list[dict], diagrams: dict[str, Path]) -> None:
Touches	filesystem
Calls detected	pdf_styles, Counter, Paragraph, pdf_paragraph, get, exists, append, PdfImage, str, pdf_table, len, join
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2216: def write_master_file_reference_pdf(file_facts: list[dict], diagrams: dict[str, Path]) -> None:
2217:     if not REPORTLAB_AVAILABLE:
2218:         return
2219:     styles = pdf_styles()
2220:     category_counts = Counter(facts["category"] for facts in file_facts)
2221:     story = [
2222:         Paragraph("OMB Portfolio Builder Important Code Reference", styles["Title"]),
2223:         pdf_paragraph("A generated PDF reference for the code and control files that drive the builder,
website, installer, Cloudflare AI worker, workflows, parser, compile workspace, and generated documentation.",
styles["SmallBody"]),
2224:     ]
2225:     file_map = diagrams.get("file-communication-map")
2226:     if file_map and file_map.exists():
2227:         story.append(PdfImage(str(file_map), width=6.6 * inch, height=3.72 * inch))
2228:         story.append(Paragraph("Coverage summary", styles["Heading1"]))
2229:         story.append(pdf_table([
2230:             ("Important files documented", str(len(file_facts))),
2231:             ("Selection rule", "Only behavior-driving code/control files are included. Favicons, binary
assets, and passive marker files are intentionally skipped here."),
2232:             ("Categories", ", ".join(f"{name}: {count}" for name, count in sorted(category_counts.items()))),
2233:         ], styles))
2234:         story.append(Paragraph("File index", styles["Heading1"]))
2235:         story.append(pdf_table([(facts["repo_file"], f"{facts['category']}. {facts['folder_role']}") for
facts in file_facts], styles))
2236:         story.append(PageBreak())
2237:         for facts in file_facts:
2238:             story.append(Paragraph(f"Important Code Reference: {facts['repo_file']}", styles["Heading1"]))
2239:             rows = [
2240:                 ("File", facts["repo_file"]),
2241:                 ("Category", facts["category"]),
2242:                 ("Folder role", facts["folder_role"]),
2243:                 ("Size", f"{facts['size_bytes']:,} bytes"),
2244:                 ("Extension", facts["suffix"]),
2245:                 ("MIME type", facts["mime_type"]),
2246:                 ("Text lines", f"{facts['line_count']:,}" if facts["is_text"] else "Not a readable text
file"),
2247:             ]
2248:             if facts["image"]:
2249:                 image = facts["image"]
2250:                 rows.append(("Image metadata", f"Could not inspect image: {image['error']}" if "error" in
image else f"{image['format']} {image['width']}x{image['height']}, mode {image['mode']}, frames
{image['frames']}"))
2251:                 story.append(pdf_table(rows, styles))
2252:                 add_pdf_file_fact_sections(story, facts, styles)
2253:                 story.append(PageBreak())
2254:             SimpleDocTemplate(str(FILE_REFERENCE_MASTER_PDF), pagesize=LETTER, rightMargin=0.55 * inch,
leftMargin=0.55 * inch, topMargin=0.55 * inch, bottomMargin=0.55 * inch).build(story)
2255:
2256:
```

write_file_reference_docs - lines 2257-2279

Item	Explanation
Source location	Lines 2257-2279 (23 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.

Item	Explanation
Declaration	def write_file_reference_docs(files: list[str], diagrams: dict[str, Path]) -> list[Path]:
Touches	filesystem
Calls detected	remove_directory, mkdir, extract_file_facts, important_code_files, code_reference_name, write_single_file_reference_docx, append, write_single_file_reference_pdf, exists, write_master_file_reference_docx, write_master_file_reference_pdf
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2257: def write_file_reference_docs(files: list[str], diagrams: dict[str, Path]) -> list[Path]:
2258:     remove_directory(FILE_REFERENCE_DOCX_DIR)
2259:     remove_directory(FILE_REFERENCE_PDF_DIR)
2260:     FILE_REFERENCE_DOCX_DIR.mkdir(parents=True, exist_ok=True)
2261:     FILE_REFERENCE_PDF_DIR.mkdir(parents=True, exist_ok=True)
2262:     file_facts = [extract_file_facts(file) for file in important_code_files(files)]
2263:     written: list[Path] = []
2264:     for facts in file_facts:
2265:         docx_path = FILE_REFERENCE_DOCX_DIR / code_reference_name(facts["repo_file"], ".docx")
2266:         write_single_file_reference_docx(facts, docx_path)
2267:         written.append(docx_path)
2268:         pdf_path = FILE_REFERENCE_PDF_DIR / code_reference_name(facts["repo_file"], ".pdf")
2269:         write_single_file_reference_pdf(facts, pdf_path)
2270:         if pdf_path.exists():
2271:             written.append(pdf_path)
2272:     write_master_file_reference_docx(file_facts, diagrams)
2273:     written.append(FILE_REFERENCE_MASTER_DOCX)
2274:     write_master_file_reference_pdf(file_facts, diagrams)
2275:     if FILE_REFERENCE_MASTER_PDF.exists():
2276:         written.append(FILE_REFERENCE_MASTER_PDF)
2277:     return written
2278:
2279:

```

write_code_reference_docs - lines 2280-2305

Item	Explanation
Source location	Lines 2280-2305 (26 source lines).
Purpose	Persists data to local state, disk, credentials, or browser storage.
Declaration	def write_code_reference_docs(files: list[str], diagrams: dict[str, Path]) -> list[Path]:
Touches	filesystem
Calls detected	remove_directory, mkdir, is_text_code_file, extract_source_facts, code_reference_name, write_single_code_reference_docx, append, write_single_code_reference_pdf, exists, write_master_code_reference_docx, write_master_code_reference_pdf
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2280: def write_code_reference_docs(files: list[str], diagrams: dict[str, Path]) -> list[Path]:
2281:     remove_directory(CODE_REFERENCE_DIR)
2282:     remove_directory(CODE_REFERENCE_DOCX_DIR)
2283:     remove_directory(CODE_REFERENCE_PDF_DIR)
2284:     CODE_REFERENCE_DOCX_DIR.mkdir(parents=True, exist_ok=True)
2285:     CODE_REFERENCE_PDF_DIR.mkdir(parents=True, exist_ok=True)
2286:     code_files = [file for file in files if is_text_code_file(file)]
2287:     facts_by_file = {repo_file: extract_source_facts(repo_file) for repo_file in code_files}
2288:     written: list[Path] = []
2289:     for repo_file in code_files:
2290:         facts = facts_by_file[repo_file]
2291:         docx_path = CODE_REFERENCE_DOCX_DIR / code_reference_name(repo_file, ".docx")
2292:         write_single_code_reference_docx(facts, docx_path)

```

```

2293:         written.append(docx_path)
2294:         pdf_path = CODE_REFERENCE_PDF_DIR / code_reference_name(repo_file, ".pdf")
2295:         write_single_code_reference_pdf(facts, pdf_path)
2296:         if pdf_path.exists():
2297:             written.append(pdf_path)
2298:         write_master_code_reference_docx(code_files, facts_by_file, diagrams)
2299:         written.append(CODE_REFERENCE_MASTER_DOCX)
2300:         write_master_code_reference_pdf(code_files, facts_by_file, diagrams)
2301:         if CODE_REFERENCE_MASTER_PDF.exists():
2302:             written.append(CODE_REFERENCE_MASTER_PDF)
2303:         return written
2304:
2305:

```

add_generated_code_reference - lines 2306-2355

Item	Explanation
Source location	Lines 2306-2355 (50 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_generated_code_reference(doc: Document, files: list[str], diagrams: dict[str, Path]) -> None:
Touches	authentication/security data
Calls detected	add_heading, write_code_reference_docs, add_paragraph, add_table, str, relative_to, exists, len, add_page_break, is_text_code_file, extract_source_facts, fact_relationship_summary
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2306: def add_generated_code_reference(doc: Document, files: list[str], diagrams: dict[str, Path]) -> None:
2307:     doc.add_heading("Generated Word And PDF Code References", level=1)
2308:     written = write_code_reference_docs(files, diagrams)
2309:     doc.add_paragraph("The repository now includes generated Word and PDF code-reference documents rather
than only Markdown notes. These documents are meant for source-level reading beside the actual files. They
explain functions, variables, API endpoints, imports, event handlers, file relationships,
security/authentication cues, compiler calls, installer behavior, and debugging paths.")
2310:     add_table(doc, [
2311:         ("Word folder", str(CODE_REFERENCE_DOCX_DIR.relative_to(REPO))),
2312:         ("PDF folder", str(CODE_REFERENCE_PDF_DIR.relative_to(REPO))),
2313:         ("Master Word reference", str(CODE_REFERENCE_MASTER_DOCX.relative_to(REPO))),
2314:         ("Master PDF reference", str(CODE_REFERENCE_MASTER_PDF.relative_to(REPO))) if
CODE_REFERENCE_MASTER_PDF.exists() else "PDF generation skipped because ReportLab is unavailable."),
2315:         ("Artifacts generated", str(len(written))),
2316:         ("Regeneration command", "python docs/build_complete_guide.py"),
2317:     ])
2318:     doc.add_page_break()
2319:
2320:     for repo_file in [file for file in files if is_text_code_file(file)]:
2321:         facts = extract_source_facts(repo_file)
2322:         doc.add_heading(f"Code Reference: {repo_file}", level=1)
2323:         doc.add_paragraph(facts["description"])
2324:         add_table(doc, [
2325:             ("Lines", f"{facts['line_count']:,}"),
2326:             ("Size", f"{facts['size_bytes']:,} bytes"),
2327:             ("Talks to", fact_relationship_summary(facts)),
2328:             ("Functions discovered", str(len(facts["functions"]))),
2329:             ("Variables discovered", str(len(facts["variables"]))),
2330:             ("API endpoints mentioned", str(len(facts["endpoints"]))),
2331:             ("Implementation signals", str(len(facts["signals"]))),
2332:         ])
2333:         if facts["signals"]:
2334:             doc.add_heading("Signals detected", level=2)
2335:             add_table(doc, [(f"{item['type']}" at line {item['line']}", f"{item['meaning']} Evidence:
{item['evidence']}") for item in facts["signals"][:20]])
2336:             if facts["endpoints"]:
2337:                 doc.add_heading("Endpoints in this file", level=2)
2338:                 add_table(doc, [(item["endpoint"], f"Line {item['line']}.
{endpoint_purpose(item['endpoint'])}") for item in facts["endpoints"][:20]])
2339:             if facts["functions"]:
2340:                 doc.add_heading("Function details", level=2)
2341:                 for item in facts["functions"][:60]:
2342:                     doc.add_heading(f"{item['name']}" (lines {item['line']}-{item['end_line']}", level=3)
2343:                     add_table(doc, function_detail_rows(item))
2344:                     add_code(doc, item["excerpt"])
2345:             if facts["variables"]:
2346:                 doc.add_heading("Variables and constants", level=2)
2347:                 add_table(doc, [(item["name"], f"Line {item['line']}: {item['statement']}") for item in

```

```

facts["variables"][:20]))
2348:         if facts["imports"]:
2349:             doc.add_heading("Imports and dependencies", level=2)
2350:             add_table(doc, [(f"Line {item['line']}", item["statement"]) for item in
facts["imports"][:20]])
2351:             doc.add_heading("Opening snippet", level=2)
2352:             add_code(doc, facts["snippet"] or "(empty file)")
2353:             doc.add_page_break()
2354:
2355:

```

add_generated_file_reference - lines 2356-2379

Item	Explanation
Source location	Lines 2356-2379 (24 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_generated_file_reference(doc: Document, files: list[str], diagrams: dict[str, Path]) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	add_heading, write_file_reference_docs, important_code_files, add_paragraph, add_table, str, relative_to, exists, len, bullets, add_page_break
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2356: def add_generated_file_reference(doc: Document, files: list[str], diagrams: dict[str, Path]) -> None:
2357:     doc.add_heading("Generated Word And PDF Important Code References", level=1)
2358:     written = write_file_reference_docs(files, diagrams)
2359:     selected_files = important_code_files(files)
2360:     doc.add_paragraph("The repository now includes generated file-specific Word and PDF documents for the
important code and control files that explain how the system actually works. This focused set avoids passive
binary assets and marker files so the documentation stays useful for learning the builder internals.")
2361:     add_table(doc, [
2362:         ("Word folder", str(FILE_REFERENCE_DOCX_DIR.relative_to(REPO))),
2363:         ("PDF folder", str(FILE_REFERENCE_PDF_DIR.relative_to(REPO))),
2364:         ("Master Word reference", str(FILE_REFERENCE_MASTER_DOCX.relative_to(REPO))),
2365:         ("Master PDF reference", str(FILE_REFERENCE_MASTER_PDF.relative_to(REPO)) if
FILE_REFERENCE_MASTER_PDF.exists() else "PDF generation skipped because ReportLab is unavailable."),
2366:         ("Important files documented", str(len(selected_files))),
2367:         ("Artifacts generated", str(len(written))),
2368:         ("Skipped here", "Favicons, app icons, binary assets, passive marker files, and generated
reference documents."),
2369:         ("Regeneration command", "python docs/build_complete_guide.py"),
2370:     ])
2371:     doc.add_heading("What this adds beyond the code reference", level=2)
2372:     bullets(doc, [
2373:         "The selected files are the places a developer actually starts when debugging app behavior.",
2374:         "Each selected file gets source excerpts, implementation signals, API/DOM/storage clues, and
safe-change notes.",
2375:         "The master reference acts like a guided map of the app internals rather than a dump of every
asset.",
2376:     ])
2377:     doc.add_page_break()
2378:
2379:

```

add_complete_function_inventory - lines 2380-2405

Item	Explanation
Source location	Lines 2380-2405 (26 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_complete_function_inventory(doc: Document) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	extract_functions, add_heading, add_paragraph, add_table, str, len, add_page_break, setdefault, append, items, range
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2380: def add_complete_function_inventory(doc: Document) -> None:
2381:     rows = extract_functions()
2382:     doc.add_heading("Generated Function Inventory", level=1)
2383:     doc.add_paragraph("This section is generated from the real source files. It does not replace reading
the code, but it gives you a map of where functions live and what their names suggest they own.")
2384:     add_table(doc, [
2385:         ("Files scanned", "main.cjs, server.mjs, template-preview.js, script.js"),
2386:         ("Functions found", str(len(rows))),
2387:         ("How descriptions are made", "Descriptions are heuristic based on file name and function naming
patterns. For exact behavior, inspect the source around the listed line."),
2388:     ])
2389:     doc.add_page_break()
2390:
2391:     by_file: dict[str, list[tuple[str, int, str, str]]] = {}
2392:     for row in rows:
2393:         by_file.setdefault(row[0], []).append(row)
2394:     for file_name, file_rows in by_file.items():
2395:         doc.add_heading(f"Function Inventory: {file_name}", level=1)
2396:         doc.add_paragraph(f"{file_name} contains {len(file_rows)} named functions discovered by the guide
generator.")
2397:         chunk_size = 45
2398:         for index in range(0, len(file_rows), chunk_size):
2399:             chunk = file_rows[index:index + chunk_size]
2400:             add_table(doc, [(f"{name} (line {line_no})", purpose) for _, line_no, name, purpose in
chunk])
2401:             if index + chunk_size < len(file_rows):
2402:                 doc.add_paragraph("Continued on the next table.")
2403:             doc.add_page_break()
2404:
2405:

```

add_deep_detail_topics - lines 2406-2419

Item	Explanation
Source location	Lines 2406-2419 (14 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_deep_detail_topics(doc: Document) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	add_heading, add_paragraph, bullets, add_page_break
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2406: def add_deep_detail_topics(doc: Document) -> None:
2407:     for title, body in DEEP_DETAIL_TOPICS:
2408:         doc.add_heading(f"Deep Detail: {title}", level=1)
2409:         doc.add_paragraph(body)
2410:         doc.add_heading("How to use this detail", level=2)
2411:         bullets(doc, [
2412:             "Start with the user-visible behavior.",
2413:             "Find the file that owns that behavior.",
2414:             "Trace the data handoff to the next file, endpoint, cache, or generated artifact.",
2415:             "Change the smallest responsible piece and test the flow end to end.",
2416:         ])
2417:         doc.add_page_break()
2418:
2419:

```

add_programming_syntax - lines 2420-2429

Item	Explanation
Source location	Lines 2420-2429 (10 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_programming_syntax(doc: Document) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	add_heading, add_code, add_paragraph, add_page_break
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2420: def add_programming_syntax(doc: Document) -> None:
2421:     for title, snippet, explanation in PROGRAMMING_SYNTAX_LESSONS:
2422:         doc.add_heading(f"Programming Syntax: {title}", level=1)
2423:         add_code(doc, snippet)
2424:         doc.add_paragraph(explanation)
2425:         doc.add_heading("How this appears in the project", level=2)
2426:         doc.add_paragraph("You will see this pattern in main.cjs, server.mjs, template-preview.js,
script.js, Cloudflare Worker code, HTML files, CSS files, JSON catalogs, and GitHub workflow YAML.")
2427:         doc.add_page_break()
2428:
2429:

```

add_shell_syntax - lines 2430-2439

Item	Explanation
Source location	Lines 2430-2439 (10 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_shell_syntax(doc: Document) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	add_heading, add_code, add_paragraph, add_page_break
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2430: def add_shell_syntax(doc: Document) -> None:
2431:     for title, snippet, explanation in SHELL_SYNTAX_LESSONS:
2432:         doc.add_heading(f"Shell Or Script Syntax: {title}", level=1)
2433:         add_code(doc, snippet)
2434:         doc.add_paragraph(explanation)
2435:         doc.add_heading("Why this matters", level=2)
2436:         doc.add_paragraph("Shell scripts glue tools together. They are not the app UI, but they install
tools, build releases, deploy workers, scan directories, and enforce rules.")
2437:         doc.add_page_break()
2438:
2439:

```

add_caching_methods - lines 2440-2464

Item	Explanation
Source location	Lines 2440-2464 (25 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_caching_methods(doc: Document, diagrams: dict[str, Path]) -> None:

Item	Explanation
Touches	filesystem, authentication/security data
Calls detected	add_heading, add_diagram, add_paragraph, add_table, add_page_break
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2440: def add_caching_methods(doc: Document, diagrams: dict[str, Path]) -> None:
2441:     doc.add_heading("Caching Methods Used In The Builder And Website", level=1)
2442:     add_diagram(doc, diagrams["compile-code-flow"], "Compile artifact caching happens inside the Compile
Code workspace.")
2443:     doc.add_paragraph("Caching means remembering a result so the app does not repeat slow or annoying
work. This project uses several different caches, each with a different safety boundary.")
2444:     add_table(doc, [(name, f"{where} Purpose: {purpose}") for name, where, purpose in CACHING_METHODS])
2445:     doc.add_heading("Important safety rule", level=2)
2446:     doc.add_paragraph("A cache must be scoped. Publishing authorization is scoped to the same repository,
branch, user/machine scope, and expiration time. Compiler caches are scoped to source code, language, compiler
path, runtime path, and file name. UI caches like update snooze are local convenience data and do not grant
publishing access.")
2447:     doc.add_page_break()
2448:     for name, where, purpose in CACHING_METHODS:
2449:         doc.add_heading(f"Cache Detail: {name}", level=1)
2450:         if "Compiler" in name or "Compile" in name or "Terminal" in name or "source" in name:
2451:             add_diagram(doc, diagrams["compile-code-flow"], f"Context diagram for {name}.")
2452:         elif "Publishing" in name or "Extended trust" in name:
2453:             add_diagram(doc, diagrams["release-pipeline"], f"Publishing authorization cache supports this
release/publish workflow.")
2454:         elif "HTTP" in name:
2455:             add_diagram(doc, diagrams["frontend-backend-cloudflare"], f"HTTP no-store protects request
freshness across frontend/backend paths.")
2456:             add_table(doc, [
2457:                 ("Where implemented", where),
2458:                 ("Why it exists", purpose),
2459:                 ("What can go stale", "Cached data can become outdated when files, tools, versions,
repositories, or user choices change."),
2460:                 ("How to refresh", "Use the explicit refresh/rebuild/authenticate action for that feature,
clear the relevant local data, or restart the builder when appropriate."),
2461:             ])
2462:             doc.add_page_break()
2463:
2464:

```

add_installer_details - lines 2465-2500

Item	Explanation
Source location	Lines 2465-2500 (36 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_installer_details(doc: Document, diagrams: dict[str, Path]) -> None:
Touches	filesystem
Calls detected	add_heading, add_diagram, add_paragraph, add_table, numbers, add_page_break
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2465: def add_installer_details(doc: Document, diagrams: dict[str, Path]) -> None:
2466:     doc.add_heading("How The Installer Works", level=1)
2467:     add_diagram(doc, diagrams["release-pipeline"], "Installer and update behavior sits at the end of the
release pipeline.")
2468:     doc.add_paragraph("The installer is produced by electron-builder and customized by NSIS script code
in build/installer.nsh. electron-builder gathers app files and creates the installer shell. NSIS controls
Windows setup pages, duplicate install checks, shortcut behavior, Git installation checks, update behavior, and
uninstall pages.")
2469:     add_table(doc, [

```

```

2470:         ("electron-builder", "Reads package.json build settings and packages the Electron app."),
2471:         ("NSIS", "Runs installer wizard logic and custom script functions."),
2472:         ("installer.nsh", "Adds OMB-specific setup pages, duplicate detection, update handling,
shortcuts, and Git checks."),
2473:         ("AppData install", "Default per-user install location avoids Program Files admin prompts."),
2474:         ("Update install", "Existing installation is updated in place instead of creating a duplicate
copy."),
2475:     ])
2476:     doc.add_heading("Installer flow", level=2)
2477:     numbers(doc, [
2478:         "customInit runs first.",
2479:         "The installer checks for an existing registered install.",
2480:         "If a current or newer install exists, setup stops.",
2481:         "If an older install exists, setup treats the run as an update and uses the existing location.",
2482:         "If no install exists, setup scans for duplicate builder copies.",
2483:         "The custom tools page asks about desktop shortcut and publishing tools.",
2484:         "Files are installed into the chosen per-user location.",
2485:         "customInstall creates/removes shortcuts and optionally installs Git for Windows.",
2486:         "The uninstaller has its own page explaining what will and will not be removed.",
2487:     ])
2488:     doc.add_page_break()
2489:     for name, description in INSTALLER_FUNCTIONS:
2490:         doc.add_heading(f"Installer Function: {name}", level=1)
2491:         doc.add_paragraph(description)
2492:         add_table(doc, [
2493:             ("File", "build/installer.nsh"),
2494:             ("Syntax family", "NSIS installer scripting"),
2495:             ("Function job", description),
2496:             ("Beginner note", "Installer functions run during setup/uninstall, not while the normal
builder UI is being used."),
2497:         ])
2498:         doc.add_page_break()
2499:
2500:

```

add_function_maps - lines 2501-2528

Item	Explanation
Source location	Lines 2501-2528 (28 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_function_maps(doc: Document, diagrams: dict[str, Path]) -> None:
Touches	filesystem
Calls detected	add_heading, add_diagram, add_paragraph, add_table, add_page_break
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2501: def add_function_maps(doc: Document, diagrams: dict[str, Path]) -> None:
2502:     for file_name, group_purpose, functions in FUNCTION_GROUPS:
2503:         doc.add_heading(f"Function Map: {file_name}", level=1)
2504:         if file_name == "main.cjs":
2505:             add_diagram(doc, diagrams["electron-runtime"], "Electron startup functions sit inside this
runtime flow.")
2506:         elif file_name == "server.mjs":
2507:             add_diagram(doc, diagrams["frontend-backend-cloudflare"], "server.mjs is the local backend
layer in this map.")
2508:         elif file_name == "template-preview.js":
2509:             add_diagram(doc, diagrams["builder-to-site-files"], "template-preview.js is the builder
frontend that sends edits toward parser output.")
2510:         elif file_name == "script.js":
2511:             add_diagram(doc, diagrams["cloudflare-ai-flow"], "script.js owns the public website behavior,
including the AI chat request path.")
2512:             doc.add_paragraph(group_purpose)
2513:             add_table(doc, [(name, description) for name, description in functions])
2514:             doc.add_heading("How to use this map", level=2)
2515:             doc.add_paragraph("When debugging, find the user action first, then find the function group that
owns it. For example, a compile terminal issue starts in template-preview.js, then calls server.mjs, then
returns output back to template-preview.js.")
2516:             doc.add_page_break()
2517:             for name, description in functions:
2518:                 doc.add_heading(f"Function Detail: {file_name} -> {name}", level=1)
2519:                 doc.add_paragraph(description)
2520:                 add_table(doc, [
2521:                     ("File", file_name),
2522:                     ("Function", name),

```



```

2523:         ("Responsibility", description),
2524:         ("Debug question", "What input does this function receive, what output or state change
does it produce, and what function calls it next?"),
2525:     ])
2526:     doc.add_page_break()
2527:
2528:

```

add_architecture - lines 2529-2548

Item	Explanation
Source location	Lines 2529-2548 (20 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_architecture(doc: Document, diagrams: dict[str, Path]) -> None:
Touches	authentication/security data
Calls detected	add_heading, add_paragraph, get, add_diagram, add_code, bullets, add_page_break
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2529: def add_architecture(doc: Document, diagrams: dict[str, Path]) -> None:
2530:     for title, body, diagram in ARCHITECTURE_PAGES:
2531:         doc.add_heading(title, level=1)
2532:         doc.add_paragraph(body)
2533:         diagram_name = ARCHITECTURE_DIAGRAM_BY_TITLE.get(title)
2534:         if diagram_name:
2535:             add_diagram(doc, diagrams[diagram_name], f"Generated block diagram: {title}.")
2536:             doc.add_heading("Text version of the block diagram", level=2)
2537:             add_code(doc, diagram)
2538:             doc.add_heading("Beginner explanation", level=2)
2539:             doc.add_paragraph("Read each arrow as 'this thing talks to the next thing.' The important rule is
that editing happens locally in the builder, while viewing happens publicly on the website after generated
static files are published.")
2540:             doc.add_heading("What can break", level=2)
2541:             bullets(doc, [
2542:                 "If local builder state is not saved, the parser may not see the latest edits.",
2543:                 "If Git authentication or branch state is wrong, Apply to site cannot safely push.",
2544:                 "If public hosting cache is stale, a phone may show an older copy until refresh or cache
expiry.",
2545:             ])
2546:             doc.add_page_break()
2547:
2548:

```

add_file_communication - lines 2549-2571

Item	Explanation
Source location	Lines 2549-2571 (23 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_file_communication(doc: Document, diagrams: dict[str, Path]) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	add_heading, add_diagram, add_paragraph, add_table, bullets, add_page_break
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2549: def add_file_communication(doc: Document, diagrams: dict[str, Path]) -> None:
2550:     doc.add_heading("How The Important Files Communicate", level=1)

```

```

2551:     add_diagram(doc, diagrams["file-communication-map"], "Generated map of the main builder files and
public website files.")
2552:     doc.add_paragraph("This is the file-level mental model. The desktop app starts in main.cjs, the
backend lives in server.mjs, the builder screen is template-preview.html with template-preview.css and template-
preview.js, and the public website is index.html with styles.css, script.js, projects.json, and assets.")
2553:     add_table(doc, [
2554:         ("main.cjs -> server.mjs", "Electron starts the backend so the builder UI can call local APIs."),
2555:         ("main.cjs -> template-preview.html", "Electron opens the builder window and loads the builder
HTML file."),
2556:         ("template-preview.html -> template-preview.js", "The builder HTML loads the JavaScript that
handles editing, saving, previewing, compile code, and publishing UI."),
2557:         ("template-preview.js -> server.mjs", "The builder uses fetch requests to ask the backend to save
files, parse projects, run Git, run compilers, or publish."),
2558:         ("server.mjs -> projects.json", "The parser writes public project data used by the website."),
2559:         ("index.html -> script.js/styles.css", "The public page loads its behavior and visual design."),
2560:         ("script.js -> projects.json", "The public website reads the generated catalog to render project
windows."),
2561:     ])
2562:     doc.add_heading("Why this communication shape is useful", level=2)
2563:     bullets(doc, [
2564:         "The public website can stay static and fast.",
2565:         "The builder can safely use local-only features without exposing them online.",
2566:         "The parser creates a clean boundary between editable state and visitor-facing output.",
2567:         "The same project data can be previewed locally before it is published.",
2568:     ])
2569:     doc.add_page_break()
2570:
2571:

```

add_tool_build_map - lines 2572-2581

Item	Explanation
Source location	Lines 2572-2581 (10 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_tool_build_map(doc: Document, diagrams: dict[str, Path]) -> None:
Touches	authentication/security data
Calls detected	add_heading, add_diagram, add_paragraph, add_table, add_page_break
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2572: def add_tool_build_map(doc: Document, diagrams: dict[str, Path]) -> None:
2573:     doc.add_heading("What Tool Builds What", level=1)
2574:     add_diagram(doc, diagrams["tool-build-map"], "Generated map showing which tool owns each part of the
build and release process.")
2575:     doc.add_paragraph("A common source of confusion is thinking one tool builds everything. In this
project, each tool has a narrow job. npm installs dependencies, Electron runs the app, electron-builder
packages the app, NSIS makes the installer, Git moves files, GitHub Actions automates release builds, Wrangler
deploys Cloudflare Worker code, and compilers run project source code.")
2576:     add_table(doc, [(tool, f"{builds} Used for: {where}") for tool, builds, where in TOOL_BUILD_ROWS])
2577:     doc.add_heading("Beginner shortcut", level=2)
2578:     doc.add_paragraph("When something fails, ask which tool owns that step. Installer failure points
toward NSIS or electron-builder. AI backend failure points toward Cloudflare Worker or Wrangler. Website publish
failure points toward Git, GitHub, branch state, or authentication.")
2579:     doc.add_page_break()
2580:
2581:

```

add_software_decisions - lines 2582-2600

Item	Explanation
Source location	Lines 2582-2600 (19 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_software_decisions(doc: Document) -> None:
Touches	authentication/security data
Calls detected	add_heading, add_paragraph, add_table, bullets, add_page_break

Item	Explanation
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2582: def add_software_decisions(doc: Document) -> None:
2583:     for title, body in SOFTWARE_DECISIONS:
2584:         doc.add_heading(f"Software Engineering Decision: {title}", level=1)
2585:         doc.add_paragraph(body)
2586:         add_table(doc, [
2587:             ("Decision", title),
2588:             ("Reason", body),
2589:             ("What it protects", "Predictable publishing, safer public/private separation, easier
updates, clearer debugging, or better user experience."),
2590:             ("Tradeoff", "The design may add more files or moving parts, but each moving part has a
defined job."),
2591:         ])
2592:         doc.add_heading("How to evaluate this later", level=2)
2593:         bullets(doc, [
2594:             "If the feature becomes hard to maintain, check whether responsibilities are mixed across too
many files.",
2595:             "If users are confused, improve the builder UI or guide rather than exposing implementation
details.",
2596:             "If publishing becomes risky, tighten the parser and authentication boundary first.",
2597:         ])
2598:         doc.add_page_break()
2599:
2600:

```

add_system_layers - lines 2601-2616

Item	Explanation
Source location	Lines 2601-2616 (16 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_system_layers(doc: Document, diagrams: dict[str, Path]) -> None:
Touches	filesystem, authentication/security data
Calls detected	add_heading, add_diagram, add_paragraph, add_page_break, add_table, add_code
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2601: def add_system_layers(doc: Document, diagrams: dict[str, Path]) -> None:
2602:     doc.add_heading("AI, Frontend, Backend, Cloudflare, And Secrets", level=1)
2603:     add_diagram(doc, diagrams["frontend-backend-cloudflare"], "Generated map of the frontend, backend,
Cloudflare Worker, AI provider, and secret boundary.")
2604:     doc.add_paragraph("This section separates the major layers so you know where to look when something
breaks. The frontend draws and captures input. The backend performs protected work. Cloudflare hosts the public
AI endpoint. Secrets stay server-side.")
2605:     doc.add_page_break()
2606:     for title, body, rows, snippet in SYSTEM_LAYER_SECTIONS:
2607:         doc.add_heading(title, level=1)
2608:         doc.add_paragraph(body)
2609:         add_table(doc, rows)
2610:         doc.add_heading("Representative code or command", level=2)
2611:         add_code(doc, snippet)
2612:         doc.add_heading("Debug question", level=2)
2613:         doc.add_paragraph("Ask which layer owns the problem: did the UI send the request, did the backend
receive it, did the service have the needed secret or tool, and did the response come back in the format the UI
expects?")
2614:         doc.add_page_break()
2615:
2616:

```

add_workflows - lines 2617-2650

Item	Explanation
Source location	Lines 2617-2650 (34 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_workflows(doc: Document, files: list[str], diagrams: dict[str, Path]) -> None:
Touches	filesystem, authentication/security data
Calls detected	add_heading, add_diagram, numbers, add_paragraph, add_page_break, startswith, read_text, get, add_code, join, splitlines
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```
2617: def add_workflows(doc: Document, files: list[str], diagrams: dict[str, Path]) -> None:
2618:     workflows = [
2619:         ("Content Creation Workflow", ["Open the builder.", "Edit profile, front-page text, portfolio
areas, or projects.", "Save project or Save all sections.", "Open preview and inspect the exact public layout.",
"Save draft.", "Apply to site after the target is authenticated."]),
2620:         ("Project Creation Workflow", ["Click Add project.", "Choose category.", "Choose appearance.",
"Enter title.", "Edit overview and sections.", "Add files, images, links, code blocks, and compile-code
evidence.", "Save project so the parser can build the project preview."]),
2621:         ("Publishing Workflow", ["Enter repository target.", "Authenticate GitHub write access.", "Load
from target if the target already has content.", "Save draft.", "Apply to site.", "Git commits and pushes
generated files.", "GitHub Pages or Cloudflare serves the updated website."]),
2622:         ("Update Workflow", ["The installed app checks GitHub Releases.", "If a newer version exists, the
app shows an update dialog.", "The updater downloads the installer.", "The app closes itself.", "The installer
updates the existing AppData install.", "The app relaunches after the installer finishes."]),
2623:         ("Compile Code Workflow", ["Create or import a source file.", "Pick the language.", "For
Verilog/SystemVerilog, mark design files as Design and create or import a Testbench file.", "Keep $dumpfile and
$dumpvars in the HDL testbench so the signal scope can draw waveforms.", "Save source.", "Run beautifier if
desired.", "Compile or run.", "Read terminal output, messages log, and HDL scope when available.", "Append the
source code to a project section when ready."]),
2624:     ]
2625:     for title, steps in workflows:
2626:         doc.add_heading(title, level=1)
2627:         if title in {"Content Creation Workflow", "Project Creation Workflow", "Publishing Workflow"}:
2628:             add_diagram(doc, diagrams["builder-to-site-files"], f"Generated flow diagram supporting
{title}.")
2629:             if title == "Update Workflow":
2630:                 add_diagram(doc, diagrams["release-pipeline"], "Generated release/update pipeline diagram.")
2631:             if title == "Compile Code Workflow":
2632:                 add_diagram(doc, diagrams["compile-code-flow"], "Generated Compile Code workflow diagram.")
2633:             numbers(doc, steps)
2634:             doc.add_heading("Plain-English mental model", level=2)
2635:             doc.add_paragraph("A workflow is just a repeatable recipe. The builder turns complicated actions
into visible buttons so you do not need to remember shell commands every time.")
2636:             doc.add_page_break()
2637:
2638:             for workflow in [f for f in files if f.startswith(".github/workflows/")]:
2639:                 text = (REPO / workflow).read_text(encoding="utf-8", errors="replace")
2640:                 doc.add_heading(f"GitHub Workflow File: {workflow}", level=1)
2641:                 doc.add_paragraph(FILE_DESCRIPTIONS.get(workflow, "GitHub Actions workflow file."))
2642:                 doc.add_heading("What a GitHub workflow is", level=2)
2643:                 doc.add_paragraph("A workflow is an automated script GitHub runs in the cloud. Instead of you
manually building installers or enforcing branch rules, GitHub reads this YAML file and executes each job on a
temporary machine.")
2644:                 doc.add_heading("Important YAML excerpt", level=2)
2645:                 add_code(doc, "\n".join(text.splitlines()[1:80]))
2646:                 doc.add_heading("Tradeoffs", level=2)
2647:                 doc.add_paragraph("Automated workflows reduce mistakes, but a bad workflow can block good merges.
That is why the main branch gate should be strict but understandable.")
2648:                 doc.add_page_break()
2649:
2650:
```

file_type - lines 2651-2658

Item	Explanation
Source location	Lines 2651-2658 (8 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.

Item	Explanation
Declaration	def file_type(path: Path, repo_path: str) -> str:
Touches	Mostly local calculations or local UI state.
Calls detected	endswith, lower, lstrip
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2651: def file_type(path: Path, repo_path: str) -> str:
2652:     if repo_path.endswith(".gitkeep"):
2653:         return "folder marker"
2654:     if path.suffix:
2655:         return path.suffix.lower().lstrip(".")
2656:     return "no extension"
2657:
2658:

```

add_files - lines 2659-2688

Item	Explanation
Source location	Lines 2659-2688 (30 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_files(doc: Document, files: list[str]) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	add_heading, add_paragraph, get, add_table, file_type, stat, exists, is_file, lower, read_text, splitlines, add_code
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2659: def add_files(doc: Document, files: list[str]) -> None:
2660:     text_exts = {".js", ".mjs", ".cjs", ".html", ".css", ".json", ".md", ".txt", ".yaml", ".yml",
".toml", ".nsh", ""}
2661:     for repo_file in files:
2662:         path = REPO / repo_file
2663:         doc.add_heading(f"File Explained: {repo_file}", level=1)
2664:         doc.add_paragraph(FILE_DESCRIPTIONS.get(repo_file, "Tracked repository file used by the builder
or public website.))
2665:         add_table(doc, [
2666:             ("Type", file_type(path, repo_file)),
2667:             ("Size", f"{path.stat().st_size:,} bytes" if path.exists() else "missing locally"),
2668:             ("Used by", "Builder app, public site, installer, Cloudflare, or GitHub workflow depending on
the path."),
2669:             ("Beginner idea", "A repository is a folder Git tracks. This file is one object Git can
version, review, and publish."),
2670:         ])
2671:         if path.exists() and path.is_file() and path.suffix.lower() in text_exts:
2672:             lines = path.read_text(encoding="utf-8", errors="replace").splitlines()
2673:             doc.add_heading("Representative snippet", level=2)
2674:             add_code(doc, "\n".join(lines[:24]) if lines else "(empty file)")
2675:             if len(lines) > 24:
2676:                 doc.add_paragraph(f"Only the first 24 lines are shown here. The full file has
{len(lines):,} lines and should be opened in the repository when editing.")
2677:             else:
2678:                 doc.add_heading("Binary or asset note", level=2)
2679:                 doc.add_paragraph("This is treated as an asset. The guide explains its purpose, but the raw
binary content is not printed because raw image/icon bytes would not be useful to read in Word.")
2680:                 doc.add_heading("Safe editing rule", level=2)
2681:                 bullets(doc, [
2682:                     "Make changes on a feature branch from development.",
2683:                     "Preview or test the behavior before merging.",
2684:                     "Do not edit generated/public files by hand if the builder parser is supposed to generate

```

```

them.",
2685:         ])
2686:         doc.add_page_break()
2687:
2688:

```

add_commands - lines 2689-2703

Item	Explanation
Source location	Lines 2689-2703 (15 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_commands(doc: Document) -> None:
Touches	filesystem, authentication/security data
Calls detected	add_heading, add_code, add_table, add_paragraph, add_page_break
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2689: def add_commands(doc: Document) -> None:
2690:     for command, explanation, why in COMMAND_LESSONS:
2691:         doc.add_heading(f"Command Explained: {command}", level=1)
2692:         add_code(doc, command)
2693:         add_table(doc, [
2694:             ("What it does", explanation),
2695:             ("Why it matters", why),
2696:             ("Where used", "Local development, GitHub Actions, builder publishing, installer testing, or
compiler execution."),
2697:             ("Beginner warning", "Run commands from the correct folder. In this project, most development
commands belong in the builder repository root."),
2698:         ])
2699:         doc.add_heading("How to read command output", level=2)
2700:         doc.add_paragraph("Command output is the program talking back to you. Success output usually
names what happened. Error output usually names a missing file, missing tool, denied permission, wrong branch,
or failed compile step.")
2701:         doc.add_page_break()
2702:
2703:

```

add_features - lines 2704-2726

Item	Explanation
Source location	Lines 2704-2726 (23 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_features(doc: Document, diagrams: dict[str, Path]) -> None:
Touches	filesystem, authentication/security data
Calls detected	add_heading, add_diagram, add_paragraph, lower, add_table, add_page_break
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2704: def add_features(doc: Document, diagrams: dict[str, Path]) -> None:
2705:     for topic in FEATURE_TOPICS:
2706:         doc.add_heading(f"Builder Feature: {topic}", level=1)
2707:         if topic in {"Compile Code workspace", "HDL testbench requirement", "HDL signal scope", "Terminal
output", "Messages log", "Append code to project"}:
2708:             add_diagram(doc, diagrams["compile-code-flow"], f"Context diagram for {topic}.")
2709:         elif topic in {"AI chat", "Cloudflare AI worker"}:
2710:             add_diagram(doc, diagrams["cloudflare-ai-flow"], f"Context diagram for {topic}.")

```

```

2711:         elif topic in {"Installer workflow", "Updater workflow", "Release tags", "Branch protection"}:
2712:             add_diagram(doc, diagrams["release-pipeline"], f"Context diagram for {topic}.")
2713:         elif topic in {"Project preview", "Full portfolio preview", "Project parser", "Asset
synchronization"}:
2714:             add_diagram(doc, diagrams["builder-to-site-files"], f"Context diagram for {topic}.")
2715:             doc.add_paragraph(f"This page explains the {topic.lower()} area of the OMB builder and website
system.")
2716:             add_table(doc, [
2717:                 ("Owner view", "The builder exposes controls for creating, editing, saving, previewing, or
publishing this area."),
2718:                 ("Visitor view", "The public website shows only parsed, approved output. Editing controls
stay private."),
2719:                 ("Parser role", "The parser converts local builder state into website-safe data and
markup."),
2720:                 ("Risk", "If this area is not saved before preview or publishing, the public site may show an
older version."),
2721:             ])
2722:             doc.add_heading("Plain-English example", level=2)
2723:             doc.add_paragraph("Think of the builder as the kitchen and the public website as the plate served
to visitors. The parser is the person plating the content neatly. Visitors should never see the prep
controls.")
2724:             doc.add_page_break()
2725:
2726:

```

add_tradeoffs - lines 2727-2738

Item	Explanation
Source location	Lines 2727-2738 (12 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_tradeoffs(doc: Document) -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	add_heading, add_paragraph, add_table, add_page_break
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2727: def add_tradeoffs(doc: Document) -> None:
2728:     for title, body in TRADEOFFS:
2729:         doc.add_heading(f"Tradeoff: {title}", level=1)
2730:         doc.add_paragraph(body)
2731:         add_table(doc, [
2732:             ("Benefit", "The chosen direction fits the current builder goal: owner-controlled, local-
first, preview-before-publish portfolio creation."),
2733:             ("Cost", "Every architecture choice gives up something. Usually the cost is complexity, setup
work, or less flexibility in another direction."),
2734:             ("Decision rule", "Prefer the option that keeps publishing safe, previews accurate, and
editing understandable."),
2735:         ])
2736:         doc.add_page_break()
2737:
2738:

```

add_troubleshooting - lines 2739-2751

Item	Explanation
Source location	Lines 2739-2751 (13 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_troubleshooting(doc: Document) -> None:
Touches	filesystem
Calls detected	add_heading, add_paragraph, numbers, add_page_break
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.

Item	Explanation
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2739: def add_troubleshooting(doc: Document) -> None:
2740:     for title, body in TROUBLESHOOTING_TOPICS:
2741:         doc.add_heading(f"Troubleshooting: {title}", level=1)
2742:         doc.add_paragraph(body)
2743:         numbers(doc, [
2744:             "Read the exact error message before changing anything.",
2745:             "Confirm whether the issue is local builder state, Git state, installer state, public website
cache, or backend service availability.",
2746:             "Fix the smallest proven cause first.",
2747:             "Save a clean checkpoint after the fix works.",
2748:         ])
2749:         doc.add_page_break()
2750:
2751:

```

add_concepts - lines 2752-2765

Item	Explanation
Source location	Lines 2752-2765 (14 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_concepts(doc: Document) -> None:
Touches	filesystem
Calls detected	add_heading, add_paragraph, add_table, add_page_break
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2752: def add_concepts(doc: Document) -> None:
2753:     for concept in CONCEPTS:
2754:         doc.add_heading(f"Beginner Concept: {concept}", level=1)
2755:         doc.add_paragraph(f"{concept} is one of the building blocks behind the OMB builder and portfolio
website. You do not need to memorize every term at once. Learn the role it plays, then come back when you see it
in an error message or file name.")
2756:         add_table(doc, [
2757:             ("Simple meaning", "This concept helps explain how the app, website, GitHub workflow,
installer, or backend behaves."),
2758:             ("Where it appears", "Look for it in README instructions, source files, GitHub Actions logs,
installer behavior, browser behavior, or terminal output."),
2759:             ("How to debug it", "Ask: what is supposed to happen, what actually happened, what file or
service owns that behavior, and what changed most recently?"),
2760:         ])
2761:         doc.add_heading("Tiny mental model", level=2)
2762:         doc.add_paragraph("When confused, draw boxes. Put the user on the left, the thing they click in
the middle, and the file or service that responds on the right. Most software problems become easier once the
boxes are visible.")
2763:         doc.add_page_break()
2764:
2765:

```

add_closing - lines 2766-2780

Item	Explanation
Source location	Lines 2766-2780 (15 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def add_closing(doc: Document) -> None:
Touches	filesystem

Item	Explanation
Calls detected	add_heading, numbers, add_note
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2766: def add_closing(doc: Document) -> None:
2767:     doc.add_heading("Final Operating Checklist", level=1)
2768:     numbers(doc, [
2769:         "Make feature changes on a feature or codex branch from development.",
2770:         "Update README and the in-app guide when behavior changes.",
2771:         "Run local syntax checks and useful smoke tests.",
2772:         "Merge feature work into development.",
2773:         "Only merge development into main for release-ready builds.",
2774:         "Bump package.json before a new main release.",
2775:         "Confirm GitHub Actions publish a fresh installer and stable latest asset.",
2776:         "Open the builder, check update behavior, and verify the public website link downloads the latest
installer.",
2777:     ])
2778:     add_note(doc, "The core rule", "The builder is the source of truth for creating content. The preview
is the truth test before publishing. GitHub is the delivery path. The public website is the final read-only
result.")
2779:
2780:

```

main - lines 2781-2819

Item	Explanation
Source location	Lines 2781-2819 (39 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	def main() -> None:
Touches	Mostly local calculations or local UI state.
Calls detected	tracked_files, load_package, make_diagrams, setup_document, add_cover, add_reading_map, add_entry_points, add_architecture, add_flow_walkthroughs, add_file_communication, add_tool_build_map, add_software_decisions
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

2781: def main() -> None:
2782:     files = tracked_files()
2783:     package = load_package()
2784:     diagrams = make_diagrams()
2785:     doc = setup_document()
2786:     add_cover(doc, package)
2787:     add_reading_map(doc)
2788:     add_entry_points(doc)
2789:     add_architecture(doc, diagrams)
2790:     add_flow_walkthroughs(doc, diagrams)
2791:     add_file_communication(doc, diagrams)
2792:     add_tool_build_map(doc, diagrams)
2793:     add_software_decisions(doc)
2794:     add_system_layers(doc, diagrams)
2795:     add_data_contracts(doc)
2796:     add_api_endpoint_catalog(doc)
2797:     add_programming_syntax(doc)
2798:     add_shell_syntax(doc)
2799:     add_caching_methods(doc, diagrams)
2800:     add_installer_details(doc, diagrams)
2801:     add_function_maps(doc, diagrams)
2802:     add_complete_function_inventory(doc)
2803:     add_generated_code_reference(doc, files, diagrams)
2804:     add_generated_file_reference(doc, files, diagrams)
2805:     add_deep_detail_topics(doc)
2806:     add_workflows(doc, files, diagrams)

```

```

2807:     add_files(doc, files)
2808:     add_commands(doc)
2809:     add_features(doc, diagrams)
2810:     add_tradeoffs(doc)
2811:     add_troubleshooting(doc)
2812:     add_concepts(doc)
2813:     add_closing(doc)
2814:     removed_breaks = compact_generated_docx(doc, keep_first_page_break=True)
2815:     print(f"Compacted generated DOCX layout by removing {removed_breaks} page breaks.")
2816:     doc.save(OUTPUT)
2817:     print(OUTPUT)
2818:
2819:

```

Representative opening snippet

```

1: from __future__ import annotations
2:
3: import html
4: import json
5: import math
6: import mimetypes
7: import re
8: import shutil
9: import subprocess
10: import textwrap
11: from collections import Counter
12: from pathlib import Path
13:
14: from PIL import Image, ImageDraw, ImageFont
15: from docx import Document
16: from docx.enum.table import WD_CELL_VERTICAL_ALIGNMENT, WD_TABLE_ALIGNMENT
17: from docx.enum.text import WD_ALIGN_PARAGRAPH
18: from docx.oxml import OxmlElement
19: from docx.oxml.ns import qn
20: from docx.shared import Inches, Pt, RGBColor
21:
22: try:
23:     from reportlab.lib import colors
24:     from reportlab.lib.pagesizes import LETTER
25:     from reportlab.lib.styles import ParagraphStyle, getSampleStyleSheet
26:     from reportlab.lib.units import inch
27:     from reportlab.platypus import Image as PdfImage
28:     from reportlab.platypus import PageBreak, Paragraph, Preformatted, SimpleDocTemplate, Spacer, Table,
TableStyle
29:
30:     REPORTLAB_AVAILABLE = True
31: except ImportError:
32:     REPORTLAB_AVAILABLE = False
33:
34:
35: REPO = Path(__file__).resolve().parents[1]
36: OUTPUT = REPO / "docs" / "OMB_Portfolio_Builder_Complete_Guide.docx"
37: DIAGRAM_DIR = REPO / "docs" / "guide-diagrams"
38: CODE_REFERENCE_DIR = REPO / "docs" / "code-reference"
39: CODE_REFERENCE_DOCX_DIR = REPO / "docs" / "code-reference-docx"
40: CODE_REFERENCE_PDF_DIR = REPO / "docs" / "code-reference-pdf"
41: CODE_REFERENCE_MASTER_DOCX = REPO / "docs" / "OMB_Portfolio_Builder_Code_Reference.docx"
42: CODE_REFERENCE_MASTER_PDF = REPO / "docs" / "OMB_Portfolio_Builder_Code_Reference.pdf"
43: FILE_REFERENCE_DOCX_DIR = REPO / "docs" / "important-code-reference-docx"
44: FILE_REFERENCE_PDF_DIR = REPO / "docs" / "important-code-reference-pdf"
45: FILE_REFERENCE_MASTER_DOCX = REPO / "docs" / "OMB_Portfolio_Builder_Important_Code_Reference.docx"
46: FILE_REFERENCE_MASTER_PDF = REPO / "docs" / "OMB_Portfolio_Builder_Important_Code_Reference.pdf"
47:
48:
49: FILE_DESCRIPTIONS = {
50:     ".github/workflows/build-windows-builder.yml": "GitHub Actions workflow that builds the Windows
installer and portable application, checks the stable download link, enforces release version bumps, uploads
artifacts, and publishes GitHub Releases.",
51:     ".github/workflows/main-branch-gate.yml": "GitHub Actions guard that rejects direct pushes to main
and rejects pull requests into main unless the source branch is development.",
52:     ".gitignore": "Git ignore rules for build outputs, temporary files, and local-only data that should
not be committed.",
53:     ".nojekyll": "Tells GitHub Pages not to process the site with Jekyll, so folders and static files
publish exactly as generated.",
54:     ".well-known/security.txt": "Public security contact file used by browsers, researchers, and scanners
to find responsible disclosure information.",
55:     "BRANCHING.md": "Human-readable branch model explaining development, main, feature branches, and
release flow.",
56:     "Backgrounds/.gitkeep": "Keeps the Backgrounds folder in Git even when no background image files are
present.",
57:     "README.md": "Main README for the standalone builder application: install, updates, workspaces, text
editing, publishing, build commands, branch model, and uninstall.",
58:     "_headers": "Cloudflare Pages style HTTP headers for security and caching behavior on the public
website.",
59:     "assets/apple-touch-icon.png": "Apple touch icon shown when the public site is saved on iOS or
compatible launch surfaces.",
60:     "assets/favicon-16.png": "Small browser favicon used by tabs and browser UI.",
61:     "assets/favicon-192.png": "Large web app icon for Android and installable browser surfaces.",
62:     "assets/favicon-32.png": "Standard browser favicon size.",

```

```
63:   "assets/favicon-48.png": "Windows/browser icon size used by some desktop surfaces.",
64:   "assets/favicon.ico": "ICO favicon bundle used by browsers that request favicon.ico.",
65:   "assets/omb-app-icon-256.png": "Builder application icon source at 256 pixels.",
66:   "assets/omb-app-icon-512.png": "Builder application icon source at 512 pixels.",
67:   "assets/omb-app-icon.ico": "Windows ICO application icon used by Electron Builder and shortcuts.",
68:   "assets/omb-mark.png": "OMB snake/mark image used for branding in the site and builder.",
69:   "build/installer.nsh": "NSIS installer customization script for update behavior, shortcut placement,
old install detection, and setup details.",
70:   "builder-rich-future-sections.js": "Builder-side helper code for richer future portfolio sections and
section defaults.",
71:   "cloudflare/README.md": "Cloudflare setup notes for deploying the AI Worker endpoint and wiring the
public site to the backend.",
72:   "cloudflare/portfolio-ai-worker.js": "Cloudflare Worker backend for the public portfolio AI assistant
and fallback intelligence path.",
```

Safe change checklist

- Confirm which app, website, installer, workflow, or document consumes this file.
- Change it on a feature branch from development.
- Regenerate docs if the file purpose, API surface, or behavior changes.
- Run the smallest relevant smoke test before merging into development.

Important Code Reference: builder-rich-future-sections.js

Item	Explanation
File	builder-rich-future-sections.js
Category	JavaScript source
Folder role	Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
Size	21,340 bytes
Extension	.js
MIME type	application/javascript
Text lines	410

Why this file exists

- Builder-side helper code for richer future portfolio sections and section defaults.

How it is used

Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.

How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

Important implementation signals

Item	Explanation
Rich text at line 82	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <button type="button" data-rich-action="font-small">Small</button>
Rich text at line 83	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <button type="button" data-rich-action="font-normal">Normal</button>
Rich text at line 84	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: <button type="button" data-rich-action="font-large">Large</button>
Rich text at line 97	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: data-placeholder="Write the overview here. You can paste images directly into this area."
Browser DOM at line 125	Builder or website interface behavior in the browser/Electron renderer. Evidence: if (editor.dataset.richEditor === "pending-description") {
Browser DOM at line 139	Builder or website interface behavior in the browser/Electron renderer. Evidence: if (editor.dataset.richEditor === "pending-description") {
Browser DOM at line 193	Builder or website interface behavior in the browser/Electron renderer. Evidence: document.querySelector("#pending-editor")?.scrollIntoView({ behavior: "smooth", block: "nearest" });
Browser DOM at line 204	Builder or website interface behavior in the browser/Electron renderer. Evidence: const type = button.dataset.openEditor;
Browser DOM at line 205	Builder or website interface behavior in the browser/Electron renderer. Evidence: const mode = button.dataset.mode "add";
Browser DOM at line 206	Builder or website interface behavior in the browser/Electron renderer. Evidence: const index = button.dataset.index;
Browser DOM at line 208	Builder or website interface behavior in the browser/Electron renderer. Evidence: const section = (project.sections []).find((item) => item.id === button.dataset.sectionId);
Browser DOM at line 213	Builder or website interface behavior in the browser/Electron renderer. Evidence: sectionId: button.dataset.sectionId,
Browser DOM at line 222	Builder or website interface behavior in the browser/Electron renderer. Evidence: const section = (project.sections []).find((item) => item.id === button.dataset.sectionId);
Browser DOM at line 226	Builder or website interface behavior in the browser/Electron renderer. Evidence: sectionId: button.dataset.sectionId,

Item	Explanation
Cloudflare or AI at line 238	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: // their richDescription field back to the project model. Fallback to
Browser DOM at line 245	Builder or website interface behavior in the browser/Electron renderer. Evidence: const richEditor = form.querySelector("[data-rich-editor='pending-description']");
Browser DOM at line 393	Builder or website interface behavior in the browser/Electron renderer. Evidence: document.addEventListener("contextmenu", (event) => {

Important constants and variables

Item	Explanation
text	Line 34: const text = plainTextFromRich(rich).trim();
label	Line 73: const label = editor?.type === "custom-section" ? "Section overview" : "Subsection overview";
folder	Line 74: const folder = pendingEditorFolder(editor);
originalPopulateRichEditor	Line 108: const originalPopulateRichEditor = typeof populateRichEditor === "function" ? populateRichEditor : null;
originalSaveRichEditorToProject	Line 109: const originalSaveRichEditorToProject = typeof saveRichEditorToProject === "function" ? saveRichEditorToProject : null;
originalOpenPendingEditor	Line 110: const originalOpenPendingEditor = typeof openPendingEditor === "function" ? openPendingEditor : null;
originalOpenEditorFromButton	Line 111: const originalOpenEditorFromButton = typeof openEditorFromButton === "function" ? openEditorFromButton : null;
originalRenderPendingEditor	Line 112: const originalRenderPendingEditor = typeof renderPendingEditor === "function" ? renderPendingEditor : null;
originalRenderSectionContent	Line 113: const originalRenderSectionContent = typeof renderSectionContent === "function" ? renderSectionContent : null;
originalSavePendingEditor	Line 114: const originalSavePendingEditor = typeof savePendingEditor === "function" ? savePendingEditor : null;
originalRenderCustomSection	Line 115: const originalRenderCustomSection = typeof renderCustomSection === "function" ? renderCustomSection : null;
originalParseProjectForPortfolio	Line 116: const originalParseProjectForPortfolio = typeof parseProjectForPortfolio === "function" ? parseProjectForPortfolio : null;
originalParsedSectionHasContent	Line 117: const originalParsedSectionHasContent = typeof parsedSectionHasContent === "function" ? parsedSectionHasContent : null;
originalPreviewSectionHasRenderableContent	Line 118: const originalPreviewSectionHasRenderableContent = typeof previewSectionHasRenderableContent === "function" ? previewSectionHasRenderableContent : null;
rich	Line 140: const rich = extractRichSummary(editor);
title	Line 154: const title = pendingEditor.title "";
description	Line 155: const description = pendingEditor.description "";
isTool	Line 156: const isTool = pendingEditor.type === "tool";
isSimpleText	Line 157: const isSimpleText = pendingEditor.type === "text-array";
isSectionDetails	Line 158: const isSectionDetails = pendingEditor.type === "custom-section";
usesRich	Line 159: const usesRich = pendingEditorUsesRichDescription();
project	Line 202: const project = selectedProject();
type	Line 204: const type = button.dataset.openEditor;
mode	Line 205: const mode = button.dataset.mode "add";
index	Line 206: const index = button.dataset.index;
section	Line 208: const section = (project.sections []).find((item) => item.id === button.dataset.sectionId);
item	Line 209: const item = mode === "edit" ? section?.items?.[Number(index)] : {};
section	Line 222: const section = (project.sections []).find((item) => item.id === button.dataset.sectionId);
project	Line 242: const project = selectedProject();
title	Line 244: const title = form.elements.title.value.trim();

Item	Explanation
richEditor	Line 245: const richEditor = form.querySelector("[data-rich-editor='pending-description']");
richDescription	Line 246: const richDescription = richEditor ? extractRichSummary(richEditor) : null;
description	Line 247: const description = richEditor ? richPlainTextOrSpacer(richDescription) : form.elements.description?.value.trim() "";
section	Line 253: const section = (project.sections []).find((item) => item.id === pendingEditor.sectionId);
nextItem	Line 256: const nextItem = {
section	Line 278: const section = (project.sections []).find((item) => item.id === pendingEditor.sectionId);
section	Line 297: const section = (project.sections []).find((item) => item.id === sectionId);
parsed	Line 352: const parsed = originalParseProjectForPortfolio(project);
sectionId	Line 355: const sectionId = parsedSection.id.replace("custom:", "");
sourceSection	Line 356: const sourceSection = (project.sections []).find((section) => section.id === sectionId);
editor	Line 394: const editor = event.target.closest("[data-rich-editor]");
block	Line 395: const block = event.target.closest(".rich-block");

Event handlers and user interactions

Item	Explanation
Line 393	document.addEventListener("contextmenu", (event) => {

Function-by-function detail

hasRichContent - lines 23-31

Item	Explanation
Source location	Lines 23-31 (9 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	function hasRichContent(rich) {
Touches	filesystem
Calls detected	some
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

23:     function hasRichContent(rich) {
24:         return Boolean(rich?.blocks?.some((block) =>
25:             (block.type === "image" && block.url) ||
26:             (block.type === "formula" && block.formula) ||
27:             block.text ||
28:             block.title ||
29:             block.caption
30:         ));
31:     }

```

richPlainTextOrSpacer - lines 33-36

Item	Explanation
Source location	Lines 33-36 (4 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.

Item	Explanation
Declaration	function richPlainTextOrSpacer(rich) {
Touches	Mostly local calculations or local UI state.
Calls detected	plainTextFromRich, trim, hasRichContent
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

33:     function richPlainTextOrSpacer(rich) {
34:         const text = plainTextFromRich(rich).trim();
35:         return text || (hasRichContent(rich) ? "\u200B" : "");
36:     }

```

pendingEditorUsesRichDescription - lines 41-43

Item	Explanation
Source location	Lines 41-43 (3 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	function pendingEditorUsesRichDescription(editor = pendingEditor) {
Touches	Mostly local calculations or local UI state.
Calls detected	includes
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

41:     function pendingEditorUsesRichDescription(editor = pendingEditor) {
42:         return Boolean(editor && ["custom", "custom-section"].includes(editor.type));
43:     }

```

pendingEditorFolder - lines 48-57

Item	Explanation
Source location	Lines 48-57 (10 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	function pendingEditorFolder(editor = pendingEditor) {
Touches	Mostly local calculations or local UI state.
Calls detected	slugify
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 4 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

48:     function pendingEditorFolder(editor = pendingEditor) {
49:         if (!editor) return "custom-overview";
50:         if (editor.type === "custom-section") {

```

```

51:         return `custom-${slugify(editor.sectionId || "section")}-overview`;
52:     }
53:     if (editor.type === "custom") {
54:         return `custom-${slugify(editor.sectionId || "section")}-subsection`;
55:     }
56:     return "custom-overview";
57: }

```

pendingEditorRichBlocks - lines 62-65

Item	Explanation
Source location	Lines 62-65 (4 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	function pendingEditorRichBlocks(editor = pendingEditor) {
Touches	Mostly local calculations or local UI state.
Calls detected	textBlocksFromPlainText, richBlocksFromValue
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 2 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

62:     function pendingEditorRichBlocks(editor = pendingEditor) {
63:         if (!editor) return textBlocksFromPlainText("");
64:         return richBlocksFromValue(editor.richDescription, editor.description || "");
65:     }

```

renderPendingRichDescriptionEditor - lines 72-103

Item	Explanation
Source location	Lines 72-103 (32 source lines).
Purpose	Renders UI, markup, or display output.
Declaration	function renderPendingRichDescriptionEditor(editor = pendingEditor) {
Touches	filesystem
Calls detected	pendingEditorFolder, escapeHtml
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

72:     function renderPendingRichDescriptionEditor(editor = pendingEditor) {
73:         const label = editor?.type === "custom-section" ? "Section overview" : "Subsection overview";
74:         const folder = pendingEditorFolder(editor);
75:         return `
76:         <div class="pending-rich-description rich-design-summary-field">
77:             <div class="summary-builder-heading">
78:                 <span>${label}</span>
79:             </div>
80:             <div class="rich-summary-panel">
81:                 <div class="rich-summary-toolbar" aria-label="${escapeHtml(label)} tools">
82:                     <button type="button" data-rich-action="font-small">Small</button>
83:                     <button type="button" data-rich-action="font-normal">Normal</button>
84:                     <button type="button" data-rich-action="font-large">Large</button>
85:                     <button type="button" data-rich-action="align-left">Left</button>
86:                     <button type="button" data-rich-action="align-center">Center</button>
87:                     <button type="button" data-rich-action="align-right">Right</button>
88:                     <button type="button" data-rich-action="add-image">Add image</button>
89:                     <button type="button" data-rich-action="add-formula">Add formula</button>
90:                     <button type="button" data-rich-action="edit-block">Edit selected</button>
91:                     <button class="danger-icon" type="button" data-rich-action="delete-block">Delete
selected</button>

```



```

92:         </div>
93:         <div
94:             class="rich-summary-editor"
95:             data-rich-editor="pending-description"
96:             data-rich-folder="${escapeHtml(folder)}"
97:             data-placeholder="Write the overview here. You can paste images directly into this area."
98:             aria-label="${escapeHtml(label)} rich editor"
99:         ></div>
100:     </div>
101: </div>
102: `;
103: }

```

section - lines 208-209

Item	Explanation
Source location	Lines 208-209 (2 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	const section = (project.sections []).find((item) => item.id === button.dataset.sectionId);
Touches	browser/Electron DOM
Calls detected	find
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

208:         const section = (project.sections || []).find((item) => item.id ===
button.dataset.sectionId);
209:         const item = mode === "edit" ? section?.items?.[Number(index)] : {};

```

section - lines 222-230

Item	Explanation
Source location	Lines 222-230 (9 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	const section = (project.sections []).find((item) => item.id === button.dataset.sectionId);
Touches	browser/Electron DOM
Calls detected	find, openPendingEditor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 0 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

222:         const section = (project.sections || []).find((item) => item.id ===
button.dataset.sectionId);
223:         openPendingEditor({
224:             type,
225:             mode: "edit",
226:             sectionId: button.dataset.sectionId,
227:             title: section?.title || "",
228:             description: section?.description || "",
229:             richDescription: section?.richDescription || null
230:         });

```

section - lines 253-262

Item	Explanation
Source location	Lines 253-262 (10 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	const section = (project.sections []).find((item) => item.id === pendingEditor.sectionId);
Touches	Mostly local calculations or local UI state.
Calls detected	find
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

253:         const section = (project.sections || []).find((item) => item.id ===
pendingEditor.sectionId);
254:         if (!section) return;
255:         section.items = section.items || [];
256:         const nextItem = {
257:             title,
258:             description,
259:             richDescription,
260:             type: "Text",
261:             status: "draft"
262:         };

```

section - lines 278-291

Item	Explanation
Source location	Lines 278-291 (14 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	const section = (project.sections []).find((item) => item.id === pendingEditor.sectionId);
Touches	filesystem
Calls detected	find, setStatus, scheduleAutosave, renderAll, originalSavePendingEditor
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 3 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

278:         const section = (project.sections || []).find((item) => item.id ===
pendingEditor.sectionId);
279:         if (!section) return;
280:         section.title = title;
281:         section.description = description;
282:         section.richDescription = richDescription;
283:         pendingEditor = null;
284:         setStatus("Saved in the project editor. Click Save project to include this version in the
portfolio preview.");
285:         scheduleAutosave();
286:         renderAll();
287:         return;
288:     }
289:     return originalSavePendingEditor(form);
290: };
291: }

```

section - lines 297-301

Item	Explanation
Source location	Lines 297-301 (5 source lines).

Item	Explanation
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	<code>const section = (project.sections []).find((item) => item.id === sectionId);</code>
Touches	Mostly local calculations or local UI state.
Calls detected	<code>find</code> , <code>escapeHtml</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 <code>await</code> expression(s), 2 <code>return</code> statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

297:         const section = (project.sections || []).find((item) => item.id === sectionId);
298:         if (!section) return `
```

sourceSection - lines 356-360

Item	Explanation
Source location	Lines 356-360 (5 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	<code>const sourceSection = (project.sections []).find((section) => section.id === sectionId);</code>
Touches	Mostly local calculations or local UI state.
Calls detected	<code>find</code> , <code>hasRichContent</code> , <code>clone</code> , <code>richPlainTextOrSpacer</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 <code>await</code> expression(s), 0 <code>return</code> statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

356:         const sourceSection = (project.sections || []).find((section) => section.id ===
sectionId);
357:         if (hasRichContent(sourceSection?.richDescription)) {
358:             parsedSection.rich = clone(sourceSection.richDescription);
359:             parsedSection.description = sourceSection.description ||
richPlainTextOrSpacer(sourceSection.richDescription);
360:         }

```

improveRichContextMenu - lines 392-405

Item	Explanation
Source location	Lines 392-405 (14 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	<code>function improveRichContextMenu() {</code>
Touches	<code>browser/Electron DOM</code>
Calls detected	<code>addEventListener</code> , <code>closest</code> , <code>preventDefault</code> , <code>selectRichBlock</code> , <code>currentRichBlock</code> , <code>syncRichContextMenuControls</code>
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 <code>await</code> expression(s), 1 <code>return</code> statement(s).

Item	Explanation
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

392:     function improveRichContextMenu() {
393:         document.addEventListener("contextmenu", (event) => {
394:             const editor = event.target.closest("[data-rich-editor]");
395:             const block = event.target.closest(".rich-block");
396:             if (!editor && !block) return;
397:             event.preventDefault();
398:             activeSummaryEditor = editor || block.closest("[data-rich-editor]");
399:             selectRichBlock(block || currentRichBlock(activeSummaryEditor));
400:             syncRichContextMenuControls(activeSummaryBlock);
401:             summaryContextMenu.style.left = `${event.clientX}px`;
402:             summaryContextMenu.style.top = `${event.clientY}px`;
403:             summaryContextMenu.hidden = false;
404:             }, true);
405:     }

```

Representative opening snippet

```

1: // JavaScript source code
2: (() => {
3:     "use strict";
4:
5:     /**
6:      * Extend the portfolio builder with rich description editors for custom
7:      * sections and subsections. This patch adds a new rich editor to the
8:      * pending editor for custom items so that images, formulas and other
9:      * formatted content can be used in overviews of future sections. It
10:     * hooks into existing helper functions such as `populateRichEditor`
11:     * and `saveRichEditorToProject` by overriding them at runtime. To
12:     * avoid interfering with built-in behaviour, everything is wrapped
13:     * inside an IIFE and existing functions are stored before being
14:     * redefined. If a function is not present in the current context
15:     * (because the builder script changed), the patch simply does
16:     * nothing.
17:     */
18:
19:     // Guard against applying the patch multiple times.
20:     if (window.builderRichFutureSectionsPatchLoaded) return;
21:
22:     // Helpers to detect whether an object has meaningful rich content.
23:     function hasRichContent(rich) {
24:         return Boolean(rich?.blocks?.some((block) =>
25:             (block.type === "image" && block.url) ||
26:             (block.type === "formula" && block.formula) ||
27:             block.text ||
28:             block.title ||
29:             block.caption
30:         ));
31:     }
32:
33:     function richPlainTextOrSpacer(rich) {
34:         const text = plainTextFromRich(rich).trim();
35:         return text || (hasRichContent(rich) ? "\u200B" : "");
36:     }
37:
38:     // Determine whether the pending editor should use a rich editor. All
39:     // custom sections and subsections use rich editors. Tools, design
40:     // objects and other built-in forms continue to use plain text areas.
41:     function pendingEditorUsesRichDescription(editor = pendingEditor) {
42:         return Boolean(editor && ["custom", "custom-section"].includes(editor.type));
43:     }
44:
45:     // Build a folder name for images uploaded via the pending editor. For
46:     // custom sections and subsections, use a prefix based on the section
47:     // id so that uploaded images are organised under docs/<project>/<folder>/.
48:     function pendingEditorFolder(editor = pendingEditor) {
49:         if (!editor) return "custom-overview";
50:         if (editor.type === "custom-section") {
51:             return `custom-${slugify(editor.sectionId || "section")}-overview`;
52:         }
53:         if (editor.type === "custom") {
54:             return `custom-${slugify(editor.sectionId || "section")}-subsection`;
55:         }
56:         return "custom-overview";
57:     }
58:
59:     // Derive blocks from a rich value or fallback plain text. When the
60:     // editor is empty, provide a single text block so that the content
61:     // area is editable.
62:     function pendingEditorRichBlocks(editor = pendingEditor) {
63:         if (!editor) return textBlocksFromPlainText("");
64:         return richBlocksFromValue(editor.richDescription, editor.description || "");
65:     }

```

```
66:
67: // Render the rich description field for the pending editor. This
68: // function builds a UI similar to other rich editors in the builder,
69: // including a toolbar and editable area with appropriate data
70: // attributes. The label varies depending on whether we're editing
71: // a section or a subsection.
72: function renderPendingRichDescriptionEditor(editor = pendingEditor) {
```

Safe change checklist

- Confirm which app, website, installer, workflow, or document consumes this file.
- Change it on a feature branch from development.
- Regenerate docs if the file purpose, API surface, or behavior changes.
- Run the smallest relevant smoke test before merging into development.

Important Code Reference: electronics-search.js

Item	Explanation
File	electronics-search.js
Category	JavaScript source
Folder role	Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
Size	9,856 bytes
Extension	.js
MIME type	application/javascript
Text lines	162

Why this file exists

- Electronics vocabulary and search support used by the website search behavior.

How it is used

Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.

How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

Important implementation signals

Item	Explanation
Rich text at line 8	Rich editor behavior, formatting, paste handling, or content sanitization. Evidence: "phase margin", "slew rate", "common mode", "common mode range", "cmrr", "psrr", "loop gain",
Security or auth at line 45	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "oscilloscope", "scope capture", "logic analyzer", "spectrum analyzer", "network analyzer", "function generator",
Compiler or simulator at line 59	Part of the Compile Code workspace or language/tool detection. Evidence: "static timing analysis", "sta", "fpga", "asic", "rtl", "verilog", "systemverilog", "vhd",
Compiler or simulator at line 61	Part of the Compile Code workspace or language/tool detection. Evidence: "modelsim", "questa", "iverilog", "yosys", "formal verification", "assertion", "coverage", "axi",
Installer at line 80	Windows setup, update, shortcut, or installation behavior. Evidence: "potentiometer", "trimmer", "thermistor", "varistor", "photodiode", "phototransistor", "hall effect sensor",
Cloudflare or AI at line 87	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "layout", "route", "solder", "assemble", "test", "verify", "model", "analyze", "optimize",
Compiler or simulator at line 100	Part of the Compile Code workspace or language/tool detection. Evidence: "ltspice", "spice", "pspice", "ngspice", "matlab", "simulink", "python", "numpy", "scipy",
Git command at line 101	Repository state, publishing, branch checks, or release automation. Evidence: "excel", "kiCad", "kicad", "vivado", "stm32cubeide", "git", "github", "markdown", "c",

Important constants and variables

Item	Explanation
analogTerms	Line 2: const analogTerms = [
powerTerms	Line 20: const powerTerms = [
pcbTerms	Line 33: const pcbTerms = [
testTerms	Line 44: const testTerms = [
digitalTerms	Line 55: const digitalTerms = [

Item	Explanation
embeddedTerms	Line 66: const embeddedTerms = [
components	Line 76: const components = [
actions	Line 85: const actions = [
metrics	Line 92: const metrics = [
toolTerms	Line 99: const toolTerms = [
phraseTerms	Line 105: const phraseTerms = [];
categoryKeywords	Line 114: const categoryKeywords = {
allKeywords	Line 120: const allKeywords = [...new Set([
haystack	Line 139: const haystack = [
buckets	Line 149: const buckets = new Set(categoryKeywords[project?.category] []);
compactTerm	Line 154: const compactTerm = term.replace(/^[^a-z0-9+#]+/g, " ").trim();

JSON/YAML-style keys detected

Item	Explanation
analog-mixed-signal	Line 115: "analog-mixed-signal": [...analogTerms, ...powerTerms, ...pcbTerms, ...testTerms, ...components, ...metrics],

Function-by-function detail

normalize - lines 134-136

Item	Explanation
Source location	Lines 134-136 (3 source lines).
Purpose	Validates or normalizes input so later code receives safe predictable values.
Declaration	function normalize(value) {
Touches	filesystem
Calls detected	toLowerCase
API endpoints	None detected inside this function body.
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

134:   function normalize(value) {
135:     return String(value || "").toLowerCase();
136:   }

```

projectSpecificKeywords - lines 138-158

Item	Explanation
Source location	Lines 138-158 (21 source lines).
Purpose	Named function in the repository. Inspect the surrounding lines to learn its exact inputs and outputs.
Declaration	function projectSpecificKeywords(project, categoryLabel = "") {
Touches	filesystem
Calls detected	map, join, Set, forEach, includes, toLowerCase, add, replace, trim
API endpoints	None detected inside this function body.

Item	Explanation
Storage keys	No local/session storage keys detected.
Async/return clues	0 await expression(s), 1 return statement(s).
Debug approach	Trigger the user action that reaches this function, inspect inputs/state before the function, then inspect the returned value, DOM update, file write, process output, or API response after it runs.

Source excerpt:

```

138:   function projectSpecificKeywords(project, categoryLabel = "") {
139:       const haystack = [
140:           project?.title,
141:           project?.summary,
142:           project?.category,
143:           categoryLabel,
144:           ...(project?.focus || []),
145:           ...(project?.tools || []).map((item) => typeof item === "string" ? item : [item.name, item.title,
item.label, item.description].join(" ")),
146:           ...(project?.languages || []),
147:       ].map(normalize).join(" ");
148:
149:       const buckets = new Set(categoryKeywords[project?.category] || []);
150:       toolTerms.forEach((term) => {
151:           if (haystack.includes(term.toLowerCase())) buckets.add(term);
152:       });
153:       allKeywords.forEach((term) => {
154:           const compactTerm = term.replace(/^[^a-z0-9+#]+/g, " ").trim();
155:           if (compactTerm && haystack.includes(compactTerm)) buckets.add(term);
156:       });
157:       return [...buckets];
158:   }

```

Representative opening snippet

```

1: (function () {
2:     const analogTerms = [
3:         "op amp", "operational amplifier", "fully differential op amp", "differential amplifier",
"instrumentation amplifier",
4:         "transimpedance amplifier", "transconductance amplifier", "current mirror", "cascode", "folded
cascode",
5:         "common source", "common gate", "common drain", "emitter follower", "source follower", "differential
pair",
6:         "active load", "gain stage", "output stage", "input stage", "bias circuit", "bandgap reference",
7:         "voltage reference", "current reference", "low noise", "offset voltage", "input bias current", "gain
bandwidth",
8:         "phase margin", "slew rate", "common mode", "common mode range", "cmrr", "psrr", "loop gain",
9:         "stability", "compensation", "miller compensation", "frequency response", "bode plot", "pole zero",
10:        "small signal", "large signal", "noise analysis", "thermal noise", "flicker noise", "shot noise",
11:        "analog filter", "low pass filter", "high pass filter", "band pass filter", "notch filter", "active
filter",
12:        "rc filter", "rlc circuit", "resonance", "q factor", "cutoff frequency", "impedance", "admittance",
13:        "matching network", "attenuator", "buffer", "comparator", "window comparator", "schmitt trigger",
"hysteresis",
14:        "oscillator", "vco", "voltage controlled oscillator", "relaxation oscillator", "ring oscillator",
"crystal oscillator",
15:        "pll", "phase locked loop", "charge pump", "sample and hold", "track and hold", "analog switch",
16:        "multiplexer", "adc", "dac", "data converter", "sigma delta", "successive approximation", "sar adc",
17:        "pipeline adc", "integrator", "differentiator", "rectifier", "precision rectifier", "peak detector"
18:    ];
19:
20:    const powerTerms = [
21:        "ac to dc", "charger", "battery charger", "power supply", "bench supply", "linear regulator", "ldo",
22:        "switching regulator", "buck converter", "boost converter", "buck boost", "flyback", "forward
converter",
23:        "sepic", "inverter", "rectifier bridge", "diode bridge", "full wave rectifier", "half wave
rectifier",
24:        "ripple", "load regulation", "line regulation", "transient response", "power stage", "mosfet driver",
25:        "gate driver", "current limit", "overvoltage", "undervoltage", "reverse polarity", "thermal
shutdown",
26:        "power path", "battery management", "bms", "cc cv", "constant current", "constant voltage", "trickle
charge",
27:        "precharge", "protection circuit", "fuse", "polyfuse", "tvS diode", "esd diode", "snubber", "clamp",
28:        "soft start", "inrush current", "efficiency", "loss budget", "heat sink", "thermal resistance",
"power dissipation",
29:        "load switch", "ideal diode", "current sense", "shunt resistor", "hall sensor", "isolation",
"transformer",
30:        "optoisolator", "dc dc", "ac dc", "emi filter", "common mode choke", "ferrite bead", "ground loop"
31:    ];
32:
33:    const pcbTerms = [
34:        "pcb", "printed circuit board", "schematic", "layout", "footprint", "symbol", "netlist", "bom",
35:        "gerber", "drill file", "pick and place", "assembly", "bring up", "board bring up", "prototype",
36:        "revision", "rev a", "dfm", "dft", "design rule", "clearance", "creepage", "trace width", "via",
37:        "microvia", "blind via", "buried via", "ground plane", "power plane", "stackup", "controlled
impedance",
38:        "differential pair routing", "return path", "star ground", "analog ground", "digital ground", "guard

```



```

ring",
39:   "decoupling capacitor", "bypass capacitor", "bulk capacitor", "test point", "connector", "header",
40:   "silkscreen", "solder mask", "copper pour", "thermal relief", "stitching via", "keepout", "fiducial",
41:   "kicad", "altium", "eagle", "orcad", "cadence allegro", "pcbway", "jlcpcb", "osh park"
42: ];
43:
44:   const testTerms = [
45:     "oscilloscope", "scope capture", "logic analyzer", "spectrum analyzer", "network analyzer", "function
generator",
46:     "signal generator", "multimeter", "dmm", "electronic load", "power analyzer", "curve tracer",
"probe",
47:     "differential probe", "current probe", "bench test", "validation", "verification",
"characterization",
48:     "measurement", "calibration", "test plan", "test report", "waveform", "transient", "dc sweep", "ac
sweep",
49:     "monte carlo", "corner analysis", "temperature sweep", "load test", "line test", "frequency sweep",
50:     "gain measurement", "phase measurement", "jitter", "rise time", "fall time", "duty cycle", "pwm",
51:     "latency", "throughput", "noise floor", "snr", "thd", "thd+n", "eye diagram", "ber", "debug",
52:     "troubleshooting", "root cause", "failure analysis", "regression test", "unit test", "integration
test"
53:   ];
54:
55:   const digitalTerms = [
56:     "digital design", "logic design", "combinational logic", "sequential logic", "finite state machine",
"fsm",
57:     "flip flop", "latch", "counter", "register", "shift register", "clock divider", "clock domain
crossing",
58:     "cdc", "metastability", "synchronizer", "debounce", "timing closure", "setup time", "hold time",
59:     "static timing analysis", "sta", "fpga", "asic", "rtl", "verilog", "systemverilog", "vhdl",
60:     "testbench", "uvm", "simulation", "synthesis", "place and route", "bitstream", "vivado", "quartus",
61:     "modelsim", "questa", "iverilog", "yosys", "formal verification", "assertion", "coverage", "axi",
62:     "apb", "ahb", "spi controller", "i2c controller", "uart", "fifo", "ram", "rom", "bram",
63:     "pll clock", "mmcm", "serdes", "lvds", "gpio", "interrupt", "dma", "bus protocol"
64:   ];
65:
66:   const embeddedTerms = [
67:     "embedded", "firmware", "microcontroller", "mcu", "stm32", "stm32cubeide", "arm cortex", "cortex m",
68:     "bare metal", "rtos", "freertos", "scheduler", "interrupt service routine", "isr", "timer",
69:     "pwm output", "adc input", "dac output", "gpio", "spi", "i2c", "uart", "usb", "can bus",
70:     "ethernet", "bootloader", "flash memory", "eeprom", "dma", "low power", "sleep mode", "watchdog",
71:     "sensor interface", "motor control", "control loop", "pid", "encoder", "hall sensor", "imu",
72:     "temperature sensor", "pressure sensor", "display", "lcd", "oled", "ble", "wifi", "esp32",

```

Safe change checklist

- Confirm which app, website, installer, workflow, or document consumes this file.
- Change it on a feature branch from development.
- Regenerate docs if the file purpose, API surface, or behavior changes.
- Run the smallest relevant smoke test before merging into development.

Important Code Reference: package.json

Item	Explanation
File	package.json
Category	Configuration or structured data
Folder role	Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
Size	2,412 bytes
Extension	.json
MIME type	application/json
Text lines	85

Why this file exists

- Node/Electron package manifest with scripts, dependencies, version, installer settings, and extra resources.

How it is used

Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.

How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

Important implementation signals

Item	Explanation
Security or auth at line 5	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "author": "Maurice Otieno",
Installer at line 10	Windows setup, update, shortcut, or installation behavior. Evidence: "pack": "electron-builder --win portable --publish never",
Installer at line 11	Windows setup, update, shortcut, or installation behavior. Evidence: "installer": "electron-builder --win nsis --publish never",
Installer at line 12	Windows setup, update, shortcut, or installation behavior. Evidence: "dist": "electron-builder --win --publish never"
Installer at line 16	Windows setup, update, shortcut, or installation behavior. Evidence: "electron-builder": "^26.0.12"
Cloudflare or AI at line 37	Public AI assistant, Worker deployment, or model-provider fallback behavior. Evidence: "cloudflare/**",
Installer at line 50	Windows setup, update, shortcut, or installation behavior. Evidence: "installer-notes.txt",
Security or auth at line 54	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: "_headers",
Installer at line 64	Windows setup, update, shortcut, or installation behavior. Evidence: "nsis"
Installer at line 70	Windows setup, update, shortcut, or installation behavior. Evidence: "nsis": {
Installer at line 76	Windows setup, update, shortcut, or installation behavior. Evidence: "include": "build/installer.nsh",
Installer at line 77	Windows setup, update, shortcut, or installation behavior. Evidence: "createDesktopShortcut": "always",
Installer at line 78	Windows setup, update, shortcut, or installation behavior. Evidence: "createStartMenuShortcut": true,
Installer at line 79	Windows setup, update, shortcut, or installation behavior. Evidence: "deleteAppDataOnUninstall": true,
Installer at line 81	Windows setup, update, shortcut, or installation behavior. Evidence: "license": "installer-notes.txt",
Installer at line 82	Windows setup, update, shortcut, or installation behavior. Evidence: "shortcutName": "OMB Portfolio Builder"

JSON/YAML-style keys detected

Item	Explanation
name	Line 2: "name": "omb-portfolio-builder",
version	Line 3: "version": "0.2.34",
private	Line 4: "private": false,
author	Line 5: "author": "Maurice Otieno",
description	Line 6: "description": "Standalone Windows desktop application for building and publishing portfolio websites.",
main	Line 7: "main": "main.cjs",
scripts	Line 8: "scripts": {
dev	Line 9: "dev": "electron .",
pack	Line 10: "pack": "electron-builder --win portable --publish never",
installer	Line 11: "installer": "electron-builder --win nsis --publish never",
dist	Line 12: "dist": "electron-builder --win --publish never"
devDependencies	Line 14: "devDependencies": {
electron	Line 15: "electron": "^39.2.7",
electron-builder	Line 16: "electron-builder": "^26.0.12"
build	Line 18: "build": {
appId	Line 19: "appId": "com.mauriceotieno.portfoliobuilder",
productName	Line 20: "productName": "OMB Portfolio Builder",
copyright	Line 21: "copyright": "Copyright (c) Maurice Otieno",
asar	Line 22: "asar": true,
directories	Line 23: "directories": {
output	Line 24: "output": "dist"
files	Line 26: "files": [
extraResources	Line 30: "extraResources": [
from	Line 32: "from": ".",
to	Line 33: "to": "site",
filter	Line 34: "filter": [
win	Line 60: "win": {
icon	Line 61: "icon": "assets/omb-app-icon.ico",
target	Line 62: "target": [
portable	Line 67: "portable": {
artifactName	Line 68: "artifactName": "OMB-Portfolio-Builder-Portable-\${version}-\${arch}.\${ext}"
nsis	Line 70: "nsis": {
artifactName	Line 71: "artifactName": "OMB-Portfolio-Builder-Setup-\${version}-\${arch}.\${ext}",
oneClick	Line 72: "oneClick": false,
perMachine	Line 73: "perMachine": false,
allowElevation	Line 74: "allowElevation": false,
allowToChangeInstallationDirectory	Line 75: "allowToChangeInstallationDirectory": true,
include	Line 76: "include": "build/installer.nsh",
createDesktopShortcut	Line 77: "createDesktopShortcut": "always",
createStartMenuShortcut	Line 78: "createStartMenuShortcut": true,

Item	Explanation
deleteAppDataOnUninstall	Line 79: "deleteAppDataOnUninstall": true,
runAfterFinish	Line 80: "runAfterFinish": true,
license	Line 81: "license": "installer-notes.txt",
shortcutName	Line 82: "shortcutName": "OMB Portfolio Builder"

Representative opening snippet

```

1: {
2:   "name": "omb-portfolio-builder",
3:   "version": "0.2.34",
4:   "private": false,
5:   "author": "Maurice Otieno",
6:   "description": "Standalone Windows desktop application for building and publishing portfolio
websites.",
7:   "main": "main.cjs",
8:   "scripts": {
9:     "dev": "electron .",
10:    "pack": "electron-builder --win portable --publish never",
11:    "installer": "electron-builder --win nsis --publish never",
12:    "dist": "electron-builder --win --publish never"
13:  },
14:  "devDependencies": {
15:    "electron": "^39.2.7",
16:    "electron-builder": "^26.0.12"
17:  },
18:  "build": {
19:    "appId": "com.mauriceotieno.portfoliobuilder",
20:    "productName": "OMB Portfolio Builder",
21:    "copyright": "Copyright (c) Maurice Otieno",
22:    "asar": true,
23:    "directories": {
24:      "output": "dist"
25:    },
26:    "files": [
27:      "main.cjs",
28:      "package.json"
29:    ],
30:    "extraResources": [
31:      {
32:        "from": ".",
33:        "to": "site",
34:        "filter": [
35:          "assets/**",
36:          "Backgrounds/**",
37:          "cloudflare/**",
38:          "docs/**",
39:          "server.mjs",
40:          "index.html",
41:          "styles.css",
42:          "script.js",
43:          "electronics-search.js",
44:          "builder-rich-future-sections.js",
45:          "projects.json",
46:          "project-templates.json",
47:          "template-preview.html",
48:          "template-preview.css",
49:          "template-preview.js",
50:          "installer-notes.txt",
51:          "README.md",
52:          "portfolio-README.md",
53:          ".nojekyll",
54:          "_headers",
55:          ".well-known/**",
56:          "robots.txt"
57:        ]
58:      }
59:    ],
60:    "win": {
61:      "icon": "assets/omb-app-icon.ico",
62:      "target": [
63:        "portable",
64:        "nsis"
65:      ]
66:    },
67:    "portable": {
68:      "artifactName": "OMB-Portfolio-Builder-Portable-${version}-${arch}.${ext}"
69:    },
70:    "nsis": {
71:      "artifactName": "OMB-Portfolio-Builder-Setup-${version}-${arch}.${ext}",
72:      "oneClick": false,

```

Safe change checklist

- Confirm which app, website, installer, workflow, or document consumes this file.
- Change it on a feature branch from development.
- Regenerate docs if the file purpose, API surface, or behavior changes.
- Run the smallest relevant smoke test before merging into development.

Important Code Reference: project-templates.json

Item	Explanation
File	project-templates.json
Category	Configuration or structured data
Folder role	Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
Size	6,428 bytes
Extension	.json
MIME type	application/json
Text lines	184

Why this file exists

- Appearance template definitions used by projects to control visual feel rather than content.

How it is used

Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.

How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

JSON/YAML-style keys detected

Item	Explanation
templates	Line 2: "templates": [
id	Line 4: "id": "appearance-light-blue-red-click",
label	Line 5: "label": "Light blue, navy hover, red click",
description	Line 6: "description": "Light blue engineering background with calm navy hover states and a sharp red click flash.",
visual	Line 7: "visual": {
style	Line 8: "style": "schematic",
palette	Line 9: "palette": ["#e9f8ff", "#1f6ed4", "#ef4444", "#ffffff"],
background	Line 10: "background": "#e9f8ff",
panel	Line 11: "panel": "#ffffff",
accent	Line 12: "accent": "#1f6ed4",
hover	Line 13: "hover": "#1f6ed4",
click	Line 14: "click": "#ef4444",
clickText	Line 15: "clickText": "#ffffff",
line	Line 16: "line": "#b8def3",
text	Line 17: "text": "#172636",
interaction	Line 18: "interaction": "Navy-blue hover with a red active click flash."
id	Line 22: "id": "appearance-white-blue-click",
label	Line 23: "label": "Clean white, blue hover, white click",
description	Line 24: "description": "Minimal white project panels with professional blue hover behavior and quiet white click feedback.",

Item	Explanation
visual	Line 25: "visual": {
style	Line 26: "style": "device",
palette	Line 27: "palette": ["#ffffff", "#2563eb", "#f8bfff"],
background	Line 28: "background": "#ffffff",
panel	Line 29: "panel": "#ffffff",
accent	Line 30: "accent": "#2563eb",
hover	Line 31: "hover": "#2563eb",
click	Line 32: "click": "#ffffff",
clickText	Line 33: "clickText": "#172636",
line	Line 34: "line": "#d8e2ef",
text	Line 35: "text": "#172636",
interaction	Line 36: "interaction": "Blue hover with a clean white click state."
id	Line 40: "id": "appearance-deep-navy-cyan-click",
label	Line 41: "label": "Deep navy, cyan hover, white click",
description	Line 42: "description": "Dark navy technical background with cyan hover states and high-contrast white click feedback.",
visual	Line 43: "visual": {
style	Line 44: "style": "chip",
palette	Line 45: "palette": ["#10263c", "#0f6f96", "#ffffff"],
background	Line 46: "background": "#10263c",
panel	Line 47: "panel": "#f8bfff",
accent	Line 48: "accent": "#0f6f96",
hover	Line 49: "hover": "#0f6f96",
click	Line 50: "click": "#ffffff",
clickText	Line 51: "clickText": "#142536",
line	Line 52: "line": "#8ccfe8",
text	Line 53: "text": "#f8bfff",
interaction	Line 54: "interaction": "Cyan hover with a white active click state."
id	Line 58: "id": "appearance-warm-red-pale-click",
label	Line 59: "label": "Warm red, wine hover, pale click",
description	Line 60: "description": "Warm signal-lab styling with wine-red hover behavior and a soft pale-red click response.",
visual	Line 61: "visual": {
style	Line 62: "style": "waveform",
palette	Line 63: "palette": ["#fff7f5", "#c2413b", "#ff4f2"],
background	Line 64: "background": "#fff7f5",
panel	Line 65: "panel": "#ffffff",
accent	Line 66: "accent": "#b9332e",
hover	Line 67: "hover": "#c2413b",
click	Line 68: "click": "#ff4f2",
clickText	Line 69: "clickText": "#2f2020",
line	Line 70: "line": "#efb0aa",
text	Line 71: "text": "#2f2020",

Item	Explanation
interaction	Line 72: "interaction": "Wine-red hover with a pale red click state."
id	Line 76: "id": "appearance-emerald-mint-click",
label	Line 77: "label": "Emerald bench, green hover, mint click",
description	Line 78: "description": "Instrumentation-style panels with emerald hover states and a light mint click response.",
visual	Line 79: "visual": {
style	Line 80: "style": "instrument",
palette	Line 81: "palette": ["#f0fdf6", "#16865a", "#efff6"],
background	Line 82: "background": "#f0fdf6",
panel	Line 83: "panel": "#ffffff",
accent	Line 84: "accent": "#16865a",
hover	Line 85: "hover": "#16865a",
click	Line 86: "click": "#efff6",
clickText	Line 87: "clickText": "#182f26",
line	Line 88: "line": "#9adfbf",
text	Line 89: "text": "#182f26",
interaction	Line 90: "interaction": "Emerald hover with a soft mint click state."
id	Line 94: "id": "appearance-amber-cream-click",
label	Line 95: "label": "Amber rail, gold hover, cream click",
description	Line 96: "description": "Power-rail inspired appearance with amber hover behavior and warm cream click feedback.",
visual	Line 97: "visual": {

Representative opening snippet

```

1: {
2:   "templates": [
3:     {
4:       "id": "appearance-light-blue-red-click",
5:       "label": "Light blue, navy hover, red click",
6:       "description": "Light blue engineering background with calm navy hover states and a sharp red click
flash.",
7:       "visual": {
8:         "style": "schematic",
9:         "palette": ["#e9f8ff", "#1f6ed4", "#ef4444", "#ffffff"],
10:        "background": "#e9f8ff",
11:        "panel": "#ffffff",
12:        "accent": "#1f6ed4",
13:        "hover": "#1f6ed4",
14:        "click": "#ef4444",
15:        "clickText": "#ffffff",
16:        "line": "#b8def3",
17:        "text": "#172636",
18:        "interaction": "Navy-blue hover with a red active click flash."
19:      }
20:    },
21:    {
22:      "id": "appearance-white-blue-click",
23:      "label": "Clean white, blue hover, white click",
24:      "description": "Minimal white project panels with professional blue hover behavior and quiet white
click feedback.",
25:      "visual": {
26:        "style": "device",
27:        "palette": ["#ffffff", "#2563eb", "#f8fbff"],
28:        "background": "#ffffff",
29:        "panel": "#ffffff",
30:        "accent": "#2563eb",
31:        "hover": "#2563eb",
32:        "click": "#ffffff",
33:        "clickText": "#172636",
34:        "line": "#d8e2ef",
35:        "text": "#172636",
36:        "interaction": "Blue hover with a clean white click state."
37:      }
38:    },
39:  ]

```



```

40:     "id": "appearance-deep-navy-cyan-click",
41:     "label": "Deep navy, cyan hover, white click",
42:     "description": "Dark navy technical background with cyan hover states and high-contrast white click
feedback.",
43:     "visual": {
44:       "style": "chip",
45:       "palette": ["#10263c", "#0f6f96", "#ffffff"],
46:       "background": "#10263c",
47:       "panel": "#f8fbff",
48:       "accent": "#0f6f96",
49:       "hover": "#0f6f96",
50:       "click": "#ffffff",
51:       "clickText": "#142536",
52:       "line": "#8ccfe8",
53:       "text": "#f8fbff",
54:       "interaction": "Cyan hover with a white active click state."
55:     }
56:   },
57:   {
58:     "id": "appearance-warm-red-pale-click",
59:     "label": "Warm red, wine hover, pale click",
60:     "description": "Warm signal-lab styling with wine-red hover behavior and a soft pale-red click
response.",
61:     "visual": {
62:       "style": "waveform",
63:       "palette": ["#fff7f5", "#c2413b", "#fff4f2"],
64:       "background": "#fff7f5",
65:       "panel": "#ffffff",
66:       "accent": "#b9332e",
67:       "hover": "#c2413b",
68:       "click": "#fff4f2",
69:       "clickText": "#2f2020",
70:       "line": "#efb0aa",
71:       "text": "#2f2020",
72:       "interaction": "Wine-red hover with a pale red click state."

```

Safe change checklist

- Confirm which app, website, installer, workflow, or document consumes this file.
- Change it on a feature branch from development.
- Regenerate docs if the file purpose, API surface, or behavior changes.
- Run the smallest relevant smoke test before merging into development.

Important Code Reference: projects.json

Item	Explanation
File	projects.json
Category	Configuration or structured data
Folder role	Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.
Size	1,490 bytes
Extension	.json
MIME type	application/json
Text lines	46

Why this file exists

- Public project catalog consumed by the website after parsing builder data.

How it is used

Repository root. Usually an entry file, app config, public website file, package manifest, or top-level documentation.

How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

JSON/YAML-style keys detected

Item	Explanation
categories	Line 2: "categories": [
id	Line 4: "id": "analog-mixed-signal",
label	Line 5: "label": "Analog and Mixed Signal",
description	Line 6: "description": "Circuit-level work including amplifiers, filters, converters, measurement front ends, power stages, and mixed-signal interfaces.",
accent	Line 7: "accent": "#117c7a"
id	Line 10: "id": "digital",
label	Line 11: "label": "Digital",
description	Line 12: "description": "Digital design projects involving logic design, FPGA workflows, verification, timing, and ASIC-oriented architecture.",
accent	Line 13: "accent": "#52606d"
id	Line 16: "id": "embedded",
label	Line 17: "label": "Embedded",
description	Line 18: "description": "MCU-based systems with firmware, board bring-up, interfaces, sensing, control, and hardware/software integration.",
accent	Line 19: "accent": "#d18b21"
funFacts	Line 22: "funFacts": [],
funFactsRich	Line 23: "funFactsRich": null,
profile	Line 24: "profile": {
brandImage	Line 25: "brandImage": "",
brandText	Line 26: "brandText": "Portfolio",

Item	Explanation
contactIntro	Line 27: "contactIntro": "",
displayName	Line 28: "displayName": "",
email	Line 29: "email": "",
githubUrl	Line 30: "githubUrl": "",
heroImage	Line 31: "heroImage": "",
linkedinUrl	Line 32: "linkedinUrl": "",
phone	Line 33: "phone": "",
portfolioLabel	Line 34: "portfolioLabel": "Portfolio",
profileImage	Line 35: "profileImage": "",
resumeUrl	Line 36: "resumeUrl": "",
websiteUrl	Line 37: "websiteUrl": ""
siteContent	Line 39: "siteContent": {
heroCopy	Line 40: "heroCopy": "Add your projects, documents, diagrams, source code, images, profile details, and links.\nSave drafts locally, preview the site, then publish when your target reposito
heroEyebrow	Line 41: "heroEyebrow": "Engineering portfolio",
heroTitle	Line 42: "heroTitle": "Build a portfolio that presents your work clearly."
projects	Line 44: "projects": [],
siteSections	Line 45: "siteSections": []

Representative opening snippet

```

1: {
2:   "categories": [
3:     {
4:       "id": "analog-mixed-signal",
5:       "label": "Analog and Mixed Signal",
6:       "description": "Circuit-level work including amplifiers, filters, converters, measurement front
ends, power stages, and mixed-signal interfaces.",
7:       "accent": "#117c7a"
8:     },
9:     {
10:      "id": "digital",
11:      "label": "Digital",
12:      "description": "Digital design projects involving logic design, FPGA workflows, verification,
timing, and ASIC-oriented architecture.",
13:      "accent": "#52606d"
14:    },
15:    {
16:      "id": "embedded",
17:      "label": "Embedded",
18:      "description": "MCU-based systems with firmware, board bring-up, interfaces, sensing, control, and
hardware/software integration.",
19:      "accent": "#d18b21"
20:    }
21:  ],
22:  "funFacts": [],
23:  "funFactsRich": null,
24:  "profile": {
25:    "brandImage": "",
26:    "brandText": "Portfolio",
27:    "contactIntro": "",
28:    "displayName": "",
29:    "email": "",
30:    "githubUrl": "",
31:    "heroImage": "",
32:    "linkedinUrl": "",
33:    "phone": "",
34:    "portfolioLabel": "Portfolio",
35:    "profileImage": "",
36:    "resumeUrl": "",
37:    "websiteUrl": ""
38:  },
39:  "siteContent": {
40:    "heroCopy": "Add your projects, documents, diagrams, source code, images, profile details, and
links.\nSave drafts locally, preview the site, then publish when your target repository is ready.",
41:    "heroEyebrow": "Engineering portfolio",
42:    "heroTitle": "Build a portfolio that presents your work clearly."
43:  },

```

```
44:   "projects": [],  
45:   "siteSections": []  
46: }
```

Safe change checklist

- Confirm which app, website, installer, workflow, or document consumes this file.
- Change it on a feature branch from development.
- Regenerate docs if the file purpose, API surface, or behavior changes.
- Run the smallest relevant smoke test before merging into development.

Important Code Reference:

[.github/workflows/build-windows-builder.yml](#)

Item	Explanation
File	.github/workflows/build-windows-builder.yml
Category	GitHub workflow
Folder role	GitHub automation. GitHub Actions reads this during CI/release/branch-gate runs.
Size	3,867 bytes
Extension	.yml
MIME type	unknown
Text lines	107

Why this file exists

- GitHub Actions workflow that builds the Windows installer and portable application, checks the stable download link, enforces release version bumps, uploads artifacts, and publishes GitHub Releases.
- Workflow changes should be tested on a feature branch because mistakes can block merges or releases.

How it is used

GitHub automation. GitHub Actions reads this during CI/release/branch-gate runs.

How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

Important implementation signals

Item	Explanation
Security or auth at line 11	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: permissions:
Installer at line 16	Windows setup, update, shortcut, or installation behavior. Evidence: name: Build Windows installer and portable app
Compiler or simulator at line 23	Part of the Compile Code workspace or language/tool detection. Evidence: - name: Set up Node.js
Compiler or simulator at line 24	Part of the Compile Code workspace or language/tool detection. Evidence: uses: actions/setup-node@v4
Compiler or simulator at line 26	Part of the Compile Code workspace or language/tool detection. Evidence: node-version: "24"
Installer at line 44	Windows setup, update, shortcut, or installation behavior. Evidence: throw "Windows setup installer was not found."
File read at line 57	Reads local builder state, project files, website assets, or configuration. Evidence: \$package = Get-Content package.json -Raw ConvertFrom-Json
Network request at line 73	Frontend-backend communication or public web/API calls. Evidence: \$latestInstallerUrl = "https://github.com/otienoma/urice/omb-portfolio-builder/releases/latest/download/OMB-Portfolio-Builder-Setup-latest-x64.exe"
File read at line 74	Reads local builder state, project files, website assets, or configuration. Evidence: \$index = Get-Content index.html -Raw
Installer at line 75	Windows setup, update, shortcut, or installation behavior. Evidence: if (\$index -notmatch [regex]::Escape(\$latestInstallerUrl)) {
Installer at line 76	Windows setup, update, shortcut, or installation behavior. Evidence: throw "The portfolio download link must use the stable latest installer URL: \$latestInstallerUrl"
Git command at line 83	Repository state, publishing, branch checks, or release automation. Evidence: git fetch --tags --force
Git command at line 85	Repository state, publishing, branch checks, or release automation. Evidence: \$tagExists = [bool](git tag --list \$tag)

Representative opening snippet

```
1: name: Build Windows Portfolio Builder
2:
3: on:
4:   workflow_dispatch:
5:     push:
6:       branches:
7:         - main
8:       tags:
9:         - "builder-v*"
10:
11: permissions:
12:   contents: write
13:
14: jobs:
15:   build-windows:
16:     name: Build Windows installer and portable app
17:     runs-on: windows-latest
18:
19:     steps:
20:       - name: Check out repository
21:         uses: actions/checkout@v4
22:
23:       - name: Set up Node.js
24:         uses: actions/setup-node@v4
25:         with:
26:           node-version: "24"
27:
28:       - name: Set up pnpm
29:         uses: pnpm/action-setup@v4
30:         with:
31:           version: "11.5.3"
32:
33:       - name: Install desktop builder dependencies
34:         run: pnpm install --frozen-lockfile
35:
36:       - name: Build Windows artifacts
37:         run: pnpm run dist
38:
39:       - name: Create stable latest download names
40:         shell: pwsh
41:         run: |
42:           $setup = Get-ChildItem dist -Filter "OMB-Portfolio-Builder-Setup-*-x64.exe" | Sort-Object
LastWriteTime -Descending | Select-Object -First 1
43:           if (-not $setup) {
44:             throw "Windows setup installer was not found."
45:           }
46:           Copy-Item $setup.FullName "dist/OMB-Portfolio-Builder-Setup-latest-x64.exe" -Force
47:
48:           $portable = Get-ChildItem dist -Filter "OMB-Portfolio-Builder-Portable-*-x64.exe" | Sort-Object
LastWriteTime -Descending | Select-Object -First 1
49:           if ($portable) {
50:             Copy-Item $portable.FullName "dist/OMB-Portfolio-Builder-Portable-latest-x64.exe" -Force
51:           }
52:
53:       - name: Resolve release version
54:         id: release_meta
55:         shell: pwsh
56:         run: |
57:           $package = Get-Content package.json -Raw | ConvertFrom-Json
58:           $version = [string]$package.version
59:           if (-not $version) {
60:             throw "package.json version is required before publishing a release."
61:           }
62:           if ($env:GITHUB_REF -like "refs/tags/*") {
63:             $tag = $env:GITHUB_REF_NAME
64:           } else {
65:             $tag = "builder-v$version"
66:           }
67:           "version=$version" >> $env:GITHUB_OUTPUT
68:           "release_tag=$tag" >> $env:GITHUB_OUTPUT
69:
70:       - name: Verify portfolio download link follows latest release
71:         shell: pwsh
72:         run: |
```

Safe change checklist

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- Change it on a feature branch from development.
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- Run the smallest relevant smoke test before merging into development.

Important Code Reference:

.github/workflows/main-branch-gate.yml

Item	Explanation
File	.github/workflows/main-branch-gate.yml
Category	GitHub workflow
Folder role	GitHub automation. GitHub Actions reads this during CI/release/branch-gate runs.
Size	1,259 bytes
Extension	.yml
MIME type	unknown
Text lines	40

Why this file exists

- GitHub Actions guard that rejects direct pushes to main and rejects pull requests into main unless the source branch is development.
- Workflow changes should be tested on a feature branch because mistakes can block merges or releases.

How it is used

GitHub automation. GitHub Actions reads this during CI/release/branch-gate runs.

How this file participates in the system

This generated section reads the actual file and points out line numbers, declarations, endpoints, events, source excerpts, and debugging cues.

Important implementation signals

Item	Explanation
Security or auth at line 11	Publishing protection, API key safety, HTTP security, or user authorization. Evidence: permissions:

Representative opening snippet

```
1: name: Main Branch Gate
2:
3: on:
4:   pull_request:
5:     branches:
6:       - main
7:   push:
8:     branches:
9:       - main
10:
11: permissions:
12:   contents: read
13:
14: jobs:
15:   require-development-source:
16:     name: Require development as the main source branch
17:     runs-on: ubuntu-latest
18:
19:     steps:
20:       - name: Verify main only receives development
21:         shell: bash
22:         env:
23:           HEAD_COMMIT_MESSAGE: ${ github.event.head_commit.message }
24:         run: |
25:           if [ "${ github.event_name }" = "push" ]; then
26:             if printf '%s' "$HEAD_COMMIT_MESSAGE" | grep -Eq '^Merge pull request #[0-9]+ from
[^/]+/development$'; then
27:               echo "main was updated by a development release PR."
28:               exit 0
29:             fi
30:
31:             echo "::error::Direct pushes to main are not allowed. Open a pull request from development"
```

```
into main."
32:         exit 1
33:     fi
34:
35:     if [ "${{ github.head_ref }}" != "development" ]; then
36:         echo "::error::Pull requests into main must come from development. Merge this branch into
development first, test development, then open the development -> main release pull request."
37:         exit 1
38:     fi
39:
40:     echo "main release source is development."
```

Safe change checklist

- Confirm which app, website, installer, workflow, or document consumes this file.
- Change it on a feature branch from development.
- Regenerate docs if the file purpose, API surface, or behavior changes.
- Run the smallest relevant smoke test before merging into development.